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The Syntax of Natural Language

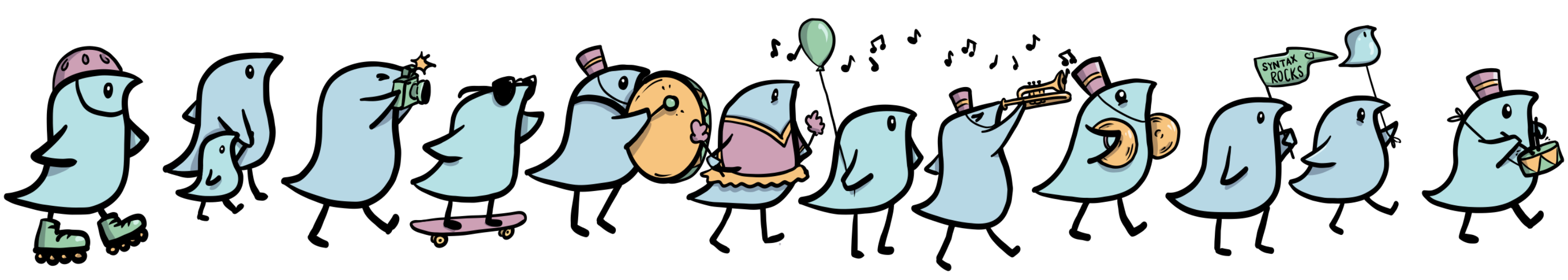
Michael Diercks
Pomona College

Beatrice Santorini
University of Pennsylvania

Anthony Kroch
University of Pennsylvania

2nd Edition

The Syntax of Natural Language



Michael Diercks, Beatrice Santorini, and Anthony Kroch

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Cover art (the wug parade) by Laura McLaren and Hannah Lippard. The wug character is a product of the Jean Berko-Gleason's (1958) language acquisition experiment.

This textbook was typeset using LaTeX.

This book is dedicated to our past, current, and future students.

Preface

July, 2024

It may be of interest to record some of how the textbook in its current form came to be. In what follows, Beatrice and Michael reminisce in turn.

Beatrice Santorini

In the early 2000s, I was assigned to teach the undergraduate introduction to syntax in the Linguistics Department at the University of Pennsylvania. My appointment with the Department provided half my salary, with the other half coming from research grants to Tony Kroch. Tony was concerned that the teaching would encroach on his research time, and the syntax course turned out to be welcome in this regard, since it was offered every year, so that any work on course preparation in one year would yield dividends in the next, and indeed for the foreseeable future. I already had notes that were online, so Tony proposed developing an online textbook that would eventually reduce my class preparation time to zero. That sounded fine to me, and so Tony and I met to discuss the content, and I made notes of those discussions.

The course at Penn was not intended to be an introduction to any particular type of syntactic theory such as Minimalism or a history of recent syntactic theory. Rather, the aim of the course—and thus of the developing textbook—was the usual one of presenting some basic results of the field, along with related argumentation. We felt free to omit commonly used arguments that we didn't regard as compelling. There were no prerequisites to the course, so we resolved trade-offs between descriptive and conceptual/explanatory simplicity in favor of the former. This was notably the case with our choice of X-bar schema, where we adopted a "classic" three-level schema relying on fairly traditional syntactic categories versus a two-level schema requiring more abstract categories. In such cases, we would briefly lay out the trade-off for the benefit of both students and instructors.

Our discussion of syntactic structure was framed in terms of so-called elementary trees and three operations for combining them: substitution, adjunction, and movement. This approach was informed by a conceptual and a pedagogical consideration. First, in the 1980s, Tony had worked with Aravind Joshi and Joshi's students on applying Joshi's Tree-Adjoining Grammar (TAG) formalism to natural language, and the derivations presented in the textbook were inspired by the formalism, though not consistent with it in every detail. Second, in connection with teaching the general undergraduate introduction to linguistics, Tony had found that students would build quite crazy structures that betrayed a continuing ignorance of basic syntactic principles. In an effort to improve their understanding, he collaborated with a graduate student by the name of Sean Christ to produce a program called the Trees program, which forced users to choose from among legal structures (say, maximal projections of various syntactic categories) and to combine them in specified constrained ways (say, using only the three operations above). Students were still able to build incorrect trees if they set their minds to it (for instance, consistently right-branching structures), but the worst excesses were checked. Since the Trees program had already proved useful, we incorporated it into the textbook by writing exercises using it. The program was actually more sophisticated than just described, and Tony wrote exercises for his graduate syntax class where the program simulated (in a very simple way) native-speaker judgments with respect to various phenomena (say, anaphor binding). The programming interface was not very developer-friendly, however, and the exercises in the textbook never exploited the program's full capabilities. At one point, Tony also produced animations of derivations using structures built with Trees. Eventually, with the transition to 64-bit technology, Trees became obsolete. Perhaps some day it will be revived in a more developer-friendly and less infrastructure-dependent incarnation.

When formulating exercises, we attempted to counteract the tendency for students to build consistently right-branching structures by focusing on structural ambiguity, especially in the early chapters. In an effort to help students to retain material from early chapters, we created exercises that built on each other across chapters. Finally, we wrote exercises illustrating variation with respect to syntactic properties like headedness and verb movement based on historical stages (and to a lesser extent, non-mainstream varieties) of English. The idea was to bring the impact of various unfamiliar language properties home by using bona fide English varieties. Using examples from other varieties of English also made it natural to enrich the discussion of syntactic variation and change with reference to language contact, social and stylistic factors, and so on.

Turning from issues to content to issues of distribution, it may be that Tony was thinking of eventually bringing the book along with the Trees program to market with a conventional publisher to offset the uncertainty associated with soft money. There were discussions at one point with an acquisitions editor from Harcout Brace. A few years later, Harcourt was acquired by Houghton Mifflin, who did not pursue the project.

In any event, it is clear that Tony preferred a much less corporate route. He was always looking for colleagues to contribute chapters on their fields of expertise, such as scrambling. Sometimes, people both in the United States and abroad would ask us for permission to use the book in their classes, and we were always happy to hear that it was useful to a broader community. We were especially thrilled and honored when Caroline Heycock at the University of Edinburgh used half of the chapters for a very professionally produced YouTube series on syntax.

Tony died in the spring of 2021, and in the fall of 2022, I began to worry a bit about what to do with the material for the book, apart from leaving it up on my Penn website to decay more or less gracefully. Less than a month later, entirely out of the blue, Michael Diercks contacted me, inquiring whether I might be interested in collaborating on updating the book and turning it into a professionally (but not corporately!) produced resource. Beyond the timing, there were synchronicities involved that might have struck even Carl Jung as noteworthy. As an Africanist, Michael was particularly interested in developing resources for his colleagues in Africa and for their students, and this fit nicely with the beginnings of Tony's academic career in Sénégal, studying the myths of the Bassari. Tony's widow, Martha, was as thrilled as I was by this African connection. She was even more thrilled by Michael's concern for the equitable distribution of intellectual resources, because anti-racism was the keystone of both Tony's and Martha's politics. Finally, it turned out that our collaboration was keeping the book in the family, so to speak, since Michael is an academic grandson of Tony's via Raffaella Zanuttini. I find it almost impossible to imagine a more suitable new home for the textbook than with Michael, his students, and his librarian colleagues at Pomona, and I thank them all for the tremendous amount of work they have put in to updating the book and the exercises. I am sure that Tony would agree wholeheartedly.

Michael Diercks

I have been troubled for years by for-profit publishing in academia: in linguistics, almost none of this profit trickles down to individual authors, so any idea of merit-driven compensation is either highly diminished or absent. And the effect (intended or not) is that our field ends up gatekeeping knowledge. Beyond this being a problem generally, it is an extraordinary problem in a field where many new insights come from research that specifically investigates the languages of under-resourced and marginalized communities around the world. The end result is that the very communities whose languages are under discussion often can't access those discussions. We are in an era of instantaneous distribution of information, but the most reliable and vetted information (peer-reviewed scientific research) is often kept behind paywalls.

My own students at Pomona College are not among those who lack access to academic resources while they are students with me, but the concerns still apply: I find it difficult to ask them to pay for textbooks that, in my opinion, should be freely available. But when I work with students and colleagues in Kenya and Africa more broadly, the paywalling of academic resources becomes a much more acute problem. In exploring how to make resources available to students within copyright laws, a college librarian gently asked me, "can you write your own textbook?" I laughed: my current life would not allow such a thing. But a week later it snuck into my mind that there already was a well-developed textbook out of UPenn; crucially, it was already Open Access, so with the bold confidence of a mediocre white man I reached out to Beatrice to inquire about what the future plans for the textbook were. After Beatrice considered the inquiry, I was ecstatic to learn that we shared a vision and that this was a beneficial situation all around: I could facilitate bringing the textbook into a new era, perpetuating its usefulness, and this would also meet a current need in the field for Open Access syntax textbooks.

Together with a team of students here at Pomona College, I undertook converting the HTML textbook to a more traditional textbook-style typesetting in 2023. At the same time Beatrice gave a first pass revision, resulting in a textbook that was used at Pomona College and the University of Pennsylvania in Fall 2023. Based on these experiences, I undertook a more significant revision process in 2024. Beatrice and Tony's main expertise is in Germanic languages; I work mainly on African languages, and my research assistants (especially Xiaoxing Yu) brought significant experience with prominent East Asian languages. A goal of the revision was to include more breadth of coverage in terms of the geography and typology of the languages that are addressed in the main text and in exercises and problem sets. The hope is that the result is a resource that can be used on its own for an entire curriculum, but which can also be used to supplement curricula run with other textbooks, or without textbooks.

We made several other significant adjustments in the revision process. I added a new chapter on locality of selection in syntax, and material was moved there from various other chapters in the text. These changes required adjustment of details in the sequence of argumentation at various point (including delaying introduction of VPISH until that chapter). Discussion of relative clauses was moved from the *wh*-movement chapter until later; this allows the *wh*-movement chapter to focus on clause-level effects, additional kinds of A′-movement, and to spend appropriate time on the theoretical shifts in that chapter (specifically, X° and XP movement). At various points (e.g. regarding X′-syntax, island effects, agreement/case, and others) the revised text takes an intermediate stance between traditional Principles and Parameters / GB approach and a more modern Minimalist approach. Some of the first edition choices (e.g. *elementary trees* and *substitution nodes*) are retained despite not being canonical theory, largely because they are quite useful pedagogically (as we explain at various points in the text). Part of the move towards more cross-linguistic coverage meant reducing material from the previous edition; in some instances this material was converted to problem sets, and in other instances we have retained the previous material to be included as supplementary readings in the Teacher's Manual, which is still in development as of this writing.

As Beatrice noted above, we share an academic history; she was trained by Tony, and I inherited similar mindsets through Raffaella. We are deeply theory-informed within the generative framework, and this textbook thoroughly reflects that. But we also recognize that current models will shift and change. Our opinion is that introductory students ought to have their attention directed to the most stable, consistent findings of our field. So the main goal of this second edition, much like the first, is not to introduce current theoretical frameworks. Rather, our aim is to provide an introduction to human language syntax, from a perspective that emphasizes mental grammars (as opposed to solely surface empirical structures). This obviously requires engaging a theoretical model, and the choice of our theoretical model is driven by our own expertise and research (broadly operating in the tradition that Culicover and Jackendoff 2005 refer to as Mainstream Generative Grammar), but the theoretical model is not the main purpose of the textbook. The result is a text that focuses on the most stable aspects of syntactic theory while de-emphasizing new theory or theory that is in active transition. So while there are certainly adjustments from the first edition, there are also various ways that the textbook also does not reflect active theorizing in the field as of 2024. There are strengths and weaknesses to such an approach, but the hope is that it makes a useful textbook for undergrads or graduate students taking a single course about syntax, but that a student having traversed this curriculum can make a fairly straightforward transition to contemporary research practices if they move on in the field.

For the foreseeable future, I am taking over stewardship of the textbook. The goal is to maintain and continue to develop an Open Access resource for education in natural language syntax, rooted in the generative tradition. Future planned developments include additional exercises/problem sets with appropriate cross-linguistic representation and a teacher's manual with an answer key, additional data handouts, and other supplementary materials for instructors. My revisions were written based on my own pedagogical materials built over 15 years at Pomona

College, as well as based on my current expertise and experience. My hope would be that future editions of the textbook (perhaps with additional co-authors?) would be able to continue the trend of decentering English and Western European languages, both in exercises and in main text. In the meantime, it's my hope that this is of use to my own students and to whoever else happens to find it along the way.

Acknowledgements

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Of course, we alone are responsible for the use we have made of their help.

Thanks also to the students at the University of Pennsylvania and Pomona College, for whom this book was developed.

Xiaoxing Yu and Charis Kim were largely responsible for converting this textbook from its original HTML format to the current typeset form (with regular participation from Diercks). They proofread carefully and made editorial and expository contributions. They also contributed to some of the problem sets and answers in the Instructor's Manual. Xuehuai He made major contributions to updating the Instructor's Manual.

Wug drawings (including cover art) are by Laura McLaren and Hannah Lippard. The wug has become somewhat of an unofficial mascot of linguistics, and is a product of the Jean Berko-Gleason's (1958) language acquisition experiment.

In Fall 2023 the syntax classes at University of Pennsylvania and Pomona College read the original reformatted and revised version of the text. Their questions and comments were very helpful to preparing the published second edition.

Publication Notes

Notes on versioning

This textbook is published Open Access via the Claremont Colleges library and should be accessible through normal networks of Open Access textbooks as well as at https://scholarship.claremont.edu/pomona_facbooks/50/. It is published as the 2nd edition of the textbook, but is specifically listed as a version that is an iteration of the second edition (2.xx). An upside of the only-digital publication is that we don't have to wait for larger publication runs in order to correct issues around typos and other kinds of copy-editing issues. We expect to be making minor revisions to the textbook with some frequency; these will be limited to copy-editing corrections for both text and data/trees/images, revisions for clarity (especially in problem sets and exercises, as we receive feedback from students from different places), and potentially adding exercises and problem sets.

We will borrow from the standard conventions of software versioning for any revisions; we will iterate the decimal, so subsequent revisions will be published as 2.02, 2.03, etc. All such changes will be annotated here with dates of the revision and with explanations of any changes. The expectation is that a syntax course should be able to run with participants using different versions, with the only changes of substance most likely being small changes to pagination or section numbering. Any such changes will be annotated with careful notes in the publication history below. But new versions will always be Open Access, so it is recommended that courses be run using the most recent version of the text.

Assuming an eventual third edition, that will be published as version 3.01, and versions will iterate from there.

Publication history

- First Edition, published online at https://www.ling.upenn.edu/~beatrice/syntax-textbook/.
- Second Edition, published online at https://scholarship.claremont.edu/pomona_facbooks/50/.
  - Version 2.01, distributed informally in June 2024
  - Version 2.02, distributed informally in July 2024
  - Version 2.03, published via Scholarship @ Claremont by the Claremont Colleges Library in January 2025

Contents

1

Foundational issues

Chapter outline

Prescriptive versus descriptive grammar 1.1

This book is an introduction to generative grammar from a Chomskyan perspective. By the time you finish this chapter, you will have a clearer understanding of what we mean by this sentence, and by the time you finish the entire book, your understanding of it will be clearer and deeper still. But for the moment, you have probably gained the impression that this book is about grammar of some sort. And right there, we have a problem. The problem is that there is an everyday sense of the term "grammar" and a quite different sense in which the term is used in linguistics.

In the everyday sense, "grammar" refers to a collection of rules concerning what counts as socially acceptable and unacceptable language use. Some of these rules, like the ones in (1), make reference to particular words and apply to both spoken and written language.

(1) a. Don't use *ain't*.
 b. Don't use *seen* as the past tense of *see* (as in *I seen him at the party last night*).

But mainly, the rules in question concern the proper composition of sentences in written language. You may recall being taught rules at school like those in (2).

(2) Selected prescriptive grammatical rules for English
 a. Don't end a sentence with a **linking verb**.
 b. Don't end a sentence with a preposition.
 c. Don't split infinitives.
 d. Don't start a sentence with a conjunction.
 e. Don't use a plural pronoun to refer back to a singular noun like *everyone, no-one, someone*, and the like.
 f. Don't use an object pronoun for a subject pronoun in a conjoined subject.
 g. Don't use contractions.
 h. Don't use dangling participles.
 i. Don't use sentence fragments.
 j. Use *whom*, not *who*, as the object of a verb or preposition.

Someone who composes sentences in accordance with rules like those in (2) is said to have "good grammar", whereas someone said to have "bad grammar" doesn't apply the rules when they ought to be applied[1] and so produces sentences like (3).

(3) a. Over there is the guy *who* I went to the party *with*. violates (2b), (2j)
 b. Bill and *me* went to the store. violates (2f)

From the amount of attention that people devote to rules like those in (1) and (2), it is easy to get the impression that they are the only linguistic rules there are. But it is also easy to see that that can't be so. The reason is that even people who don't follow the rules in (1) and (2) don't produce rampantly variable, confusing word salad. For instance, even people who invariably produce sentences like (3) do not produce the likes of (4).

(4) a. Over there is guy the who I went to party the with.
b. Over there is the who I went to the party with guy.
c. Bill and me the store to went.

The sentences in (3) may be instances of "bad grammar" in the sense of the term "grammar" that is often used in English classes in elementary and secondary school. But they are nonetheless still English sentences that people will readily say in conversation and will accept as possible sentences in their language. By contrast, we don't need to rely on school rules to tell us that the examples in (4) are not English sentences—even though they contain exactly the same English words as the sentences in (3).

Since native speakers of English do not produce a variable mishmash of words of the sort in (4), there must be another type of rules according to which sentences are composed. We can determine what some of them are by taking a closer look at the sequences in (4). Why exactly is it that they are so clearly not an English sentence? In (4a), the article *the* is in the wrong order with respect to the nouns that it belongs with, *guy* and *party.* In (4b), the relative clause (*who I went to the party with*) is in the wrong order with respect to the noun that it modifies (*guy*). In (4c), the preposition *to* is in the wrong order with respect to its object (*the store*). In other words, the sentences in (4) do not follow the rules in (5).

(5) a. Articles precede nouns.
b. Relative clauses follow the noun that they modify.
c. Prepositions precede their objects.

Rules like those in (5) have a different intention than those in (2). The rules in (2) are **prescriptive;** those in (5) are **descriptive.** Rules of prescriptive grammar have the same status as rules of etiquette (like table manners or dress codes) or the laws of society, which divide the spectrum of possible human behavior into socially acceptable or legal behavior, on the one hand, and socially unacceptable or illegal behavior, on the other. Rules of prescriptive grammar make statements about how people ought to use language, and often they are framed as negative injunctions ("Thou shalt not split infinitives", on a par with "Thou shalt not steal"). In contrast, rules of descriptive grammar have the status of scientific observations, and they are intended as insightful generalizations about the way that speakers use language in fact, rather than about the way that someone says that they ought to use it. Descriptive rules are more general and more fundamental than prescriptive rules in the sense that all sentences of a language are formed in accordance with them. Beyond this, most prescriptive rules have a classist and often racist foundation in the U.S., in that only certain dominant groups' language counts as "standard" or "acceptable", whereas the language of socially marginalized communities is marginalized as "bad" grammar. More on that later.

A useful way to think about the descriptive rules of a language (to which we return in more detail below) is that they produce, or **generate**, all the sentences of a language. The prescriptive rules can then be thought of social control: attempts to filter out as socially unacceptable some (relatively minute) portion of the entire output of the descriptive rules. In syntax, as in modern linguistics more generally, we adopt a resolutely descriptive perspective concerning language. In particular, when linguists say that a sentence is **grammatical**, we don't mean that it is correct from a prescriptive point of view, but rather that it conforms to descriptive rules like those in (5). In order to indicate that a sequence of words or morphemes is ungrammatical in this descriptive sense, we prefix it with an asterisk. Grammatical sentences are usually not specially marked, but sometimes we prefix them with a checkmark (✓) for clarity. These conventions are illustrated in (6) and (7).

(6) a. * Over there is guy the who I went to party the with. (= (4a))
b. * Over there is the who I went to the party with guy. (= (4b))

(7) a. ✓ Over there is the guy who I went to the party with. (= (3a))
b. ✓ Over there is the guy with whom I went to the party.

That is to say, by saying that the sentences in (7) are grammatical, we are describing a psychological phenomenon that is automatic. Nobody who speaks Mainstream U.S. English as a first language needs to be taught that the sentences in (7) are grammatical but the sentences in (6) are not: everyone already knows this.

Prescriptive grammar is based on the idea that there is a single right way to speak. When there is more than one way of saying something, prescriptive grammar is generally concerned with declaring one (and only one) of the variants to be correct. The favored variant is usually justified as being better (whether more logical, more euphonious, or more desirable on some other grounds) than the marginalized variant that is being discriminated against. In the same situation of linguistic variability, descriptive grammar is content simply to document the variants—without passing judgment on them. For instance, consider the variable subject-verb agreement pattern in (8).

(8) a. **There's** *some boxes* left on the porch.
b. **There** *are some boxes* left on the porch.

In (8a), the singular verb *is* (contracted to *'s*) agrees in number with the preverbal expletive subject *there* (in **bold**), whereas in (8b), the plural verb *are* agrees with the postverbal logical subject *some boxes* (in *italics*). The color and font of the verb indicates which of the two subjects it agrees with. The prescriptive and descriptive rules concerning this pattern are given in (9). The differences between the two rules are emphasized by underlining.

(9) In a sentence containing both the singular expletive subject *there* and a plural logical subject …
a. Prescriptive rule: … the verb <u>should</u> agree in number with <u>the logical subject</u>.
b. Descriptive rule: … the verb <u>can</u> agree in number with <u>either the expletive subject or the logical subject</u>.

To take another example, let's consider the prescriptive rule that says, "Don't end a sentence with a preposition."[2] A prescriptivist might argue that keeping the preposition (in *italics*) together with its object (in **boldface**), as in (10a), makes sentences easier to understand than does separating the two, as in (10b).

(10) a. *With* **which friend** did you go to the party?
b. **Which friend** did you go to the party *with*?

But verbs have objects in the same way that prepositions have objects. By the reasoning expressed above, verbs and objects should be resistant to being separated as well. But in (11a), where the verb and its object are adjacent, is clearly preferable to (11b), where they are not. In fact, however, (11a) is completely ungrammatical in English.

(11) a. * Adopt which cat did your friend?
b. ✓ Which cat did your friend adopt?

Beyond that, the prescriptive rule against sentences ending with prepositions is quite clearly not a rule that is part of our mental grammars. This is illustrated by the (perhaps apocryphal) story about Winston Churchill's response to this rule against preposition stranding:

(12) This is the type of arrant pedantry *up with which I will not put.*

(12) is an absurdly weird sentence: clearly, it's not a rule of mental grammars of English that prepositions can't end sentences. The only way we can express that sentence is by stranding *both* prepositions: *a rule which I will not put up with.*

It is important to understand that there is no semantic or other conceptual reason that prepositions can be separated from their objects in English, but that verbs can't. From a descriptive perspective, the grammaticality contrast between (10a) and (11a) is simply a matter of fact about mental grammars. (13) highlights the difference between the relevant prescriptive and descriptive rule.

(13) When the object of a preposition appears in a position other than its ordinary one (as in a question), ...
a. Prescriptive rule: ... it <u>should</u> be preceded by the preposition.
b. Descriptive rule: ... it <u>can either</u> be preceded by the preposition, <u>or stand alone, with the preposition remaining in its ordinary position</u>.

The contrasting attitude of prescriptive and descriptive grammar towards linguistic variation has a quasi-paradoxical consequence: namely, that prescriptive rules are never descriptive rules. The reason for this has to do with the way that social systems (not just language) work. If everyone in a community consistently behaves in a way that is socially acceptable in some respect, then there is no need for explicit prescriptive rules to ensure the behavior in question. It is only when behavior that is perceived as socially unacceptable becomes common that prescriptive rules come to be formulated to keep the unacceptable behavior in check. For example, if every customer entering a store invariably wears both a shirt and shoes, there is no need for the store owner to put up a sign that says "No shirt, no shoes, no service." Conversely, it is precisely at illegal dump sites that we observe "No dumping" signs. In an analogous way, in the domain of language use, prescriptive rules are only ever necessary to attempt to get people to stop using language in a particular way. But if people are using language in a particular way, that means those properties of their languages are real and exist, and can be described with a descriptive grammar approach.

Notably, the rules of prescriptive grammar are only ever formulated in situations where linguistic variation is common. But being prescriptive, they do not treat all of the occurring variants as equally acceptable—with the result that they can't ever be descriptive. So prescriptive rules only ever exist to attempt to outlaw grammatical constructions that **are** in the mental grammars of many speakers of a language. We can't emphasize enough that the modern basis of a vast amount of prescriptive grammar teaching around English in the United States is largely classist and often racist: certain kinds of language have been deemed acceptable, and those deemed unacceptable are those of social groups that have been marginalized in society. This is not to impugn the motives of any particular English teacher: they are just doing their job as they have been trained to do it. But a system has arisen that centers a particular variety of English and marginalizes a host of other perfectly legitimate varieties of English. Lippi-Green (2012) is a good first resource on this topic, though students very interested in these social phenomena should take a sociolinguistics course, which will cover these issues in much more depth. In the realm of syntax specifically, the Yale Grammatical Diversity Project (https://ygdp.yale.edu) is an excellent resource, as it provides linguistic descriptions of grammatical constructions that are often the target of prescriptive rules.

Rule formation and syntactic structure in language acquisition 1.2

As we have just seen, prescriptive and descriptive rules of grammar differ in their goals. In addition, they differ in how they come to be part of a speaker's knowledge. Prescriptive rules are taught at school, and because they are taught, people tend to be conscious of them, even if they don't actually follow them. By contrast, we follow the rules of descriptive grammar consistently[3] and effortlessly, yet without learning them at school. In fact, children have essentially mastered most of these rules on their own by first grade. Ordinarily, we are completely unconscious of the descriptive rules of language. If we do become conscious of them, it tends to be in connection with learning a foreign language whose descriptive grammar differs from that of our native language. In order to emphasize the difference between the unconscious way that we learn a native language (or several) in early childhood and the conscious way that we learn a foreign language later on in life, the first process is often called **language acquisition** rather than language learning.

As you consider descriptive rules like those in (5), you might not find it all that surprising that a child raised in an English-speaking community would acquire, say, the rule that articles precede nouns. After all, you might say, all the child ever hears are articles and nouns in that order. (Hmmm, though, actually… see the note.)[4] So why would it ever occur to such a child to put the article and the noun in the other order? Isn't it just common sense that children learn their native language by imitating older speakers around them?

Well, yes and no. It is true that children learn some aspects of their native language by imitation and memorization. Children in English-speaking communities learn English words, children in Navajo-speaking communities learn Navajo words, children in Swahili-speaking communities learn Swahili words, and so on. But language acquisition isn't purely a process of memorization. In fact, given current human life spans, it couldn't possibly be!

1.2.1 A thought experiment

To see this, let's consider a toy version of English that contains three-word sentences consisting of a noun, a transitive verb, and another noun. The toy version contains sentences like (14) that are sensible given the real world as well as sentences like (15) that aren't, but that might be useful in fairy tale or science fiction contexts.

(14) a. Cats detest lemons.
b. Children eat tomatoes.
c. Cheetahs chase gazelles.

(15) a. Lemons detest cats.
("Secret life of citrus fruits")
b. Tomatoes eat children.
("Attack of the genetically modified tomatoes")
c. Gazelles chase cheetahs.
("Avenger gazelle")

Again for the sake of argument, let's assume a (small) vocabulary of 1,000 nouns and 100 verbs. This gives us a list of $1,000 \times 100 \times 1,000$ (= 100 million) three-word sentences of the type in (14) and (15). Numbers of this magnitude are difficult to put in human perspective, so let's estimate how long it would take a child to learn all the sentences on the list. Again, for the sake of argument, let's assume that children can memorize sentences quickly, at a rate of one sentence a second. The entire list of three-word sentences could then be memorized in 100 million seconds, which comes to 3.17 years. So far, so good. However, the minute we start adding complexity to Toy English, the number of sentences and the time it would take to memorize them quickly mushrooms. For instance, adding only 10 adjectives to the child's vocabulary would cause the number of five-word sentences of the form in (16) to grow to 10 billion (100 milion$\times 10 \times 10$).

(16) a. Black cats detest green peas.
b. Happy children eat ripe tomatoes.
c. Hungry cheetahs chase speedy gazelles.

Even at the quick rate of one sentence per second that we're assuming, the list of all such five-word sentences would take a bit over 317 years to learn. Clearly, this is an absurd consequence. For instance, how could our memorious child ever come to know, as every English speaker plainly does, that the sentence in (17) is ungrammatical? If grammatical knowledge were based purely on rote memorization, the only way to determine this would be to compare (17) to all of the 10 billion five-word sentences and to find that it matches none of them.

(17) * Cats black detest peas green.

And even after performing the comparison, our fictitious language learner still wouldn't have the faintest clue as to why (17) is ungrammatical!

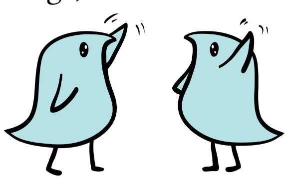
In addition to this thought experiment with its absurd consequences, there is another reason to think that language acquisition isn't entirely based on rote memorization—namely, that children use what they hear of language as raw material to construct linguistic rules. How do we know this? We know because children sometimes produce rule-based forms that they have never heard before.

1.2.2 Rule-based word formation

One of the earliest demonstrations that children acquire linguistic rules, rather than simply imitating the forms of adult language, is the wug experiment (Berko 1958). In it, the psycholinguist Jean Berko Gleason used invented words to examine (among other things) how children between the ages of 4 and 7 form plurals in English. She showed the children cards with simple line drawings of objects and animals and elicited plurals from them by reading them accompanying texts like (18).[5]

(18) This is a wug. Now there is another one. There are two of them. There are two ___ .

(19) Wugs, for reference:



More than 75% of the children pluralized the invented words *cra*, *lun*, *tor*, and *wug* in exactly the same way that adults did in a control group: they added the sound *-z* to the word (Berko 1958, 159–162).[6] Since none of the children had encountered the invented words before the experiment, their response clearly indicates that they had acquired a plural rule and were using it to produce the novel forms.

Children are also observed to produce novel rule-based forms instead of existing irregular adult forms (for instance, *comed* or *goed* instead of *came* or *went*). This process, which is known as **overregularization**, is further illustrated in (20) (Marcus et al. 1992, 148–149; based on Brown 1973).

(20) a. beated, blowed, catched, cutted, doed, drawed, drived, falled, feeled, growed, holded, maked, sleeped, standed, sticked, taked, teached, throwed, waked, winned
(Adam, between the ages of 2 and 5)
b. drinked, seed, weared (Eve, between the ages of 1½ and 2)

Overregularized forms don't amount to a large fraction of the forms that children produce overall (less than 5% in the case of past tense forms, according to Marcus et al. 1992, 35), but they are important because they clearly show that even the acquisition of words can't be completely reduced to rote memorization.

1.2.3 Question formation

In addition to morphological rules (which concern the structure of words), children also acquire syntactic rules (which concern the structure of sentences). Some of these rules are of particular interest because they differ from the corresponding adult rules that the children eventually acquire. At the same time, however, the children's novel rules don't differ from the rules of the adult grammar in completely arbitrary ways. Rather, the children's rules share certain abstract properties with the adult rules, even when they differ from them.

To see this, let's consider how young children form yes-no questions. Some 3- to 5-year-olds form such questions from declarative sentences by copying the auxiliary element to the beginning of the sentence, as in (21) (Crain and Nakayama 1987, 536). (We use the term "auxiliary element" as a convenient cover term for elements that invert with the subject in (adult) English questions, like forms of the verb *to be* or modals like *can*. See § 2.5.1 of Chapter 2 for more details.)

(21) a. The girl is tall. → Is the girl is tall?
b. The red pig can stand on the house. → Can the red pig can stand on the house?

In the course of language acquisition, the questions in (21) are eventually replaced by those in (22), where we can think of the auxiliary element as having been moved rather than copied.

(22) a. Is the girl ___ tall?
b. Can the red pig ___ stand on the house?

But now notice a striking indeterminacy, first pointed out by Chomsky (1971, 26–27). When children produce questions like those in (22), there is no way of telling whether they are using the adult rule for question formation in (23a) or the logically possible alternative rule in (23b).

(23) a. Adult question formation rule:
To form a question from a declarative sentence containing an auxiliary element, find the subject of the sentence, and invert the subject and the auxiliary.
b. Logically possible alternative:
To form a question from a declarative sentence containing an auxiliary element, find the first auxiliary element, and move it to the beginning of the sentence.

Both rules in (23) give the same result for simple sentences, which are likely to form most of the data that young children attend to. Both rules also require children to identify auxiliary elements. However, the adult rule additionally requires children to identify the subject of the sentence by grouping together sequences of words like *the girl* or *the red pig* into a single abstract structural unit. Because of this grouping requirement, the adult rule is called **structure-dependent**. By contrast, the alternative rule in (23b) is not structure-dependent, since it requires the child only to classify words according to their grammatical category (Is this word an auxiliary element?), but not to group the words into structural units. The rule in (23b) is simpler in the sense that it relies on fewer, as well as computationally less complex, cognitive operations, and children might reasonably be expected to experiment with it in the course of acquiring question formation. Nevertheless, Chomsky (1971) predicted that children would use only structure-dependent rules in the course of acquisition.

As we mentioned, both rules give the same result for simple sentences. So how could we possibly tell which of the two rules a child was actually using? Well, forming yes-no questions is not restricted to simple sentences. So although we can't tell which rule a child is using in the case of simple sentences like (21), the rules in (23) give different results for a complex sentence like (24), which contains a relative clause (*who was holding the plate*).

(24) The boy who was holding the plate is crying.

In particular, the sentence in (24) contains two auxiliary elements—one (*was*) for the relative clause, and another one (*is*) for the entire sentence (the so-called matrix sentence, which contains the relative clause). A child applying the structure-dependent question formation rule to (24) would first identify the subject of the matrix sentence (*the boy who was holding the plate*) and then invert the entire subject—including the relative clause and the auxiliary contained within it (*was*)—with the matrix auxiliary (*is*). On the other hand, a child applying the structure-independent rule would identify the first auxiliary (*was*) and move it to the beginning of the sentence. As shown in (25), the two rules have very different results:

(25) a. Structure-dependent rule:
[The boy who was running] <u>is</u> crying.
<u>Is</u> [the boy who was running] ___ crying?

b. Structure-independent rule:
The boy who <u>was</u> running is crying.
<u>Was</u> the boy who ___ running is crying?

Recall that Chomsky predicted that children would not use structure-independent rules, even though they are (in some sense) simpler than structure-dependent ones. This prediction was tested in an experiment with 3- to 5-year-old children by Crain and Nakayama (1987). In the experiment, the experimenter had the children pose yes-no questions to a doll (Jabba the Hut from *Star Wars*). For instance, the experimenter would say to each child *Ask Jabba if the boy who was holding the plate is crying.* This task elicited various responses. Some children produced the adult question in (25a), whereas others produced the copy question in (26a) or the restart question in (26b).

(26) a. Is [the boy who was holding the plate] is crying?
b. Is [the boy who was holding the plate], is he crying?

Although neither of the questions in (26) uses the adult rule in (23a), the rules that the children used to produce them are structure-dependent in the same way that the adult rule is. This is because children who produced (26a) or (26b) must have identified the subject of the sentence, just like the children who produced (25a). Out of the 155 questions that the children produced, none were of the structure-independent type in (25b). Moreover, no child produced the structure-independent counterpart of (26a), shown in (27), which results from copying (rather than moving) the first auxiliary element in the sentence.

(27) Was the boy who was holding the plate is crying?

In other words, regardless of whether a child succeeded in producing the adult question in (25a), every child in the experiment treated the sequence *the boy who was holding the plate* as a structural unit, thus confirming Chomsky's prediction.

1.3 More evidence for syntactic structure

We have seen that young children are capable of forming and applying both morphological and syntactic rules. Moreover, as we have seen in connection with question formation, children do not immediately acquire the rules of the adult grammar. Nevertheless, the syntactic rules that children are observed to use in the course of acquisition are a subset of the logically possible rules that they might postulate in principle. In particular, as we have just seen, children's syntactic rules are structure-dependent, even when they differ from the target adult rules. Another way of putting this is that the objects that syntactic rules operate on (declarative sentences in the case of the question formation rule) are not simply sequences of words (as if they were beads on a string), but rather groups of words that belong together, so-called **syntactic constituents.**

1.3.1 Intuitions about words belonging together

Evidence for **syntactic structure** isn't restricted to data from child language acquisition. Further evidence comes from the intuitions that speakers, whether adults or (older) children, have that certain words in a sentence belong together, whereas others do not. For instance, in a sentence like (28), we have the strong intuition that the first *the* belongs with *dog*, but not with *did*, even though *the* is adjacent to both.

(28) Did the dog chase the cat?

Similarly, the second *the* in (28) belongs with *cat* and not with *chase.* But a word doesn't always belong with the following word. For instance, in (29), *dog* belongs with the preceding *the*, not with the following *the*.

(29) Did the dog the children like chase the cat?

Words that belong together can sometimes be replaced by placeholder elements such as pronouns, as illustrated in (30).

(30) a. Did the dog chase the cat? → Did she chase him?
b. Did the dog the children like chase the cat? → Did the dog they like chase him?

Extra Info

Strictly speaking, the term 'pronoun' is misleading since it suggests that pronouns substitute for nouns regardless of syntactic context. In fact, pronouns substitute for noun phrases (as will be discussed in more detail in Chapter 3). A less confusing term would be 'pro-noun phrase', but we'll continue to use the traditional term.

It's important to recognize that pronouns don't simply replace strings of words regardless of context. Just because a string like *the dog* is a constituent in (30a) doesn't mean that it's always a constituent. We can see this by replacing *the dog* by a pronoun in (30b), which leads to the ungrammatical result in (31).

(31) Did the dog the children like chase the cat?
↓
* Did she the children like chase the cat?

The ungrammaticality in (31) is evidence that *the* and *dog* belong together less closely in (30b) than in (30a). In particular, in (30b), *dog* combines with the relative clause, and *the* combines with the result of this combination, not with *dog* directly, as it does in (30a). (We will consider the internal structure of noun phrases more closely in Chapter 6.)

In some sentences, we have the intuition that words belong together even when they are not adjacent. For instance, *see* and *who* in (32a) belong together in much the same way as *see* and *Bill* do in (32b). That is, *who* in (32a) and *Bill* in (32b) both refer to the entity that is being seen.

(32) a. Who will they see?
b. They will see Bill.

Finally, we can observe that there are various sorts of ways that words can belong together. For instance, in a phrase like *the big dog*, *big* belongs with *dog*, and we have the intuition that *big* **modifies** *dog*. On the other hand, the relation between *see* and *Bill* in (32b) isn't one of modification. Rather, we have the intuition that Bill is a **participant** in a seeing event.

In the course of this book, we will introduce more precise ways of expressing and representing intuitions like the ones just discussed. For the moment, what is important is that we have strong intuitions that words belong together in ways that go beyond adjacency.

1.3.2 Structural ambiguity

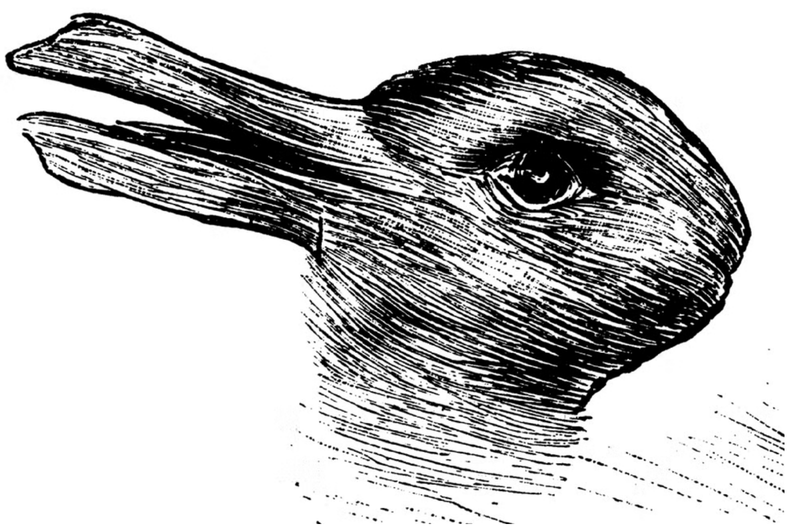
A particularly striking piece of evidence for the existence of syntactic structure is the phenomenon of **structural ambiguity.** The classified advertisement in (33) is a humorous illustration.

(33) Wanted: Man to take care of cow [that does not smoke or drink] .

The bracketed phrase is something known as a relative clause: a sentence that is embedded inside a noun phrase, serving as a modifer of that noun phrase. Here, the ambiguity is about which noun the relative clause modifies. World knowledge tells us that the intent of the advertiser is to hire a clean-living man to take care of a cow. But because of the way the advertisement is formulated, it also has an unintentionally comical interpretation—namely, that the advertiser has a clean-living cow and that the advertiser wants a man (possibly a chain-smoking alcoholic) to take care of this animal. The intended and unintended interpretations describe sharply different situations; that is why we say that (33) is ambiguous, and not vague. A vague meaning is one that is non-specific and perhaps hard to identify with precision: an ambiguous sentence is one that has two or more interpretations, but they are all specific and identifiable.

There are parallels from other aspects of cognition distinguishing between ambiguity and vagueness. A famous optical illusion is the famous duck/rabbit in (34), which can be perceived as either a duck or a rabbit (the long narrow part is either the duck's bill, or the rabbit's ears).

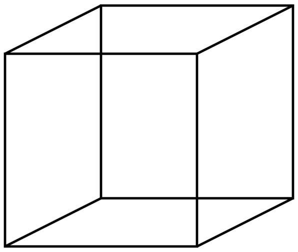
(34)



This drawing originated from a German humor magazine in 1892, and was later brought to broader awareness by Wittgenstein.

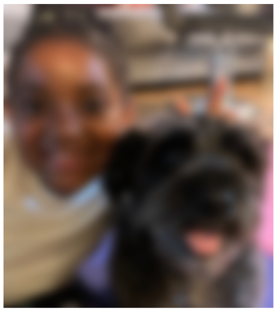
A similar example is a Necker cube, illustrated in (35). This image is visually ambiguous as to it's orientation; it can be perceived as having the lower left corner in the front (and the upper right corner in the back), or the upper right corner in the front, and the lower left corner in the back.

(35)



Source: Wikipedia, Necker Cube

Both of these examples are instances where the image is perceptually uncertain, but not because it is indiscernable—instead it flip-flops between two different images in our perception. This is notably quite different than blurry/vague image as in (36a), wherein the perception of the image is simply imprecise as compared to the unblurred image in (36b).

(36) a. 

b. 

Ambiguous sentences are like the two mutually exclusive perceptions of the optical illusions in (34) and (35): the meanings themselves are precise, but there are multiple possible meanings. Vague and imprecise meanings in sentences (like a blurry image) do exist, but are a separate phenomenon (one residing in the domain of semantics/pragmatics, and not in syntax).[7] So in (33) the meanings aren't mysterious, there are just two of them: a clean-living man or a clean-living cow. Moreover, the ambiguity of the announcement in (33) can't be pinned on a particular word, as is possible in ambiguous sentences like those in (37).

(37) a. As far as I'm concerned, being any gender is a **drag.** (Patti Smith)
b. Our bikinis are exciting. They are simply the **tops.**
c. Atheism is a **non-prophet** organization.
d. They told me all of my cages were mental; so I got **wasted** like all my potential (Taylor Swift, *this is me trying*)

Sentences like those in (37) are examples of lexical ambiguity; their ambiguity is based on a lexeme (that is, a vocabulary item) with two distinct meanings. In (33), on the other hand, the words themselves have the same meanings in each of the two interpretations, and the ambiguity derives from the possibility of grouping the words in distinct ways. In the intended interpretation, the relative clause *that does not smoke or drink* modifies *man*; in the unintended interpretation, it modifies *cow.*

To avoid any confusion, we should emphasize that we are here considering structural ambiguity from a purely descriptive perspective, focusing on what it tells us about the design features of human language and disregarding the practical issue of effective communication. As writers of advertisements ourselves, we would take care not to use (33), but to **disambiguate** it by means of an appropriate **paraphrase**. For the ordinary interpretation of (33), where the relative clause modifies *man*, we might move the relative clause next to the intended modifier, as in (38a). The comical interpretation of (33), on the other hand, cannot be expressed unambiguously by moving the relative clause. If it were the desired interpretation, we would have to resort to a more drastic reformulation, such as (38b).

(38) a. Wanted: Man that does not smoke or drink to take care of cow.
b. Wanted: Man to take care of nonsmoking, nondrinking cow.

The topic of our investigation in this book is not about editorial questions like when sentences like (38) should replace sentences like (33), though the skills you build here will surely help with editorial precision. Rather, our focus is on the more foundational issues: why does human language have the characteristics that it does? For example, why do ambiguities like these only arise in certain contexts and not others?

1.4 Universal Grammar

1.4.1 Introducing Universal Grammar

The field of syntax within linguistics is not simply interested in understanding the structure of individual languages (though we spend a lot of time on that!). One of the reasons that linguistics is situated in **cognitive sciences** (and not simply in the humanities, or in anthropology, or in the social sciences) is that the structure of languages can give us insight into how human cognition is structured more generally. It is not sufficient to describe the properties of a single language in trying to understand the human **language faculty** (the part(s) of the mind/brain that is devoted to language): instead we must attempt to understand what the universal properties of human cognition are that allow *any* human language to arise.

Very obvious facts about the world can raise important questions about human nature. Any neurotypical child (and neurodiverse children of many sorts, though not all) will learn languages that are spoken by the community they are raised in. Particular languages are not genetic: I may be genetically European, but if I am raised in South Korea among a community of Korean speakers, I will grow up speaking Korean as a first language as fluently as any other member of the community. If instead I was raised in the townships of South Africa, I would probably be speaking Xhosa or Zulu. Hearing children of deaf adults will grow up fluent signers of the sign language of their household, despite not being deaf themselves. Again, the point is an obvious fact about the world, but it is highly significant: there are different languages, and children can (and will!) learn any language that is used by the community that they are raised as a member of. What this means is that the ability to acquire any human language is a genetic property of humans. But what is the precise nature of that genetic property? Can humans learn literally any communication system? Or do all human languages share specific properties and lack others?

A central puzzle of human language is that we appear to know more than we should about language. Consider the examples in (39): in each of the first three examples, the bolded pronoun can refer to the aforementioned *Mike*:[8]

(39) a. Mike loves **his** dog.
b. Mike's dog loves **him**.
c. Mike believes that Lilly adores **him**.
d. Mike loves **him**.

Children are not explicitly instructed that the pronoun in examples like (39d) can't refer to the individual previously mentioned in the sentence. So why is it that all native speakers of English know that the pronoun cannot mean *Mike* in (39d)?

This is a version of what Noam Chomsky refers to as **Plato's problem**: how can we "explain how we know so much, given that the evidence available to us is so sparse?"[9] (Chomsky 1986b, xxvii) This has been described as the **poverty of the stimulus**: the argument is that the language data that children are exposed to while they are acquiring language is insufficient to explain the level of detailed knowledge that adults have about language. Of course, children are exposed to a vast amount of language data. But the point is that there are many aspects of mental grammar that seem near-impossible for a child to arrive at based on the evidence they are exposed to.

Noam Chomsky is an American linguist whose work in the 1950s and 1960s launched a new kind of approach to studying language (generative grammar): this textbook describes a model of human language that grew out of his work. Chomsky's solution to the poverty of the stimulus was to posit that all humans are born with a genetic endowment called **Universal Grammar**, the genetic predisposition to develop a human language that has specific properties. Universal Grammar obviously cannot be a particular language itself, but rather is the underlying ability to acquire a language. The idea is that Universal Grammar has specific properties, so children's grammatical knowledge of language is in part driven by the particular language that they are exposed to, but is also driven by Universal Grammar (UG).

(40) **Universal Grammar (UG)**
The universal innate cognitive mechanisms that generate human language, as discoverable through language grammars.[10]

Universal Grammar is thought to facilitate language acquisition: by giving children innate expectations about what language structures should look like, this effectively constrains the range of possible hypotheses that children have to consider when acquiring a language, making it easier to learn language. But this means that UG also predetermines human languages to some extent: by hypothesis, the theory of UG says that human languages don't vary endlessly, but rather there are shared properties among all languages.[11]

The big project of syntax from this perspective, then, is to understand the properties of Universal Grammar. First, we work to understand the grammars of particular languages, but then we ask what properties of language grammars do we see recurring in ways that are much too common to be a coincidence? These are signs that these properties of language are driven by the properties of human cognition in some way. Universal Grammar is the model, the framework, that we theorize about in an attempt to arrive at specific claims about what the requirements of human cognition are on human language grammars. We can therefore refer to the approach described in this text as "Chomskyan" because it is describes the findings of the research community that grew out of Chomsky's proposal of UG.

1.4.2 Formal universals

We have noted above that the descriptive grammatical rules of syntax are structure-dependent, that is, relying on hierarchical structure and not simply linear order. The structure-dependent character of syntactic rules is a general property of UG, especially when considered in abstraction from any particular language. There are two sources of evidence for this. First, as we have seen, the syntactic rules that children acquire are structure-dependent (even when they are not the rules that adults use). Second, even though structure-independent rules are logically possible and computationally tractable, no known human language actually has rules that disregard syntactic structure as a matter of course. For instance, (41) gives two examples of computationally very simple rules for question formation, but no known human language has rules of this type.

(41) a. To form a question, switch the order of the first and second words in the corresponding declarative sentence.

The girl is tall.	⟶	Girl the is tall?
The blond girl is tall.	⟶	Blond the girl is tall?

b. To form a question, reverse the order of the words in the corresponding declarative sentence.

The girl is tall.	⟶	Tall is girl the?
The blond girl is tall.	⟶	Tall is girl blond the?

The structure-dependent character of syntactic rules (often referred to more briefly as **structure dependence**) is what is known as a **formal universal** of human language—a property common to all human languages that is independent of the meanings of words. Formal universals are distinguished from **substantive universals**, which concern the substance, or meaning, of linguistic elements. An example of a substantive universal is the fact that all languages have **indexical** elements such as *I*, *here*, and *now*. These words have the special property that their meanings are predictable in the sense that they denote the speaker, the speaker's location, and the time of speaking, but that what exactly they refer to depends on the identity of the speaker.

1.4.3 Recursion

Another formal universal is the property of **recursion**. A simple illustration of this property is the fact that it is possible for one sentence to contain another. For instance, the simple sentence in (42a) forms part of the complex sentence in (42b), and the resulting sentence can form part of a still more complex sentence. Recursive embedding is illustrated in (42) up to a level of five embeddings.

(42) a. She won.
b. The Times reported that
  [she won].
c. John told me that
  [the Times reported that
    [she won]].
d. I remember distinctly that
  [John told me that
    [the Times reported that
      [she won]]].
e. They don't believe that
  [I remember distinctly that
    [John told me that
      [the Times reported that
        [she won]]]].
f. I suspect that
  [they don't believe that
    [I remember distinctly that
      [John told me that
        [the Times reported that
          [she won]]]]].

1.4.4 Parameters

Formal universals like structure dependence and recursion are of particular interest to linguists in the Chomskyan tradition. This is not to deny, however, that individual languages differ from one another, and not just in the sense that their vocabularies differ. In other words, Universal Grammar is not completely fixed, but allows some variation. The ways in which grammars can differ are called **parameters**.

One simple parameter concerns the order of verbs and their objects. In principle, two orders are possible: verb-object (VO) or object-verb (OV), and different human languages use either one or the other. As illustrated in (43) and (44), English and French are languages of the VO type, whereas Hindi, Japanese, and Korean are languages of the OV type.

(43) a. Peter **read** <u>the book.</u> **English**

b. Pierre lisait le livre. **French**
Pierre was.reading the book
'Pierre was reading the book.'

(44) a. पवित्र ने किताब **पढ़ी ।** **Hindi**
Pavitr-ne kitaab **parh-ii.**
Pavitr book read
'Pavitr read the book.'

b. 漣が 本を **読んだ。** **Japanese**
Ren-ga hon-o **yon-da.**
Ren book read
'Ren read the book.'

c. 현서가 책을 **읽었다.** **Korean**
Hyense-ka chayk-ul **ilk-ess-ta.**
Hyense book read
'Hyense read the book.'

Another parameter of Universal Grammar concerns the possibility, mentioned earlier in connection with prescriptive rules, of separating a preposition from its object, or **preposition stranding.** (The idea behind the metaphor is that the movement of the object of the preposition away from its ordinary position leaves the preposition stranded high and dry.) The parametric alternative to preposition stranding goes by the name of **pied piping**,[12] by analogy to the Pied Piper of Hamelin, who took revenge on the citizens of Hamelin for mistreating him by luring the town's children away with him. In pied piping of the syntactic sort, the object of the preposition moves away from its usual position, just as in preposition stranding, but it takes the preposition along with it. The two parametric options are illustrated in (45).

(45) a. Preposition stranding: ✓ **Which house** does your friend live *in*?
b. Pied piping: ✓ *In* **which house** does your friend live?

Just as in English, preposition stranding and pied piping are both grammatical in Swedish. (In Swedish, it is preposition stranding that counts as prescriptively correct! Pied piping is frowned upon, on the grounds that it sounds stiff and artificial.)

(46) Swedish
a. ✓ **Vilket hus** bor din kompis *i*?
which house lives your friend in
'Which house does your friend live in?'
b. ✓ *I* **vilket hus** bor din kompis?

In other languages, such as French and Italian, preposition stranding is actually ungrammatical. Speakers of these languages reject examples like (47b) and (48b), and accept only the corresponding pied-piping examples in (47a) and (48a).

(47) a. ✓ *Dans* **quelle maison** est-ce que ton ami habite? **French**
in which house is it that your friend lives
'Which house does your friend live in?'

b. * **Quelle maison** est-ce que ton ami habite *dans*?
which house is it that your friend lives in
Intended meaning: 'Which house does your friend live in?'

(48) a. ✓ *In* **quale casa** abita il tuo amico? **Italian**
in which house lives the your friend
'Which house does your friend live in?'

b. * **Quale casa** abita il tuo amico *in*?
which house lives the your friend in
Intended meaning: 'Which house does your friend live in?'

Therefore we can say that the order of verbs and objects is parameterized between languages; likewise, whether a language has pied-piping or not is parameterized between languages. UG clearly cannot designate the order of verbs and objects, because this varies between languages (similarly for pied-piping). Therefore UG is our theoretical notion that represents/captures the principles of language grammar that are universal, and any point where there can be variation, we claim that parameters of UG designate those options.

Generative grammar 1.5

At the beginning of this chapter, we said that this book was an introduction to generative grammar from a Chomskyan perspective. Until now, we have clarified our use of the term "grammar", and we have explained that a Chomskyan perspective on grammar is concerned with the formal principles that all languages share as well as with the parameters that distinguish them. Let's now turn to the notion of a generative grammar.

(49) A **generative grammar** is an **algorithm** for specifying, or **generating**, all and only the grammatical sentences in a language.

What's an algorithm? It is any finite, explicit procedure for accomplishing some task, beginning in some initial state and terminating in a defined end state. Computer programs are the algorithms par excellence. More concrete examples of algorithms include recipes, knitting patterns, or the instructions for assembling an IKEA bookcase.

An important point to keep in mind is that it is often difficult to construct an algorithm for even trivial tasks. A quick way to gain an appreciation for this is to describe how to tie a bow. Like speaking a language, tying a bow is a skill that most of us master around school age and that we perform more or less unconsciously thereafter. But describing (not demonstrating!) how to do it is not that easy, especially if we're not familiar with the technical terminology of knot-tying. In an analogous way, constructing a generative grammar of English is a completely different task than speaking the language, and much more difficult (or at least difficult in a different way)!

Take an English example as illustration: in English there are verb-particle sequences that look highly similar, here *run up* and *ring up*, the latter meaning something like 'to calculate and process expenses at a register'.

(50) a. The children ran up the hill.
b. The server rang up the bill.

These two sentences appear surface similar (containing verbs following by the preposition-like element *up*). But only in one of these sentences can the preposition-like particle be moved to a position after the object:

(51) a. * The children ran the hill up.
b. ✓ The server rang the bill up.

Native speakers of Mainstream U.S. English know the facts in (51), though for most people this knowledge is unconscious. The goal of generative grammar is to specify how and why it is that (51a) is grammatical but (51a) is not. We do this via the rules, principles, and structures of human language broadly (via principles and parameters of UG) and of the English language specifically. Just like a cooking recipe, a generative grammar needs to specify the ingredients and procedures that are necessary for generating grammatical sentences. We won't introduce all of these in this first chapter, but in the remainder of the section, we'll introduce enough ingredients and procedures to give a flavor of what's to come.

1.5.1 Elementary trees and substitution

The raw ingredients that sentences consist of are **vocabulary items**. These belong to various **grammatical categories**, like noun, adjective, transitive verb, preposition, and so forth. Depending on their category, vocabulary items combine with one another to form constituents, which in turn belong to syntactic categories of their own. For instance, determiners (a category that includes the articles *a* and *the* and the demonstratives *this, that, these* and *those*) can combine with nouns to form noun phrases, but they can't (or at least don't ordinarily) combine with other syntactic categories like adverbs, verbs, or prepositions.

(52) a. ✓ a house
b. ✓ the cats
c. ✓ those books

(53) a. * a slowly
b. * the went
c. * those of

It's possible to represent the information contained in a constituent by using **labeled bracketing.** Each vocabulary item is enclosed in brackets that are labeled with the appropriate grammatical category. The constituent that results from combining vocabulary items is in turn enclosed in brackets that are labeled with the constituent's grammatical category. The labeled bracketings for the noun phrases in (52) are given in (54).

(54) a. $[_{\text{NounPhr}}\ [_{\text{Det}}\ \text{a}\ ][_{\text{Noun}}\ \text{house}\ ]\ ]$
b. $[_{\text{NounPhr}}\ [_{\text{Det}}\ \text{the}\ ][_{\text{Noun}}\ \text{cats}\ ]\ ]$
c. $[_{\text{NounPhr}}\ [_{\text{Det}}\ \text{those}\ ][_{\text{Noun}}\ \text{books}\ ]\ ]$

Noun phrases can combine with other syntactic categories, such as prepositions or transitive verbs. Prepositions combine with a noun phrase to form prepositional phrases. A transitive verb combines with one noun phrase to form a verb phrase, which in turn combines with a second noun phrase to form a complete sentence.

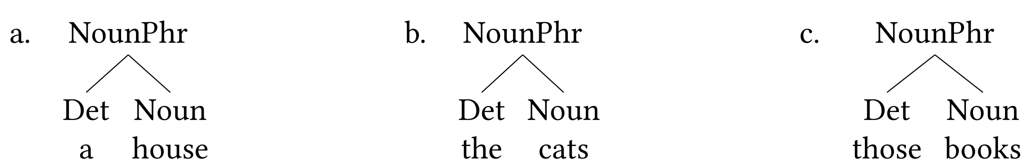
(55) a. $[_{\text{PrepPhr}}\ [_{\text{Prep}}\ \text{on}\ ]\ [_{\text{NounPhr}}\ [_{\text{Det}}\ \text{the}]\ [_{\text{Noun}}\ \text{table}\ ]\ ]\ ]$
b. $[_{\text{VerbPhr}}\ [_{\text{TrVerb}}\ \text{drafted}\ ]\ [_{\text{NounPhr}}\ [_{\text{Det}}\ \text{a}\ ]\ [_{\text{Noun}}\ \text{letter}\ ]\ ]\ ]$
c. $[_{\text{Sentence}}\ [_{\text{NounPhr}}\ [_{\text{Det}}\ \text{the}\ ]\ [_{\text{Noun}}\ \text{secretary}\ ]\ ]\ [_{\text{VerbPhr}}\ [_{\text{TrVerb}}\ \text{drafted}\ ]\ [_{\text{NounPhr}}\ [_{\text{Det}}\ \text{a}\ ]\ [_{\text{Noun}}\ \text{letter}\ ]\ ]\ ]\ ]$

Noun phrases don't, however, combine with any and all syntactic categories. For instance, they can't combine with determiners (at least not in English).

(56) * the this letter

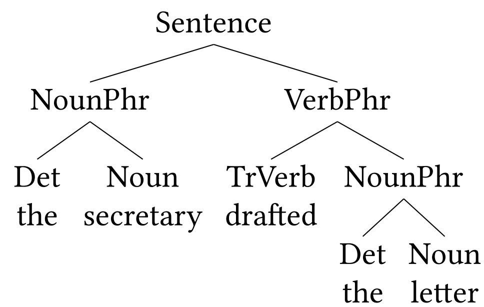
As constituent structure grows more complex, labeled bracketings very quickly grow difficult for humans to visually process, as you may have experienced while looking at (55c). It's often more convenient to represent constituent structure with **tree diagrams.** Tree diagrams, or trees for short, convey exactly the same information as labeled bracketings, but the information is presented differently. Instead of enclosing an element in brackets that are labeled with a grammatical category, the category is placed immediately above the element and connected to it with a line or **branch.** The labeled bracketings that we have seen so far translate into the trees in (57) and (58).[13]

(57)



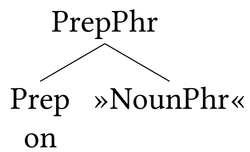
(58)

c.

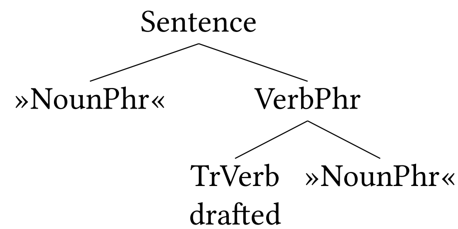


Continuing with the metaphor of ingredients and recipes, trees like those in (57) and (58) resemble dishes that are ready to serve; they don't provide a record of how they were brought into being. We can provide such a record by representing vocabulary items themselves in the form of trees that include combinatorial information. For example, prepositions and transitive verbs can be represented as trees with empty slots for noun phrases to fit into, as shown in (59).

(59) a.

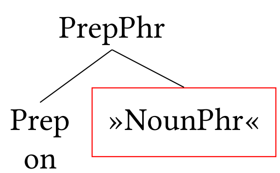


b.

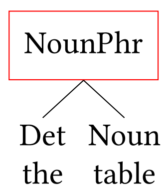


We'll refer to trees for vocabulary items like those in (59) as **elementary trees.** The purpose of elementary trees is to represent a vocabulary item's combinatorial possibilities, and so they ordinarily contain unfilled nodes. We refer to such nodes as **substitution nodes**, and they are filled by a **substitution** operation, as shown in (60). In this text, we will indicate substitution nodes by enclosing them in inverted guillemets (» «). The guillemets "point" to the spot where a node needs to be filled.

(60) a.



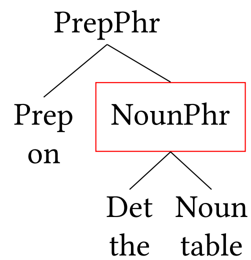
b.



(60a) has a substitution node of some grammatical category.

The root (topmost) node in (60b) has the same grammatical category as the substitution node in (60a).

c.

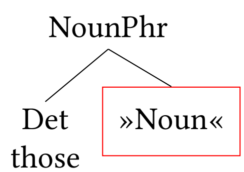


Substitution occurs when the root node of (60b) is identified with the substitution node in (60a).

Elementary trees don't necessarily contain substitution nodes, though; ones that invariably play the role of (60b) in the substitution operation don't. The elementary tree for the noun in (61b) is an example.

Notice, by the way, that there are two conceivable ways to arrive at trees for noun phrases like *those cats*, depending on whether it is the noun that is taken as the substitution node, as in (61), or the determiner, as in (62). At this point, there is no reason to prefer one way over the other, but in Chapter 6, we will adopt a variant of (61).

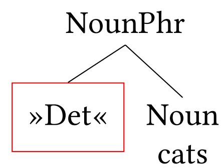
(61) a.



b. Noun
cats

(62) a. Det
those

b.



Elementary trees and substitution nodes won't be part of our final theory, but they are very helpful in the early stages of building the theory. And as we will discuss in Chapter 7, they are very useful ways to visualize the lexical (memorized) properties of specific words and morphemes.

In summary, a generative grammar as we've constructed it so far consists of a set of elementary trees, which represent the vocabulary items in a language and the range of their combinatorial possibilities, and a substitution operation, by means of which the elementary trees combine into larger constituents and ultimately into grammatical sentences. There is a lot more to be said to create a generative grammar that can correctly predict all the grammatical and ungrammatical sentences for any particular language, let alone all languages, but these are the pieces we will start building from.

1.5.2 Grammaticality

As we mentioned earlier, the aim of a generative grammar is to generate all and only the grammatical sentences of a language. Since the notion of grammaticality is basic to syntactic theory, it is important to distinguish it from notions with which it is easily confused. First and foremost, 'is grammatical' is not the same thing as 'makes sense.' The sentences in (63) all 'make sense' in the sense that it is easy to interpret them. Nevertheless, as indicated by the asterisks, they are not grammatical in Mainstream U.S. English.[14]

(63) a. * Is our children learning?
b. * Me wants fabric.
c. * To where are we be taking thou, sir?
d. * The introduction explained that "the Genoese people, *besides of hard worker, are good eater too, and even 'gourmand',* of that honest gourmandise which will not drive a man to hell but which is, after all, one of the few pleasures that mankind can enjoy in this often sorrowful world."

Conversely, sentences can be grammatical, but not 'make sense.' The 'fairy tale' or 'science fiction' sentences in (15) are of this type. Two further examples are given in (64). Since the sentences are grammatical, they aren't preceded by an asterisk. Their semantic anomaly is indicated by a prefixed pound sign (hash mark). Each of these sentences are presented with a parallel (non-anomalous) sentence that has the same structure but is semantically acceptable.

(64) a. # Colorless green ideas sleep furiously. (Chomsky 1965, 149) — cf. Revolutionary new ideas appear infrequently.
b. # I plan to travel there last year. — cf. I plan to travel there next year.

Second, 'grammatical' must be distinguished from 'acceptable' or 'easily processable by human beings'. This is because it turns out that certain well-motivated simple grammatical operations can be applied in ways that result in sentences that are virtually impossible for human beings to process. For instance, it is possible in English to modify a noun with a relative clause, and sentences containing nouns that are modified in this way, like those in (65), are ordinarily perfectly acceptable and easily understood. (Here and in the following examples, the relative clauses are bracketed and the modified noun is underlined.)

(65) a. The <u>mouse</u> [that the cat chased] escaped.
b. The <u>cat</u> [that the dog scared] jumped out the window.

But now notice what happens when we modify the noun within the relative clause in (65a) with a relative clause of its own.

(66) The <u>mouse</u> [that the <u>cat</u> [that the dog scared] chased] escaped.

Even though (66) differs from (65a) by only four additional words and a single additional level of embedding, the result is virtually uninterpretable without pencil and paper. The reason is not that relative clause modification can't apply more than once, since the variant of (65a) in

(67), which contains exactly the same words and is exactly as long, is perfectly fine (or at any rate much more acceptable than (66)).

(67) The mouse escaped [that the cat chased] [that the dog scared].

The reason that (66) is virtually uninterpretable is also not that it contains recursive structure (the relative clause that modifies *mouse* contains the relative clause that modifies *cat*). After all, the structures in (42) are recursive, yet they don't throw us for a loop the way that (66) does.

(66) is unacceptable not because it is ungrammatical, but because of certain limitations on human short-term memory (Chomsky and Miller 1963, 286, 471). Specifically, notice that in the (relatively) acceptable (67), the subject of the main clause *the mouse* doesn't have to "wait" (that is, be kept active in short-term memory) for its verb *escaped* since the verb is immediately adjacent to the subject. The same is true for the subjects and verbs of each of the relative clauses (*the cat* and *chased*, and *the dog* and *scared*). In (66), on the other hand, *the mouse* must be kept active in memory, waiting for its verb *escaped*, for the length of the entire sentence. What is even worse, however, is that the period during which *the mouse* is waiting for its verb *escaped* overlaps the period during which *the cat* must be kept active, waiting for its verb *chased*. What makes (66) so difficult, then, is not the mere fact of recursion, but that two relations of exactly the same sort (the subject-verb relation) must be kept active in memory at the same time. In none of the other relative clause sentences is such double activation necessary. For instance, in (65a), *the mouse* must be kept active for the length of the relative clause, but the subject of the relative clause (*the cat*) needn't be kept active since it immediately precedes its verb *chased*.

Extra Info

Sentences like (65) and (66) are often referred to as **center-embedding** structures, and the dependencies between the subjects and their verbs are said to be **nested**.

The mouse that the cat chased escaped.

The mouse that the cat that the dog scared chased escaped.

By contrast, the corresponding dependencies in (67) are not nested.

The mouse escaped that the cat chased.

The mouse escaped that the cat chased that the dog scared.

This remains true even if we focus on the dependencies between the relative clauses and the nouns that they modify.

The mouse escaped that the cat chased that the dog scared.

A final important point to bear in mind is that any sentence is an expression that is paired with a particular interpretation. Grammaticality is always determined with respect to a pairing of form and meaning. This means that a particular string can be grammatical under one interpretation, but not under another. For instance, (68) is ungrammatical under an subject-object-verb (SOV) interpretation (that is, when the sentence is interpreted as *Sue hired Tom*).

(68) Sue Tom hired.

(68) is grammatical, however, under an object-subject-verb (OSV) interpretation (that is, when it is interpreted as *Tom hired Sue*). On this interpretation, *Sue* receives a special intonation marking contrast, which would ordinarily be indicated in writing by setting off *Sue* from the rest of the sentence by a comma. In other words, the grammaticality of (68) depends on whether its interpretation is analogous to (69a) or (69b).

(69) a. * She them hired.
b. ✓ HER, they hired. (The other job candidates, they didn't even call back.)

1.5.3 Grammar versus language

We conclude this chapter by considering the relationship between the concepts of grammar and language. The notion of language seems straightforward because we are used to thinking and speaking of "the English language," "the Hindi language," "the Swahili language," and so forth. But these terms are actually much vaguer than they seem at first glance because they cover a multitude of varieties, including ones that differ enough to be mutually unintelligible. For instance, Ethnologue distinguishes 32 dialects of English in the United Kingdom alone. In addition, distinct dialects of English are spoken in former British colonies, including Australia, Canada, India, New Zealand, South Africa, the United States, and many other African, Asian, and Caribbean nations, and many of these dialects have subdialects of their own. Similarly, Ethnologue distinguishes 11 dialects of French in France and 10 dialects of Swahili in Kenya, and there are further dialects in other countries in which these languages are spoken. Moreover, we use terms like "the English language" to refer to historical varieties that differ as profoundly as present-day English does from Old English, which is about as intelligible to a speaker of modern English as German is—that is to say, it is not.

Although the most salient differences between dialects are often phonological (that is, speakers of different dialects often have different accents), dialects of a so-called single language can differ syntactically as well. For instance, in standard French, as in the Romance languages more generally, adjectives ordinarily follow the noun that they modify. But that order is reversed in Walloon, a variety of French spoken in Belgium. The two parametric options are illustrated in (70) (Bernstein 1993, 25–26).

(70) a. un chapeau noir [Standard French]
a hat black
b. on neûr tchapê [Walloon]
a black hat
'a black hat'

Another example of the same sort concerns multiple negation in sentences like (71a) in varieties of English.

(71) a. The kids didn't eat nothing.
b. The kids didn't eat anything.

In present-day Mainstream U.S. English, *didn't* and *nothing* each contribute their negative force to the sentence, and the overall force of (71a) isn't negative; rather, the sentence means that the kids ate something. In many varieties of English, however, (71a) conveys exactly the same meaning as in Mainstream U.S. English (71b); that is, the sentence as a whole has negative force. In these varieties, the negation in *nothing* can be thought of as agreeing with (and reinforcing) the negation in *didn't* rather than cancelling it; hence the term **negative concord** for this phenomenon ('concord' is a variant term for 'agreement'). Negative concord is routinely characterized as "illogical" by prescriptivists,[15] and it is one of the most heavily stigmatized features in present-day U.S. English.[16] However, it was productive in earlier forms of English, and it is attested in renowned masters of the language such as Chaucer and Shakespeare. Moreover, negative concord is part of the mainstream forms of languages like Afrikaans, French, modern Greek, Hausa, Italian, Lubukusu, Moroccan Arabic, Nweh, Spanish, and Tarifit Berber (just to name a few: see Bell 2004). From a descriptive and generative point of view, negative concord is simply a parametric option of Universal Grammar just like any other, and negative concord is no more illogical than the noun-adjective order in (70a) or preposition stranding/pied-piping. If anything, marking negation multiple times in a sentence could have a clear functional (i.e. communicative) motivation, as it is important for a listener to note whether a sentence was negative or affirmative.

In both of the examples just discussed, we have dialects of "the same language" (English and French, respectively) differing with respect to a parameter. The converse is also possible: two "different languages" that are parametrically (all but) indistinguishable. For example, the same linguistic variety spoken on the Dutch-German border may count as a dialect of Dutch or German depending on which side of the political border it is spoken, and the same is true of many other border dialects as well. According to Max Weinreich, "a language is a dialect with an army and a navy." A striking confirmation of this aphorism concerns the recent terminological history of Serbo-Croatian. As long as Yugoslavia was a federal state, Serbo-Croatian was considered a single language with a number of regional dialects. The 14th edition of Ethnologue, published in 2000, still has a single entry for Serbo-Croatian. In the 15th edition, published in 2005, the single entry is replaced by three new entries for Bosnian, Croatian, and Serbian. What changed in the meantime wasn't the languages, but the political situation.

The notion of "a language" is based more on sociopolitical considerations than on strictly linguistic ones. By contrast, the term "grammar" refers to a particular set of parametric options that a speaker acquires. For this reason, the distinction between language and grammar that we have been drawing is also referred to as the distinction between E-language and I-language (Chomsky 1986b). E-language (external language) refers to the actual utterances and discourses of individuals and communities—language out in the world. I-language (internal language), in contrast, are the mental representations of grammatical knowledge that speakers possess. Generative grammar is concerned with I-language, not E-language; the theory of Universal Grammar is an attempt to model what the possible and impossible I-languages are.

As we have seen, the same language label can be associated with more than one grammar (the label "English" is associated with grammars both with and without negative concord), and a single grammar can be associated with more than one language label (as in the case of border dialects). It is important to distinguish the concept of shared grammar from mutual intelligibility. To a large extent, Mainstream U.S. English and many other varieties of American English are mutually intelligible even where their grammars differ with respect to one parameter or another: that is to say, two speakers of two different varieties of American English can often understand each other reasonably well. On the other hand, it is perfectly possible for two or more varieties that are mutually unintelligible to share a single grammar. For instance, in the Indian village of Kupwar (Gumperz and Wilson 1971), the three languages Marathi, Urdu, and Kannada, each spoken by a different ethnic group, have been in contact for about 400 years, and most of the men in the village are bi- or trilingual. Like the standard varieties of these languages, their Kupwar varieties have distinct vocabularies, thus rendering them mutually unintelligible to monolingual speakers, but in Kupwar, the considerable grammatical differences that exist among the languages as spoken in other parts of India have been virtually eliminated. The difference between standard French and Walloon with respect to prenominal adjectives is another instance of this same convergence phenomenon. Here, too, the adjective-noun order in Walloon is due to language contact and bilingualism, in this case between French and Flemish, the other language spoken in Belgium; in Flemish, as in the Germanic languages more generally, adjectives ordinarily precede the nouns that they modify.

Finally, we should point out that it is perfectly possible for a single speaker to acquire more than one grammar. This is most strikingly evident in balanced bilinguals. Speakers can also acquire more than one grammar in situations of syntactic change. For instance, in the course of its history, English changed from an OV to a VO language, and individual speakers during the transition period (which began in late Old English and continued into Middle English) acquired and used both parametric options. Speakers can also acquire more than one grammar in situations of diglossia or stable syntactic variation. For instance, English speakers whose vernacular grammar has negative concord or preposition stranding might acquire the parametric variants without negative concord or with pied piping in the course of formal education.

New Empirical Observations

Each chapter of this textbook introduces a range of new empirical properties of language, that is, factual information about how human language syntax works. At the end of each chapter we will summarize that chapter's empirical contributions.

- All human languages (in both their formal and informal instantiations) follow consistent rules without apparent effort, despite the fact that these rules often conflict with externally taught "school grammar" rules (motivates the idea that we have innate grammatical knowledge)
- Children acquire productive linguistic rules, rather than simply memorizing all the linguistic knowledge they have.
- Fluent users of a language have intuitions (a.k.a. judgments) about whether a given sentence or phrase adheres to the grammatical patterns of their language.

- Linguistic rules (both adults' and children's) seem to be structure-dependent.
- Sentences may be structurally ambiguous, that is, the string of morphemes in a sentence may be consistent with multiple different syntactic structures.

Components of the Theoretical Model

Throughout this textbook we will be building a ***scientific model*** *of syntax. At the end of each chapter, we will summarize what the current state of our model is and highlight which concepts were newly added.*

We are building generative grammars of languages, meaning that we are attempting to determine the set of rules/structures that correctly predicts the grammatical and ungrammatical sentences in any particular language.

The larger task is that we are modeling Universal Grammar, the hypothesized principles and parameters in (universal) human cognition that allow the grammatical structures that occur in the world's languages, and disallow the structures that do not occur in the world's languages. Universal Grammar is hypothesized to be innate (i.e. genetically-endowed in all humans), and a language-specific domain of cognition.[17]

In this chapter we added these notions, which play a central role in our model:

- generative grammar
- Universal Grammar
- Principles of Universal Grammar (formal universals)
- Parameters of Universal Grammar (points of variation between languages)
- E-language vs. I-language

Notes 1.6

1. It is also possible to overzealously apply rules like those in (2), even in cases where they shouldn't be applied. This phenomenon is known as hypercorrection. Two common instances are illustrated in (i).

(i)

Hypercorrect example	Mainstream target	Explanation
Over there is the guy whom I think took her to the party.	the guy *who* I think took her to the party	the relative pronoun *who* is the subject of the relative clause, not the object (cf. the guy *who, *whom* took her to the party)
This is strictly between you and I.	between you and *me*	the second pronoun is part of the conjoined object of the preposition *between*, not part of a subject

2. The prescriptive rule is actually better stated as "Don't separate a preposition from its object," since the traditional formulation invites exchanges like (i).

(i) A: Who are you going to the party with?
B: Didn't they teach you never to end a sentence with a preposition?
A: Sorry, let me rephrase that. Who are you going to the party with, Mr. Know-it-all?

3. As William Labov has often pointed out, everyday speech (apart from false starts and other self-editing phenomena) hardly ever violates the rules of descriptive grammar.

4. Actually, that's an oversimplification. Not all the articles and nouns an English-speaking child hears appear in the article-noun order. To see why, carefully consider the underlined sentence in this note.

5. Since then, wugs have become something of an unofficial mascot for linguistics; it is wugs that adorn the cover of this book.

6. When children didn't respond this way, they either repeated the original invented word, or they didn't respond at all. It's not clear what to make of these responses. Either response might indicate that the children were stumped by the experimental task. Alternatively, repetition might have been intended as an irregular plural (cf. *deer* and *sheep*), and silence might indicate that some of the invented words (for instance, *cra*) struck the children as phonologically strange.

7. Examples of vagueness in this respect are words like *bald* or *heap*, which participate in what is known as the Sorites Paradox. Imagine someone with a full head of hair, and imagine that you pluck out their hairs one single strand of hair at a time. If you do that for long enough, they will eventually be bald. But it is wholly unclear when, in that process, they become bald; and most likely there is no single instance when a single hair being pulled changes someone from *not-bald* to being actually *bald*. This is an example of vagueness: we know what it means to be bald, but there are fuzzy boundaries around the notion, so it is difficult to draw firm and discrete boundaries between *bald* vs. *not-bald*. Many words in language display this kind of vagueness. But this is notably different from the ambiguity that we are discussing in the main text, wherein there are multiple (precise) meanings that are all available in a given sentence.

8. The examples in (39) and the discussion in this section are modeled on the discussion in Lightfoot (2021).

9. This echoes Plato's philosophy positing innate human knowledge in a number of contexts.

10. We will note that the definition of Universal Grammar here is our own; it is relatively consistent with the original claims about UG, but leaves wiggle room to accommodate some modern approaches to the concept.

11. The theory of Universal Grammar is not the only logical conclusion: there are camps within linguistics that take human language to be no different than any other cognitive ability, and thus understand human language to be learned in precisely the same way that anything is learned, and there are no universal contraints on what grammatical forms can be apart from those imposed by general cognitive abilities like limits on memory and processing. Approaches

like this are often called **usage-based**, and many such approaches adopt an approach often called **construction grammar** (Goldberg 2006, 2019; Tomasello 2003; Goldberg 2013; Hoffmann and Trousdale 2013).

12. The term 'pied piping' was invented in the 1960s by John Robert Ross (1967), a syntactician with a penchant for metaphorical terminology.

13. Online corpora that are annotated with syntactic structure, such as the Penn Treebank, the Penn Parsed Corpora of Historical English, and others like them, tend to use labeled bracketing because the resulting files are computationally extremely tractable. The readability of such corpora for humans can be improved by suitable formatting of the labeled bracketing or by providing an interface that translates the bracketed structures into tree diagrams.

14. (63a) is from a speech by George W. Bush (here). (63b) was the subject line of an email message in response to an offer of free fabric; the author is humorously attempting to imitate the language of a child greedy for goodies. (63c) is from "Pardon my French" (Calvin Trillin. 1990. Enough's enough (and other rules of life). p. 169). (63d) is from "Connoisseurs and patriots" (Joseph Wechsberg. 1948. Blue trout and black truffles: The peregrinations of an epicure. p. 127).

15. Two important references concerning the supposed illogicality of negative concord (and of marginalized varieties of English more generally) are Labov (1972a, 1972b).

    Those who argue that negative concord is illogical often liken the rules of grammar to those of formal logic or arithmetic, where one negation operator or subtraction operation cancels out another; that is, (NOT (NOT A)) is identical to A, and $(-(-5)) = +5$. Such prescriptivists never distinguish between sentences containing even and odd numbers of negative expressions. By their own reasoning, (i.a) should have a completely different status than (i.b)—not illogical, but at worst redundant.

    (i) a. They never told nobody nothing.
        b. They never told nobody.

16. Because of the social stigma associated with it, it is quite challenging to study negative concord in present-day English. This is because even for those speakers of negative concord varieties who don't productively control Mainstream U.S. English as a second dialect, the influence of prescriptive grammar is so pervasive that if such speakers reject negative concord sentences as unacceptable, we don't know whether they are rejecting them for grammatical or for social reasons.

17. Reasonable people can disagree about some of these details while still operating within a UG framework.

Exercises and problems 1.7

Exercise 1.1: Descriptive and prescriptive grammar A

For each of the sentences below, provide an acceptability judgment (i.e. put an asterisk or a checkmark next to the sentence, or rate it as somewhere in between). Also for each rule state whether or not the sentence violates a prescriptive rule or some kind of descriptive rule (or neither, or both).

(1)
a. Up what did the roofer climb?
b. What did the roofer climb up?
c. i. I really messed my project up.
ii. What did you really mess up?
iii. Up what did you really mess?
d. Jay says they are a gym bro but I don't think they are.
e. Jay says they are a gym bro but I don't think are.
f. i. I didn't do nothing.
ii. I not did anything.
g. Me and Alex are gonna get sushi for lunch.
h. i. Q: What did you eat for lunch?
ii. A: a sandwich
i. i. Q: What did you eat for lunch?
ii. A: I ate a sandwich for lunch.
j. i. Who did you visit last week?
ii. Whom did you visit last week?
k. At my new job I want to hit the ground running.
l. i. I want to quickly grab lunch before work picks up again.
ii. Quickly I want to grab lunch. (with no pauses in pronunciation)

Exercise 1.2: Descriptive and prescriptive grammar B

Can you come up with examples of sentences that are not mentioned in the text above but which fall into each of these categories? The purpose is to think through the overlap and non-overlap of descriptive and descriptive grammatical intuitions.

1. One sentence that doesn't violate a prescriptive rule and is grammatical;
2. One sentence that violates a prescriptive rule but is grammatical;
3. One sentence that doesn't violate a prescriptive rule (that you know of) but is ungrammatical;
4. One sentence that violates a prescriptive rule and is ungrammatical.

In each instance, specify the prescriptive rule as you originally learned it (it doesn't need to be the same rule for each category above). This can be done for any language where you have native speaker intuitions and received education about language prescription. If any of the categories prove challenging to find an example for (i.e. you can't think of something), say so in your response and give 1-2 sentences of thoughts explaining why you think it might have been hard to think of something.

Exercise 1.3: Descriptive and prescriptive grammar C

The sentences in (4) violate several descriptive rules of English, three of which were given in (5). As mentioned in the text, there is a fourth descriptive rule that is violated in (4). Formulate the rule (you shouldn't need more than a sentence).

Exercise 1.4: Belfast English subject agreement

(1)–(4) illustrate the facts of subject-verb agreement in the variety of English spoken in Belfast, Ireland (data from Henry 1995, chapter 2). Describe the data as clearly and briefly as you can.

Take Note

In order to avoid conflating morphological form with semantic function, you can refer to "is" and "are" as "the *i-* form" and "the *a-* form", rather than as "singular" and "plural".

(1) a. ✓ The girl is late.
b. ✓ She is late.
c. ✓ Is {the girl, she} late?

(2) a. * The girl are late.
b. * She are late.
c. * Are {the girl, she} late?

(3) a. ✓ The girls are late.
b. ✓ They are late.
c. ✓ Are {the girls, they} late?

(4) a. ✓ The girls is late.
b. * They is late.
c. * Is {the girls, they} late?

Exercise 1.5: Identifying ambiguity

A. Which of the newspaper headlines in (1) are lexically ambiguous, which are structurally ambiguous, and which are a mixture of both types of ambiguity? Explain.

(1) a. Beating witness provides names
b. Child teaching expert to speak
c. Drunk gets nine months in violin case
d. Enraged cow injures farmer with ax
e. Prostitutes appeal to pope
f. Teacher strikes idle kids
g. Teller stuns man with stolen check

B. At least two of the examples form a subgroup even within their type (lexical, structural, mixed). Can you find them and explain their similarity?

Exercise 1.6: Identifying recursive phrases

In the chapter, we showed that sentences are recursive categories. In other words, one instance of the grammatical category 'Sentence' can contain another instance of the same category. Provide evidence that noun phrases and prepositional phrases are recursive categories, too.

Word of Caution

Be careful to give examples that are recursive, and not just ones in which the grammatical category in question occurs more than once. For instance, (1) does not provide the evidence required in this exercise, because the second prepositional phrase is not contained in the first. This is clearly shown by the fact that the order of the prepositional phrases can be switched.

(1) a. The cat jumped [$_{PP}$ onto the table] [$_{PP}$ without the slightest hesitation].
b. The cat jumped [$_{PP}$ without the slightest hesitation] [$_{PP}$ onto the table].

Exercise 1.7: Ungrammaticality and semantic anomaly

Which, if any, of the sentences in (1)–(6) are ungrammatical? Which, if any, are semantically or otherwise anomalous? Briefly explain.

(1) a. They decided to go tomorrow yesterday.
b. They decided to go yesterday tomorrow.

(2) a. They decided yesterday to go tomorrow.
b. They decided tomorrow to go yesterday.

(3) a. Yesterday, they decided to go tomorrow.
b. Tomorrow, they decided to go yesterday.

(4) They decided to go yesterday yesterday.

(5) How long didn't Rebecca wait?

(6) Pelé came from a much poor family.

Exercise 1.8: Structural ambiguity

A. The following expressions are structurally ambiguous. For each reading (i.e. interpretation), provide a paraphrase that is itself unambiguous or a diagnostic scenario (that is, a scenario that is compatible only with the reading in question).

(1) a. chocolate cake icing
b. clever boys and girls
c. Charis will answer the question precisely at noon.
d. Laila will watch the man from across the street.

e. They should decide if they will come tomorrow.

B. For each reading, provide as much of a labeled bracketing as you can, focusing on distinguishing between the readings. In other words, you may not be able to give a full labeled bracketing, but for each reading, determine which words go together more or less closely.

Exercise 1.9: Outcomes of language prescription

In (2b) of §1.1 we saw this prescriptive rule:

(1) Don't end a sentence with a preposition.

In general, this is meant to capture the difference between the sentences in (2), where (2a) is prescriptively unacceptable (since the sentence ends with the preposition *with*) and instead we are taught that (2b) is the preferable sentence.

(2) a. Who did you go to the store with?
b. With whom did you go to the store?

To our ears, (2a) is comfortably natural in contemporary Mainstream U.S. English, whereas (2b) sounds awkward/stilted. But both are presumably possible for many U.S. English speakers.

A. Explain what relevance the facts in the examples below have for the prescriptive rule in (1).

B. Briefly discuss how syntacticians ought to engage with prescriptive rules.

(3) a. I ran into Sarah at the grocery store this afternoon.
b. Who did you run into?
c. * Into who(m) did you run?

(4) a. I really messed up the scheduling for this week.
b. What did you mess up?
c. * Up what did you mess?
d. I really messed it up.
e. * I really messed up it.

(5) a. I fell down the stairs yesterday.
b. I fell down them last week!
c. * I fell them down last week!
d. What did you fall down?
e. ? Down what did you fall?

(6) a. I burned down the house last week.
b. * I burned down it last week.
c. I burned it down last week.
d. What did you burn down?
e. * Down what did you burn?

Exercise 1.10: Distinguishing linguistic intuitions

As discussed in this chapter, linguistic intuitions can be tricky to perceive and describe.[1] This exercise is practice in doing this. Each of the pairs/sets of sentences below show a grammatical and natural sentence, and an anomalous sentence that is marked with a %. Your task is to specify (as much as possible) the nature of the anomaly: try to identify whether the issue is semantic (a problem with a meaning), pragmatic, or syntactic.

(1) a. Maya ate an apple and an orange.
 b. What did Maya eat?
 c. What did Maya eat an apple and?

(2) a. The children ate crackers for snack.
 b. % The children ate juice for snack.

(3) a. I baked and ate a cake for dessert.
 b. % I ate and baked a cake for dessert.

(4) a. He gave a gift to his mom.
 b. % He gave to his mom.
 c. % He gave a gift.
 d. % He gave to his mom a gift.

(5) a. I wonder why Alex likes Oscar so much.
 b. Who wonders why Alex likes Oscar so much?
 c. % Who do you wonder why Alex likes so much?

(6) a. I am happy man but I am not a strong man.
 b. % I am a happy man but I am not a man.

(7) a. It seems that the children want to devour the cake.
 b. The children seem to want to devour the cake.
 c. % The cake seems that the children want to devour.
 d. %The cake seems the children to want to devour.

(8) a. The community members gathered to protest.
 b. % Mike gathered to protest.

(9) % "She was entirely dressed in articles of clothing and had nothing on her feet except a pair of socks and two shoes." [Excerpt from Lemony Snicket: *The Unauthorized Autobiography* by Lemony Snicket (a.k.a. Daniel Handler), via Gretchen McCulloch's *All Things Linguistic*.]

1. This exercise was inspired by an exercise from Matt Pearson.

Problem 1.1: An article puzzle

Can you solve the riddle posed in endnote 4 of this chapter?

Problem 1.2: Logical dependencies in our theory

Syntactic structure and recursion are both formal universals of human language. Are they both equally basic properties of human language? Explain in a (possibly very) brief paragraph.

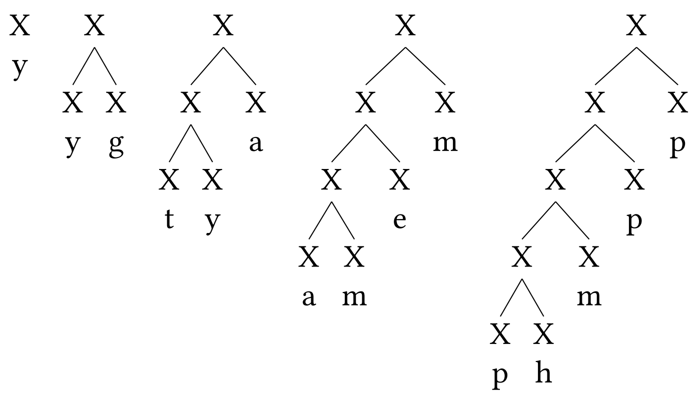
Problem 1.3: Constructing ambiguities

A. Can you come up with a sentence (or other expression) that is structurally ambiguous more than two ways? For each reading, provide an unambiguous paraphrase or a diagnostic scenario.

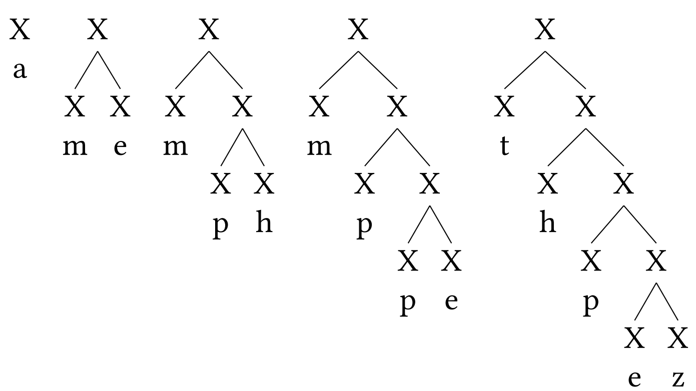
B. For each reading, which words go together? Provide a labeled bracketing, with a focus on distinguishing the interpretations (that is, you don't necessarily need to label each word).

Problem 1.4: Evaluating grammars

Assume a toy grammar G1 with a vocabulary of lowercase letters and a single grammatical category X. G1 generates structures like the following:

(1)


Another toy grammar G2 generates structures like this:

(2)


Briefly answer the following questions:

A. What is the difference between G1 and G2?

B. If presented with a string of lowercase letters (for example, "`abc`"), is it possible to tell which grammar has generated the string?

Problem 1.5: Language change

The grammars of Early Modern English (1500–1710) and present-day English differ enough for certain Early Modern English sentences to be ungrammatical today. Find several such sentences, and briefly describe the source of the ungrammaticality as best as you can. Early Modern English texts that are easily accessible on the Web include Shakespeare's plays and the Authorized Version of the Bible (also known as the King James Bible).

2

Grammatical categories

Chapter outline

As we will discuss in more detail in Chapters 3, 5 and 6, phrases come in various "flavors" like noun phrases, verb phrases, adjective phrases, adverb phrases, and several others. The reason is that the words that make up the phrases themselves belong to different grammatical categories. Traditionally, these are known as parts of speech.

However, the traditional notion of part of speech is flawed in two ways. First, the concept of "word" seems simple and straightforward, but isn't. Second, parts of speech are traditionally defined in terms of meaning, even though these definitions quickly break down. Our solution to these problems will be to allow grammatical categories to be associated with morphemes (rather than words) and to define them in terms of **morphological form** and **morphosyntactic distribution** (rather than semantic function).

The chapter is organized as follows. We begin by clarifying the notion of syntactic category itself, as just discussed, and presenting the four main lexical categories, that is, grammatical categories of lexical elements. We briefly discuss grammatical categories for functional elements, and then discuss the idea of abstract grammatical categories: grammatical categories that are composed of various kinds of subcategories. In this introduction we focus on **Inflection**, but there are others. We conclude with a discussion of whether grammatical categories are universal or not.

2.1 Introduction to grammatical categories (parts of speech)

In the U.S. education system, many people's first experiences with so-called grammar involves identifying parts of speech as part of the task of "diagramming sentences." So in an example like (1), we can label the words in the sentence based on the category they fall into.

(1) Dolphins play endlessly.
noun verb adverb

As we will show in this chapter, there is definitely a mental reality to this kind of categorization, but it also diverges significantly from the ways that we are taught in school.

2.1.1 Lexical and functional categories

Linguists tend to divide **lexical items** (memorized elements in a language) into two broad sets of categories. First are lexical categories, which tend to be open-class elements with real-world semantic content (words like *bicycle*, *pterodactyl*, *beautiful*, *fashionable*, etc; see § 2.1.2). Second are functional categories, which tend to mainly serve functions within grammar itself: to make sentences fit together, as it were. Functional categories have meaning that is less tangibly describable in the real world: words like *the*, *might*, and *that*, among many others. Rather than using the term **parts of speech**, we will use the term **grammatical category** to capture the same notion (the term 'syntactic category' is sometimes used for the same).

Lexical categories:	
Category	**Examples**
noun	*kumquat, Darius, happiness*
verb	*synthesize, endure, appear*
adjective	*glad, big, unequivocal*
adverb	*quickly, simply, luckily*

Functional categories:	
Category	**Examples**
preposition	*with, under, beside*
article	*the, a, an*
demonstrative	*this, that, yonder*
quantifier	*every, any, no, few*
pronoun	*me, us, I, who*
possessive pronoun	*hers, mine, theirs*
complementizer/subordinator	*that, whether, if*
auxiliary verb	*be, have, do*

Table 2.1: Selected English grammatical categories, both lexical and functional

In some sense, grammatical categories might seem simple or fundamental, and they are very useful in understanding grammatical patterns. But they can get somewhat complicated to define, especially given how we're accustomed to thinking about them.

2.1.2 Open and closed classes

Lexical categories tend to be open classes, meaning that new words are readily added to them consciously (for example, inventing a new word for a new technology). Closed classes, on the other hand, are much harder to add new words/morphemes to, especially intentionally. Most functional categories are closed-class, as noted in Table 2.1. So the inventory of prepositions in English is large, but it's uncommon for a preposition to be added to the language. On the other hand, new nouns and verbs are added to language all the time. Technology is a frequent driver of this in the modern world: it was not that long ago that words like *google* and *iPhone*, and phrases like *upload it to the web*, would have been meaningless in the 1980s. Slang and language change are also drivers of innovation in open classes: for example, *bae* as a term of endearment, *cap* to mean 'lie', *drip* to refer to someone's sense of style, *bet* as an interjection as a positive affirmation, etc.[1] What slang generally does not do is change verb conjugation paradigms, add new prepositions or complementizers to a language, etc.

The history of popular discussion about pronouns in U.S. culture is a good example of the psychological reality of open and closed classes; we report this social history here as described in Baron (1981). Unlike many languages, English has traditionally lacked a third person singular pronoun that is non-specified for gender.[2] So Mainstream U.S. English had *she* and *he*, and *it* for inanimate referents, but no singular pronoun for humans that was unspecified for gender. This has led to both the systematic exclusion of the female gender (use of default masculine pronouns such as the so-called generic *he*), but also to a history of nightmarish circumlocutions in attempts to be inclusive: sentences like *Each student should bring his or her assignment with him or her to class.*[3] But even these attempts at inclusion nonetheless still exclude anyone not within that gender binary. There is an extraordinarily long history of academics and others attempting to introduce a new gender-neutral third-person singular (also called **epicene**) pronoun: as far back as 1878 an anonymous column appears in the *Atlantic Monthly* magazine asking for a gender-neutral third-person pronoun and that “the eminent linguists leave the spelling reform and such trifles long enough to coin us a word”. In the second half of the 19th century gender-neutral pronoun proposals proliferated, producing a positively dizzying list of alternatives both lighthearted and serious through borrowings, blends, clippings, and root creations, among other strategies: *ne/nis/nim, hiser/himer, en, hi/hes/hem, le/lis/lim, unus, talis, ip/ips, ir/iro/im, tha/thar/them, ons, e/es/em, hizer.* The earliest of these date back to around 1850, while the most famous proposal is likely *thon/thons*, a blend of *that one.* Another explosion of proposals came in the 1970s on the tails of the feminist movement: *she/heris/herm, co/cos, ve/vis/ver, tey/ter/tem, fm, shis/shim/shims/shimself, ze/zim/zees/zeeself, per/pers, na/nan, shem, herm, ne/nis/ner, hir/herim, she/herm/hs, po/xe/jhe, e/e’s/em, sheme/shis/shem, heshe/hisher/himmer, em/ems, ae, hir.*

Alongside these neological proposals, the grammatically plural pronoun *they* has been used as a singular gender-neutral pronoun the way it is today since at least the time of Shakespeare (Bjorkman 2017). So instead of the awkward circumlocution above, for many speakers it is grammatical to say *Each student should bring their homework with them to class.* Uses of *they* like this, in tandem with words like *each* and *every*, are attested as far back as Middle English with Chaucer’s late-fourteenth century *Canterbury Tales* (Balhorn 2004). These uses tend to focus on instances where the identity of the referent is unknown, though still a singular usage.

A natural extension of this usage has happened much more recently on a broad scale, using *they* for singular referents who have a known identity. This would be a sentence like *Max said that they are bringing their friend*, where *they* refers to *Max.* This is the precise usage for which the long list of unsuccessful novel pronouns two paragraphs up have been proposed since 1850 or so.[4] This kind of historical shift is already familiar from the history of English: we used to have separate pronouns for 2nd person singular and plural (*thou* and *you*, respectively), which over time merged, and we now use the same pronoun (you) for an addressee whether they are a singular individual or multiple individuals.[5]

The point, of course, is that it is not impossible for changes to occur in closed classes like pronouns, but the tendency is for those changes to occur using forms already existing internal to the grammatical system in some way, borrowing either from other pronoun forms, or from other nominal forms in the language. Such changes are in fact not altogether uncommon: examples include *usted* ‘you (formal)’ < *vuestra merced* ‘your mercy’ in Spanish, *saya* ‘I (formal) < Sanskrit *sahāya* ‘servant’ in Indonesian, *kimi* ‘you (familiar)’ shifting meaning from ‘ruler’ in Japanese, and *ce* ‘I (humble)’ shifting meaning from ‘that one’ in Korean (Heine and Song 2011). This social history of pronouns in the U.S. is an interesting illustration of the nature of open and closed classes, as inside the U.S. this was a long recognized grammatical restriction with unpleasant social consequences: speakers were well aware of the need to refer to an individual without referencing gender, and also aware that English lacked a broadly-accepted mechanism to do so. But despite that need, no attempts to introduce new pronouns ever caught on: pronouns are closed classes and highly resistant to novel words. A borrowing/extension from within the pronoun system has been much more successful, however.

The trouble with "words" and "parts of speech" 2.2

What kinds of elements are able to be assigned a part of speech? Colloquially we think of a "word" as having a part of speech, but the concept of a "word" falls apart surprisingly quickly when you investigate it. The elements we call "words" generally have some degree of phonological and grammatical independence. So, phonologically, we can see a word standing alone, not requiring other content to be said; this is illustrated in the answer to the question below.

(2) Q: Which do you like better, coffee or tea?
A: Tea.

Likewise, many "words" tend to be able to combine with other "words" in flexible ways. So in (3) *tea* can be used as the subject or the object of the sentence, or as part of a prepositional phrase.

(3) a. Tea is good for you.
b. She doesn't drink tea.
c. There are beneficial antioxidants in tea.

But these notions start to fall apart fairly quickly once you poke at them a little. The sentences in (4) include the phonetically transcribed [t^hiz].

(4) a. Please do not [t^hiz] your younger sibling. (tease)
b. That shop sells [t^hiz] from around the world. (teas)
c. I think the antioxidants are [t^hiz] best property. (tea's = tea POSSESSIVE)
d. [t^hiz] good for you. (tea's = tea is)

Most of us would say without hesitation that *tease* in (4a) is a word. And even though *teas* in (4b) is made up of two morphemes (*tea* + PLURAL), we'd guess that most of you would still be inclined to call that a word. But are the instances of [t^hiz] *tea's* in (4c) and (4d) words? For many native English speakers, intuitions about what a 'word' is start to fall apart here. Despite being phonetically identical to the other elements we readily call words, in (4d) we have a sense that two different grammatical elements are being combined, a noun *tea* and a verbal element *is* (a similar argument can be applied to (4c)). But at the same time, most of us would also hesitate to call [tʰiz] in (4d) two words. So though we are taught in primary school (we can call it "4th grade grammar") that "words" have parts of speech, it's not clear what a word even is.

The issue becomes even more salient when looking at agglutinative languages, where entire sentences can be expressed with a single "word". Swahili is an example here, where what is one morphological and phonological string (*waliwaona*) contains all the components of a fully grammatical sentence (subject, tense, object, verb).

(5) Wa-li-wa-ona.
AGR-PST-AGR-see
'They (plural) saw them (plural).'

So perhaps at bare minimum we need to say that **morphemes** are assigned parts of speech, and not "words". But even a monomorphemic word can be trouble. Consider our 4th grade grammar definitions of major parts of speech, which tend to rely on semantic classifications.

(6)
a. **Noun** person place thing (or idea!)
b. **Verb** an action (or state!)
c. **Adjective** modifies a noun
d. **Adverb** modifies a verb or adjective

These classifications fall apart fairly quickly on closer inspection. A word like *yesterday* in (7a) we might want to say is an adverb, since it modifies a verb (yesterday is when the going happened). But *yesterday* in (7b) is clearly not an adverb, being used as a possessor (which is a position that only noun phrases occur in). But *yesterday* means the same thing in both instances.

(7)
a. We went to the beach yesterday.
b. Yesterday's beach trip was fun.

Likewise, *beer* is clearly a noun in (8a) (and typically would be considered a noun), but is used as a verb in (8b) to mean 'give (someone) a beer.'

(8)
a. Give me a beer!
b. Beer me! (Colloquial Mainstream English, = "give me a beer")[6]

The morphological properties of "words" further complicates things. *Reading* in (9a) behaves like a verb, yet in (9b) it behaves like a noun. But in each case it's referring to something we'd reasonably call the action of reading.

(9)
a. Maisha is reading an Elephant and Piggie book.
b. Daily reading is recommended for kindergarteners.

The canonical example of this from the literature is *destroy*, as in (10). A verbal use of *destroy* is unambiguous in (10a), but even though in (10b) *destruction* is a noun, it transparently describes an action.

(10)
a. The cat destroyed the couch.

b. the cat's destruction of the couch

So what do we do about our definitions from (6) above? Clearly they're not sufficient. In fact, a linguist is much more inclined to write definitions like those in (11).

(11)
a. **Noun** "words" that do nouny things
b. **Verb** "words" that do verby things
c. **Adjective** "words" that do adjective-y things
d. **Adverb** "words" that do adverby things

The definitions in (11) are of course silly, but they're actually not far off. What (11a) presupposes, for instance, is that there are such things as "nouny things". What this is saying is that grammatical categories are defined by their *grammatical distribution*, not by *semantic properties*. So what are "nouny" sorts of things to do? We can consider both morphological distribution and syntactic distribution. Syntactically, for example, nouns often serve as the subject or object of a sentence and they commonly follow determiners and adjectives. Morphologically, they can be marked as plural and take various derivational affixes that only attach to nouns (like *-tion* and *-er*, for example). In English, elements that behave in this way are consistently nouns. The tables below illustrate (non-exhaustively) some of the distributional properties of major grammatical categories in English.

Category	**Syntactic distribution**	**Examples**
Noun	▶ subject/object of sentence ▶ object of preposition ▶ often follow determiners/adjectives	*The small **dog** ran away.* *The chew toy is with **Lilly**.*
Verb	▶ takes a subject/object ▶ can follow auxiliary verbs like *be* and *have* ▶ can follow modals like *might* and *should* ▶ can follow the infinitive marker *to*	*Lilly might **run** away.* *Cooper wants Lilly to **run** away.*
Adjective	▶ often follow a determiner and precede a noun ▶ often follow linking verbs like *be* and *seem* ▶ can follow intensifiers like *very* and *so*	*The very **small** dog ran away.* *Lilly looks so **small**!*
Adverb	▶ often precede adjectives ▶ often precede or follow verb phrases ▶ some appear at the beginning of a sentence	*The small dog ran away **briefly**.* ***Luckily** it won't take a week.*

Table 2.2: Syntactic distribution of selected English grammatical categories (Nagelhout et al. 2018)

Category	Morphological distribution	Examples
Noun	▶ bears plural morphology ▶ some derivational affixes are only on nouns (e.g. *-tion*, *-er*)	*cat**s**, wish**es**, child**ren*** *predic**tion**, driv**er***
Verb	▶ take the third-person singular *-s* suffix ▶ bear past tense morphology ▶ can bear the affixes *-ize*, *-ate*, *-ify*	*loathe**s**, say**s**, trie**s*** *jump**ed**, dream**t**, froze* *pixel**ize**, formul**ate**, beaut**ify***
Adjective	▶ can bear the affixes *-ish*, *-like* ▶ can occur in comparative (*-er*) and superlative (*-est*) forms	*child**ish**, child**like*** *bigg**er**, bigg**est***
Adverb	▶ often bear the affix *-ly* ▶ can occur in comparative (*-er*/*more*) and superlative (*-est*/*most*) forms	*brief**ly**, cool**ly**, equal**ly*** ***more** briefly, cool**er**, cool**est***

Table 2.3: Morphological properties of selected English grammatical categories (Nagelhout et al. 2018)

The fact that distributional properties are primary in our knowledge of grammatical categories is evident in literature, especially children's literature, where invented nonsense words sometimes play a major role. The following is a brief selection from Dr. Seuss's *The Lorax*:

> You won't see the **Once-ler**. Don't knock at his door. He stays in his **Lerkim** on top of his store. He stays in his Lerkim, cold under the floor, where he makes his own clothes out of **miff-muffered moof**. And on special dank midnights in August, he peeks out of the shutters and sometimes he speaks and tells how the **Lorax** was lifted away. He'll tell you, perhaps… if you're willing to pay.
>
> …
>
> Then he pulls up the pail, makes a most careful count to see if you've paid him the proper amount. Then he hides what you paid him away in his **Snuvv**, his secret strange hole in his **gruvvulous** glove. Then he grunts. "I will call you by **Whisper-ma-Phone**, for the secrets I tell you are for your ears alone."
>
> *SLUPP!*
>
> Down **slupps** the Whisper-ma-Phone to your ear and the old Once-ler's whispers are not very clear, since they have to come down through a **snergely** hose, and he sounds as if he had smallish bees up his nose. "Now I'll tell you", he says, with his teeth sounding gray, "how the Lorax got lifted and taken away… It all started way back… such a long, long time back…

Each of the bolded words in the selection above were invented by Theodor Geisel (a.k.a. Dr. Seuss), and if you've never read *The Lorax*, this may be the first time you've encountered them. Sometimes we can get an interpretation: *"Down slupps the Whisper-ma-Phone"* invokes an onomatopoeic interpretation of *slupp* as the sound the phone is making while being lowered. Others, like *gruvvulous*, don't suggest much of an interpretation at all, instead achieving a stylistic and rhythmic effect. Yet in all cases we have a very clear sense of their syntactic category: *slupps* is a verb describing the motion of the Whisper-ma-phone, *gruvvulous* and *snergelly* are adjectives describing *glove* and *hose*, respectively, and *Lerkim*, *Snuvv*, and *Lorax* are all nouns. For each, our ability to readily identify their syntactic category is due to the fact that their use falls within the distributional criteria for each category, like those illustrated in Tables 2.2 and 2.3.

Again we are left to conclude that the proper defining characteristics of grammatical categories are not based in semantics: even nonsense words have grammatical categories, despite not meaning anything. Instead, grammatical categories are defined by grammatical criteria: their morphological and syntactic distributions. In the sections that follow, we provide additional description of the nuances around some prominent grammatical categories.

Major lexical categories 2.3

In this section we briefly introduce major lexical categories that are common across languages, as well as briefly describing relevant properties and sub-categories that arise within each. The goal here is not to be comprehensive, but rather to provide some common ground for the ways that these categories are referenced in this text.

2.3.1 Nouns

In the traditional school definition, a noun "refers to a person, place, or thing". But as has often been pointed out, this definition incorrectly excludes nouns like those in (12) (unless the concept of *thing* is reduced to be nearly vacuous).

(12) explosion, glint, mind, moment, thunder

As a result, modern linguists define nouns not in semantic (meaning-based) terms, but in distributional terms—with reference to their occurrence relative to other grammatical categories in the language. In English, for instance, a useful criterion for whether a word is a noun is whether it can form a phrase with the determiner *the*. According to this criterion, the words in curly brackets in (13a) are nouns, but those in (13b) are not.

(13) a. the { explosion, glint, mind, moment, thunder }
b. * the { explodes, glinted, minded, momentous, thundered }

Grammatical gender, number, and person

It is common for nouns and noun phrases to bear **features** for gender, number, and person. Perhaps the most familiar of these are number features: in English, a noun phrase can be either singular (ex., *cat*) denoting a single individual, or plural (ex., *cats*) denoting more than one individual. Singular vs. plural is the most common number distinction, but languages do display others. Many varieties of Arabic feature a **dual**: dual marking on a count noun means that there are two of those things. This occurs along with singular and plural numbers—Emirati Arabic is demonstrated here (data from Leung, Ntelitheos, and Al Kaabi 2020).

(14) a. مهندس مهندسين
muhandəs muhandəs-iin
engineer engineer-PL
‘engineer’, ‘engineers’

b. مُعلّم معلّمين
mʕalləm mʕallm-een
teacher teacher-DU
‘teacher’, ‘two teachers’

Many varieties of Arabic also feature a **singulative number**, which exists for so-called collective nouns. As their names imply, a collective noun indicates a collection and in some ways functions like a mass noun. A singulative noun indicates a countable unit of a collective noun. The singulative can in turn be pluralized, sometimes with the meaning of ‘a few’ or ‘a small amount’ (a **paucal** use). However, the meaning of this kind of plural is not always straightforward and can carry a wide range of other meanings as well. The data below, concerning Levantine Arabic, is from Brustad (2008), which discusses this kind of plural in more detail.

(15) a. سمك سمكة سمكات
samak samk-e samk-āt
fish fish-SINGULATIVE fish-PAUCAL
‘fish’, ‘a fish’, ‘fishes’

b. وسخ وسخة وسخات
wasaḵ wasḵ-e wasḵ-āt
dirt dirt-SINGULATIVE dirt-PAUCAL
‘dirt’, ‘a spot of dirt’, ‘dirt’

“Person” features may also be familiar from grade school grammar classes. “Person” refers to the role that an individual does or does not play in the current conversation. First person is the speaker, second person is the addressee, and third person is an independent party. Pronouns are commonly marked for person. Table 2.4 gives a subset of the English personal pronouns.

Person	Number	Subject forms	Non-subject forms
1	SG	I	me
2		you	you
3		he, she, they, it	him, her, them, it
1	PL	we	us
2		you	you
3		they	them

Table 2.4: Selected English personal pronouns

The pronouns above make clear another common property of nominals: grammatical gender, or more accurately, noun classes. English only shows this in third person singular pronouns, having masculine and feminine pronouns, along with a non-gendered pronoun (singular *they*). English also has an inanimate third-person singular pronoun (*it*). But some languages have pervasive marking of gender on nominals, such that all nominals (whether human/animate or not) are specified as being either masculine (M) or feminine (F). Examples from Bulgarian and Amharic are given below.

(16) a. дроб *drób*.M, ‘liver’, дрóбът *dróbât*.M ‘the liver’ **Bulgarian** (Nicolova 2017)
b. дроб *drób*.F, ‘fraction’, дробтá *drobtá*.F ‘the fraction’

(17) a. ፈረስ *färäs*.M, ‘horse’, ፈረሱ *färäsu*.M ‘the horse’ **Amharic** (Leslau 2000)
b. ጸሐይ *ṣähay*.F, ‘sun’, ጸሐይዋ *ṣähaywa*.F ‘the sun’

(18) a. Bul. лунá *luná* ‘moon.F’, Amh. ጨረቃ *č̣äräḳa* ‘moon.M’

As has commonly been noted, the word for ‘moon’ is masculine in some languages (such as Amharic), and feminine in others (such as Bulgarian). This of course is striking, since all humans are looking at the same moon. Clearly, then, gender in this sense is not about human sex or gender at all, but is a categorical grouping of nominals that is relatively arbitrary: essentially, a classification system for nouns that happens to correspond to human sex/gender in some languages. This is why some languages have only masculine and feminine (like French and Arabic) whereas others (like German and Albanian) have three genders: masculine, feminine, and neuter.

This is even more evident in looking at noun class systems cross-linguistically. We will use Lubukusu, a Bantu language spoken in western Kenya as an example. All nominals in Lubukusu fall into some noun class, but these classes do not correspond in any way to the gendered classification systems of Indo-European languages. Table 2.5 gives the Lubukusu noun classes, which are labeled with cardinal numbers (1, 2, 3, etc). Odd numbers are singular classes, and (usually) the immediately following even numbers are the plural form for those nouns. Class 1-2 is for humans, a fairly strong semantic restriction. But otherwise there are limited strict semantic associations with particular noun classes. For example, the Lubukusu terms for arm/hand (*kumukhono* class 3) and foot/leg (*siikele*, class 7) both describe human appendages but are in different noun classes.

In Lubukusu there are kinds of nominals that fall into the noun class system that don’t have direct parallels in gender-based noun class systems. For example, most nominals (like *ku-*

Table 2.5: Lubukusu Noun Classes (Mutonyi 2000, 6)

Class	Pre-Prefix	Prefix	Example	Gloss
1	o-	mu-	ómukhasi	‘woman’
2	ba-	ba-	babaana	‘children
3	ku-	mu-	kúmukhono	‘arm/hand’
4	ki-	mi-	kímikhono	‘arms/hands’
5	li-	li-	lilyaanda	‘ember’
6	ka-	ma-	kamaanda	‘embers’
7	si-	si-	sisyaangu	‘sponge’
8	bi-	bi-	bibyaangu	‘sponges’
9	e-	N-	eendubi	‘basket’
10	chi-	N-	chiindubi	‘baskets’
11	lu-	lu-	lúlwiika	‘horn’
12 (Diminutive)	kha-	kha-	khákhaana	‘small child’
14	bu-	bu-	búbwíino	‘ink’
15	khu-	khu-	khukhwaanja	‘to begin’
16 (‘at’)	a-		amulyaango	‘at/near the door’
16a (‘towards’)	sya-		syamulyaango	‘towards the door’
17 (‘on’)	khu-		khumulyaango	‘on the door’
18 (‘in’)	mu-		mumulyaango	‘in the door’
20 (Augm)	ku-	ku-	kúkwaana	‘big child’
(4) (Aug)	ki-	mi-	kímyaana	‘big children’
23 (‘in the vicinity of’)	e-		énaaróbi	‘at Nairobi’

mulyaango 'door' in (19a)) can be transformed into a location by putting them in either class 16, 17, 18.

(19) Lubukusu locative forms (Mutonyi 2000, 25–27)

a. ku-mu-lyaango
3-3-door
'door'

b. a-mu-lyaango
16-3-door
'near the door'

c. khu-mu-lyaango
17-3-door
'on the door'

d. mu-mu-lyaango
18-3-door
'in the door'

The features of noun class (grammatical gender), number, and person are commonly referred to collectively as **phi-features** (or written using the Greek letter, φ-features).

Count nouns and mass nouns

Nouns are traditionally divided into two major subclasses: **count nouns** and **mass nouns**. Count nouns (in Mainstream U.S. English) can be accompanied by the indefinite article, and they have a plural form, whereas mass nouns have neither of these properties.

(20) Count noun:
  a. a(n) { book, child, couch, explosion, invitation, job, problem, vegetable }
  b. books, children, couches, explosions, invitations, jobs, problems, vegetables

(21) Mass noun:
  a. * a(n) { advice, furniture, rice }
  b. * advices, furnitures, rices

Conversely, singular count nouns cannot follow expressions of quantity like *a lot of, much*, and so on, whereas mass nouns can.[7]

(22) Count noun:
  a. * a lot of { book, child, couch, explosion, invitation, job, problem, vegetable }
  b. * much { book, child, couch, explosion, invitation, job, problem, vegetable }

(23) Mass noun:
  a. a lot of { advice, furniture, rice }
  b. much { advice, furniture, rice }

Proper nouns and common nouns

A further distinction is traditionally drawn between **proper nouns** (also known as **proper names**) and **common nouns**. In contrast to common nouns, proper nouns are conventionally capitalized in English.

(24) a. Common nouns:
city, collie, country, planet, state

b. Proper nouns:
Philadelphia, Lassie, France, Jupiter, Maryland

Proper nouns refer to particular individual entities. The relation between proper nouns and individuals is not necessarily one-to-one, however. For instance, the proper noun *Sandra* may refer to many different individuals of that name. Individuals with the same proper name needn't share any distinguishing properties (other than having the same name). So Athens, Greece and Athens, Georgia needn't have any substantive property in common that sets them apart from other cities, and there is nothing to stop us from giving the name *Lassie* to a pet without any of the prototypical qualities of Lassie from the television show (intelligence, loyalty, and so on). Our Lassie, in fact, needn't even be a dog. In short, proper nouns function like pure pointers. In the same way that one and the same pointer can be used to point to different and unrelated items in a presentation, the same proper noun can be used to refer to different and unrelated individuals. In contrast, common nouns refer to sets of entities that are related by sharing certain properties. That is, common nouns have intrinsic semantic content and cannot be used in the relatively arbitrary way that proper nouns can be.[8]

2.3.2 Verbs

As we will discuss in Chapter 4, in most sentences a predication relationship is the heart of the sentence. What is predication? Speaking informally, it is the attribution of some property to an individual. So in the sentence *Lilly is a dog*, *dog-ness* is being attributed to Lilly. The sentence *Lilly is gray* attributes the property of being gray to Lilly. The sentence *Lilly ate dinner* attributes the property of eating dinner to Lilly. In each sentence, the sentence says *something* (the predicate) *about* an entity (*Lilly*, in this instance). We will mention predicative uses of various categories in this chapter, but see § 4.1 of Chapter 4 for a more comprehensive discussion of the notion of a predicate.

Verbs are language's main tool for forming predicates of this sort. They serve to communicate either that some state holds, or some event is taking place. They are therefore identifed in most languages as the grammatical category that communicates the state or event being reported, and which co-occurs with **arguments** of various sorts. Arguments are the central participants in an event.[9] The verbs are in bold in the examples below.

(25) a. It is **raining**.

b. The children will **jump** in the puddles.

c. The dog **rolled** in a mud puddle.

d. We **drank** hot chocolate to warm up afterward.

Many languages have one or more special verbs, termed copulas, which allow other entities (like nouns or adjectives) to serve as predicates. In Mainstream U.S. English the copula is *be*.

(26) a. Lilly **is** gray.
b. Lilly **is** sleepy.
c. Lilly **is** a dog.
d. Maisha **is** a student.

The particular identifying aspects of grammatical categories vary across languages, but there are also consistent and persistent patterns cross-linguistically. For verbs, it is quite common for them to inflect in various ways: verbs tend to inflect for the tense or aspect of the construction they occur in, for example. **Tense** marks whether an event is in the past, present, or future, and **aspect** communicates a perspective on the event, namely, whether an event is ongoing (ex., imperfective: *Lilly is sleeping*) or an event is completed (ex., perfective: *Lilly slept*).

(27) Mainstream U.S. English tense
a. The children ate lunch. **Past tense**
b. The children eat lunch (every day). **Present/habitual tense**
c. The children will eat lunch. **Future tense**

(28) Mainstream U.S. English aspect
a. The children are eating lunch. **imperfective/progressive aspect**
b. The children have eaten lunch. **perfective aspect**

Likewise, it is quite common cross-linguistically for verbs to agree with their arguments, particularly their subjects. We only see this in Mainstream U.S. English with third person singular:

(29) a. A horse walk-**s** into a bar.
b. Horses walk-∅ into a bar.

Many other languages show robust agreement on verbs; Bantu languages are highly agglutinative, with morphology on verbs inflecting with the features of both subjects and objects. The example in (30) shows the subject marker (SM) agreeing in noun class 1 with the class 1 subject *mama* 'mother', with an object marker (OM) also on the verb agreeing with the class 2 object *watoto* 'children.'

(30) a. Mama a-li-wa-ona wa-toto. **Swahili**
1mother 1SM-PST-2OM-see 2-children
'Mother saw the children.'
b. Wa-toto wa-li-mw-ona mama w-ao.
2-children 2SM-PST-1OM-see 1mother 1-their
'The children saw their mother.'

This example also illustrates the tense-marking on finite verbs that occurs in Swahili, the past tense *li-* prefix. The sentences in (31) are the same as (30a) but in different tenses:

(31) a. Mama a-na-wa-ona wa-toto. **Swahili**
1Mother 1SM-PRES-2OM-see 2-children
'Mother is seeing the children.'

b. Mama a-ta-wa-ona wa-toto. **Swahili**
1Mother 1SM-FUT-2OM-see 2-children
'Mother will see the children.'

Being inflected with tense and agreement is not a condition for being a verb, of course. It is just a typical property in many languages. There are some languages that don't inflect verbs at all, such as Mandarin; the verbs in (32) all take the same form despite being in different tenses. Distinctions in time are communicated in other ways, such as the temporal adverbs in (32).

(32) Verbs lack inflection in Mandarin

a. 我 今天 去 看 电影。
wǒ jīntiān qù kàn diànyíng.
I today go see movie
'I'm going to see a movie today.'

b. 我 昨天 去 看 电影。
wǒ zuótiān qù kàn diànyíng.
I yesterday go see movie
'I went to see a movie yesterday.'

c. 我 明天 去 看 电影。
wǒ míngtiān qù kàn diànyíng.
I tomorrow go see movie
'I'm going to see a movie tomorrow.'

Even in languages like English and Swahili there are constructions as in (33) where there is no tense or agreement inflected on the verb.

(33) a. I let [my mom **watch** the children]

b. N-a-taka [**ku-leta** zawadi]
1SG.SM-PRES-want INF-bring gifts
'I want to bring gifts.'

Tables 2.2 and 2.3 offer some summaries of the syntactic and morphological distributional properties of verbs in English. The discussion here just calls out some major cross-linguistic patterns, but the particulars of what distinguishes a verb from a noun (or something else) will vary from language to language. This introduction should serve our current purposes, though, of establishing some baseline expectations regarding properties of verbs.

2.3.3 Adjectives

Adjectives vs. modifiers

In the teaching of high school English in the U.S., the terms “adjective” and “modifier” seem to be used relatively interchangeably. From the point of view of linguistic theory, this usage reflects a lamentable lack of appreciation for the central linguistic distinction between function and form. **Modifiers** are linguistic expressions that serve a certain function—namely, to restrict or qualify some other expression. **Adjectives**, on the other hand, are members of a syntactic category that is defined by certain formal properties. For instance, it is possible to derive adverbs from many adjectives (*heavy, heavily*; *mere, merely*; *rough, roughly*; *sweet, sweetly*).

There is no one-to-one relation between modifiers and adjective phrases. Modifiers are not necessarily adjective phrases, as illustrated in (34): each of the italicized phrases are modifers, but only the first is an adjective phrase.

(34) a. Adjective phrase:
a *very aggressive* driver
b. Adverb phrase:
They drive *very aggressively*.
c. Prepositional phrase:
the desk *next to the window*
They arrived *on time*.
d. Noun phrase:
They will arrive *Monday night*

Conversely, adjective phrases are not necessarily modifiers. For instance, the adjective phrase in (35) is the predicate of a simple clause.

(35) This candidate is *very suitable*.

Adnominal versus predicative use

In English, adjective phrases that modify a noun typically occur in **prenominal** position.

(36) a. a *very big* box, this *blue* book
b. * a box *very big*, this book *blue*

However, there are other languages that typically require adjectival modifiers of nouns to be **postnominal**. The examples in (37) are from French.[10]

(37) a.		une	chaise	*très*	*confortable*,	ce	livre	*bleu*
		a	chair	very	comfortable	this	book	blue
b.	*	une	*très*	*confortable*	chaise,	ce	*bleu*	livre
		a	very	comfortable	chair	this	blue	book

A convenient cover term for 'prenominal' and 'postnominal' is **adnominal**. It is worth noting that certain adjectives can only be used adnominally, and not predicatively.

(38) a. the mere fact, her utter surprise
b. * The fact is mere, Her surprise was utter

The postnominal use of adjective phrases should not be confused with their predicative use, as illustrated above in (35).

In many languages, adjectives inflect to match the features of the nominal that they modify. In the Spanish (39) and Swahili (40) examples below, the adjectives *small* and *big* inflect to match the gender/noun class of the nominal.

(39) sill-a pequeñ-a ; libr-o pequeñ-o — Spanish
chair-F small-F book-M small-M
'small chair; small book'

(40) m-tu m-kubwa ; ki-tabu ki-kubwa ; ∅-nyumba ∅-kubwa — Swahili
1-person 1-big 7-book 7-big 9-house 9-big
'big person; big book; big house'

Despite adjectives sometimes bearing phi-features, they don't inherently bear phi-features: the phi-features they bear are dependent on the nominal they are modifying. Therefore we still view phi-features as lexical properties of nominals, not adjectives.

2.3.4 Adverbs

The last major lexical category we will cover are **adverbs**, though we will also address other kinds of adverbial modifiers here too. Roughly speaking, adverbs modify verb phrases and sentences. In English 4th grade grammar these are often identified as words that end in *-ly*, as listed in (41a), but English is also rich in adverbs that do not end in *-ly* (41b):

(41) Selected English adverbs:
a. *quickly*, *slowly*, *poorly*, *happily*, *sadly*, *early*, *accidentally*, *cheerfully*, *usually*, *frequently*, *honestly*, *frankly*
b. *well*, *today*, *tomorrow*, *yesterday*, *always*, *never*, *fast*, *hard*, *late*, *soon*, *often*, *sometimes*

In the same way we referred to forms of adnominal modification above, there are forms of adverbial modification that don't themselves involve adverbs. So we will at times refer to **adverbial modifiers** that are not themselves adverbs: prepositional phrases are common (42a), as are noun phrases (42b):

(42) a. I will go to town [$_{PP}$ **in the morning**] .
b. I will go to town [$_{NP}$ **the day after tomorrow**] .

So we can see that not all elements participating in adverbial modification are themselves from the grammatical category that we call adverbs. In fact, adverbs as a class are in fact somewhat poorly named, as they often modify *sentences*, not verbs themselves. Arguably some adverbs do modify verbs or verb phrases. But many other kinds of adverbs clearly modify sentences as a whole. The adverbs in (43) are generally thought to be modifying verb phrases.

(43) a. I read the book **quickly** last week.
b. I built that table **poorly**.
c. I made enchiladas **yesterday**.

In contrast, the adverbs in (44) all do not directly modify a verb phrase (e.g. telling the time or manner or place that something happened), and instead describe a speaker's stance toward's the sentence that they uttered: it's relative likelihood (modal adverbs), the strength of evidence supporting the claim (evidential), or a speaker's evaluation of the proposition being reported.

(44) Selected sentential adverbs (Ernst 2014)
a. Epistemic adverbs:
i. modal: *probably*, *certainly*, *possibly*, *maybe*, *definitely*, *necessarily*, *perhaps*
ii. evidential: *obviously*, *clearly*, *evidently*, *allegedly*
b. Evaluative adverbs: *(un)fortunately*, *mysteriously*, *tragically*, *appropriately*, *significantly*

Adverbs play a highly significant role in syntactic analysis, and in unexpected ways. It turns out that the syntactic position of adverbs varies, but is highly predictable based on their semantics. So adverbs describing the manner that an event happened in tend to modify the verb phrase (in English, this means they occur towards the right of a sentence). Adverbs that are speaker-oriented, however, modify the sentence as whole—in English, they appear towards the left edge of the sentence.

(45) Lilly **obviously** needs a bath **today**.

Understanding the position of adverbs can then be very helpful in analyzing syntactic puzzles: if we know where an adverb is expected to occur, we can use that knowledge to analyze the structure of a construction we are curious about. We will see adverbs used in precisely this way in Chapter 12.

Major functional categories 2.4

In contrast to lexical categories that are those normally considered rich in real-world meaning, functional categories either have minimal semantic content or have much more restricted interpretations. Functional categories are normally closed classes, and therefore have much smaller inventories within each class. We will spend less time on functional categories than we did for lexical categories, as much of the course will be spent clarifying their properties. But it will be useful to be on the same page regarding basic terminology. We will focus our discussion on English at the moment.

Determiners are elements that introduce noun phrases. This includes the traditional so-called 'articles' like *a(n)* and *the*, but in English also includes demonstratives like *this* and *that*.

(46) **a** book; **the** book; **these** books

Quantifiers are grammatical elements that quantify nominals—that is to say, they communicate something about the amount of a thing (for mass nouns) or number of things (for count nouns).

(47) Selected English quantifiers:
every, *any*, *no*, *few*, *many*, *some*

(48) Every student attended class today.

Pronouns are a category of elements that can substitute for noun phrases. Regular pronouns like *me*, *they*, and *it* substitute for noun phrases. Possessive pronouns represent possessors or entire possessive noun phrases: these are forms like *her(s)*, *my*, *mine*, and *their(s)*.

We use the term **complementizer** for morphemes that introduce subordinate clauses (they are sometimes referred to as *subordinators*).

(49) Selected English complementizers:
that, *whether*, *if*, *since*, *until*, *after*

(50) a. I know **that** Liverpool will win the Cup.
b. I wonder **whether** Chelsea will ever be good again.

One final functional category that we'll mention at the moment are prepositions, which are used in many languages to describe locations (among other functions).

(51) Selected English prepositions:
with, *under*, *beside*, *in*, *next to*, *up*, *down*, *across*, *around*, *at*, *off*, *toward*

(52) Sample English prepositional phrases:
a. **up** the hill
b. **down** the mountain
c. **at** the house
d. **across** the ocean

These are just a small selection of grammatical categories that appear in English, let alone the world's languages. We discuss auxiliary verbs and modals at length in the next section, which are common functional categories in syntax. But in fact, there is vast variation between languages in the kinds of functional grammatical categories they possess, and what their specific properties are. We will expand our understanding of the functional categories of human languages throughout the book.

Abstract categories: Inflection as a case study 2.5

We discuss some additional grammatical categories in this section, again focused on English for our present purposes as it allows us to use a familiar language to motivate the more abstract concepts. We introduce a group of grammatical categories that are connected with a more abstract category, that of finite inflection of a clause. We show how varied properties of verbs, auxiliaries, and modals in English all point to an important underlying property of the clause, the presence of the abstract category of Inflection (Infl).

2.5.1 Modals

Historically, English modals, which are listed in (53), derive from a special class of verbs in Proto-Germanic (the ancestor of English and the other Germanic languages).

(53) English modals:
can, *could*, *may*, *might*, *must*, *shall*, *should*, *will*, *would*

Modals have always differed from ordinary verbs in Germanic, and over the history of English, they have diverged even further, to the point where they now behave as a grammatical category of their own. Because the modals have meanings that are often expressed in other languages by verbal inflections, this category has been called I(nflection).

Morphology of modals

Modals differ markedly from verbs in the range of forms that they exhibit. English verbs appear in a number of distinct forms (see § 2.9.1 for details), whereas modals have a single, invariant form.

Modals never end in *-s.* even in sentences with third person singular subjects.

(54) a. ✓ She { can, may } play the piano.
b. * She { can-s, may-s } play the piano.

Modals also lack productive past tense forms. It is true that *could, might, should,* and *would* originated in Germanic as past tense forms of *can, may, shall,* and *will.* But today, only *could* can serve as the past tense of *can,* and that only in certain contexts.[11]

Example		Potential paraphrase
Nowadays, you can get one for a dollar.	=	Nowadays, it is possible to get one for a dollar.
Back then, you could get one for a dollar.	=	Back then, it was possible to get one for a dollar.
We can go there tomorrow.	=	It is possible for us to go there tomorrow.
We could go there tomorrow.	≠	It was possible for us to go there tomorrow.
You may ask the boss.	=	You are allowed to ask the boss.
You might ask the boss.	≠	You were allowed to ask the boss.
Shall I pick up some bread?	=	Is it a good idea for me to pick up some bread?
Should I pick up some bread?	≠	Was it a good idea for me to pick up some bread?

Table 2.6: Lack of productive past tense forms for English modals

English modals also do not have nonfinite forms the way that verbs do (infinitives, *-ing* forms, past participles). We attempt (ungrammatical) paraphrases here to show what they possibly could have been, to illustrate the differences with verbs. We also provide a paraphrase with a verbal form (*be able to*) that shares meaning with the modal considered here (*can*) to show that the meaning itself is unproblematic.

English verbs can appear (untensed) after a modal, but modals cannot follow other modals in the same way in Mainstream U.S. English:

(55) Infinitive after (another) modal:
  a. * They must can speak Spanish. (modal)
  b. ✓ They must be able to speak Spanish. (paraphrase of modal)
  c. ✓ They must speak Spanish. (verb)

Similarly, the infinitive marker *to* can precede untensed verbs, but cannot precede modals.

(56) Infinitive after infinitival *to*:
  a. * They want to can speak Spanish. (modal)
  b. ✓ They want to be able to speak Spanish. (paraphrase of modal)
  c. ✓ They want to speak Spanish. (verb)

The *-ing* morphology that makes a gerund (a nominal form of a verb) is acceptable on verbs, but unacceptable on modals.

(57) *-ing* as gerund:
  a. * They like { cann-ing, may-ing } play the piano. (modal)
  b. ✓ They like { being able, being allowed } to play the piano. (paraphrase of modal)
  c. ✓ They like learning to play the piano. (verb)

Finally, the past participle form that appears with the auxiliary verb *have* (*have … V-ed*) is grammatical with verbs, but ungrammatical with modals.

(58) Past participle:
 a. * They have { cann-ed, may-ed } play the piano. (modal)
 b. ✓ They have { been able, been allowed } to play the piano. (paraphrase of modal)
 c. ✓ They have learned to play the piano. (verb)

In short, despite deriving from more verb-like forms historically, in contemporary English (specifically Mainstream U.S. English) modals are a distinct grammatical category from verbs.

Syntax of modals

The loss of nonfinite forms for modals gives rise to three further differences between verbs and modals, all of them manifestations of an important phenomenon in the grammar of modern English called ***do* support**.

In the simplest case, emphatic *do* appears in affirmative sentences whose truth is being emphasized. It involves replacing the finite verb by the verb's bare form and adding a form of auxiliary *do* to the sentence in the appropriate tense (either present or past tense). This form of *do* then receives emphatic stress, as indicated by underlining in (59).

(59) a. ✓ He dances; she sang. (unemphatic, without *do* support)
 b. ✓ He does dance; she did sing. (emphatic, with *do* support)

By contrast, emphasizing the truth of a sentence with a modal is achieved by simply stressing the modal, and emphatic *do* is ungrammatical with modals.

(60) a. ✓ He can dance; she will sing. (emphasis without *do* support)
 b. * He does can dance; she does will sing. (emphasis with *do* support)

Emphatic *do* therefore is another construction where differences emerge between modals and verbs.

Another such difference emerges in negated sentences. Negated sentences without modals require *do* support in modern English, i.e. the addition of the auxiliary *do* that does not appear in declarative sentences. As in the case of emphasis above, the verb appears in its bare form, and an appropriately tensed form of auxiliary *do* is added to the sentence, followed by negation.

(61) a. ✓ She { does, did } not dance. (negation with *do* support)
 b. * She not { dances, danced }. (negation without *do* support)
 * She { dances, danced } not.

By contrast, English sentences containing modals are negated by simply adding *not* (or its contracted form *-n't*) after the modal. *Do* support is ungrammatical.

(62) a. ✓ He { may, must, should, will, would } not dance. (negation without *do* support)
 b. * He does not { may, must, should, will, would } dance. (negation with *do* support)

The final syntactic difference between modals and verbs concerns question formation. If a declarative sentence contains a modal, the corresponding question is formed by inverting the modal with the subject. *Do* support is ungrammatical.

(63) a. ✓ { Can, may, must, should, will, would } he dance? (question without *do* support)
b. * Does he { can, may, must, should, will, would } dance? (question with *do* support)

In a sentence without a modal, question formation requires *do* support. That is, it is an appropriately tensed form of *do,* rather than the verb itself, that inverts with the subject.

(64) a. ✓ { Does, did } she dance? (question with *do* support)
b. * { Dances, danced } she? (question without *do* support)

Across the board, then, we can see that modals are not verb-like in many ways in modern Mainstream U.S. English. It is reasonable to describe them as their own grammatical category, and linguists often do. But what we will see in what follows here is that there are multiple ways that modals are a part of a superceding grammatical category as well.

2.5.2 Auxiliary verbs in English

English has three auxiliary verbs, each **homophonous** with an ordinary verb: *be*, *do*, and *have.*

Auxiliary *do*

The goal of the previous section was to establish the special status of modals, and we used the facts of *do* support as a criterion for distinguishing modals from verbs. In this section, we consider some of the same facts, but with a different focus. Rather than focusing on the distinctive properties of modals, we focus on the morphological and syntactic properties of auxiliary *do* itself.

A major difference between auxiliary *do* and the modals is that it has an *-s* form. In this respect, it patterns with ordinary verbs, including its main verb counterpart.

(65) Modal:
a. I can dance the polka.
b. He { can, * can-s } dance the polka.

(66) Auxiliary *do* (emphasis, negation, question):
a. I do dance the polka. I do not dance the polka. Do you dance the polka?
b. She do-es dance the polka. She do-es not dance the polka. Do-es she dance the polka?

It's important to distinguish another use of *do* in English, that of a main verb: we will see the same for other auxiliaries in English as well. When *do* is a main verb, it can be the only verb in a sentence:

(67) Main verb *do*:
  a. I do the dishes.
  b. He do-es the dishes.

(68) Ordinary lexical verb:
  a. I dance the polka.
  b. She dance-s the polka.

In this main verb usage, *do* can co-occur with a modal, just like other lexical verbs can.

(69) a. I can do the dishes.
  b. She can dance the polka.

We will treat main verb *do* and auxiliary *do* as separate grammatical elements that are **homophonous** (are pronounced the same way).

In many other respects, auxiliary *do* behaves (syntactically) like a modal rather than an ordinary verb. In particular, it does not occur in nonfinite (untensed) contexts. Notice the clear contrast between the judgments for auxiliary *do* in (71) and main verb *do* in (72).[12]

(70) Modal:
  a. * They will can dance the polka. (after another modal)
  b. * They want to can dance the polka. (after infinitival *to*)
  c. * Their canning dance the polka while blindfolded is unusual. (gerund)

(71) Auxiliary *do* (emphasis):
  a. * They will do dance the polka. (after another modal)
    Intended meaning: It will be the case they do dance the polka.
  b. * They claim to do dance the polka. (after infinitival *to*)
    Intended meaning: They claim that they do dance the polka.
  c. * Their doing dance the polka while blindfolded is unwise. (gerund)
    Intended meaning: That they do dance the polka while blindfolded is unwise.

(72) Main verb *do*:
  a. ✓ They will do the dishes. (after another modal)
  b. ✓ They want to do the dishes. (after infinitival *to*)
  c. ✓ Their doing the dishes was considerate. (gerund)

(73) Ordinary verb:
  a. ✓ They will dance the polka. (after another modal)
  b. ✓ They want to dance the polka. (after infinitival *to*)
  c. ✓ Their dancing the polka while blindfolded is unwise. (gerund)

The main lesson here is that auxiliary *do* is not a main verb, and is not a modal, though it shares a number of properties with modals. The next two auxiliaries that we consider are also distinct from main verbs but show further distinctions from modals as well.

Auxiliary *have*

Let's now turn to auxiliary *have,* which combines with past participles to form the **perfective** forms of verbs, for example, a sentence like (74). Perfective aspect takes an external viewpoint on the event being reported, that is to say, in (74) the speaker's viewpoint on the event is external to the event, considering the event to be complete (and here, in the past).

(74) The children **have** eat**en** lunch.

Auxiliary *have* behaves like a V with respect to its morphology and its occurrence in nonfinite contexts. (75) and (76) show that auxiliary *have*, like auxiliary *do* (cf. (66)), behaves morphologically like its main verb counterpart in having an *-s* form.

(75) Auxiliary *have*:
- a. I have adopted two cats.
- b. He ha-s adopted two cats.

(76) Main verb *have*:
- a. I have two cats.
- b. He ha-s two cats.

Auxiliary *have* differs from auxiliary *do* (compare (71)) and resembles main verbs in being able to appear in uninflected (nonfinite) contexts, ex., selected by a modal (a examples), in infinitive contexts (b examples) and as a gerund selected by a verb like *regret.*

(77) Auxiliary *have*:
- a. ✓ They must have adopted two cats. (after modal)
- b. ✓ They claim to have adopted two cats. (after infinitival *to*)
- c. ✓ They do not regret having adopted two cats. (gerund)

(78) Main verb *have*:
- a. ✓ They must have two cats. (after modal)
- b. ✓ They claim to have two cats. (after infinitival *to*)
- c. ✓ They do not regret having two cats. (gerund)

However, just like modals (cf. (ii)) and in contrast to main verb *have*, auxiliary *have* is ruled out in *do* support contexts, i.e. the contexts where *do* is inserted.

(79) Auxiliary *have*:
- a. ✓ He HAS adopted two cats. (emphasis)
- b. ✓ He hasn't adopted two cats. (negation)
- c. ✓ Has he adopted two cats? (question)

(80) Auxiliary *have* incompatible with do-support:
- a. * He does have adopted two cats. (emphasis)
- b. * He doesn't have adopted two cats. (negation)

c. * Does he have adopted two cats? (question)

(81) Main verb *have*:

a. ✓ She does have two cats. (emphasis)

b. ✓ She doesn't have two cats. (negation)

c. ✓ Does she have two cats? (question)

Therefore we can see that auxiliary *have* behaves similarly to modals and auxiliary *do*, inverting in questions and being stressed in emphatic contexts. Likewise, auxiliary *have* is incompatible with adding do-support to a sentence in negation and questions. But it is also fundamentally different from modals and auxiliary *do*, in that there is an uninflected version of the auxiliary that can co-occur with modals, similar to a main verb. So *inflected* auxiliary *have* behaves like modals and auxiliary *do*, but uninflected *have* behaves more like a main verb.

Auxiliary *be*

The auxiliary *be* is used to form the progressive (*is coming, was dancing*).[13] Auxiliary *be* behaves just like auxiliary *have* with respect to the properties we have been considering. In particular, it has an inflected form (irregular though that form is), it can appear in nonfinite (uninflected) contexts, but it is excluded from *do* support contexts. Note, all of these examples are specifically about Mainstream U.S. English; some varieties of English have another *be* auxiliary with additional functions.

(82) and (83) show that auxiliary *be* has inflected forms (unlike modals, and like auxiliary *do*) but also has uninflected forms (like auxiliary *have* and main verbs, but unlike auxiliary *do*).

(82) Auxiliary *be* has inflected forms:

a. ✓ They are learning Spanish. They are invited to the ceremony. (non-third person)

b. ✓ He is learning Spanish. He is invited to the ceremony. (third person)

(83) Auxiliary *be* has uninflected forms:

a. ✓ They must be learning Spanish. (after modal)

b. ✓ They must be invited to the ceremony.

c. ✓ They claim to be learning Spanish. (after infinitival *to*)

d. ✓ They claim to be invited to the ceremony.

e. ✓ They don't regret being invited to the ceremony. (gerund)

(84) Auxiliary verb *be*, with emphasis, negation, and questions:

a. ✓ He IS learning Spanish. They ARE invited to the ceremony. (emphasis)

b. ✓ He isn't learning Spanish. They aren't invited to the ceremony. (negation)

c. ✓ Is he learning Spanish? Are they invited to the ceremony? (question)

(85) Auxiliary verb *be* without emphatic *do* or *do*-support:

a. * He DOES be learning Spanish. They DO be invited to the ceremony. (emphasis)

b. * He doesn't be learning Spanish. They aren't be invited to the ceremony. (negation)

c. * Does he be learning Spanish? Do they be invited to the ceremony? (question)

These properties are essentially the same as auxiliary *have*. Main verb *be* in English behaves differently from other main verbs: see Problem 2.7 for an exploration of this.

Summary

Table 2.7 provides a synopsis of the morphological and syntactic properties of the items that we have discussed, arranged from most to least verb-like. As is evident from the table, the syntactic category of an item depends on whether it is the verb-like or the modal-like properties that predominate.

	Ordinary verb	Auxiliary *have*/*be*	Auxiliary *do*	Modal
Has -*s* form	yes	yes	yes	no
Occurs in nonfinite contexts	yes	yes	no	no
Occurs with emphatic *do*	yes	no	no	no
Occurs with *do* in negation	yes	no	no	no
inverts with subjects in questions	no	yes	yes	yes
Syntactic category	V	V	I	I

Table 2.7: Summary of morphological and syntactic properties of ordinary verbs, auxiliaries verbs, and modals in English

Clearly there is a lot of complexity to the syntactic and morphological properties of auxiliaries in Mainstream U.S. English. There is a line in the table above (syntactic category) that is thus far unexplained: we will discuss this in the next section.

2.5.3 Inflection as a grammatical category

This section previews a conclusion that will be impactful in our analysis of sentences throughout the rest of the book: Inflection is a syntactically significant grammatical category, and plays a significant role in the structure of clauses.

Even though Table 2.7 summarizes the properties explained in the previous section, it can still be hard to see an overarching pattern emerge. But in fact, a very strong pattern emerges: modals, auxiliary *do*, and inflected *be*/*have* all share the same properties. Uninflected auxiliaries behave like main verbs.

To bring us there, first notice that modals are in **complementary distribution** with inflection, whether that inflection is on an auxiliary (86c) or on a main verb (87c).

(86) a. The child is running. **tensed auxiliary**

b. The child can be running. **modal**

c. * The child can is running. **modal and tensed aux.**

(87) a. The child ran. **tensed main verb**

b. The child can run. **modal**

c. * The child can ran. **modal and tensed main verb**

Likewise, inflection on an auxiliary rules out inflection on a main verb.

(88) a. The child ran. **tensed main verb**

b. The child has run. **tensed aux.**

c. * The child has ran. **tensed aux. and tensed main verb**

The complementary distribution diagnostic alone is a strong argument that inflection and modals are underlying the same category. This is the same diagnostic that is used in phonology (and learned in many introductory linguistics classes) to identify two phones as realizations of the same phoneme. Here, the complementary distribution of modals and inflection suggests that they both realize the same (abstract) category.

We see a similar pattern with our other diagnostics. What is it that inverts to sentence initial position in questions? We can see that it is either modals (89) or inflected auxiliaries (90): uninflected auxiliaries cannot invert (89c).

(89) a. The child should be running.

b. Should the child be running?

c. * Be the child should running.

(90) a. The child is running.

b. Is the child running?

Both inversion and inflection suggest that modals and inflected auxiliaries behave the same. So it is no surprise, perhaps, that auxiliary *do* (which is *always* inflected) cannot co-occur with inflection on other verbs/auxiliaries, and cannot co-occur with modals.

(91) a. The child did run.

b. * The child did ran.

c. The child might run.

d. * The child might did run.

The conclusion here is that auxiliary *do*, inflection, and modals all behave similarly in Mainstream U.S. English. They are in complementary distribution (only one instance of any of them may occur in a sentence) and they share properties like appearing before negation and inverting with subjects in questions, properties that are not shared by main verbs or uninflected auxiliaries.

Syntacticians have concluded that all of these elements are instances of the same abstract grammatical category, which we will refer to as **Inflection**, or **Infl** for short. In chapters that are upcoming we will see how this is realized in a clause structure (and there are definitely open questions for which solutions aren't obvious from our discussion thus far).[14] The takeaway lesson here, however, is the one focused on grammatical categories: grammatical categories can be abstract, and not every significant grammatical category in a language will be one that is consciously obvious to native speakers, or one which is taught in schools (for languages that have formal instruction in schools). And some categories subsume others: it is reasonable to talk about modals and auxiliaries as categories of their own in some ways (modals and auxiliaries are not identical, even modals and tensed auxiliaries are not identical). But in Mainstream U.S. English they do all behave syntactically as a single category, which we can call Infl.

On the (non-)universality of grammatical categories 2.6

Generative syntax as we describe in this book is focused on universal questions: what is Universal Grammar? Put more descriptively, what are the universal, innate properties of human cognition that result in a child learning any human language? We explore this by exploring the grammatical properties of human languages and seeing what is consistent among them, and what varies among them.

One potential universal is that all human languages appear to have something like verbs and something like nouns: this has been explicitly proposed at times, or simply assumed in many analyses. But the individual properties of verbs and nouns can vary a lot between particular languages, such that it's difficult to pin down specific properties that can dependably diagnose a noun and verb cross-linguistically. For example, some concepts that are expressed by adjectives in English are expressed by verbs in Bantu languages. The Swahili sentences in (92) are verbs (with subject agreement and tense), whereas their translations in English are adjectives.

(92) a. Ni-me-furahi.
1SG.AGR-PRF-be.happy
'I am happy.'

b. M-zigo u-li-tosha.
3-load 3AGR-PST-be.sufficient
'The load was sufficient.'

Therefore, translations from one language to another can be highly unreliable to reveal categorical properties of language without additional consideration: grammatical categories must be determined according to the systematic properties of each language itself. Likewise, it is relatively uncontroversial, insofar as we can tell, that the specific morphological and syntactic properties of categories will vary based on languages. So some languages inflect verbs highly, some not at all; some languages inflect nominals robustly for noun class, gender, or case-marking (see Chapter 9), but others don't. So the specific meanings and specific grammatical properties that identify a category will vary language by language.

But it has even been claimed at many points that some languages lack clear categorical distinctions between major categories, even between nouns and verbs. Even in English it is possible to convert nouns to verbs with no overt morphology. In the current age of technology, names of products often get converted to verbs, as in (93a) and (93b). This is of course a quite general process in English, such as the use of *address* in (93c); the examples listed in (94) are an entirely non-exhaustive list of English words that function as both verbs and nouns. The comic strip artist Bill Watterson offered his commentary on this process in a Calvin and Hobbes comic strip with the sentence in (93d).

(93) a. Let's **google** that.
b. We need to **uber** to the airport.
c. We used to have to **address** envelopes and affix a stamp.
d. Verbing weirds language. (from a *Calvin and Hobbes* cartoon)

(94)

a. access	e. silence	i. mine	m. fall
b. mail	f. arrest	j. slice	n. move
c. shelter	g. milk	k. slide	o. smile
d. dust	h. exhibit	l. bandage	p. grill

In addition, some linguists have argued that there are languages that lack these underlying grammatical categories altogether. In the Turkish the same lexical item can be used as a nominal (95a), an attributive adjective (95b), or an adverbial (95c).

(95) Turkish (Hengeveld 2013, 33, original examples from Göksel and Kerslake 2005)
a. güzel-im
beauty-1.POSS
'my beauty' (head of referential phrase)
b. güzel bir köpek
beauty art dog
'a beautiful dog' (modifier of head of referential phrase)
c. Güzel konuş-tu-Ø.
beauty speak-PST-3SG
'S/he spoke well.' (modifier of head of predicate phrase)

Some linguists take the strongest stance that there are no universal categories (ex., Haspelmath 2007, 2010). Some such linguists are typological and/or descriptive linguists who are skeptical of the notion of Universal Grammar, but even linguists working within the framework of Universal Grammar have posited that grammatical categories of various sorts are emergent, that is, that they arise from a combination of other aspects of human cognition and language experience, rather than an innate inventory of categories. To our knowledge, all linguists affirm the existence of categories within the grammar of individual human languages, which means that what must be universal, on this approach, is the ability to form and use grammatical categories, rather than any particular categories themselves (Biberauer 2019a, 2019b; Wiltschko 2014; Croft 2001).

When functional categories are considered, much more variation appears across languages. Even if English, for example, has a category of elements called *modals* that has specific properties, this does not mean that all languages have a similar category (despite all languages being able to communicate modal contexts like probability and obligation). For example, Korean systematically integrates politeness and social hierarchy into its grammar via grammatical morphemes known as honorifics (HON in the examples below). Notice two different kinds of honorific marking in the examples below. The nominal *sensayng* 'teacher' bears an honorific recognizing teachers' socially elevated position in (96b) and (96d). But the verbs also bear honorific marking in each example, marking that varies based on the addressee (who the speaker is talking to). Speaking to a close friend is marked as an informal declarative (statement); speaking to a coworker is labeled similarly, but a politeness morpheme (POL) is added; speaking to a superior at work requires an even more polite morpheme we have glossed as FORMAL here.[15]

(96) a. 민지가 나에게 빵을 줬어.
Minci-ka na-eykey ppang-ul cw-ess-e.
Minji-NOM I-DAT bread-ACC give-PST-DEC.INFORMAL
'Minji gave me bread.' (to a close friend)

b. 선생님께서 나에게 빵을 주셨어.
sensayng-nim-kkeyse na-eykey ppang-ul cwu-sy-ess-e.
teacher-HON-NOM.HON I-DAT bread-ACC give-HON-PST-DEC.INFORMAL
'The teacher gave me bread.' (to a close friend)

c. 민지씨가 저에게 빵을 줬어요.
Minci-ssi-ka ce-eykey ppang-ul cw-ess-e-yo.
Minji-POL-NOM I.HUMBLE-DAT bread-ACC give-PST-DEC.INFORMAL-POL
'Minji gave me bread.' (to a coworker)

d. 선생님께서 저에게 빵을 주셨어요.
sensayng-nim-kkeyse ce-eykey ppang-ul cwu-sy-ess-e-yo.
teacher-HON-NOM.HON I.HUMBLE-DAT bread-ACC give-HON-PST-DEC.INFORMAL-POL
'The teacher gave me bread.' (to a coworker)

e. 민지씨가 저에게 빵을 주었습니다.
Minci-ssi-ka ce-eykey ppang-ul cwu-ess-supni-ta.
Minji-POL-NOM I.HUMBLE-DAT bread-ACC give-PST-FORMAL-DEC
'Minji gave me bread.' (to a superior at work)

f. 선생님께서 저에게 빵을 주셨습니다.
sensayng-nim-kkeyse ce-eykey ppang-ul cwu-sy-ess-supni-ta.
teacher-HON-NOM.HON I.HUMBLE-DAT bread-ACC give-HON-PST-FORMAL-DEC
'The teacher gave me bread.' (to a superior at work)

A completely different example comes from **evidentials**: many languages have grammatical morphemes attached to sentences that indicate the source of evidence for the statement being uttered: how did the speaker come to learn the thing that they are saying? Evidential systems can be relatively simple, such as distinguishing direct evidence from indirect evidence, but they can also be relatively complex. Aikhenvald presents the minimal set from Tariana (Arawak, northwest Amazonia) below, which varies just in the evidential marking on the verb.

(97) Tariana evidentials Aikhenvald (2004, 2–3)

a. Juse iɾida di-manika-ka
José football 3SG.NF-play-RECP.PST.VIS
'José has played football (we saw it)'

b. Juse iɾida di-manika-mahka
José football 3SG.NF-play-RECP.PST.NONVIS
'José has played football (we heard it)'

c. Juse iɾida di-manika-nihka
José football 3SG.NF-play-RECP.PST.INFR
'José has played football (we infer it from visual evidence)'

d. Juse iɾida di-manika-sika
José football 3SG.NF-play-RECP.PST.ASSUM
'José has played football (we assume this on the basis of what we already know)'

e. Juse iɾida di-manika-pidaka
José football 3SG.NF-play-RECP.PST.REP
'José has played football (we were told)'

In each case above, a different choice of suffix on the verb communicates a different source of evidence for the statement being uttered. Evidentiality in Tariana therefore has a grammaticalized implementation in Tariana, realizing its own grammatical category, that is not present in all languages. All languages are of course capable of allowing a speaker to communicate the source of their evidence, but not all languages have a specific grammatical category of EVIDENTIAL dedicated to marking the distinction systematically (though many languages do have grammaticalized evidentiality similar to Tariana).

The takeaway lesson here is important not just for understanding how languages work, but also for theorizing about the nature of human language in cognition, that is, the nature of Universal Grammar. It is tempting to overgeneralize from the small set of languages we may have experience with. But there are approximately 10,000 languages in the world (the exact number depends on how you count, and is always changing); we have detailed descriptions of a fraction of those languages. Furthermore, there are robust research communities working in only a handful of those well-described languages (by that we mean, communities of researchers consistently generating new documentation and analysis of the languages). We don't need to be embarrassed by attempts to investigate what is universal about language: it is a major question about cognition and human nature and is worthy of inspection. But we do need to be humble in recognizing (1) the extent of language variation that is already documented, and (2) the number of linguistic facts that are yet to be documented in the humanistic and scientific literature. The kind of variation noted in this section is not the central purpose of this book, but there are textbooks that emphasize this kind of morphosyntactic variation to a greater degree: we recommend the interested reader to Kroeger (2005) and Croft (2022). But the perpetual exercise of the syntactician is to investigate whether the theory we have built thus far readily extends (or not) to all human languages, and we ought not be distressed when we learn that it doesn't: we just adjust the model and continue our work.

Insofar as grammatical categories go, in this textbook we assume labels for the commonly attested categories (N, V, Adj, Adv, P, Comp, etc.) uncritically for the most part, for the purposes of an introductory textbook. They are useful for quickly capturing the general properties of a category, and allow for cross-linguistic comparison readily. But we don't ever mean to imply these are necessarily part of UG, or that they cannot be deconstructed to find deeper underlying generalizations, or critiqued to find instances of variation in properties across languages. Both research exercises should be expected by students continuing in the field of syntax.

New Empirical Observations

- Groups of morphosyntactic elements behave similarly to each other, showing consistent patterns: we call these groupings **grammatical categories.**
- Morphemes have grammatical categories:
  - functional vs. lexical morphemes
  - open vs. closed classes
- Major lexical categories include:
  - nouns
  - verbs
  - adjectives
  - adverbs
- Inflection is a grammatical category.

Components of the Theoretical Model

- Morphemes, words, and phrases bear features, which may manifest as overt morphological forms.
- Some grammatical categories could be universal (as we treat them to be here) but they may alternatively simply be emergent.
- Grammatical categories could also potentially emerge from deeper properties of language and cognition.

Notes 2.7

1. The language varieties and segments of societies that drive language innovation in this way vary. In the U.S., African American language is a common driver of linguistic innovation, with words first used in Black communities often ending up as mainstream youth slang years later. In Kenya, the prominent use of Sheng by youth drives a lot of linguistic innovation; Sheng is a language contact phenomenon that is highly locale-specific, as youth in a specific area of a specific generation combine aspects of English, Swahili, and other local languages to generate a new vocabulary (and sometimes arguably a new language), some aspects of which can end up in mainstream use of English

and Swahili.

2. To be clear, this is a language-particular sort of problem; in the Bantu language family containing approximately 800 languages, for example, the usual pattern is for all 3rd person pronouns to be unmarked for human gender (ex., masculine vs. feminine vs. other). It is not universal (or necessary, from a linguistic perspective) to mark human gender on pronouns, but the fact that English does has created this particular social history.

3. Back in the bad old days when singular *they* was not widely accepted in academic writing (or anywhere), my (Diercks) literature professor in college (James Wardwell) used to rant about the *he or she* and *him or her* circumlocutions as horrible writing. His recommendation was, for each sentence/sequence, to choose either *he/him* or *she/her*. As he said it, you should generally vary the choice of pronoun in your writing and hope that in the great cosmic reckoning it ends up that you were equitable, choosing masculine and feminine at roughly equal rates. Needless to say, the broader acceptance of singular *they* is very welcome, for a multitude of reasons.

4. Neological pronouns (more often 'neopronouns') like *ze/zim* still have some currency among certain communities and online spaces, but don't have the same widespread traction singular *they* has garnered; in 2019 Merriam-Webster anointed singular *they* as its Word of the Year, demonstrating the mainstreaming of its use in that function.

5. This in turn has spawned a large collection of regionalisms and variations to distinguish singular and plural *you*, such as *you guys*, *y'all*, *y'inz*, *youse*, and *you-uns*, among others.

6. Credit with gratitude to *The Office*'s Andy Bernard.

7. Although the distinction between count and mass nouns is generally clear-cut, under special circumstances, what are ordinarily mass nouns in English can be used as count nouns—for instance, when it is possible to impose an interpretation of *a kind of X* or *a salient quantity of X* on the mass noun. Notice, incidentally, that the interpretation of the plural mass nouns in (i.b) depends on whether *we* refers to a group of coffee shop customers (salient quantity reading) or to a business owner (salient kind reading).

   (i) a. Different *rices* have different cooking times.
   b. We ordered four *coffees* and two *teas*.

   For some mass nouns, the conversion to count noun is very natural.

   (ii) a. Mass noun use (basic):
   a lot of *wine*, *wine* from Burgundy, more *wine* than we can drink tonight
   b. Count noun use:
   a *wine* from Alsace, *wines* from Burgundy, more *wines* than I had ever heard of

   Conversely, it is possible for what are ordinarily count nouns to be pressed into service as mass nouns.

   (iii) a. ?This recipe for carrot cake calls for { a lot of *carrot*, more *carrot* than I have on hand }
   b. ?There's just not enough *couch* for all ten of you.
   c. 2nd Servant: Pray heauen it be not full of *Knight* againe.
   1st Servant: I hope not, I had liefe as beare so much lead.
   (William Shakespeare. *Merry Wives of Windsor*, Act 4, Scene 2.)
   d. The article says that [Mr. Mangetout] once consumed fifteen pounds of *bicycle* in twelve days … Just as an example, I have never eaten so much as a pound of *bicycle*. … I can see myself acting with considerable restraint at a dinner party at which the main course, is, say, *queen-size waterbed*.
   (Calvin Trillin. 1983. Third helpings. 3.)

   Although the match between count and mass nouns across languages is reasonably good, mismatches occur. Some examples of nouns that are mass nouns in English, but count nouns in other languages are given in (iv)–(vi).

(iv) French:

a. un meuble, des meuble-s
a furniture.SG, of.the furniture-PL
'a piece of furniture, furniture'

b. un renseignement, des renseignement-s
an information.SG, of.the information-PL
'a piece of information, information'

(v) German:

a. ein Möbel, Möbel
a furniture.SG, furniture.PL
'a piece of furniture, furniture'

b. ein Ratschlag, Ratschläg-e
an advice.SG, advice-PL
'a piece of advice, advice'

c. eine Nachricht, Nachricht-en
a news.SG, news-PL
'a piece of news, news'

(vi) Italian:

a. un consigli-o, consigl-i
an advice.SG, advice-PL
'a piece of advice, advice'

b. una notizia, notizi-e
a news.SG, news-PL
'a piece of news, news'

Not surprisingly, when learning English as a foreign language, speakers from these languages often produce ungrammatical examples like (vii).

(vii) a. *They gave me some useful *advices*.
b. *The room was crowded with too many *furnitures*.
c. *Before I decide, I need some further *informations*.

8. A note on proper nouns with and without determiners: First, although proper nouns in English typically appear without a determiner, certain proper nouns must be accompanied by the definite article. Some examples are given in (i).

(i) the Bronx, the Hague, the Thames, the Titanic, the Soviet Union, the United States, the White House

Pronouns, on the other hand, can't be accompanied by an article, as shown in (ii).

(ii) *the { you, he, she, it, we, they }

Given the contrast between (i) and (ii), the proper nouns in (i) can't themselves be determiners, but must be nouns, as English does not tolerate sequences of determiners. Even the misnomer view must therefore admit the existence of nouns that refer to particular individual entities. But if the proper nouns in (i) are in fact nouns, then there is no conceptual advantage in assuming that those in (24b) are determiners. Instead, we could simply take the view that proper nouns are invariably nouns, but that some are complements of the ordinary overt definite article *the*, whereas others are complements of a silent counterpart of it. In other words, the proper nouns in (24b) would be structurally parallel to those in (i), as indicated in (iii).

(iii) a. Overt determiner:
the Bronx, the Hague, the Thames, the Titanic, ...

b. Silent determiner:
[DEF] Philadelphia, [DEF] Lassie, [DEF] France, …

Several additional pieces of evidence, all of the same type, point in the same direction. Before the breakup of the Soviet Union, the Soviet republic whose capital is Kiev was called the Ukraine, but after the breakup, the newly independent country began to be called Ukraine, without the article. Similarly, the United States is generally referred to in Spanish as *los Estados Unidos* 'the United States', but in recent years, Latin American newspapers have begun to omit the article *los*. According to the misnomer view, *Ukraine* and *Estados Unidos* would be nouns in the old usage and turn into determiners in the new usage. Treating the change as affecting the pronunciation of the determiner is clearly more straightforward.

In French, names of countries and regions must be accompanied by the definite article, whereas in English, they generally aren't (the change from *the Ukraine* to *Ukraine* thus eliminated its exceptional status in English). Some examples are shown in (iv); *la* and *le* are the feminine and masculine forms of the French definite article, respectively.

(iv) a. French:
la Bolivie, la Bourgogne, le Brésil, le Canada, le Danemark, la France, le Pérou

b. English:
Bolivia, Burgundy, Brazil, Canada, Denmark, France, Peru

According to the misnomer view, the names of countries and regions would be nouns in French but determiners in English. Again, it seems more reasonable to pin the difference between the two languages not on the nouns themselves, but on the status of the determiner.

In standard German, personal names stand alone in the standard language. But in the vernacular, they can be accompanied by the definite article, as shown in (v); *der* and *die* are masculine and feminine forms of the German definite article, respectively.

(v) a. Standard German:
Annemarie, Eva, Lukas, Peter

b. Vernacular German:
die Annemarie, die Eva, der Lukas, der Peter

According to the misnomer view, proper names would change syntactic category depending on the speaker's register, whereas the alternative view again more straightforwardly pins the variation on the determiner.

Finally, in modern Greek, personal names are normally accompanied by the definite article. This is shown for the **nominative** in (vi); *o/o* and *η/i* are masculine and feminine forms of the Greek definite article, respectively. ("Nominative" is the form in which subjects of sentences appear: see Chapter 9.)

(vi) Greek, nominative:
η Ελένη, η Βάλια, ο Παύλος, ο Γιάννης
i Eleni, i Valia, o Pavlos, o Yannis

Modern Greek retains from its ancestor language Indo-European a special **vocative** case form that is used when addressing someone. Masculine nouns lose their final *-s* in the vocative, and other nouns remain unchanged, but for all nouns, the article is obligatorily absent in the vocative, as shown in (vii).[16]

(vii) Greek, vocative:
(*η) Ελένη, (*η) Βάλια, (*ο) Παύλος, (*ο) Γιάννης
(*i) Eleni, (*i) Valia, (*o) Pavlos, (*o) Yannis

Once again, under the misnomer view, names in Greek would change their syntactic category depending on their case form—a bizarre consequence.

A second type of argument against the misnomer view of proper nouns is based on linguistic borrowing. Speakers of one language often borrow words from another language, either because no native words exist for certain concepts or because the borrowed word is perceived as having a cachet that the native counterparts lack. Proper nouns are easily borrowed in this way; we need only to think of the many geographical names in English derived from various indigenous languages. The prolific borrowing of proper nouns is not surprising if proper nouns are a subcategory of nouns, since nouns are precisely the category that is most readily borrowed. But it is unexpected under the misnomer view, since pronouns are ordinarily not borrowed at all.

On the basis of the considerations just discussed, we conclude, then, that proper nouns are a semantically (and often syntactically) special subtype of noun.

9. This is an oversimplification (for many reasons): the most significant reason it is a simplification is that there are stative predicates that are hard to describe as "events" that still take arguments. For example, in the sentence *Lilly is short* the predicate *short* is stative, describe a state (not an event) and it still takes *Lilly* as an argument.

10. But even in English, adjective phrases can occur postnominally if they are 'heavy' enough, and even relatively short adjectives are sometimes used postnominally to convey elegance or high style.

(i) a. ✓ an *unlikely* story
b. * an *unlikely to be true* story

(ii) a. *a story *unlikely*
b. ✓ a story *unlikely to be true*

(iii) *House Beautiful* (upscale interior decorating magazine)

Originally, English, like the other Germanic languages, allowed adjective phrase modifiers only in prenominal position. After the Norman Conquest in 1066, Norman French became the official language of England and remained so for several centuries. In Norman French, as in the other Romance languages, adjective phrase modifiers were primarily postnominal. The word order variation illustrated in (36) and (i) and the stylistic connotations of the postnominal order are thus the result of language contact. See also Note .

The situation in English is the reverse of that in Walloon, a variety of French spoken in Belgium. Unlike in standard French, adjective phrases modifying nouns are prenominal in Walloon. This is the result of language contact with Flemish, a variety of Germanic.

11. It is useful to distinguish between form (morphology) and meaning (reference). Ordinarily, forms with past-tense morphology are used to refer to an event or state prior to the time of speaking. However, it is possible in English to use past-tense forms to refer to events or states contemporaneous with a reported time of speaking; this is the so-called sequence-of-tense phenomenon in reported speech, illustrated in (i).

(i) a. He said, "They know a lot". (direct speech)
b. He said (that) they knew a lot. (reported speech)

As is evident from (ii) and (iii), *can, may, shall,* and *will* continue to maintain a productive morphological relationship with *could, might, should,* and *would,* respectively, in sequence-of-tense contexts.

(ii) Direct speech:
a. She said, "That may be so".
b. He told us, "You won't last long".

(iii) Reported speech:
a. She said (that) that might be so.
b. He told us (that) we wouldn't last long.

Nevertheless, in keeping with the point made in the body of the text, the morphological relationship between the modals in (ii) and their counterparts in (iii) is purely formal, lacking the referential underpinning evident in (2.7).

(iv) a. They know a lot { ✓ now, * then }.
b. They knew a lot { * now, ✓ then }.

We thank Aaron Dinkin for drawing our attention to the sequence-of-tense phenomenon.

12. Auxiliary *do* also behaves like a modal in *do* support contexts. Double instances of auxiliary *do* are ruled out, just like double modals are (see (58a)). Once again, auxiliary *do* and main verb *do* differ sharply, as shown in (ii) and (iii).

(i) Modal:
a. *They does can dance the polka. (emphasis)
b. *They doesn't can dance the polka. (negation)
c. *Does he can dance the polka? (question)

(ii) Auxiliary *do*:
a. *She does do dance the polka. (emphasis)
Intended meaning: It is the case that she does dance the polka.
b. *She doesn't do dance the polka. (negation)
Intended meaning: It isn't the case that she does dance the polka.
c. *Does she do dance the polka? (question)
Intended meaning: Is it the case that she does dance the polka?

(iii) Main verb *do*:
a. ✓ He does do the dishes. (emphasis)
b. ✓ He doesn't do the dishes. (negation)
c. ✓ Does he do the dishes? (question)

(iv) Ordinary verb:
a. ✓ She does dance the polka. (emphasis)
b. ✓ She doesn't dance the polka. (negation)
c. ✓ Does she dance the polka? (question)

13. A *be* auxiliary is also used to form passive sentences (*is abandoned, was sold*) in English. We ignore passives for now; with regard to the properties we are currently discussing, the passive *be* auxiliary behaves the same as the progressive *be* auxiliary. But they are not the same thing, which is evident in the fact that they can co-occur in the same sentence: *the salad* ***is being*** *prepared right now.*

14. One of these is how it is that a modal—which is a free morpheme—and a morphological affix on a verb (like past tense) can be the same grammatical category when they behave so differently morphologically-speaking. This is not lost on linguists, of course—we will see discussion of this throughout the textbook, but especially in Chapters 9 and 12.

15. Readers familiar with Korean may notice other kinds of honorific marking in (96) that we have neglected to discuss here (for example, the verbal subject honorific morpheme *-sy-*). Korean has a particularly rich system of honorifics and politeness that, while fascinating, is too much to discuss fully here.

16. For the (*) notation, see the glossary entry asterisk (*).

2.8 Exercises and problems

Exercise 2.1: Identifying grammatical categories A

Using the properties of English grammatical categories as described in this chapter, identify the grammatical category of each of the bolded words below. For each one, describe the distributional properties (syntactic or morphological) that lead to that conclusion.

(1) a. The **children** will hopefully **sleep** all night.
b. I poured the coffee **into** the cup.
c. I really want **an** apple.
d. I am going to **clean** the kitchen **while** the children are napping.
e. I am **from** New Jersey.
f. I am frightened **because** thunder is scary.

Exercise 2.2: Identifying grammatical categories B

Using the properties of English grammatical categories as described in this chapter, identify the grammatical category of each of the bolded words below. For each one, describe the distributional properties (syntactic or morphological) that lead to that conclusion.

Don't just consider the sentences offered here: investigate these words used in similar functions in other kinds of sentences also.

(1) I plan to order myself **some** french fries.

(2) I just need to put on my shoes **before** leaving.

(3) Nonetheless, I **still** left on time.

Exercise 2.3: Identifying grammatical categories C

It is possible for words that are not the same category to carry similar meanings. In Mainstream U.S. English, the words *a my*, *particular*, and *that* all carry a meaning something like 'a specific one' in the sentences in (1). This is not the entirety of the meaning of each, but each of them do carry a notion of specificity and each sentence in (1) could be used to describe the same situation.

(1) a. I met **a** friend.
b. I met **my** friend.
c. I met a **particular** friend.
d. I met **that** friend.

Using distributional evidence from syntax and/or morphology, argue for *a*, *my*, *particular*, and *that* in English are the same grammatical categories or different grammatical categories.

Exercise 2.4: Count nouns and mass nouns

Is *mail* a count noun or a mass noun? And how about *email*? Ask a couple of friends: do your intuitions match, especially about *email*?

Exercise 2.5: English adverbial modifiers

What is the grammatical category of English adverbial modifiers *yesterday* and *tomorrow*? Provide evidence to make an arguments for your conclusion based.

Problem 2.1: Grammatical categories of nonce words

As we mentioned in this chapter, children's stories and poetry are often excellent instances of nonce words (nonsense words) being used creatively for expressive effect. Lewis Carroll's poem "Jabberwocky" is another such instance, and gives an opportunity to explore our intuitions about these invented words.

Jabberwocky, by Lewis Carroll

'Twas brillig, and the **slithy** toves
    Did gyre and gimble in the wabe:
All mimsy were the borogoves,
    And the **mome raths outgrabe**.

"Beware the Jabberwock, my son!
    The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
    The frumious Bandersnatch!"

He took his vorpal sword in hand;
    Long time the **manxome** foe he sought—
So rested he by the **Tumtum** tree
    And stood awhile in thought.

And, as in uffish thought he stood,
    The Jabberwock, with eyes of flame,
Came whiffling through the tulgey wood,
    And burbled as it came!

One, two! One, two! And through and through
    The vorpal blade went **snicker-snack**!

He left it dead, and with its head
    He went **galumphing** back.

"And hast thou slain the Jabberwock?
    Come to my arms, my **beamish** boy!
O **frabjous** day! Callooh! Callay!"
    He chortled in his joy.

'Twas brillig, and the slithy toves
    Did gyre and gimble in the wabe:
All mimsy were the borogoves,
    And the mome raths outgrabe.

(*Through the Looking Glass*, Lewis Carroll, 1871)

Your task in this exercise is to identify the grammatical categories of each of the words bolded in the poem. For list items with phrases containing multiple words, identify the syntactic category for each. For each word, explain the diagnostic evidence that supports your conclusion. This will include evidence from both morphological and syntactic distribution, but can also include additional intuitions that you have, just be as specific as possible about where you think the intuition comes from.

There are no specifically correct answers here, in the sense that these words are clearly made up. But for some of them, native English speakers will have very clear judgments about what syntactic category they must be (which arise from specific properties of their distribution). For other words here, there may be multiple different ("right") answers, in the sense that there are multiple parses/structures of the sentence that are possible. The point is not to get a right answer, but to build a strong argument, based on specific evidence.

Problem 2.2: Grammatical categories of quantifiers

From a semantic perspective, both *all* and *every* can be described as **quantifers**, because they quantify over (that is, count) a group of entities. In fact, they are both universal quantifiers: they both capture the idea the each relevant individual is included in a statement. The two sentences in (1) are roughly synonymous, meaning that the entire group of relevant children left.

(1) a. Every child left.
    b. All the children left.

The question: do *all* and *every* in Mainstream U.S. English fall into the same grammatical category? What is the grammatical category of each of them? Consider the following evidence in drawing your conclusion, along with the examples in (1):

(2) a. * Every the child left.
    b. * The every child left.
    c. * All child left.
    d. ? All children left.

e. * Every children left.
f. ✓ All these children left.
g. * Every this child left.
h. * This every child left.

Problem 2.3: Adjective and verb modifiers in English

In elementary school in the U.S., students are taught that words like *very*, *rather*, *quite*, and *really* are adverbs, and are taught that adverbs can modify either verbs or adjectives. One of the motivations for this is that many of them end in *-ly* (as is typical for English adverbs).

(1) a. I am **rather** happy.
b. The children are **extremely** content.

When used in this way, they tend to intensify the interpretation of the word they modify.

(2) Additional words from this category:

- absolutely
- completely
- exceptionally
- particularly
- really
- quite
- totally
- utterly
- quite
- too

Are these words adverbs or some other grammatical category? Provide a rationale for your response using empirical evidence to make your case. This will require native English intuitions, so rely on your own judgments or ask a native speaker of English for their judgments.

Problem 2.4: Grammatical category of the English possessive morpheme

Background

Possessive constructions are complex, as they allow for a noun phrase (the possessor) embedded inside another noun phrase that refers to the possessed element (the possessee). In (1) the possessor is *Lilly* and the possessee is *ball*.

(1) Lilly**'s** ball

English allows for quite complex possessors to occur in possessive noun phrases. So instead of *Lilly* the possessor can be noun phrases of almost any size.

(2) a. [The dog]**'s** ball.
b. [The aging dog]**'s** ball.
c. [The dog that is always sleeping]**'s** ball

We can see a pattern, namely,

(3) [POSSESSOR] **'s** POSSESSEE

The puzzle

What is the grammatical category of the English possessive morpheme *'s*? One plausible answer is that it forms its own category. But the data below suggest a different answer. In each possessive construction, the possessor is in brackets the same way as above.

(4)
a. the ball
b. the big ball
c. this ball
d. [Lilly]'s ball
e. * the [Lilly]'s ball
f. * this [Lilly]'s ball
g. * [Lilly]'s the ball
h. * [the dog]'s the ball
i. * [the dog]'s this ball
j. * the [the dog]'s ball
k. * this [the dog]'s ball

Use the data given above to answer the following questions.

A. The data in (4) show a complementary distribution pattern. What is it?

B. What grammatical category is the English possessive behaving like?

Problem 2.5: Tiriki grammatical categories

Tiriki is a Bantu language of the Luyia subgroup that is spoken in Western Kenya.

A. In the example below, outline what the morphosyntactic properties of nouns, adjectives, and verbs are.

B. In English, *happy* and *sad* are adjectives. What grammatical category are they in Tiriki?

Background: Cardinal numbers in glosses for Bantu languages (like Tiriki) signify noun class, which is arbitrarily organized by numbers by linguists. Noun class is like grammatical gender (masculine, feminine) in Romance languages like Spanish and Italian, but it is not linked with human sex/gender.

Glosses: AGR = agreement; cardinal numbers (1, 2, 3...) = noun class; PST = past tense; FUT = future tense; *some glossing is simplified for the purposes of this exercise.*

(1) Mw-ana mu-hulili a-a-rhula.
1-child 1AGR-polite 1SM-PST-leave
'A polite child left.'

(2) Va-na va-hulili va-a-rhula.
2-child 2AGR-polite 2AGR-PST-leave
'Polite children left.'

(3) Mw-ana mu-hindila na-a-rhule.
1-child 1AGR-big FUT-1AGR-leave
'A big child will leave.'

(4) Va-na va-tunyi va-a-rhula.
2-child 2AGR-sad 2AGR-PST-leave
'Sad children left.'

(5) Va-na va-sangali na-va-rhule.
2-child 2AGR-happy FUT-2AGR-leave
'Happy children will leave.'

(6) Va-na va-sangali va-a-rhula.
2-child 2AGR-happy 2AGR-PST-leave
'Happy children left.'

(7) Mw-ana mu-sangali na-a-rhule.
1-child 1AGR-happy FUT-1AGR-leave
'A happy child will leave.'

(8) Mw-ana mu-tunyi na-a-rhule.
1-child 1AGR-sad FUT-1AGR-leave
'A sad child will leave.'

(9) Mu-linga mu-kali kw-a-kwaa.
3-beehive 3AGR-big 3AGR-PST-fall
'A big beehive fell.'

(10) Mu-linga mu-tii na-ku-kwii.
3-beehive 3AGR-small FUT-3AGR-fall
'A small beehive will fall.

(11) Mi-linga mi-ti ch-a-kwaa.
4-beehive 4AGR-small 4AGR-PST-fall
'Small beehives fell.

(12) Mi-linga mi-kali na-chi-kwi.
4-beehive 4AGR-big FUT-4AGR-fall
'Big beehives will fall.'

(13) Mu-toka mu-khulu kw-onekha.
3-car 3AGR-old 3AGR-PST.break
'An old car broke.'

(14) Mu-toka mu-hya kw-onekha.
3-car 3AGR-new 3AGR-PST.break
'A new mutoka (car) broke.'

(15) Mi-toka mi-khulu ch-onekha.
4-car 4AGR-old 4AGR-PST.break
'Old cars broke.'

(16) Mi-toka mi-hya ch-onekha.
3-car 3AGR-new 3AGR-PST.break
'New cars broke.'

(17) Mw-ana mu-ti a-sangali.
1-child 1AGR-small 1AGR-happy
'A small child is happy.'

(18) Mw-ana mu-kali na-a-sangale.
1-child 1AGR-big FUT-1AGR-happy
'A big child will be happy.'

(19) Va-na va-kali va-tunyi.
2-child 2AGR-big 2AGR-sad
'Big children are sad.'

(20) Va-na va-ti va-tunyi.
2-child 2AGR-small 2AGR-sad
'Small children are sad.'

(21) Tsi-ng'ombe tsi-ngali tsy-a-ng'wa ma-tsi.
10-cow 10AGR-big 10AGR-PST-drink 6-water
'Big cows drank water.'

(22) Tsi-ng'ombe tsi-ndi na-tsi-ng'wi ma-tsi.
10-cow 10AGR-small FUT-10AGR-drink 6-water
'Small cows will drink water.'

(23) I-ng'ombe i-ndi y-a-ng'wa ma-tsi.
9-cow 9AGR-small 9AGR-PST-drink 6-water
'A small cow drank water.'

(24) I-ng'ombe i-ngali n-i-ng'wi ma-tsi.
9-cow 9AGR-small FUT-9AGR-drink 6-water
'A big cow will drink water.'

Problem 2.6: Grammatical category of English pronouns

English pronouns in some ways behave as a category of their own, as we can readily describe them and organize them in a chart like (1), repeated from Table 2.4 of the chapter.

(1)

Person	Number	Subject forms	Non-subject forms
1	SG	I	me
2		you	you
3		he, she, they, it	him, her, them, it
1	PL	we	us
2		you	you
3		they	them

But we have a sense that pronouns stand in for some category in a sentence. What is the grammatical category of pronouns? Put another way, what kinds of elements can pronouns replace in a sentence? In the data examples below, we include various attempts to replace content with a pronoun.

(2) The enthusiastic child ran down the hill.
- a. * The enthusiastic she ran down the hill.
- b. * The she ran down the hill.
- c. ✓ She ran down the hill.

(3) Three yellow cars drove out of the parking lot.
- a. * Three yellow they drove out of the parking lot.
- b. * Three they drove out of the parking lot.
- c. ✓ They drove out of the parking lot.

Based on the data, answer the following questions.

A. Are pronouns nouns?

B. If not, what category do pronouns behave as?

Problem 2.7: English main verb *be*

In Mainstream U.S. English, main verb *be* poses some interesting puzzles. The task here is to articulate what the pattern is that is a puzzle, something to be solved by a syntactican. So you will identify a pattern, but won't have a full solution yet (that requires more syntax).

Background

Auxiliary *be* and main verb *be* inflect in the same way:

(1) Main verb *be* inflection:
  a. ✓ They are happy.
  b. ✓ She is happy.

(2) Auxiliary *be* inflection:
  a. ✓ They are learning Spanish. They are invited to the ceremony. (non-third person)
  b. ✓ He is learning Spanish. He is invited to the ceremony. (third person)

Other main verbs do bear inflection, though they are not all irregular in the same way that *be* is.

(3) Main verb *run* inflection:
  a. ✓ They run every day.
  b. ✓ She run-s every day.

The puzzle

The question you are investigating: syntactically-speaking, does main verb *be* behave like an auxiliary or a main verb? How do you know? Be specific. The judgments here are given for Mainstream U.S. English (MUSE)—some judgments may differ for different varieties of English.

(4) Main verb *be*:
  a. ✓ They must be happy.
  b. ✓ They claim to be happy.
  c. ✓ They don't regret being syntacticians.

(5) Auxiliary *be*:
  a. ✓ They must be learning Spanish.
  b. ✓ They claim to be learning Spanish.
  c. ✓ They don't regret being invited to the ceremony.

(6) Main verb *run*:
  a. ✓ They must run every day.
  b. ✓ They claim to run every day.
  c. ✓ They don't regret running ever day.

(7) Main verb *be*:
  a. * She DOES be happy.
  b. * She doesn't be happy.
  c. * Does she be happy?

(8) Main verb *be*:
   a. ✓ She IS happy.
   b. ✓ She isn't happy.
   c. ✓ Is she happy?

(9) Auxiliary verb *be*:
   a. ✓ He IS learning Spanish.
   b. ✓ He isn't learning Spanish.
   c. ✓ Is he learning Spanish?

(10) Auxiliary verb *be*:
   a. * He DOES be learning Spanish. They DO be invited to the ceremony.
   b. * He doesn't be learning Spanish. They aren't be invited to the ceremony.
   c. * Does he be learning Spanish? Do they be invited to the ceremony?

(11) Main verb *run*:
   a. ✓ They DO run every day.
   b. ✓ They don't run every day.
   c. ✓ Do they run every day?

Problem 2.8: Grammatical categories in Korean

Use the following sentences to practice identifying grammatical categories in Korean. For parts (A)–(B) assume distributional criteria for identifying grammatical categories in Korean are largely similar (but perhaps not identical) to those in English.

A. Which items are nouns? How can you tell?

B. Which items are verbs? How can you tell?

C. What are some ways in which distributional criteria for Korean differ from those for English?

D. Are there any words here that are ambiguous or difficult to assign to a category?

Background Info

In the following glosses, NOM indicates the subject of a sentence and ACC indicates the object of a sentence. DECL indicates a sentence is just a statement, while Q indicates a sentence is a question. ADN is short for 'adnominal', and indicates a special morphological form used to modify nouns. NFUT means 'non-future tense' (that is, present or past tense). CI is a suffix you can safely ignore for this exercise.

(1) Minci-ka ta-n ttalki-lul cohaha-nta.
Minji-NOM sweet-ADN.NFUT strawberry-ACC like-DECL
'Minji likes sweet food.'

(2) Cohaha-nun salam-i cikum yeki iss-ni?
like-ADN.PRS person-NOM now here exist-Q
'Is the person you like here right now?'

(3) I ttalki-ka acwu tal-ass-ta.
this strawberry-NOM very sweet-PST-DECL
'This strawberry was very sweet.'

(4) Salam-tul-i pothong ttalki-lul silheha-ci anh-nunta.
person-PL-NOM usually strawberry-ACC dislike-CI NEG-DECL
'People usually don't dislike strawberries.'

(5) Ku eli-n ai-ka wul-ess-ta.
that young-ADN.NFUT child-NOM cry-PST-DECL
'That young child cried.'

(6) Wul-ci anh-nun ai-man hakkyo-ey ka-nta.
cry-CI NEG-ADN.PRS child-only school-to go-DECL
'Only kids who don't cry go to school.'

Problem 2.9: Variation in English categories

What is the syntactic difference between mainstream U.S. English *them* and the variant illustrated in (1), which is a part of many varieties of English?

(1) This is definitely one of *them* jobs, man, if you're one of *them* worriers…
(Overheard on the southwest corner of 34th Street and Walnut Street, Philadelphia, PA, 31 August 1999)

Appendix: Finiteness in English 2.9

Take Note

A number of points in the constituency discussion in the next chapter run into questions about how finite verbs pattern differently from nonfinite verbs in English. This appendix details finiteness in English in additional detail.

2.9.1 Verb forms

English verbs can take the various forms listed in Table 2.8.

Name	Description	Examples
bare	Default form in present tense sentences. Also appears in various nonfinite contexts, such as in *to* infinitive clauses,	They *play* together. I *see*. I want to *play*. They need to *see* you.
	after modals,	They may *play*. We will *see*.
	and in connection with *do* support.	They don't *play* lacrosse. Do you *see*?
-s	Special form used in the present tense to mark agreement with a third person singular subject.	Lukas *runs* for miles. The cat *enjoys* treats.
-ing	As **present participle**, combines with auxiliary *be* to express various aspectual nuances.	The cat is *playing* with the yarn. I was *seeing* her until a week ago.
	Also occurs on its own as the **gerund**.	*Playing* with landmines is dangerous. We always enjoy *seeing* you.
past tense	The morphologically simplest way of expressing past tense.	The cat *played* with the yarn. We *saw* a deer.
past participle	Combines with auxiliary *be* to form **passive** forms.	Baseball is *played* all over the world. She was last *seen* off Mozambique.
	Combines with auxiliary *have* to form **perfect** forms.	They have never *played* lacrosse. I had never *seen* it before.

Table 2.8: Verb forms in English

- For all verbs, the *-ing* form is predictable from the bare form, being derived from it by the affixation of *-ing* (*play-ing, see-ing, hav-ing, be-ing*).
- The *-s* form is similarly predictable for most verbs, with major (*be, is*) or minor (*have, has*) exceptions.
- The past tense and past participle forms are predictable from the bare form in some cases, but not in others. With regular verbs, the past tense and past participle forms are homonymous and are formed by affixing *-ed* to the bare form. Why bother distinguishing between the two forms? That is, why not just posit a single past form? The reason is that the past tense and the past participle are distinct for irregular verbs such as *go* (*went, gone*), *see* (*saw, seen*), *sing* (*sang, sung*), or *write* (*wrote, written*).

- The past tense form is sometimes called the **-ed form** (read "ee dee form") because *-ed* is the default expression of the past tense morpheme.
- The past participle form is sometimes called the **-en form** (read "ee enn form") because *-en* is a common expression of the past participle morpheme (though not the default expression, which, as just mentioned, is homonymous with the default past tense morpheme).
- A verb's bare form, past tense, and past participle (in other words, exactly the forms that aren't generally predictable) are known as its **principal parts**. (2) shows some examples.

(2) a. play, played, played
  b. come, came, come
  c. stand, stood, stood

2.9.2 Finiteness

Finiteness of verbs

The verb forms just discussed are classified into two categories: **finite** and **nonfinite**. Finite verbs can function on their own as the core of an independent sentence, whereas nonfinite verbs cannot. Rather, nonfinite verbs must ordinarily combine with a modal, an auxiliary verb, or the infinitival particle *to.*

A verb's *-s* form and past tense form are always finite, and the two participles (the *-ing* and *-en* forms) are always nonfinite.

(3) Finite verb:
  a. ✓ He *gives* a performance every week.
  b. ✓ He *gave* a performance every week.

(4) Nonfinite verb:
  a. ✓ She is *giving* a performance here tonight.
  b. ✓ She has *given* a performance here in the past.

To complicate matters a bit, a verb's bare form can be either finite or nonfinite. Bare forms that occur on their own are finite and express present tense. Otherwise, bare forms are nonfinite. Examples are given in (5) and (6).

(5) Finite verb:
  a. ✓ We *give* performances all the time.

(6) Nonfinite verb:
  a. ✓ We will *give* a performance tomorrow.
  b. ✓ We promised to *give* a performance soon.

Table 2.9 summarizes the above discussion.

Example	Verb form	Finite?
play-s, see-s	3SG present tense	yes
play-ed, saw	past tense	
play, see	bare (present tense)	
	bare (otherwise)	no
play-ing, see-ing	*-ing*	
play-ed, see-n	past participle	

Table 2.9: Verb forms and finiteness in English

Finiteness of clauses

Finiteness is a property not only of verbs, but also of clauses. Ordinary clauses—ones that can stand alone as an utterance—are finite. All finite clauses contain exactly one finite element per clause. In the simplest case, this finite element is a finite verb, as illustrated in (7).

(7) They helped us. (finite clause, finite main verb)

It is also possible for a finite clause's finite element to be a finite auxiliary verb or a modal (finite by definition). In this case, illustrated in (8), the entire clause is then finite, the auxiliary verb or modal is finite, but the main verb is nonfinite.

(8) They did help us. (finite clause, finite auxiliary verb, nonfinite main verb.)

Clauses can also be nonfinite. The verb of a nonfinite clause is always nonfinite. (9) illustrates the case that is easiest to recognize—the case where the nonfinite clause (enclosed in square brackets) contains the particle *to.*

(9) They agreed [to help us]. (nonfinite clause, nonfinite main verb.)

You might be wondering why we classify the subordinate-clause verb in (9)—*help*—as nonfinite. After all, *help* is a bare verb form, and so in principle might be finite (recall Table 2.9). The reason is that the verb neither expresses present tense nor agrees with a third-person singular subject. We can see this clearly by changing the past tense verb of the main clause to present tense and the main-clause subject to third-person singular. As shown in (10), *help* does not change to *help-s,* as we would expect if it were finite.

(10) He always agrees [to { help, *help-s } us].

The same reasoning extends to cases where the nonfinite clause has a subject of its own. The examples in (11) and (12) show that neither a subordinate-clause nor a main-clause third-person subject causes the bare verb to change form.

(11) a. We expect [her to { help, *help-s } us].
b. She expects [him to { help, *help-s } us].

(12) a. We make [her { help, *help-s } us].
b. She makes [him { help, *help-s } us].

We conclude from this that the bare forms are nonfinite, as are the clauses containing them.

Table 2.10 summarizes the above discussion. The relevant clauses are uniformly delimited by square brackets. In the general case, the yes/no values agree across columns; it is only the second row that contains a mismatch.

Example	**Is the clause finite?**	**Does the clause contain a finite element?**	**Is the main verb finite?**
[They helped us] .	yes	yes	yes
[They did help us] .	yes	yes	no
He always agrees [to help us] .	no	no	no
She expects [him to help us] .	no	no	no
He makes [her help us] .	no	no	no

Table 2.10: Clauses and finiteness in English

3

Constituency and hierarchical structure

Chapter outline

In Chapter 1, we provided evidence for the idea that sentences aren't just strings of words, but that they consist of **syntactic constituents**—groups of words that belong together. In many cases, native speakers have clear intuitions about whether a particular string is a constituent. But not all cases are clear, and it is useful to have diagnostic tests at our disposal for determining constituenthood in such cases. It also sometimes happens that native speakers disagree over whether a string is a constituent. In such cases, the tests can help us to clearly establish that fact so that we have a firm basis for further exploring the differences between the alternative mental grammars. The chapter is organized as follows. We first present the constituenthood tests and some complications in applying them in connection with certain syntactic categories. Finally, we introduce the graphic representation of constituenthood by means of syntactic trees.

Tests for determining syntactic constituenthood 3.1

At first glance, a sentence consists of a string of words arranged in a single dimension—that of linear order. However, we presented evidence in Chapter 1 for a second dimension that is less obvious (though no less real!) than linear order—the dimension of **syntactic structure**. Whether a particular string of words is a syntactic constituent isn't always self-evident, and so several tests have been developed as diagnostics for constituenthood. In this chapter, we review these tests, along with some of the complications that arise in applying them. We also discuss in more detail how syntactic structure is represented in tree diagrams of the sort introduced in § 1.5.1 of Chapter 1.

3.1.1 Substitution

The most basic test for syntactic constituenthood is the **substitution** test. The reasoning behind the test is simple. A constituent is any syntactic unit, regardless of length or syntactic category. A single word is the smallest free-standing constituent belonging to a particular syntactic category. So if a single word can substitute for a string of several words, that's evidence that the string is a constituent.

We mentioned in Chapter 1, § 1.3.1 that pronouns can substitute for noun phrases. Some examples are given in (1).

(1) a. The little boy fed the cat. → ✓ He fed her.
    b. Black cats detest green peas. → ✓ They detest them.

As already mentioned in Chapter 1, it's important to understand that a particular string of words can be a noun phrase in one syntactic context, but not in another. For instance, the substitution test tells us that the underlined strings are noun phrases in (1), but not in (2).

(2) a. The little boy from next door fed the cat without a tail.
    ↓ ↓
    * He from next door fed her without a tail.

  b. These black cats detest those green peas.
    ↓ ↓
    * These they detest those them.

Rather, in these sentences, the noun phrases are the longer underlined strings in (3).

(3) a. The little boy from next door fed the cat without a tail. → ✓ He fed her.
    b. These black cats detest those green peas. → ✓ They detest them.

Pronouns are not the only placeholder elements, or **pro-forms**. For instance, the adverbs *here* or *there* can substitute for constituents that refer to locations or directions. As in the case of noun phrases, whether a particular string is a constituent depends on its syntactic context.[1]

(4) a. Put it on the table. → ✓ Put it there.
b. Put it over on the table. → ✓ Put it over there.
c. Put it over on the table. → ✓ Put it there.

(5) a. Put it on the table that's already set. → * Put it there that's already set.
b. Put it over on the table that's already set. → * Put it over there that's already set.
c. Put it over on the table that's already set. → * Put it there that's already set.

The word *so* can substitute for adjective phrases (here, the most natural-sounding results are obtained in contexts of comparison or contrast). As usual, the same string sometimes is a constituent and sometimes isn't. (The judgments in (6) are ours; for some speakers, (6c) is grammatical.)

(6) a. ✓ I am very happy, and Linda is so, too.
b. ✓ I am very fond of Lukas, and Linda is so, too.
c. * I am very fond of my nephew, and Linda is so of her niece.

Finally, pronouns and sometimes the word *so* can substitute for subordinate clauses introduced by *that,* as in (7) and (8).

(7) a. I { know, suspect } that they're invited.
b. ✓ I { know, suspect } it.

(8) a. I { imagine, think } that they're invited.
b. ✓ I { imagine, think } so.

We conclude our discussion of substitution by noting a complicating factor. It is not the case that every constituent has a corresponding pro-form. For instance, although some prepositional phrases can be replaced by the pro-forms *here* or *there,* others—for instance, ones referring to purposes or reasons—can't. As a result, it is perfectly possible for the substitution test to give false negative results, and for the results of the substitution to disagree with the results of the other tests presented in what follows. We return to the issue of false negatives later on in § 3.2.2.

3.1.2 Movement

Substitution by pro-forms is not the only diagnostic for whether a string is a constituent. If it is possible to move a particular string from its ordinary position to another position—typically, at least in English, to the beginning of the sentence—that, too, is evidence that the string is a constituent (because the string is behaving together as a unit). In order to make the result of movement completely acceptable, especially in the case of noun phrases, it's sometimes necessary to use a special intonation or to invoke a special discourse context. In the examples that follow, empty underlining ("___") indicates the ordinary position that a constituent has moved from, and appropriate discourse material (enclosed in parentheses) may be added to make the examples sound natural.

(9) a. I fed the cats.
b. ✓ The cats, I fed ___. (The dogs, I didn't.)

(10) a. I fed the cats with long, fluffy tails.
b. ✓ The cats with long, fluffy tails, I fed ___. (The other cats, I didn't.)

Movement of constituents other than noun phrases is illustrated in (11)–(13).

(11) Adjective phrase emphasis movement:
a. Ali Baba returned from his travels wiser than before.
b. ✓ Wiser than before, Ali Baba returned from his travels ___.

(12) Adverb phrase emphasis movement:
a. They arrived at the concert hall more quickly than they had expected.
b. ✓ More quickly than they had expected, they arrived at the concert hall ___.

(13) Prepositional phrase emphasis movement:
a. The cat strolled across the porch with a confident air.
b. ✓ With a confident air, the cat strolled across the porch ___.

3.1.3 Questions and short answers

Another diagnostic for constituenthood is whether the string can function as a short answer to a question. The question itself also functions as a diagnostic test, since we can think of it as being derived by substituting a question word for a string and subsequently moving the question word to the beginning of the sentence.[2] (14)–(18) illustrate this pair of tests for a variety of constituent types.

Noun phrase:

(14) What do you see?
  a. The cats.
  b. Cats with long, fluffy tails.
  c. The cats with long, fluffy tails.

Adjective phrase:

(15) How did the traveler return?
  a. Wiser than before.
  b. Fairly jet-lagged.

Adverb phrase:

(16) How did they do?
  a. Not badly.
  b. Surprisingly well.
  c. Much better than they had expected.

Prepositional phrase:

(17) How did the cat stroll across the porch?
  a. With a confident air.
  b. quickly.

(18) Where did Ali Baba go?
  a. On a long journey.
  b. To New York.

Notice, incidentally, that *so* substitution, discussed earlier, has a variant that is reminiscent of questions. In addition to just substituting for the string of interest, *so* can subsequently move to the beginning of the sentence, triggering subject-aux inversion—the same process that turns declarative sentences into yes-no questions. This variant of *so* substitution is illustrated in (19) and (20). (The judgment for (19c) varies across speakers in the same way as the one for (6c).)

(19) a. ✓ I am very happy, and so is Linda.
  b. ✓ I am very fond of Lukas, and so is Linda.
  c. * I am very fond of my nephew, and so is Linda of her niece.

(20) ✓ I { imagine, think } that they're invited, and so do they.

3.1.4 *It*-clefts

The final constituent test that we'll consider is based on a special sentence type known as ***it*-clefts**. We begin by noting that ordinary sentences can often be divided into two parts: a part that contains background information that is presupposed, the **ground**, and a part that is intended to be particularly informative, the **focus**. In spoken language, this focus-ground partition (also known as its **information structure**) is often conveyed by intonation alone in English.[3] But English can also express the focus/ground distinction via a syntactic frame consisting of *it*, a form of the copula *to be*, and the subordinating conjunction *that*. In the examples in (21)–(22), the frame is in standard font, the ground is in *italics*, and the focus is in **bold**. Notice that a single sentence can be partitioned into focus and ground in more than one way, giving rise to more than one *it*-cleft.

(21) a. **Ordinary cats** *detest the smell of citrus fruits.*
   b. ✓ It is **ordinary cats** that *detest the smell of citrus fruits*.

(22) a. *Ordinary cats detest* **the smell of citrus fruits.**
   b. ✓ It is **the smell of citrus fruits** that *ordinary cats detest*.

If a string can appear as the focus of an *it*-cleft, then it is a constituent. Some examples for various constituent types other than noun phrase are given in (23)–(25).

(23) Adjective phrase *it*-cleft:
   a. Ali Baba returned from his travels wiser than before.
   b. ✓ It was wiser than before that Ali Baba returned from his travels ___.

(24) Adverb phrase *it*-cleft:
   a. They arrived at the concert hall more quickly than expected.
   b. ✓ It was more quickly than expected that they arrived at the concert hall ___.

(25) Prepositional phrase *it*-cleft:
   a. The cat strolled across the porch with a confident air.
   b. ✓ It was with a confident air that the cat strolled across the porch ___.

Some complications for constituency diagnostics 3.2

3.2.1 Mismatches between syntactic structure and other structure

We mentioned earlier that it is not always self-evident whether a particular sequence of words is a syntactic constituent. For instance, in reading a sentence like (26) out loud, we can perceive an intonation break between *cat* and *that* (indicated by the slash).

(26) This is the cat / that chased the rat.

Because the intonation break is clearly audible, it is very tempting to equate the sentence's abstract syntactic structure with its relatively concrete prosodic structure. Specifically, because *the* and *cat* belong to the same prosodic constituent, it is tempting to treat *the cat* as a syntactic constituent.

There are two pieces of evidence against doing so. First, as we have already seen in similar examples, substituting a pronoun for the string *the cat* is ungrammatical in the context of (26) (though not in other contexts).

(27) a. This is the cat that chased the rat.
b. * This is it that chased the rat.

(28) a. I petted the cat.
b. ✓ I petted it.

Second, the string *cat that chased the rat* is shown to be a constituent by the grammaticality of substituting the pro-form *one* (*one* substitution is discussed in more detail in section 6.1.4 of Chapter 6).

(29) a. This is the cat that chased the rat.
b. ✓ This is the one.

The facts in (27) and (29) converge to tell us that the word *cat* first combines with the relative clause, not with *the.* Thus, (26) exhibits a **mismatch** between two types of linguistic structure: syntactic and prosodic.

It is worth noting that the syntactic structure just described corresponds to the way that the interpretation of the entire expression *the cat that chased the rat* is composed from the interpretation of smaller expressions. In a simple semantics, the term *cat* denotes the set of all cats. Combining *cat* with the relative clause yields *cat that chased the rat,* which denotes a subset of all cats—namely, those with the property of having chased the rat. Further combining *cat that chased the rat* with the definite article *the* yields as a denotation some unique individual within the rat-chasing subset of cats (exactly which individual this is depends on the discourse context).[4]

This correspondence of syntactic structure and semantic structure (the step-by-step composition of the expression's meaning, as just illustrated for (26)) holds up as a first approximation, and it is consistent with the correspondence between noun phrases and individuals, between adjective phrases and properties, between prepositional phrases and locations, directions, etc., between verb phrases and events, states, etc., and so on. Nevertheless, mismatches between syntactic structure and semantic structure are possible. For instance, the sentence in (30) has two distinct meanings, which can be paraphrased as in (31).

(30) Every student knows two languages.

(31) a. For every student, it is the case that they know two languages.
(Langston knows Arabic and Basque, Maisha knows Chinese and Danish, Lucca knows English and French, ...)

b. There are two languages that every student knows.
(Arabic and Basque are known by Langston, Maisha, Lucca, ...)

In the interpretation in (31a), the universal quantifier *every* is said to take **scope** over the number *two* (`EVERY > TWO`). In the interpretation in (31b), the number takes scope over the universal quantifier (`TWO > EVERY`). In either case, though, the ambiguous sentence itself (not the paraphrases!) has a single syntactic structure. This is evident from the syntactic constituenthood tests in (32)–(33), where the question and short answer pair are compatible with either scope interpretation.

(32) a. Every student knows two languages.

b. Who knows two languages? Every student.

(33) a. Every student knows two languages.

b. What does every student know? Two languages.

Other mismatches are also possible. Recall from the section on *it*-clefts that one and the same sentence can be associated with more than one information structure. Finally, mismatches between syntactic and morphological structure are common (we discuss two important cases in Chapters 12 and 13).

3.2.2 False negative results

In a perfect world for syntacticians, the relation between a string being a constituent and it passing the constituenthood tests would be biconditional, as indicated in the "Dream world" column in Table 3.1. The world, however, is not made to order for syntacticians, and it turns out to be possible for constituents to fail one or more constituenthood tests. The actual state of affairs is thus as indicated in the "Real world" column of Table 3.1.[5]

If a string...	then it...	Dream world	Real world
passes the constituency tests,	is a constituent.	TRUE	TRUE
is a constituent,	passes the constituency tests.	TRUE	sometimes FALSE

Table 3.1: False negatives in constituency tests

In other words, the failure of a string to pass a constituenthood test can be a false negative result. In what follows, we present three such cases—constituents that fail at least some of the constituenthood tests

- because they are individual syntactic heads rather than phrases,
- because they contain inflected verbs, or
- because they are contained within so-called syntactic islands.

- because the diagnostic as constructed is pragmatically infelicitous.

The unifying issue is that the grammatical constructions that we use to diagnose constituency are not magical tests: they are themselves grammatical constructions within the grammatical system, subject to constraints that are perhaps independent of the constituency question. The end of this section discusses this in more depth.

Phrasal versus lexical constituents

Since single lexical items are minimal (often indivisible) units, they are constituents by definition.[6] Nevertheless, they don't necessarily behave on a par with multiword constituents. For instance, *cats* passes the constituenthood tests reviewed earlier in (34), but not in (35).

(34) Cats are not social animals. → They are not social animals.

(35) a. *The* cats are hungry. → * *The* they are hungry.
b. *Tabby* cats are quite common. → * *Tabby* they are quite common.
c. Cats *without tails* are relatively rare. → * They *without tails* are relatively rare.

The reason for the grammaticality contrast in (34) and (35) is a systematic difference between the syntactic contexts in these examples. In (35), the word *cats* is accompanied by a determiner or a modifier of some sort, indicated by *italics*. In such contexts, *cats* combines with these other words to form a noun phrase, but it isn't a noun phrase in its own right. By contrast, *cats* in (34) is a bare, or unmodified, noun. As such, it functions as a noun and as a noun phrase at the same time. In other words, there are two levels of constituenthood: the **lexical** level, where single words are constituents by definition, and the **phrasal** level, where single words don't necessarily behave on a par with multiword constituents.

The constituenthood tests reviewed earlier turn out to be diagnostic only for phrasal constituents. Moving, questioning, and *it*-clefting lexical constituents, rather than phrasal ones, yields ungrammatical results, as illustrated in (36)–(47). As before, the relevant lexical constituent is underlined, and any material belonging with it to the same phrasal constituent is in italics.

No emphasis movement of individual heads:

(36) a. I fed *the* cats.
b. * Cats, I fed *the* ___.

(37) a. Ali Baba returned from his travels wiser *than before.*
b. * Wiser, Ali Baba returned from his travels ___ *than before.*

(38) a. They arrived at the concert hall *more* quickly *than they had expected.*
b. * Quickly, they arrived at the concert hall *more* ___ *than they had expected.*

(39) a. The cat strolled across the porch with *a confident air.*
b. * With, the cat strolled across the porch ___ *a confident air.*

No questions and answers with individual heads:

(40) a. * What did you see *the* ___?
b. * Cats.

(41) a. * How did Ali Baba return from his travels ___ *than before*?.
b. * Wiser.

(42) a. * How did they arrive at the concert hall *more* ___ *than they expected*?
b. * Quickly.

(43) a. * How did the cat stroll across the porch ___ *a confident air*?
b. * With.

No *it*-clefts focusing individual heads:

(44) a. Ordinary cats detest *the* smell *of citrus fruits*.
b. * It is smell that ordinary cats detest *the* ___ *of citrus fruits*.

(45) a. Ali Baba returned from his travels wiser *than before*.
b. * It was wiser that Ali Baba returned from his travels ___ *than before*.

(46) a. They arrived at the concert hall *more* quickly *than they had expected*.
b. * It was quickly that they arrived at the concert hall *more* ___ *than they had expected*.

(47) a. The cat strolled across the porch with *a confident air*.
b. * It was with that the cat strolled across the porch ___ *a confident air*.

By contrast, the examples in (48)–(56) illustrate the grammatical results of moving, questioning, and *it*-clefting phrasal constituents that happen to consist of a single word (notice the absence of italicized material in this case). Examples for prepositional phrases are missing because prepositions, at least the ones in the examples that we have been using, require an object.

Emphasis movement of a single word that makes up an entire phrase:

(48) a. I like cats.
b. ✓ Cats, I like ___.

(49) a. Ali Baba returned from his travels wiser.
b. ✓ Wiser, Ali Baba returned from his travels ___.

(50) a. They arrived at the concert hall quickly.
b. ✓ Quickly, they arrived at the concert hall ___.

Questions and answers with a single word that makes up an entire phrase:

(51) a. ✓ What do you like ___?
b. ✓ Cats.

(52) a. ✓ How did Ali Baba return from his travels ___ ?.
b. ✓ Wiser.

(53) a. ✓ How did they arrive at the concert hall ___ ?
b. ✓ Quickly.

***It*-clefts focusing a single word that makes up an entire phrase**:

(54) a. Ordinary cats detest citrus.
b. ✓ It is citrus that ordinary cats detest ___.

(55) a. Ali Baba returned from his travels wiser.
b. ✓ It was wiser that Ali Baba returned from his travels ___.

(56) a. They arrived at the concert hall quickly.
b. ✓ It was quickly that they arrived at the concert hall ___.

What this shows is that the results from (36)-(47) do not mean that individual words cannot be constituents; instead, in these instances the diagnostic is looking for phrasal constituents, and in (36)-(47) (where the diagnostics fail for individual words), those individual words are not also entire phrases.

Inflected verb phrases

Testing for the constituenthood of verb phrases is a bit more complicated than is testing for the constituenthood of other syntactic categories. First, there are no simple pro-forms for verb phrases in Mainstream U.S. English. The best we can do is to use the periphrastic forms *do so* for substitution and *do what* for questions.[7] (Notice that it's only *what,* rather than the entire pro-form *do what,* that moves to the beginning of a question.)

(57) Substitution:
a. She will write a book. → ✓ She will do so.
b. The two boys could order tuna salad sandwiches. → ✓ The two boys could do so.

(58) Question/short answer:
a. What will she do? → ✓ Write a book.
b. What could the two boys do? → ✓ Order tuna salad sandwiches.

Second, and more importantly (given our present focus on false negative results), verbs and the verb phrases that contain them are sometimes overtly inflected for tense and/or agreement. In general, diagnostics that isolate the uninflected verb phrase show a verb phrase constituent, but inflected verbs give different results. (59) gives grammatical results for with the verb doesn't inflect for the tense (future tense in English) and also when the verb does inflect for tense (past tense in English), though notably in the latter the substituted form is also inflecting for tense.

(59) a. Substitution, uninflected verb:
She will write a book. → ✓ She will do so.
b. Substitution, inflected verb:
She wrote a book. → ✓ She did so.

In contrast, an attempt to use the Q&A diagnostic for a verb phrase is fine when the verb is uninflected (again, English future) but gives anomalous results in the past tense when the answer contains tense, as in (60b). The past tense question can be answered, but the verb phrase must be uninflected, as in (60c).

(60) a. Question/short answer, uninflected verb:
What will she do? → ✓ Write a book.
b. Question/short answer, inflected verb:
What did she do? → *? Wrote a book.
c. Question/short answer, inflected verb:
What did she do? → ✓ Write a book.

Similar patterns appear in emphasis movement. Movement of uninflected verb phrases is grammatical,[8] but movement of inflected ones is not.

(61) Movement, uninflected verb phrase:
a. (She says that) she will write a book, → ✓ (and) write a book, she will ___.
b. though she may write a book → ✓ write a book though she may ___

(62) Movement, inflected verb phrase:
a. (She says that) she wrote a book, → * (and) wrote a book, she ___.
b. though she wrote a book → * wrote a book though she ___

These facts don't have an obvious solution based on our model so far; surely constituency of a verb phrase does not depend on whether the phrase is in past or future tense. These facts do in fact have very natural interpretations based on the model that we develop, but it will be easier to lay out the rationale after more of the model is established (see Exercise 5.3).

The differences between inflected and uninflected verb phrases is not always the root cause of unacceptable diagnostics testing verb phrases. In *it*-clefts, even uninflected verb phrases are largely unacceptable in focus.

(63) *It*-cleft, uninflected verb phrase:
She will write a book. → ?? It is write a book that she will ___.

(64) *It*-cleft, inflected verb phrase:

She wrote a book. ⟶ * It is wrote a book that she ___.

This instead appears to be a more general restriction about *it*-clefts: they are useful for diagnosing constituency of arguments and adjuncts of a sentence, but verb phrases can never be the focus of an *it*-cleft in Mainstream U.S. English. Because this is always the case, we interpret this not as an instance of non-constituency of verb phrases, but rather as an independent restriction on *it*-clefts, such that *it*-clefts can't be relied on to test constituency of verb phrases.

Apart from this wrinkle concerning inflection, verb phrases behave just as we have come to expect from other constituent types. The tests yield grammatical results for verb phrases, but not for verbs alone.

(65) Substitution:

She will write a book. ⟶ * She will do so a book.

(66) Movement:

a. (She says that) she will write a book, ⟶ * and write, she will ___ a book.
b. though she may write a book ⟶ * write though she may ___ a book

(67) Question/short answer:

a. * What will she do a book?
b. * Write.

(68) *It*-cleft:

She will write a book. ⟶ * It is write that she will ___ a book.

And once again, particular strings can be phrasal constituents in one syntactic context, but not in another. For instance, *write* isn't a phrasal constituent when it combines with a direct object, but it is when used on its own. This is the source of the grammaticality contrast between (65)–(68) and (69)–(72).

(69) Substitution:

She will write. ⟶ ✓ She will do so.

(70) Movement:

a. (She says that) she will write, ⟶ ✓ and write, she will ___.
b. though she may write ⟶ ✓ write though she may ___

(71) Question/short answer:

a. What will she do?
b. ✓ Write.

(72) *It*-cleft:

She will write. ⟶ ? It is write that she will ___.

Chapter 5 develops a model of phrase structure that readily explains the apparently variable behavior of (un)inflected verb phrases as constituents (or not). The conclusions for our purposes right now are twofold: first, we need to be aware that overt inflection on verb phrases can affect our constituency diagnostics. And second, the larger lesson is that an ungrammatical attempt to construct a constituency diagnostic does not mean that the string is necessarily not a constituent: such conclusions must be drawn based on converging evidence from multiple sources. Furthermore, this also points to the idea that diagnostics are not unreliable when they conflict, it is a reminder that language is complex and these diagnostics do multiple jobs (for example, themselves being emphatic constructions with their own requirements, in the case of emphasis movement and *it*-clefts). Interpreting the evidence needs to take this state of affairs into consideration.

Islands

In (73a), *the doctors* is a constituent, as is evident from the possibility of substituting a pronoun for the string, as in (73b).

(73) a. We should invite the lawyers and the doctors.
     b. We should invite the lawyers and them. (pointing to the doctors)

But although *the doctors* passes the substitution test, the other three tests yield ungrammatical results.

(74) * The doctors, we should invite the lawyers and ___.

(75) a. * Who should we invite the lawyers and ___?
     b. * The doctors.

(76) * It is the doctors that we should invite the lawyers and ___.

Take Note

Earlier, we pointed out that question formation can be thought of as a combination of substitution and movement. The parallel between movement and question formation in (74), (75) and *it*-clefting in (76) suggests that the latter, too, involves movement in some way. In the remainder of this section, we will therefore use the term 'movement' in a broad sense to include all three constituenthood tests in (74)–(76).

Notice that there is nothing semantically or conceptually ill-formed about (74)–(76), since it is possible to paraphrase the intended meaning grammatically as in (77)–(79).

(77) ✓ The doctors, we should invite ___ together with the lawyers.

(78) a. ✓ Who should we invite ___ together with the lawyers?
     b. ✓ The doctors.

(79) ✓ It is the doctors that we should invite ___ together with the lawyers.

Taken together with the grammaticality of (73b), the contrast between (74)–(76) and (77)–(79) shows that movement of the noun phrase *the doctors* is somehow prevented by the specific syntactic configuration in (74)–(76). Ross (1967) introduced the metaphorical term **island** for configurations in which movement is blocked where it is expected to be possible. The conceit underlying the term is that the constituents that might be expected to move, but can't, are stranded on an island like castaways.[9]

Ross identified several types of islands, including conjoined phrases like those in (74)–(76), and his work has given rise to an enormous body of literature. Our purpose here is neither to catalog the types of islands nor to pursue the proper linguistic analysis of them (we return to the topic in Chapter 14), but simply to point out that constituenthood tests based on movement will yield false negative results for phrasal constituents if they happen to be contained within islands. In these contexts, a sentence failing to pass a movement-based diagnostic is not evidence against constituency; the sentence is ungrammatical for an independent reason (movement out of an island).

Why false negatives?

We've shown that there can be false negatives for constiteuncy diagnostics: instances where a string consistently behaves like a constituency by some diagnostics, but is not grammatical with other constituency diagnostics. Why exactly does this happen? Are diagnostics haphazard? The answer to that is a definitive 'no', but it's important to understand why.

There are no litmus tests for constituenthood that are separate from the grammar itself. That is to say, constituency tests are not like a medical diagnostic test for COVID or the flu, where some instrument or form of measurement that is external to the body can be used to gain information on what's happening inside the body. Rather, constituency diagnostics are more like medical symptoms—they provide evidence for what's going on inside the body, despite being themselves a part of the bodily system.

So consider emphasis movement in a sentence like *The DOCUMENTS, Maisha delivered.* If someone walks up to you at a work event and says that out of nowhere, you might be confused. What's more, on its own the sentence sounds unnatural (compare *Maisha delivered the documents*, which simply reports information). It might be easy to think that the sentence *The DOCUMENTS, Maisha delivered* is ungrammatical because it attempts to move a non-constituent. But the noun phrase *the documents* is in fact a constituent. So what's unnatural about the emphasis movement? In this example, it's the pragmatics of the construction. Emphasis movement creates emphasis, and using emphasis without relevant context can sound unnatural (we use the word **infelicitous** for this kind of pragmatic unnaturalness). So if we were in a conversation and someone brings up the fact that Maisha delivered the relevant work, they could say, *The DOCUMENTS, Maisha delivered; the equipment, she did not.* The sentence now sounds much more natural. The point, of course, is that when we're using a construction to test constituency (which it can do!) we have to be sure to use the construction itself naturally, the way it prefers to be used. Otherwise, we might generate unnatural results for reasons which have nothing to do with constituency.

Likewise, we saw above that *it*-clefts are ungrammatical when attempting to focus a single lexical head (like a noun out of a noun phrase) or when attempting to focus a verb phrase. Does this mean that the *it*-cleft has "failed?" No—it is just that an *it*-cleft is a construction with a specific use, namely, focusing phrases that are themselves arguments or adjuncts of a sentence. So if a string can appear in an *it*-cleft, there is good evidence that it is a constituent. Failing to appear in an *it*-cleft could be for many reasons, including just that the string is not the sort of thing that *it*-clefts are used to focus. This is because it is not an *it*-cleft's sole function to diagnose constituency; the cleft is itself part of the grammar of Mainstream U.S. English and serves its own function in the language. The *it*-cleft systematically targets a subset of constituents, so if a phrase *can* be *it*-clefted, it is good evidence that it is a constituent. If a string cannot be *it*-clefted, it could be for a variety of reasons (one of those reasons *is* that the string is not a constituent, but there are also others).

This is why no single piece of evidence is sufficient when analyzing syntax; we always rely on multiple sources of evidence to draw reliable conclusions. It is not that diagnostics are themselves unreliable: they are reliable when it comes to doing what they do. *It*-clefts and focus movement are excellent at creating emphasis on certain kinds of constituents. But it is not as if their main job is to diagnose constituency—they perform their own functions in the English language, and just so happen to operate on a subset of English constituents. We can draw conclusions about constituency based on diagnostics, but we rely on them only with full context; we look for converging evidence from other sources and consider ungrammatical sentences to indicate non-constituency only potentially, not definitively (in isolation, at least).

3.2.3 Relevant diagnostics, to be used with caution

A couple of other diagnostics are also very helpful for diagnosing constituents, and we summarize them briefly here. We have isolated these into their own section because both are easy to misuse, so we want to introduce them with special instructions to help avoid their misapplication.

Ellipsis is a kind of grammatical construction that uses silence to specifically refer to a familiar constituent from context. Take the English example in (80) for example:

(80) Maisha will play board games this weekend, and Lucca will too.

In this example, even though it is not said, all native English speakers know exactly what it is that Lucca will do: *play board games this weekend*. Only constituents can be elided in this fashion (*elide* = undergo ellipsis), and as such ellipsis constructions can be used as constituency diagnostics. Ellipsis can appear in a conjoined sentence (as in (80), or using other conjunctions like *but*, *although*, etc). Ellipsis can also appear in question and answer contexts as in (81), where the answer is silent on the activity but the interpretation clearly supplies *sell Girl Scout cookies this weekend* from the context.

(81) a. Will Anabella sell Girl Scout cookies this weekend?
   b. Yes, she will.

Ellipsis is also possible in other kinds of conversation or narrative discourse where the silent information is supplied from discourse context.

What is tricky about ellipsis is that proper ellipsis constructions are those where the specific elided material can be reconstructed (in full detail) from discourse. Ellipsis does not include other ways in which speakers may just not mention additional information. The example in (82) can be easily mistaken for an ellipsis construction.

(82) a. Maisha will read on the couch the entire weekend.
b. I will read, too.

The main reason why (82b) is not in fact an ellipsis construction is the silence on where and when the speaker will be reading *could* be interpreted as saying the speaker will read with Maisha in the same way, but it does not necessarily mean that. So (82b) could be followed by the clarification *Except I plan to read in bed, just for Saturday* and there would be no contradiction. But any attempt to follow up (81b) to say that Anabella is not selling cookies would be interpreted as a contradiction. The unmentioned material in (81) is necessarily supplied by context and is therefore ellipsis. The sentence in (82b) is complete, and while the speaker could be intending to imply parallelism with the previous discussion, it's not obligatory, and therefore not ellipsis.

Another highly useful constituency diagnostic is **conjunction**, also known as **coordination**. We will focus on the simplest version of these, the conjunction *and*. Conjunction/coordination necessarily conjoins two constituents of the same grammatical category.

(83) Her dad packed [cut-up apples] and [sliced oranges] in Maisha's lunchbox.

Conjunction of the bracketed phrases in (83) suggests that 1) each of them are constituents in their own right, and 2) they are constituents of the same type. (84c) is ungrammatical; despite both constituents being viable to participate in conjunction, they cannot be conjoined together because they are not constituents of the same category.

(84) a. I like [apples] and [oranges].
b. I like [to run] and [to swim].
c. *I like [apples] and [to swim].

One of the reasons that conjunction has to be applied with caution is that conjunction at times facilitates and interacts with other grammatical processes. So for example, conjunction in (80) above was a viable way to license ellipsis. But without taking ellipsis into account, it would be tempting to suggest that *Lucca will* is a constituent on its own that is of the same type as *Maisha will play board games this weekend* (which, we will see in what follows, it is not).

Both ellipsis and conjunction are in fact quite reliable constituency diagnostics, and deserve to be included in analytical discussions. But both have potential confounds to be mindful of. Specifically, when using them to diagnose constituency, you want to be attentive to whether either of the constructions is perhaps more complicated than originally intended. Are any other constructions (like ellipsis) co-occurring? Is the pronunciation (prosody) normal or are there obligatory pauses or emphases that you might not have expected? Sometimes an altered prosody is a sign of a complicated syntactic construction at play. As always, it is for reasons like this that we consider more than a single constituency diagnostic: it can be all too easy to misinterpret the result of a single diagnostic when each construction comes with its own quirks and specific usages. A collective and recurring pattern of constituency diagnostic results, on the other hand, is much more reliable.

Representing syntactic constituenthood 3.3

In Chapter 1, § 1.5.1, we will introduce tree diagrams as a convenient way of representing syntactic structure. For mathematicians working in the field of graph theory, the formal properties of tree diagrams are interesting in their own right, but for syntacticians, the interest of trees lies in the fact that they are **representations**, or **models**, of constituent structure. In other words, the graphic structure of a tree on the page is intended as a statement (or at least, a hypothesis) about the way that speakers group together syntactic elements in their minds. In a good model, the properties of the model correspond straightforwardly to the properties of the domain of inquiry. Such a close correspondence allows us to state observations and generalizations about the domain of inquiry without undue complication. Moreover, if we're lucky, we might even be able to use our understanding of the model's formal properties as a sort of conceptual lever to generate hypotheses and to discover facts and generalizations about the domain of inquiry that would otherwise escape notice.

3.3.1 A case study: The existence of verb phrases

In light of these considerations, let's consider the sentence in (85), focusing particularly on the constituenthood of the underlined string.

(85) The *secretary* will <u>draft the letter</u>.

According to the two tests that apply to verb phrases, the string *draft the letter* is a constituent.

(86) Substitution:

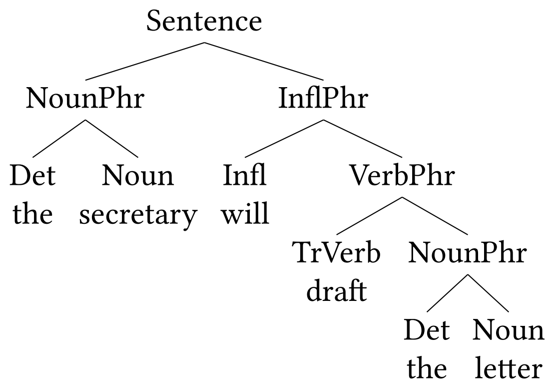
The *secretary* will <u>draft the letter</u>. → The secretary will <u>do so</u>.

(87) Question/short answer:

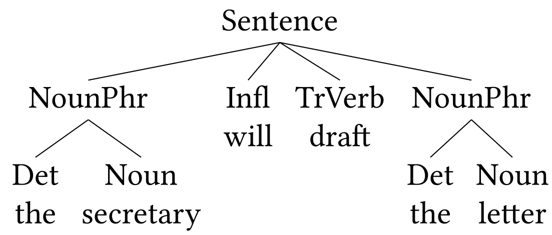
a. <u>What</u> will the secretary <u>do</u>?
b. <u>Draft the letter</u>.

Having established this fact, let's now consider two alternative representations of the sentence. (88a) proposes more structural hierarchy (including an InflPhr and a VerbPhr), whereas (88b) is an alternative, 'flatter' tree where most constituents are on the same level as each other.

(88) a.



b.



At first glance, the flatter tree might be argued to be preferable on the grounds that it is simpler in the sense of containing fewer nodes. But let's focus on the question of which tree is a better representation of the sentence as it exists in an English speaker's mind. Specifically, let's ask whether either of the trees in (88) has some graphic property that corresponds to the results of the constituenthood tests in (86) and (87). In (88a), the answer is 'yes', since there is a single node (the one labeled VerbPhr) that dominates all and only the words in the string *draft the letter*, a string which we showed in (86)-(87) behaves like a constituent in this sentence. We say that the node **exhaustively dominates** the string (for more details, see § 11). By contrast, the tree in (88b) lacks such a node and has no other graphic property corresponding to the string's constituenthood. Clearly, then, (88a) is a more useful model of the sentence, because it follows the natural convention in (89).

(89) Syntactic constituents are represented graphically as nodes in a tree.
(*This description will be refined below*)

As is always the case, if (88a) is an accurate representation of our mental grammar of the sentence, we expect multiple kinds of evidence to converge on that conclusion. We already see this with the substitution evidence in (86) and the Q&A/sentence fragment diagnostic in (87). (90)-(92) all confirm the presence of a verb phrase constituent in this sentence.

(90) Conjunction:

The secretary will [draft the letter] and [appear before Congress].

(91) Ellipsis:

Q: Will the secretary draft the letter?

A: He will ___.

gap is interpreted as: draft the letter

(92) Emphasis movement:
The secretary said he will draft the letter, and draft the letter, he will ___.

At the same time, some diagnostics show a different result.

(93) * It is draft the letter that the secretary will ___.

But as we noted above, all *it*-clefts for verb phrases fail, so (93) doesn't actually tell us anything informative about whether there is a verb phrase constituent. Therefore, these tests confirm the finding that the sentence behaves as if there is a verb phrase constituent independent of the other components of the sentence (including tense), as drawn in (88a).

3.3.2 Trees versus brackets

Syntacticians use both trees and bracket structures to show the exact same things. A node on a tree represents a constituent, whereas square brackets are used to isolate a constituent. So the noun phrase from (88a) can also be represented as in (94):

(94) $[_{\text{NounPhr}}\ [_{\text{Det}}$ the $]\ [_{\text{Noun}}$ letter $]\ ]$

The verb phrase from (88a) can be drawn with a tree as in that example, or with a bracket structure as in (95).

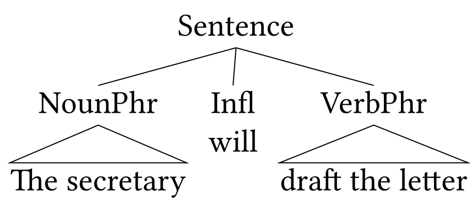
(95) $[_{\text{VerbPhr}}\ [_{\text{Verb}}$ draft $]\ [_{\text{NounPhr}}\ [_{\text{Det}}$ the $]\ [_{\text{Noun}}$ letter $]\ ]\ ]$

You may well find (95) harder to read/interpret (we do!). And you can imagine how visually complicated larger structures would be in bracket notation. This is a reason that we sometimes avoid bracket structures—because they only demarcate structures in one visual dimension (horizontal), they make less use of visual differentiation to communicate structure, which can make it harder to interpret them. So why use them at all? One way they are especially helpful is when communicating underspecified structure. (96) gives the sentence we've been considering and shows that there exists a verb phrase constituent in it.

(96) The secretary will $[_{\text{VerbPhr}}$ draft the letter $]$.

We have tools in trees for underspecification also; we use **triangles** for constituents where we don't want to show substructure. So (97) shows essentially the same thing as (96) in a tree (adding a node and triangle for *the secretary*: trees force somewhat more precision than brackets do).

(97)



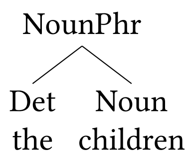
When should you use brackets, and when should you use a tree? In general the guiding principle is to use whichever tool best communicates what you are trying to tell your audience with the visualization. In our estimation, (96) shows this particular underspecified structure more naturally than the tree in (97) does, but the full verb phrase structure is much more readable as a tree in (88a) than it is as a bracket structure in (95).

3.3.3 Basic terms and relations

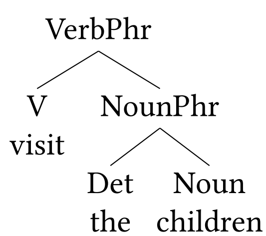
We have seen above that the graphic structures that we refer to as **trees** are very useful for modeling constituency and phrase structure in natural language. This section discusses a number of basic definitions about the components of trees (and relationships between those components) that are useful for us to analyze grammar.

Dominance

Trees consist of a set of **nodes** connected by **branches**. It is sometimes useful to distinguish between two types of nodes: **terminal nodes**, which are labeled with vocabulary items, and **nonterminal nodes**, which are labeled with syntactic categories. In a very simple tree like (98), the terminal nodes are the Det(erminer) node (*the*) and the N(oun) node (*children*), and NounPhr is a nonterminal Node.

(98)


The NounPhr node is a branching node, with more than one branch emanating from it. All terminal nodes are non-branching by definition, but at times there are non-terminal nodes that are also non-branching (meaning they have only one element below them in the tree). The NounPhr node is related to Det and N in this tree by the **dominance** relation. Dominance is a theoretical primitive; in other words, it is an irreducibly basic notion, comparable to a mathematical concept like "point". Dominance is represented graphically in terms of top-to-bottom order. That is, if a node A dominates a node B, A appears above B in the tree. In (98), for instance, NounPhr dominates Det and N. The node that dominates all other nodes in a tree, and is itself dominated by none, is called the **root node**. In (98), NounPhr is the root node. In a slightly more complex tree like (99) that shows the verb phrase *visit the children*, the NounPhr node is no longer the root node: instead, VerbPhr is the root node in (99).

(99)


Dominance is a **transitive** relation (in the logical sense of the term, not the grammatical one). In other words, if A dominates B, and if B dominates C, then it is necessarily the case that A dominates C. So in (99) the VerbPhr node dominates the NounPhr node, and the NounPhr node dominates the N node, so VerbPhr also necessarily dominates N.[10] An important subcase of dominance is **immediate dominance**. This is the case where the two nodes in question are connected by a single branch without any intervening nodes. More formally, immediate dominance is defined as in (100).

(100) A **immediately dominates** B if and only if

a. A dominates B, and

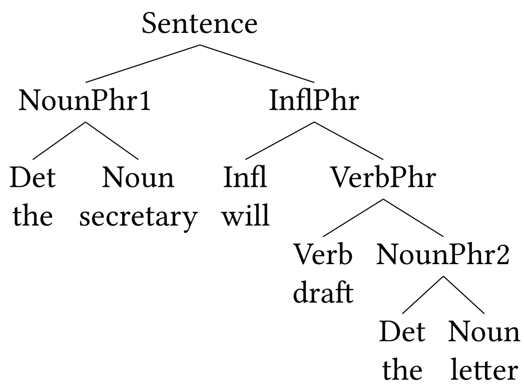
b. there is no node C, distinct from A and B, such that A dominates C and C dominates B.

Unlike dominance, immediate dominance is not a transitive relation. This is apparent from even a simple structure like (99), where VerbPhr immediately dominates NounPhr, and NounPhr immediately dominates N, but VerbPhr does not immediately dominate N because NounPhr is distinct from VerbPhr and N and intervenes between the two.

Precedence

In general, trees are more complex than the very simple case in (98), and they contain multiple nodes that have more than one branch emanating from them, as in (101).

(101)



In such trees, two nodes are related either by dominance or by a second primitive relation, **precedence**. Precedence is represented graphically in terms of left-to-right order. Dominance and precedence are mutually exclusive. That is, if A dominates B, A cannot precede B, and conversely, if A precedes B, A cannot dominate B. Like dominance, precedence is a transitive relation, and just as with dominance, there is a nontransitive subcase called **immediate precedence**. The definition of immediate precedence is analogous to that of immediate dominance; the term *dominates* in (100) is simply replaced by *precedes*. The difference between precedence, which is transitive, and immediate precedence, which isn't, can be illustrated in connection with (101). The first instance of Noun (secretary) both precedes and immediately precedes Verb, and Verb in turn both precedes and immediately precedes the second instance of NounPhr (the letter). The first instance of Noun precedes the second instance of NounPhr, but not immediately.

3.3.4 Derived terms and relations

Kinship terminology

Certain relations among nodes are often expressed by using kinship terms. If A dominates B, then A is the **ancestor** of B, and B is the **descendant** of A. If A immediately dominates B, then A is the **parent** of B, and B is the **child** of A. If A immediately dominates B and C, then B and C are **siblings**.[11] In (101), Sentence is the root node; in the kinship metaphor, it is the ancestor of every other node in the tree. The terminal *secretary* is the child of NounPhr1. NounPhr1 and VerbPhr are siblings, and so are Verb and NounPhr2, but *draft* and the second instance of *the* are not (they don't have the same parent node). Notice, incidentally, that syntactic trees are single-parent families. Most theories of syntax do not allow nodes with more than one parent.

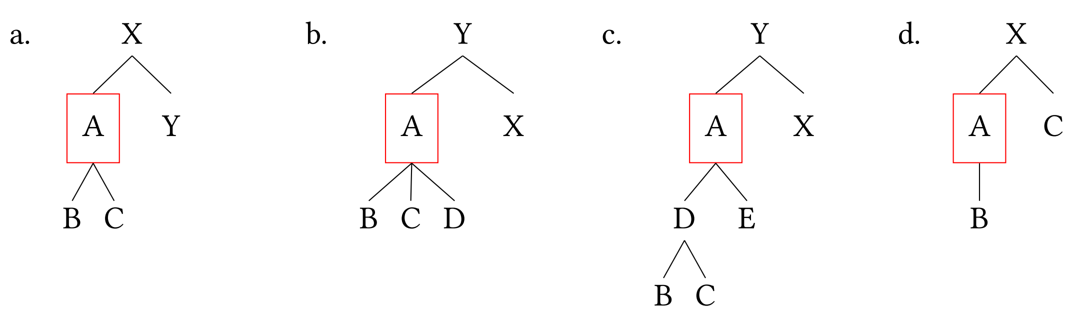
Branching

Depending on the number of children, nodes are classified as either **nonbranching** (one child) or **branching** (more than one child). A more detailed system of terminology distinguishes nodes that are **unary-branching** (one child), **binary-branching** (two children), and **ternary-branching** (three children). Nodes with more than three children are hardly ever posited in syntactic theory. Indeed, according to an influential hypothesis (Kayne 1984), Universal Grammar allows at most binary-branching nodes. According to this hypothesis, it is a formal universal of human language that no node can have more than two branches (a parent node has no more than two children).

Exhaustive dominance

Some node A **exhaustively dominates** two or more nodes B, C, … if and only if A dominates all and only B, C, … For instance, A dominates the string B C in (102a)–(102c), but exhaustively dominates it only in (102a). In (102b) and (102c), A fails to exhaustively dominate B and C because it runs afoul of the *only* condition (it dominates too much material). In (102d), A fails to exhaustively dominate B and C for the opposite reason. A doesn't dominate C, because it runs afoul of the *all* condition (it dominates too little material).

(102)



As is evident from (102b)–(102d), dominance is a necessary but not a sufficient condition for exhaustive dominance. This also puts us in a position to create a more precise definition of constituency than we've been able to up until this point.

(103) **Constituent** (final definition)
A constituent is a string of terminals that is exhaustively dominated by a single node.

C-command

An important derived relation in syntactic theory is **c-command**,[12] which is defined as follows.

(104) A **c-commands** B if and only if
  a. neither A nor B dominates the other, and
  b. the lowest branching node that dominates A also dominates B.

Notice that the notion of c-command is defined in terms of dominance and makes no mention of precedence. It is tempting to assume that c-command logically implies precedence, or vice versa, but it is a temptation to be firmly resisted.[13] Notice further that c-command is not necessarily a **symmetric** relation. In other words, a node A can c-command a node B without B c-commanding A. For instance, in (101), InflPhr c-commands *secretary* (N) (because the first branching node dominating InflPhr, namely Sentence, dominates *secretary*), but not vice versa (because the first branching node that dominates *secretary*, namely NounPhr1, doesn't dominate VerbPhr).

Although c-command isn't necessarily a symmetric relation, it is possible for two nodes to c-command each other. This is the case when the two nodes are siblings. Syntactic siblinghood is also known as **mutual c-command** or **symmetric c-command**. So in (101) NounPhr1 and InflPhr c-command each other (the first node that dominates each is Sentence, and Sentence dominates each of them).

An inquisitive reader may look at the c-command relation and feel that it appears to be rather arbitrary: why would this particular configuration matter? We could specify any number of arbitrary relations between the various nodes on a tree. The short answer is that human language seems to care—a variety of patterns in language seem to be dependent on this notion of c-command to describe them accurately. Binding effects are one major one (see Chapter 15), but there are others, as some of the exercises at the end of this chapter explore.

Regarding scientific models 3.4

We conclude this discussion of the model character of syntactic representations by emphasizing that models are just that—models, and not the actual domain of inquiry itself. The purpose of any model is to help us understand some part of reality that is too complex to understand in all of its detail, at least all at once. This means that models are partial in two respects. A first way is that they leave out properties of a phenomenon that aren't relevant from a particular point of view. This fact is often stated as the maxim "Don't mistake the map for the territory".

For instance, a mountaineer's map might show topographical information in great detail, but completely ignore political boundaries, whereas just the reverse might be true of a diplomat's map. Both topographic information and political boundaries are relevant to someone walking over those spaces, and are equally real, but are not equally relevant to any given map: maps can serve a particular purpose.

Analogously, in linguistics, syntactic models leave out many important properties of language, such as real-world plausibility, pragmatic felicity, the location of intonation breaks, variation based on social factors like race, gender, age, and other aspects of identity, and so on. These are the focus of other subdisciplines of linguistics, which in turn often ignore syntactic structure. Our attention on the aspects of language we are working on should not be interpreted as trivializing the aspects we are not working on. But these choices, of course, are not always inconsequential: sometimes we choose not to build a factor into our model that turns out to be relevant to understanding the phenomenon, and we have to change what we include in our model.

A second way that models are partial is what we just alluded to: models are subject to revision as our understanding of a particular domain improves and deepens. Another way of putting this—which sounds more impressive—is to say that scientific progress is possible. You will experience this as this textbook proceeds; some of what we present in initial chapters is the product of simplifying assumptions that are very good for introducing new concepts, but aren't good enough as final descriptions/analyses. But even in a longer-term mindset, theories of syntax are not yet at the same stage of stability as, say, the periodical table of elements in chemistry, or the theory of evolution in biology. Syntax is a younger science (and, along with other aspects of human cognition, still part of the big, hard problem of human consciousness). There are many things that we can report with high confidence about natural language syntax, and this textbook focuses on those things. But a student continuing in syntax can expect details of our model to adjust and shift as they get deeper into the field. The model we are building is helpful in many ways, but it is a model, and not the final truth about syntax. So you should also expect adjustments and changes to the model.

New Empirical Observations

- Constituents are strings within a sentence that consistently behave as if they are a unit.
- Constituency tests are grammatical constructions that target constituents: they can be leveraged to help determine whether a string is a syntactic constituent.
- Constituency tests in English include:
    - substitution
    - movement
    - questions and short answers
    - *it*-clefts
    - ellipsis
    - coordination
- All constituency tests are their own grammatical constructions with their own requirements and restrictions, which can be violated separately from constituency concerns. Therefore so-called constituency tests can be ungrammatical for reasons

separate from constituency questions (i.e., false negatives exist).
- Syntactic constituents may not align with other kinds of linguistic constituents (for example, prosodic constituents).
- Phrase structure is hierarchical (larger constituents contain smaller constituents).

Components of the Theoretical Model

- Constituenthood may be graphically modeled by trees.
- A constituent is a node in a tree.
- A parent node may have no more than two children (binary-branching hypothesis, Kayne 1984).
- We use trees to define structural relationships between components of a sentence.
  - dominate
  - immediately dominate
  - exhaustively dominate
  - precede
  - immediate precede
  - ancestor of
  - descendant of
  - sibling of
  - parent of
  - child of
  - c-command

Notes 3.5

1. It's worth noting that the syntactic category of the pro-forms in (4) and (5) differs from the category of the underlined strings. As just mentioned, the pro-forms are adverbs, whereas the underlined strings are prepositional phrases. The mismatch in syntactic category doesn't affect the validity of the test, however. All that is necessary is that the pro-form can replace the string, preserving (at least roughly) the original meaning.

2. Under certain discourse conditions, English allows the question word to remain **in situ** (that is, in the place where it substitutes before it moves), as illustrated in (i). See Chapter 8 for more discussion.

   (i) a. Who did you see? (information question or echo question)
   b. You saw who? (only echo question)

3. It is worth pointing out that focus-ground partitioning is relevant not just for *it*-clefts, but also for questions and (short) answers. The focus in a question is the unknown information expressed by the question word. A short answer to a question consists of a focus and no other material. Repeating the ground of the question yields a full sentence (that is, includes material apart from the focus).

   (i) a. **What animals** *detest the smell of citrus fruits?* (question)
   b. **Ordinary cats.** (short answer)

c. **Ordinary cats** *detest the smell of citrus fruits.* (full answer)

(ii) a. **What** *do ordinary cats detest?* (question)

b. **The smell of citrus fruits.** (short answer)

c. *Ordinary cats detest* **the smell of citrus fruits.** (full answer)

4. Notice that the resultant interpretation is distinct from the one that would result from first combining *cat* and *the* and then combining *the cat* with the relative clause. The denotation of *the cat* is a unique member of the set of cats. Combining *the cat* with the relative clause would attribute to this unique entity the property of having chased the rat. Given that *the cat* already denotes a unique entity, the property of having chased the rat wouldn't be a defining property of the cat in question; it would simply be an additional, more or less accidental one. The interpretation in question is possible semantically, and it can be expressed by using a non-restrictive relative clause, as in (i).

   (i) This is the cat, which (by the way) chased the rat.

   But (i) is not synonymous with (26), where the rat-chasing property is restrictive, that is, defining.

5. In contrast to the statement in row 2 of Table 3.1, the statement in (i), derived from the statement in row one of Table 3.1 by the *modus tollens* rule of propositional logic, is true.

   (i) If a string isn't a constituent, then it doesn't pass the constituenthood tests.

6. For simplicity, we ignore syntax-morphology mismatches of the sort alluded to earlier.

7. For completeness, we should mention that *do so* substitution and the question test for verb phrases are subject to a semantic restriction. Specifically, *do so* and *do what* cannot substitute for verb phrases with so-called stative verbs like *know* or *want.*

   (i) a. They { know her parents, want the cookies }.

   b. * They do so.

   (ii) a. What do they do?

   b. * { Know her parents, want the cookies }.

   As their name implies, stative verbs refer to states (rather than to activities or accomplishments), and a reasonably reliable diagnostic for them is their inability to appear in the progressive construction.

   (iii) a. * They are { knowing her parents; wanting the cookies }. (stative verb)

   b. ✓ They are { meeting her parents; eating the cookies }. (nonstative verb)

   Since *do* is the prototypical activity verb, it is not surprising that expressions containing it, like *do so* and *do what,* give rise to a semantic clash when they substitute for verb phrases containing stative verbs.

8. It is true that movement of verb phrases in out-of-the-blue contexts, as in (i), is not very felicitous:

   (i) Out of the blue (i.e. with no preceding discourse context):
   Write a book, she will ___.

   But it is clearly grammatical given appropriate discourse contexts, as the examples in the text show. This is another demonstration of how constituency diagnostics are themselves part of language, not magical tests that are divorced from language. The emphasis movement here does diagnose constituents, but it is also an emphasis construction, and will sound odd if used in a context where emphasis is unnatural.

9. The island metaphor is not perfect. Although constituents can't move out of an island, islands as a whole are able to move, as shown in (i)–(iii).

(i) ✓ The doctors and lawyers, we should invite ___.

(ii) a. ✓ Who should we invite ___?

b. ✓ The lawyers and the doctors.

(iii) ✓ It is the lawyers and the doctors that we should invite ___.

10. Does a node A dominate itself? If the answer to this question is defined to be yes, then the dominance relation is **reflexive** (again, in the logical sense of the term, not the grammatical one); if not, then it is **irreflexive**. In principle, it is possible to build a coherent formal system based on either answer. From the point of view of syntactic theory, it is preferable to define dominance as reflexive because it simplifies the definitions of linguistically relevant derived relations such as c-command and binding (Chapter 15 discusses binding in detail).

11. The female kinship terms **mother**, **daughter**, and **sister** are typically used for the corresponding sex-/gender-neutral ones in work on syntax; this textbook defaults to the gender-neutral terms.

12. The odd name c-command is short for 'constituent-command' and reflects the fact that the c-command relation is a generalization of a relation (now obsolete) called command, defined as in (i) (Langacker 1969, 167).

(i) A **commands** B if and only if

a. neither A nor B dominates the other, and

b. the S(entence) node that most immediately dominates A also dominates B.

13. The Linear Correspondence Axiom (LCA) proposes that c-command relations are always expressed as precedence relations (Kayne 1994). Contrary to what is sometimes believed, this proposal underscores the logical independence of c-command and precedence rather than eliminating it. If precedence were logically derivable from c-command, there would be no need for an axiom postulating a correspondence between them (indeed, given the meaning of the term 'axiom', such an axiom would be a contradiction in terms). Kayne's proposal has been highly influential in various ways, but is by no means universally accepted.

Exercises and problems 3.6

Exercise 3.1: Identifying constituents A

Using the constituenthood tests reviewed in this chapter, determine whether the underlined strings in the following sentences are (phrasal) constituents. Some of the sentences below have more than one interpretation. Begin by focusing on the ordinary interpretation, and then consider any more unusual readings.

Take Note

Here and throughout the exercises in the book, your answer should include not just your conclusions, but also the supporting evidence. What we mean by the term evidence is one or more linguistic expressions (sentences or phrases) together with an indication of their grammaticality (checkmark or asterisk). Sometimes providing the evidence is all that is necessary. At other times, a bit of discussion is required to explain the relevance of the evidence to the question at hand.

(1) a. They put <u>the car</u> in the garage.
   b. They put the car <u>in the garage</u>.
   c. They put <u>the car in the garage</u>.

(2) a. They know <u>the guy</u> with the fedora.
   b. They know <u>the guy with the fedora</u>.

(3) a. They threw the towel <u>in the closet</u>.
   b. They threw <u>in the towel</u>.

Exercise 3.2: Identifying constituents B

For the two sentences below, give a tree structure of the sentence, and give evidence (in the form of acceptable or unacceptable sentences) that the structure you propose is accurate. That is to say, for all constituents in your tree structure, give at least one constituency diagnostic showing that it is, in fact, a constituent.

(1) This little piggy went to the market.

(2) The doting grandma gave the overstimulated children big hugs after Thanksgiving dinner.

Exercise 3.3: Evaluating structures for accuracy

How well does each of the trees in (1) and (2) represent the syntactic structure of the sentence it is intended to represent? Here are key questions to consider:

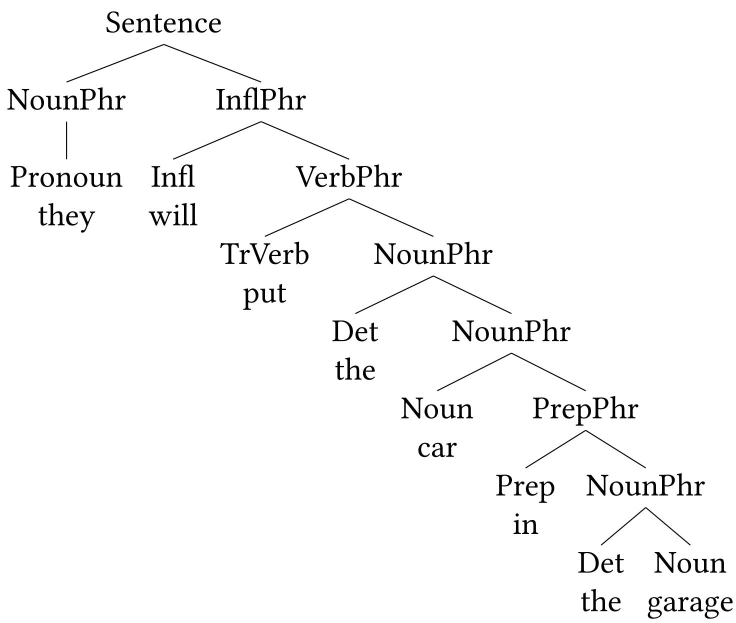
- Are any strings represented as constituents that shouldn't be?
- Are any strings not represented as constituents that should be?
- Are any of the trees misleading in other respects?

Background Info

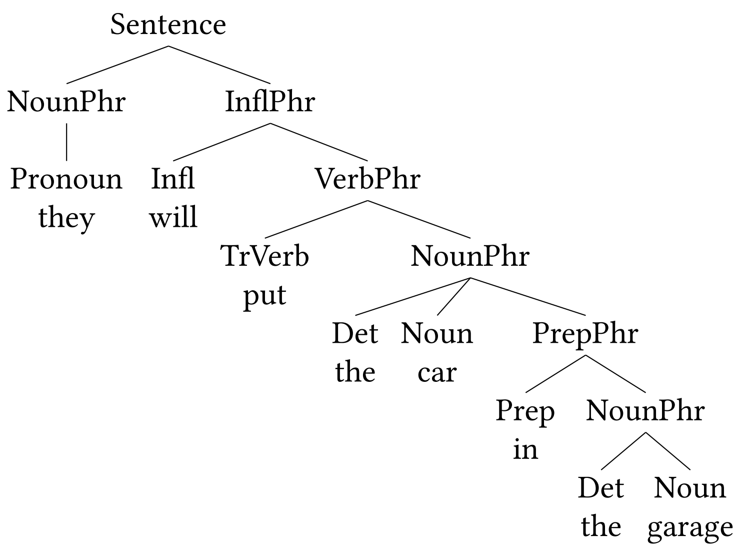
Abbreviations for syntactic categories:

- Det - determiner (roughly speaking, article or demonstrative pronoun)
- NounPhr - noun phrase
- PrepP - prepositional phrase
- TrVerb - transitive verb
- VerbPhr - verb phrase

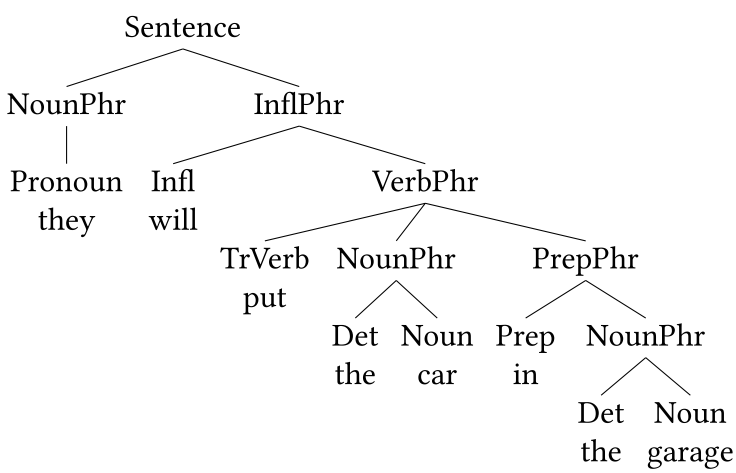
(1) a.


Sentence
NounPhr
InflPhr
Pronoun
they
Infl
will
VerbPhr
TrVerb
put
NounPhr
Det
the
NounPhr
Noun
car
PrepPhr
Prep
in
NounPhr
Det
the
Noun
garage

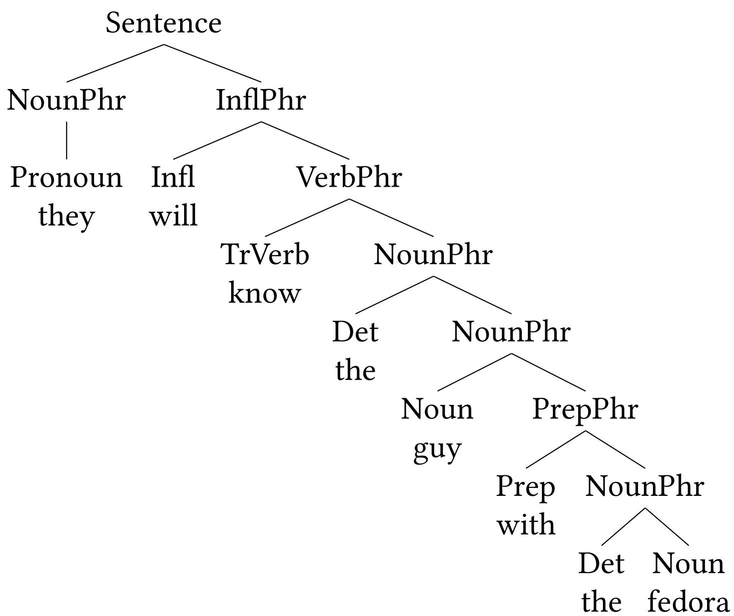
b.


Sentence
NounPhr
InflPhr
Pronoun
they
Infl
will
VerbPhr
TrVerb
put
NounPhr
Det
the
Noun
car
PrepPhr
Prep
in
NounPhr
Det
the
Noun
garage

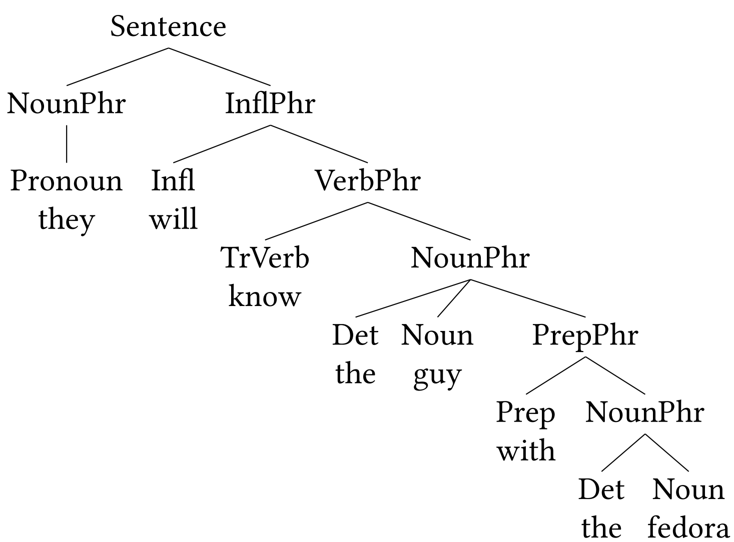
c.


Sentence
NounPhr
InflPhr
Pronoun
they
Infl
will
VerbPhr
TrVerb
put
NounPhr
Det
the
Noun
car
PrepPhr
Prep
in
NounPhr
Det
the
Noun
garage

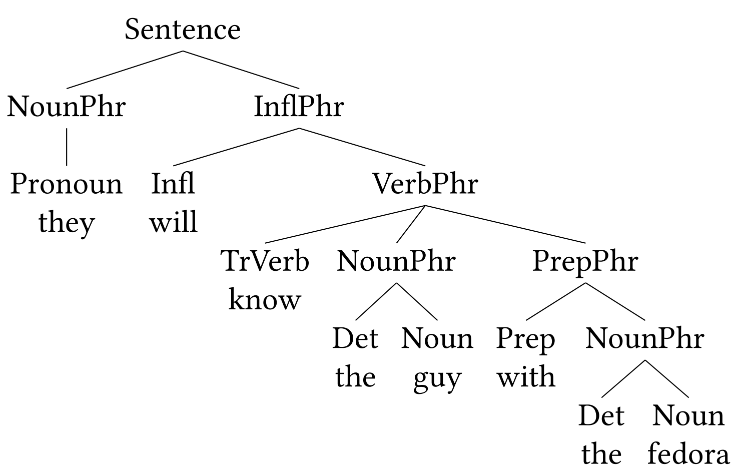
(2) a.


Sentence
NounPhr
InflPhr
Pronoun
they
Infl
will
VerbPhr
TrVerb
know
NounPhr
Det
the
NounPhr
Noun
guy
PrepPhr
Prep
with
NounPhr
Det
the
Noun
fedora

b.


Sentence
NounPhr
InflPhr
Pronoun
they
Infl
will
VerbPhr
TrVerb
know
NounPhr
Det
the
Noun
guy
PrepPhr
Prep
with
NounPhr
Det
the
Noun
fedora

c.


Sentence
NounPhr
InflPhr
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Infl
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VerbPhr
TrVerb
know
NounPhr
Det
the
Noun
guy
PrepPhr
Prep
with
NounPhr
Det
the
Noun
fedora

Exercise 3.4: Translating from brackets to trees

For each of the bracket structures below, draw a tree structure that exactly matches the bracket structure in its constituency. If a portion of the bracket structure is underspecified, leave it underspecified in the tree as well.

(1) a. The children will read three books this weekend.

b. [$_{\text{Sentence}}$ [$_{\text{NounPhr}}$ [$_{\text{Det}}$ The] [$_{\text{N}}$ children]] [$_{\text{Tense}}$ will] [$_{\text{VerbPhr}}$ [$_{\text{V}}$ read] [$_{\text{NounPhr}}$ three books] [$_{\text{NounPhr}}$ this weekend]]]

(2) a. All the tired teachers will take a long vacation.

b. [$_{\text{Sentence}}$ [$_{\text{NounPhr}}$ All the tired teachers] [$_{\text{Tense}}$ will] [$_{\text{VerbPhr}}$ [$_{\text{V}}$ take] [$_{\text{NounPhr}}$ [$_{\text{Det}}$ a] [$_{\text{Adj}}$ long] [$_{\text{N}}$ vacation]]]

(3) a. The guests put their names in the visitors log.

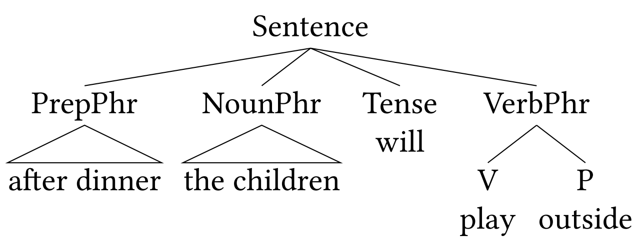
b. [$_{\text{Sentence}}$ [$_{\text{NounPhr}}$ The guests] [$_{\text{VerbPhr}}$ [$_{\text{V}}$ put] [$_{\text{NounPhr}}$ their names] [$_{\text{PrepPhr}}$ [$_{\text{P}}$ in] [$_{\text{NounPhr}}$ the visitors log]]]].

Exercise 3.5: Translating from tree structures to bracket structures

For each of the tree structures below, draw a bracket structure that exactly matches the tree structure in its constituency. If a portion of the tree is underspecified, leave it underspecified in the bracket structure as well.

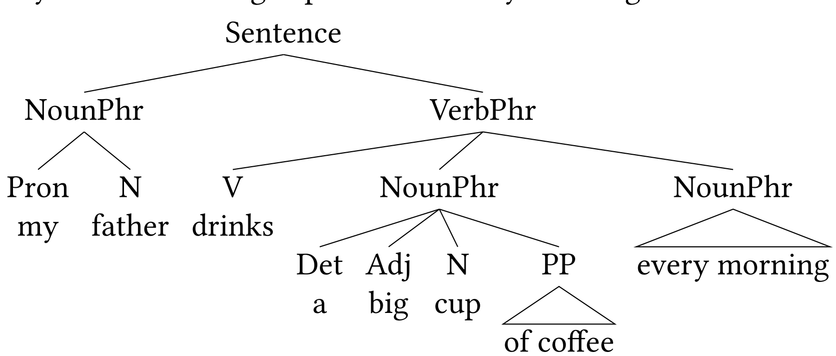
(1) a. After dinner the children will play outside.

b.

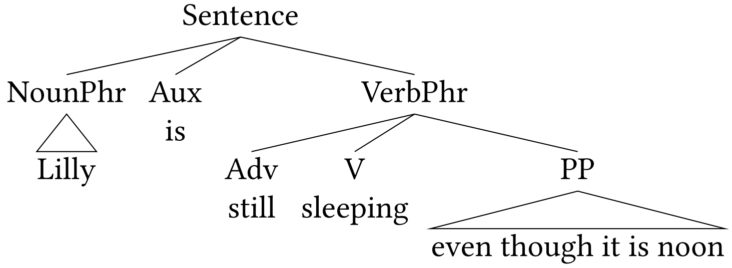


(2) a. My dad drinks a big cup of coffee every morning.

b.

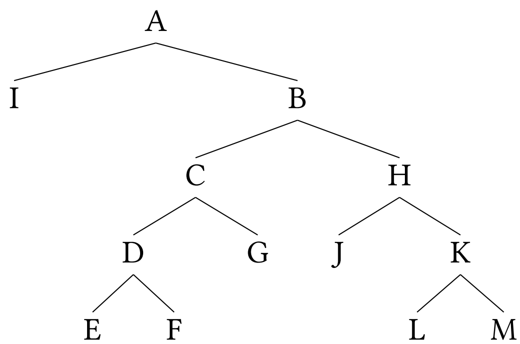


(3) a. Lilly is still sleeping even though it is noon.

b. 

Exercise 3.6: Identifying structural relationships

The questions you will answer below are based on the abstract tree structure given in (1).

(1) 

1. What nodes dominate G?
2. What nodes (if any) does G dominate?
3. What nodes (if any) dominate H?
4. What nodes (if any) does H dominate?
5. What nodes (if any) does C exhaustively dominate?
6. What nodes (if any) does H c-command?
7. What nodes (if any) does G c-command?
8. What nodes (if any) are the sibling of B?
9. What node (if any) is the parent of B?
10. What nodes (if any) are children of C?
11. what terminal nodes (if any) does H exhaustively dominate?
12. What node (if any) exhaustively dominates the nodes E, F, L, and M?

Exercise 3.7: Substitution

The substitution test discussed in detail in this chapter and the substitution operation introduced in Chapter 1 are not identical, but they are related. In a few sentences, explain how.

Exercise 3.8: Dealing with conflicting diagnostics A

The examples below attempt to apply constituency diagnostics for the potential constituent that is bracketed in (1). The constituency diagnostics do not all show the same result. Your task is to assess whether or not the bracketed string is in fact a constituent, and to discuss (and, preferably, explain) why the diagnostics do not all agree with each other. Judgments are provided for each sentence based on the intuitions of a Mainstream U.S. English speaker.

(1) The defense will [defend the goal aggressively.]

(2) Attempted constituency diagnostics:
   a. ? Defend the goal aggressively, the defense will.
   b. ✓ The defense will defend the goal aggressively, and the goalkeeper will, too. (*interpretation: the goalkeeper will defend the goal aggressively*).
   c. ✓ The defense will defend the goal aggressively, and the goalkeeper will do so also. (*interpretation of do so = defend the goal aggressively*).
   d. * It is defend the goal aggressively that the defense will.
   e. ✓ The defense will defend the goal aggressively and stop the offense.
   f. ✓ The coach said that the defense will defend the goal aggressively, and defend the goal aggressively, they will.
   g. ✓ The defense will defend the goal aggressively and immediately.
      i. Q: What will the defense do?
      ii. A: ✓ defend the goal aggressively.

Problem 3.1: Constituency of (potential) PPs

The two examples in (1) bracket a potential PP constituent.

(1) a. The children jumped [in the water].
    b. The open window let [in the cold air].

Using constituency diagnostics as introduced in this chapter, argue for each example whether this bracketed string is or is not a constituent. This exercise requires native speaker judgments about English; you can use your own if you are a speaker of English, or construct sentences and ask any native English speaker for judgments.

Problem 3.2: Phrasal and prepositional verbs: *turn on*

Particle verbs in Germanic languages pose a number of interesting syntactic puzzles. This exercise investigates a limited component of those puzzles. In English, like other Germanic languages, many constructions have a preposition-like particle following the verb.

(1) a. I messed up the assignment.
    b. I climbed up the stairs.
    c. I looked after my friend.

As it turns out, there are a number of sub-categories of these particle verbs, some of which can be identified based on their constituent structure. Specifically looking at the constituency of the potential prepositional phrase following the verb, note the difference between *climb up* and *mess up* in how they behave when testing for a PP constituent using the *it*-cleft constituency diagnostic.

(2) a. It is up the stairs that I climbed.
    b. * It is up the assignment that I messed.

This suggests that in an example like (1b) there is a PP constituent, but there is no PP constituent in an example like (1a).

(3) a. Climb [up the stairs]
    b. * Mess [up the assignment]

For the sake of this exercise, we can use the term "prepositional verb" for a particle verb construction where the P forms a PP constituent with the following noun phrase, and the term "phrasal verb" for a particle verb construction where the P does not form a constituent with the following noun phrase. The structure in (4b) is underspecified, so we won't be trying to offer exhaustive structures for phrasal verbs; rather, we are working to distinguish phrasal verbs from prepositional verbs.

(4) a. Prepositional Verb: $[_{\text{VP}}$ V $[_{\text{PP}}$ P NP]]
    b. Phrasal Verb: $[_{\text{VP}}$ V P NP]]

For this exercise, you will look at the English phrase *turn on.* This phrase is multiply-ambiguous, as we illustrate below.

(5) a. Alex turned on College Avenue. (physical turning onto a road)
    b. Alex turned on the light. (activation)
    c. Alex turned on Jay.
        i. sexual arousal—Alex aroused Jay
        ii. betrayal—Alex betrayed Jay

Your task is to use constituency diagnostics (or other differences you note) to say whether the ambiguities here are lexical ambiguities or structural ambiguities. That is, do these interpretations all have the same syntactic structure and just mean different things (lexical ambiguity)? Or are there different syntactic structures involved (syntactic ambiguity)? Or is it some of both?

If you are a native speaker of English, you can provide the judgments for the sentences you create on your own. If you are not a native speaker of English, you will need to work with someone who is, or ask someone who is for judgments on the diagnostic sentences you construct.

We specifically recommend testing for the potential PP constituent, though you can note other patterns that you find as well.

Problem 3.3: Distribution of English *any*

Examine the distribution of *any* in English based on the data below. What licenses the acceptability of *any* in a sentence? What structural relationship must *any* be in with this licensor?

Take Note

No judgments are marked on these examples. You will need to mark the judgments yourself.

(1) a. Jay read seven books last night.
    b. Seven books were read last night.

(2) Jay read any books last night.

(3) a. Jay didn't read any books last night
    b. Any books weren't read last night.

(4) a. The person that didn't like reading read 7 books last night.
    b. The person that didn't like reading read any books last night.

(5) a. The Sagehens never lost any games before today.
    b. The Sagehens always lost any games before today.

(6) a. The never-beaten Sagehens lost any games before today.
    b. The always-beaten Sagehens lost any games before today.

List of structural relationships between nodes from § 3.3 (for reference):

- dominate
- immediately dominate
- exhaustively dominate
- precede
- immediate precede
- ancestor of
- descendant of
- sibling of
- parent of
- child of
- c-command

Problem 3.4: English *-self* phrases

Section 3.3 above introduced various structural relationships that are represented in tree structures that appear to be significant in natural language syntax.

Self-phrases are morphemes known as *reflexives* (also called *anaphors*) which do not refer to an individual on their own—instead they pick up their meaning from another element in the sentence (the *antecedent)*. In the examples below, **coreference** (phrases referring to the same individual) is marked with subscripted numbers. What structural relationship does the anaphor and the antecedent need to be in for a grammatical sentence to result?

(1) a. Jamal$_1$ saw himself$_1$ in the mirror.
b. * Himself$_1$ saw Jamal$_1$.

(2) a. [Jamal's mother]$_1$ saw herself$_1$ in the mirror.
b. * [[Jamal$_1$]'s mother] saw himself$_1$ in the mirror.

(3) a. [The folks sitting next to Jamal]$_1$ saw themselves$_1$ in the mirror.
b. * [The folks sitting next to [Jamal$_1$]] saw himself$_1$ in the mirror.

List of structural relationships between nodes from § 3.3 (for reference):

- dominate
- immediately dominate
- exhaustively dominate
- precede
- immediate precede
- ancestor of
- descendant of
- sibling of
- parent of
- child of
- c-command

Problem 3.5: Pied-piping and preposition stranding in English

Movement operations are excellent constituency diagnostics. In English we can draw on two kinds of movements: movements of *wh*-words in questions, and movements of non-*wh*-words for emphasis. Likewise, *it*-clefts can be used to diagnose constituents.

(1) a. The children read the lengthy book.
b. Which book did the children read?
c. The lengthy book, the children read (not the short one).
d. It was the lengthy book that the children read.

In the example below there is question about whether or not there is a prepositional phrase (PrepPhr) constituent.

(2) The children ran down the big hill.

(3) a. Which hill did the children run down?
b. The big hill, the children ran down (not the small one).
c. It was the big hill that the children ran down.

(4) a. Where did the children run?
b. Down the big hill, the children ran (not up the small hill).
c. It was down the big hill that the children ran.

Discuss how this evidence affects your analysis of the constituency of (2). Is there a prepositional phrase constituent, or not? How is it possible for all of the examples in (3) and (4) to be grammatical?

Identifying Puzzles 3.1: Dealing with conflicting diagnostics B

The examples below attempt to apply constituency diagnostics for the potential constituent that is bracketed in (1). The constituency diagnostics do not all show the same result. Your task is to assess whether or not the bracketed string is in fact a constituent, and to discuss (and, preferably, explain) why the diagnostics do not all agree with each other. Judgments are provided for each sentence based on the intuitions of a Mainstream U.S. English speaker.

(1) The guardian dog always looks [after his goats].

(2) <u>Attempted constituency diagnostics</u>:
a. The guardian dog always looks after his goats and after his sheep.
b. * The guardian dog always looks there. (* on the intended interpretation of looking after the goats)
c. * It is after his goats that the guardian dog always looks.
d. * After his goats, the guardian dog always looks.
e. Q: Who does the guardian dog always look after?
(*different possible answers considered below*)
A1: his goats
A2: after his goats
(*dispreferred to A1, but still grammatical*)
A3: his goats and his sheep
A4: after his goats and after his sheep
(*dispreferred to A3, but still grammatical*)
f. The guardian dog looks after his goats, and the farmer does too. (*interpretation = the farmer looks after his goats*)
g. The guardian dog looks after his goats and chases coyotes.

The diagnostics do not all point in the same direction, and the material in this chapter does not provide an easy explanation. Explain what diagnostics are inconsistent with each other, and postulate a rationale for why the diagnostics do not reach the same conclusion.

4

Some basic linguistic relations

Chapter outline

While notions of grammatical categories and constituency are central to our model of syntactic structure, these are not the only foundational concepts that we will rely on to build our model. In this chapter, we introduce a number of fundamental linguistic relationships that underlie syntactic structure, which will lay the foundation for discussions that follow. We first discuss predication: the division of sentences into predicates and arguments. We then discuss modifiers, that is, components of sentences that are neither predicates nor arguments. The rest of the chapter discusses more fine-grained concepts that we use to describe arguments and modifiers: thematic roles (the semantic role that non-predicates play in sentences) and grammatical relations, the syntactic implementation of notions like *subject* and *object*.

Predicates and arguments 4.1

The first basic linguistic property we address here has to do with one of the necessary components to make a complete sentence: a **predicate**, and its **arguments**. As a very basic introduction, predication is essentially the attribution of a property to some individual/element (the argument). The predicates are bracketed in (1), where "dog-ness" is attributed to Lilly in (1a) and happiness is attributed to Lilly in (1b).

(1) a. Lilly is [a dog].
b. Lilly is [happy].

So in (1a) the predicate is *a dog* and the argument is *Lilly*; in (1b) the predicate is *happy* and the argument is *Lilly*. We need to get more detailed than this (and there are fine-grained notions of predication that are non-identical to each other), so here we go!

4.1.1 Semantic arguments of predicates

The most obvious factor that determines how vocabulary items combine in sentences has to do with their meaning, a point most conveniently illustrated with verbs. For the purposes of this section (and just this section!) we are going to ignore the role of tense, not because it's unimportant but because it adds complications that are independent of our immediate concerns.

Consider the basic sentence in (2):

(2) The children devoured dessert.

We have clear judgments that a devouring event requires both the one doing the consuming, and the thing being consumed.

(3) a. * Devoured dessert.
b. * The children devoured.

This is a syntactic fact, but it's also very closely linked with semantic properties of *devour*. And we're going to put our attention on the semantics for the moment, albeit informally. When we think about (2) and (3), there's a clear sense that a *devouring* event requires someone or something to do the devouring, and something to be devoured.

Semanticists (and philosophers of language) use the term **proposition** to refer to a full sentence that is a complete thought. More specifically, a proposition is a potential fact: something that can possibly be true or false. So look at the examples of nouns, noun phrases, prepositional phrases, and verb phrases in (4)-(6). Notably, it makes no sense to label any of these examples as "true" or "false", because none of these are propositions.

(4) a. dog
 b. the dog

(5) a. park
 b. the park
 c. in the park

(6) a. saw the dog
 b. saw the dog in the park

However, if we add a subject to the examples in (6), as in (7), then we have full sentences that are propositions, which can be labeled as either "true" or "false" based on the real world circumstances they are stated in.

(7) a. Maisha saw the dog
 b. Maisha saw the dog in the park

So there is some sense in which the constituents in (4)–(6) are incomplete. And this is the key intuition that we rely on here in developing clarity about what predicates and arguments are. We can think of propositions, in part, as describing an **event**. An event is composed of an action or a state and the participants in that action/state.[1] We use the term **predicate** to refer to the part of the event that describes the action, state, or relationship between the participants in the event and the real world. We use the term **argument** to refer to the participants in the event. So in (2) above, the event being described is an act of devouring, i.e., rapidly consuming something. A devouring event has two arguments—the devour-er (the one doing the consuming), and the devour-ee (the thing being consumed). Likewise, in our examples in (7), what is being described is a seeing event. Maisha is the one experiencing the seeing, and the dog is the one being seen.

Part of having a complete event in a sentence is including all participants in the event. A devouring event necessarily requires the one doing the devouring and the thing being devoured, and leaving either out is unacceptable: hence (3). Similarly with a seeing event: it is downright odd to say something like (8) where the element being perceived is not included.

(8) *? Maisha saw in the park.

Notably, there is not a prespecified number of arguments that all predicates must have. So while it is odd to say *Maisha saw* or *the children devoured*, it is perfectly natural to say *Maisha* *walked* or *The children laughed*. So some predicates require only a single argument, others requires two (or more): we discuss this with more precision in § 4.1.3.

So we can think of a predicate, then, as a syntactic (and semantic) component of a sentence that is combined with arguments to make a complete event. A complete event, then, is a central component of a proposition. Specifically, an event needs to be anchored in time and space (via **tense** in many languages) to become a complete proposition, but we continue to set aside the issue of tense for the moment.

4.1.2 Aristotelian versus Fregean predicates

On one way of thinking about the structure of sentences, essentially every clause consists of a subject and a predicate. The term "predicate" as used here has a different sense than in our earlier discussion concerning argumenthood; it refers to what remains of a clause when its subject is removed. On this understanding of a predicate, the subject is bolded in the examples below and the predicate is underlined.

(9) a. **The tabby cat** enjoys catnip immensely.
  b. **They** have found that the tabby cat enjoys catnip immensely.
  c. **My downstairs neighbor** suspects that they have found that the tabby cat enjoys catnip immensely.

For clarity, we can use the term "Aristotelian predicate" for this sense, since the observation that all sentences consist of a subject and a predicate goes back (at least) to Aristotle (384–322 BCE). **Predication** is the relation between a subject and an Aristotelian predicate.

The sense of "predicate" that we used earlier, to refer to a single vocabulary item, is much more recent and can be attributed to one of the founders of modern logic, the mathematician and philosopher Gottlob Frege (1848–1925). Accordingly, we can use the term "Fregean predicate" for this sense. What Frege recognized is that Aristotle's division of a clause into subject and predicate is simply the first of a potential series of such bifurcations. Just as it is possible to peel off, as it were, the subject of a clause, leaving the Aristotelian predicate, it is possible to further peel off any arguments (and modifiers) contained within the Aristotelian predicate, yielding in the final instance a single vocabulary item, the Fregean predicate. So in a sentence like *the children chased the dogs with water balloons*, the Aristotelian predicate is *chased the dogs with water balloons*, whereas the Fregean predicate is *chase*.[2]

In (10), the Aristotelian predicate of the largest clause is in italics, and its Fregean predicate is underlined. As the increasingly complex sentences show, Aristotelian predicates are recursive categories. Fregean predicates, on the other hand, not being phrases, are not.

(10) a. The tabby cat *enjoys catnip immensely.*
  b. They *have found that the tabby cat enjoys catnip immensely.*
  c. My downstairs neighbor *suspects that they have found that the tabby cat enjoys catnip immensely.*

4.1.3 Valency

Similar to how it is used in chemistry, the term *valency* refers to how a component of grammar combines with other grammatical elements. So for verbs, the term "valency" refers to how many arguments a verb occurs with in sentences. In principle, a predicate's valency might completely determine the syntactic structure that it appears in. The ungrammaticality of the sentences in (11) would fall out directly from such a system: a one-place predicate like *laugh* cannot have a distinct object argument, and a two-place predicate like *invite* requires the object argument.

(11) a. * Lukas laughed the train. (one-place predicate; superfluous argument)
b. * Andy invited. (two-place predicate; missing theme argument)

The actual situation, however, is more complex. For instance, *eat* denotes a relation between eaters and food. It is therefore a two-place predicate, like *invite*. However, unlike *invite*, *eat* has both a **transitive** and an **intransitive** use, as illustrated in (12).

(12) a. Transitive:
The children have eaten their supper.
b. Intransitive:
The children have eaten.

Notice that the semantic properties of *eat* remain constant in (12). In other words, (12a) and (12b) are both interpreted as involving the ingestion of food, even though there is no explicit mention of food in (12b). In view of the mismatch between the semantic and syntactic properties of *eat* in sentences like (12b), it is useful to distinguish between semantic and syntactic arguments. As mentioned earlier, we can think of semantic arguments as central participants in a situation. Syntactic arguments, on the other hand, are constituents that appear in particular syntactic positions (see Chapter 5 for further discussion). Semantic arguments are typically expressed as syntactic arguments, but the correspondence between the two is not perfect, as (12b) shows.

We will use the term **valency** to refer to the number of syntactic arguments that a verb combines with, and we can then divide verbs into three subcategories as in Table 4.1.

Valency	**Number of syntactic arguments**
Intransitive	1
Transitive	2
Ditransitive	3

Table 4.1: Valency is the number of syntactic arguments a verb takes

Because of mismatches as in (12), it turns out to be quite rare in English for verbs to belong to just one syntactic subcategory (and it is quite common in the world's languages to use a single lexical verb, plus derivational morphology, to create related verbs of different valency). (13)–(15) shows some two-place verbs besides *eat* that can be used either transitively or intransitively. The slashes separate the arguments from the predicate and each other.

(13) a. He / *interrupted* / the meeting. (basically transitive)
b. He / *interrupted*. (intransitive use)

(14) a. Amy / *knits* / sweaters. (basically transitive)
b. Amy / *knits*. (intransitive use)

(15) a. They / are *reading* / a book. (basically transitive)
b. They / are *reading*. (intransitive use)

Conversely, certain one-place verbs can be used not only intransitively, but also transitively, as illustrated in (16)–(18). Notice that the verb and its object in the transitive examples are etymologically related, or cognate. For this reason, the transitive use of one-place verbs as in (16)–(18) is known as the **cognate object construction**.

(16) a. Charlie / *died*. (basically intransitive)
b. Charlie / *died* / a peaceful death. (transitive use)

(17) a. Maisha / *laughed*. (basically intransitive)
b. Maisha / *laughed* / an infectious laugh. (transitive use)

(18) a. Mona Lisa / was *smiling*. (basically intransitive)
b. Mona Lisa / was *smiling* / a mysterious smile. (transitive use)

Further, it is possible to use some basically three-place verbs not just ditransitively, but transitively and even intransitively.

(19) a. We / *teach* / college students / syntax. (basically ditransitive)
b. We / *teach* / college students. (transitive use)
c. We / *teach* / syntax. (transitive use)
d. We / *teach*. (intransitive use)

(20) a. He / *told* / me / the whole story. (basically ditransitive)
b. He / *told* / me. (transitive use)
c. He / *told* / the whole story. (transitive use)
d. He / better not *tell*. (intransitive use)

Finally, it is possible to use two-place transitive verbs in ditransitive contexts, essentially adding an additional object.

(21) a. I / *baked* / a delicious cake. (basically transitive)
b. I / *baked* / my friends / a delicious cake. (ditransitive use)

(22) a. She / *sang* / a lullaby. (basically transitive)
b. She / *sang* / her baby / a lullaby. (ditransitive use)

Many transitive predicates in Mainstream U.S. English allow for the addition of a recipient of the event, as in (21) and (22), which is always placed between the verb and the theme object. See § 4.3 for more discussion of what kinds of thematic roles are relevant for syntax.

English implements all of these adjustments to valencies with no overt morphological change to the verb. But in many of the world's languages, such adjustments are accomplished via derivational morphology on the verb form. The Swahili example in (23) shows how a transitive predicate can be changed to a ditransitive predicate (adding a recipient to the sentence) with the addition of an **applicative morpheme** to the verb.

(23) a. ku-jibu
INF-answer/respond
'to answer'

b. ku-jib-i-a
INF-answer-APPL-FV
'to answer for (someone)'

Likewise, a transitive can be made intransitive via the stative suffix.

(24) ku-jib-ik-a
INF-answer-STAT-FV
'to be answerable'

Finally, a causative morpheme can be used to add a second agent to a predicate, with a reading something like "to make someone else do something".

(25) ku-jib-iz-a
INF-answer-CAUS-FV
'to make (someone) answer'

Many languages have verbal derivational morphology like this, with a lot of interesting patterns resulting. At present, we simply mention this to illustrate valency of predicates; the English patterns are attested in many other languages, but often with overt morphology showing the patterns more explicitly.

4.2 Modification

Events are associated with more or less central participants and properties. The central participants are the semantic arguments just discussed. Properties of a situation typically taken to be less central, such as manner, place (location, origin, destination, path), time (point in time, duration, frequency), reason (cause, purpose), and so on, can be expressed by **modifiers**. Semantic arguments are closely associated with specific predicates; modifiers, less so. For examples, not all situations are associated with themes, so themes are arguments. But situations are all located in time, so temporal expressions are modifiers.

Take Note

Arguments and modifiers both introduce restrictions on the denotation (i.e., the formal semantic definition) of a predicate, and the relationships of argumenthood and modification do not differ in this respect. For instance, the situations denoted by *invite Langston* (where *Langston* is an argument) are a subset of those denoted by *invite*, just as the situations denoted by *laugh uproariously* (where *uproariously* is a modifier) are a subset of those denoted by *laugh*. So there are shared properties between arguments and modifiers as well, despite the syntactic distinctions between them.

Modifiers of verbs (and verb phrases) are typically adverb phrases or prepositional phrases, but noun phrases can serve as verbal modifiers as well. In the following examples, the modifier is in italics, and the verb that it modifies is underlined.

(26) a. Manner:
He <u>read</u> the letter *carefully*.

b. Location:
We <u>met</u> the students *in my office*.

c. Origin:
We <u>departed</u> *from Bangalore*.

d. Destination:
We <u>arrived</u> *in Benares*.

e. Path:
We <u>followed</u> *along the path*.

f. Point in time:
They <u>discussed</u> the proposal *in the afternoon*.

g. Duration:
She <u>kept</u> their books *for five years*.

h. Frequency:
They <u>read</u> the Times *quite often*.

i. Cause:
He <u>threw</u> it away *out of spite*.

j. Purpose:
You should <u>send</u> a message *(in order) to warn everyone*.

Because of their semantically peripheral character, modifiers are syntactically optional. The converse is not true, however. Not all syntactically optional constituents are modifiers; recall from (12b) that semantic arguments aren't always expressed.

Verbs are not the only category that can be modified. For instance, nouns are often modified by adjective phrases, prepositional phrases, and relative clauses, and contrary to what the name might suggest, even adverb phrases can modify nouns.

(27) a. a *very important* period
b. a period *of great import*
c. the car *that just turned the corner*
d. the instructions *below*

Moreover, adjective phrases, adverb phrases, and preposition phrases, which typically serve as modifiers (though they are not restricted to that function), can themselves contain modifiers.

(28) a. *very* proud of the children
b. *surprisingly* good to eat

(29) a. *almost* there
b. *right* below

(30) a. *very much* in the dark
b. *two feet* under the shed

There are cases where it is difficult to tell which constituent is the modifier and which the modifiee. For instance, in an expression like *early in the morning*, is the adverb modifying the preposition, as indicated in (31a), or is the prepositional phrase modifying the adverb, as in (31b), or is either of these possibilities grammatical (that is, is the expression structurally ambiguous)?

(31) a. *early* in the morning
b. early *in the morning*

4.3 Thematic roles

This section provides an elementary overview of concepts and terminology related to **thematic roles**. We talked about valency of predicates above: *laugh* is a one-place predicate, and *invite* is a two-place predicate. But it is not enough to just count the arguments of a verb, because the nature of the arguments is different. Thinking about *invite* for the moment: in the sentence *Maisha invited her classmates*, we know that *Maisha* is the one who initiated the invitation, and *her classmates* are the recipients of the invitation, the ones who are invited. If we say *Her classmates invited Maisha* the sentence is still grammatical, but the roles of the participants in the event have switched around: now the classmates are the ones extending the invitation, and Maisha is the one who received the invitation.

We can see, then, that it is not sufficient to simply describe the number of arguments a predicate takes: different arguments in a sentence have specific (and predictable) roles in the event being described. The preverbal argument of *invite* in the sentences above is the inviter, and the postverbal argument is the invited participant. As it turns out, languages systematically capture specific kind of roles in events, such that different predicates take arguments representing different kinds of roles. We call these **thematic roles**, which enter into our theory in different ways (we will see in Chapter 7 that these are called θ-roles in some iterations of generative theory). In this section, we outline the different kinds of thematic roles within events that languages regularly incorporate into their grammar in systematic ways.

4.3.1 Agent, cause, and instrument

Agents are arguments that bring about a state of affairs.[3] The line between agents, on the one hand, and **causes** or **instruments**, on the other, can be fuzzy, but agents are (or are perceived to be) conscious or sentient, in a way that causes or instruments aren't. Some examples are given in (32)–(34).

(32) Agent:
  a. *The lions* devoured the wildebeest.
  b. *The boys* caught some fish.
  c. *My mother* wrote me a letter.

(33) Cause:
  a. *Hurricane-force winds* demolished much of the town.
  b. *An epidemic* killed off all of the tomatoes.
  c. *An economic downturn* put thousands of workers out of work.

(34) Instrument:
  a. *This key* opens the door to the main office.
  b. They must have used *indelible ink*.

4.3.2 Experiencer

Experiencers are arguments that undergo a sensory, cognitive, or emotional experience.

(35) Experiencer:
  a. *The rhesus monkey* had never seen snow before.
  b. *Many people* fear snakes.
  c. Their resourcefulness struck *her* as admirable.

4.3.3 Recipient

Recipients are arguments that receive something (whether good or bad) in a situation.

(36) Recipient:
  a. They gave *the workers* a raise.
  b. I paid *my landlord* the rent.
  c. He spared *me* his usual sob story.

4.3.4 Location, path, and goal

Locations are points (or regions) in space.

(37) Location:
  a. We always eat breakfast *in the kitchen.*
  b. The cork has been bobbing *under the bridge* for an hour.

Paths connect locations.

(38) Path:
  a. Lucky raced *down the driveway.*
  b. The boat passed *under the bridge* so quickly I missed seeing it.
  c. We drove *the scenic route.*

When locations serve as endpoints of paths, we generally refer to them as **goals**.

(39) Goal:
  a. We traveled to *Paris* quite a bit in those days.
  b. Lucky raced to *the edge of the woods.*
  c. I'd like to send this package to *France.*

Recipients can also serve as the endpoint of paths, and the distinction between goals and recipients can be difficult.

(40) I'd like to send this package to *my cousin.*

Locations and paths can be spatial or temporal.

(41) a. Point in time:
  Let's start the meeting at two.
  b. Goal in time:
  The meeting will last until two.

4.3.5 Measure

Measure (also called amount) arguments express extensions along some dimension (length, duration, cost, and so on).

(42) Measure:
  a. They rowed *for three days.*
  b. The book costs *ten dollars.*

4.3.6 Theme

Finally, the thematic role of **theme** is something of a catch-all. According to one definition, "theme" refers to an argument undergoing motion of some sort, including motion in a metaphorical sense, such as a change of state. As is usual in the syntactic literature, we will also use the term for arguments that are most "affected" in a situation or for arguments that refer to the content of an experience.

(43) Theme:
  a. The lions devoured *the wildebeest.*
  b. This key opens *the front door.*
  c. Hurricane-force winds demolished *much of the town.*
  d. They gave the workers *a raise.*
  e. I'd like to send *this package* to France.
  f. Many people fear *snakes.*

4.3.7 Intermediate summary

In this section we gave a very brief introduction to thematic roles, and the idea that arguments of predicates tend to have specific, pre-specified roles that are determined by the predicate.

A good question is why we have described these roles and not others. Real world human experience is almost impossibly complex, and there are many consistent human experiences that we might expect to be encoded in grammatical roles, but which are not. (44) imagines kinds of universal human experiences that one could plausibly think might be encoded in thematic roles in language, but which are not.

(44) a. Human connections (and dissolution of human connections) are universal experiences. But there are no thematic roles for predicates that specifically describing CONNECTION OR BETRAYAL.
  b. Social approval/honoring (or shaming, in the inverse) are both major tools of social control (both virtuous and malicious) across societies. But languages don't have thematic roles for predicates that specify something like "TO X'S HONOR" or "TO X'S SHAME."

c. Humans have positive and negative valence emotional responses to all kinds of experiences in life. But to our knowledge predicates in human language don't encode the notion of "pleasing to X" or "distressing to X" across predicates in any systematic way.

So why have we reported the thematic roles that we did, and not roles like these? In principle we might expect (based on human experience) that languages would have grammatical properties encoding roles like these, too: but on a whole, languages do not. Why not? We don't have an easy answer to that question here, though it presumably does tell us something foundational about human cognition that languages consistently encode certain thematic roles but not concepts that could at least potentially be conceived of as thematic roles. The roles described in this section are much too common across languages for it to be a coincidence.

Grammatical relations 4.4

This section shows that **grammatical relations** must be carefully distinguished from thematic roles. While a thematic role describes the role that a participant plays in an event, a grammatical relation (sometimes called a grammatical role) describes the role that a phrase plays in a sentence. In what follows, we illustrate three basic grammatical relations: subject, indirect object, and direct object.

4.4.1 Subjects

Subjects are ordinarily the only argument to precede the predicate in English. As the examples in (45) illustrate, subjects can express many different thematic roles.

Word of Caution

Take care not to confuse the grammatical relation of subject with the thematic role of agent. The existence of passive sentences is a clear indication that the two notions are not synonymous (compare (45a) with (45h) and also (45i)).

(45) a. *The lions* devoured the wildebeest. (agent)
b. *Hurricane-force winds* demolished much of the town. (cause)
c. *This key* opens the door to the main office. (instrument)
d. *The children* liked all their Christmas presents. (experiencer)
e. *The workers* got a raise. (recipient)
f. *An unpaved road* led up to the shanty. (path)
g. *The summit* wasn't attained until years later. (goal)

h. *The wildebeest* was devoured by the lions. (theme)
i. *The ball* rolled down the hill. (theme)

4.4.2 Indirect objects

Indirect objects occur in conjunction with direct objects, as illustrated in (46).

(46) They gave *the workers* a raise. (recipient)

Indirect objects in Mainstream U.S. English are thematically very restricted—namely, to recipients. Nevertheless, there is no one-to-one correspondence between the grammatical relation of indirect object and the thematic role of recipient. This is because recipients needn't be indirect objects, but can also be expressed as subjects, as in (45e).

4.4.3 Direct objects

Direct objects include a verb's remaining noun phrase arguments. Thematically, they are almost as flexible as subjects, as shown in (47).

(47) a. You should use *this key* for the door to the main office. (instrument)
b. The children's drawings pleased *their parents* to no end. (experiencer)
c. We drove *the scenic route*. (path)
d. We reached *our hotel* after a short subway ride. (goal)
e. The performance lasted *two hours*. (measure)
f. The lions devoured *the wildebeest*. (theme)
g. We rolled *the ball* down the hill. (theme)
h. They gave the workers *a raise*. (theme)

4.4.4 Distinguishing properties of grammatical relations

Linear order

In many languages, word order correlates to some degree with grammatical relations, in that a language will place subjects and objects in some typical position. In most of the world's languages, subjects precede their predicates, regardless of whether a language is verb-initial (SVO) or verb-final (SOV). The ordinary position for direct objects is adjacent to the verb—immediately following it in verb-initial languages (V-DO) and immediately preceding it in verb-final ones (DO-V). However, indirect objects ordinarily intervene between the verb and the direct object in verb-initial languages, yielding V-IO-DO order. In verb-final languages, the corresponding order often is not, as might be expected, the mirror image order. Rather, the indirect object continues to precedes the direct object, yielding the order IO-DO-V.

These are of course drastic oversimplifications: there are exceptions to these generalizations, and there is a lot of nuance within languages affecting canonical word orders (as reported above) vs. non-canonical word orders that are often used for various kinds of discourse functions. The point, of course, is that in many languages word order has a relatively direct correlation with grammatical relations.

Case

However, in many languages, word order is not a diagnostic property of grammatical roles. In many such languages, it is often instead a property called **case** that is a more consistent identifier of which noun phrases represent which grammatical relation in the sentence. We discuss this in much more depth in Chapter 9, but we will introduce it briefly here as is relevant for our current concerns.

Basic aspects of case are already familiar to English speakers, which has case-marking of nominals in its pronominal system. In English, pronominal subjects of a sentence appear in a distinct form from pronouns that appear elsewhere.

(48) a. They like her.
b. She likes them.

(49) a. * Them like she.
b. * Her likes they.

We use the term **nominative** to refer to the case form that sentence subjects appear in, and **accusative** to refer to the form that sentence objects appear in. So the nominative forms of the pronouns above are *they* and *she*, and the accusative forms are *them* and *her*.[4]

There are many languages, however, where case-marking is more thorough, appearing on most or all nominals in the language, and differentiating much more than subjects vs. non-subjects. The Turkish example in (50) distinguishes the morphological form of direct objects (accusative case) from indirect objects (**dative case**)

(50) Mehmet-∅ adam-a elma-lar-ı ver-di
Mehmet-NOM man-DAT apple-PL-ACC give-PST.3SG
'Mehmet gave the apples to the man.' (Blake 2004, 1)

Turkish marks nominals with six distinct cases based on their grammatical function in a sentence. These forms (and their functions) are described by the table in (51).

(51) Case-marking in Turkish (Blake 2004, 2)

Case	**Grammatical function**	**Form of the word *adam-* 'man'**
nominative	subject	*adam*
accusative	direct object	*adamı*
genitive	possessor	*adamın*
dative	indirect object	*adama*
locative	location	*adamda*
ablative	source location	*adamdan*

There are two main (distinct) systems of case-marking that appear across human languages: what is known as nominative-accusative case alignment (as illustrated above), and ergative-absolutive alignment, which slices up the case-marking pie in a different sort of way (marking agentive subjects differently from non-agentive subjects, for example). Case-marking in human languages is extraordinarily complex cross-linguistically, so this is only a rudimentary introduction: the point is that case-marking is one of the ways that grammatical relations are morphosyntactically represented in language.

Verbal inflectional morphology

Another way that morphology encodes grammatical relations in syntax is via **agreement** and other kinds of inflectional morphology, where the morphological form of one element changes based on the properties of some other element. The version of this that most language learners are relatively familiar with is verbal agreement (i.e., verb conjugation). Inflection on Swahili verbs appears in the template in (52).

(52) Subject.Agreement - Tense - Object.Agreement - Verb

So in an example like (53a), the plural subject agreement agrees with *wazazi* 'parents', and the singular object agreement agrees with *mtoto* 'child.' In Swahili it is possible to leave the subject and object unpronounced as in (53b) so that just the agreement morphemes themselves specify the relevant grammatical relations of subject and object.

(53) a. Wazazi wa- li- mu- ona mtoto.
parents subj.agr.pl- pst- obj.agr.sg- see child
'The parents saw the child.'

b. Wa- li- mu- ona.
subj.agr.pl- pst- obj.agr.sg- see
'They.pl (parents) saw her/him/them.sg (child)

c. * Mu- li- wa- ona.
obj.agr.sg- pst- subj.agr.pl- see

Example (53c) attempts to switch the position of the subject and object agreement, but this is ungrammatical. As shown in (54), it is possible to switch the singular and plural agreements (subject agreement for the singular form has a different form), but this switches the grammatical functions: now the child is the one seeing the parents.

(54) A- li- wa- ona.
subj.agr.sg- pst- obj.agr.pl- see
'She/he/they.sg (child) saw them.pl (parents).'

So we see that languages may encode grammatical relations in multiple ways: word order, nominal inflection (case), and verbal inflection (agreement). And many languages use more than one of these strategies, at times marking the same grammatical relation repeatedly, such as marking subjecthood with word order, nominative case-marking, and subject agreement on the verb. Each of these strategies is itself a large area of inquiry, but for our purposes we are simply using these to understand grammatical relations.

Special cases of predicates and arguments 4.5

Before concluding our discussion of these foundational concepts we do want to note some specific special cases relevant to predicates and arguments. We focus our discussion on English at the moment.

4.5.1 The subject requirement

Evidence for a subject requirement: Expletive *it*

It is not obvious that predication can be reduced to semantic considerations; this is demonstrated by a requirement for subjects in sentences that appears to be independent of semantics (we can call it the **subject requirement**). We begin by distinguishing between two kinds of *it*: **ordinary referential** *it* and **expletive** *it*. As the name implies, referential *it* has some referent (for more on referents and reference, see § 15.1 of Chapter 15). As a result, it can be replaced by a more complex description of that referent, as in (55).

(55) a. It bit the zebu.
b. The tsetse fly bit the zebu.

Referential *it* can also receive stress (indicated by underlining) in answers to questions:

(56) a. What bit the zebu?
b. (pointing) <u>It</u> did.

By contrast, expletive *it* doesn't refer, so it can't be replaced by a description:

(57) a. It seems that the manuscript has been found.
b. * The fact seems that the manuscript has been found.

This lack of content of the expletive is also evident in the lack of a content question that replaces the expletive with *what*:

(58) * What seems that the manuscript has been found?

Expletive *it* is unable to receive stress. In fact, as (59) shows, a situation designed to elicit an answer analogous to (56b) is ungrammatical. So if (59a) is uttered based on some evidence, it's not possible to utter (59b) in response in order to invoke the idea that some different sort of evidence is relevant.

(59) a. It seems that the manuscript has been found.
b. * (pointing) No, IT does.

The point is that expletive *it* in examples like (57a) is a different sort of element from referential *it* in examples like (55a).

Returning now to the main thread of the discussion, we note that the italicized *that* clause in (60a) functions as the sole syntactic argument of the adjective *evident*, on a par with the noun phrase in (60b). (For simplicity, we ignore the copula as semantically vacuous.)

(60) a. *That they are corrupt* is evident.
b. *Their corruption* is evident.

An indication of the semantic equivalence of the two expressions is the fact that they can both serve as a short answer to the question in (61a).

(61) a. What is evident?
b. That they are corrupt.
Their corruption.

(62) shows a syntactic variant of (60a) where the *that* clause appears at the end of the entire sentence. The original preverbal position of the *that* clause is occupied by expletive *it*.

(62) It is evident that they are corrupt.

Given that the *that* clause satisfies the semantic requirement of *evident* for an argument in both (60a) and (62), the presence of the expletive pronoun in (62) is apparently superfluous. From a semantic point of view, one might therefore expect it to be optional. But this is not the case, as the ungrammaticality of (63) shows.

(63) * Is evident that they are corrupt.

The ungrammaticality of (63) leads us to conclude that there exists a purely syntactic well-formedness condition requiring all clauses to have a **subject**.

(64) **The Subject Requirement**:
All clauses must have a subject.

Earlier, we saw that it is possible for arguments to be semantically motivated and yet not appear overtly in the syntax. Expletive subjects represent roughly the converse of this situation, being cases where an expression that is not motivated by semantic considerations is nevertheless obligatory in the syntax.

More evidence for the subject requirement: Expletive *there*

In English, further evidence for the purely syntactic character of the subject requirement comes from the expletive *there* construction. As with *it,* English *there* comes in two variants: locative and expletive. The pro-form *there* has inherently locative meaning in (65); this is evident in the fact that the adjuncts *(right) here* and *(over) there* render (65a) and (65b) self-contradictory and redundant, respectively.

(65) a. # There comes the train (right) here.
  b. ✓ There comes the train (over) there.

It is possible to stress locative *there*, regardless of its position in a sentence.

Expletive *there* has no locative meaning, and so both sentences in (66) are completely acceptable. In contrast to locative *there,* expletive *there* can never bear stress.

(66) a. ✓ There is a clean shirt (right) here.
  b. ✓ There is a clean shirt (over) there.

Expletive *there* behaves like an ordinary subject in ways that clause-initial locative *there* doesn't. For instance, it inverts with the verb in direct questions, and it is grammatical in so-called small clauses (in this instance, selected by *despite*: see § 4.5.3 for more on small clauses).

(67) a. ✓ Is there a problem?
  b. * Is (over) there our train?

(68) a. ✓ despite [there being a problem]
  b. * despite (over) [there being our train]

(69) illustrates an ordinary sentence and its expletive *there* counterpart.

(69) a. Several vexing questions remain.
  b. There remain several vexing questions.

Just as expletive *it* occupies the position that would otherwise be occupied by a clausal subject, expletive *there* occupies the position that would otherwise be occupied by a noun phrase subject. And just as with expletive *it*, omitting expletive *there* results in ungrammaticality.

(70) * Remain several vexing questions.

It should be pointed out that not every English sentence has an expletive *there* counterpart. Rather, expletive *there* is subject to a **licensing condition** (a necessary condition for its occurrence) that can be stated roughly as in (71).

(71) <u>Licensing condition on expletive *there*</u>:

For expletive *there* to be grammatical as the subject of a clause, the (Fregean) predicate of that same clause must be a verb of existence or coming into existence.

In the following examples, the (Fregean) predicate licensing the expletive *there* is in blue boldface.

(72) a. After their military defeat, *there* **arose** among the Plains tribes a powerful spiritual movement.
b. *There* **is** a problem.
c. *There* **began** a reign of terror.
d. In the end, *there* **emerged** a new caudillo.
e. *There* **ensued** a period of unrest and lawlessness.
f. *There* **exists** an antidote.
g. *There* has **occurred** an unfortunate incident.
h. *There* **remains** a single course of action.

Predicates that aren't verbs of (coming into) existence don't license expletive *there*. This is the reason for the ungrammaticality of the following examples; the non-licensing (Fregean) predicates are underlined in red.

(73) a. * *There* came more than sixty dignitaries.
b. * *There* continued the same problem.
c. * *There* rang the mail carrier.
d. * *There* sang an impressive choir from Russia.
e. * *There* strutted a poodle into the room.

4.5.2 Nonfinite clauses

The instances of predication provided so far have all been finite clauses like those in (74) (i.e. clauses that bear tense).

(74) a. *He* laughed uproariously.
b. *It* will seem that they won the game.
c. *There* is a problem.

Nonfinite clauses like those in (75) are also instances of predication; the clauses at issue are set off by brackets.

(75) a. We expected [*him* to laugh uproariously].
b. We expected [*it* to seem that they won the game].
c. We expected [*there* to be a problem].

At first glance, it might seem preferable to treat the italicized noun phrases in (76) as objects of *expected*, rather than as subjects of the embedded nonfinite clause the way we have done. However, such an approach faces at least two difficulties. First, the relation between the italicized and underlined constituents in the all of nonfinite embedded clauses in (75) is the same as the relation between the undoubted subjects and predicates of the finite clauses in (74) and their embedded counterparts in (76).

(76) a. We expected that [*he* would laugh uproariously].
b. We expected that [*it* would seem that they won the game].
c. We expected that [*there* would be a problem].

Second, in (75a), the thematic relation of agent that the noun phrase *him* bears to *laugh* is exactly the same as that between the subject *he* and *laughed* in (74a) and (76a). If *him* were the object of *expected* rather than the subject of the nonfinite clause, that fact would not be captured directly. Moreover, we would be forced to admit the otherwise unprecedented pairing of the grammatical relation of direct object with the thematic role of agent (see § 4.3 and § 4.4).

4.5.3 Small clauses

Because of the parallel between nonfinite and finite embedded clauses illustrated in (75) and (76), it makes sense to assign *to* in *to*-infinitive clauses to the same category as modals (see Chapter 5, § 5.2.2). There also exist instances of predication without any inflection whatsoever (inflection being the marking of tense and agreement on the clause, normally on the verb). These are called **small clauses** (the idea behind the name is that the absence of inflection makes them smaller than an ordinary clause).[5] (77)–(80) provide some examples; the captions indicate the syntactic category of the small clause's (Aristotelian) predicate, which is underlined.

(77) Adjective phrase:
a. We consider [*the proposed solution* completely inadequate].
b. They proved [*the theory* false].

(78) Noun phrase:
a. They called [*the actor* a traitor].
b. I consider [*Mark* Judy's closest collaborator].

(79) Prepositional phrase:
a. They made [*him* into a star].
b. I want [*everyone* off the boat].

(80) Verb phrase:
a. God let [*there* be light]. (bare verb)
b. I hear [*the cat* scratching at the door]. (gerund)

Small clauses are typically arguments of verbs, but they can also be arguments of (certain) prepositions—notably *with*—as illustrated in (81).

(81) a. Adjective phrase:
With *the weather* much less turbulent, flights were able to resume for the first time in days.
b. Noun phrase:
With *her husband* an airline industry lobbyist, the senator's support for the bailout was hardly surprising.

c. Prepositional phrase:
With *all three of their kids* in college, their budget is pretty tight.

d. Verb phrase (gerund):
With *the parade* passing right in front of her house, Jenny could not have asked for a better view of it.

Small clauses will play a significant role in some of our theorizing in Chapter 7, so we will see these come back eventually.

4.5.4 Imperatives

Imperative sentences like (82) appear to lack a subject.

(82) Come over here.

There is reason to believe, however, that they contain a second-person subject comparable to the pronoun *you* except that it is silent (the "you understood" of traditional grammar). For one thing, (82) has a variant in (83) where the subject is explicitly expressed.

(83) You come over here.

Another reason to assume that all imperatives contain a silent, yet syntactically active subject is that the grammaticality pattern in (84), where the subject is overt, has an exact counterpart in (85).

(84) a. You shave { yourself, yourselves }.
b. * You shave you.
c. * You shave themselves.

(85) a. Shave { yourself, yourselves }.
b. * Shave you.
c. * Shave themselves.

New Empirical Observations

- expletive *it* and *there*
- small clauses

Components of the Theoretical Model

- predicates
- arguments

- thematic roles
  - agent
  - cause
  - instrument
  - experiencer
  - recipient/benefactive
  - location
  - path
  - goal
  - measure
  - theme
- grammatical functions
  - subject
  - direct object
  - indirect object
- Subject Requirement

Notes 4.6

1. Formal semanticists reserve the term *event*/*eventive* for actions, and use the term *eventuality* to collectively refer to actions and states together. While more precise, the term *eventuality* is a little less transparent to non-semanticists, so with apologies to those folks we will use the term *event* here to describe eventualities.

2. Fruitful as Frege's analytic insight is, we should not let it obscure the key difference between subjects and other constituents of a clause: namely, that constituents contained within an Aristotelian predicate must be **licensed** by semantic considerations (in other words, these constituents are present because of semantic considerations), whereas the subject, which is external to the Aristotelian predicate and combines with it, is required independently of semantic considerations.

3. Here and in the rest of the section, our usage is a bit sloppy. Specifically, we ignore the distinction between linguistic expressions and the discourse entities that they refer to (for some discussion, see § 15.1 of Chapter 15). For instance, in (32a), the agent of the devouring is a group of live animals, not the two words referring to them). So a more precise formulation would be something like "We use the term 'agent' to refer to linguistic expressions that (in turn) refer to entities that bring about a state of affairs". This mild imprecision is hopefully tolerable in service of a more natural exposition.

4. In fact, those forms are probably default forms rather than specifically accusative forms, since they are used much more broadly than just accusative contexts.

5. Small clauses are exceptional in another regard: they are a further instance of constituents where the constituenthood tests of Chapter 3 yield false negative results (at least for most speakers).

   (i) a. What do you like?

   b. ?? My coffee hot.

Exercises and problems 4.7

Exercise 4.1: Practice with predicates and arguments

For each sentence given in (1), identify the predicate and the arguments of the predicate. If doing the exercise by hand, circle the predicates and underline the arguments. Otherwise mark/identify them however is consistent and convenient.

(1)
a. The children chased butterflies yesterday in the park.
b. The footballers practiced their free kick routines the day before the game.
c. The announcers narrated the events of the game.
d. The dog slept peacefully in his bed.
e. The student read the entire textbook before the semester started.
f. Professors assign their students too much homework.
g. AI doesn't do syntax assignments well. Yet.
h. It is raining heavily outside.
i. The rain is falling gently.
j. Link saves the kingdom of Hyrule every time.
k. I paid the cashier $7 for this sandwich.
l. I bet my friend $100 that Liverpool will win the title this year.

Exercise 4.2: Practice with arguments and modifiers (adjuncts)

For each sentence in (1), identify which constituents are arguments of the predicate (which is bolded) and which constituents are modifiers/adjuncts. Some constituents of the sentence include predicates of their own, but we focus on the predicates/arguments/modifiers of the main clause.

(1)
a. The defender **tackled** the striker forcefully before she ran past.
b. The defender unsurprisingly **received** a yellow card for her hard tackle.
c. The referee **gave** the defender a yellow card for her hard tackle.
d. I **made** tacos for dinner last night.
e. The broth **simmered** on the stove for hours.
f. The line chef skillfully **chopped** the onions quickly with a large knife.
g. I would **kill** for a cheesesteak from South Philly.
h. I **watch** videos about cooking more than I actually cook.
i. The rain **poured** down all night.
j. We **arrived** at class early today.
k. Alex said they are **leaving** class early today.
l. Lilly always **snores** loudly.

Exercise 4.3: Practice with thematic roles

For each of the underlined arguments, identify the thematic role that it holds for the (bolded) predicate it is occurring with.

(1)
  a. My mom sent me the recipe for chicken curry that I had been asking for.
  b. I miss soft pretzels from Philly.
  c. My coffee spilled on my shirt this morning.
  d. The goalkeeper almost dropped the ball after being hit.
  e. The midfielder handed her coach the ball.
  f. The referee placed the ball on the penalty spot.
  g. The final score of the match shocked me, to be honest.
  h. We could hear the crowd from outside the stadium.
  i. The wind opened the door.
  j. The children go to school every morning.
  k. I send my kiddo to school every morning.
  l. My fingerprint unlocks my computer.
  m. The pandemic inflicted hardship and suffering on many people.

Exercise 4.4: Clarifying core concepts

A. In your own words, discuss the difference between "modify" and "predicate". Feel free to use illustrative examples, but be as concise as you can.

B. This part of the exercise focuses on a fundamental distinction in linguistics—that of form vs. function. In everyday usage, the term "modifier", which refers to any expression that adds information about some other expression, is sometimes used interchangeably with the term "adjective". By contrast, linguists distinguish carefully between the two terms and define adjectives on the basis of morphosyntactic form rather than on the basis of function (see Chapter 2, § 2.3.3 for details). Assuming the linguistic definition of "adjective", give an example of your own of 1) an adjective (or adjective phrase) that is not a modifier, and 2) a modifier that is not an adjective.

C. This part of the exercise is intended to help you get a better feel for the concept of "small clause", which many students find difficult. Make up some example sentences that you think might contain small clauses, but that you're not sure about. Clearly indicate the string that you think is the small clause.

Exercise 4.5: Form versus function

A. This part of the exercise focuses on a fundamental distinction in linguistics—that of form vs. function. The function of modification is typically expressed by constituents of a certain form—namely, adverb phrases or prepositional phrases. But modifiers can also take the form of noun phrases, as in (1).

(1) Manner: They solved the problem *another way*.

Provide examples of noun phrase modifiers for the other types of modification illustrated in the chapter in (26), repeated for convenience as (2). (This is not necessarily possible for every type.)

(2)
a. Manner:
He read the letter *carefully*.
b. Location:
We met the students *in my office*.
c. Origin:
We departed *from Bangalore*.
d. Destination:
We arrived *in Benares*.
e. Path:
We followed *along the path*.
f. Point in time:
They discussed the proposal *in the afternoon*.
g. Duration:
She kept their books *for five years*.
h. Frequency:
They read the Times *quite often*.
i. Cause:
He threw it away *out of spite*.
j. Purpose:
You should send a message *(in order) to warn everyone*.

B. Here are some questions concerning the relationship between thematic roles and grammatical functions.

- Is there a one-to-one relations between thematic roles and grammatical relations?
- Are there any thematic roles that must be expressed by a particular grammatical relation? For instance, are agents invariably expressed as subjects?
- Conversely, are there grammatical relations that are restricted to expressing a particular thematic role? For instance, do direct objects always express themes?

- Are there typical or common mappings between thematic roles and grammatical relations?
- Can you imagine a prepositional phrase expressing the thematic role of theme (not necessarily in English)?

Problem 4.1: Grammatical functions across languages

In this exercise you will compare Mainstream U.S. English, Japanese (Japonic, Japan), Mandarin Chinese (Sinitic, China), Korean (Koreanic, Korea), and Tiriki (a Bantu language of the Luyia subgroup spoken in western Kenya). Grammatical functions are represented similarly between the languages in some ways, and differently in others. Instructors may specify whether students should work on each language included here, or only selected languages.

A. For each language: how are the grammatical functions of subject and object encoded in the language? List all grammatical correlates of those grammatical functions that you note.

B. Discuss the similarities and differences in representing the grammatical functions of subject and object between the languages.

Background Info

Tiriki nominals are all designated with some noun class, which is annotated with a cardinal number (1, 2, 3, etc). You don't need to understand details about noun class except that noun classes exist, and different categories in Tiriki inflect (agree) based on noun class. All verbs in Tiriki necessarily end in a vowel—sometimes this vowel is part of another affix, at other times we simply gloss it as FV (final vowel). PASS = passive; AGR= agreement; PRON= pronoun. Some details about the phonological processes in preverbal verb length are adjusted here in order to normalize the data for the purposes of the pedagogical exercise.

For Japanese, Korean, and Chinese, PST indicates past tense, DECL declarative mood, and PFV perfective aspect (you may safely ignore tense-aspect-mood for this exercise). DIM indicates 'diminutive' (here, it's a prefix attached to a name to show affection or familiarity) and may also be safely ignored. The functions of GA, O, I/KA, (L)UL, and BA are for you to discover! Finally, the keen-eyed student may notice some verb-stem allomorphy, but they and their classmates should feel free to ignore it.

English

(1) a. The window breaks.
    b. The ball breaks the window.
    c. The balls break the window.
    d. The window was broken.

(2) a. She greeted them.
    b. They greeted her.

c. *Them greeted she.
d. *Her greeted they.
e. They were greeted.
f. *Them were greeted.

Tiriki

(3) a. I-nyungu y-a-tikha.
9-pot 9AGR-PST-be.broken
'The pot broke.'

b. Tsi-nyungu tsy-a-tikha.
10-pots 10AGR-PST-be.broken
'The pots broke.'

c. Mw-ana y-a-rhana i-nyungu.
1-child 1AGR-PST-break 9-pot
'The child broke the pot.'

d. Va-na va-a-rhana i-nyungu.
2-children 2AGR-PST-break 9-pot
'The children broke the pot.'

e. I-nyungu y-a-rhan-w-a.
9-pot 9AGR-PST-break-PASS-FV
'The pot was broken.'

f. Tsi-nyungu tsy-a-rhan-w-a.
10-pots 10AGR-PST-break-PASS-FV
'The pots were broken.'

Each of the examples in (4) are different ways of saying the same thing. As a reminder, third person singular human noun phrases are class 1 in Tiriki; third person plural human noun phrases are class 2. Third person is not differentiated by human gender in Tiriki, but we translate examples only as feminine here for readability (rather than listing all the options).

(4) a. Y-a-va-shelits-a.
1AGR-PST-2AGR-greet-FV
'She greeted them.' (*They greeted her.)

b. Ye y-a-va-shelits-a.
1PRON 1AGR-PST-2AGR-greet-FV
'She greeted them.' (*They greeted her.)

c. Ye y-a-shelits-a vo.
1PRON 1AGR-PST-greet-FV 2PRON
'She greeted them.' (*They greeted her.)

(5) a. Va-a-shelits-w-a.
2AGR-PST-greet-PASS-FV
'They were greeted.'

b. Vo va-a-shelits-w-a.
2PRON 2AGR-PST-greet-PASS-FV
'They were greeted.'

Each of the examples in (6) are different ways of saying the same thing. As a reminder, third person singular human noun phrases are class 1 in Tiriki; third person plural human noun phrases are class 2. Third person is not differentiated by human gender in Tiriki, but we translate examples only as feminine here for readability (rather than listing all the options).

(6) a. Va-a-mu-shelits-a.
2AGR-PST-1AGR-greet-FV
'They greeted her.' (*She greeted them.)

b. Vo va-a-mu-shelitsa.
2PRON 2AGR-PST-1AGR-greet-FV
'They greeted her.' (*She greeted them.)

c. Vo va-a-shelits-a ye.
2PRON 2AGR-PST-greet-FV 1PRON
'They greeted her.' (*She greeted them.)

(7) a. Y-a-shelits-w-a.
1AGR-PST-greet-PASS-FV
'She was greeted.'

b. Ye y-a-shelits-w-a.
1PRON 1AGR-PST-greet-PASS-FV
'She was greeted.'

Japanese

(8) a. 窓が 開いた。
Mado-ga ai-ta.
window-GA open-PST
'The window opened.'

b. すべての 窓が 開いた。
Subeteno mado-ga ai-ta.
all window-GA open-PST
'All the windows opened.'

c. 子供が 窓を 開けた。
Kodomo-ga mado-o ake-ta.
child-GA window-O open-PST
'The child opened the window.'

d. 子供たちが 窓を 開けた。
Kodomo-tachi-ga mado-o ake-ta.
child-PL-GA window-O open-PST
'The children opened the window.'

e. 窓が 開けられた。
Mado-ga ake-rare-ta.
window-GA open-PASS-PST
'The window was opened.'

f. すべての 窓が 開けられた。
Subeteno mado-ga ake-rare-ta.
all window-GA open-PASS-PST
'All the windows were opened.'

g. # 窓たちが 開いた。
Mado-tachi-ga ai-ta.
window-PL-GA open-PST
'The windows opened.' (sounds like a fairytale with living windows)

h. # 窓たちが 開けられた。
Mado-tachi-ga ake-rare-ta.
window-PL-GA open-PASS-PST
'The windows were opened.' (sounds like a fairytale with living windows)

Each of the examples in (9) are different ways of saying the same thing.

(9) a. 彼女が 彼を 起こした。
Kanojo-ga kare-o okoshi-ta.
she-GA he-O wake.up-PST
'She woke him up.'

b. 彼を 彼女が 起こした。
Kare-o kanojo-ga okoshi-ta.
he-O she-GA wake.up-PST
'She woke him up.'

Each of the examples in (10) are different ways of saying the same thing.

(10) a. 彼が 彼女を 起こした。
Kare-ga kanojo-o okoshi-ta.
he-GA she-O wake.up-PST
'He woke her up.'

b. 彼女を 彼が 起こした。
Kanojo-o kare-ga okoshi-ta.
she-O he-GA wake.up-PST
'He woke her up.'

Korean

(11) a.

물이	끓었다.
Mul-i	kkulh-ess-ta.
water-I/KA	boil-PST-DECL

‘The water boiled.’

b.

성희가	물을	끓였다.
Senghuy-ka	mul-ul	kkulhy-ess-ta.
Sunghee-I/KA	water-(L)UL	boil-PST-DECL

‘Sunghee boiled the water.’

c.

물을	성희가	끓였다.
Mul-ul	Senghuy-ka	kkulhy-ess-ta.
water-(L)UL	Sunghee-I/KA	boil-PST-DECL

‘Sunghee boiled the water.’

d.

# 물이	성희를	끓였다.
Mul-i	Senghuy-lul	kkulhy-ess-ta.
water-I/KA	Sunghee-(L)UL	boil-PST-DECL

‘#The water boiled Sunghee.’

Each of the examples in (12) are different ways of saying the same thing.

(12) a.

그녀가	그들을	그리워했다.
Kunye-ka	ku-dul-ul	kuliwehayss-ta.
she-I/KA	3-PL-(L)UL	miss-PST-DECL

‘She missed them.’

b.

그들을	그녀가	그리워했다.
Ku-dul-ul	kunye-ka	kuliwehayss-ta.
3-PL-(L)UL	she-KA	miss-PST-DECL

‘They missed her.’

Each of the examples in (13) are different ways of saying the same thing.

(13) a.

그들이	그녀를	그리워했다.
Ku-dul-i	kunye-lul	kuliwehayss-ta.
3-PL-I/KA	she-(L)UL	miss-PST-DECL

‘They missed her.’

b.

그녀를	그들이	그리워했다.
Kunye-lul	ku-dul-i	kuliwehayss-ta.
she-(L)UL	3-PL-I/KA	miss-PST-DECL

‘They missed her.’

Mandarin Chinese

(14) a. 实分析作业 终于 写完了。
Shí-fēnxī-zuòyè zhōngyú xiě-wán-le.
real-analysis-homework finally write-done-PFV
'The real analysis homework is finally finished.'

b. 小明 终于 写完了 实分析作业。
Xiǎo-Míng zhōngyú xiě-wán-le shí-fēnxī-zuòyè.
DIM-Ming finally write-done-PFV real-analysis-homework
'Ming finally finished the real analysis homework.'

c. 小明 终于 把 实分析作业 写完了。
Xiǎo-Mínğ zhōngyú bǎ shí-fēnxī-zuòyè xiě-wán-le.
DIM-Ming finally BA real-analysis-homework write-done-PFV
'Ming finally finished the real analysis homework.'

d. # 实分析作业 终于 写完了 小明。
Shí-fēnxī-zuòyè zhōngyú xiě-wán-le Xiǎo-Míng.
real-analysis-homework finally write-done-PFV DIM-Ming
'#The real analysis homework finally finished Ming.' (*nonsensical or humorous*)

e. # 实分析作业 终于 把 小明 写完了。
Shí-fēnxī-zuòyè zhōngyú bǎ Xiǎo-Míng xiě-wán-le.
real-analysis-homework finally BA DIM-Ming write-done-PFV
'#The real analysis homework finally finished Ming.' (*nonsensical or humorous*)

Each of the examples in (15) are different ways of saying the same thing. Third person is not differentiated by human gender in spoken Mandarin, but we translate examples only as feminine here for readability (rather than listing all the options). COS indicates a sentence-final change-of-state particle (but you may ignore it).

(15) a. 她 带 他们 去 看 电影 了。
Tā dài tā-men qù kàn diànyǐng le.
3SG take 3-PL go see movie COS
'She took them to go see a movie.' (*They took her to watch a movie.)

b. 她 把 他们 带 去 看 电影 了。
Tā bǎ tā-men dài qù kàn diànyǐng le.
3SG BA 3-PL take go see movie COS
'She took them to go see a movie.' (*They took her to watch a movie.)

(16) a. 他们 被 带 去 看 电影 了。
Tā-men bèi dài qù kàn diànyǐng le.
3-PL PASS take go see movie COS
'They were taken to go see a movie.'

b. * 他们 被 把 带 去 看 电影 了。
Tā-men bèi bǎ dài qù kàn diànyǐng le.
3-PL PASS BA take go see movie COS
'They were taken to go see a movie.'

c. ?* 他们 把 被 带 去 看 电影 了。
Tā-men bǎ bèi dài qù kàn diànyǐng le.
3-PL BA PASS take go see movie COS
Intended: 'They were taken to go see a movie.'

d. * 把 他们 被 带 去 看 电影 了。
Bǎ tā-men bèi dài qù kàn diànyǐng le.
BA 3-PL PASS take go see movie COS
Intended: 'They were taken to go see a movie.'

Each of the examples in (17) are different ways of saying the same thing. Third person is not differentiated by human gender in spoken Mandarin, but we translate examples only as feminine here for readability (rather than listing all the options). COS indicates a sentence-final change-of-state particle (but you may ignore it).

(17) a. 他们 带 她 去 看 电影 了。
Tā-men dài tā qù kàn diànyǐng le.
3-PL take 3SG go see movie COS
'They took her to go see a movie.' (*She took them to watch a movie.)

b. 他们 把 她 带 去 看 电影 了。
Tā-men bǎ ta dài qù kàn diànyǐng le.
3-PL BA 3SG take go see movie COS
'They took her to go see a movie.' (*She took them to watch a movie.)

(18) a. 她 被 带 去 看 电影 了。
Tā bèi dài qù kàn diànyǐng le.
3SG PASS take go see movie COS
'She was taken to go see a movie.'

Problem 4.2: Adjusting thematic structure of predicates in Tiriki

Tiriki is a Bantu language of the Luyia subgroup that is spoken in the western parts of Kenya. Like many Bantu languages, predicates can be adjusted in terms of what thematic roles are referenced using derivational morphology on the verb form. Your task is to identify what function each morpheme accomplish which functions in Tiriki. The relevant morphemes are glossed with question marks in the examples below. You don't have to explain any other aspects of the morphology of Tiriki.

Background: Tiriki has a robust noun class system: noun classes are glossed with cardinal numbers (1, 2, 3, 4, etc). You don't have to explain anything about noun classes, though they do show what is agreeing with that, which is sometimes helpful to know. SM = subject marker (i.e., subject agreement on verbs), SM glosses are accompanied with a noun class gloss based on what the noun class is of the subject that they agree in; PST = past tense; PFV = perfective aspect (translated similar to past tense); all verbs in Tiriki necessarily end in a vowel—sometimes this vowel is part of another affix, at other times we simply gloss it as FV (final vowel).

(1) a. Va-cheni va-rerh-i vi-hanwa.
2-guests 2sm-bring-pst 2-gifts
'The guests brought gifts.'

b. Va-cheni va-rerh-er-e va-na vi-hanwa.
2-guests 2sm-bring-?-pst 2-children 2-gifts
'The guests brought the children gifts.'

c. Va-na va-rerh-er-w-e vi-hanwa.
2-children 2sm-bring-?-?-pst 2-gifts
'The children were brought gifts.'

(2) a. Shi-kombe shi-itsul-e.
7-cup 7sm-fill-pst
'The cup is filled.'

b. Mu-hinzili y-itsul-its-e shi-kombe.
1-server 1sm-fill-?-pst 7-cup
'The server filled the cup.'

c. Shi-kombe shi-itsul-its-w-e na mu-hinzili.
7-cup 7sm-fill-?-?-pst by 1-server
'The cup was filled by the server.'

(3) a. Kukhu y-a-fung-a mu-lyango.
1Grandma 1sm-pst-close-fv 3-door
'Grandma closed the door.'

b. Mu-lyango kw-a-fung-ikh-a (*na kukhu).
3-door 3sm-pst-close-?-fv (*by 1grandma)
'The door closed.'

c. Mu-lyango kw-a-fung-w-a na kukhu.
3-door 3sm-pst-close-?-fv by 1grandma
'The door was closed by grandma.'

(4) a. Mw-ana a-khalak-e vi-tungu.
1-child 1sm-cut-pst 8-onions
'The child cut the onions.'

b. Mw-ana a-khalak-il-e mama vi-tungu.
1-child 1sm-cut-?-pst 8-onions
'The child cut the onions for Mama.'

c. Mama a-khalak-il-w-e vi-tungu na mw-ana.
1mama 1sm-cut-?-?-pst 8-onions by 1-child
'Mama had onions cut for her by the child.'

(5) a. Azangu a-kalukh-e i-ngo.
1Azangu 1sm-return-pst 9-home
'Azangu returned home.'

b. Azangu a-kalukh-il-e va-cheni i-ngo.
1Azangu 1sm-return-?-pst 2-guests 9-home
'Azangu returned home for the guests.'

(6) a. Mu-tekhi a-shukhanyiny-e mu-nyu.
1-cook 1sm-stir-pst 3-soup
'The cook stirred the soup.'

b. Mu-tekhi a-shukhany-il-e mu-nyu shi-chiko.
1-cook 1sm-stir-?-pst 3-soup 7-spoon
'The cook stirred the soup with a spoon.'

c. Mu-nyu khu-shukhany-il-w-e shi-chiko na mu-tekhi.
3-soup 3sm-stir-?-?-pst 7-spoon by 1-cook
'The soup was stirred with a spoon by the cook.'

(7) a. Mu-tekhi y-a-samb-a ma-tuma.
1-cook 1sm-pst-roast-fv 6-maize
'The cook roasted the maize.'

b. Ma-tuma ka-a-samb-ikh-a (*na mu-tekhi).
6-maize 6sm-pst-roast-?-fv (*by 1-cook)
'The maize roasted.'

c. Ma-tuma ka-a-samb-w-a na mu-tekhi.
6-maize 6sm-pst-roast-?-fv by 1-cook
'The maize was roasted by the cook.'

(8) a. I-simbwa i-vukh-i.
9-dog 9sm-wake.up-pst
'The dog woke up.'

b. Li-yoko li-vukh-its-e i-simbwa.
5-loud.noise 5sm-wake.up-?-pst 9-dog
'The loud noise woke up the dog.'

c. I-simbwa i-vukh-its-w-e na li-yoko.
9-dog 9sm-wake.up-?-?-pst by 5-loud.noise
'The dog was woken by the loud noise.'

(9) a. V-ivuli va-chot-i.
2-parents 2sm-be.tired-pst
'The parents were tired.'

b. Va-na va-chot-its-e v-ivuli.
2-children 2sm-be.tired-?-pst 2-parents
'The children tired the parents.'

c. V-ivuli va-chot-its-w-e na va-na.
2-parents 2sm-be.tired-?-?-pst by 2-children
'The parents are tired by the children.'

(10) a. Mu-hinzili a-khalak-e lu-pao.
1-worker 1SM-chop-PST 11-wood
'The worker chopped the wood.'

b. Mu-hinzili a-khalak-il-e lu-pao i-haywa.
1-worker 1SM-chop-?-PST 11-wood 9-axe
'The worker chopped the wood with an axe.'

c. Lu-pao lu-khalak-il-w-e i-haywa na mu-hinzili.
11-wood 11SM-chop-?-?-PST 9-axe by 1-worker
'The wood was chopped with an axe by the worker.'

(11) a. Vi-vuli va-som-i vi-tapu.
2-parents 2SM-read-PST 8-books
'The parents read the books.'

b. Vi-vuli va-som-el-e va-na vi-tapu.
2-parents 2SM-read-?-PST 2-children 8-books
'The parents read the children the books.'

c. Va-na va-som-el-w-e vi-tapu na vi-vuli.
2-children 2SM-read-?-?-PST 8-books 2-children by 2-parents
'The children were read books by the parents.'

(12) a. Mu-tekhi y-a-khalak-a vi-tungu.
1-cook 1SM-PST-cut-FV 8-onions
'The cook cut the onions.'

b. Vi-tungu vy-a-khal-ikh-a (*na mu-tekhi).
8-onions 8SM-PST-cut-?-FV (*by 1-cook)
'The onions (got) cut.'

c. Vi-tungu vy-a-khal-w-a na mu-tekhi.
8-onions 8SM-PST-cut-?-FV by 1-cook
'The onions were cut by the cook.'

Problem 4.3: Identifying structural differences

Using concepts introduced in this chapter, discuss the syntactic difference(s) between the two sentences in (1).

(1) a. Winston considered the judges careful.

b. Winston considered the judges carefully.

(The sentences come from this source, but the discussion there of (1a) is confusing. Thanks to Rosemary George for tracking down the archived link.)

Problem 4.4: Ambiguity

A. The following headlines each have an intended reading and (possibly more than one) unintended humorous reading. Using the concepts introduced in this chapter, explain the differences between the intended and unintended readings.

(1) a. Greeks Fine Hookers
    b. Lawmen from Mexico Barbecue Guests
    c. Lawyers Give Poor Free Legal Advice
    d. Lung Cancer in Women Mushrooms

(2) Q: What did the Zen master say to the guy at the hot dog stand?
    A: Make me one with everything.

(3) The comedian and civil rights activist Dick Gregory tells of walking up to a lunch counter in Mississippi during the days of racial segregation. The waitress said to him, "We don't serve colored people". "That's fine," he replied, "I don't eat colored people. I'd like a piece of chicken."

B. Recall Exercise 1.5. (Re)formulate (and possibly improve/generalize) your answer to Part B in terms of the concepts introduced in this chapter.

Problem 4.5: Lubukusu locative inversion

Background: Lubukusu is canonically an SVO language, so (2) displays the canonical (i.e., usual, regular) word order. It has noun classes, and lots of agreement. This puzzle is specifically about how verbs inflect for noun class in a particular grammatical construction known as **locative inversion**, where a locative phrase appears in preverbal position and the subject is in a non-canonical postverbal position. English has similar patterns; (1a) shows canonical English word order, and (1b) shows a locative inversion construction.

(1) a. The ball rolled down the hill.
    b. Down the hill rolled the ball.

Instructions: There are two different kinds of locative inversion in this Lubukusu data set. Identify what the two patterns are from the "basic examples" below. Then, consider the variety of verbs in the locative inversion constructions in the next section. What patterns do you notice? What is the generalization with respect to which version of locative inversion is available with which verbs?

Glosses: SM = subject marker, i.e., verbal agreement with the subject, PRF = perfect tense, PRS = present tense, LOC = locative

Notes:

- Lubukusu is a tone language, but tone is not relevant for these patterns and is not marked.
- The numbers on nouns and agreement morphemes signal noun class. For the purposes of this assignment, they are mainly helpful to see what is agreeing with what.

- Locative phrases (i.e., locations) are formed by taking some noun and replacing the first noun class marker with a locative class marker. The locative noun classes are 16, 17, 18.

Basic examples

(2) Ku-mu-saala **kw**-a-kwa mu-mu-siiru.
3-3-tree **3SM**-PST-fall 18-3-forest
'A tree fell in the forest.'

(3) Mu-mu-siiru **kw**-a-kwa-**mo** ku-mu-saala.
18-3-forest **3SM**-PST-fall-**18LOC** 3-3-tree
'In the forest fell a tree.'

(4) Mu-mu-siiru **mw**-a-kwa-**mo** ku-mu-saala.
18-3-forest **18SM**-PST-fall-**18LOC** 3-3-tree
'In the forest fell a tree.'

(5) Ku-m-pira **kw**-a-biringikha khu-si-kulu.
3-3-ball **3SM**-PST-roll 17-7-hill
'The ball rolled on/down the hill.'

(6) Khu-si-kulu **kw**-a-biringikha-**kho** ku-m-pira.
17-7-hill **3SM**-PST-roll-**17LOC** 3-3-ball
'On/down the hill rolled the ball.'

(7) Khu-si-kulu **khw**-a-biringikha-**kho** ku-m-pira.
17-7-hill **17SM**-PST-roll-**17LOC** 3-3-ball
'On the hill rolled the ball.'

Examples using different verbs

(8) a. Ba-ba-ana **b**-ola e-engo.
2-2-child **2SM**-PST.arrive 9-home
'The children arrived home.'

b. Mu-nju **b**-ola-**mo** ba-ba-ana.
18-home **2SM**-PST.arrive-**18LOC** 2-2-child
'Inside/at home arrived the children.'

c. Mu-nju **mw**-ola-**mo** ba-ba-ana.
18-home **18SM**-PST.arrive-**18LOC** 2-2-child
'Inside/at home arrived the children.'

(9) a. E-nyuni **y**-embela khu-mu-saala.
9-bird **9SM**-sang 17-3-tree
'A bird sang on the tree.'

b. Khu-mu-saala **y**-embela-**kho** e-nyuni.
17-3-tree **9SM**-sang-**17LOC** 9-bird
'On the tree sang a bird.'

c. * Khu-mu-saala **khw**-embela-**kho** e-nyuni.
17-3-tree **17SM**-sang-**17LOC** 9-bird
'On the tree sang a bird.'

(10) a. Ba-ba-ana **ba**-a-kona e-ngo.
2-2-child **2sm**-pst-sleep 16-house
'The children slept at home.'

b. E-ngo **ba**-a-kona-**o** ba-ba-ana.
16-house **2sm**-pst-sleep-**16loc** 2-2-child
'At home slept the children.'

c. E-ngo **e**-kona-**o** ba-ba-ana.
16-house **16sm**.pst-sleep-**16loc** 2-2-child
'At home slept the children.'

(11) a. Ba-ba-ndu **b**-engila mu-kanisa.
2-2-person **2sm**-pst.enter 18-9church
'People entered the church.'

b. Mu-kanisa **b**-engila-**mo** ba-ba-andu.
18-9church **2**-pst.enter-**18loc** 2-2-person
'In the church entered people.'

c. * Mu-kanisa mw-engila-mo ba-ba-andu.
18-9church **18sm**-pst.enter-**18loc** 2-2-person

(12) a. Mw-i-duka **ka**-a-chekhela-**mo** Moses.
18-9-store **1sm**-pst-laugh-**18loc** 1Moses
'Inside the store laughed Moses.'

b. * Mw-i-duka **mw**-a-chekhela-**mo** Moses.
18-9-store **18sm**-pst-laugh-**18loc** 1Moses

(13) a. Ku-m-pira **kw**-a-biringila khu-si-kulu.
3-3-ball **3sm**-pst-roll 17-7-hill
'The ball rolled on/down the hill.'

b. Khu-si-kulu **kw**-a-biringila-**kho** ku-m-pira.
17-7-hill **3sm**-pst-roll-**17loc** 3-3-ball
'On/down the hill rolled the ball.'

c. Khu-si-kulu **khw**-a-biringila-**kho** ku-m-pira.
17-7-hill **17sm**-pst-roll-**17loc** 3-3-ball
'On the hill rolled the ball.'

(14) a. O-mw-ana **a**-a-suna mu-si-wanja.
1-1-child **1sm**-pst-jump 18-7-field
'A child jumped into the field.'

b. Mu-si-wanja **a**-a-suna-**mo** o-mw-ana.
18-7-field **1sm**-pst-jump-**18loc** 1-1-child
'Into the field jumped a child.'

c. * Mu-si-wanja **mw**-a-suna-**mo** o-mw-ana.
18-7-field **18sm**-pst-jump-**18loc** 1-1-child
'Into the field jumped a child.'

Problem 4.6: MUSE subject requirement

The acceptability of (1) in vernacular usage of Mainstream U.S. English might tempt one to conclude that the subject requirement is not absolute in English. (See § 4.5 of this chapter for relevant evidence, diagnostics, and discussion.)

(1) a. Seems like they're finally getting somewhere.
    b. Ran to the store for milk this morning.

Provide as much evidence as you can bearing on this conclusion. What contexts can sentences like those in (1) be used in? Are they highly productive, or highly restricted? What restrictions do you find, if any? Can you find evidence for the presence (or absence) of subjects in these instances?

Identifying Puzzles 4.1: Licensing expletives A

The sentences in (1) apparently violate the licensing condition on expletive *there*, yet they are still acceptable (though quite formal).

(1) a. At the end of the intermission, there sounded a silvery bell.
    b. Then the curtain rose, and there waltzed onto the stage a strangely, but exquisitely dressed apparition.

Explain what the challenge for the licensing condition is here.

Identifying Puzzles 4.2: Licensing expletives B

Given the licensing condition on expletive *there*, repeated in (1), determine whether each of the grammaticality judgments in (2) is expected. The brackets indicate clause boundaries (of *to* infinitive clauses or small clauses).

(1) For expletive *there* to be grammatical as the subject of a clause, the (Fregean) predicate of that same clause must be a verb of existence or coming into existence.

(2) a. ✓ Feynman suspected [there to be a problem with the O-ring].
    b. * Feynman suspected [there a problem with the O-ring].
    c. ✓ There was suspected [to be a problem with the O-ring].
    d. * There was suspected [a problem with the O-ring].

Identifying Puzzles 4.3: Nonfinite complement clauses

Nonfinite clauses like the bracketed sequence in (1) are prima facie counterexamples to the subject requirement.

(1) I promised [to come on time].

Provide as much evidence as you can for the existence of a silent subject in (1) and nonfinite clauses like it.

Introducing the X′ schema of phrase structure

5

Chapter outline

As was mentioned in Chapter 1 we can represent the individual vocabulary items of a language as small pieces of syntactic structure, or **elementary trees**. The idea is to **generate** phrases and sentences by composing (and possibly otherwise manipulating) these elementary trees in well-defined ways. In this view, vocabulary items are comparable to the atoms of physical matter. Atoms do not combine into molecules just because they happen to be next to each other; rather, their combinatorial possibilities are governed by their internal structure (for instance, the number of electrons on an atom's outermost shell and the relative number of protons and electrons).

Accordingly, in the first part of this chapter, we consider the internal structure of elementary trees. As in the last chapter, we begin by focusing on how verbs combine with their arguments to form larger phrases. For the time being, we will treat noun phrases and prepositional phrases as unanalyzed chunks, postponing discussion of their internal structure until Chapter 6.

We then generalize the approach developed for verbs and their arguments to the point where we can build sentences—both simple ones and complex ones containing subordinate clauses. In order to derive sentences, we will find it necessary to introduce a formal operation called **movement**, which allows us to represent the fact that constituents can have more than one function in a sentence.

In the second part of the chapter, we turn to the representation of the modification relation (already familiar from Chapter 4, § 4.2). As we will show, it is not possible to combine modifiers with elementary trees by the substitution operation introduced in Chapter 1. Besides substitution and movement, we therefore introduce a third and final formal operation called **adjunction**.

This chapter repeatedly references structural relationships in trees. If the jargon becomes difficult to follow, you may find yourself wanting to revisit Chapter 3, § 3.3, where the terms are introduced.

The X′ schema for elementary trees 5.1

5.1.1 Transitive elementary trees

We begin our investigation of the internal structure of elementary trees by considering how a transitive verb like *eat* and future tense *will* combine with the two arguments of *eat* in a sentence like (1).

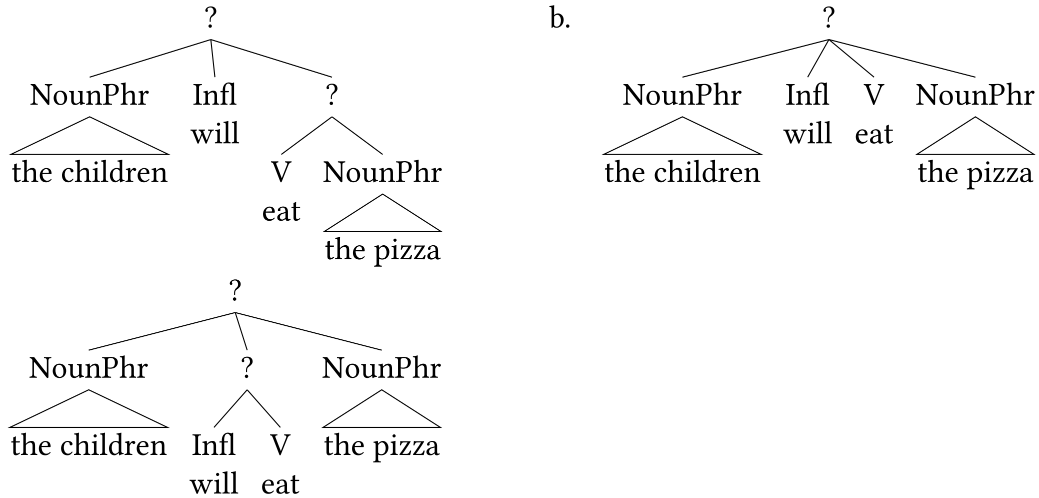
(1) The children will eat the pizza.

From the possibility of pronoun substitution, as in (2), we know that the two arguments are constituents (specifically, noun phrases).

(2) They will eat it.

We have previously established that English tense and modals are part of the same grammatical category, called inflection (I°). In principle, the verb could combine with its two noun phrase arguments and tense in either order, or all at once. Three such possibilities are represented by the structures in (3) (we address the question of which syntactic category to assign to the nodes labeled by question marks in a moment).

(3) a. b. c.



However, as we already know from a variety of diagnostics introduced in Chapter 3, § 3.3, transitive verbs form a constituent with their object. We illustrate this with the substitution diagnostic shown in (4).

(4) The children will eat the pizza. ⟶ ✓ The children will do so.

In other words, only (3a) is an adequate representation of the sentence.

What is the syntactic category of the constituents that result from these combinations? In principle, the result of combining a verb with a noun phrase might be a phrase with either verbal or nominal properties. But clearly, a phrase like *eat the pizza* doesn't have the distribution of a noun phrase. For instance, it can't function as the object of a preposition (even a semantically vacuous one like *of*). Nor does it pattern like a noun phrase in other respects. As we have just seen, the appropriate pro-form for it is not a pronoun, but a form of *do so*, just as would be the case if the predicate of the sentence were an intransitive verb. In other words, for the purposes of *do so* substitution, the combination of a verb and its object is equivalent to an intransitive verb (say, intransitive *eat*); cf. (4) with (5).

(5) The children will eat. ⟶ ✓ The children will do so.

So how about if we assign the syntactic category V to the verb-object combination? In that case, we would be saying that *do so* can substitute for V. But that won't do, because then *do so* should be able to substitute for *eat* regardless of the presence of an object. But that isn't the case, as shown in (6), which contrasts with (5).

(6) The children will eat the pizza. ⟶ * The children will do so the pizza.

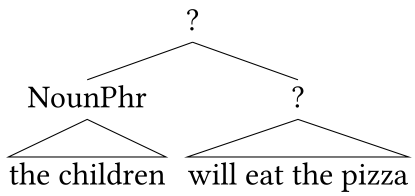
Our conclusion from above that these are phrases, characterized as verbal phrases of some sort, seems most natural. Therefore we will call this a verb phrase (VP).

We can see that future tense (or modals) are outside of this verb phrase by virtue of the constituency diagnostics isolating the verb phrase. But are subjects and tense at equivalent levels of structure outside of VP? Evidence like (7) suggests that this is not so: because *will eat* *pizza* and *will love it* are conjoined, it suggests that these are constituents to the exclusion of the subject.

(7) The children [will eat the pizza] and [will love it].

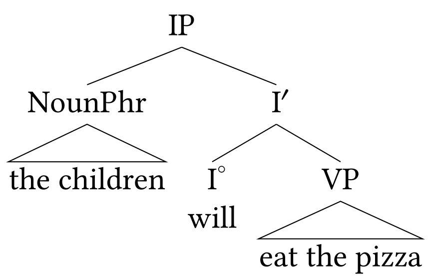
This suggests that a structure like (8) exists, where tense forms a constituent with the verb phrase, not including the subject.

(8)



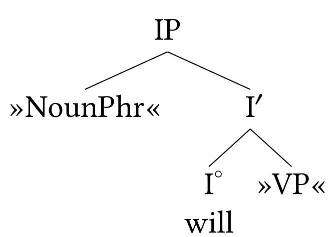
In order to represent the facts in (4)–(7), the following notation has been developed. Heads like Inflection (I) are said to **project** three **levels**. Therefore, various higher nodes connected to inflection (I) are said to be **projections** of I. As we will see, a central assumption is that characteristics of phrases are defined by the lexical item that projects. The lowest level, I°, is a syntactic category for vocabulary items; it is often indicated simply by I without a superscript, or as I°. These are referred to as syntactic **heads**. The next level is I′ (read as “I-bar”),[1] which contains an I head (I°) and its sibling (VP). The highest level is the maximal projection (the last level of the inflection phrase), which is marked as IP.[2] The fully labeled structure for (1), with the standard labels for the three verbal projections, is given in (9).

(9)



Given (9), we can “un-substitute” the two arguments, marking these instead as substitution nodes. This yields the elementary tree for future tense *will* in English in (10).

(10)

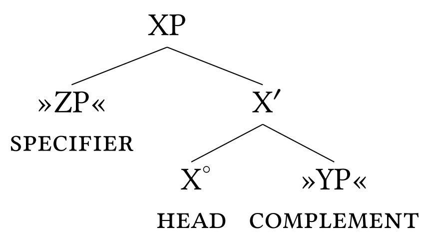


What this communicates is that the inflection head (I°) will require a noun phrase under its maximal projection, and will require a verb phrase as a sibling. And it is conclusions like this that led to a more abstract and general formalization of what phrase structures look like, which we introduce in the next section.

5.1.2 The X′ schema

As we show in § 5.2 later on in this chapter and in Chapter 6, the basic form of the elementary tree in (10) can be extended to other syntactic categories. In other words, (10) is an instantiation of a general phrase structure template, shown in (11) and known as the **X′ schema** (read: X-bar schema) of phrase structure, also known as X′-syntax (read: X-bar syntax). Here, X, Y, and Z are variables over syntactic categories. X′-syntax is stated abstractly like this so that any kind of syntactic category can have a structure like in (11).

(11)



A number of standard terms are used in connection with the X′ schema. The lexical item in X°—the element that projects the entire syntactic structure—is known, following traditional terminology, as the **head**. X° (sometimes just written X) is the head's **lexical projection**, X′ the **intermediate projection**, and XP is the **maximal projection** (sometimes also called **phrasal projection**).

Word of Caution

The terms "intermediate" and "phrasal" are somewhat misleading, since they suggest that the syntactic status of intermediate projections is somehow intermediate between lexical and phrasal constituents. This is not the case. Intermediate projections are full-fledged phrases, and "intermediate" simply refers to the position of the projection in the tree structure—that is to say, the fact that it is not the maximal projection of that XP.

The three projections of the head form the minimal structure that can make up an elementary tree. Crucially, these projections/levels are all necessary for there to be a well-formed X′-syntax structure. So a structure without an intermediate projection, and a maximal projection is not a well-formed X′-syntax structure.

Label	Projection
X°	Lexical
X′	Intermediate
XP	Maximal (phrasal)

Table 5.1: Terminology and labels for the X′-schema

Again following traditional terminology, the sibling of the head—YP in (11)—is called its **complement**. As we discuss in the next subsection, elementary trees need not include a complement position. The child of the maximal projection that is not the bar-level node—ZP in (11)—is called the **specifier**. Each elementary tree has at most one specifier, and elementary trees can lack a specifier altogether, as we will see below in § 5.2.3 in this chapter. The specifier and complement positions of a head are its (syntactic) **argument** positions. In summary, an elementary tree consists of a spine (X°-X′-XP) and from zero to two substitution nodes, which are potential positions for arguments.

Word of Caution

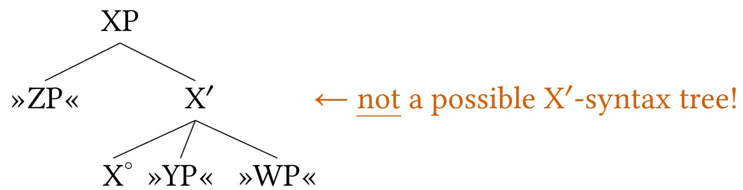
Note the spelling of *complement* with *e* (not *i*). The idea is that complements complete the meaning of the head.

Word of Caution

The term "specifier" suggests that constituents in that position somehow specify the remainder of the tree, and this might lead you to confuse specifiers with modifiers (discussed later on in this chapter). At one point, constituents in specifier position were indeed thought to have this function, and that is how the name arose. We no longer believe this, but the name has stuck (what we might call terminological inertia). In order to minimize this confusion, syntacticians often use the abbreviation "spec" (pronounced "speck" and sometimes spelled with an initial capital).

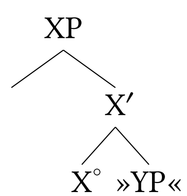
Before ending this section we want to clarify some basic assumptions about what form X′-syntax structures can take. First, it is widely assumed that syntactic structure is at most **binary-branching**, meaning that a node can have at most two children. That is to say, X′-syntax disallows structures like (12).

(12) Unacceptable ternary branching

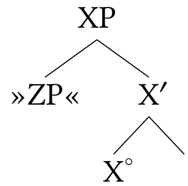


Second, while it is possible for phrases to be in complement and specifier positions, it is not strictly necessary. So it is possible for those positions to be empty. (13) shows a structure with no phrase in the specifier position, and (14) shows a structure with no phrase in complement position.

(13) X′-syntax tree without a phrase in specifier position



(14) X′-syntax tree without a phrase in complement position



We will see examples of both of these in what follows.

Take Note

The terms "specifier", "complement", and "argument" can be used to refer to the structural positions just defined or to the constituents that substitute into those positions. (This is analogous to the way we can use a nontechnical term like "bowl" to refer either to the container itself (*Hand me the bowl*) or to its contents (*I'd like a bowl (of soup)*). If it is necessary to avoid confusion between the two senses, we can distinguish between "specifier position" and "constituent in specifier position" (and analogously for "complement" and "argument").

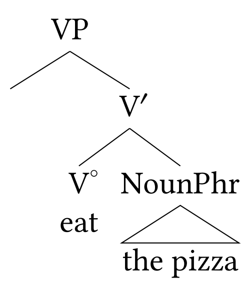
Deriving sentences 5.2

In this section we walk through the structures that are necessary to construct basic sentences in English (and many other languages). We start with verb phrases, then discuss inflection, and then discuss complementizers.

5.2.1 Verb phrases, transitive and intransitive

The X′-schema, on the approach developed here, is the structure of all phrases in human language, as specified by Universal Grammar. This means that all phrases, not just IPs, will follow the X′-schema generally. Take the verb phrase from our tree in (9) above (*eat the pizza*), which we expect to have the structure in (15), where the THEME object of the verb is represented as the complement of the verb.

(15)

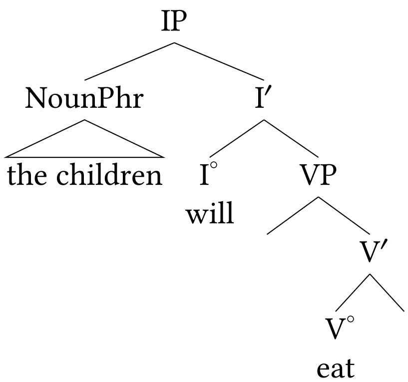


What is the structure of the verb phrase when there is no object? The verb *eat* gives us the opportunity to look at this, as it's possible in English to leave objects of *eat* unspoken.

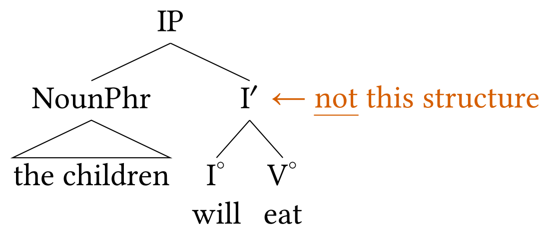
(16) The children will eat.

What is the structure of a sentence like this? In principle we could imagine a structure like (17a) or a structure like (17b) that just includes the lexical items *will* and *eat*.

(17) a.



b.



While (17b) is obviously simpler in some ways (because it contains fewer projections), it is not in fact a well-formed syntactic structure in the X′ schema because the verb does not project an intermediate projection and a maximal projection. So instead, the correct structure here is (17a), because it conforms to our formal constraints for phrase structure within the X′-syntax framework.

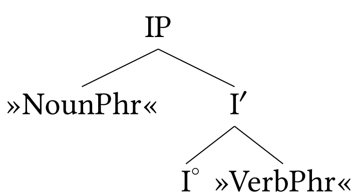
5.2.2 Simple sentences with Inflection

The sentences above place future tense in the head of an IP, the category where inflection is introduced. As we concluded previously, there are a range of subcategories of I° (modals, emphatic *do*) which are all compatible with structures like this.

(18) a. The children will eat the pizza.
b. The children might eat the pizza.
c. The children could eat the pizza.
d. The children DID eat the pizza.

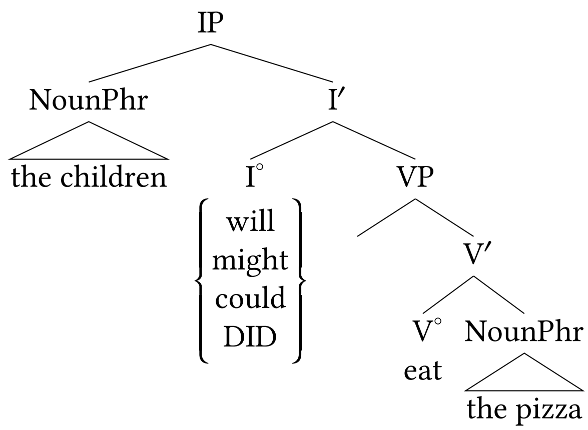
As we might expect at this point, our elementary tree for inflection from above (and repeated here) is applicable.

(19)



We therefore expect a full IP structure for these:

(20)



A point that is likely not lost on English speakers is that past tense and present tense behave quite differently, in ways that are not obvious to explain based on our discussion thus far. Focusing on past tense for simplicity: in many past tense sentences, the past tense is inflected directly on the verb.

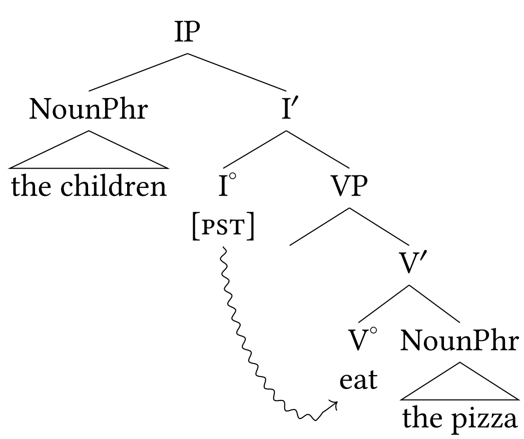
(21) The children ate the pizza.

How does the past tense come to be inflected on the verb? First we want to note that in certain past tense constructions, the tense appears just where we would expect it to. Emphatic *do* in (22a) and negated sentences in (22b) are two such examples.

(22) a. The children DID eat the pizza!
  b. The children did not eat the pizza.

These give us reason to think that past tense sentences in English are not fundamentally different in some way: in some instances, the tense nonetheless appears just where we would expect based on our conclusions about IP thus far. So what happens in examples like (21)? For now, we'll just assume that past tense lowers in the structure to attach to the verb. We draw this lowering with a squiggly arrow to distinguish it from other arrows we will be using.

(23)



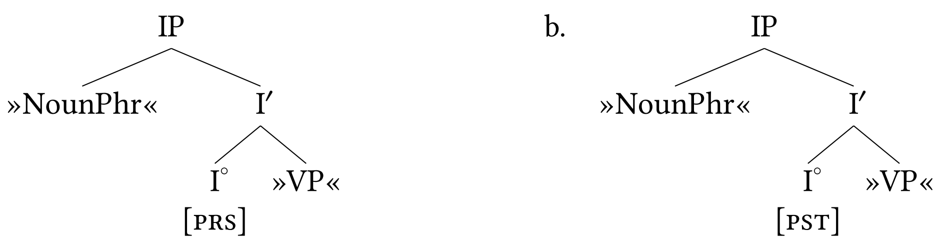
Extra Info

In this book, we give the syntactic category for modals, auxiliary *do*, and the silent tense morphemes that we introduce presently the name "I", short for "inflection". (The auxiliary verbs *be* and *have* are introduced in connection with the passive in Chapter 10.)

Nowadays, the usual name for this category is "T", short for "tense". Both names designate exactly the same category. We prefer I (or Infl), because the category includes modals, which do not productively express tense in English, and also includes agreement (conjugation to match the subject). The reason for the other name is historical. The abbreviation I was current at one point, but then there were thought to be reasons to "split" it into two separate heads: T(ense) and Agr(eement). This split was later abandoned, and Agr(eement) became obsolete. Although there was no reason not to bring back the old name I, the new label T stuck. Another instance of terminological inertia!

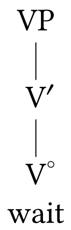
We are now in a position to answer the second question posed earlier—namely, how can sentences be represented in a syntactically uniform way regardless of the morphological expression of tense? A simple answer to this question is possible if we assume that English has tense elements that are structurally analogous to auxiliary *do* and modals, but not pronounced independently, as shown in (24). In this book, as is conventional in linguistics we adopt the convention of enclosing such silent elements in square brackets. As in phonology and morphology, we call such designations **features**: properties of a head that are linguistically-real, whether or not they are directly pronounced.

(24) a.



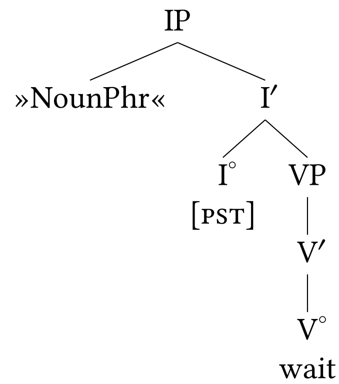
Elementary trees like (24) make it possible to derive structures for sentences in which tense is expressed as a bound morpheme along the same lines as for sentences containing a modal or auxiliary *do*. In (25), we illustrate the derivation of *They waited*.

(25) a.



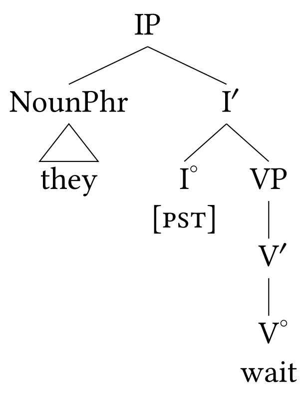
Select elementary tree for verb

b.



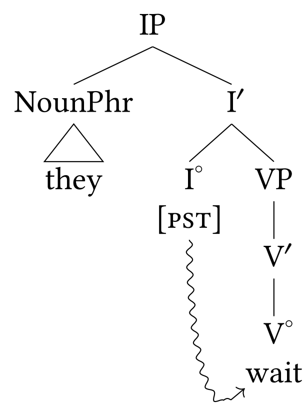
Substitute structure in (25a) in elementary tree for inflection

c.



Substitute subject argument

d.



Lower tense onto verb

As a final note, notice in (25a) that we have shown yet another way to draw the simplest sort of X′-syntax trees: here the binary branches are not drawn because there are no phrases in those positions. This is equivalent to drawing empty binary branches.

5.2.3 Complex sentences and complementizers

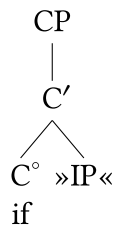
This section is devoted to the derivation of sentences that contain complement clauses (also known as **clausal complements**). Some examples are given in (26); the complement clauses are in italics.

(26) a. We will ask *if they might leave.*

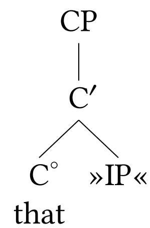
b. We believe *that they might come.*

Although sentences with complement clauses can become unboundedly long (recall the instances of recursion in Chapter 1), deriving structures for them proceeds straightforwardly along the lines already laid out. *If* and *that* are both **complementizers**, so called because they have the effect of turning independent sentences into the complements of a matrix verb. Complementizers are a grammatical category (as we introduced in Chapter 2), and as their own lexical heads, per X′-syntax they project full X′-syntax structures, as shown in the elementary trees in (27).

(27) a.

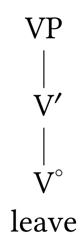


b.

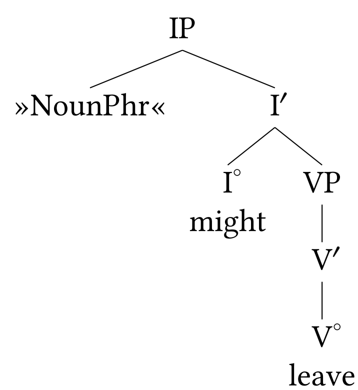


Given elementary trees like (27), we can derive the italicized complement clause in (26a) as in (28).

(28) a.



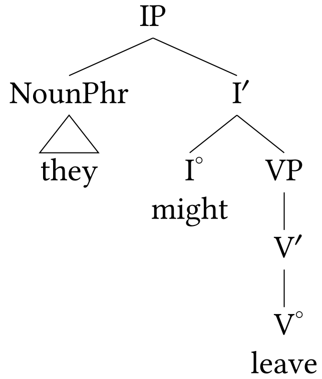
b.



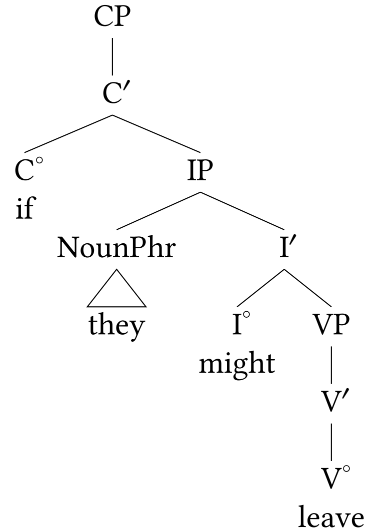
Select elementary tree for verb

Substitute structure in (28a) in elementary tree for inflection

c.



d.



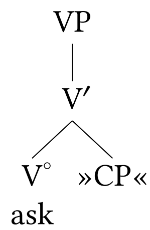
Substitute argument

Add complementizer layer

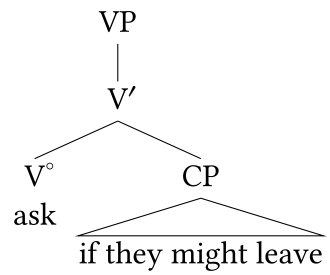
The structure in (28d) in turn allows us to derive the entire matrix clause, as in (29). For readability, we collapse the internal structure of the complement clause in what follows.

(29)

a.



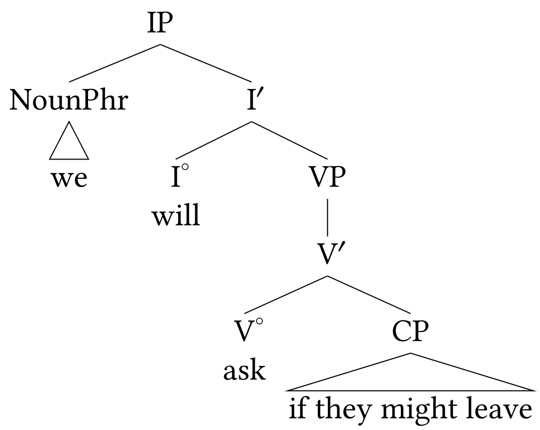
b.



Elementary tree for matrix clause verb

Substitute clausal complement (28d)

c.



Substitute (29b) and subject noun phrase into elementary tree for English *will*

Given representations like (29c), we can now formally characterize **recursive structures** as in (30).

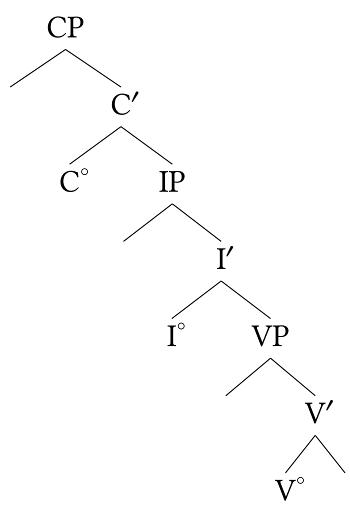
(30) a. A structure is **recursive** if and only if it contains at least one recursive node.
b. A node is recursive if and only if it dominates a node distinct from it, but with the same label.

The VP node in (29c) is recursive because it dominates another VP node below it (inside the embedded clause). Some recursive nodes are indirectly recursive as in this case: here, a VP node is contained within the main clause VP but the main clause node does not directly dominate the embedded VP. Below we will see an instance of directly recursive nodes as well (for adjunction).

5.2.4 Expected structures of sentences

There is much more to be said even about the structure of basic sentences, let alone more detailed aspects of specific more complicated constructions. But with VP, IP, and CP, we have the structure of most basic clauses.

(31)



This is what is often called the main "spine" of the clause; various constituents will be added to these structures, but trees will often display a main skeleton like this that is the backbone of the clause. As we will see, different kinds of syntactic constructions will use different subparts of this structure, and will also add different projections. But as a baseline expectation, this will be the typical structure of full sentences, which we will motivate more in what follows.

5.3 The adjunct relation

5.3.1 Modification is different

The elementary trees introduced so far allow us to represent two of the three basic linguistic relations discussed in Chapter 4—namely, argumenthood and predication. As we have seen, objects of a verb are represented by substituting the complement argument positions in the verb's elementary tree. VPs and IPs can be treated as arguments (specifically, as complements) of I° and C°, respectively: VP is typically a complement of I°, and IP is typically a complement of C°. Subjects in our model so far appear in Spec,IP. But recall from Chapter 4 that there is another major fundamental linguistic relation: modification. While arguments are the significant, semantically required (and often syntactically required) participants of an event, many sentences include additional information in the form of adjectives, adverbs, prepositional phrases, and the like. Modifiers like these are not required by a predicate and are usually optional in a sentence, but they nonetheless have to be accounted for in our theory.

Therefore, an important remaining question is how to represent the modification relation using the X′ schema developed so far. Neither of the two head-argument relations (head-specifier, head-complement) adequately represents the relation between a head and its modifier. As we have seen, when a verb combines with a complement, the category of the resulting constituent

(V′) is distinct from that of the verb (V°) (recall the contrast between (4) and (6)). By contrast, modifying a verb-complement combination like *eat* in (32) does not change the syntactic category of the resulting constituent, which remains V′ (the modifier is in italics).

(32) a. The children will eat the pizza.
  b. The children will eat the pizza *with gusto*.

This is evident from the *do so* substitution facts in (33), where *do so* can replace either the unmodified or the modified verb-complement combination.

(33) a. The children will eat the pizza with gusto. ⟶ ✓ The children will do so with gusto.
  b. The children will eat the pizza with gusto. ⟶ ✓ The children will do so.

The same pattern holds for intransitive verbs that combine with a modifier.

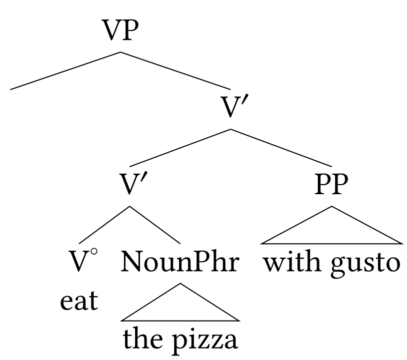
(34) a. The children will eat with gusto. ⟶ ✓ The children will do so with gusto.
  b. The children will eat with gusto. ⟶ ✓ The children will do so.

Let us assume a definition of *do so* substitution that says that *do so* substitutes for V′ constituents (as articulated in (35)).

(35) *Do so* substitutes for instances of V′.

The *do so* substitution facts in (33) and (34) motivate the syntactic structure for (32b) that is given in (36) (for clarity, we focus on the internal structure of the VP).

(36)

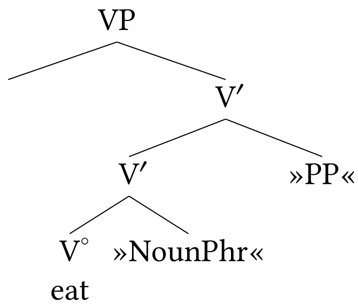


Because *do so* can substitute for a verbal constituent without a complement (*eat*), with a complement (*eat the pizza*), and with a complement and a modifier (*eat the pizza with gusto*), it is reason to think that all of these phrases are the same (structural) sort of phrase: here we represent them as instances of intermediate projections of V (V′). The structural relation of the modifier *with gusto* to the spine of the V projection is known as the **adjunct** relation, and the modifier itself is said to be an **adjunct**. Modifiers are always represented as adjuncts. As a result, "modifier" and "adjunct" tend to be used somewhat interchangeably.[3]

5.3.2 The need for an adjunction operation

The structure in (36) raises the question of what elementary tree for transitive *eat* is involved in its derivation. "Unsubstituting" both the object and the modifier, as we did in (10) for trees containing only arguments, yields the structure in (37).

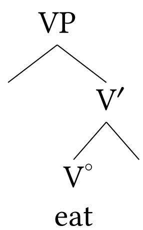
(37)



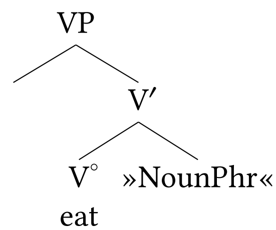
← not a possible elementary tree!

Is the structure in (37) a satisfactory elementary tree? Clearly, allowing it means that our grammar now contains two elementary trees for transitive *eat* because there must exist an elementary tree without the modifier as well (since the modifier is optional). At first glance, this doesn't seem like a serious problem, since we already allow two elementary trees for *eat*, the intransitive and transitive ones in (38).

(38) a.



b.



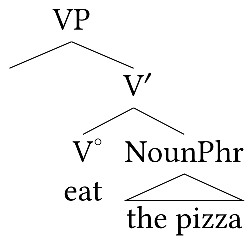
But (37) differs in a crucial respect from the structures in (38): it is a recursive structure: a V′ node contains another V′ node. This has an extremely undesirable consequence: namely, that if we were to derive structures like (36) by means of elementary trees like those in (37), there would be no principled way to avoid an unlimited number of such elementary trees as part of English speaker's knowledge of their language. For instance, the derivations of the sentences in (39), with their increasing number of modifiers, would each require a distinct elementary tree for *drink*, and each additional modifier would require an additional elementary tree.

(39)
a. We would drink lemonade.
b. We would drink lemonade / in summer.
c. We would drink lemonade / in summer / on the porch.
d. We would drink lemonade / in summer / on the porch / with friends.
e. We would drink lemonade / in summer / on the porch / with friends / for fun.

But the whole point of a generative grammar is to generate an unbounded set of sentences from a finite set of elementary expressions and operations. Given this aim, elementary trees must be non-recursive structures, with the consequence that adjuncts cannot be integrated into larger syntactic structures by substitution. Accordingly, we introduce a further tree operation called **adjunction**.[4]

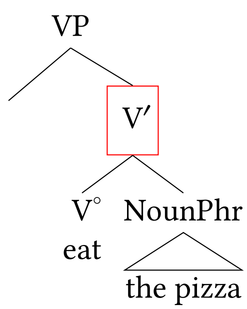
For the moment, we will use the adjunction operation to integrate modifiers into syntactic structures. As we will see in Chapter 8, the adjunction operation is also used for other purposes. Whatever its linguistic purpose, however, it is always the same formal operation. We will show how this works with our sentence in (40a). First we substitute the object argument into the elementary tree for eat from (38b).

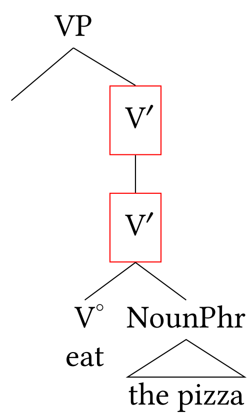
(40) a. The children will eat the pizza with gusto.

b. 

Substitute argument

We can think of adjunction as a two-step process that targets a particular node. When the purpose of adjunction is to integrate a modifier into a larger structure, as it is here, the target of adjunction is an intermediate projection, indicated by the red box in (41a). The first step in carrying out adjunction is to make a clone of the target of adjunction that immediately dominates the original node, as in (41b). The second step is to attach the tree for the modifier as a child of this higher clone, as in (41c).

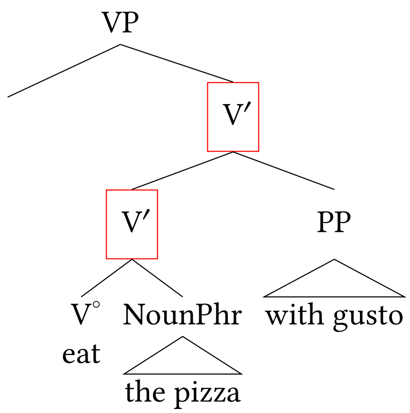
(41) a. 

b. 

Select target of adjunction

Clone target of adjunction

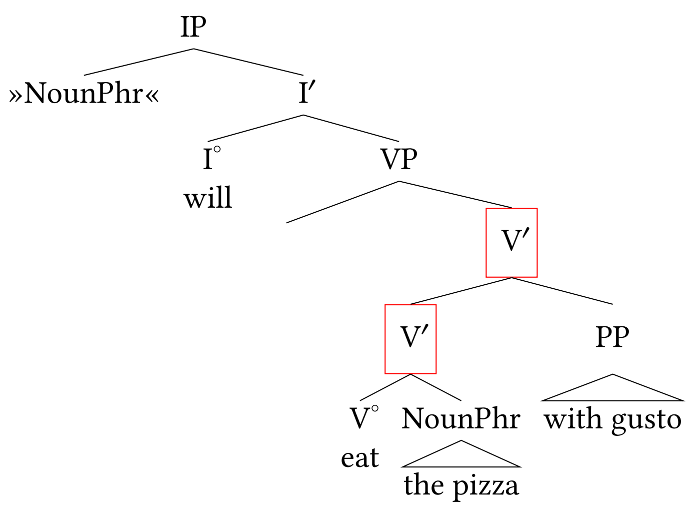
c.



Attach modifier as child of higher clone

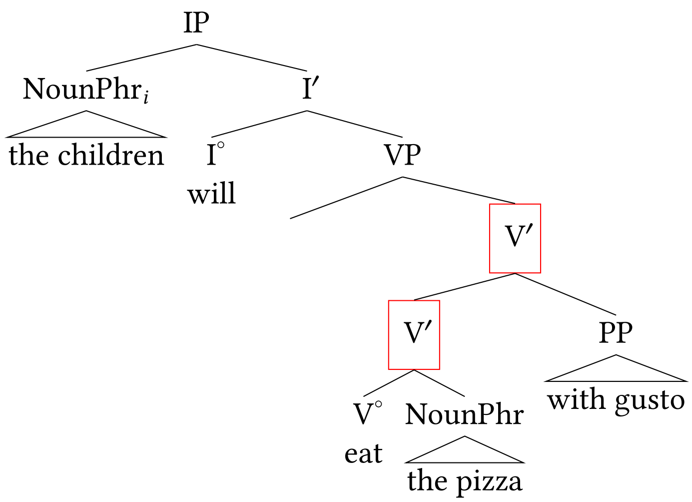
Deriving the rest of the structure for the entire sentence proceeds as outlined in § 5.2.2 earlier, as shown in (42).

(42) a.



Substitute (40b) in elementary tree for tense

b.



Substitute subject into Spec,IP

Distinguishing syntactic dependents 5.4

5.4.1 A typology of syntactic dependents

A convenient cover term for complements, specifiers, and adjuncts is **syntactic dependent**. Each type of syntactic dependent stands in a unique structural relation to the head and the spine that it projects. Complements and adjuncts are both children of intermediate projections, but they differ in that complements are siblings of heads, whereas adjuncts are siblings of intermediate projections. As siblings of intermediate projections, adjuncts resemble specifiers. But adjuncts are children of intermediate projections, whereas a specifier is the child of a maximal projection. These structural relations and distinctions are summarized in Table 5.2, which also includes the formal operations that fill or create the positions in question. The relevance here is that complements and specifiers can realize substitution nodes in elementary trees—that is to say, they can be **selected** by heads (see Chapter 7). Adjuncts, on the other hand, are not selected by heads, and are instead optional additions to structure.

Relation to head	Sibling of...	Child of...	Formal operation
Complement	Head	Intermediate projection	Substitution
Adjunct	Intermediate projection	Intermediate projection	Adjunction
Specifier	Intermediate projection	Maximal projection	Substitution

Table 5.2: Summary of structural relations and formal operations

5.4.2 Diagnosing complements and adjuncts

Given Table 5.2, it is relatively straightforward to tell whether a constituent is represented in a particular tree diagram as a complement or as an adjunct. However, it is not always self-evident whether a phrase is a complement or an adjunct as a matter of linguistic fact (that is, in speakers' mental representations of their language's grammar).

Word of Caution

Remember that tree structures are models of linguistic facts, that can be correct or incorrect. Just because it is possible to build a tree that represents a certain phrase as a complement doesn't mean that the phrase actually is a complement in the mental grammar of speakers. So learning to draw trees correctly according to our theory is just the first step towards *using* that theory and its trees to accurately model linguistic knowledge.

Distinguishing verbal complements and adjuncts with *do so*

A fairly reliable way to determine the relation of a particular phrase to a verb in English is to use *do so* substitution. If a phrase need not be included as part of the sequence being replaced by *do so*, then it is an adjunct. If it must be included, then it is a complement. Using this test, we find that phrases specifying cause or rationale, time, location, or manner are generally adjuncts, even if they are noun phrases. Some examples, including the results of *do so* substitution, are given in (43); the adjuncts are in italics.

(43) a. Rationale:
They will wait *for no good reason*, but we will do so *for a very good one.*

b. Duration:
They will wait *(for) a day*, but we will do so *(for) a month.*

c. Location:
They will wait for her *in the parking lot*, but we will do so *across the street.*

d. Manner:
They will wait for her *patiently*, but we will do so *impatiently.*

In the examples that we have seen in this book so far, semantic arguments are expressed as syntactic arguments (or not at all). It is possible, however, for semantic arguments to be expressed in the syntax as adjuncts. For example, the predicate *rent* (from a semantic point of view) is a five-place predicate, with arguments denoting property owner, tenant, rental property, amount of money, and lease term. Some of these semantic arguments are expressed as syntactic arguments. For instance, in (44), the phrase denoting the rental property is a complement, as is evident from the results of *do so* substitution.

(44) Dennis rented the apartment to Lois. ⟶ * Dennis did so the apartment to Lois.

By contrast, *do so* substitution shows that the phrase denoting the lease term is an adjunct, even though lease terms are arguably semantic arguments of *rent* on a par with rental properties.

(45) Dennis rented Lois the apartment for two months. ⟶ ✓ Dennis did so for two months.

Locality as a diagnostic

The nature of complements and adjuncts in the X′-schema requires that complements have a more local relationship with heads than adjuncts do. But the X′-schema says nothing about whether adjuncts (to the same phrase) are ordered with respect to each other. (46) show that two adjuncts to the verb phrase can appear in either order:

(46) a. We would drink **lemonade** in summer on the porch.
b. We would drink **lemonade** on the porch in summer.

(47) a. * We would drink in summer **lemonade** on the porch.
b. * We would drink on the porch **lemonade** in summer.
c. * We would drink in summer on the porch **lemonade**.
d. * We would drink on the porch in summer **lemonade**.

It is facts like this that lead to the conclusion that semantic arguments of predicates are complements (or specifiers, in the case of subjects), and adjuncts are modifiers that have distinct properties. Here, the THEME object of *drink* (*lemonade*) must be closer to the verb than adjuncts are: complements are siblings of heads and adjuncts are siblings of bar level nodes.

Obligatoriness as a diagnostic

A final word should be said about the correlation between a syntactic dependent's obligatory or optional character and its status as a complement or adjunct. It is tempting to assume the biconditional relationship in the "Dream world" column in Table 5.3.[5]

If a syntactic dependent is...	then it is...	Dream world	Real world
obligatory	a complement.	TRUE	TRUE
a complement	obligatory.	TRUE	sometimes FALSE

Table 5.3: Correlation between obligatoriness and complement status

But as the rightmost column indicates, the biconditional relationship doesn't hold. It is true that obligatory syntactic dependents are complements. For instance, the contrast in (48) is evidence that the noun phrase following *devour* is a complement, a conclusion that is borne out by *do so* substitution in (49).

(48) Every time I see him...
a. * ...he's devouring.
b. ✓ ...he's devouring a six-inch steak.

(49) He devoured a hamburger and french fries, ...
a. * ...and she did so six samosas.

b. ✓ …and she did so, too.

But not all complements are obligatory. The grammaticality of both (50a) and (50b) shows that the phrase *French fries* in (50b) is optional. But the ungrammaticality of (50c) shows that it is nevertheless a complement.

(50) a. ✓ She ate, and he did so, too.
b. ✓ She ate French fries, and he did so, too.
c. * She ate French fries, and he did so three samosas.

Although the second row of Table 5.3 is false, the first row does have the valid consequence in (51). We exclude mention of specifiers here, as this discussion is specifically addressing the distinction between complements and adjuncts.

(51) If a non-specifier syntactic dependent is not a complement, it is not obligatory.

The two valid generalizations in the first row of Table 5.3 and (51) can be summarized succinctly as in (52); these are useful for distinguishing complements and adjuncts.

(52) a. Obligatory syntactic dependents are complements.
b. Adjuncts are optional.

Therefore, optionality and obligatoriness are both useful properties to help us diagnose complements and adjuncts. But there is not a one-to-one relationship between the structures and the empirical patterns, so structures must be analyzed with precision, and utilizing multiple diagnostics when they are available.

New Empirical Observations

- Verb phrases form a constituent distinct from inflection.
- Complements are different from adjuncts.
  - Complements are selected by heads, and are positioned close to heads.
  - Adjuncts are not selected by heads, and are outside of the head-complement constituent.

Components of the Theoretical Model

Stable concepts:

- All nodes in a tree are binary-branching (some branches may be empty)
- X′-syntax
- heads
- complements
- specifiers

- adjuncts
- adjunction (operation)
- new phrase types, and the sentential "spine":
  - VP
  - IP
  - CP

Concepts to be developed in more detail later:
- tense-lowering

Notes 5.5

1. Why is X′ read as X-bar when it contains not a bar ($\overline{\mathrm{X}}$), but a prime symbol? The reason is that when the idea of bar levels was introduced in the 1970s, the various levels were distinguished by horizontal bars over a syntactic category. The lowest level had no bars, the first level one, and the second two. But back in the days of typewriters, such overbars were cumbersome to type (you typed the symbol –*, rolled up the platen a bit, backspaced, typed an overbar *–, repeated from –* to *– for each overbar, and then rolled the platen down again the right amount). Overbars are also expensive to typeset, and even today, they aren't part of many standard character sets. Therefore, it was and continues to be convenient to substitute prime symbols for overbars. However, linguists haven't updated their terminology (terminological inertia!), and so the old term "bar" is still with us.

2. At earlier stages of the theory development the highest bar level is I″ (read as "V double bar"), which is the result of combining a I′ with a subject. Likewise, the lowest level was the first "bar-level". For a long time, now, the tradition has been to only refer to the intermediate level as a bar-level, referring to the lowest as the head, and the highest level as the maximal projection, or the phrase-level.

3. In this book, we will usually use *modifier*/*modification* to refer to the semantic role of modifiers, and will generally use *adjunct* to refer to the syntactic relationship. But in most instances that we encounter in this book, these terms will consistently refer to the same entities. For instance, as we mentioned in Chapter 4, a verb like *laugh* denotes the set of entities that laugh. Combining the verb with a modifier like *uproariously* yields the expression *laugh uproariously*, which denotes a subset of the set denoted by *laugh*. We will use the term 'adjunct' when focusing on a constituent's structural position in a tree. As we will see later on in this chapter, it is possible for semantic arguments to be represented as syntactic adjuncts. This does not change the semantic argument into a modifier, however!

4. The specific operation of interest to us is sometimes called Chomsky-adjunction, to distinguish it from Joshi-adjunction, a different formal operation that plays a central role in Tree-Adjoining Grammar (Joshi, Levy, and Takahashi 1975).

5. A similarly tempting biconditional relationship (and false there, too) was discussed in § 3.2.2 of Chapter 3.

5.6 Exercises and problems

Exercise 5.1: Trees with X′-syntax A

Draw trees for the sentences below using X′-syntax. Draw triangles over all noun phrases (NounPhr or NP) and prepositional phrases (PrepPhr or PP) and adverb phrases (AdvP), since we have not gotten to the internal structure of those phrases yet.

1. The very clever student will study every night in the library.
2. Tired students will always take naps before class.
3. Khydence found a book about Maya Angelou in the park.
4. Babies run rather slowly.
5. Lucca's incessant talking always annoys Maisha.

Exercise 5.2: Trees with X′-syntax B

Draw trees for the sentences below using X′-syntax. Draw triangles over all noun phrases (NounPhr or NP) and prepositional phrases (PrepPhr or PP) and adverb phrases (AdvP), since we have not gotten to the internal structure of those phrases yet.

1. Langston wants Maisha to come to school today. (assume *to* is Infl)
2. Preston said that Maisha will eat a hamburger for dinner.
3. That Maisha worked for the whole night might intimidate Preston.
4. Anabella thinks that Lucca should thoroughly clean the kitchen.
5. Preston suggested that Victor should write a joke on the blackboard.
6. Many culinary students believe that their exams should be eaten.
7. The baker kneaded the dough with impressive force.

Exercise 5.3: Trees with X′-syntax C

Draw trees for the sentences below using X′-syntax. Draw triangles over all noun phrases (NounPhr, or NP) and prepositional phrases (PrepPhr, or PP) and adverb phrases (AdvP), since we have not gotten to the internal structure of those phrases yet.

1. The dog jumped over the ditch.
2. An aspiring doctor should study hard every day.
3. That the faucet is dripping might upset your father.
4. It might anger your mother that you turned up the thermostat.
5. The chef carefully stirred the sauce.

6. Every dog knows the location of their treats.
7. Lucca might never play his music in public.
8. The students might play games at recess today.
9. The latest article in the New York Times explains the current international crisis.

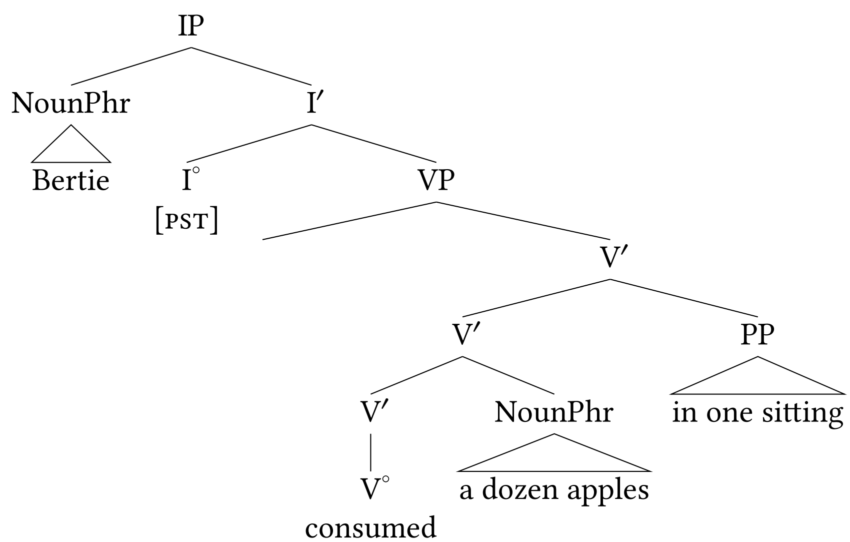
Exercise 5.4: Fregean and of Aristotelian predicates

What is the X′ status of Fregean and of Aristotelian predicates? That is, what kind of node in an X′-syntax tree corresponds to a Fregean predicate, and what kind of node corresponds to an Aristotelian predicate? You should be able to answer in one or two brief sentences.

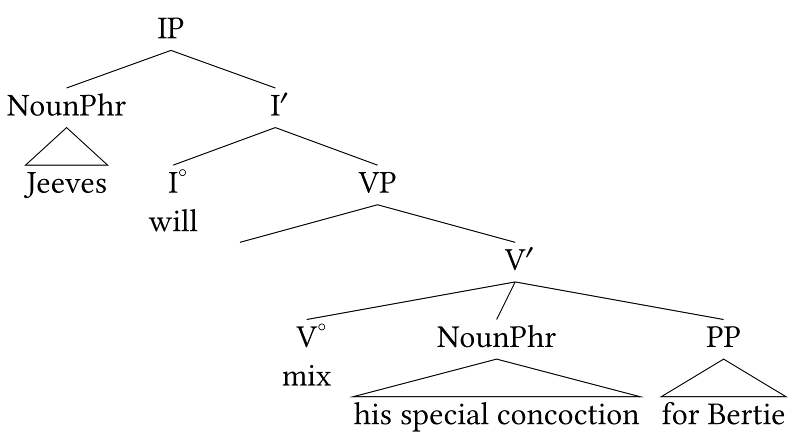
Exercise 5.5: Evaluating tree structures

The trees in (1) fail to correctly account for certain grammaticality judgments. That is to say, they make wrong predictions about the structure of these sentences. What are these wrong predictions? Provide evidence that suggests that each of these trees are inaccurate analyses.

(1) a.



b.



Exercise 5.6: Distinguishing arguments and adjuncts A

A. Are the italicized phrases in (1) syntactic arguments or adjuncts? Explain. There is no need for extensive discussion beyond your one-word conclusion and the evidence from *do so* substitution on which you base it.

(1)
a. They waited *for us*.
b. This program costs *twenty dollars*.
c. We drove *to Denver*.
d. We worded the letter *carefully*.
e. They may behave *very inconsiderately*.
f. This volcano might erupt *any minute*.

B. Build structures for the sentences in (1). The structures you build should be consistent with the evidence you gave in (A).

Exercise 5.7: Distinguishing arguments and adjuncts B

A. Build structures for the sentences in (1). Motivate your attachment of the lowest argument or adjunct in each sentence (in other words, provide the evidence that leads you to attach the phrase in question the way that you do).

(1)
a. They demolished the house.
b. Mona Lisa called the other neighbor.
c. Mona Lisa called the other day.
d. You will recall that her smile amazed everyone.
e. Most people doubt that Mona Lisa lives in Kansas.
f. My friend wondered if Mona Lisa would come to his party.

B. Indicate all recursive nodes in the structures that you build for (1).

Exercise 5.8: Structures for ambiguous sentences

A. (1) is structurally ambiguous. Paraphrase (or otherwise distinguish) the two relevant interpretations. (Focus on the structural ambiguity, ignoring the referential ambiguity of *they*.)

(1) They claimed that they paid on the 15^{th}.

B. Build a structure for each of the interpretations, indicating which structure goes with which interpretation. We encourage you to build chunks and indicate how they fit together differently for the two interpretations.

C. Indicate all recursive nodes in the structures that you build for (1).

Exercise 5.9: Distinguishing arguments and adjuncts C

A. Make up a sentence with two adjuncts. Provide syntactic evidence that the adjuncts are adjuncts rather than syntactic arguments. Then build the structure for the sentence. Finally, switch the linear order of the adjuncts, and build the structure for the resulting word order variant of your original sentence.

B. Make up a simple sentence in which one of the semantic arguments of the verb is expressed in the syntax as an adjunct. Provide evidence that the adjunct is an adjunct and not a complement. Finally, build the structure for your sentence.

Exercise 5.10: Core X′ structures

Is it possible for two adjuncts to be siblings? Explain why or why not.

Exercise 5.11: Basic phrase structure of Swahili

Below are some simple sentences from Swahili. Consider each of them in order to inform your analysis, and draw a tree for the bracketed portion of (1d).

You can ignore subject agreement (glossed here as SM ‘subject marker’); we generally assume it arises on Inflection, and we will develop ideas about how this works later in the textbook. If you want to draw it into the tree, put it in Infl along with tense.

Background: Swahili has a robust noun class system: noun classes are glossed with cardinal numbers (1, 2, 3, 4, etc). You don’t have to explain anything about noun classes, though they do show what is agreeing with what, which can be helpful to know. Glossing conventions: SM = subject marker (i.e., subject agreement on verbs), SM glosses are accompanied with a noun class gloss based on what noun class the subject they agree in belongs to; PST = past tense; PRF = perfect aspect (translated similar to past tense); FUT = future tense; PRS = present tense.

(1) a. Watoto wa-me-ondoka.
2-children 2SM-PRF-leave
‘The children left.’

b. M-toto a-ta-tembelea wa-zazi wao
1-child 1SM-FUT-visit 2-parents 2.their
‘The child will visit their parents.’

c. Wa-zazi wetu wa-li-kula maharagwe.
2-parents our 2SM-PST-eat beans
‘Our parents ate beans.’

d. Wa-toto wa-na-jua [kwamba wa-zazi wao wa-li-kula maharagwe.]
2-children 2SM-PRS-know that 2SM-parents 2.their 2SM-PST-eat beans
‘The children know that their parents ate beans.’

Problem 5.1: Thinking about the model

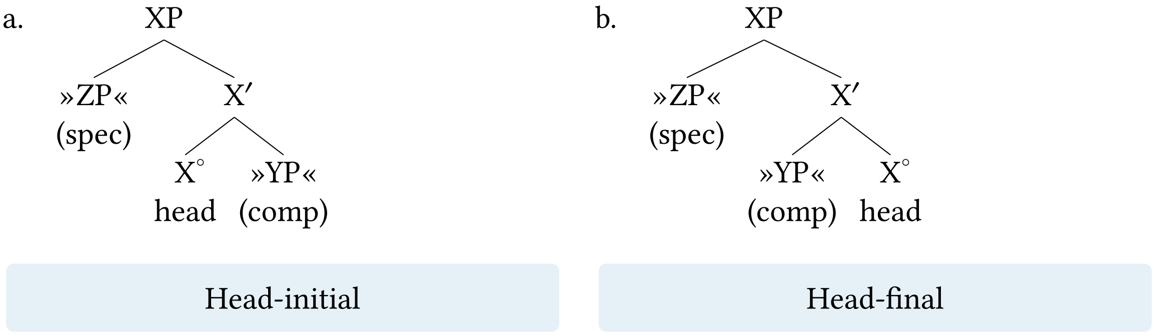
In connection with introducing the formal operation of adjunction, we insisted that "the whole point of a generative grammar is to generate an unbounded set of sentences from a <u>finite</u> set of elementary expressions and operations". What problem arises if we allow a generative grammar to include a potentially unbounded set of elementary trees or operations? That is to say, why restrict our theory to certain kinds of trees and operations? Why not just invent whatever we need for new puzzles ad infinitum?

Problem 5.2: Linear order and the X′ schema

Is it possible for an adjunct to precede a complement? How about immediately precede?

What if you are allowed to "swivel" at X′ in the X′ schema, so that YP precedes X? (The resulting head-final structures are discussed in more detail in Chapter 6, § 6.4.)

(1) a. b.



Problem 5.3: Finite verb phrases in English

Uninflected verb phrases in English are generally consistently diagnosable as constituents using our diagnostics, but inflected verb phrases (past tense, in English) consistently fail constituency diagnostics.

(1) Substitution:
   a. She will <u>write a book</u>. → ✓ She will <u>do so</u>.
   b. The two boys could <u>order tuna salad sandwiches</u>. → ✓ The two boys could <u>do so</u>.

(2) Question/short answer:
   a. <u>What</u> will she <u>do</u>? → ✓ <u>Write a book</u>.
   b. <u>What</u> could the two boys <u>do</u>? → ✓ <u>Order tuna salad sandwiches</u>.

(3) a. Substitution, uninflected verb:
   She will <u>write a book</u>. → ✓ She will <u>do so</u>.
   b. Substitution, inflected verb:
   She <u>wrote a book</u>. → ✓ She <u>did so</u>.

(4) a. Question/short answer, uninflected verb:
What will she do? → ✓ Write a book.

b. Question/short answer, inflected verb:
What did she do? → *?Wrote a book.

c. Question/short answer, inflected verb:
What did she do? → ✓ Write a book.

(5) Movement, uninflected verb phrase:

a. (She says that) she will write a book, → ✓ (and) write a book, she will ___.

b. though she may write a book → ✓ write a book though she may ___

(6) Movement, inflected verb phrase:

a. (She says that) she wrote a book, → * (and) wrote a book, she ___.

b. though she wrote a book → * wrote a book though she ___

(7) *It* cleft, uninflected verb phrase:
She will write a book. → ?? It is write a book that she will ___.

(8) *It* cleft, inflected verb phrase:
She wrote a book. → * It is wrote a book that she ___.

Based on the structure of the clause that we established in this chapter:

1. Why do uninflected verb phrases in English (verb phrases in the future tense or with modals in which the verb doesn't bear inflection) consistently pass constituency diagnostics?
2. Why do inflected verb phrases in English (past tense) consistently fail constituency diagnostics?
3. What does this tell us about the relationship of verb inflection and constituency diagnostics?

Identifying Puzzles 5.1: Ditransitive verbs

Consider the individual examples from a range of different languages below which are ditransitive verbs, that is, verbs with two objects. The examples below are drawn from a wide variety of languages from different families (and with different word orders) to demonstrate the generality of the pattern.

Glossing conventions: PST = past tense; PRF = perfect aspect; FUT = future tense; PRS = present tense; STAT = stative aspect; DAT = dative case. Glosses for agreements are simplified in each language as the details are irrelevant for the purposes of this exercise.

(1) a. The children gave their parents a present.

b. M-toto a-li-wa-pa wa-zazi wao zawadi.
child AGR-PST-AGR-give parents their gift
'The child gave their parents a gift.'

c. Amharic (Afroasiatic, Ethiopa; Amberger 2009, 747)
Ləmmɑ lə-lɨǰ-u məs'hɑf sət't'-ə-(w).
Lemma to-child-DEF book give-AGR-(AGR)
'Lemma gave the book to the child.'

d. Tlapanec (Otomanguean, Mexico; Wichman 2010)
Ni-ʃn-û marìa dudî adà.
PFV-give-AGR Maria avocado boy
'María gave the boy an avocado.'

e. Tima (Kordofanian, Sudan; G. J. Dimmendaal 2010)
Kúúkù à-ŋ́-kéél címìì íí-hamid.
Kuuku STAT-AGR-buy goat DAT-Hamid
'Kuuku has bought a goat for Hamid.'

All of the examples we dealt with in the main text of this chapter had (at most) one object. This is intentional, and the text will maintain this stance until Chapter 13. The reason for this is because ditransitive verbs (which are common across a wide range of languages) pose a central puzzle for our theory. What is that puzzle?

Don't attempt to give a solution within the X′ schema! Just identify what the theoretical challenge is that is posed by the data. We will get to solutions later in the textbook.

Identifying Puzzles 5.2: The formal constraints on adjunction

In the chapter, we define adjunction as an operation that clones a target node and attaches a phrase as the child of the higher clone. Imagine an operation that clones the target node and attaches a phrase as the child of the lower clone. Would such an operation be useful? Explain.

6

Extending the X′ schema

Chapter outline

In Chapter 5, we introduced a normal form for phrase structure, the X′ schema, according to which lexical items project an elementary tree consisting of a spine and up to two argument positions. In this chapter, we extend the X′ schema to syntactic categories other than V, I, or C, to include N(oun), D(eterminer), A(djective), and P(reposition). The final section of the chapter illustrates crosslinguistic variation with regard to the order of heads and complements.

Noun phrases 6.1

6.1.1 Parallels and differences between noun phrases and sentences

The X′ schema of phrase structure that we introduced in Chapter 5 is a specific expression of a more general idea—namely, that lexical items of different syntactic categories show significant **cross-categorial** parallels. In the history of generative grammar, this idea was primarily based on the cross-categorial parallels between noun phrases and sentences (Chomsky 1970). In what follows, we review these parallels as well as some differences between the two categories.

Argument structure

Early in the history of generative grammar (Lees 1960), it was observed that sentences like (1a) and noun phrases like (1b) share several important properties.

(1) a. The cat <u>destroyed</u> the couch.
    b. the cat's <u>destruction</u> of the couch

The semantically central element of the sentence in (1a) is the verb *destroyed.* Its semantic arguments, the agent *the cat* and the theme *the couch*, are both expressed as syntactic arguments of the sentence. In a parallel way, the semantically central element in the noun phrase in (1b) is the nominal counterpart of *destroy*, the noun *destruction.* Like the verb, the noun is associated with an agent argument and a theme argument that are both overtly expressed—in this case, as the possessive expression *the cat's* and the prepositional phrase *of the couch.*

The correspondence in (1) is supported by a similar one between the passive sentence in (2a) and its passive-like noun phrase counterpart in (2b).

(2) a. The couch was <u>destroyed</u> (by the cat).
    b. the couch's <u>destruction</u> (by the cat)

In both of these examples, the argument preceding the head is now the theme *the couch('s)*, and the agent argument is expressed by an optional *by* phrase.

A further parallel between sentences and noun phrases is that in both categories, the semantically central element—the verb or the noun—can be modified in similar ways, as illustrated in (3) and (4).[1]

(3)	Adverb:	She *regularly* <u>gives</u> money to the organization.
	Prepositional phrase:	She <u>gives</u> money to the organization *on a regular basis.*
(4)	Adjective:	her *regular* <u>gifts</u> of money to the organization.
	Prepositional phrase:	her <u>gifts</u> of money to the organization *on a regular basis*

(5) Adverb: The cat *frequently* destroys the couch.
Prepositional phrase: The cat destroys the couch *with high frequency*.

(6) Adjective: the cat's *frequent* destruction of the couch.
Prepositional phrase: the cat's destruction of the couch *with high frequency*

There are of course differences between sentences and noun phrases: for example, agents within noun phrases (in English) are typically introduced as possessors and not directly; likewise, noun phrase object arguments of nouns are generally inside PPs and cannot be directly complements of the N heads themselves. So there are certainly important differences, but also intriguing similarities.

6.1.2 Noun phrases as DPs

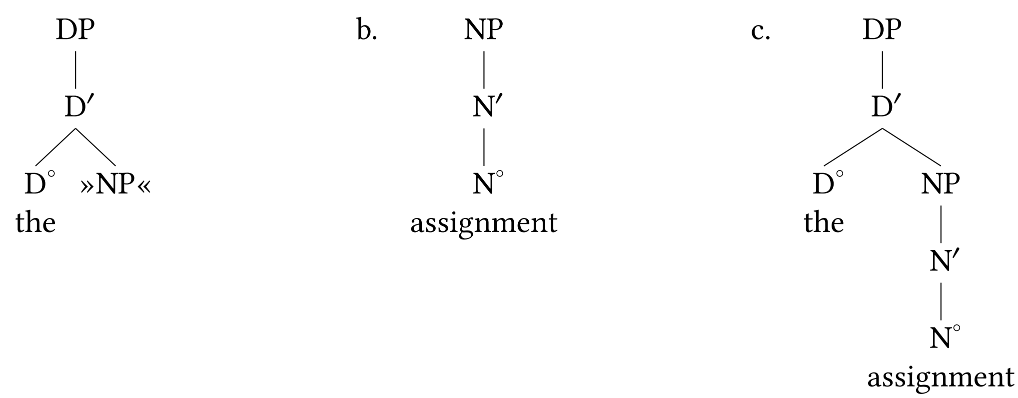
A striking fact about nouns is that, in many constructions in many languages, they cannot in general function as arguments on their own, but must be accompanied by a determiner.

(7) a. * *Assignment* is not difficult.
b. * You should hand in *assignment*.

(8) a. { *The, this, that* } *assignment* is not difficult.
b. You should hand in { *the, this, that* } *assignment*.

A perhaps-surprising conclusion of generative syntacticians from facts like these (and others) is that noun phrases are the result of composing two projections, one headed by the noun and the other by the determiner, as shown in (9).

(9) a. b. c.



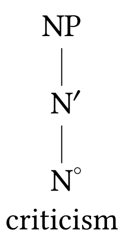
Take Note

Given the structure in (9c), the traditional term "noun phrase" is a misnomer since noun phrases are maximal projections of D rather than of N. Because the term "noun phrase" is firmly established in usage, we continue to use it as an informal synonym for "DP". However, in order to avoid confusion, we will use the term "NP" only to refer to the subconstituent of a noun phrase that is the complement of a determiner. We will never use "NP" to refer to an entire noun phrase (that is, a DP).

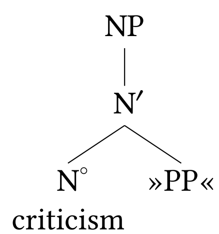
In the simplest case, the elementary tree for a noun consists of just the NP projection, as in (9b). But as we have just seen, nouns, like verbs, can have both complements and specifiers. For instance, depending on which of the noun phrases in (10) it appears in, *criticism* is associated with one of the elementary trees in (11).

(10) a. They refuted *the criticism.*
b. They refuted *the criticism of the proposal.*
c. They refuted *the committee's criticism.*
d. They refuted *the committee's criticism of the proposal.*

(11) a.

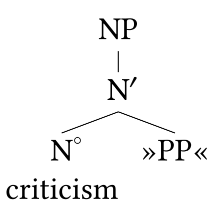


b.

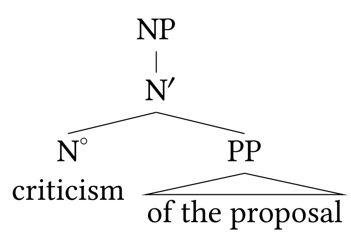


In (10a), the phrase *the criticism* is derived in exactly the same way as *the assignment* in (9)—by substituting the NP in (11a) as the complement of the determiner. In (11b), the noun phrase containing *criticism* is derived as in (12). (The internal structure of prepositional phrases is covered later on in the chapter; for the moment, we ignore it.)

(12) a.



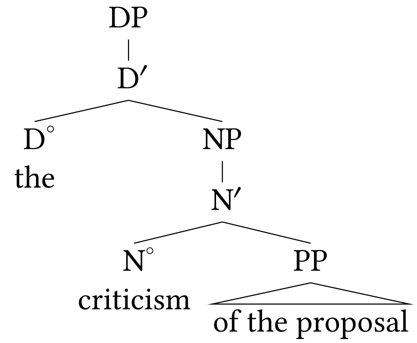
b.



Elementary tree for N (11b)

Substitute theme argument

c.



Substitute (12b) in elementary tree for D

One of the reasons that it is compelling to treat traditional noun phrases[2] as DPs is that it allows for a natural way to explain the pattern that emerges in possessive noun phrases. A central aspect of possessive noun phrases in Mainstream U.S. English is that the possessor (italicized in the examples below) is itself a complete noun phrase: it can be a proper name, it can be a DP, and it can even be a complex and informative DP.

(13) a. *Langston*'s criticism of the proposal.
b. *the committee*'s criticism of the proposal
c. *the current and potential ongoing leader*'s criticism of the proposal

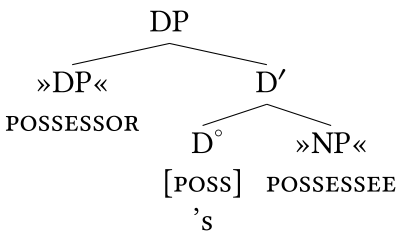
So a possessive noun phrase requires a structure that embeds a full DP inside another DP. But it also requires an approach that captures the fact that a possessive DP rules out the possibility of determiner (e.g. a definite determiner or a demonstrative) to cooccur with the possessive.

(14) a. *Langston*'s criticism of the proposal.
b. * The *Langston*'s criticism of the proposal.
c. * That *Langston*'s criticism of the proposal.
d. * *Langston*'s the criticism of the proposal.
e. * *Langston*'s that criticism of the proposal.

We saw above that the possessor itself can contain a determiner, but here we use a proper name as the possessor to demonstrate that the larger possessive DP itself has no way to incorporate a determiner. Non-possessive noun phrases have no such problem incorporating determiner elements:

(15) a. The criticism of the proposal
b. This criticism of the proposal

We therefore see a complementary distribution of determiners (like *a*, *the*, *this/that*, etc) with the possessive morpheme, suggesting that they are instances of the same grammatical category. We can arrive at an analysis where English noun phrases containing possessive noun phrases, like (10c), require a possessive head *’s* that is of the category D°. The elementary tree for this head is shown in (16), containing two argument positions: one for the possessee (the element being possessed) that is the NP complement of the possessive D°, and a position for the possessor, the DP embedded in the specifier of the possessive DP.

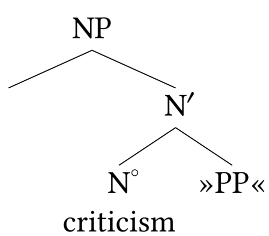
(16)


This also readily allows for recursive possessive structures: the DP possessor in (16) can itself be a possessive DP.

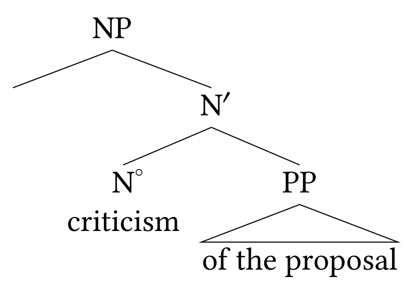
(17) a. [$_{\text{DP}}$ [$_{\text{DP}}$ The children’s mother]’s sister] is a good friend of mine.
b. [$_{\text{DP}}$ [$_{\text{DP}}$ Lilly’s fur]’s texture] is finer than I expected.

Notice that the possessive head is not a free morpheme. Recall from Chapter 5 that we posited elementary trees headed by silent bound tense morphemes in order to allow us to represent English past and present tense sentences analogously to their future tense counterparts. In the same spirit, we allow elementary trees headed by overt bound morphemes like *’s*. In general, modern syntactic theory is not terribly concerned with whether the heads of elementary trees are bound or free morphemes, or silent or overt, as long as the trees allow us to provide maximally similar and accurate representations for linguistically related phenomena.

Deriving (10d) involves substituting both the relevant arguments into the elementary trees for the NP/DP, as outlined in (18).

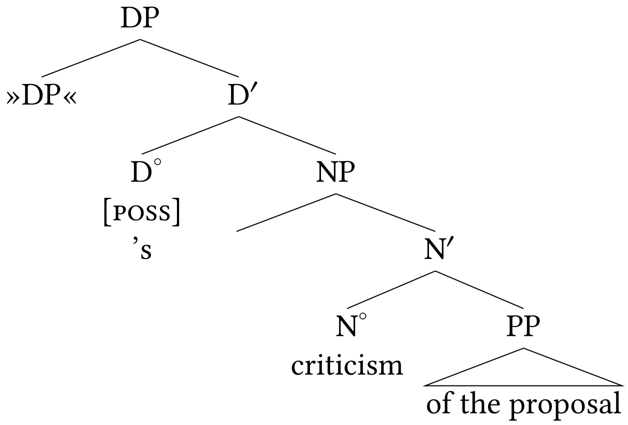
(18) a.


Elementary tree for N
criticism

b.


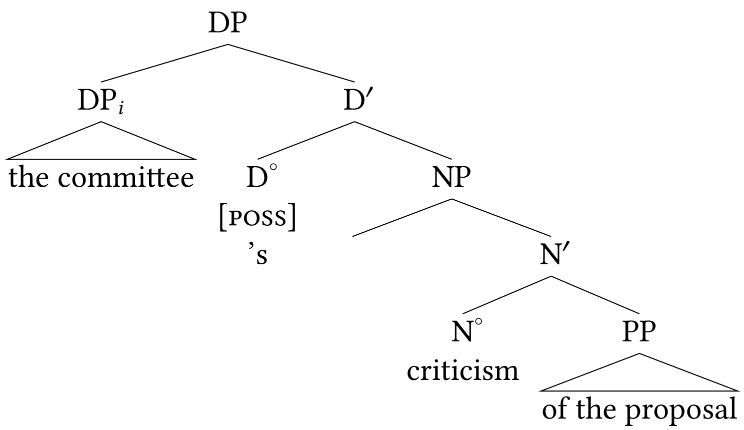
Substitute theme argument

c.



Substitute (18b) into the elementary tree for a possessive DP (16)

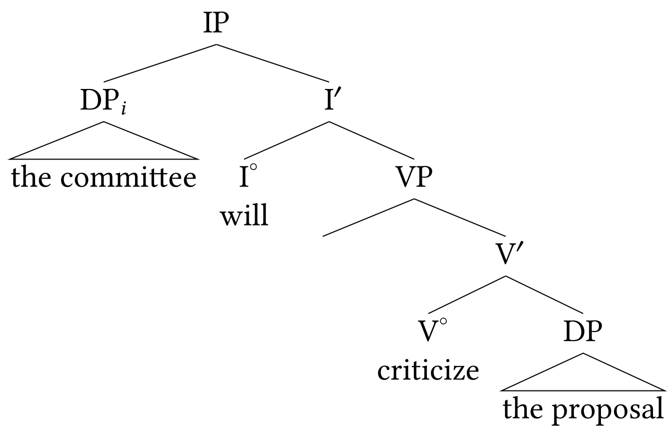
d.



Substitute agent/possessor into Spec(DP)

The sentential counterpart of the noun phrase in (18d) is given in (19). As is evident, apart from the labels for the syntactic categories, the two-layered structure for noun phrases (NP, DP) presented here is analogous to the two-layered structure for sentences from Chapter 5 (VP, IP).

(19)



6.1.3 More on determiners

Subcategories of determiners

Like verbs and nouns, determiners have different selectional properties—they can be transitive or intransitive (select for an object or not). For instance, the definite article *the* and the indefinite article *a(n)* are obligatorily transitive, whereas the demonstratives *this* and *that* are optionally so.

(20) a. I'll buy { the, a } book.
b. * I'll buy { the, a }.

(21) a. I'll buy { this, that } book.
b. I'll buy { this, that }.

Certain ordinary pronouns pattern just like demonstratives, as shown in (22), and so we will treat them, too, as optionally transitive determiners.

(22) a. we Americans, you fool(s)
b. we, you

Finally, some ordinary pronouns behave like obligatorily intransitive determiners, as shown in (23).

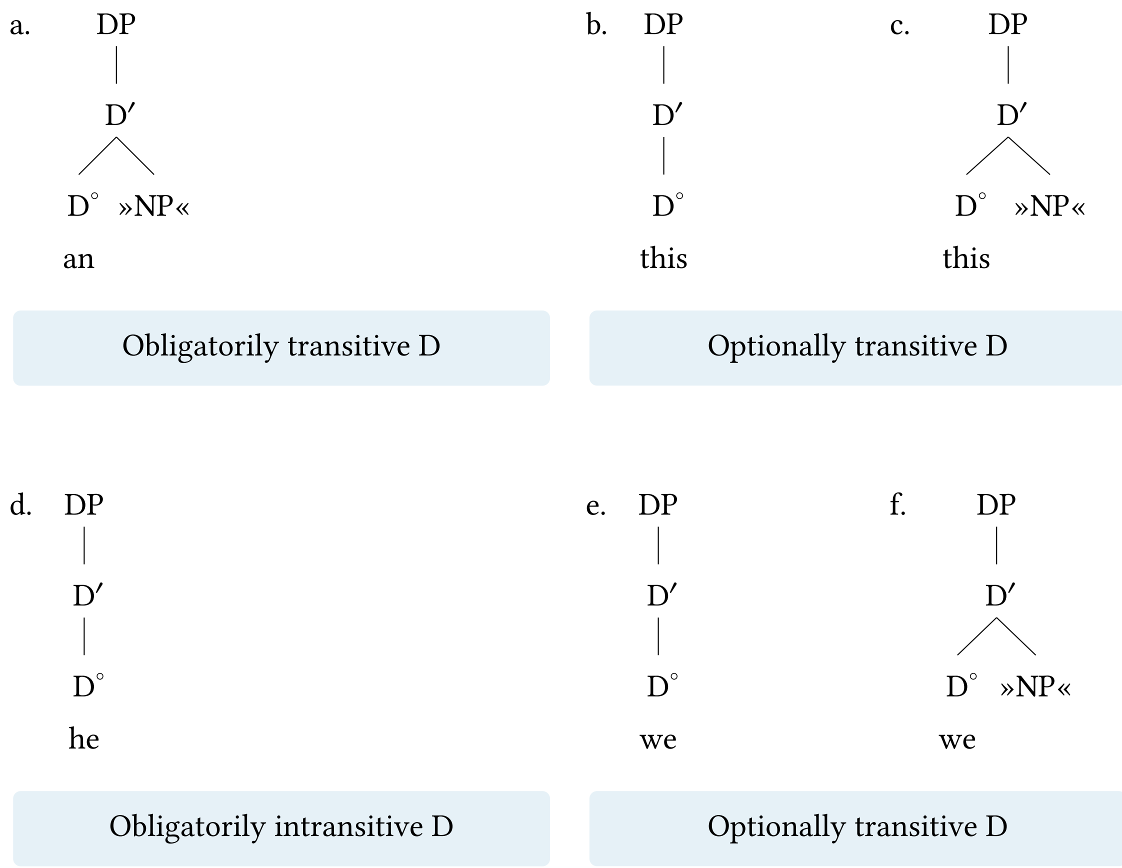
(23) a. I, he, she, it, they
b. * I idiot, he fool, she linguist, it piece of junk, they traitors

In this connection, recall the warning in Chapter 1 that the term "pronoun" is potentially misleading. It suggests that pronouns are a subclass of nouns. If that were so, then pronouns should combine with articles and demonstratives in the same way that other nouns do. In fact, however, pronouns behave exactly like complete noun phrases in this regard, as shown in (24). The facts in (24) thus provide strong evidence for the analysis of pronouns as determiners just presented.

(24) a. D + noun: the people, this woman, that addressee
b. D + noun phrase (= DP): * the these people, this the woman, that the addressee
c. D + pronoun (= pro-DP): * the they, this she, that you

The argument is from simple reasoning based on distribution: pronouns have the distribution of DPs, not of NPs: NPs can co-occur with determiners, but pronouns cannot. Elementary trees for the various types of determiners that we have just discussed are given in (25).

(25)



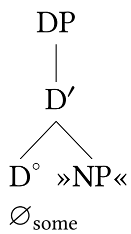
Silent determiners

As shown in (26), plural indefinite count nouns and indefinite mass nouns are apparently not accompanied by an article in English, in contrast to their singular or definite counterparts.

(26) a. ___ cars, ___ apples; ___ rice
b. a car, an apple; the rice

However, we assume for conceptual reasons that the examples in (26a) contain a silent article that is semantically roughly comparable to the unstressed *some* in *I would like some apples and some rice.* The elementary tree is shown in (27).

(27)



We have two reasons for assuming the existence of silent determiners. First, this assumption allows us to minimize the difference between English and a language like Spanish, where the indefinite article is overt in the same circumstances. The resulting correspondence between English and Spanish determiners is shown in Table 6.1; the plural indefinite articles are in boldface. For simplicity, we give only the masculine forms of the Spanish determiners.

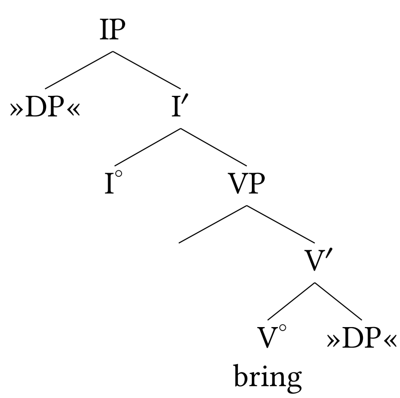
	English		Spanish	
	SG	PL	SG	PL
Demonstrative	this	these	este	estos
	that	those	ese	esos
Definite article	the	the	el	los
Indefinite article	a(n)	**∅INDF.PL**	un	**unos**

Table 6.1: English and Spanish determiners

Second, assuming the silent determiner allows us to maintain that all noun phrases are DPs. Sentences like (28) can then all be derived using the elementary trees in (29).

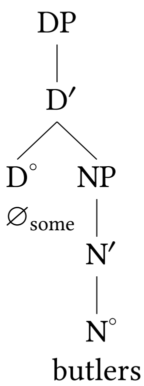
(28) a. Some butlers might bring some tea.
b. Some butlers might bring tea.
c. Butlers might bring some tea.
d. Butlers might bring tea.

(29)



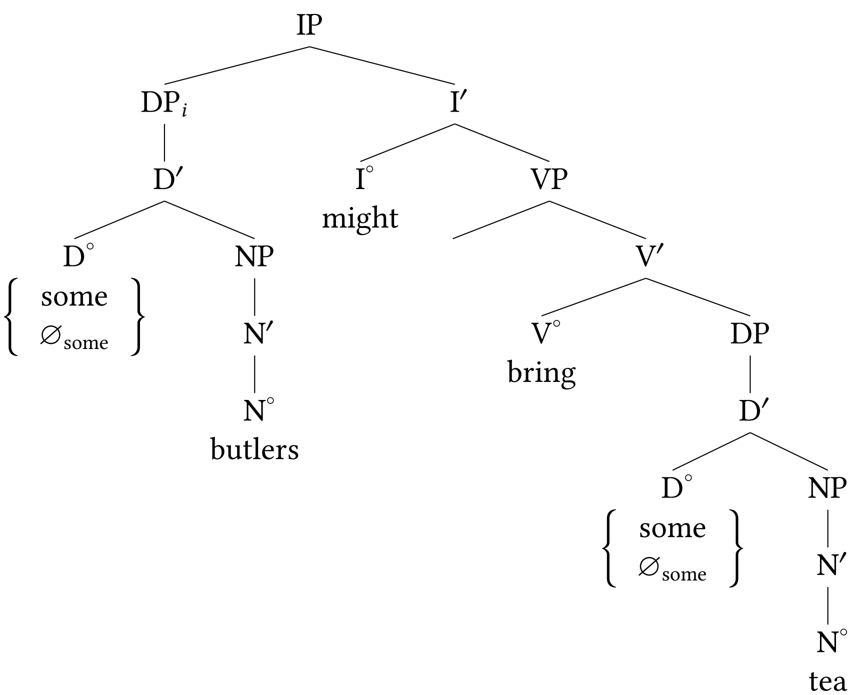
We show the full structure for the apparently articleless noun phrase *butlers* in (30). The structure for *tea* is analogous.

(30)



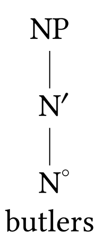
(31) combines the trees for all four sentences in (28) into a single representation.

(31)



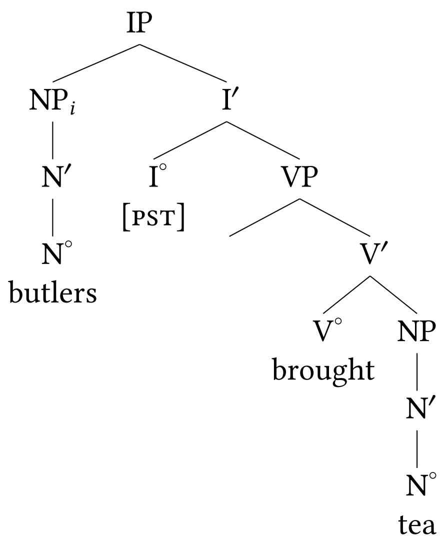
In principle, we could take an alternative tack. If it were our goal to assign the least possible amount of structure (that is, the structures with the fewest nodes) to each sentence in (28), we would reject the silent determiner in (27) and represent *butlers* as a bare NP, as in (32) (and analogously for *tea*).

(32)



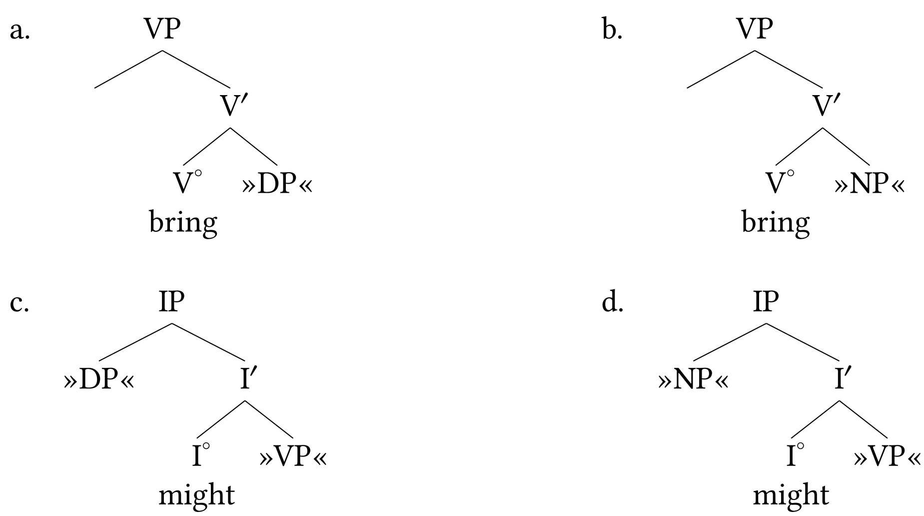
The alternative structure under discussion for (28d) is given in (33).

(33)



Clearly, the tree in (33) is simpler than its counterpart in (31) in the sense of containing fewer nodes. However, this simplicity comes at the price of a veritable explosion in the number of elementary trees in the grammar, since every argument position that can be filled by a noun phrase would need to be associated with two elementary trees (one with a DP substitution node, and one with an NP substitution node). For instance, instead of the single elementary trees for *might* and *bring*, we would need an additional elementary tree for each that has an NP substitution node and not just a DP substitution node. Every head that selects for an argument would therefore need to select for *both* an NP and a DP.

(34)



This result seems unappealing on computational grounds, as every instance of a noun phrase requires additional elementary trees. We also miss an insight in doing this—noun phrases without determiners share a syntactic distribution with noun phrases with determiners, and our model naturally accommodates this better by treating them all as DPs.

More fundamentally, requiring that we simplify representations of individual sentences can end up with results that are inconsistent with the Chomskyan paradigm of language. Why? From a Chomskyan perspective, what syntactic theory attempts to model and understand is the mental capacity to generate sentences: what is Universal Grammar? A reasonable working hypothesis is that the best model for this capacity is the simplest possible grammar. So striving to cut down on the numbers of nodes in the representations of sentences can come at the expense of complicating our model of UG itself, trading simpler trees for a more complex theory overall (which we deem an undesirable result).

6.1.4 Modification and related issues

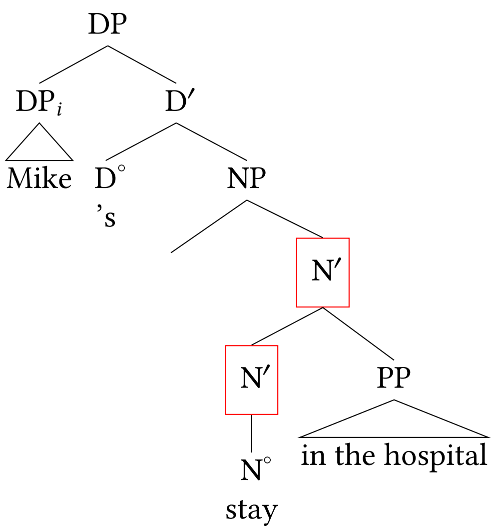
N′ as target of adjunction

As we noted in our introductory review of the parallels between noun phrases and sentences in § 6.1.1, nouns and verbs can be modified in similar ways. In (35), for instance, the same prepositional phrase *in the hospital* modifies the noun *stay* and the morphologically related verb *stayed*.

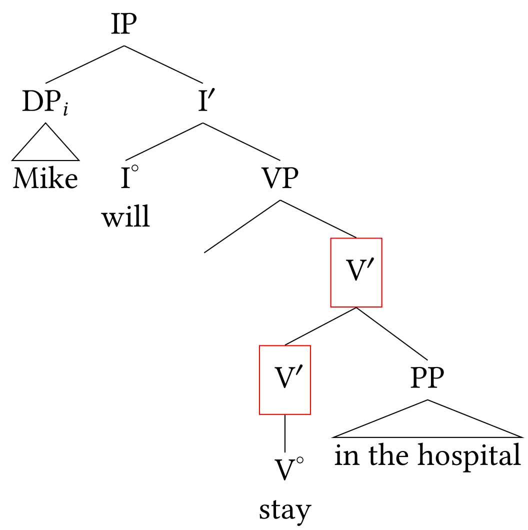
(35) a. Mike's stay *in the hospital*
     b. Mike will stay *in the hospital.*

Extending the approach to representing modification introduced in Chapter 5, (§ 5.3.1) we arrive at the structures of nominal modification shown in (36a). This structure is clearly analogous to the structure for the corresponding sentence in (36b).

(36) a.



b.

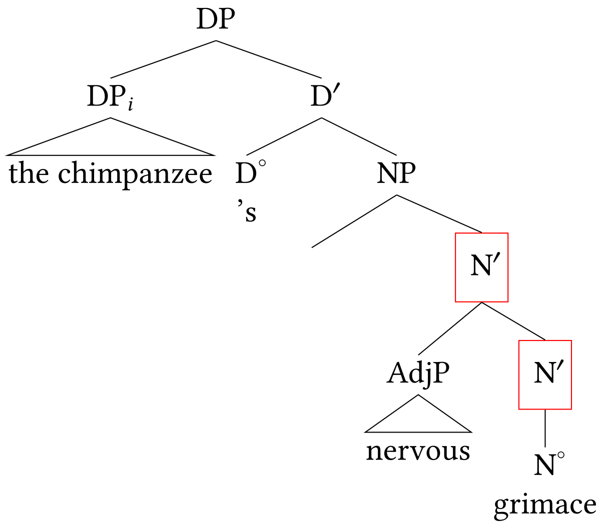


Leftward adjunction

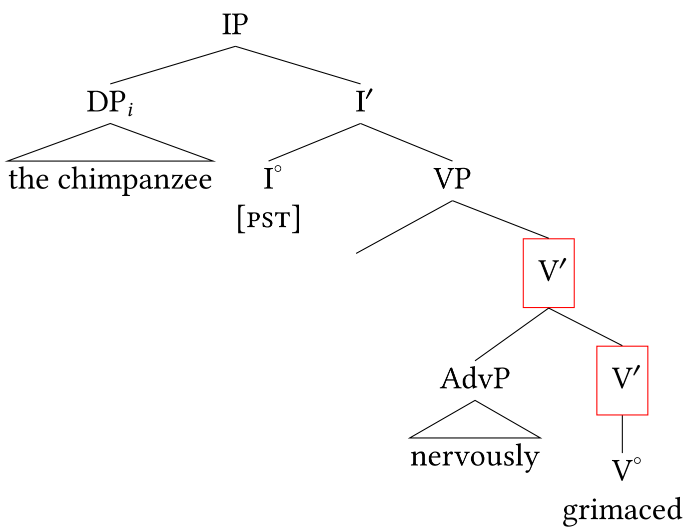
So far, we have discussed modifiers that follow the head, whose representation involves rightward adjunction. Structures for examples like (37), where the modifier precedes the head it modifies, can be derived by leftward adjunction, as shown in (38).

(37) a. the chimpanzee's *nervous* grimace
b. The chimpanzee *nervously* grimaced.

(38) a.



b.



As an aside, notice that in both trees in (38) the complement position of the main lexical head (V°/N°) is empty. Part of what makes the AdjP/AdvP an adjunct in our model is that it is **not** the sibling of the head, so it is crucial to leave the complement position empty in order for the modifier to be appropriately represented as an adjunct.

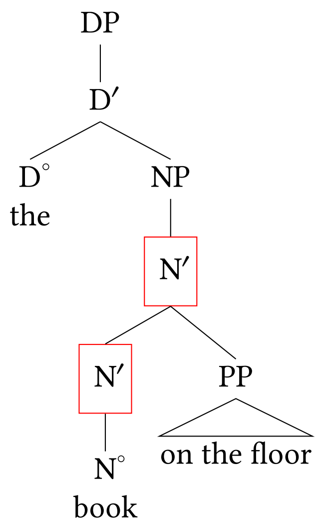
One substitution

As discussed in Chapter 5, *do so* substitution allows us to distinguish between complements and adjuncts in the verbal system. A similar diagnostic is available in the nominal system—*one* substitution, which is illustrated in (39).

(39) a. this book on the floor and that one
b. this book on the floor and that one on the table

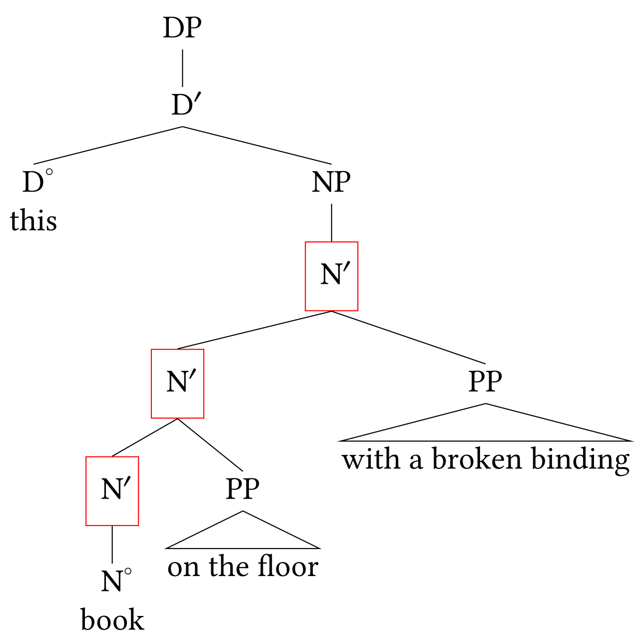
In the most natural interpretation of (39a), *one* is interpreted as *book on the floor.* In (39b), on the other hand, *one* is interpreted as simply *book.* We can represent these facts by assuming that the first conjunct in both cases has the structure in (40).

(40)



According to (40), the noun *book* has no complement, and the PP *on the floor* is an adjunct. The pro-form *one* substitutes for instances of N′, just as *do so* substitutes for instances of V′. *One* substitutes for the higher N′ in (39a), and for the lower N′ in (39b).

As in the case of V′, adjunction to N′ can apply more than once, yielding multiply recursive structures like (41).

(41) 

Restrictions on *one* substitution

A cautionary note is in order about *one* substitution. Although in principle *one* can substitute for any instance of N′, it is subject to two restrictions, which are important to keep in mind when using *one* substitution as a diagnostic for syntactic structure. The first restriction, which makes sense given the etymological connection of the pro-form to the homonymous number word, is that *one* can substitute only for count nouns, as illustrated in (42).

(42)
  a. I prefer this { bed, chair, sofa, table }, and you prefer that one.
  b. * I prefer this furniture, and you prefer that one.

A second restriction is that *one* cannot immediately follow the indefinite article, a cardinal number, a possessive noun phrase, or (for many speakers) the plural demonstratives *these* and *those*. Whatever the exact source of this restriction is (in some cases it is difficult to derive from the etymological connection just mentioned), the restriction is very superficial, since an intervening word renders the ungrammatical (a) examples in (43)–(46) grammatical.[3]

(43) Indefinite article
  a. * I bought a book, and you bought a one, too.
  b. I bought a blue book, and you bought a red one.

(44) Cardinal number
  a. * I bought { two , ten } books, and you bought { two, ten } ones, too.
  b. I bought { two, ten } blue books, and you bought { two, ten } red ones.

(45) Possessive
  a. * I like { Mary's, her } book, and you like { John's, his } one.
  b. I like { Mary's, her } blue shirt, and you like { Mary's, her } red one.

(46) Plural demonstrative
  a. * I like these books, and you like those ones.
  b. I like these blue books, and you like those red ones.

Structural ambiguity

Having introduced N′ as a possible target of modification, we are now in a position to associate structurally ambiguous sentences like (47) with two distinct syntactic representations.

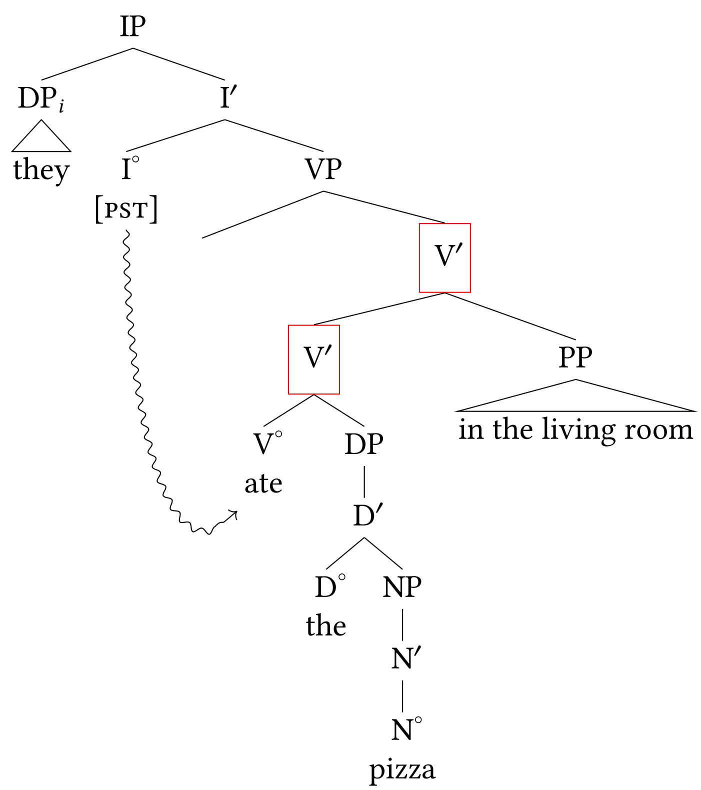
(47) They ate the pizza in the living room.

(47) has two interpretations, which can be paraphrased as in (48).

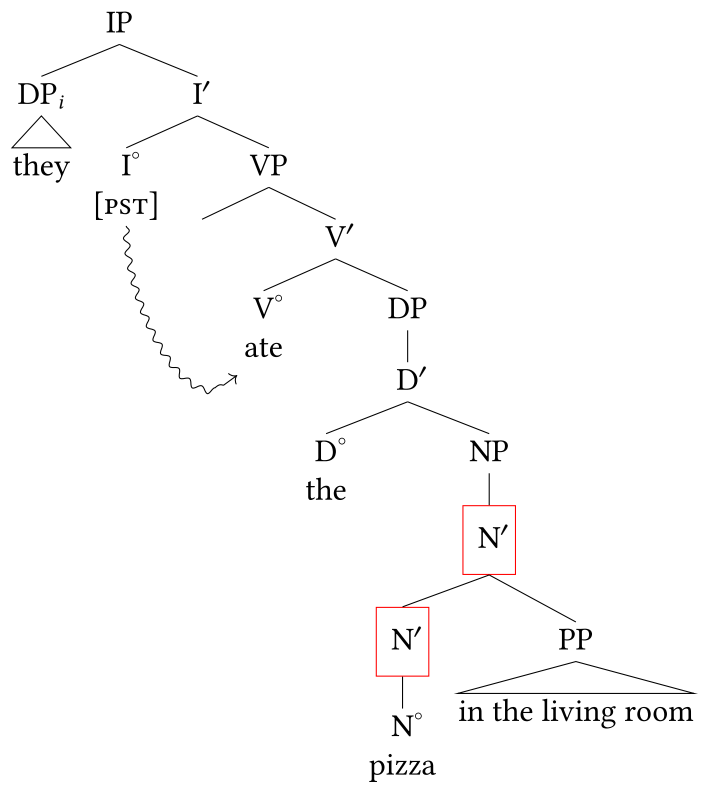
(48) a. Verbal modifier interpretation:
  It was in the living room that they ate the pizza (though the pizza may have started out elsewhere).
  b. Nominal modifier interpretation:
  It was the pizza in the living room that they ate (though perhaps they then took it and ate it elsewhere).

On the interpretation in (48a), the prepositional phrase *in the living room* modifies the verb *ate*, and (47) has the structure in (49a). On the interpretation in (48b), the prepositional phrase modifies the noun *pizza*, and the sentence has the structure in (49b).

(49) a.



b.



The structures in (49) are consistent with the results of relevant constituenthood tests. For instance, substituting the ordinary pronoun *it* for *the pizza* and substituting *did so* for *ate the pizza* yields (50a) and (50b), respectively.

(50) a. They ate it in the living room.
b. They did so in the living room.

In both sentences, the prepositional phrase is unambiguously interpreted as a verbal modifier, as expected given that *the pizza* and *ate the pizza* are represented as constituents in (49a), but not in (49b).

Conversely, in the question-answer pair in (51), the prepositional phrase is unambiguously associated with a nominal modifier interpretation. Again, this is expected, since *the pizza in the living room* is represented as a constituent in (49b), but not in (49a).

(51) What did they eat? The pizza in the living room.

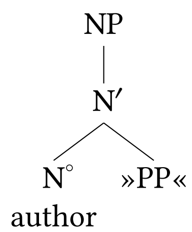
The complement-adjunct distinction in the nominal system.

Given the semantic parallel between the sentence in (52a) and the noun phrase in (52b), it is reasonable to treat the *of* phrase in (52b) as a complement of the noun *author*.

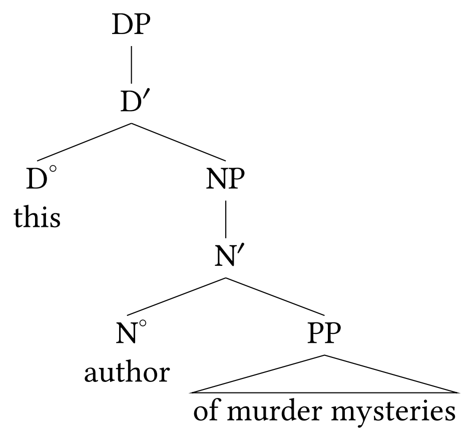
(52) a. This woman *authors* murder mysteries.
b. this *author* of murder mysteries

That is, the elementary tree for *author* needed to derive (52b) is as in (53a), and the structure for the entire noun phrase is (53b).

(53) a.



b.



Since *one* is analogous to *do so* in substituting for intermediate rather than for lexical projections, we expect the contrast between (54) and (55), and this accurately reflects the judgment of many speakers.[4]

(54) a. ✓ This woman *authors murder mysteries*, and that man *does so*, too.
b. ✓ this *author of murder mysteries* and that *one*

(55) a. * This woman *authors* murder mysteries, and that man *does so* nature guides.
b. * this *author* of murder mysteries and that *one* of nature guides

Both complements and adjuncts function semantically as restrictors, that is, content that constrains what/who the DP refers to (the set of authors of murder mysteries is a subset of the set of authors). Therefore, there won't be an obvious semantic clue for speakers whether their grammar differs from that of other speakers. The only clue will come from the difference with respect to *one* substitution judgments, and any such differences are not going to be salient in everyday life.

Adjective phrases 6.2

In this section, we discuss the structure of adjective phrases, beginning with examples like those in (56) and (57), where the prepositional phrase following the adjective is optional.

(56) a. They are proud.
b. They are proud of their grandchild.

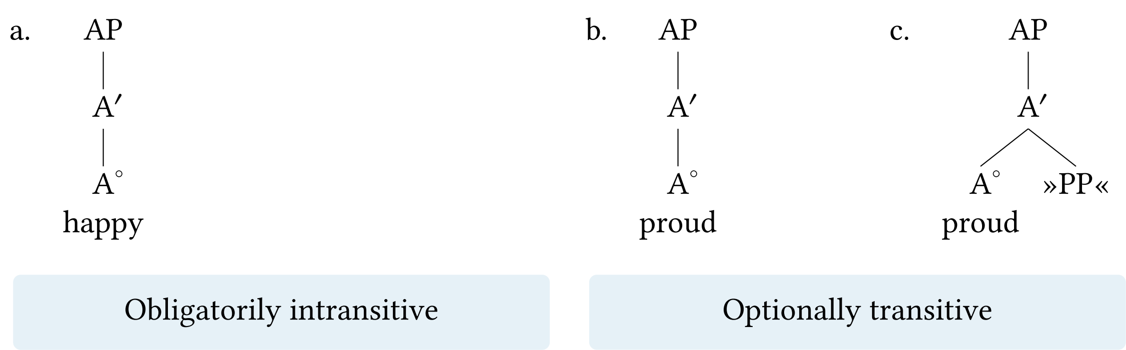
(57) a. They are happy.
b. They are happy with their car.

Recall from Chapter 3 that the pro-form *so* substitutes for adjective phrases. More specifically, examples like those in (58) and (59) allow us to conclude that the *of* phrase is a complement of *proud* in (58), but that the *with* phrase is an adjunct of *happy* in (59).

(58) a. ✓ They are *proud*, and we are *so*, too.
b. ✓ They are *proud of their grandchild*, and we are *so*, too.
c. * They are *proud* of their grandchild, and we are *so* of him, too.

(59) a. ✓ They are *happy*, and we are *so*, too.
b. ✓ They are *happy with their car*, and we are *so*, too.
c. ✓ They are *happy* with their car, and we are *so* with our bikes.

We can represent these facts by associating the two adjectives with the elementary trees in (60) and by stating that *so* substitutes for instances of A′. (We generally use "A" for "adjective". We always use "Adv" for "adverb", and we use "Adj" where the distinction between the two categories is at issue.)

(60)


Most adjectives in English, like the two just discussed, are optionally or obligatorily intransitive. A rare case of an obligatorily transitive adjective is *fond*.[5] The contrast in (61) is evidence for the complement status of the *of* phrase (recall from Chapter 5 that obligatory syntactic dependents are complements), and that status is confirmed by the results of *so* substitution.

(61) a. * They are fond.

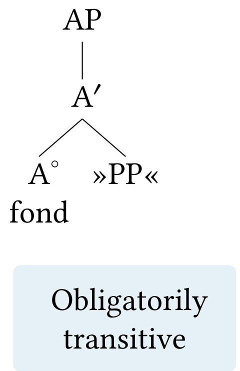
b. ✓ They are fond of their grandchild.

(62) a. * They are *fond* of their grandchild, and we are *so* of him, too.

b. ✓ They are *fond of their grandchild*, and we are *so*, too.

In view of the facts in (61) and (62), *fond* is associated with the single elementary tree in (63).

(63)



Prepositional phrases 6.3

The syntactic category P corresponds closely to the traditional part of speech called "preposition", but is not identical to it. We address two differences between the syntactic category and the traditional part of speech in the next two subsections.

Take Note

Following standard usage in the syntax literature, we sometimes use the term "preposition" to refer to the syntactic category P in contexts where the difference is either clear or immaterial.

6.3.1 Transitivity

The etymology of the term "preposition" (< Latin *prae* 'before' and *positio* 'position') implies that all prepositions should precede a complement, and English does in fact have a number of obligatorily transitive Ps, some of which are illustrated in (64). The asterisk (*) outside the parenthesized material is a convention to indicate that the parenthesized material is obligatory.

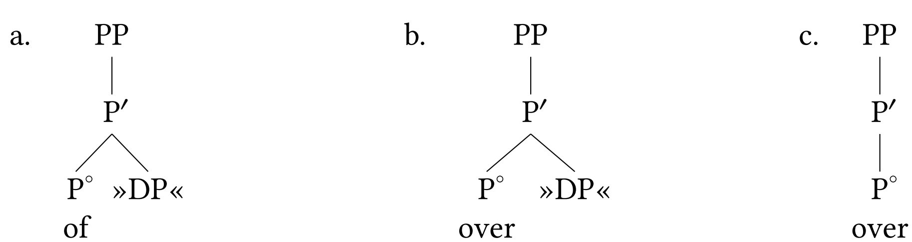
(64) a. They drove from *(Boston).
b. He's the inventor of *(that gizmo).
c. She dove into *(the water).
d. They jumped onto *(the bandwagon).

But X′ theory leads us to expect that there should also be intransitive Ps, and as the examples in (65) show, this expectation is fulfilled.

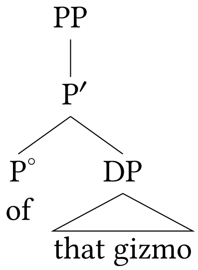
(65) a. I've never seen him *before* (this meeting).
b. Are you *for* (the proposal) or *against* (it)?
c. The bird flew { *in, out* } (the window).
d. It's time to get { *off, on* } (the train).
e. They jumped *over* (the ditch).
f. We've been fast friends ever *since* (that time).
g. She came *to* (her senses).
h. Have you looked *underneath* (the sombrero)?

In traditional grammar, Ps that are used intransitively are known as adverbs or particles, rather than as prepositions, but this terminology goes against the spirit of X′ theory, which seeks to maximize the parallels among categories. From our point of view, there is as little reason for the syntactic category of a lexical item to depend on its transitivity in the case of a P like *since* as there is in the case of a V like *eat*. In both cases, the intransitive variant has a semantic argument that is not expressed in the syntax, but is supplied in the course of interpretation, based on the discourse context.

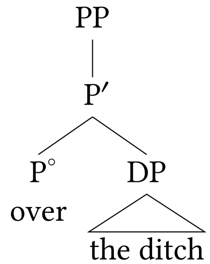
The elementary trees for *of* and *over* are shown in (66), and the full structures for the PPs headed by them in (64) and (65) are shown in (67). (67c) comes out as identical to (66c).

(66)


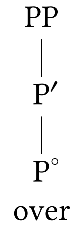
(67) a.



b.



c.



6.3.2 Clausal complements of prepositions

As we saw in Chapter 5, verbs can take either noun phrase complements or clausal complements. (68) gives a further example.

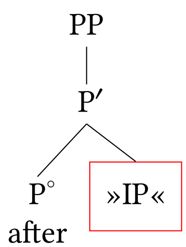
(68) a. They reported the monkey's dislike of camphor.
b. They reported that the monkey dislikes camphor.

The examples of transitive Ps discussed so far have all had noun phrase complements, but given the parallel between verbs and prepositions concerning transitivity, we might expect Ps to allow clausal complements as well. Once again, this expectation is borne out, as shown in (69).[6]

(69) a. Noun phrase complement: { *after, before, since* } the war
b. Clausal complement: { *after, before, since* } the war ended

The question now arises of what syntactic category clausal complements of prepositions belong to. At first glance, examples like (69b) suggest that the answer to this question is IP. The elementary tree for *after* in (69b) would then be as in (70), and the elementary trees for *before* and *since* would be analogous.

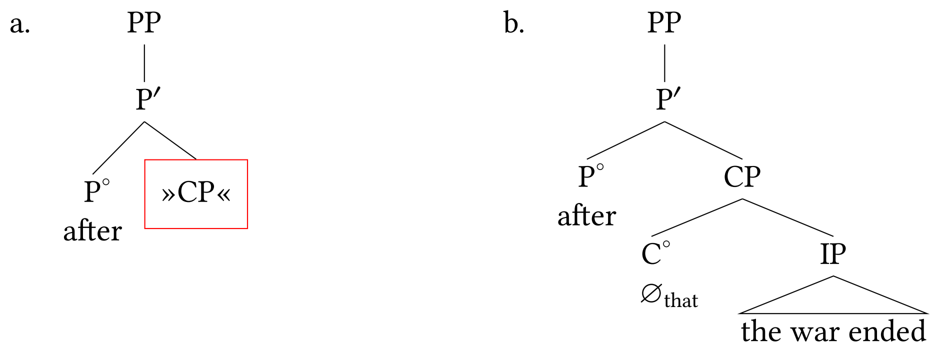
(70)



Not like this!

There is good reason to believe, however, that clausal complements (specifically, finite clausal complements) of P are CPs rather than IPs. As illustrated in (71a), the clausal complement of *after* and prepositions like it would be headed by a silent counterpart of the complementizer *that*, resulting in (71b) as the structure for *after the war ended* (for simplicity, the internal structure of IP is omitted).

(71)



There are several empirical arguments for preferring the elementary tree in (71a) over the one in (70). First, at least one preposition in English allows—indeed, requires—CP complements headed by an overt complementizer, as shown in (72).

(72) They differ *in* *(that) they hold sharply opposing views on educational reform.

A second reason for preferring (71a) over (70) is that sentences like (73), with an overt complementizer, occurred freely in Middle English (and are still acceptable for some speakers of Modern English). Some naturally-occurring examples are given in (74).

Extra Info

The thorn character (þ) was borrowed from Old Norse (it is still used in Icelandic) and used in Old and Middle English where we use <th> today. The yogh character (ȝ) was used in Middle English where we use <g> or <y>.

(73) * { after, before, since } that the war ended

(74) a. And *after þat* þis bataile was done, þe Britons assemblede ham
‘And after this battle was over, the Britons assembled (themselves).’
(PPCME2, cmbrut3,100.3080)

b. Ȝit *bifore that* Dauith cam to Jerusalem, a new debate roos bitwixe the men of Israel and the men of Juda
‘Yet before David came to Jerusalem, a new debate arose between the men of Israel and the men of Judah.’ (PPCME2, cmpurvey,I,11.415)

c. Now, *sith that* i have toold yow of which folk ye sholde been conseilled, now wol I teche yow which conseil ye oghte to eschewe.
‘Now, since I have told you what kind of people you should be counseled by, now I will teach you which advice you ought to eschew.’ (PPCME2, cmctmeli,223.C1.247)

Finally, analyzing clausal complements of prepositions as CPs allows us to treat prepositions in English in the same way as prepositions in other languages such as French, where it is clear that the clausal complements are CP complements.[7]

(75) a. ✓ {*après*, *depuis*, *pendant*} la danse
after since during the dance
‘{ after, since, during } the dance’

b. ✓{*après*, *depuis*} que l' enfant a dansé; *pendant* que l' enfant dansait
after since that the child has danced while that the child was.dancing
'{ after, since } the child danced; while the child was dancing'

c. * { *après, depuis* } l'enfant a dansé; *pendant* l'enfant dansait

A broader conclusion that we don't have time to defend here is that all finite IPs are embedded in CPs, generally.

6.4 Crosslinguistic variation in headedness

Recall the overall project: we are building a model of Universal Grammar.

(76) **Universal Grammar (UG)**

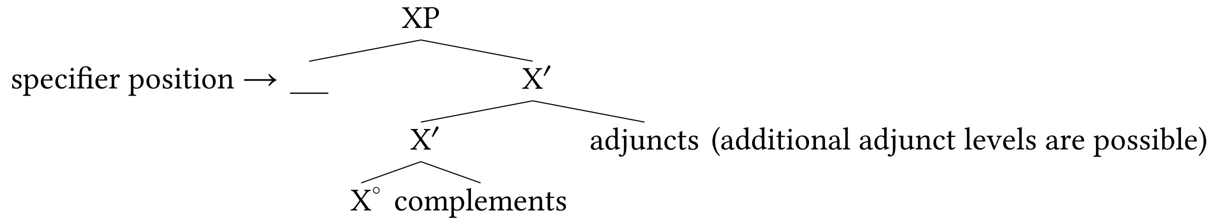
The universal cognitive mechanisms that generate human language, as discoverable through language grammars.

So while we investigate the mental grammars of individual speakers' individual languages, what we are attempting to do overall is use those grammars to theorize about what is universal about human language. What are the universal properties of human cognition that allow for all human languages to exist?

There is of course a very important point that is thankfully not lost on most linguists: languages are in fact different from each other. Syntacticians use the notion of a **parameter** to capture this. Parameters are points of variation between language. At earlier stages of the generative syntax tradition, the model was referred to as "Principles and Parameters", where the principles were the properties of Universal Grammar that were universal, and parameters were the properties of universal grammar that varied between languages.

Traditionally parameters have been thought of in the metaphor of a switch, like a light switch—they are binary (on or off) and all existed cognitively in some kind of literal sense. Currently there are many people who conceive of parameters as something more like points of underspecification. That is to say: principles say what must be the case for human language, and wherever the principles don't have an opinion, languages can evolve in any way that is otherwise consistent with UG. No matter how parameters are conceived of from a cognitive perspective, within syntactic theory parameters are ways that we state (with precision) what kinds of syntactic variation occur between languages.

(77) X′-schema



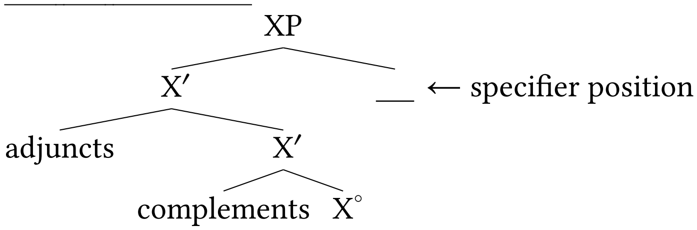
The components of X′-syntax are defined hierarchically, not by their linear order. So we can summarize the defining characteristics of complements, adjuncts, and specifiers as in (78).

(78) X′-schema

- Complements: siblings of heads
- Adjuncts: siblings of intermediate projections and children of intermediate projections
- Specifiers: siblings of intermediate projections and children of maximal projections

Notably, trees that adhere to (78) can look quite different from (77). So the schematic in (79) is just as faithful to the X′ schema, but produces a very different word order.

(79) Still the X′-schema



And any permutation of these orderings is consistent with the X′ schema as well. The model we are building has strict restrictions on the hierarchical relationships between different kinds of syntactic elements, but says nothing specific about linear order. So one of the first areas of human language to be understood in the context of parameterization (though certainly not the last) is the relative order of components of the X′ schema. Here we illustrate with heads and their complements.

6.4.1 Head-final and head-initial patterns

As illustrated in (80)–(86), heads in English precede their complements, and English is therefore said to be a **head-initial** language. The **headedness** of a lexical item (or a category or an entire language) always refers to the order of heads and complements. In other words, headedness is determined with respect to the intermediate projection of elementary trees, not with respect to the maximal projection. For instance, English determiners can be medial in their maximal projection (the possessive morpheme *'s* must be preceded by a DP in Spec(DP)), and English verbs and modals must be medial in their maximal projections, but they all count as head-initial because they are all instances where heads precede their complements.

(80) English head-initial verbs

a. They [$_{\text{V}'}$ **pursued** [$_{\text{DP}}$ their goal]].
b. She [$_{\text{V}'}$ **submitted** [$_{\text{DP}}$ her application]].

(81) English head-initial Inflection
a. They [$_{\text{I}'}$ **should** [$_{\text{VP}}$ pursue their goal]].
b. She [$_{\text{I}'}$ **could** [$_{\text{VP}}$ submit her application]].

(82) English head-initial complementizers
a. They agreed [$_{\text{C}'}$ **that** [$_{\text{IP}}$ they should pursue their goal]].
b. She wondered [$_{\text{C}'}$ **if** [$_{\text{I}'}$ she could submit her application]].

(83) English nouns precede their complements
a. the [$_{\text{N}'}$ **pursuit** [$_{\text{PP}}$ of their goal]]
b. the [$_{\text{N}'}$ **submission** [$_{\text{PP}}$ of her application]]
c. Lisa's [$_{\text{N}'}$ **pride** [$_{\text{PP}}$ in her work]]

(84) English determiners precede associated NPs
a. [$_{\text{D}'}$ **the** [$_{\text{NP}}$ pursuit of their goal]]
b. [$_{\text{D}'}$ **the** [$_{\text{NP}}$ submission of her application]]
c. Lisa [$_{\text{D}'}$ **'s** [$_{\text{NP}}$ pride in her work]]

(85) English adjectives precede their complements
a. She is [$_{\text{A}'}$ **proud** [$_{\text{PP}}$ of her work]].
b. He is [$_{\text{A}'}$ **fond** [$_{\text{PP}}$ of his children]].

(86) English prepositions precede their complements
a. [$_{\text{P}'}$ **over** [$_{\text{DP}}$ the next five years]]
b. [$_{\text{P}'}$ **with** [$_{\text{DP}}$ great fanfare]]

Universal Grammar by no means prescribes head-initial phrase structure. Rather, many languages exhibit consistently **head-final** phrase structure; two such languages are Japanese and Korean. The examples in (87)–(92) are from Korean. In order to avoid using "head-final preposition", which is an etymological contradiction in terms, linguists have coined the term **postposition** for the Ps in (92). The term adposition is a cover term for prepositions and postpositions (that is, for Ps regardless of headedness). Examples for I and D are missing because Korean has neither overt modals of the English sort nor overt articles; the abbreviations in the glosses are explained in the notes,[8] but are not crucial for present purposes.

(87) Korean head-final verb phrases
a. 그들 -은 목적 -을 **추구하** -였 -다.
kutul -un [$_{\text{V}'}$ [$_{\text{DP}}$ mokcek -ul] **chukwuha**] -yess -ta.
3PL -TOP goal -ACC pursue -PST -DECL
'They pursued their goal.'

b. 그 -는 지원서 -를 **제출하** -였 -다.
ku -nun $[_{V'}$ $[_{DP}$ ciwonse -lul] **ceychwulha**] -yess -ta.
3SG -TOP application -ACC submit -PST -DECL
'They.SG submitted their.SG application.'

(88) Korean head-final complementizer
그들 -은 목적 -을 추구해야 한 **-다고** 통의하 -였 -다.
$[_{C'}$ $[_{IP}$ kutul -un mokcek -ul chukwuhayya ha-n] **-tako**] tonguyha -yess -ta.
3PL -TOP goal -ACC pursue must-PRS that agree -PST -DECL
'They agreed that they should pursue their goal.'

(89) Another Korean head-final complementizer
그 -는 지원서 -를 제출해도 되 **-는지**
$[_{C'}$ $[_{IP}$ ku -nun ciwonse -lul ceychwulhayto toy] **-nunci**]
3.SG -TOP application -ACC submit can if
궁금하 -였 -다.
kwungkumha -yess -ta
wonder -PST -DECL
'They.SG wondered if they.SG could submit their.SG application.'

(90) Korean nouns following complements
a. 그들 -의 목적의 **추구**
kutul -uy $[_{N'}$ $[_{DP}$ mokcek -uy] chukwu]
they -GEN goal.GEN pursuit
'their pursuit of their goal'
b. 그 -의 지원서 -의 **제출**
ku -uy $[_{N'}$ $[_{DP}$ ciwonse -uy] **ceychwul**]
3.SG -GEN application -GEN submission
'their.SG submission of their.SG application'
c. 경림 -의 일 -에 대한 **자부심**
kyenglim -uy $[_{N'}$ $[_{PP}$ il -ey tayhan] **capwusim**]
Kyenglim -GEN work -in regarding pride
'Kyenglim's pride in their.SG work'

(91) Korean head-final PPs
a. 다음 오 년 **동안**
$[_{P'}$ $[_{DP}$ taum o nyen] **tongan**]
next five years over
'over the next five years'
b. 대 팡파르 **-와 함께**
$[_{P'}$ $[_{DP}$ tay phangphalu] **-wa hamkkey**]
great fanfare -with
'with great fanfare'

(92) Korean adjectives following head nouns

a. 일 -이 **자랑스러** **-운** 사람
[$_{\text{A}'}$ [$_{\text{DP}}$ il -i] **calangsulew** **-un**] saram
work -NOM proud -MOD person
'a person proud of their.SG work'

b. 아이들 -이 **좋** **-은** 사람
[$_{\text{A}'}$ [$_{\text{DP}}$ aitul -i] **coh** **-un**] saram
children -NOM fond -MOD person
'a person fond of their.SG children'

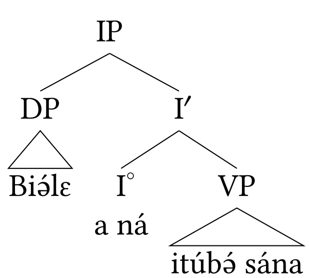
6.4.2 Harmonic and disharmonic headedness

Languages tend to be **harmonic** with respect to headedness; that is, they tend to be consistently head-initial or head-final across syntactic categories. This is what is exemplified by the English and Korean examples above, and it is very common for languages to pattern like this. However, in certain languages, some syntactic categories project head-initial trees and others project head-final ones.

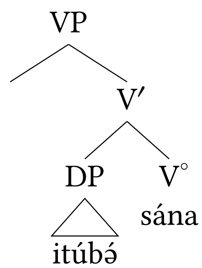
Such mixed phrase structure is found, for instance, in many languages in West and Central Africa. One example is Tunen, a Bantu language spoken in Cameroon (Zone A).[9]

(93) *Context: You enter the room and see a broken window. Someone announces…*
Biə́lɛ a ná [$_{\text{VP}}$ itúbə́ sána] Tunen
1.Pierre SBJ.AGR PST window break
'Pierre broke the window.' (Kerr 2024b, 212)

Even this single example gives a very good illustration of mixed-headedness. We have bracketed the verb phrase in the example above. If we assume that the inflection of past tense emerges on I°, we can see that the verb phrase follows tense: that is, IP is head-initial, where the head (bearing tense) precedes its complement (the verb phrase).

(94)


At the same time, the story is different inside the verb phrase. Clearly in (93), the head (the verb *sána* 'break') follows its complement (the object *itúbə́* 'window'). The Tunen verb phrase therefore appears to be head-final:

(95)


Future tense illustrates the same pattern, with the future tense morpheme preceding its VP complement.

(96) Samuɛ́lɛ a ŋɔ $[_{VP}$ ɔ-lɛ́sa nɛ́á-aka] Tunen
1.Samuel AGR FUT 3-rice eat-DUR
'Samuel will eat rice tomorrow.' (this future tense is the 'tomorrow' future)
(Kerr 2024a, 326)

Both demonstratives and prepositions precede their respective complements, so we see Dem-N word order (97) and P-DP word order (98).

(97) tɔ́ɔyɛ tɔ-banána tɔ-tɛ́^{L}tɛ́á tɔ-fítitiə tɔ́-fandɛ́
DEM NC-banana AGR-small AGR-black AGR-two
'these two small black bananas' (Kerr 2024b, 246)

(98) *Context: Where are you?*
Mɛ lɛa $[_{PP}$ ɔ nɛ-oní].
AGR be PREP NC-market
'I am at the market.' (Kerr 2024a, 102)

Head nouns, however, tend to precede modifiers of those nouns, as we can see in (97). Head nouns likewise precede their complements in Tunen, distinct from verbs following their complements.

(99) *Context: "They—they are—I see three people here."*
Mmh. Bá lɛa ɔ ɛ-ndínə́ yɛ bɔ-lɛa.
mmh AGR be PREP NC-foot ASSOC NC-tree
'Mmh. They're at the foot of the tree.' (Elisabeth Kerr, personal communication)

Complement clauses introduce some additional complexity. Inside the complement clause, we see the familiar head-initial IP and head-final VP. But in the main clause, the CP complement of *manya* 'know' follows the verb, in an exception to the head-final VP pattern we saw above.

(100) Mɛ ^{H}ndɔ **manya** $[_{CP}$ ɔwá Matɛ́ŋɛ a ka hɛ-əfulə **fana-aka**].
SM.1SG PRES know COMP Martin AGR PAST book read-DUR
'I know that Martin has read the book.' (Kerr 2024b, 208)

And in the embedded CP itself (bracketed in (100)), we see that the complementizer precedes the IP it is associated with, suggesting that CP itself is head initial, in contrast to Korean CPs in (88) and (89) above.

Tunen shows much more complexity in word order based on what is being emphasized and de-emphasized in a sentence (as is the case for many languages). But as an initial description, it is clear that headedness need not be completely harmonic among all categories in a language. It is certainly common to see patterns like Korean or English that are relatively consistent in their headedness, but it is also not surprising among human languages to find patterns like in Tunen, where some projections are head-initial but others are head-final.

Encountering new grammatical categories 6.5

As mentioned in Chapter 2, languages can vary in the particular inventory (and properties) of grammatical categories that they possess. What was presented so far here has a very universalist bias ("languages work like this"); how do we deal with potentially unique kinds of syntactic properties in languages we are investigating? First, we will start with some relevant assumptions:

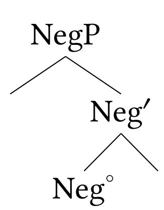
(101) Important X′-theoretic assumptions:
   a. All heads (X°) project to phrases (XP)
   b. All category information resides on a syntactic head.
   c. All categories project full X′-structure.
   d. Where an XP is located in the syntactic structure is determined empirically—where do the data suggest it is located?

Given these assumptions, we are given a toolkit for building an analysis for a novel (to us) grammatical category. As an illustration, we work through this process with a category that is novel to us thus far in this textbook: negation.

As is common across languages, English has a morphosyntactic construction—negation—that inverts the truth conditions of a sentence. So (102a) is true in exactly the opposite circumstances that (102b) is true in.

(102) a. The children will leave tomorrow.
   b. The children will not leave tomorrow.

As a baseline assumption, negation is a grammatical category with predictable properties of its own, and should be its own syntactic head. We can call it Neg°, which projects to a NegP.

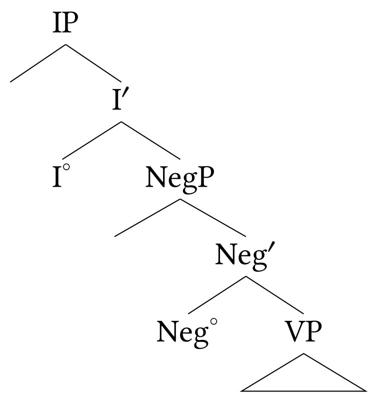
(103)


Where does it sit syntactically, though? We can look at it empirically. (102) suggests that English negation sits right between IP and the VP; if this is true, other elements that are in IP should precede negation as well. (104) suggests that this is true.

(104) a. The children might not leave tomorrow.
b. The children should not leave tomorrow.

We can therefore represent the structure of English negation in an abstract sentence structure as in (105):

(105)



We don't mean to imply that any initial analysis based on basic empirical patterns will turn out to be correct; it is the pattern of normal science for a researcher to propose a structure that is later contested/revised as additional researchers identify additional relevant data. So the appropriate analysis in the end is a matter of what holds up to empirical scrutiny. But this is the general analytical approach that the X′ schema makes available: we expect any new grammatical category X to project a full XP, and where that XP sits in a sentence depends on what the empirical evidence suggests.

New Empirical Observations

- head-initial and head-final patterns
- harmonic and disharmonic headedness

Components of the Theoretical Model

- New phrase types:
  - NP
  - AdjP
  - PP
  - NegP

- ▶ Important X′-theoretic assumptions:
  - All heads (X°) project to phrases (XP)
  - All category information resides on a syntactic head.
  - All categories project full X′-structure.

Notes 6.6

1. Sentences and noun phrases also exhibit certain differences. First, arguments and modifiers are not always expressed in exactly the same way across the two categories. For instance, the agent argument is expressed as an ordinary noun phrase in a sentence like (1a), but as a possessive noun phrase in a noun phrase like (1b). In a sentence, the theme argument is expressed as a noun phrase, but in a noun phrase, it must be part of a prepositional phrase, usually an *of*-phrase in English. Finally, although verbs and nouns can both be modified by prepositional phrases, verbs are modified by adverbs, whereas nouns are modified by adjectives (with some exceptions in both directions). In connection with this last difference, notice that adverbs can precede or follow the verb they modify, whereas adjectives (in English) are ordinarily restricted to prenominal position.

(i) Adverb

a. The kids *regularly* <u>donate</u> their old toys.

b. The kids <u>donate</u> their old toys *regularly*.

(ii) Adjective

a. the kids' *regular* <u>donation</u> of their old toys.

b. *The kids' <u>donation</u> their old toys *regular*.

A further and even more fundamental difference between sentences and noun phrases concerns the subject requirement. As we saw in Chapter 4, all sentences require a syntactic subject, even when it does not correspond to a semantic argument, as is evident from the contrast between (iii) and (iv).

(iii) a. ✓ *It* appears that the manuscript has been found.

b. ✓*There* exists a solution.

(iv) a. *Appears that the manuscript has been found.

b. * Exists a solution.

By contrast, noun phrases never require a subject. For instance, the agent argument of a noun can be expressed, but it needn't be, as shown in (v).[10]

(v) a. ✓ the committee's criticism of the proposal

b. ✓ the criticism of the proposal

What is even more striking is that sentences with expletive subjects have no noun phrase counterparts. As (vii) shows, the very expletive expressions that are obligatory in (iii) are ungrammatical in noun phrases. Notice also the related contrast in (vi); the construction in (vi.a)—so-called raising—is discussed in Chapter 11.

(vi) a. ✓The manuscript appears to have been found.
b. * the manuscript's appearance to have been found.

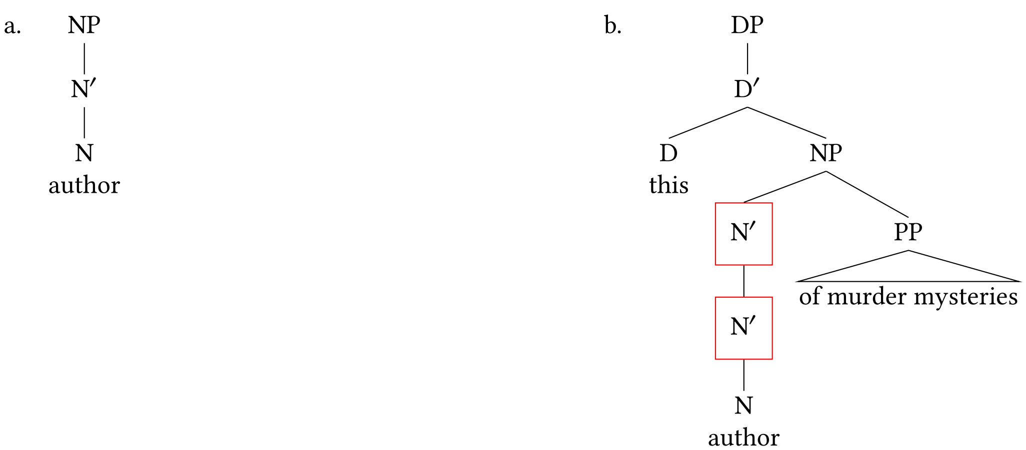
(vii) a. **it(s)* appearance that the manuscript has been found
b. **there('s)* existence of a solution

2. We borrow the term "traditional noun phrase" from Željko Bošković to describe the constituent commonly referred to as a noun phrase irrespective of its underlying analysis.

3. More evidence for the idiosyncratic character of the constraint against (49a) comes from the acceptability of (6.6) (at least in formal registers) (thanks to Sonali Mishra for drawing our attention to such examples).

(i) such a one

4. Some speakers, however, accept (55b), or at least do not completely reject it. How can we make sense of this variation among speakers' judgments? Recall that complements of nouns, unlike those of verbs, are always expressed as prepositional phrases. This means that the evidence whether a particular phrase is a complement or an adjunct is murkier in the case of nouns than in the case of verbs, both for children acquiring the language and for adult speakers. A further, probably related, complication is that even nouns that are morphologically derived from obligatorily transitive verbs are themselves optionally intransitive (for instance, compare *consume, destroy, employ* with *consumer, destroyer, employer*). Moreover, the intransitive use of these nouns might be more frequent than their transitive use. As a result, the mental grammar of some speakers might include only the intransitive elementary tree in (i), and not the transitive elementary tree in (53a). Such speakers would have no way of deriving the structure in (53b), but they would be able to derive the alternative structure in (105b) by adjoining the *of* phrase, rather than by substituting it.

(i)


For such speakers, *author* in (55b) would be an N′, rather than an N, and so they would accordingly accept (55b).

Notice furthermore that the intransitive elementary tree in (i) is available even for speakers whose mental grammar includes the transitive elementary tree in (53a), since all speakers of English accept (ii).

(ii) ✓ this *author* and that *one*

If some of these speakers allow the *of* phrase to adjoin into the intransitive elementary tree in addition to substituting into the transitive one, then they, too, would judge (55b) to be acceptable (at least marginally so).

5. Strictly speaking, this statement is true only of *fond* in predicative position, not in prenominal position.

(i) Predicative: *Their parents are fond.

(ii) Prenominal: ✓their fond parents

6. In traditional grammar, Ps that take clausal complements are classified as subordinating conjunctions (along with *if* and *that*), but as in the case of intransitive Ps, we again reject the traditional approach. First, it is conceptually uneconomical. Specifically, it expresses the difference between (69a) and (69b) in terms of the syntactic category of the heads (preposition versus subordinating conjunction), which redundantly encodes the difference in the syntactic category of the complement (noun phrase versus clause). Second, the items in (69) share roughly the same semantic content, regardless of the categorial status of their complement. In contrast, *if* and *that* are relatively contentless and give the impression of functioning purely as "grammatical glue". In the approach that we are advocating, the distinction between P and C corresponds to the distinction between contentful and contentless subordinating conjunctions.

7. In (73), some French speakers prefer or require the subjunctive form of the auxiliary (*ait*) rather than the indicative form (*a*). For present purposes, this variation, which is comparable to that found in English between *If I was a rich man* and *If I were a rich man*, is irrelevant.

8. ACC = accusative (case of direct object),
GEN = genitive (possessive case),
DECL = declarative clause,
MOD = modifier,
PRS = present tense,
PS = person,
SG = singular,
TOP = topic of sentence.

9. The data discussed in this section come from Kerr (2024a), with the foundational work on Tunen coming from Dugast (1971) and Mous (1997, 2003, 2005, 2014). Additional overview discussion of the word order of languages of this region can be found in Sande, Baier, and Jenks (2019). Glosses have been simplified for expository purposes; see original sources for more precise glossing. The pattern reported here was first addressed in the generative literature by Hilda Koopman for Vata and other Kru languages (Koopman 1984).

10. It is not just the expression of agent arguments that is freer in noun phrases than in sentences. As the contrast between (i) and (ii) shows, the same is true of theme arguments.

(i) a. ✓ The mills employed thousands; their practices damaged the environment.
b. * The mills employed; their practices damaged.

(ii) a. ✓ an employer of thousands; the damage to the environment
b. ✓ an employer; the damage

This has potential consequences for the acquisition of *one* substitution, as we discuss later on in the chapter.

6.7 Exercises and problems

Exercise 6.1: Tree-drawing practice: adverb phrases

Draw trees for the adverb phrases below using X′-syntax, using no triangles. You will need to decide on a grammatical category for morphemes like *very* and *too*.

(1) a. very quietly
  b. too loudly
  c. much too loudly
  d. very much too loudly

Exercise 6.2: Tree-drawing practice: noun phrases (DPs)

Draw trees for the noun phrases (DPs) below using X′-syntax, using no triangles. Like the previous exercise, you will need to come up with an analysis for words like *very*.

(1) a. the fuzzy fleece
  b. the old red ugly Christmas sweater
  c. this very spicy salsa
  d. the frying pan with a loose handle
  e. the cookbook with notes in the margins
  f. the bottle of hoisin sauce in the back of the fridge.
  g. a bright yellow bottle of mustard
  h. the very fluffy puppy

Exercise 6.3: Tree-drawing practice: possessive noun phrases (DPs)

Draw trees for the noun phrases below using X′-syntax, using no triangles.

(1) a. the puppy
  b. the puppy’s collar
  c. my mother
  d. my mother’s puppy’s collar
  e. the puppy’s right ear

Exercise 6.4: Tree-drawing practice with sentences A

The examples here are the same as were given in Chapter 5, Exercise 5.1. Revisit these sentences, drawing trees using no triangles.

1. The very clever student will study every night in the library.
2. Tired students will always take naps before class.
3. Khydence found a book about Maya Angelou in the park.
4. Babies run rather slowly.
5. Lucca's incessant talking always annoys Maisha.

Exercise 6.5: Tree-drawing practice with sentences B

The examples here are the same as were given in Chapter 5, Exercise 5.2. Revisit these sentences, drawing trees using no triangles.

1. Langston wants Maisha to come to school today. (assume *to* is Infl)
2. Preston said that Maisha will eat a hamburger for dinner.
3. That Maisha worked for the whole night might intimidate Preston.
4. Anabella thinks that Lucca should thoroughly clean the kitchen.
5. Preston suggested that Victor should write a joke on the blackboard.
6. Many culinary students believe that their exams should be eaten.
7. The baker kneaded the dough with impressive force.

Exercise 6.6: Tree-drawing practice with sentences C

The examples here are the same as were given in Chapter 5, Exercise 5.3. Revisit these sentences, drawing trees using no triangles.

1. The dog jumped over the ditch.
2. An aspiring doctor should study hard every day.
3. That the faucet is dripping might upset your father.
4. It might anger your mother that you turned up the thermostat.
5. The chef carefully stirred the sauce.
6. Every dog knows the location of their treats.
7. Lucca might never play his music in public.
8. The students might play games at recess today.
9. The latest article in the *New York Times* explains the current international crisis.

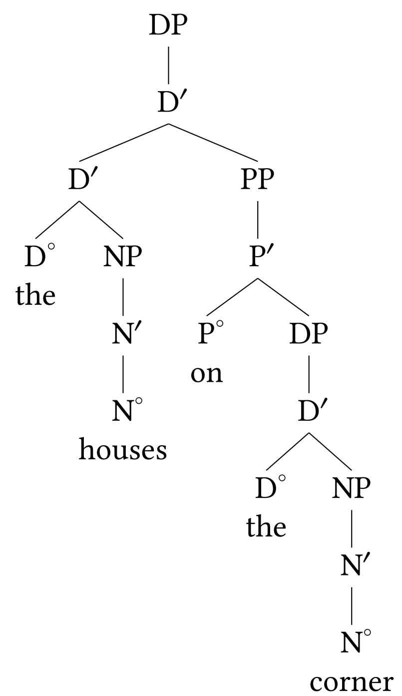
Exercise 6.7: Examining predictions structures make

Formally, the structures in (1) are consistent with X′ theory. Empirically, however, they are unsatisfactory representations because they are inconsistent with certain linguistic judgments. What are those judgments?

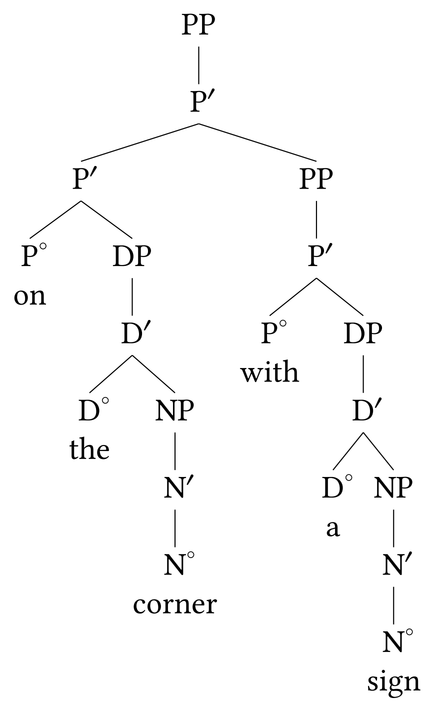
Take Note

In the evidence that you provide, you should feel free to replace *the* with other determiners and plural nouns with singular nouns.

(1) a.



b.



Exercise 6.8: Structure of possessive noun phrases

A. Build structures for the (a) examples in (1)–(3). The other examples are there to guide you; you're not being asked to build separate structures for them.

(1) a. the monster's mother's lair
 b. the monster
 c. the monster's mother

(2) a. the hero of the poem's name
 b. the hero
 c. the hero of the poem

(3) a. the mother of the monster's dislike of the poem's hero

b. the monster
c. the mother of the monster
d. the poem
e. the poem's hero

B. Build structures for the noun phrases in (4).

(4) a. yesterday's lecture
b. this week's unseasonably high temperatures

C. Build structures for the noun phrases in (5), and indicate the recursive nodes. (Treat *my* as *I* plus *'s*.)

(5) a. the neighbor's doctor's dog
b. my father's mother's watch

Exercise 6.9: Identifying grammatical categories

A. What is the syntactic category of *hiring* in (1a) and (1b)? Provide evidence for your conclusions.

(1) a. We dislike Kim's impulsive hiring of incompetents.
b. We dislike Kim impulsively hiring incompetents.

B. Build structures for the underlined gerund phrases in (1). Give the full structure for *incompetents*, but feel free to use triangle notation for the proper noun *Kim*.

Exercise 6.10: Structures for ambiguous noun phrases

A. The noun phrase in (1) has two distinct interpretations, which can be paraphrased as in (2).

(1) the houses on the corner with a sign

(2) a. the houses on the corner that have a sign
b. the houses on the corner that has a sign

Build two structures for (1), clearly indicating which structure goes with which interpretation in (2).

B. Give paraphrases for (or otherwise distinguish between) the two interpretations available for (3), and then build the relevant structures, clearly indicating which structure goes with which interpretation.

(3) I enthusiastically recommend this candidate with no qualifications.

C. Repeat (B) for (4).

(4) the mother of the bride's father

Exercise 6.11: Structures for ambiguous sentences A

Thanks to Amy Forsyth for the template for the examples in Exercise 6.11.

A. The sentences in (1) are many-ways ambiguous. Find as many interpretations as you can, clearly describing the relevant situations you have in mind (see (2) for example descriptions). Build trees for each interpretation that you find, clearly indicating which tree is associated with which interpretation.

Hint: Work backwards. First construct possible trees, and then figure out the interpretation from the syntactic structure.

(1) a. The trainer tapped the seal with the ball on its nose.
b. The woman chased the cat with the mouse on the bicycle.

B. Many-ways ambiguous though the sentences in (1) are, they cannot be used to describe the situations in (2), respectively.

(2) a. There is a seal balancing on its nose, and the trainer uses a ball to tap this seal.
b. There is a cat riding a bicycle, and the woman is using a mouse to chase this cat.

Exercise 6.12: Structures for ambiguous sentences B

This exercise harks back to Exercise 1.8 from Chapter 1. Each of these sentences is ambiguous. Relying on your now more sophisticated knowledge of phrase structure, build structures for the possible interpretations of the sentences in (1).

(1) a. Charis will answer the question precisely at noon.
b. Laila will watch the man from across the street.
c. They should decide if they will come tomorrow.

Exercise 6.13: Structures for complement clauses

Build subtrees for the matrix and the subordinate clauses in (1), and indicate how they fit together.

(1) a. The students will solve the problem, though it seems difficult.
 b. The students will solve the problem, though we acknowledge that it seems difficult.

Exercise 6.14: Tuki phrase structure

The data in this problem set are drawn from Biloa (2013).

Tuki is a Bantu language spoken in Cameroon, distantly related to more familiar languages like Swahili and Zulu. Given our model to this point, use the data here to develop an initial analysis of Tuki phrase structure.

- Draw a tree for each of the following sentences from the data set:
  - Example (2a)
  - Example (4a)
- With respect to headedness, is Tuki harmonic, or disharmonic? Explain, referencing specific examples.

Background Info

Tuki has a noun class system, so all nominals fall into a noun class, and agreement with nominals is agreement in noun class. Noun class is not glossed in this data set in order to limit our focus on clause structure, and all agreement in noun class is simply glossed as AGR. Tuki has multiple past tenses and future tenses, which can be assumed to behave similarly syntactically insofar as we are concerned here. All data here are from Biloa (2013), with some glosses adjusted for our current purposes.

(1) a. Atangana a-mu-nyá cuya ibísi aye.
 Atangana AGR-PST$_1$-eat fish morning this
 'Atangana ate fish this morning.'
 b. Atangana a-tó-o-nyá cwí ibísi aye.
 Atangana AGR-NEG-PST$_1$-eat fish morning this
 'Atangana did not eat fish this morning.'
 c. Ngono a-má-súwa tsono ídzo.
 Ngono AGR-PST$_2$-wash clothes yesterday
 'Ngono washed clothes yesterday.'
 d. Pasa a-nû-námbam Vibúfa anénga aye.
 Pasa AGR-FUT$_1$-cook Vegetables evening this
 'Pasa will cook vegetables this evening.'

(2) a. Iyere a-mú-éná vâdzú.
teacher AGR-PST$_1$-see children
'(The) teacher saw the children'.'

b. Vâdzú vá-mú-eńá iyere.
Children AGR-PST$_1$-see teacher
'(The) children saw a/ the teacher.'

(3) a. ì sutú n-aámɛ
AGR belly AGR-my
'my belly'

b. ì-ngùrú aádzíí
AGR-feet those
'those feet'

c. vì-kúndá iívì
AGR-beds these
'these beds.'

(4) a. Mbárá a-mu-údzá éé Putá a-nu-nambám cwí.
Mbara AGR-PST$_1$-say that Puta AGR-FUT$_1$-cook fish
'Mbara said that Puta would cook fish'

b. íyére a-dingám éé vǎdzú va súkúrú váá va-yére.
teacher AGR-loves that children of school his AGR-teach
'The teacher wants his students to study.'

Exercise 6.15: Attachment of embedded clauses

This exercise assumes that you are familiar with the terms "relative clause" and "noun complement clause". If not, see the end of the exercise for a tutorial.

A. Provide evidence whether relative clauses, illustrated in (1), are arguments or adjuncts of the noun they modify.

(1) Relative clauses
  a. The **idea** that Columbus was working with was incorrect.
  b. The **fact** that they have discovered is important.

B. Subordinate clauses of the type illustrated in (2) are traditionally called noun complement clauses.

(2) Noun complement clauses
  a. The **idea** that Columbus was the first European to discover America is incorrect.
  b. The **fact** that they are wrong is lost on them.

Provide evidence whether noun complement clauses are syntactic arguments of the noun or adjuncts. In other words, given the way that the term "complement" is used in X′ theory, is the term "noun complement clause" for these clauses a misnomer?

Background Info

If you are not familiar with the distinction between noun complement clauses and *that* relative clauses, here are two simple diagnostics for distinguishing between them. Our baseline examples of noun complement clauses and relative clauses are listed here:

(i) Relative clauses

(3) The **idea** that Columbus was working with was incorrect.

(4) The **fact** that they have discovered is important.

(ii) Noun complement clauses

(5) The **idea** that Columbus was the first European to discover America is incorrect.

(6) The **fact** that they are wrong is lost on them.

- First, stripping away the complementizer *that* leaves a complete sentence in the case of a noun complement clause, but something incomplete in the case of a relative clause (it feels like there is a gap, as indicated by the underlining).

(iii) a. ✓ Columbus was the first European to discover America.

b. ✓ They are wrong.

(iv) a. * Columbus was working with ___.

b. * They have discovered ___.

- Second, the complementizer *that* can generally be replaced by a *wh*-word in a relative clause, but never in a noun complement clause.

(v) a. * The idea which Columbus was the first European to discover America is incorrect.

b. * The fact which they are wrong is lost on them.

(vi) a. ✓ The idea which Columbus was working with was incorrect.

b. ✓ The fact which they have discovered is important.

Problem 6.1: English prenominal dependents

Noun phrases like those in (1) and (2) show an ambiguity, reported in these examples.

(1) *a Russian teacher*
ambiguous: either someone who teaches Russian, or who is a teacher (of anything) who is of Russian origin/nationality

(2) *a Spanish teacher*
ambiguous: either someone who teaches Spanish, or who is a teacher (of anything) who is of Spanish origin/nationality

That ambiguity disappears in more complex noun phrases like those in (3) and (4).

(3) *a Russian Spanish teacher*
someone who teaches the Spanish language, who is of Russian origin/nationality

(4) *a Spanish Russian teacher*
someone who teaches the Russian language, who is of Spanish origin/nationality

Instructions:

1. Why are the noun phrases in (1) and (2) ambiguous, and why are the noun phrases in (3) and (4) not ambiguous?
2. Draw a tree for the noun phrase in (4).

Problem 6.2: Ambiguity and structure

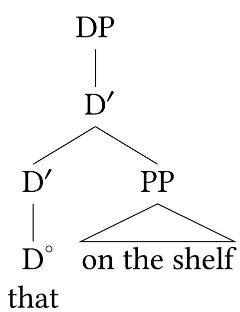
If a given string is structurally ambiguous (that is, it has more than possible structural representation), is it necessarily associated with more than one meaning? Explain, giving examples.

Hint: Exercise 6.10, B.

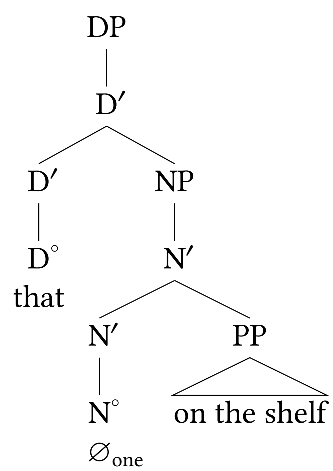
Problem 6.3: DP structure

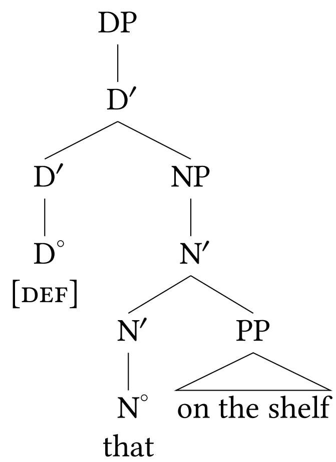
The structures in (1) are intended to represent the second conjunct in (2). DEF in (1c) stands for "definite".

(1) a.



b.



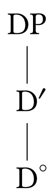
c. 

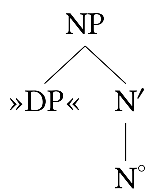
(2) the book on the table, and <u>that on the shelf</u>

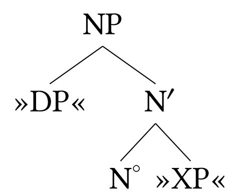
Discuss the relative merits of the three structures in (1). In other words, what considerations (whether empirical or conceptual) make each of the structures attractive or unattractive?

Problem 6.4: DPs or NPs

In (62) of Chapter 1, we mentioned an alternative approach to noun phrase structure different from the one presented in this chapter. According to the alternative approach (updated to be in accordance with the X′ schema), all determiners are intransitive, and all nouns have substitution nodes in the specifier position for (possibly silent) determiners, as shown in (1).

(1) a. 

b. 

c. 

Discuss the relative merits of this approach compared to the one presented in the text. In other words, what considerations (whether empirical or conceptual) would lead you to adopt or reject this alternative approach to noun phrase structure?

Problem 6.5: Determiner phrases

Wouldn't it be simpler to say that determiners (and possessive pronouns) are modifiers, just like adjectives? Then we could continue say that noun phrases are the maximal projections of nouns, just like in the olden days in Chapters 3 and 5. What are the costs and benefits of this approach?

Problem 6.6: Malagasy word order (Austronesian, Madagascar)

Data set provided by Matt Pearson.

Evaluate the evidence below and describe what parameters must exist within our current model in order to accommodate the word orders illustrated here.

- Include a list of precisely-stated parameters that you deem necessary for X′-syntax in order to account for the data set. Parameters can simply be stated in prose (e.g. *complements either precede or follow heads*).
- For each parameter, state what specification the language has (e.g. *In English, complements follow heads*).
- Your response should point out any data that your parameters fail to cover.
  - The idea is that between your list of parameters and your list of patterns/data that your parameters you don't cover, all the data patterns from the data set are addressed.
  - Ideally, your parameters explain everything, but our analyses don't always accomplish that.
  - But you need to be specific in *how* it falls short.
- Draw a tree for each of the following sentences from the data set:
  - Example (4)
  - Example (8)
- Abbreviations: 1s = first person singular, 2s = second person singular, 3 = third person (singular or plural), 1ex = first person plural exclusive, Acc = accusative (i.e. the case form of objects), Irr = irrealis/future tense/mood, Nom = nominative (i.e. the case form of subjects), Pfx = prefix used to form verb stems from roots, Pres = present tense, Pst = past tense.
- Note: Embedded CPs always appear at the end of the sentence they are embedded in; this is a common cross-linguistic pattern, and you are not required to explain that. They are included here to provide you with evidence of the word order of complementizers.

(1) Tonga ilay zazavavy
arrived that girl
'That girl (has) arrived'

(2) M-a-tory ny mpamboly
Pres-Pfx-sleep the farmer
'The farmer is sleeping'

(3) N-am-angy ny ankizy ny mpamboly
Pst-Pfx-visit the children the farmer
'The farmer visited the children'

(4) N-am-angy anay tamin' ny alakamisy ilay mpamboly
Pst-Pfx-visit us on the Thursday that farmer
'That farmer visited us on Thursday'

(5) M-i-hinana ahitra ny omby
Pres-Pfx-eat grass the cow
'Cows eat grass'

(6) N-an-droso ny sakafo ny vahiny izahay
Pst-Pfx-serve the meal the guest we
'We served the guests the meal'

(7) N-an-droso sakafo ny vahiny t-amin' ny lovia vaovao izahay
Pst-Pfx-serve meal the guest Pst-on the plate new we
'We served the guests a meal on the new plates'

(8) Tonga taorian' ny mpamboly ny mpampianatra antitra
arrived after the farmer the teacher old
'The old teacher arrived after the farmer'

(9) N-am-ono ny akoho ny vehivavy
Pst-Pfx-kill the chicken the woman
'The woman killed the chicken'

(10) N-am-ono ny akoho tamin' ny antsy lehibe ny vehivavy
Pst-Pfx-kill the chicken with the knife big the woman
'The woman killed the chicken with the big knife'

(11) M-a-tahotra bibilava ilay lehilahy
Pres-Pfx-fear snake that man
'That man is afraid of snakes'

(12) M-a-tahotra anay ilay lehilahy
Pres-Pfx-fear us that man
'That man is afraid of us'

Problem 6.7: Word order in Japanese (Japonic, Japan)

Data set provided by Xiaoxing Yu

- Instructions:
  1. Describe the linear ordering of the following, specifying whether they precede or follow their siblings:
     - complements
     - adjuncts
     - specifiers
  2. Draw trees for the following sentences:
     - Example (2)
     - Example (9)
     - Example (4)
  3. Are there any data that don't readily fit the overall patterns? If so, specify them here.

- Some background:
  - You can generally assume that the lexical category of a Japanese word is **the same** as its English equivalent. For instance, Shiori-ga 'Shiori-NOM' is a noun. If you want to propose a specific difference, that's fine, just be specific.
  - The suffixes -*ga*, -*o*, and -*no* are case-markers, morphemes that correlate to the grammatical function/role of a noun phrase. **You may ignore these nominal affixes for this problem set.** For example, treat *Shiori-ga* as a noun (N).
    - For reference: the affix -*ga* marks a noun as nominative, which is the form of subjects (NOM); -*o* marks a noun as accusative, which is the form of objects (ACC); -*wa* marks a noun as the topic of the sentence; -*no* marks a noun as genitive (GEN, possessor, like English -*s* or *of*); -*ni* marks a phrase as dative (DAT, indirect objects and other goals).
  - The suffix -*mai* is a classifier (CLF), which is usually obligatory after numbers in Japanese. **You may ignore it for this problem set.** That is, treat *nana-mai* as just the number "seven".

(1) Shiori-wa warat-ta.
Shiori-TOP laugh-PST
'Shiori laughed.'

(2) Masaki-wa Shiori-o mi-ta.
Masaki-TOP Shiori-ACC see-PST
'Masaki saw Shiori.'

(3) Shiori-wa dorayaki-o tabe-ta.
Shiori-TOP dorayaki-ACC eat-PST
'Shiori ate dorayaki.'

(4) Shiori-wa Yuuto-ga dorayaki-o tabe-ta to omotte-i-ru
Shiori-TOP Yuuto-NOM dorayaki-ACC eat-PST that think-PROG-PRS
'Shiori thinks that Yuuto ate dorayaki.'

(5) Tohru-ga sono aoi ie-o mi-ta
Tohru-NOM that blue house-ACC see-PST
'Tohru saw that blue house.'

(6) Kazumi-wa hayaku tabe-ta.
Kazumi-TOP quickly eat-PST
'Kazumi ate quickly.'

(7) Kazumi-wa basutei ni aruk-u.
Kazumi-TOP bus.stop to walk-PRS
'Kazumi walks to the bus stop.'

(8) Kazumi-wa kinou odot-ta.
Kazumi-TOP yesterday dance-PST
'Kazumi danced yesterday.'

(9) Kazumi-wa gakkou de juusu-o nom-u.
Kazumi-TOP school at juice-ACC drink-PRS
'Kazumi drinks juice at school.'

(10) Kazumi-wa Ryōsuke-ga Isao-to kissu shita ato kare-o shikat-ta.
Kazumi-TOP Ryōsuke-NOM Isao-ACC kiss do-PST after him-ACC scold-PST
'Kazumi told off Ryōsuke after he kissed Isao.'

(11) Ano hito-wa Kazumi-ni raburetaa-o nana-mai okut-ta.
That person-TOP Kazumi-DAT love.letter-ACC seven-CLF send-PST
'That person sent Kazumi seven love letters.'

(12) kono osharena wanpiisu
this fashionable dress
'this fashionable dress'

(13) Kazumi-wa zakuro-ga daisuki.
Kazumi-TOP pomegranate-NOM loves
'Kazumi loves pomegranates.'

Problem 6.8: Uyghur word order

Uyghur is a Southeastern Turkic language spoken by about 10 million native speakers in northwestern China, Kazakhstan, Kyrgyzstan, and Uzbekistan, among other places (Yakup 2022).

Examine the Uyghur sentences from Major, Mayer, and Eziz (2023) below, and draw trees for (1), (4), (6), (7), (10), and (16). You may assume the case ending *-ni*/ACC is part of the DP it attaches to and abbreviate all DPs with triangles. You can also assume that tense and agreement arise on the same syntactic head.

Background Info

In what follows, PTCP indicates a participle (a kind of verbal form analogous to 'breaking' in 'The vase is breaking' or 'broken' in 'The vase was broken' in English). FML stands for 'formal'.

(1) Men oqu-d-um.
1SG read-PST-1SG
'I read.'

(2) Men Tursun-ni kör-d-üm.
1SG Tursun-ACC see-PST-1SG
'I saw Tursun.'

(3) Men Tursun-ni ut-t-um.
1SG Tursun-ACC win-PST -1SG
'I beat Tursun.'

(4) Men Tursun-ni kes-t-im.
1SG Tursun-ACC cut-PST-1SG
'I cut Tursun.'

(5) Men Tursun-ni al-d-im.
1SG Tursun-ACC take-PST-1SG
'I took Tursun.'

(6) Tursun bar-i-du.
Tursun go-NPST-3
'Tursun will go.'

(7) Men oqu-ghan i-d-im.
1SG read-PTCP.PST AUX-PST-1SG
'I had read.'

(8) Men ut-qan i-d-im.
1SG win-PTCP.PST AUX-PST-1SG
'I had won.'

(9) Mahinur bar-ghan i-d-i.
Mahinur read-PTCP.PST AUX-PST-3
'Mahinur had gone.'

(10) Men bar-ghan e-mes i-d-im.
1SG read-PTCP.PST AUX-NEG AUX-PST-1SG
'I had not gone.'

(11) Men kél-er i-d-im.
1SG come-PTCP.FUT AUX-PST-1SG
'I would come.'

(12) Men oqughuchi.
1SG student
'I am a student.'

(13) Men oqughuchi e-mes.
1SG student AUX-NEG
'I am not a student.'

(14) Men oqughuchi i-d-im.
1SG student AUX-PST-1SG
'I was a student.'

(15) Mahinur oqughuchi i-d-i.
1SG student AUX-PST-3
'I was a student.'

(16) Men oqughuchi e-mes i-d-im.
1SG student AUX-NEG AUX-PST-1SG
'I was not a student.'

Identifying Puzzles 6.1: Scottish Gaelic word order

Data set provided by Jason Ostrove

Your task is to articulate the puzzle, or the challenge, that is posed to our model thus far by the data provided here. Explain in prose, using relevant jargon and/or tree structures as is helpful, why the data from Scottish Gaelic below don't have an account based on our model thus far.

(1) Phòg Eòghainn Seamus.
kissed Eòghainn Seamus
'Eòghainn kissed Seamus.'
cannot mean 'Seamus kissed Eòghainn.'

(2) Phòg Seamus Eòghainn.
kissed Seamus Eòghainn
'Seamus kissed Eòghainn.'
cannot mean 'Eòghainn kissed Seamus.'

(3) Phòg am fear a' mhuc.
kissed the man the pig
'The man kissed the pig.'

Don't attempt to give a solution! Just identify what the theoretical challenge is that is posed by the data. We will get to solutions later in the textbook.

Identifying Puzzles 6.2: English *one*-replacement

For all speakers of English, the sentence in (1) can have either of the interpretations in (2).

(1) Jane has a big black dog, and Jean has a small *one*.

(2) a. Jane has a big black dog, and Jean has a small *dog*.
b. Jane has a big black dog, and Jean has a small *black dog*.

On the other hand, (3) means only (4a) for most speakers of English. However, some speakers are able to interpret (3) as (4b) (Radford 1988) (such variable judgments among different speakers are conventionally indicated by a percent symbol).

(3) Jane has a big black dog, and Jean has a brown *one*.

(4) a. Jane has a big black dog, and Jean has a brown *dog*.
b. % Jane has a big black dog, and Jean has a *big* brown *dog*.

A. Which of the interpretations of (1) and (3) is problematic? Explain.

B. Can you think of a way of resolving the problem you laid out in your answer to (A)?

Identifying Puzzles 6.3: Tiriki word order

Your task is to articulate the puzzle, or the challenge, that is posed to our model thus far by the data provided here. Explain in prose, using relevant jargon and/or tree structures as is helpful, why the data from Tiriki below don't have an account based on our model thus far. That is, explain the puzzle that these data create for our current theory.

Background:

- Tiriki is a Bantu language spoken in Western Kenya.
- Tiriki's canonical word order is SVO.
- The cardinal numbers in glosses represent noun classes (a form of grammatical gender). They show you what is agreeing with what, but aside from that you can ignore them.
- Canonical word order of Tiriki is SVO.
- SM stands for "subject marker" and is the subject agreement form on the verb.

Assumptions: Assume that subject agreement and tense morphemes arise from I°.

(1) Tsi-ngómbe tsi-ng'w-ere ma-tsi
10-cows 10SM-drink-PFV 6-water
'The cows drank water.'

(2) a. Musa a-tuy-i mu-saala khavere
1Musa 1SM-hit-PST 3-tree twice
'Moses hit the tree twice.'

b. Musa a-tuy-i khavere mu-saala
1Musa 1SM-hit-PST twice 3-tree
'Moses hit the tree twice.'

(3) a. Amukotso a-tekh-i ma-kanda mwisaangaala
1Amukotso 1SM-cook-PST 6-beans in.happiness
'Amukotso cooked beans happily.'

b. Amukotso a-tekh-i mwisaangaala ma-kanda
1Amukotso 1SM-cook-PST in.happiness 6-beans
'Amukotso cooked beans happily.'

Don't attempt to give a solution! Just identify what the theoretical challenge is that is posed by the data. We will get to solutions later in the textbook.

7

Locality of selection

Chapter outline

In this chapter we explore some additional foundational assumptions, taking a break from expanding our conclusions about X′-syntax to explore how selection works in human syntax. After an initial discussion of **c-selection** and **s-selection**, we discuss the VP-Internal-Subject-

Hypothesis (VPISH), which is a significant instance of selectional properties in syntax leading to conclusions that are not always surface evident. The notion of syntactic **movement** is introduced to explain why this structure of the verb phrase is not always surface-evident. After discussing VPISH we establish some foundational assumptions about selection: it is **lexical**, and it is **local**. We then discuss how movement operations can be used to resolve many different kinds of apparently non-local selectional relationships. Both VPISH and movement will be foundational in the chapters to come.

7.1 What is selection?

A central component of syntactic theory is that some elements in syntax are choosy about what they co-occur with. Linguists talk about this in terms of "selection": lexical elements can select for the kinds of linguistic elements they co-occur with (in various ways). We have already been discussing different kinds of selection in the preceding chapters, and in this section we clarify and expand on these notions a bit.

Some examples of selectional restrictions appear to be driven by syntactic considerations, rather than semantic considerations. We will refer to these as instance of category selection, also known as c-selection.

(1) a. The children will eat [$_{NP}$ breakfast] tomorrow morning.
b. The children will eat [$_{NP}$ three bananas] this morning.
c. * The children will eat [$_{CP}$ that the oatmeal is ready.]

Examples like (1c) are clearly ungrammatical: the verb *eat* in English selects for noun phrases and not for whole clauses. But in an example like (1c) it's not straightforward that the problem is a syntactic one, as opposed to a semantic one (perhaps a meaning simply cannot be constructed for the sentence). Examples like (2) help us to address this.

(2) a. I saw happiness everywhere.
b. * I saw happy everywhere.

In (2) the ungrammatical example is much more semantically plausible, but it is still clearly ungrammatical. Why? The verb *see* in English selects for noun phrases (among other things), but it does not select for adjectives. So even though a listener might well be able to discern what is meant by a sentence like (2b), it is nonetheless ungrammatical in English.

There are also selectional restrictions that are semantic in nature, which we refer to as semantic selection, or **s-selection**. Predicates (specifically Fregean predicates) often impose selectional restrictions on one or more of their arguments, depending on the meaning of the predicate. For instance, *drink* imposes a selectional restriction on its theme argument to the effect that the theme argument must refer to a kind or amount of liquid.

(3) a. Amy drank the { lemonade, #sandwich }.
b. Lukas drank a whole { quart, #piece }.

The predicate *elapse* selects a subject that refers to an amount of time.

(4) { Two hours, the remainder of the shift, #two liters, #Preston } elapsed without further incident.

The felicitous use of the verb *murder* requires (among other conditions) that both the AGENT and the THEME arguments refer to humans (and arguably, also other species—see below for related discussion). By contrast, *kill* imposes weaker selectional restrictions, requiring only that the THEME argument refer to a living being, and does not require the cause of the death to be an intentional, volitional agent.

(5) a. The fall from the roof killed the squirrel.
b. # The fall from the roof murdered the squirrel.

(6) The { ✓ paramilitary, #bomb, #avalanche } murdered { ✓their parents, #the olive tree, #their house }.

(7) The { ✓ paramilitary, ✓bomb, ✓avalanche } killed { ✓their parents, ✓the olive tree, #their house }.

Similarly, many people have said or heard the equivalent of (8) at some point (perhaps from a parent, or from a significant other). The predicate *hear* generally only requires a perceiver as subject, but doesn't attribute any active participation/agency. The verb *listen*, on the other hand, requires an agentive, participatory perceiver.

(8) You might be hearing me, but you're not listening to me.

Two points are important to keep in mind concerning selectional restrictions. First, notice that we are using pound signs, rather than asterisks, in (3)–(7); in other words, we are treating the ill-formedness of sentences that violate selectional restrictions as semantic/pragmatic deviance, not as ungrammaticality. This approach is consistent with the fact that selectional restrictions can be flouted for special effect. For example, the ordinary (literal) meaning of *lap up* is 'to drink, usually eagerly, using the tongue, hence especially of animals'. Based on this meaning, we would expect *lap up* to select a nonhuman animate agent and a liquid theme. But although both restrictions are violated in (9), the sentence does not come across as deviant.

(9) The teachers lapped up their students' praise.

Rather, the violation of the selectional restrictions signals to the hearer that the sentence is intended to be taken not literally, but figuratively—as an instance of metaphor. (10) summarizes the kind of reasoning that a hearer of (9) would go through; the reasoning process itself is ordinarily not explicit, but subconscious and lightning-quick.

(10) The teachers lapped up their students' praise?? Whoa there, that's complete nonsense!

It's only nonhuman animals that lap up things. And even then, whatever they're lapping up has to be liquid, not something abstract like praise.

But the speaker seems to know English and be *compos mentis*, so what could they have possibly meant by what they said?

Oh, maybe what they mean is that the attitude of the teachers towards their students' praise resembles the eagerness with which a thirsty animal laps up some welcome liquid.

Word of Caution

In distinguishing figurative from literal uses of language, don't let yourself be confused by the fact that in the vernacular, the adverb *literally* is routinely used to draw attention to a statement's *figurative* character, as when people say things like *My boss literally hit the roof.* In other words, *literally* has come to include the meaning of *figuratively*!

Don't, by the way, conclude from examples like (9) that selectional restrictions are in force only intermittently (in force when language is used literally, but not in force when language is used figuratively). Rather, it is precisely the fact that selectional restrictions are always in force that prompts a hearer of (9) to go through the reasoning process outlined in (10) and to come up with an interpretation in which the selectional restrictions are satisfied at the level of the metaphor.

A second point to keep in mind concerning selectional restrictions is that the criteria for set membership that the restrictions are based on are not always crystal-clear. In other words, sets like liquid things, animate beings, murderers, or murder victims, and so on, are somewhat fuzzy around the edges. Speakers might disagree, for instance, about whether the sentences in (11) are deviant; the disagreement would concern whether the selectional restrictions on *murder* might, on the basis of recent advances in the understanding of animal intelligence, be broadened to include members of species other than *Homo sapiens*.

(11) a. The { chimpanzee, dolphin } murdered the explorer.
 b. The explorer murdered the { chimpanzee, dolphin }.

Fortunately, for our purposes in this chapter, locating the exact boundary between cases that meet selectional restrictions and ones that violate them will not be necessary. (And, in fact, fuzzy boundaries to conceptual categories is a general human phenomenon and is not restricted to s-selectional properties.) The important thing is that selectional restrictions exist, and that there are sentences in which they are clearly met and ones in which they are clearly violated.

The discussion above shows that while it's not always straightforward to identify the source of marginal judgments, there are ways to sort out what is causing the marginality. One final diagnostic does not always apply, but where it does, it is useful for making the point clear. We can call this the *cartoon test*: the idea is that if a sentence sounds odd but you can imagine it happening in a cartoon, then the sentence isn't ungrammatical, but there's something about the lexical items being used that is simply incompatible with the actual world. Consider (12), where (12a) is acceptable, but (12b) is anomalous.

(12) a. The cat jumped on the trampoline.

b. # The flower jumped on the sound wave.

The source of the anomaly here is about semantic selection: flowers are not the kinds of things that can jump (which generally must be volitional and animal/animate), and sound waves are not the kind of things that can be jumped on (we expect a location of some sort based on the semantics of *jump*). But these are also facts about the real world: one can easily imagine a cartoon depicting a scenario like (12b). If a sentence sounds odd in general, but could be a cartoon, it means the sentence is itself grammatical, but there is a problem with s-selection wherein some selectional restrictions have not been met. The cartoon test won't be compatible with all issues with s-selection, but if it does work (like (12b) does), then this is good evidence that there is an s-selectional anomaly in the sentence.

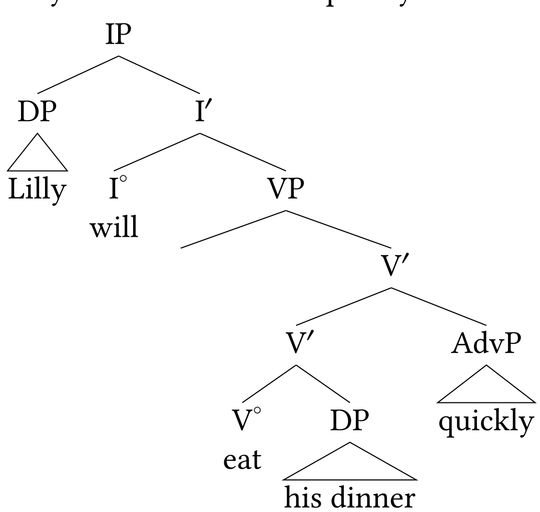
To recap: looking at verbs specifically, we can see that they place selectional restrictions on their arguments. These restrictions can be either syntactic or semantic in nature. **C-selection** (category-selection) refers to the ways in which a verb may require only complements of a specific grammatical category. **S-selection** (semantic selection) refers to how verbs can require certain kinds of semantic properties from their arguments. And in fact, s-selection and c-selection are much broader properties of syntax than just about verbs: any head can place selectional restrictions on phrases it selects (that is, any phrase introduced in its complement or its specifier).

VP-Internal Subject Hypothesis (VPISH) 7.2

Up until this point, we have been placing subjects in sentences where we see them overtly in English: Spec,IP.

(13) a. Lilly will eat his dinner quickly.

b.



What we will see, however, is that there are good reasons to think that subjects (in English and in other languages) don't originate in Spec,IP, even if they are pronounced in that position. This will also address what may have been a nagging worry for some students, that being the empty Spec,VP position in (13b).

7.2.1 Verbs place selectional restrictions on subjects

First, consider the difference between (14a) and (14b):

(14) a. The customers will drink the lemonade.
b. The children will hear the train when it arrives.

The point of the following discussion may seem obvious, but we will build important conclusions from it. In (14a) the subject is an AGENT, as the customers are the volitional, intentional drinkers of the lemonade. The role of the subject in (14b) is different: the children are not an AGENT, but instead are the EXPERIENCER—they may have no choice but to hear the train, and have likely done nothing to cause the sound it is making, so the subject here does not fit the profile of AGENT or CAUSE. This is relatively straightforward because of the verbs in each sentence: *drink* takes an AGENT subject, and *hear* takes an EXPERIENCER subject.

What this makes clear, though, is that the thematic role of the subject comes not from I° but from V°. This also helps explain the observation that the inflection can change without affecting the acceptability of the subject:

(15) a. The customers **will** drink the lemonade.
b. The customers **might** drink the lemonade.
c. The customers **could** drink the lemonade.
d. The customers **are** drinking the lemonade.
e. The customers **drank** the lemonade.

What's more, changing what occupies I° has no effect on the thematic properties of the subjects: in each example in (15), *the customers* remains the AGENT of the sentence. The takeaway here is that despite the overt appearance of subjects in English in Spec,IP, they are clearly *selected* by verbs. And this matches our intuitions as well: when we think about what in (14a) makes the subject an AGENT, we instinctively start discussing the nature of drinking, not the nature of future tense. So already we see reason to think that there is a closer relationship between subjects and verbs than between subjects and tense or inflection (in spite of the fact that subjects in these examples appear adjacent to inflection and not to the verb).

7.2.2 Subjects inside of small clauses

We can see this close relationship between subjects and verbs from a construction that syntacticians call a “small clause”: an instance of predication without the other components of a typical clause structure. In English we can see this with the causative-*make* construction, as in (16):

(16) a. The server made [$_{\text{VP}}$ the customers drink the lemonade.]
     b. The parents made [$_{\text{VP}}$ the children buckle their seatbelts.]

The bracketed phrase selected by English causative *make* is a verb phrase (VP): it contains a verb and its arguments, but crucially not higher sentential material like inflection. So the complement of *make* is ungrammatical if it contains inflection (tense/agreement), as (17) shows.

(17) a. * The server made [$_{\text{IP}}$ the customers will drink the lemonade.]
     b. * The parents made [$_{\text{IP}}$ the children will buckle their seatbelts.]

Small clause complements of *make* show us something crucial about the structure of verb phrases. English causative-*make* selects for VPs, but those VPs necessarily include subjects. So the examples in (16) include the subject inside the VP introduced by *make*, and attempting to leave subjects out is unacceptable.

(18) a. * The server made drink the lemonade.
     b. * The parents made buckle their seatbelts.

The conclusion, then, is that verb phrases contain subjects. So what do we do about instances where the thematic subject of the verb does not appear overtly inside the verb phrase, as with all of the verb phrases we used in the preceding chapter? Here we will use the syntactic concept of **movement**: we will say that the subject is first positioned inside the verb phrase, and **moves** to Spec,IP.

(19) The customers will [$_{\text{VP}}$ ~~the customers~~ drink the lemonade.]

This notion of syntactic movement is highly significant, but at this point very nonspecific and potentially unconstrained. If elements of a sentence can move from one position to another, can anything move anywhere? How do we know where anything is located, structurally speaking? These are important questions and we get much more precise about it as the book proceeds. For now, we will continue to focus our attention on subjects themselves.

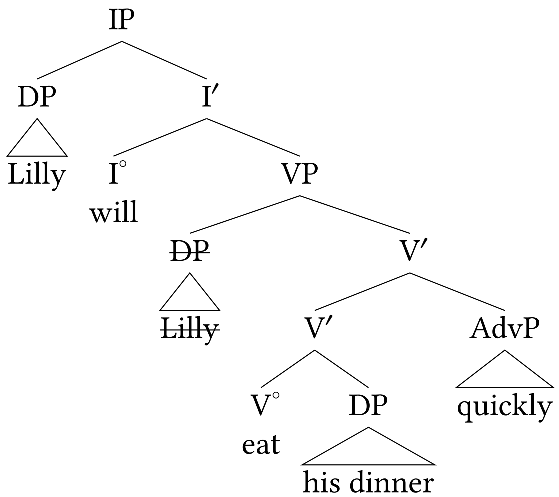
The idea that subjects originate inside the verb phrase (even if they move to higher positions) has become known as the VP-Internal Subject Hypothesis (VPISH).

(20) **VP-Internal Subject Hypothesis** (**VPISH**)
     Subjects originate inside the verb phrase.

This means that in trees for basic sentences with subjects, those subjects have an original position inside the verb phrase, moving from Spec,VP to Spec,IP (in English). So our structure for (13) above will now be as in (21b).

(21) a. Lilly will eat his dinner quickly.

b.



As with any new proposal, we want to see converging evidence from different constructions that this is true. This was already initially suggested by the presence of selectional restrictions from verbs and the structure of small clauses, but more evidence is desirable for an abstract proposal like this. We can see this evidence most explicitly in two different constructions in English: floating quantifiers and low subjects in progressive sentences.

7.2.3 Floating quantifiers show a low position for subjects

Looking at (22) and (23), we can see that the quantifier *all* that modifies the subject can appear "floating" away from the subject (a floating quantifier). As a reminder, a quantifier is a grammatical category that modifies a nominal by quantifying over a set (essentially, counting the members of the set). So *all the customers* means that out of the relevant set of customers, this sentence refers to all of them.

(22) a. All the customers will drink the lemonade.

b. The customers will all drink the lemonade.

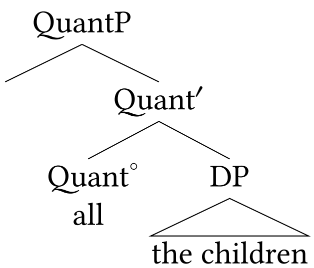
(23) a. All the children will hear the train.

b. The children will all hear the train.

Even in instances where the quantifier doesn't appear adjacent to the subject, as in (22b) and (23b), the quantifier *all* still modifies the subject (that is, the sentence in (23b) is saying that all the children will hear the train, not that the children will hear all the trains, or do all the hearing).

In English, the quantifer *all* co-occurs with a full DP, so we can say that there is a **quantifier phrase** that selects for a DP in these instances.

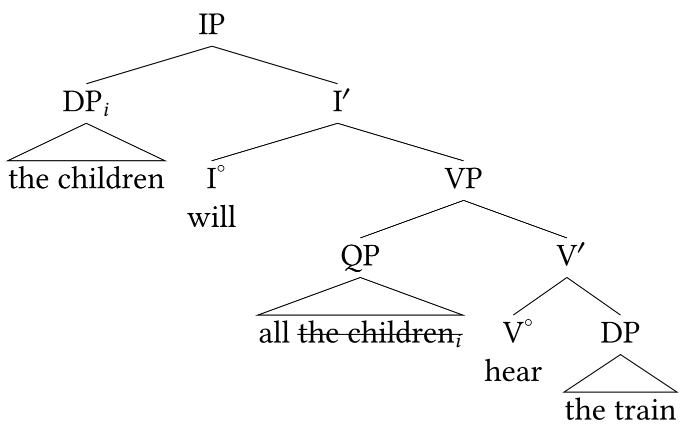
(24)



So what do we say about a sentence like (23b) where the quantifier and the noun phrase it modifies are separated? We can say that the DP has moved out of the Quantifier Phrase, with the result that the quantifier is stranded in its initial position.

(25) a. The children will all hear the train.

b.



This allows us to explain why *all* modifies *the children* (they originate as a constituent), and it also explains why *all* appears in the particular position that it does: it is precisely in Spec,VP, which we now take to be the initial position of subjects. Attempting to position the floating quantifier elsewhere, as in (26), is ungrammatical.

(26) * **The children** will hear **all** the train.

7.2.4 Low subjects in progressive presentational constructions

Another English argument in favor of VPISH comes from progressive sentences, which exceptionally (for English, at least) allow subjects to stay in a low position in the clause. Presentational constructions in English use the expletive *there* as a subject, and the thematic subject can stay lower in the clause. In progressive (*-ing*) constructions, this is available more generally than otherwise is typical for English.

(27) a. the children are eating lunch in the park.

b. [$_{\text{IP}}$ There are [$_{\text{VP}}$ children eating lunch in the park.]]

(28) a. The customers are drinking the lemonade.

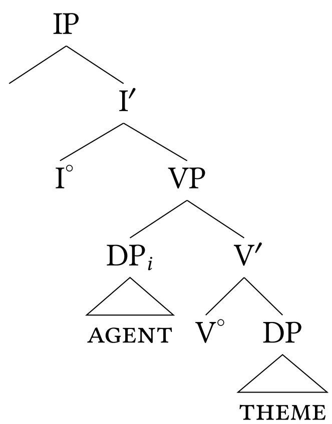
b. [$_{\text{IP}}$ There are [$_{\text{VP}}$ customers drinking the lemonade.]]

Constructions like these might be mysterious if we thought subjects always belonged in Spec,IP. But if subjects of sentences start inside the VP, it is more straightforward to understand why/how subjects can occasionally appear structurally lower than Spec,IP. Here, we insert the expletive *there* into Spec,IP as part of the presentational construction, leaving the subject in Spec,VP.

7.2.5 A summary of VPISH

What we are left with is a perhaps-unexpected conclusion: subjects, which appear in Spec,IP in most instances in English, seem to have a grammatical representation inside the VP. And what we will see in what follows is that this is not a quirk of English: rather, it seems to be a fundamental feature of all human languages.

(29)



We see, then, that the arguments that a verb selects appear to be represented (at least, initially) in a close, local relationship to the verb itself. We will see below in § 7.4 that selection is consistently a local property of syntax.

This raises another point that we will make more clear in what follows below: there are consistent thematic roles assigned to the consistent positions we're noting inside the VP for subjects and objects. More on this below in § 7.6. Notably, this consistency is not a quirk about English, but is a broad crosslinguistic property that has been repeatedly affirmed in a broad range of research, suggesting that VPISH and the structure of thematic roles inside the VP is a significant lesson to be learned about the nature of Universal Grammar.

7.3 Selection is a lexical property

7.3.1 What makes up lexical knowledge?

At a basic level, language connects some linguistic form (a sequence of sounds or signs) with some meaning. This is simplest to talk about at the level of lexical morphemes: for example, *cat*, *dog*, *rain*, *pumpkin pie*. Each of these "words" is made up of a sequence of sounds that correlates with some meaning. So *cat* is pronounced /k^hæt/ and refers to the set of creatures that share certain physical, behavioral, and genetic properties (house cats, wild cats, big cats, and so on).

We know that this knowledge, at some level, simply has to be memorized about a language. We know this because languages can have completely different sequences of sounds or signs to refer to the same creature.

(30)	*cat*	English
	gato	Spanish
	paka	Swahili
	猫 *māo*	Mandarin Chinese
	póéso	Cheyenne
	猫 *neko*	Japanese
	고양이 *koyang'i*	Korean
	kucing	Malay
	बिल्ली *billī*	Hindi
	misto̱n	Mecayapan/Tatahuicapan Nahuatl

Facts like "how does my language refer to small cranky furry creatures with four legs, a tail, and knives on their paws" are simply things that a child has to memorize about the language they are acquiring. So a child learning Swahili will learn that the word for a cat is *paka* whereas a child learning Korean will learn that the word for cat is 고양이 *koyang'i*. These are not complicated facts, and recognizing them doesn't require deep insights about human cognition: children will learn the lexical items that are spoken around them, associating some linguistic form with some meaning. We refer to memorized knowledge about a language as **lexical knowledge**, something which belongs in our mental **lexicon**.

The mental lexicon is the theoretical construct we use to describe the knowledge that must simply be memorized by children. So lexical knowledge is something about a word or morpheme that cannot be explained by a broader pattern of grammar, but is rather something that is learned about an individual **lexical item** like 'cat.'[1] The questions about what makes up lexical meaning become much more complicated when talking about grammatical morphemes, grammatical rules/structures, and other kinds of more complicated lexical structure.

7.3.2 Syntactic heads make up lexical knowledge

In the generative model we are developing, how does lexical (memorized) knowledge compare to the other properties that make up syntax? The baseline assumption that will work for this text is to assume that syntactic heads are lexical knowledge, and X′-syntax and other aspects of the principles and parameters of syntax are predictable based on the general properties of syntax, which is to say Universal Grammar. X′-syntax, by design, predicts what phrase structures will look like based on the general principles of the X′ schema—all heads project to phrases, complements are different from adjuncts and specifiers, and so on. The properties of complements and adjuncts don't need to be specified for individual verbs, because they are consistent across languages.

But what information is stored in a particular head **does** vary from language to language. So many languages may have a syntactic head that holds information about tense inflection (we are using the category Infl, with the head I°). But a given language can show very different kinds of tense distinctions. For example, English displays two basic non-present tense distinctions (past vs. future) but ChiBemba in (32) displays eight such distinctions: four past tenses and four future tenses.[2]

(31) English tense distinctions

a. Past: They worked

b. Future: They will work

(32) ChiBemba tense distinctions (Bantu) (Givón 1972)

a. Remote past
ba-**àlí**-bomb-**ele**
'they worked (before yesterday)'

b. Removed past
ba-**àlíí**-bomba
'they worked (yesterday)'

c. Near past
ba-**àcí**-bomba
'they worked (today)'

d. Immediate past
ba-**á**-bomba
'they worked (within the last 3 hours)'

e. Immediate future
ba-**áláá**-bomba
'they'll work (within the next 3 hours)'

f. Near future
ba-**léé**-bomba
'they'll work (later today)'

g. Removed future
ba-**kà**-bomba
'they'll work (tomorrow)'

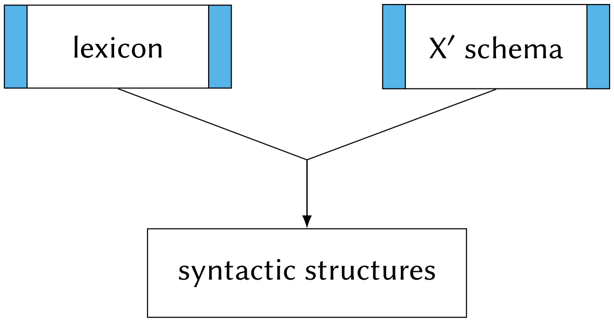
h. Remote future
ba-**ká**-bomba
'they'll work (after tomorrow)'

There is robust variation across languages in the particular time distinctions that tense-marking picks out—so nothing about general principles of grammar explain that. Instead, a child learning a language will learn to associate a specific morpheme with a specific timeline of a particular event: a piece of lexical knowledge. Here, therefore, we expect that the content of I° heads is lexical knowledge, varying by language.

This conclusion generalizes. So not only is the content of N°s and V°s lexical knowledge, but the content of functional projections like I° and C° are likewise lexical knowledge, because these are the parts of syntax that are not predictable. This contrasts with the structures generated by the X′ schema and the operations (like movement) that enter into our model of syntax, which behave consistently and predictably. So syntactic heads are necessarily components of our mental lexicon, but the various structures that are created by them are products of syntax.

This division of labor among parts of our theory is often referred to as the T-model (or Y-model) of syntax. We can illustrate these observations as in (33).

(33) T-model of syntax (to be revised)



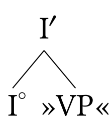
This is essentially a conceptual map, similar to a flow chart or an organizational chart of a corporation/institution. Here we are saying that lexical knowledge and our toolkit for building phrase structure (the X′ schema) combine to produce syntactic structures.

Notably, this is a model of competence, not of performance. So when we say that one thing occurs "before" another, we are not describing the sequence of mental operations needed to say or perceive a sentence. Rather, we are describing the logical structure of a conceptual architecture. That is to say, the model given above describes how different aspects of human language relevant to syntax (core phrase structure rules, lexical knowledge, pronunciation of sentences, interpretation of sentences) must fit together, and what the logical dependencies are between parts of the system. So the lexicon "precedes" syntactic structures because the existence of syntactic structures presupposes the existence of lexical items: you need lexical items to build the syntactic structures. So while we often talk about the "timing" of elements in syntax, within the model this largely correlates to logical dependencies: what elements logically depend on other elements for their existence.

7.3.3 Elementary trees as a tool to represent lexical knowledge

In this textbook we have represented lexical knowledge via elementary trees: the predicted structures based on the particular lexical items under consideration. But crucially, in each case, it is the syntactic head that carries the relevant information: the rest of the structure is predictable based on the properties of X′-syntax. As we've been using them so far, they are displaying c-selectional properties of heads. So our partial elementary tree of I° in English given in (34) shows that I° expects VP to be its complement.

(34)



Elementary trees are not a part of generative syntax theory per se: they are a pedagogically useful way of presenting the c-selectional properties of heads (how those heads will combine with other elements in the syntax). We will continue to use them as they are useful for presenting selectional properties of heads!

Selection is local 7.4

The conclusions we arrived at above (language shows properties of selection, and selection is lexical) lead to another important conclusion: selection is a *local* property. Here we discuss the conceptual and empirical reasons to think this is true.

7.4.1 Conceptual rationale for locality of selection

First is the main conceptual reasons for thinking that selection is local. By conceptual reason, we mean to say that, within the model of syntax we are building, locality of selection is internally consistent with the rest of the conclusions that we have been reaching. If heads are the only "substantial" part of grammar that can be memorized as part of speaker's mental lexicons, then they are the only piece of phrase structure in which we can encode selectional restrictions. All other syntactic structures are generated either by core principles of syntax (ex., the X′-schema) or by operations like movement that affect the foundational structures.

Our model of syntax holds that phrase structures are endocentric, meaning that the properties of phrases are determined from within the phrase itself, and in more precise terms, they are determined by the properties of the head. So, the properties of a VP are defined by V°, the properties of IP are determined by I°, and so on. But correlating with this is the idea that the properties of a given phrase are not defined by some *other* head.[3] So we don't expect the properties of the object of a verb to be determined by I° or C°, for example. The model has no natural way of accommodating that.

So: most consistent with our model as constructed thus far is the assumption that the properties of selection are determined by heads, and therefore expected to be effective locally, inside the projections of that head. Specifically, we expect heads to assert selectional restrictions over complements and specifiers (with adjuncts by definition being optional, unselected elements).

7.4.2 Empirical motivations for locality of selection

And as we would expect if our model is on the right track, the approach to selection that is consistent with our model does match the empirical facts. That is to say, empirically, it just flat-out looks like selection is local. So, for example, it's very common for verbs to place selectional restrictions on their objects, as illustrated above. What we don't see, though, is a pattern like (35) where a verb in present tense can take either a liquid or a solid object, but a verb in future tense can only take a liquid object and not a solid object. In contrast, the selectional properties of verbs often determine the semantic properties of their objects, as (36) illustrates.

(35) Hypothetical (unattested) selection of objects by I°:

a. I always consume sweet tea at picnics.

b. I always consume chicken at picnics.

c. I will always consume sweet tea at picnics.
d. * I will consume chicken at picnics.

(36) Attested selection of objects by V°:
a. I always drink sweet tea at picnics.
b. ?? I always drink chicken at picnics.
c. I will always drink sweet tea at picnics.
d. ?? I will always drink chicken at picnics.

The point being: verbs are choosy about the properties of their objects, as illustrated in (36), and this selection is not affected by changing the tense. But inflection (I°) cannot and does not place selectional restrictions on objects, so patterns like (35) are unattested in English (or elsewhere). We can describe our conclusion as in (37):

(37) **Principle of Local Selection**
Syntactic heads (may) place selectional restrictions on their complements and on their specifiers.

We see this principle playing out in many ways in different places, everything from the existence of VPISH to the empirical patterns where complements (selected) are adjacent to heads but adjuncts (unselected) are outside of complements. And we will see in what follows that these conclusions end up being helpful for us diagnostically as well.

7.4.3 Lessons about selection from idioms

Idiom chunks

Notably, it is not only simple morphemes that can be lexical items. More complex chunks of language can *also* be memorized lexically, and this is transparently evident in the widespread availability of **idioms** in natural language: sequences of lexical items that take on a specific, predictable, non-literal meaning. So *kick the bucket* in (38) can literally mean to use your feet to knock over a pail, but in Mainstream American English it can also be a crass way to refer to someone dying.

(38) Grandpa kicked the bucket.

All languages have idioms, which carry a figurative meaning (like 'die') along with their literal meaning ('kick an actual bucket'). Notably, phrasal idioms are memorized in just the same way that individual morphemes and words are memorized, requiring the exact phrasing for the idiomatic interpretation to be present. So (39) can only mean the literal event of someone kicking an actual physical bucket.

(39) Grandpa kicked the pail.

Idioms are necessarily lexical knowledge—they have to simply be memorized about a specific language. Something that is an idiom in one language is no more or less likely to be an idiom in another. So ‘kick the bucket’ means something about dying in English, but in Mandarin Chinese it has no such figurative meaning, it just means to kick an actual bucket.

(40) 老张 踹 桶子 了。
lǎo-zhāng chuài tǒngzi le.
old-Zhang kick.with.sole bucket LE
‘Old Zhang kicked the bucket’. (He has physically laid his foot on that bucket.)

All languages do have idioms, of course; this is illustrated here for Mandarin Chinese:

(41) a. 我家 老爸, 一点儿 都 不 靠谱。
wǒ-jiā lǎo-bà, yìdiǎnr dōu bǔ kào-pǔ.
my-family old-dad, a.little even NEG lean.on-musical.score
‘My dad isn’t the least bit reliable.’ (*lit.* ‘My dad doesn’t lean on the score at all.’)

b. 你 吃了 熊心豹子胆 啊?
nǐ chī-le xiǒng-xīn-bàozi-dǎn a?
you eat-PFV bear-heart-leopard-gallbladder A
‘How could you be so recklessly fearless?’ (*lit.* ‘Have you eaten a bear’s heart or a leopard’s gallbladder?’)

c. 兰姐 她 是 唯 恐 天下 不 乱。
Lán-jiě tā shì wéi kǒng tiānxià bù luàn.
Lan-elder.sister she be only fear all.under.heaven NEG be.in.disorder
‘Lan just wants to watch the world burn.’ (*lit.* ‘As for Lan, she fears only that all under heaven is not in chaos.’)

Notably, none of the literal translations in (41) have idiomatic interpretations in English. Figurative meanings are lexical (memorized) meanings associated with phrases, and are therefore language-specific.

Even though idioms are language-specific, lexical (memorized) properties of language, they can in fact teach us a fair bit about general properties of phrase structure. If an idiom is a memorized chunk that is phrasal in nature, we still expect it to be a “chunk”—or in more precise terms, a constituent. So a broad pattern across languages is that we see plenty of idioms that are composed of a verb and an object, to the exclusion of the subject.

(42) a. kick the bucket (*die*)
b. spill the beans (*reveal a secret*)
c. screw the pooch (*make a bad mistake*)
d. bite the bullet (*do something hard/challenging*)
e. break the ice (*initiate something*)
f. beat around the bush (*avoid addressing the actual main issue*)

(43) Mandarin Chinese idioms:
a. 靠谱 *kàopǔ*
lean on the musical score (*be reliable*)

b. 摸鱼 *mōyú*
stroke fish (*be deliberately unproductive at work or school*)
c. 出轨 *chū guǐ*
leave the rails (*be unfaithful to a romantic partner*)
d. 大出血 *dà chū xiě*
profusely bleed out (*spend a lot of money at once*)
e. 吃熊心豹子胆 *chī xiǒngxīn-bàozidǎn*
eat a bear's heart or a leopard's gallbladder (*be recklessly fearless*)
f. 唯恐天下不乱 *wéi kǒng tiānxià bù luàn*
fear only that all under heaven is not in chaos (*desire to see drama*)

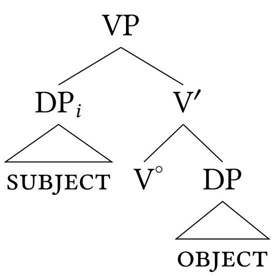
All of these idioms are composed of verb phrase material without reference to subjects—the subject of the idiom can change readily without affecting the idiomatic interpretation (*Mike spilled the beans*, *Maisha spilled the beans*, *Lilly spilled the beans*, etc). So on our model thus far, these would be V′ constituents.

Notably, idioms of this sort do not exist where the subject and the verb are the idiom, and the theme object varies. So something like (44) is unattested cross-linguistically as an idiom.

(44) Imaginary English idiom that doesn't exist:
the dog bit X (to mean *X got tired*)

It is possible for subjects to be included in idioms (as we will see in § 7.5.2), but not in a transitive verb where the object varies.

This pattern of idioms where there are XVO idioms but not SVX idioms is predicted on the model we have been building, where objects and verbs are more closely related to each other than they are to subjects.

(45)


So because there is a verb-object consituent that excludes the subject in our model, it is possible for verbs and objects (and possible adjuncts and other VP-internal material) to be memorized constituents with arbitrary meanings. But with this model there is no subject-verb constituent that doesn't also include the object, so SVX idioms do not occur.

Similar lessons from lexical idiosyncracies

We can draw similar lessons from so-called “”ight verb” constructions—these are constructions that are semi-idiomatic, where a single verb combines with a variety of different kinds of objects to produce many different kinds of meanings. The verb in these instances doesn’t add much to the meaning of the sentence, and in fact means quite different things based on the kind of object it appears with. Examples in English come from verbs like *take* and *throw*.[4]

(46) a. take a bus (= *ride a bus*)
  b. take a nap (= *(to) nap*)
  c. take a pill (= *consume medicine*)
  d. take a chance (= *opportunistically attempt something*)
  e. take a bath (= *bathe oneself in a bath*)
  f. take a shower (= *bathe oneself in a shower*)
  g. take a break (= *(to) rest*)

(47) a. throw a party (= *host a party*)
  b. throw a punch (= *(to) punch*)
  c. throw a fit (= *get very angry, rage about something*)
  d. throw a tantrum (= *immaturely lose control of one’s anger and frustration*)
  e. throw a match/game (= *intentionally lose*)
  f. throw a switch (= *toggle/flip the switch*)
  g. throw an error/exception (= *(of computer code) fail due to an error*)
  h. throw support (behind someone) (= *support someone*)

In each instance, the interpretation of the verb phrase is highly dependent on the choice of the object in a way that is different from non-light-verbs. So a predicate like *drink* will almost always describe consumption of a liquid. But it is difficult to describe the meaning of *take* or *throw* in English without reference to the object. Light verb constructions where there are lexical idiosyncrasies in verb phrase interpretation again suggest a close relationship between verbs and objects, because at times you don’t even know the interpretation of the verb without knowing what its object is.

But, notably, there are no light verbs where the meaning of the verb phrase is determined by the subject and the light verb, with the object varying. (48) imagines an unattested pattern where *get* is the light verb that varies in meaning based on the subject.

(48) Imaginary subject-verb light verb (unattested)
  a. The elf got X. (to mean *X celebrated Christmas*)
  b. The bunny got X. (to mean *X celebrated Easter*)
  c. The flag got X. (to mean *X celebrated Independence Day*)
  d. The tree got X. (to mean *X celebrated Arbor Day*)
  e. The turkey got X. (to mean *X celebrated Thanksgiving*)
  f. The leprechaun got X. (to mean *X celebrated St. Patrick’s Day*)

This of course matches the structure of the verb phrase we've been adopting up until this point. Verbs and objects are more closely related to each other than either of them are related to subjects. Lexical properties of verbs consistently reflect this pattern (including idiosyncratic interpretations in light verb constructions, and idiomatic interpreations). If idiosyncratic information is memorized as a lexical property of language, it is memorized in chunks, in constituents. And so we don't find non-constituent idioms, and the kinds of idioms and lexical idiosyncracy that we do see matches the properties of syntactic structures more broadly.

Why do we care

It's easy to lose the plot amidst all these grammatical patterns. What is the lesson to be learned from these idiosyncratic patterns of light verbs, idioms, and selectional restrictions? There are two connected conclusions. One is what we just said above: verbs and objects appear to be more closely related to each other than either are to subjects. But this reflects the larger lesson: selection, in the sense of lexical heads being choosy about what they combine with, is quite generally a *local* phenomenon. Heads are selective about their complements. Heads can also be selective about their specifiers, but not to the exclusion of their complements: any memorized VP idiom in the lexicon that includes a specifier necessarily includes a specification for the complement also, by virtue of how the X′ schema works.

Resolving apparent non-locality 7.5

We've already seen examples (and we will see more below) of how two elements of syntax that are in a selectional relationship can end up **not** syntactically adjacent. How can we simultaneously claim that selection is local, but selected elements are not always adjacent? As we noted above, the answer is movement, that is, the claim that human syntactic knowledge contains representations of elements in multiple positions. So verbs select subjects and therefore subjects start in VP, but subjects can move to a structurally higher position, creating apparent non-locality. Notably, this is not a psycholinguistic or neurolinguistic claim that constituents physically move around in people's heads. Rather, what we observe is that a model that has elements of syntax represented in multiple positions captures the empirical facts well. In this framework, "movement" is what we call this representation of elements in multiple positions.

7.5.1 *Wh*-movement

One kind of movement that we will outline in much detail later on in the book is what commonly occurs in questions in the world's languages, which we refer to as "*wh*-movement" based on the spelling of question words in English. All speakers of English know that the question word *what* in (49) is interpreted as the THEME object of the verb devour: the answer to the question will be whatever the children consumed.

(49) What did the children devour ___ ?

We know that *what* is selected by *devour* for various reasons. One, *devour* requires an object (cf. **The children devoured.*); there is a gap where *devour* usually takes an object in (49), suggesting that this is where *what* originates. Likewise, the answers to the questions are judged on naturalness in part based on the requirements of *devour*: if we answered the question in (49) with *lunch* it would be unremarkable, but answering *chairs* would be received with confusion, because humans don't devour chairs. The point is that *what* is interpreted as the object of *devour* despite not appearing adjacent to *devour*. We capture this fact with movement in our model—*devour* locally selects for its object (*what* in this instance) and then *what* moves to another position in the sentence.

(50) What$_k$ did the children devour ~~what~~$_k$?

There are various ways that the original position of a moved phrase is annotated. One is to use strikeout of the moved phrase, as we do in (50). Another common way is to represent the lower position with a **trace** of movement, represented with a lowercase italicized *t*.

(51) What$_k$ did the children devour t_k ?

Both (50) and (51) use a subscripted letter (an **index**) to show that the trace/base position is the same phrase that is represented higher: the moved element and its trace are **co-indexed** (are assigned the same index).

There are a host of additional questions about movement, such as how it works and how it is constrained. In earlier theories of generative syntax, for example, the annotation strategies for movement shown above were linked with very different underlying theories of movement. Traces and strikeout are now generally used to annotate the same thing—we tend to choose whichever one best visually illustrates our point in any given discussion. We will engage all of these questions in later chapters: at present we focus our attention on the question of selection.

7.5.2 The positions of subjects

We saw reasons in § 7.2 above to think English subjects are locally selected by the verb (inside the VP) despite canonically occurring in Spec,IP in English. Idioms give us yet another reason to think so. While some idioms (as noted above) are composed of only a verb and an object, some include a whole transitive VP (subject-V-object).

(52) a. All hell broke loose.
Figurative interpretation: Circumstances all of a sudden become chaotic, noisy, or violent.

b. The shit hit the fan.
Figurative interpretation: There is suddenly a lot of trouble.

Our assumptions from above tell us that idioms are constituents in our mental lexicons. Our small clauses from above show us that these are contained in a verb phrase:

(53) a. The Supreme Court decision made [$_{\mathrm{VP}}$ all hell break loose in the states.]
b. The Supreme Court decision made [$_{\mathrm{VP}}$ the shit hit the fan in the states.]

Nonetheless, these idioms can appear in apparently discontinuous contexts, separated by tense and aspect morphology.

(54) a. All hell might break loose.
b. All hell will break loose.
c. All hell could break loose.
d. All hell broke loose.
e. All hell could have been breaking loose, for all we knew.

(55) a. The shit will hit the fan.
b. The shit might hit the fan.
c. The shit could hit the fan.
d. The shit hit the fan.
e. The shit could have been hitting the fan, for all we knew.

This is of course consistent with our conclusions about VPISH above, which is explained if subjects are first generated inside VP, and then move to a higher position (in English).

(56) a. All hell might [$_{\mathrm{VP}}$ ~~all hell~~ break loose.]
b. The shit will [$_{\mathrm{VP}}$ ~~the shit~~ hit the fan.]

So this is another illustration that apparent non-locality of selection is nonetheless compatible with locality of selection, if we assume a theory of movement. Idioms must be constituents in our mental lexicons, and even when they are not surface constituents in a given sentence they can retain their idiomatic interpretations.

7.5.3 Passive sentences involve movement

As we discussed in Chapter 4, the notion of thematic role is distinct from the notion of grammatical function. This is clearly shown by the existence of passive sentences. For instance, in the active sentence in (57a), it is the agent argument that functions as the subject, whereas in its passive counterpart in (57b), it is the theme argument (and the *by*-phrase, a prepositional phrase, is optionally used to introduce the agent).

Background Info

In (57), grammatical functions are indicated above the boxed element and thematic roles are indicated below. The following abbreviations are used:

SBJ - subject, OBJ - object, PP - prepositional phrase, A - agent, THM - theme

(57) a. $\underset{\text{AGENT}}{\overset{\text{SBJ}}{\boxed{\text{The committee}}}}$ drafted $\underset{\text{THM}}{\overset{\text{OBJ}}{\boxed{\text{the letter.}}}}$

b. $\underset{\text{THM}}{\overset{\text{SBJ}}{\boxed{\text{The letter}}}}$ was drafted $\underset{\text{AGENT}}{\overset{\text{PP}}{\boxed{\text{(by the committee).}}}}$

Because the theme of the predicate (the element being drafted) is serving the grammatical role of subject, we can analyze the theme object in this instance as moving to subject position.

(58) The letter was drafted ~~the letter~~.

Chapter 10 gets into passives in much more depth. At present, we want to focus on how passives once again illustrate the same properties of selection that we have been noting in other constructions as well.

Selectional restrictions in passives

As saw above in (6), subjects or objects can violate a verb's selectional restrictions. In what follows, we focus on the objects. For convenience, the relevant facts are repeated in (59).

(59) a. The paramilitary murdered { ✓their parents, #the olive trees, #their houses }.

b. The paramilitary killed { ✓their parents, ✓the olive trees, #their houses }.

Example (60) gives the passive versions of these sentences. As is evident, the acceptability of the subjects in (60) matches that of the objects in (59).

(60) a. { ✓their parents, #the olive trees, #their houses } were murdered by the paramilitary.

b. { ✓their parents, ✓the olive trees, #their houses } were killed by the paramilitary.

This fact would follow straightforwardly if the theme subject of the passive sentence started out as the complement of the verb, where it would have to satisfy the verb's selectional restrictions in the same way as its counterpart does in the equivalent active sentence. Having been cleared, so to speak, it could then move to the subject position, where it appears overtly.

Object idiom chunks in passives

Additional evidence pointing in the same direction comes from so-called **object idiom chunks**, which generally appear as the complements of the verbs that license them (the examples in this section are taken from or modeled on Radford 1988, 422–423). Some examples are given in (61). The object idiom chunks are underlined, and the licensing verbs are *italicized in green*.

(61) a. They { *gave*, *paid* } hardly any heed to our proposal.
b. The Prime Minister *paid* due homage to the dead.
c. The government *keeps* close tabs on their operations.

The restriction of object idiom chunks to the complement position of the licensing verb is thrown into striking relief by the contrast between nearly synonymous expressions, such as *heed* and *attention*. The variants with the ordinary expressions are grammatical, but those with the idiom chunks are not, since they are not licensed by the verbs **bolded in red**.

(62) a. The younger one is always trying to **attract** my { *heed, ✓attention }.
b. According to the government, the situation **requires** { *tabs on, ✓monitoring of } the activists.

Given their licensing requirements, it isn't surprising that object idiom chunks are generally ungrammatical in subject position.

(63) a. * More heed to maintenance would result in lower repair bills.
b. * Tabs won't affect me.

They are, however, able to occur in subject position under one condition—in passive sentences where the passive participle is that of the licensing verb. This is illustrated by the contrast between (64) and the passive counterparts of (61).

(64) a. ✓ Hardly any heed was { *given*, *paid* } to our proposal.
b. ✓ Due homage was *paid* by the Prime Minister to the dead.
c. ✓ Close tabs were *kept* on their operations by the government.

These passive examples show that the same interpretive restrictions that typically apply to objects (whether as part of idioms, or s-selectional restrictions) apply to derived subjects in passives. These are additional instances of apparent non-locality of selection. The syntactic solution to this apparent non-locality is that there is an underlying locality, but that the relevant elements have moved to another syntactic position. That is to say, our mental representations of those sentences include representations of the relevant entity in two separate positions.

As we noted above, we are effectively soft-launching the idea of movement by introducing some basic contexts relevant to selection where movement offers some clarification. But we still need a constrained and precise theory of how movement works (and doesn't work): we will develop this as the book continues.

VSO: An additional argument for VPISH (and UTAH) 7.6

An additional argument for VPISH comes from the existence of languages with VSO word order. The argument here is slightly more complex as it requires a new form of movement not mentioned above; all of these movements will be discussed in more depth throughout the rest of this text.

We saw in Chapter 5 that X′-syntax naturally accommodates many major word order variations crosslinguistically. For example, SOV languages like Japanese can be analyzed with a head-final VP, and VOS languages like Malagasy can be analyzed with a right-facing specifier of IP.[5]

(65) パンダは 竹を 食べる。 **Japanese**
panda-wa take-o tabe-ru.
panda-TOP bamboo-ACC eat-NPST
'Pandas eat bamboo.'

(66) N-am-angy ny ankizy ny mpamboly **Malagasy**
Pst-Pfx-visit the children the farmer
'The farmer visited the children'

Many languages around the world have a canonical word order where the first major constituent in the sentence is the verb, followed by the subject and the object (VSO). VSO is one of the less common "basic" word orders (compared to SVO or SOV, for example). But it is attested regularly across a wide variety of unrelated languages, such as Niuean (Austronesian, Niue Island and New Zealand), Irish (Celtic, Ireland), Turkana (Nilotic, Kenya) , Nandi (Nilotic, Kenya) and the Mayan languages Q'anjob'al (Guatemala/Mexico) and Ch'ol (Mexico). In the following examples, the verb is **bolded in red**, the subject *italicized in green*, and the object underlined in black.

(67) a. **To lagomatai** *he ekekafo* a ia. **Niuean**
FUT help *ERG doctor* ABS him
'The doctor will help him.' (Seiter 1980, 29)

b. **D'oscail** *Seán* an doras. **Irish**
open.PST *Sean* the door
'Sean opened the door.' (Guilfoyle 2000, 66)

c. **ɛ-mɪn-a** *ŋesì* ŋa-atuk' **Turkana**
3-love-A *he* cows
'He loves cows.' (G. Dimmendaal 1983, 62)

d. **ki:ri:p** *la:kwɛ:t* ɑrtɛ̂:t **Nandi**
watched *child* lamb
'The child watched the lamb.' (Creider and Creider 1983, 3)

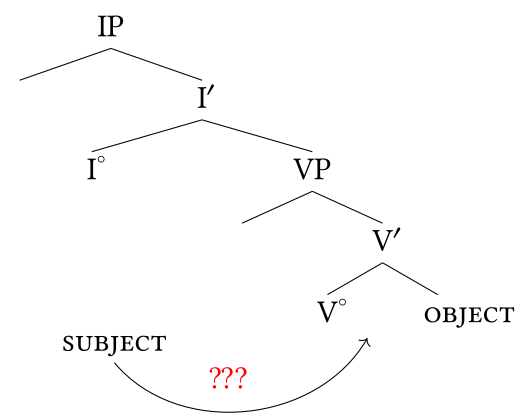
e. **Max y-il** *no' tx'i'* naq Lwin. **Q'anjob'al**
PFV A3-see *CLF dog* CLF Pedro
'The dog saw Pedro.'
(Baquiax Barreno et al. 2005, 169, via Clemens and Coon 2018, 240)

f. **Tyi i-kuch-u** *aj-Maria* jiñi si'. **Ch'ol**
PFV A3-carry-SS *CLF-Maria* DET wood
'Maria carried the wood.' (Clemens and Coon 2018, 240)

7.6.1 Deriving VSO word order

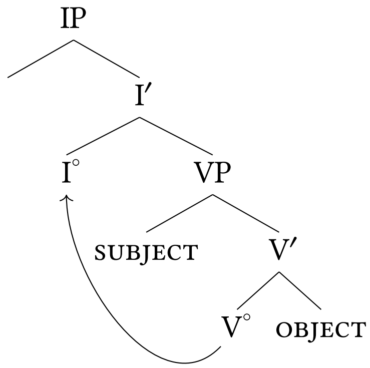
In the absence of additional components of a model of syntactic knowledge, VSO word order would fundamentally pose a challenge to X′-syntax. If THEME objects are complements of verbs (selected by verbs), within X′-syntax we expect them to be siblings. But if our structures are binary-branching, this leaves no position for the subject between the verb and its object.

(68)



In the sections that preceded, we introduced the idea above that local selection can sometimes be obscured by movement of an element of syntax to another position, and movement will again be useful to us here, especially in combination with the conclusion reached about the subjects originate inside the verb phrase (VPISH). If the verb is taken to move to a position higher than the subject, we can generate VSO word order without abandoning core generalizations about the structure of VP, or about X′-syntax more generally. A common analysis of VSO is that verbs move to I°, generating VSO word order.

(69) Schematic derivation of VSO word order:



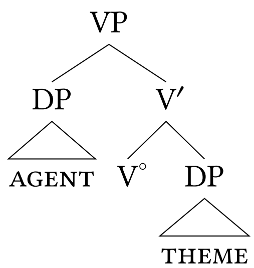
Patterns like these VSO word orders were prominent in the field drawing the conclusion that languages possess VP-internal positions for subjects (VPISH), which has come to be accepted as a universal property of human language syntax. Some languages (like English) obscure this low position of subjects by moving subjects to a higher position. But some languages (like Irish, Turkana, Ch'ol, and others) make it more obvious by generally leaving subjects in this low position. And as we saw above, even non-VSO like English languages possess constructions (like small clauses, and others) where subjects may appear inside a VP.

But this movement is another novel one to us at this point—can heads (like a verb) move to another head position (like I°)? This is a kind of movement we will refer to as **head movement**, which we will detail in much more depth in what comes, especially in Chapters 8 and 12.

7.6.2 Theta roles have consistent positions in VP

Syntacticians have concluded from patterns like these across a broad range of languages that VPISH is in fact universal—in all languages, the original position of subjects is inside the verb phrase. In some languages (like many VSO languages) the subject regularly arises in that position, whereas in other languages it regularly moves out of that position (such as English). This fits with our broader conceptual conclusion from this chapter: selection is local, and therefore the connection of a verb with its subject is encoded syntactically as well, in a structurally low initial position of the subject (inside the verb phrase).

This conversation also presupposes a point that is worth making explicit: not only do subjects start inside the VP, but thematic roles within the VP continue to appear in syntactic structures in very predictable ways. That is, in a transitive verb with an agentive subject, the agentive subject in fact arises from a consistent position inside VP: the specifier of VP (Spec,VP). And THEME objects continue to be introduced as complements of the verb.

(70) 

This observation has become formalized in a wide variety of work under different names, but the finding is clear: there are consistent structural positions for specific thematic roles in syntax.[6] These specific thematic roles therefore have a grammatical role that is more than simply descriptive of a particular sentence or predicate; instead, they are predictable based on systematic grammatical structures for different kinds of predicates. These roles have come to be known as theta roles (θ-roles), which are in essence the syntactic realization of these semantic roles.

(71) **Theta role (θ-role)**: a thematic role assigned to particular arguments of a predicate

The theory that θ-roles are assigned to particular positions is known as the Universal Alignment Hypothesis (Perlmutter and Postal 1984) or the Uniform Theta Assignment Hypothesis (Baker 1988).

(72) **Uniform Theta Assignment Hypothesis (UTAH)**:
Universal Grammar specifies that each theta role is assigned to a particular structural position.

The **argument structure** of predicates refers to what arguments (possessing what thematic roles) are associated with specific predicates. UTAH, as formulated in (70), offers specific claims about the interface of the semantic properties of argument structure with their syntactic properties. We already know that (70) is not sufficiently specific as a final theory of UTAH or argument structure: there are many more thematic roles for predicates than just AGENT and THEME, and we know that there exist predicates (for instance, ditransitive predicates like *give* and *send*) that take more than one object, which (70) cannot yet explain. So we will need to further develop our understanding of UTAH as we go, but the present formulation is sufficient for our current purposes.

Conclusion: Selection is local 7.7

Like Chapter 4, this chapter doesn't necessary add to the complexity that we draw our trees with, but instead clarifies and refines a number of concepts and theoretical constructs that will constrain the model that we are building as we move forward.

The overall claim of the chapter is that selection is syntactically local: heads place restrictions either on their complements or on their specifiers. We saw this play out in a number of empirical domains: a low position of subjects explaining selectional relationships between verbs and subjects, but likewise instances (ex., in passives) where objects are moved but nonetheless show selectional relationships with verbs in their original positions. At the end of this chapter we introduce the notions of θ-roles and UTAH to allow us to speak with precision about argument structure of predicates.

For this approach to selection as a local property of syntax to hold up, we need a theory of movement: elements in syntax can be represented in more than one position, moving from one position to another. This theory needs refinement as we move along, but part of the core motivation for movement in syntax has been provided in this chapter.

New Empirical Observations

- Heads select for specifical syntactic and semantic properties of their specifiers and complements.
- Selection is local.
- Idiom chunks and light verbs only occur in verb-object configurations, never in subject-verb configurations.
- Theta roles appear in consistent positions in VPs

Components of the Theoretical Model

- This chapter added a large amount of theoretical constructs:
  - c-selection
  - s-selection
  - locality of selection
  - Theta Criterion
  - UTAH
  - VPISH
  - existence of movement (we do not yet have a full theory of movement)
  - T-model (Y-model)
- We know that what is built in syntactic structure is dependent on lexical knowledge and on our core phrase-structure-building capacity.

Notes 7.8

1. This description of lexical knowledge can certainly be made more nuanced, as a wealth of work continues to explore whether or not lexical knowledge has structure (Beavers and Koontz-Garboden 2020). The description of lexical knowledge here, though, is relatively close to how most syntacticians use the term.

2. The ChiBemba examples come via Kroeger (2022, 416–417), via Chung and Timberlake (1985, 208), from Givón (1972).

3. There are ways in which this statement oversimplifies, because some operations (movement is one example, but there are others like Agree) extend over larger stretches of intervening material. But insofar as selection goes, this statement is sufficient and accurate.

4. These patterns are widely discussed in generative syntax, with the key observations and theoretical consequences of the patterns typically credited to Marantz (1984) and Kratzer (1994).

5. Japanese data provided by Xiaoxing Yu, Malagasy data provided by Matt Pearson.

6. There is a long history of work on this kind of thematic structure: an entirely non-exhaustive list of relevant references includes Perlmutter and Postal (1984), Baker (1988, 1997), and Ramchand (2008, 2018).

Exercises and problems 7.9

Exercise 7.1: Practicing with VPISH

We repeat sentences from exercises in Chapter 5 and 6 here. Draw trees for these sentences that are consistent with VPISH.

1. The very clever student will study every night in the library.
2. Tired students will always take naps before class.
3. Khydence found a book about Maya Angelou in the park.
4. Babies run rather slowly.
5. Lucca's incessant talking always annoys Maisha.
6. The dog jumped over the ditch.
7. An aspiring doctor should study hard every day.
8. That the faucet is dripping might upset your father.
9. It might anger your mother that you turned up the thermostat.

10. The chef carefully stirred the sauce.
11. Every dog knows the location of their treats.
12. Lucca might never play his music in public.
13. The students might play games at recess today.
14. The latest article in the New York Times explains the current international crisis.

Exercise 7.2: C-selection and s-selection

Each of the sentences below are either ungrammatical, anomalous, marginal, or flout selectional restrictions for dramatic effect. For each, annotate a judgment and explain whether the anomalousness is from a problem with c-selection, a problem with s-selection, or something else.

(1) a. The playground slept all morning.
b. My computer is mad at me.
c. I agree the conditions in the contract.
d. We ate three gallons of milk this week.
e. I want that the children should leave.
f. The children deliberately liked their pizza.
g. The professor persuaded the chairs to be arranged in a circle.
h. I wonder that Jay arrived.
i. The professor gathered in the seminar room.
j. I am angry the outcome.
k. The storm announced its arrival.
l. The weekend announced its arrival.

Exercise 7.3: Kipsigis VSO

Kipsigis is a Kalenjin language belonging to the southern Nilotic languages and is spoken in Kenya. Based on the discussion in § 7.6, draw a tree structure of (2).

(1) Kii-∅-geeɾ **dʒiitɑ** tɛɛta.
PST-3SG-see person cow
'A person saw a cow (long ago).'

(2) Koo-∅-e **tuugɑ** peek *amut*.
PST-3PL-drink cows water yesterday
'The cows drank the water yesterday.'

More work on Kipsigis would be necessary to confirm this (two sentences is not enough to know the structure of a language), but this chapter gives you enough background to offer a first hypothesis.

Problem 7.1: Welsh word order

These data are drawn from Roberts (2005), Awbery (1976), Sproat (1985), and Borsley, Tallerman, and Willis (2007). Some glosses are simplified for the purposes of this problem set.

Welsh is a Celtic language that is spoken in Wales (a part of the United Kingdom). Based on the initial proposal for VSO word order proposed in § 7.6, come to an analysis of Welsh word order based on the examples below. Draw trees for (3), (5), and (10).

(1) Diflanodd y dyn.
disappeared the man
'The man disappeared.'

(2) Gwelodd y dyn y ci.
saw the man the dog
'The man saw the dog.'

(3) Gwêl y dyn y ci.
sees the man the dog
'The man sees the dog.'

(4) Gwelai y dyn y ci.
was.seeing the man the dog
'The man was seeing the dog.'

(5) Gall Ifor ddarllen llyfrau.
can Ifor reading books
'Ifor can read books.'

(6) Dylai Gwyn ddisgrifio'r llun.
ought Gwyn describe the picture
'Gwyn ought to describe the picture.'

(7) Prynodd Elin dorth o fara.
buy.PST.3S Elin loaf of bread
'Elin bought a loaf of bread.'

(8) Gwnaeth Elin brynu torth o fara.
do.PST.3S Elin buy loaf of bread
'Elin bought a loaf of bread.'

(9) Ddaru Elin brynu torth o fara.
PST Elin buy loaf of bread
'Elin bought a loaf of bread.'

(10) Ddaru Megan fynd adre.
PST Megan go home
'Megan went home.'

Problem 7.2: *Fuck all*

McCloskey (1993) discusses varieties of English spoken in the British Isles, noting that there is an emphatic negative quantifier that has an interesting set of properties. (1a) is roughly equivalent in interpretation to (1b). All examples in this problem are drawn from McCloskey (1993).

(1) a. They wrote fuck all this year.
    b. They've written absolutely nothing this year.

In general, *fuck all* is unacceptable in subject position, despite there being paraphrases that are naturally acceptable.

(2) a. * Fuck all would make us turn back now.
    b. * Fuck all surrounds this house.
    c. * Fuck all could destroy these walls.
    d. * Fuck all would control this mob.
    e. * Fuck all will ever increase my wealth.
    f. * Fuck all could ever make me trust this government.
    g. * Fuck all would refute this hypothesis.
    h. * Fuck all around here reminds me of home.
    i. * Fuck all supports this roof but a couple of planks.
    j. * Fuck all could refute that argument.

(3) a. There's fuck all (that) would make us turn back now.
    b. There's fuck all (that) could destroy these walls.
    c. There's fuck all (that) could control this mob.
    d. There's fuck all (that) supports this roof but a couple of planks.

Notably, in passive sentences, the derived subject can readily host *fuck all*.

(4) a. Fuck all has been done about this problem.
    b. Absolutely sweet fuck all was achieved by this action.
    c. Fuck all has been said about unemployment in the campaign so far.
    d. Fuck all was conceded during that strike.

There are also select non-passive verbs that also allow *fuck all* in subject position.

(5) a. Fuck all ever happens around here.
    b. Fuck all else grows in my garden but dandelions.
    c. Fuck all emerged from those discussions that would make a body optimistic.
    d. Fuck all ever changes around here.
    e. Fuck all lasts around here.
    f. Fuck all else came my way, so I took the job as a lavatory cleaner.

g. Fuck all ever starts on time around here.

Questions to answer:

1. What is the natural class that unifies these predicates in (5), as compared to the predicates in (2)? (A natural class is a grouping based on some similar property of the elements in the group.)
2. What kinds of arguments can *fuck all* appear with (and not appear with)? That is, what is the property of an argument that allows *fuck all* to occur as that argument?

Problem 7.3: Spanish transitive predicates

Translating predicates from one language to another does not always generate parallel syntactic structures. Spanish, a Romance language, is one of the most commonly spoken languages across the world, originating in Spain (Europe). Spanish is an SVO language.

(1) El perro mordió a la niña
the dog bit A the girl
'The dog bit the girl.'

(2) La niña leyó el libro
the girl read the book
'The girl read the book.'

Spanish is what is called a null subject language, where discourse-familiar subjects do not require overt pronominal forms.

(3) Mordió a la niña
bit A the girl
'It bit the girl.'

(4) Leyó el libro
read the book
'She/he/they.SG read the book.'

In certain contexts (for example, contrasting or emphasizing subjects) overt pronominal forms can be used in subject position.

(5) Ella leyó el libro
she read the book
'SHE read the book.'

Pronominal forms are different for subjects, indirect objects, and direct objects.

Consider the Spanish examples in (6)–(8): there are different pronominal forms for subjects and objects in Spanish. (Spanish is a null subject language so pronominal subjects can often be unpronounced; we use the pronounced versions here to demonstrate the relevant grammatical pattern.)

(6) Spanish subject pronoun forms

a. **Yo** como la manzana.
I eat the apple
'I eat the apple.'

b. **Tú** ves el libro azul.
You see the book blue
'You see the blue book.'

c. **Ella** va a la playa.
She goes to the beach
'She goes to the beach.'

(7) Spanish indirect object clitic forms

a. La profe **me** dio la manzana.
The teacher to.me gave the apple
'The teacher gave me the apple.'

b. Abuelo **te** dijo la verdad.
Grandfather to.you told the truth
'Grandpa told you the truth.'

c. Inez **le** pidió una explicación.
Inez to.her asked.for an explanation
'Inez asked her for an explanation.'

(8) Spanish direct object clitic forms

a. El zorro **me** engañó.
The fox me tricked
'The fox tricked me.'

b. Gloria **te** busca.
Gloria you look.for
'Gloria's looking for you.'

c. Nico **la** sorprendió con un anillo.
Nico her surprised with a ring
'Nico surprised her with a ring.'

With that background in place, we can consider the examples below.

We can see examples of typical transitive predicates in Spanish, which can be compared to their English parallels in the translations.

(9) a. Elena toca la guitarra.
Elena plays the guitar
'Elena plays the guitar.'

b. Bebo jugo de naranja.
I.drink orange juice
'I drink orange juice.'

c. Mi abuelo escribía novelas.
my grandfather wrote novels

'My grandfather wrote novels.'

Now, consider the sentences below, which show a different pattern.

(10) a. Me gusta el té.
to.me likes the tea
'I like tea.'

* Yo gusto el té.

b. Me gustan las manzanas.
to.me like the apples
'I like apples.'

* Yo gusto las manzanas.

c. Me caen bien mis compañeros de trabajo.
to.me like my workmates
'I like my workmates.'

* Yo cayo bien a mis compañeros de trabajo.

d. Me encantan los perros.
to.me love the dogs
'I love dogs.'

* Yo encanto los perros.

e. Le duele la cabeza.
to.3SG hurts the head
'His/her/their.SG head hurts.'

f. Le duelen los pies.
to.3SG hurts the feet
'His/her/their.SG feet hurt.'

(Examples drawn from here.)

Instructions:

- In terms of thematic roles, what are the similarities and differences between Spanish and English for the sentences provided here? (English parallels are available in the translation of each Spanish sentence.)
- Focusing now on the predicates—what similarities do you see between the predicates that consistently behave differently in Spanish than they do in English? You can answer descriptively using concepts from this chapter, not using complex formal syntax or semantics.

Problem 7.4: More *fuck all*

This problem builds on Problem 7.2 and 7.3, so do those first before coming here.

Why can *fuck all* appear in the sentences below? Your answer does not need to draw or reference syntactic trees: you can answer with other concepts from this chapter.

(1) a. Fuck all worries me anymore.
  b. Fuck all interests him anymore.
  c. Fuck all frightens them.
  d. Fuck all bothers them.
  e. Fuck all annoys her.
  f. Fuck all amuses her.
  g. Fuck all surprises her.
  h. Fuck all embarrasses her.
  (McCloskey 1993)

Problem 7.5: Selection in nonverbal categories

A. Can you think of obligatorily transitive adjectives other than *fond*?

B. Can you think of plausible candidates for obligatorily intransitive prepositions?

C. What is the syntactic difference between the prepositions in (1a) and (1b)? A <u>syntactic</u> difference, not a semantic one.

  (1) a. at, despite, from, of
    b. along, besides, between, by, under

D. Does *with* belong with the prepositions in (1a) or in (1b)? (For fun, you might ask a few of your English-speaking peers whether they agree with you.)

Problem 7.6: The causative alternation

The following exercise is adapted from a class handout from Jason Merchant, which is based on Levin and Hovav (1995)

Many predicates allow for an alternation where the predicate has both an intransitive use and a transitive use. This alternation is not available for all intransitive predicates, however. What kinds of predicates are eligible to participate in this alternation, and what kinds are not?

(1) a. The window broke.
  b. The child broke the window.

(2) a. The door opened.
  b. The wind opened the door.

(3) a. The ship sank.

b. The torpedo sank the ship.

(4) a. The popsicle melted.
b. The sun melted the popsicle.

(5) a. The crowd laughed.
b. * The comedian laughed the crowd.

(6) a. The actor spoke.
b. * The director spoke the actor.

(7) a. The boy giggled.
b. * The joke giggled the boy.

(8) a. The orchestra played.
b. * The conductor played the orchestra.

(9) a. I jumped.
b. * The scary movie jumped me. (intended: made me jump)

Problem 7.7: Lubukusu locative inversion

We don't repeat the problem set here, but Problem 4.5 from Chapter 4 is a good exercise that can be tackled with more precision based on the lessons in this chapter.

Wh-questions: An instance of A′-movement

8

Chapter outline

In this chapter, we discuss the syntactic structure of questions. We will pursue an insight already mentioned there and in Chapter 7—namely, that forming ***wh*-questions** is a combination of replacing a constituent by a suitable *wh*-phrase and moving it to the beginning of the sentence (***wh*-movement**). The first section of the chapter gives a basic introduction to the syntax of questions. The next section presents the syntactic derivation of *wh*-questions, which extends straightforwardly to **yes-no questions**. We use the observations about these construction to propose that the Force of a clause (whether it is a statement or a question) is explained by clause-typing features on C°. We conclude by showing that the properties of *wh*-movement are not special to *wh*-movement, but are instead a particular instance of a more general category of movement that we call A′-movement. Additional types of A′-movement include topicalization and focus constructions.

8.1 Basic descriptive terminology for types of questions

8.1.1 Yes-no versus *wh*-questions

Questions can be divided into **yes-no questions** (also known as polar questions) and ***wh*-questions** (also known as constituent questions, or content questions), according to the expected answer. As the name implies, the answer to a yes-no question is either “yes” or “no”. And again, as the name implies, *wh*-questions utilize a *wh*-expression (*who*, *what*, *which*, *where*, *when*, *why*) in English. *How* counts as a *wh*-expression by virtue of its meaning (questioning a constituent), even though it doesn't begin with *wh*-.

The distinction between the two question types is illustrated in (1) and (2).

(1) Yes-no question:
  Q: Has he called?
  A: { Yes, no }.

(2) *Wh*-question:
  a. Q: <u>Who</u> just came in?
     A: <u>The boy from next door</u>.
  b. Q: <u>Who(m)</u> did you invite?
     A: <u>All my friends</u>.
  c. Q: <u>When</u> did she call?
     A: <u>After dinner</u>.
  d. Q: <u>Why</u> did he do that?
     A: <u>Out of ignorance</u>.
  e. Q: <u>How</u> did you fix it?
     A: <u>With the right tool</u>.

8.1.2 Direct versus indirect questions

Another distinction that can be drawn is between **direct questions** and **indirect questions**. Direct questions are main clauses, whereas indirect questions are part of a larger matrix sentence (that may or may not itself be a question). Direct questions are generally used to elicit information. They are associated with characteristic intonation contours, which are represented in standard orthography in many languages by a question mark. Indirect questions are generally used to report about direct questions and are not associated with a special intonation.

The questions in (1) and (2) are all direct questions. The corresponding indirect questions are given in (3) and (4), enclosed in square brackets.[1]

(3) Indirect yes-no question:
  I can't remember [{ whether, if } he has called].

(4) Indirect *wh*-question:
  a. I can't remember [who just came in].
  b. I can't remember [who(m) you invited].
  c. I can't remember [when she called].
  d. I can't remember [why he did that].
  e. I can't remember [how you fixed it].

8.1.3 Information questions versus echo questions

Ordinarily, speakers use questions to elicit information, potentially initiating a discourse with them, as in (5).

(5) a. How do you do?
  b. What's up?

But questions can also be used to signal a failure to understand the previous move in a conversation. The failure to understand can be genuine or feigned (calculated to express surprise, disapproval, outrage, and so on). Accordingly, we can distinguish between **information questions** and **echo questions** (also known as **reprise questions**); the latter two terms underline the response character of the second type of question. Echo questions can have the same syntactic form as information questions, but they are associated with a melody that is quite distinct from that of information questions. Speakers can also further mark the special discourse function of echo questions by giving them a special syntactic form in which the *wh*-phrase does not undergo *wh*-movement. The *wh*-phrase is said to remain **in situ**. (6) and (7) illustrates the two forms that echo questions can take (with or without *wh*-movement).

(6) A: Over break, I ended up visiting my [unintelligible]
  B: Who did you end up visiting? (*wh*-movement)
  You ended up visiting who? (*wh*-in situ)

(7) A: Her parents burned all of her clothes.
  B: What did they do?! (*wh*-movement)
  They did what?! (*wh*-in situ)

The association of the *wh*-in situ form with the echo function is not universal. Some languages do not have *wh*-movement at all see Chapter 14 for discussion), and some *wh*-movement languages allow *wh*-in situ questions to serve as either information questions or echo questions. This is illustrated for French in (8) (Engdahl 2006, 104, (33)–(34)).

(8) a. A: Ton fils, il lit quoi? (information question)
  your son he reads what
  'What does your son read?'
  B: Des bandes desinées.
  of.the comics
  'Comics.'

b. A: Mon fils, il lit [inaudible]? (echo question)
my son he reads
'My son reads …'
B: Il lit quoi?
he reads what
'He reads what?'

8.2 A movement analysis of questions

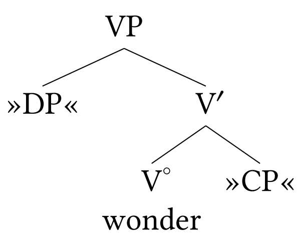
8.2.1 Indirect questions

We begin our analysis of questions by considering indirect questions. This may seem counter-intuitive, but it is actually a more direct path to understanding the structure of questions than if we were to begin with direct questions. As the presence of the complementizer *if* in (9) shows, the verb *wonder* takes a CP complement.

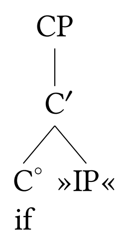
(9) They wonder *if* the lions will devour the zebra.

The elementary trees for *wonder* and *if* are given in (10a)–(10b), and the entire tree for (9) is given in (10c). Notice that our DP subject is now a part of the VP elementary tree, a product of local selection/VPISH.

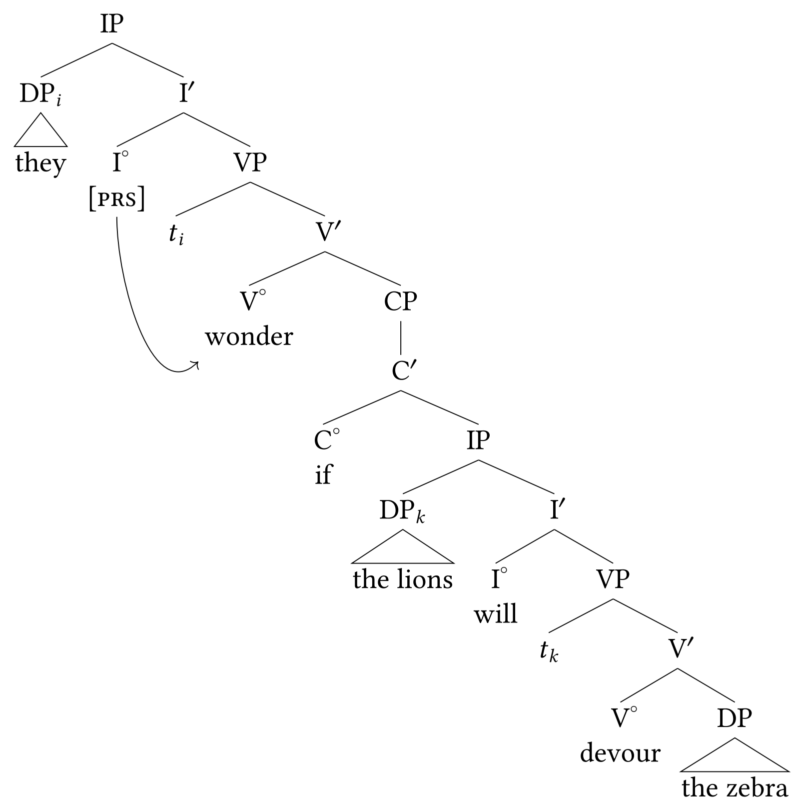
(10) a.



b.



c.

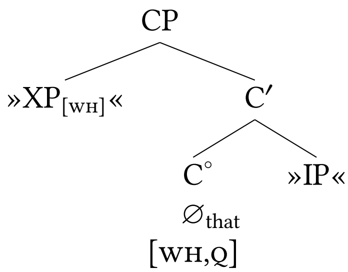


Now consider the indirect question in (11), which begins with a *wh*-phrase (a maximal projection) rather than with a complementizer (a head).

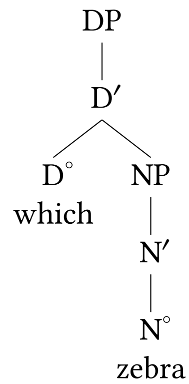
(11) They wonder *which zebra* the lions will devour.

Let's adopt the null hypothesis that *wonder* is associated with the same elementary tree in (11) as it is in (9)—namely, with (10a). Since (11) contains no overt complementizer, the CP that substitutes into the complement node of *wonder* must then be the projection of a silent complementizer. For reasons to be given shortly, we take this complementizer to be a silent counterpart of *that*. In deriving the tree for (11), a further difficulty arises concerning the *wh*-phrase *which zebra*. On the one hand, the *wh*-phrase must be the object of *devour*, just as in (9), because *devour* is obligatorily transitive. But on the other hand, the *wh*-phrase precedes the subject of the subordinate clause rather than following the verb. As usual when we are confronted with a mismatch of this sort, we invoke movement in order to allow a single phrase to simultaneously play more than one role in a sentence. Specifically, we will first substitute the *wh*-phrase into the complement node of *devour* and then move it to Spec(CP). This allows us to accommodate the word order in (11), while maintaining the status of *devour* as a transitive verb regardless of which clause type (declarative or interrogative) it happens to occur in. The requisite elementary tree for the silent complementizer is shown in (12); by contrast to the elementary tree for *if*, it contains a specifier node for the *wh*-phrase. The full structure of the *wh*-phrase that moves to that specifier node is shown in (12b). The resulting structure for (11) is shown in (12c).

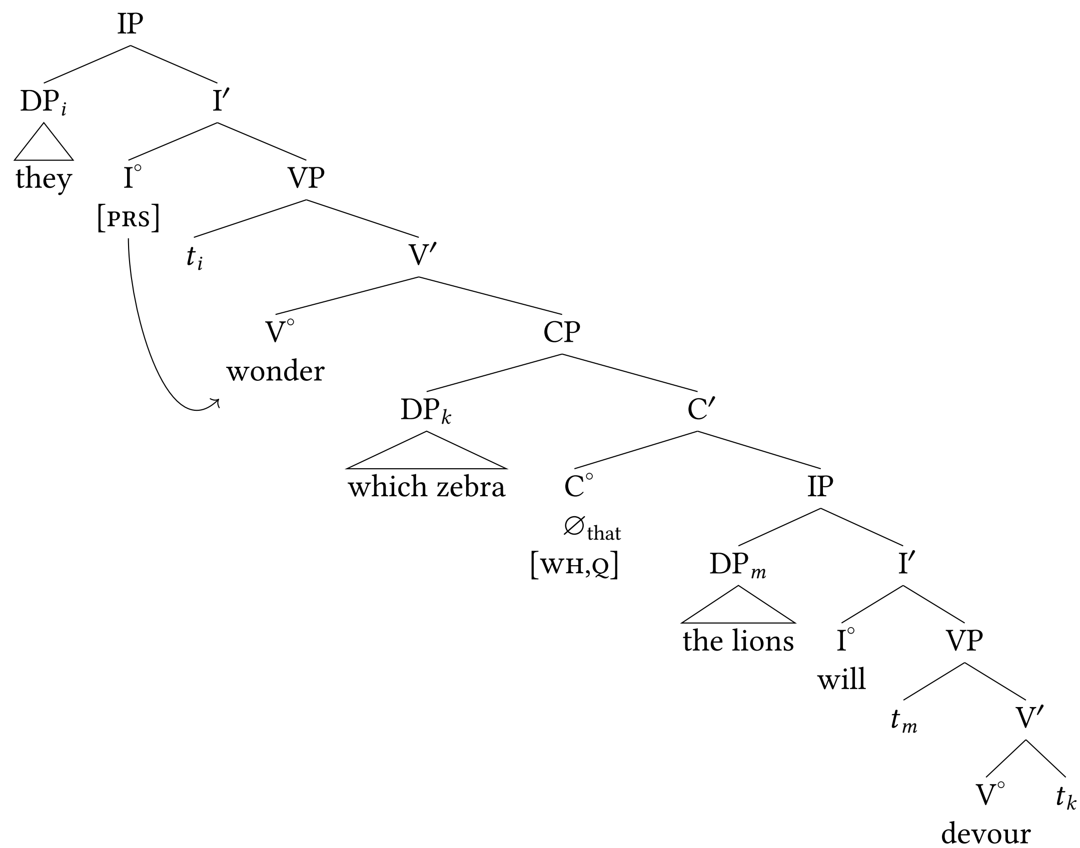
(12) a.



b.



c.



We annotate C° in these structures with Q features (marking it as interrogative) and WH features (marking it as a *wh*-construction), distinguishing it from non-*wh*-constructions). We will comment more on these annotations in § 8.3.2.

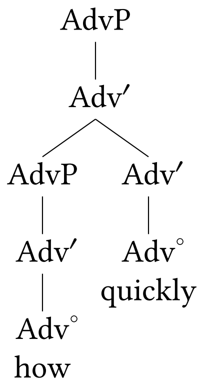
The argument just presented is based on the obligatorily transitive character of *devour*, but it extends straightforwardly to other syntactic relations—for instance, modification. Recall that the way that we represent the modification relation is to adjoin the modifier at the intermediate projection of the head that is being modified. In (13a), the adverb phrase *unbelievably quickly* modifies the verb *devour*, and so it adjoins at V′. In (13b), the corresponding *wh*-phrase *how quickly* also modifies *devour*, so it needs to adjoin to V′ as well, but it precedes the subject.

(13) a. The lions will devour the zebra *unbelievably quickly*.

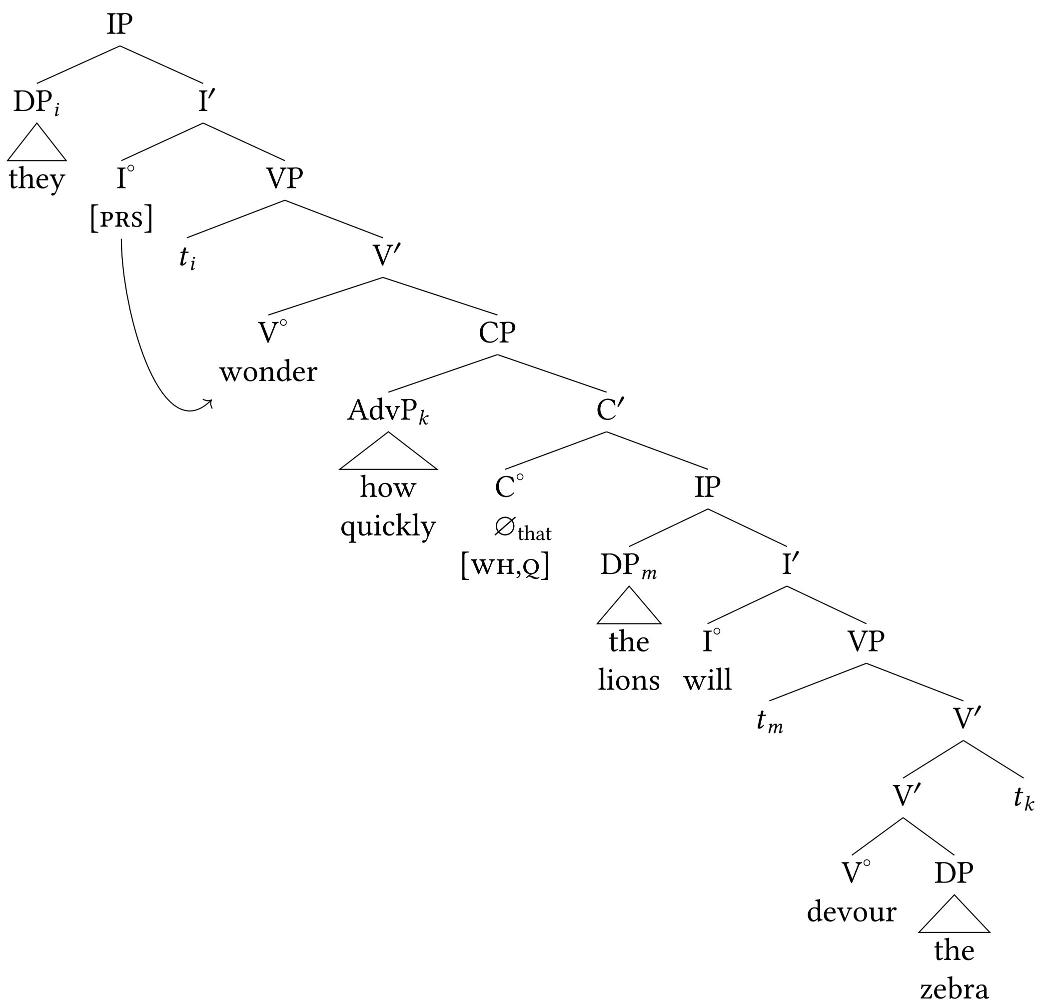
b. They wonder *how quickly* the lions will devour the zebra.

Again, we resolve the mismatch between the position where the phrase is interpreted and where it is pronounced by first adjoining the modifier to the embedded VP and then moving it to the embedded Spec(CP). (14a) shows the full structure of the phrase that undergoes movement, and (14b) shows the structure of the entire question.

(14) a.



b.



Why a silent complementizer?

Let's delve a bit deeper into why we treat the complementizer in (12) and in (14) as a silent counterpart of *that*. There are several reasons. First, Middle English (1150–1500) routinely allowed (though it did not require) overt *that* as the head of indirect *wh*-questions. The examples in (15) are all from texts by Chaucer that are part of the Penn-Helsinki Parsed Corpus of Middle English; (a)–(b) show that *that* alternated with its silent counterpart even within a single sentence by a single author.[2]

(15) a. he wiste wel hymself *what that* he wolde answere (cmctmeli.m3, 219.C1.75)
'he himself knew well what he would answer'

b. for ye han ful ofte assayed … *how wel that* I kan hyde and hele thynges (cmctmeli.m3, 221.C1.149)
'for you have very often determined how well … I can hide and conceal things'

Second, contemporary Belfast English (the variety of English that is the focus of Exercise 1.4) resembles Middle English in this respect (Henry 1995, 107).

(16) a. I wonder *which dish that* they picked.

b. They didn't know *which model that* we had discussed.

Third, *wh*-phrases followed by *that* continue to be attested in the unplanned usage of speakers of modern Mainstream U.S. English (Radford 1988, 500). A few of the examples that we have collected over the years are shown in (17).

(17) a. I realized *how interesting that* it was.
(Clara Orsitti, interviewed by Vicky Barker, World Update, National Public Radio, 25 January 1999)

b. Most of my colleagues were amazed *how quickly that* I recovered.
(advertisement for Temple University Hospital, WRTI, 24 November 1999)

c. "These recounts will determine *how much of a pick-up that* we will have", said Democratic National Committee Chairman Joe Andrew.
(https://www.cnn.com/2000/ALLPOLITICS/stories/11/08/house.races/)

Finally, sequences of *wh*-phrase and overt complementizer in indirect questions occur in languages other than English. The complementizer in question is generally the counterpart of *that*, but the counterpart of *if* is attested as well.

(18) Bavarian (Bayer 1983, 212, (8a–d)):

a. I woass ned *wer* *dass* des toa hod.
I know not who that that done has
'I don't know who did that.'

b. … *wos* *dass* ma toa soin.
what that we do should
'…what we should do.'

c. … *wann* dass da Xaver kumt.
when that the comes
'…when Xaver is coming.'

d. … *wiavui* dass a kriagt.
how.much that he gets
'…how much he gets.'

(19) Dutch (Besten 1989, 23, (21b)):

a. *welk boek* (*of*) hij wil lezen
which book if he wants read
'…which book he wants to read'

Many African languages display overt complementizer material alongside moved elements in both embedded and matrix contexts, as the normal state of things. We will see more examples of this as we go along, but we illustrate here from Lubukusu. Both the examples in (20) show the *wh*-word *siina* 'what' appearing at the left edge of a clause (in green, underlined): an embedded clause in (20a) and a main clause in (20b). In both instances, a complementizer that agrees with the *wh*-phrase (in red bold) appears adjacent, exactly where we would expect if the *wh*-phrase is moving to Spec,CP.

(20) Lubukusu complementizers alongside *wh*-words (Wasike 2007, 166)[3]

a. Wafula a-a-loma ali siina **ni-syo** Nekesa a-kha-a-rema ta?
1Wafula 1-PST-say saying what COMP-AGR 1Nekesa 1AGR-NEG-PST-cut NEG
"What did Wafula say Nekesa didn't cut?"

b. siina **ni-syo** Wafula a-kha-a-loma a-li Nekesa a-a-rema ta?
What COMP-AGR 1Wafula 1-neg-pst-say-FV saying 1Nekesa 1AGR-PST-cut NEG
"What didn't Wafula say Nekesa cut?"

We therefore see a similar pattern across a broad range of languages: *wh*-movement is targeting a specifier position of CP. Sometimes C° is overt (that is, pronounced), sometimes it is not.

8.2.2 Direct questions

Direct *wh*-questions

Having argued that *wh*-phrases move to Spec(CP) in indirect questions, we assume for uniformity that the same is true for direct questions like (21).

(21) a. [$_{DP}$ Which zebra] will the lions devour?
b. [$_{AdvP}$ How quickly] will the lions devour the zebra?
c. [$_{AdjP}$ How experienced] should they be?
d. [$_{PP}$ Under which shell] will they hide the pea?

As is evident from comparing the direct questions in (21) with their indirect question counterparts, *wh*-movement in direct questions is accompanied by movement of the modal from its ordinary position after the subject to a position immediately preceding the subject. Because this is movement of a head, we will refer to this type of movement as **head movement**. The particular instance of head movement at issue here is often called **subject-aux inversion**.[4] In what follows, we will focus on questions with modals. Questions with auxiliary and ordinary verbs involve complications that are not relevant here; see Chapter 12.

We will begin by assuming that the CP involved in the derivation of direct questions is headed by a morpheme expressing interrogative (= information-seeking) force. Despite the formal (= structural) similarity between direct and indirect questions, indirect questions do not have this same basic pragmatic force. In some languages, such as Japanese, Swahili, and Shona, the question morpheme is overt, as illustrated in (22). (The NOM and ACC morphemes in Japanese are called *case markers* and can be ignored for the moment: Chapter 9 discusses case.) The different positions of the question particles in these languages suggests the C head that hosts them is head-final in Japanese and Shona and is head-initial in Swahili.

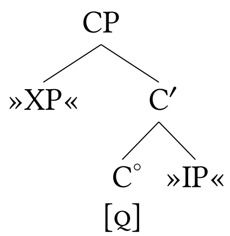
(22) Japanese declarative and interrogative
  a. 誰かが 寿司を 食べました。 **Japanese**
     Dareka-ga sushi-o tabe-mashi-ta.
     someone-NOM sushi-ACC eat-POL-PST
     'Someone ate the sushi.'
  b. 誰が 寿司を 食べました か？
     Dare-ga sushi-o tabe-mashi-ta ka?
     who-NOM sushi-ACC eat-POL-PST Q
     'Who ate the sushi?'

(23) Swahili declarative and interrogative
  a. U-na-enda nyumba-ni. **Swahili**
     2SG-PRES-go home-LOC
     'You are going home.'
  b. Je, u-na-enda wapi?
     Q 2SG-PRES-go where
     'Where are you going?'

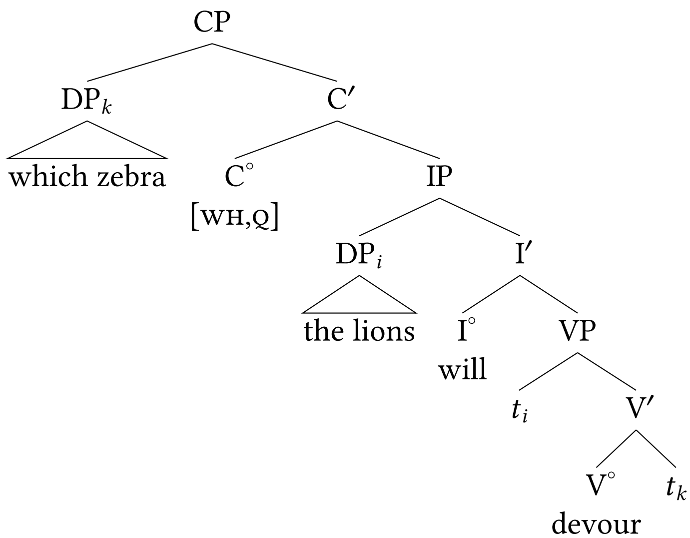
(24) Shona declarative vs. interrogative (König and Siemund 2007a)
  a. ndì-nó-tàùrà **Shona**
     1.SG-PRS-speak
     'I speak.'
  b. ndì-nó-tàùr-à hèré?
     1.SG-PRS-speak Q
     'Do I speak?'

In contrast to these languages, the question morpheme in English is silent. This gives us (25a) as the counterpart to *if* in (10b), and (25b) as the direct-question counterpart to the indirect question in (10c), pending subject-aux inversion (in Mainstream U.S. English; English varieties differ in their subject-aux inversion properties).

(25) a.



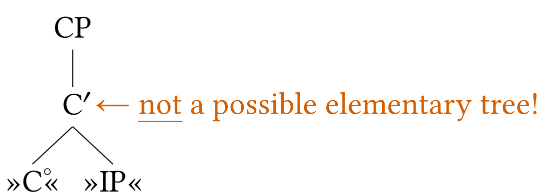
b.



Head movement

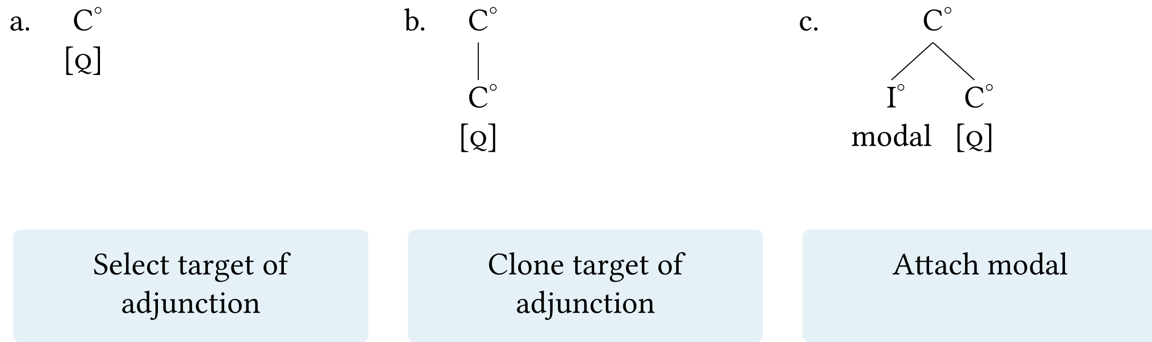
However, in mainstream varieties of English, direct questions require inversion (in contrast to (25b)). In the two cases of movement that we have considered so far (subject movement in Chapter 7 and *wh*-movement in this chapter), the "landing site" of movement has been a substitution node in an elementary tree. But this option is not available in the case of subject-aux inversion. The modal in I° cannot substitute in C°, because C° is already filled by the question morpheme. We can't simply allow the modal to overwrite the question morpheme, since then the tree would not contain the information necessary to generate the proper question intonation. We also want to continue to say that elementary trees are projections of lexical items. In other words, we do not want to allow elementary trees as in (26), which consist solely of substitution nodes.

(26)



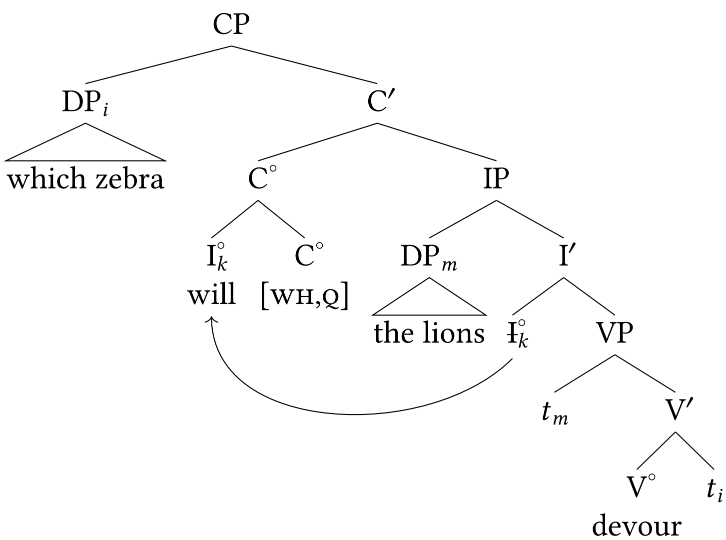
As a result of these considerations, we will implement movement of the modal to the pre-subject position via adjunction rather than with substitution. We use the same formal definition of adjunction in Chapter 5, § 5.3. That is, we first select the target of adjunction, then clone that target, and finally attach the adjoining element (here, the modal undergoing movement) as the daughter of the higher clone. (27) shows the operation on the relevant heads in isolation.

(27)



(28) shows the final structure for the direct question. Just like the moved phrases, the moved head bears a unique index linking it to its trace.

(28)



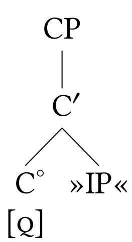
Although the formal operation of adjunction is defined in exactly the same way regardless of the use to which we put it, the two uses are associated with differences. First and foremost, the operation represents different linguistic relations in the two cases. As we already know from Chapter 5, § 5.3, adjunction can represent the modification relation, which corresponds to a semantic subset relation. In the case of head movement, adjunction represents an entirely different linguistic relation. In this case, we can think of the structure formed by adjunction as a morphologically complex head—or to put it more colloquially, as a compound word. In Chapter 12, we will invoke head movement to combine verb stems with tense suffixes to form finite verbs. In that case, the position of the tense suffix tells us that the verb left-adjoins to I°. By analogy, we assume that I° in the case at hand left-adjoins also; in other words, we are treating the silent question morpheme as a suffix, just like any other inflectional morpheme in English. Second, the projection level of the target node differs; it is an intermediate projection for modification versus a head for head movement. Third, the projection level of the adjoining element differs. It is a maximal projection for modification versus a head for head movement. Finally, for modification, the adjoining element is not yet part of the structure, and adjunction is "pure" in the sense that it doesn't involve concomitant movement. For head movement, the adjoining element is already part of the structure, and hence movement is involved. Table 8.1 summarizes the above discussion.

Use of adjunction for ...	**Modification**	**Head movement**
Adjunction structure represents ...	semantic restriction	morphological relation
Target of adjunction	intermediate projection	head
Adjoining node	maximal projection	head
Adjoining node already part of structure? (= Movement involved?)	no	yes

Table 8.1: Comparison of adjunction in modification and head movement

Yes-no questions

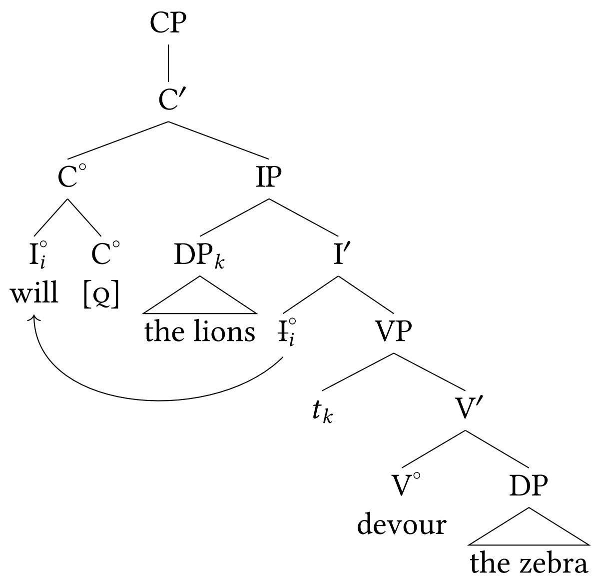
As we have already seen, the elementary trees required to derive indirect questions can either include a specifier, as in (12a), or not, as in (10b). As expected given the generality of the X′ schema, the same is true of the elementary trees for direct questions. In particular, the question morpheme in (25a) has a variant without a specifier, as shown in (29). Here, there is a [Q] feature marking the CP as interrogative, but no [WH] feature as it is not a *wh*-question. There is likewise no position for a *wh*-specifier in this instance, as it is not a [WH]-marked CP.

(29)


Using this elementary tree allows us to derive yes-no questions like (30), whose structure is shown in (31).

(30) Will the lions devour the zebra?

(31)



The direct question in (30) corresponds to the indirect yes-no question in (9) that we began our investigation with. So we now have structures for all four question types that result from crossing direct versus indirect questions with *wh-* versus yes-no questions.

8.3 Features in syntax

8.3.1 C heads and the force of clauses

Traditional grammar distinguishes three main types of **illocutionary force**, that is, three different things that a speaker can be doing with a sentence. These kinds of illocutionary force are 1) declarative (statements), 2) interrogative (questions), and 3) imperative (commands). As we noted above, the word order of English questions forces us to add a CP projection to the IP projection that we have been using to represent declaratives: this allows us to capture subject-auxiliary inversion via movement of I° to C°. To distinguish this kind of C° we posited a [Q] feature on C to mark it as a question.

In the interests of uniformity, it is sensible to represent ordinary declarative sentences as CPs also, with a head encoding the declarative illocutionary force. That is, if the feature marking a phrase structure as a question arises on C, and we think that CP is the locus of defining a clause's illocutionary force, we would expect the marking of declaratives to also be accomplished at C. And in fact, this is the conclusion of most syntacticians, that all main clauses are in fact CPs even when we don't see an overt morpheme or phrase residing in CP.

Generally we will mark the C head as bearing the feature [DECL], designating a declarative sentence (paired with [Q] for interrogatives. These are arbitrary choices of course—we could even use punctuation marks to represent force—[.], [?], and [!] (for imperatives), for example. The point is that what we call the features isn't magical, it is an annotation strategy that formalizes and systematizes our theory. We don't always draw the CP when drawing the tree of a sentence; as is typical, trees are communicative devices, and we draw the portion of a tree that is relevant to what we are trying to communicate. But the baseline assumption at this point is that all main clauses are in fact CPs.

(32) **CP-hypothesis**:
All main clauses are CPs.

These distinctions are marked overtly in a number of languages, making more clear why we might want to encode these notions of clause force in our syntactic structures. For example, West Greenlandic clauses bear a morpheme that communicates whether the sentence is a declarative (33a), an interrogative (33b), or an imperative (33c).[5]

(33) West Greenlandic (Sadock 1984 and Sadock and Zwicky 1985, via König and Siemund 2007b)

a. Iga -voq.
cook -DECL.AGR
'He cooks.'

b. Iga -va?
cook -Q.AGR
'Does he cook?'

c. Iga -git!
cook -IMP.AGR
'Cook (something)!'

We see that same pattern appear in Northern Mao, an Afroasiatic language in the Omotic family spoken in western Ethiopia. Northern Mao is a verb-final language that marks clause type on the verb forms with a declarative marker (/-á/), an interrogative marker (/-àː/), or an imperative marker that agrees with the addressee, we show the 2nd singular form here.

(34) Northern Mao declarative, interrogative, and imperative (Ahland 2012)

a. Míːm-íʃ tí-ná ha-k'ots'-á.
mosquito-SBJ 1SG-OBJ AFF-bite-DECL
'A mosquito bit me.'

b. Kó-ʃ hì-nà k'ots'-àː?
what-SBJ 2SG-OBJ bite-Q
'What bit you?'

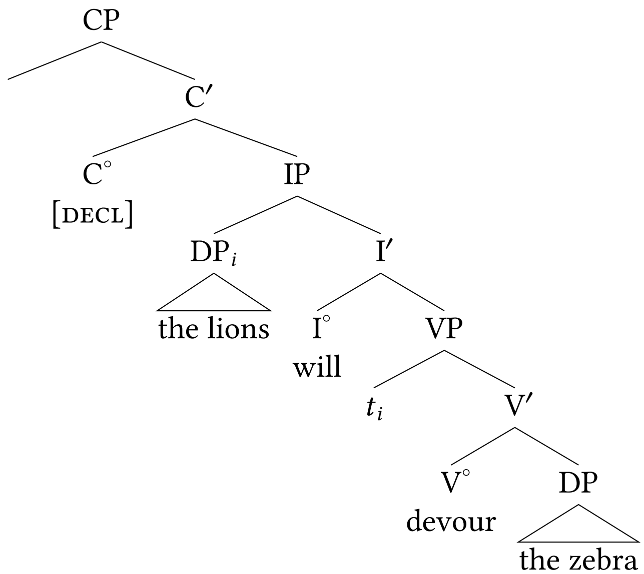
c. Kiːm-na tjám-í!
money-OBJ count:INF-IMP.AGR
'Count the money!'

While not all languages have overt morphology in C° marking illocutionary force, all languages *do* have declarative sentences, interrogative sentences, and imperative sentences. Assuming that these interpretations are part of the syntactic structure, we mark clause type on C heads in all instances.

The CP hypothesis means that even in sentences where we might not overtly observe a CP, we nonetheless assume a CP which is at minimum marking the force of the sentence. So we assume that a basic declarative sentence like (35a) has a structure as in (35b), headed by a CP that bears a feature marking the sentence as a declarative.

(35) a. The lions will devour the zebra.

b.



Whether or not we specifically annotate the [DECL] feature depends on what the point of any specific discussion is. But all sentences need to have their illocutionary force specified, which means that all sentences are CPs at their root node, headed by a C° that has relevant force features.

8.3.2 Feature-driven operations

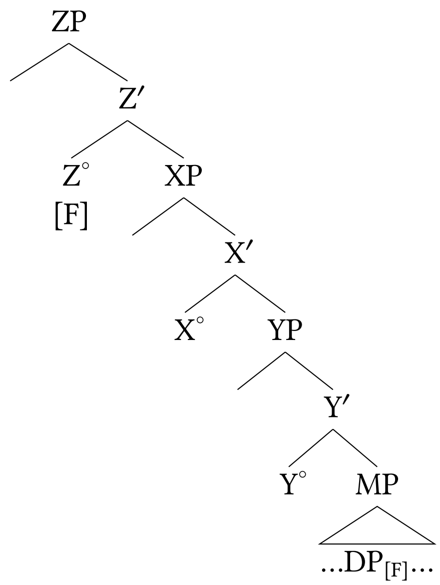
Implicit in the preceding discussion is that, in the model we are developing, properties of syntax/semantics are denoted by features, which are essentially annotations in our model that make precise where a property of grammar resides syntactically. Specifically, operations in syntax like movement and agreement are assumed to be feature-driven. So in the discussion above, we don't simply assert that heads move to C°, or that *wh*-phrases move to Spec,CP. This is annotated in the model via features on C°: [Q] and [WH] drive each of these movements, respectively.

The goal is for the structures that we adopt to generate the resulting sentences with all relevant information included in those structures. The structures we arrive at in an analysis are of course based on our knowledge of the language data, but we are trying to use those data to reverse engineer what the system must be—the system we arrive at can't also require personal analyst knowledge of the sort like "oh yeah and then X happens." Instead, we want our analyses to essentially be like a computer algorithm—specifying all the information necessary to generate the resulting grammatical or ungrammatical sentences.

This is why we assume that any operations that occur in syntax are feature-driven. So we don't simply assert that *wh*-movement moves to Spec,CP when we happen to want it to: as part of our analysis, we put a [WH] feature on C° that specifies that movement of a *wh*-phrase occurs to the specifier of CP. So our analysis also requires a notion that the system can check to confirm that a feature has been satisfied in the course of building our sentence.

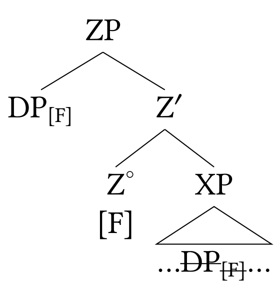
In our model, this is the normal course of things. In instances of dependencies where two positions in the syntax are connected in some way (like in instance of movement) we assume that both positions have relevant features that encode this relationship. So in the case of movement, the moved phrase is expected to bear some feature—we can call it F for the schematic in (36)—and the position where the phrase will move to is marked with feature [F] as well.[6]

(36)



The result of a configuration like (36) (in the instance of movement) therefore looks like (37), where the two F-marked phrases end up in a close structural configuration, here a so-called "spec-head relationship" where the [F]-marked DP is in the specifier of the head that bears the relevant feature [F].

(37)



We will leave the nature of the connection between the two annotations of feature F here relatively vague for the time being, but in the next chapter we get more specific about the behavior of features in syntax. The lesson to take away for now is that we are starting to annotate movements like this via features in the syntax.

Wh-like movement that is not *wh*- 8.4

Wh-movement in English (and in many other languages!) consists of movement of a question word to the left edge of the sentence (a domain that syntacticians often call the left periphery). But it is not the only such movement: there are other kinds of movement to the left edge of clauses that show the same properties as *wh*-movement, but are not instances of *wh*-questions. Syntacticians have concluded that *wh*-movement is one instance of a broader class of movements that we call A′-movements (read: A-bar movements); we discuss the terminology more below.

8.4.1 Focus constructions

English has some remnants of movements like this, but they are simply not as robustly present in English as they are in other languages.

Take Note

A possibly-new notation:

- *(X) = X is obligatorily present (the sentence is ungrammatical without X)
- (*X) = X is obligatorily absent (the sentence is ungrammatical with X)

Gungbe focus and *wh*-

Gungbe is a Kwa language from the Niger-Congo family, spoken in West Africa. The variety reported here is from Porto-Novo, Benin, with all of these data from (Aboh 2004). Gungbe is canonically an SVO language.[7]

(38) a. Sɛ́ná xìá wémà.
Sena read-PRF book
'Sena read a book.'

b. Sɛ́ná xìá wémà lɔ́.
Sena read-PRF book the
'Sena read the specific book.'

Wh-questions in Gungbe are unsurprising based on what we have covered thus far for English. In an object question as in (39a), for example, the *wh*-object does not appear in its canonical postverbal position and instead is in the left periphery, adjacent to a complementizer element (which we gloss as COMP for the moment).

(39) a. Ɛ̃tɛ́$_k$ *(wɛ̀) Sɛ́ná xìá t_k? **Object information question**
what *(COMP) Sena read-PRF
'What did Sena read?'

b. [Wémà tɛ́]$_k$ *(wɛ̀) Séná xìá t_k? **Object information question**
book which *(COMP) Sena read-PRF
'Which book did Sena read?'

c. Mɛ́nú$_k$ *(wɛ̀) t_k xìá wémà lɔ́? **Subject information question**
who *(COMP) read-PRF book SPF$_{[+DEF]}$
'Who read the specific book?'

Now, consider the constructions in (40), which are instances of focus constructions. Focus is a grammatical tool in human languages that highlights/emphasizes new or noteworthy information.[8]

(40) a. Sɛ́ná$_k$ *(wɛ̀) t_k xìá wémà. **Subject focus**
Sena *(COMP) read-PRF book
'SENA read a book.'

b. Wémàk *(wɛ̀) Sɛ́ná xìá t_k. **Object focus**
book *(COMP) Sena read-PRF
'Sena read A BOOK.'

Mainstream U.S. English communicates focus in a variety of ways—sometimes this is done purely with intonation, saying a word louder, longer, and higher pitched (this is what the all-caps denotes in the translations in (40)). But we also have syntactic constructions, like our it-clefts that we were using as a constituency diagnostics (*It is Sena that read a book*). And like with English focus, in Gungbe it's possible to focus many different kinds of constituents, like an adverb in (41a) and a PP in (41b).

(41) a. [Bléún] wɛ̀ Sɛ́ná gbá xwé étɔ̀n. **Focus on an adverb**
quickly COMP Sena build-PRF house his
'Sena QUICKLY built his house.'

b. [Távò lɔ́ jí] wɛ̀ Sɛ́ná zé gò lɛ́ ɖó.
table SPF$_{[+DEF]}$ on COMP Sena put-PRF bottle NUM LOC
Focus on a location phrase
'Sena put the bottles ON THE SPECIFIC TABLE.'

Lubukusu focus and *wh*-

Lubukusu is a Bantu language of the Luyia subgroup that is spoken in Western Kenya.[9] Lubukusu shows a pattern quite similar to Gungbe, where *wh*-phrases can appear in the left periphery of an interrogative sentence. In each instance, a complementizer element appears directly to the right of the fronted *wh*-phrase.

(42) a. Nafula a-a-siima Wafula (Wasike 2007, 234)
1Nafula 1SM-PRS-love 1Wafula
'Nafula loves Wafula.'

b. naanu *(ni-ye) Nafula a-a-sima?
1who COMP-1AGR 1Nafula 1SM-PRS-love-
'Who is it that Nafula loves?'

(43) ba-ba-ana ba-a-funa lu-u-saala **Declarative**
2-2-child 2sm-pst-break 11-11-stick
'The children broke the stick.'

(44) Siina ni-syo Wamalwa a-a-funa? ***Wh*-object cleft**
7what comp-7 1Wamalwa 1sm-pst-break
'What was it that Wamalwa broke?'

(45) naanu ni-bo ba-ba-a-funa lu-u-saala? ***Wh*-subject cleft**
2who comp-2 2agr-2sm-pst-break 11-11-stick
'Who were they that broke the stick?'
(Wasike 2007, 117–119)

And as we can see from (46), focus constructions show the same syntax: a focused element can appear at the left edge of the sentence immediately preceding the same complementizer element.

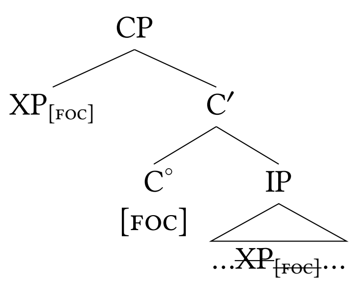
(46) a. lu-u-saala ni-lwo Wamalwa a-a-funa **Object cleft**
11-11-stick comp-11 1Wamalwa 1sm-pst-break
'It was a stick that Wamalwa broke.'

b. ba-ba-ana ni-bo ba-ba-a-funa lu-u-saala **Subject cleft**
2-2-children comp-2 2agr-2sm-past-break 11-11-stick
'It was children who broke the stick.'

Modeling left-peripheral focus constructions

We can model focus constructions like these in exactly the same way that we did for *wh*-movement in English: a feature of the relevant sort (a focus feature, in this instance) on a CP-level head.

(47)



The parallels with *wh*-movement should be evident: a feature on C° attracts a phrase with the same feature. When C° bears a [wh] feature it attracts a *wh*-phrase, and when C° bears a focus feature, it attracts a focused phrase.

8.4.2 Topicalization

We turn now to **topicalization**, another kind of emphatic construction which, instead of marking new/noteworthy information like focus, is used to mark old/familiar information, or is used to describe what the sentence is about. In English topicalization is at times accomplished by fronting a phrase, or by using an *as for*-construction as in (49).

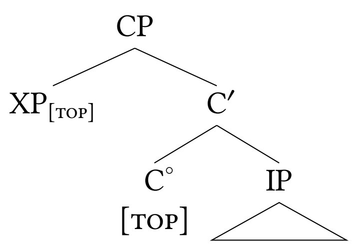
(48) a. [$_{\text{AdvP}}$ Unbelievably quickly,] the lions will devour the zebra.
b. [$_{\text{AdjP}}$ Quick as a wink,] the cat hid under the covers.
c. [$_{\text{PP}}$ Over the next few days,] the snow will melt.

(49) a. As for the zebras, the lions will devour them quickly.
b. As for the lions, they will devour the zebras quickly.

Based on what we've been concluding to this point, it is natural here to assume that these topicalized phrases are in Spec,CP.

It is quite common for languages to also show complementizer heads in topic positions in the same way we have concluded for focused phrases and *wh*-phrases above, relating the left-peripheral topic position to some other position of the topicalized element inside the clause.

(50)



As we see in the English examples in (48) and (49), topicalization constructions sometimes differ in whether they relate to some position inside the clause or not. It is common for topicalization to have a **resumptive pronoun** inside the clause, as in (49), where there is a pronoun that refers to the topicalized phrase. We see a similar kind of pattern in Gungbe: in the same place that a focus marker appears in focused constructions, a topic marker appears in topic positions. But a key difference between topic and focus in Gungbe is that topic constructions have a resumptive pronoun inside the clause inside the sentence in what would be considered the "base" position of the topic (circled in (51)).

(51) Dàn lɔ́ yà, kòfí hù-ì
snake the TOP Kofi kill.PFV-3SG
'As for the snake, Kofi killed it.'

A large area of research is whether topics in sentences like (51) consist of movement of the topic to the left periphery (in which case the resumptive pronoun would be considered an overt realization of the trace of movement) or whether the topic is base-generated (i.e., first enters the structure) in CP, and is not connected to the lower position by movement. It appears that languages can choose between these options; it's a highly interesting domain of research, but goes beyond what we will cover in an introductory text.

Some languages don't have an overt C head marking Topics, but nonetheless clearly mark topicalized phrases. Just as Japanese has an overt question morpheme *ka*, so, too, does it have an overt topic morpheme -*wa*. However, the topic morpheme isn't attached to the verb, but is instead marking the topicalized element itself. Some examples are given in (52). (Subjects and direct objects are marked in Japanese with the particles -*ga* and -*o*, respectively, but both can be overriden by -*wa*.)

(52) a. 子供は りんごを 食べました。
Kodomo-wa ringo-o tabe-mashi-ta.
child-TOP apple-ACC eat-POL-PST
'As for the child, they.SG ate the apple.'

b. りんごは 子供が 食べました。
Ringo-wa kodomo-ga tabe-mashi-ta.
apple-TOP child-NOM eat-POL-PST
'As for the apple, the child ate it.'

8.4.3 Movement to non-argument positions (A′-movement)

We don't mean to imply here that topicalization, focus, and *wh*-constructions are all identical—they clearly have differences. But they *do* share crucial properties. It quite common cross-linguistically for them to target left-peripheral position of the clause, which we model as them appearing in Spec,CP. For each, we can assume that the relevant feature resides on C° in each instance: [WH], [TOP], or [FOC]. Not all languages do topicalization, focus, or even content questions via movement to Spec,CP, but it is an extraordinarily common pattern cross-linguistically across completely unrelated languages. That is to say, it is much too common to be a coincidence that these properties are marked at the CP level, which points to these properties of CP being something quite general about human languages as a whole.

Beyond typically targeting Spec,CP, these movements often share a variety of other properties too. Some of these we don't have the background to cover yet, but we begin to introduce their shared properties around binding effects in Chapter 15. Other effects we can observe based on our current knowledge: for example, movements in syntax tend to be highly local—for example, heads move to the immediate next head: we have seen instances of V° moving to I°, and of I° moving to C°. But *wh*-movement, topicalization, and focus movements appear to be much more long-distance. In many of our examples here, the relevant phrase moved from object position or adjunct position in the VP all the way to CP. So this relatively unbounded distance of movement is typical of these movements but not of head movements.

Because there are predictable shared properties between these kinds of movements, syntacticians have grouped them together, calling them all instances of a single kind of movement that we call A′-movement (pronounced "A-bar movement"). The terminology is unfortunate because this does not relate to bar levels in X′-syntax. It is meant to contrast with a kind of movement we'll talk about in Chapter 10 that we call A-movement, that has different properties. So rather than A′-movement having anything to do with bar-levels in X′ syntax, it is better to think of it as "non-A-movement." A-movement is movement to an argument position (A-position); our movement of subjects to Spec,IP is an instance of A-movement. Since Spec,IP is the typical subject position (an argument position, or A-position), movement to that position are A-movements. The movements to Spec,CP that we have discussed in this chapter, in contrast, are not designated for a particular sort of argument (or for arguments at all: adjuncts readily participate in *wh*-movement). So these are movements to a non-argument position, i.e. an A′-position, and hence are considered A′-movement.

The main point is that *wh*-movement is not the only kind of movement that has these properties: focalization and topicalization often behave similarly, both utilizing A′-movement in many languages.

New Empirical Observations

- There is a mismatch between where *wh*-phrase is pronounced and where it is interpreted in *wh*-questions.
- Questions in English require inversion of inflection (auxiliaries) and subjects.

Components of the Theoretical Model

- movement
  - A′-movement (movement to non-argument positions)
    - *wh*-movement
    - focus movement
    - topicalization
  - head-movement (via head-adjunction)
- feature-driven operations
  - FOC, TOP, WH features on C°
  - Q features driving I°-to-C° movement
- traces/unpronounced copies
- CP-hypothesis: main clauses are CPs
- illocutionary force marked on C heads (declarative, interrogative, imperative)

Notes 8.5

1. Examples of various syntactic contexts in which indirect questions occur are given in (i).

   (i) a. Complement of verb:
   She *asked* [whether they are coming].

   b. Complement of adjective:
   I'm not *sure* [whether they are coming].

   c. Complement of preposition:
   The question *of* [whether they are coming] remains unresolved.

   d. Subject:
   [Whether they are coming] remains up in the air.

2. Additional examples of the same are provided in (i).

   (i) a. I wolde fayn knowe *how* <u>*that*</u> *ye* understonde thilke wordes and *what* is youre sentence (cmctmeli.m3, 227.C2.408)
   'I would like to know how you understand these same words and what your judgment is'

   b. And forther over, it is necessarie to understonde *whennes* <u>*that*</u> synnes spryngen, and *how* they encreessen (cmctpars.m3, 296.C1b.355)
   'And moreover, it is necessary to understand where sins come from and how they increase'

   c. Now shal ye understonde *in what manere* <u>*that*</u> synne wexeth or encreesseth in man. (cmctpars.m3, 297.C2.393)
   'Now you shall understand in what manner sin grows or increases in man.'

   d. The fifthe circumstaunce is *how manye tymes* <u>*that*</u> he hath synned … and *how ofte* <u>*that*</u> he hath falle. (cmctpars.m3, 323.C1.1502)
   'The fifth circumstance is how many times he has sinned … and how often he has fallen.'

3. Glosses for these Lubukusu examples are adjusted for our present purposes.

4. From our point of view, the term is slightly misleading since subject-aux inversion affects not just the auxiliary verbs *be*, *do*, and *have*, but also modals. Nevertheless, we will sometimes use the term when it is useful for expository purposes.

5. Glosses are slightly simplified from the original sources to better serve our current purposes.

6. It's worth pointing out that not literally every working syntactician believes this is true for every kind of syntactic operation (one exception is a lot of analyses of Germanic scrambling operations). But the feature-driven mindset is 1) assumed for most kinds of operations, and 2) predominant in the field as a requirement for the model overall.

7. Glossing in these examples is slightly modified from the original source to fit our current purposes.

8. This definition of focus is definitely a simplification: the diversity and complexity of focus constructions in human language is marvelous and somewhat stunning, so the attentive reader should be prepared for this to be complexified upon additional investigation.

9. The data reported here are from Wasike (2007), Justine Sikuku (personal communication), and Diercks' field notes.

Exercises and problems 8.6

Exercise 8.1: Tree-drawing practice with *wh*-questions

Draw trees of the following *wh*-questions in English. Draw triangles over all DPs, PPs, and AdvPs, but annotate all relevant features in the tree (especially on C°).

1. Who will study in the library tonight?
2. What did Khydence find in the park?
3. Where did Khydence find this book?
4. How fast do babies run?
5. Lucca's incessant talking always annoys Maisha.

Exercise 8.2: Tree-drawing practice with yes/no-questions

Draw trees of the following yes/no-questions in English. Draw triangles over all DPs, PPs, and AdvPs, but annotate all relevant features in the tree (especially on C°).

1. Will the dog jump over the ditch?
2. Should medical students study hard every day?
3. Will the chef carefully stir the sauce?
4. Could Lucca play his guitar for us tonight?
5. Would the students play games at recess today?

Exercise 8.3: Direct and indirect questions

A. Build structures for both of the direct questions in (9). Treat auxiliary *do* as a modal. You can use a triangle for personal pronouns, but build full structures for the *wh*-phrases.

(1) a. In which house do you live? (pied piping)
  b. Which house do you live in? (preposition stranding)

B. In your trees for (A), you have attached the PP as either a complement or an adjunct. Provide the relevant evidence.

C. Build structures for the complex sentences in (2), which contain indirect questions. From now on, you can also collapse the *wh*-phrases using triangle notation.

(2) a. I forget [in which house you live]. (pied piping)
b. I forget [which house you live in]. (preposition stranding)

Exercise 8.4: Ambiguities in *wh*-questions

A. The question in the reporter's telegram in (1a) is structurally ambiguous. Paraphrase the two interpretations.

(1) a. Reporter's telegram: How old Cary Grant?
b. Cary Grant's reply: Old Cary Grant fine.

B. Build a structure for the interpretation of (1a) that the reporter intended. Here and in (C), treat *Cary Grant* as a compound noun with a silent determiner, and do not attempt to implement verb movement to C. (The absence of *do* support is unexpected. The issue is addressed in Chapter 12.)

C. Build the structure for the other interpretation—the one that Cary Grant cleverly exploits.

Exercise 8.5: Topicalization structures

A. Using (14a) in the text as a model, build full structures for the topicalized phrases in (1a)–(1c) (= (48a)–(48c)). Treat *as* as P, and *next* and *few* as Adj. Treat *quick as a wink* as a modifier.

(1) a. [$_{\text{AdvP}}$ Unbelievably quickly] the lions will devour the zebra.
b. [$_{\text{AdjP}}$ Quick as a wink], the cat hid under the covers.
c. [$_{\text{PP}}$ Over the next few days], the snow will melt.

B. Using (14b) in the text as a model for when to use triangles, build full structures for the entire sentences in (1a)–(1c).

Exercise 8.6: Subject-auxiliary inversion in conditionals

English conditional clauses are ordinarily introduced by *if*, as in (1), but some varieties of English also allow such clauses to be marked by subject-aux inversion, as in (2).

(1) a. I will call if I have time.
b. If I have time, I will call.

(2) a. I will call should I have time.
b. Should I have time, I will call.

A. Build two structures for (1a)—a full structure for the conditional clause (no triangles) and a structure for the entire sentence. For simplicity, use triangles for noun phrases and for the conditional clause in the second structure.

B. Build a structure for the full sentence in (1b). Use triangles for the noun phrases and use a triangle for the conditional clause itself as well.

C. Repeat (A) and (B) for (2). Propose a suitable C head to accommodate the inversion of the subject with the modal.

Exercise 8.7: *Though*-preposing in English

English has two sorts of *though* clauses: ordinary ones that do not involve movement, as in (1a), and ones that do, as in (1b). We will refer to the construction in (1b) as the *though* preposing construction.

Word of Caution

The term *though* preposing is potentially confusing. What is preposed is not *though* itself, but some constituent of the *though* clause. In other words, *though* preposing is XP preposing that is licensed by *though*.

(1) a. We will solve the problem, though it seems difficult. (ordinary)
b. We will solve the problem, difficult though it seems. (*though* preposing)

A. Build two structures for (1a)—the full structure for the *though* clause (no triangles), and a structure for the entire sentence. For simplicity, collapse the noun phrases and the *though* clause in the second structure. Treat *though* as a P, as argued in Chapter 6, § 6.3.2.

B. Build the full structure for the subordinate clause in (1b).

C. How do the elementary trees for *though* needed in (A) and (B) differ?

D. *Though* preposing is grammatical in (2a), but not in (2b). Explain. You should not need more than two sentences.

(2) a. ✓ Difficult though the problem seems, ...
b. * Difficult though the problem seems very, ...

Exercise 8.8: English variation in question formation

A. Based on the data in (1) and (2), how do *why* questions differ from *how come* questions in Mainstream English?

(1) a. ✓ Why are they making such a fuss?
b. * Why they are making such a fuss?

(2) a. * How come are they making such a fuss?
b. ✓ How come they are making such a fuss?

B. Based on the data in (3), how would you characterize the difference between direct questions in the Philadelphia variety of African American English (AAE) reported here versus in Mainstream U.S. English (MUSE), as discussed in the chapter?

(3) a. Why you didn't tell me that?
(overheard at Market and 22nd Street, Philadelphia, PA, 26 September 1998)

b. What you bought tickets for?
(overheard at 30th Street Station, Philadelphia, PA, 26 November 1998)

c. Where you was at?
(overheard at Rittenhouse Square, Philadelphia, PA, 20 July 2001)

d. Where you went?
(overheard at Chestnut and 36th Street, Philadelphia, PA, 13 August 2001)

e. What I told you?
(Willie Perdomo. From where a nickel cost a dime. *Real News.* April 2002. 28)

f. What you're doing that for?
(exasperated mother to child, overheard at Primark, Fashion District, Philadelphia, PA, 21 January 2023)

g. Well, how much it weighs?
(R S, in conversation, 2 January 2023)

h. When I woked up, I said to myself, "Oh my God, where I'm at?"
(overheard at 21st Street and Fairmount Avenue, Philadelphia, PA, 14 July 2013)

i. Where he's at?
(overheard on Megabus, Philadelphia to DC, 23 April 2012)

j. Why you didn't come here? Why you didn't come here?
(overheard at Market and 12th Street Philadelphia, PA, 6 April 2012)

Problem 8.1: *Wh*-questions and focus in Guébie

Data provided by Hannah Sande based on work with Sylvain Bodji and Agodio Badiba Olivier.

The data set below is drawn from Guébie, a Kru language spoken in Côte d'Ivoire. Consider the baseline word order of declaratives in (1), and come to an analysis for the *wh*-questions in (2) and focus constructions in (3). Draw elementary trees for the CPs that plausibly occur in (1), (2), and (3); give a rationale for your analysis.

Glossing: SG = singular; DEF = definite; Tone is marked with a four-number system where the tonal marking follows the word. Here 4 represents the highest tone and 1 the lowest. Contour tones are marked with two tones within a syllable boundary, and syllable boundaries are marked with periods. Tone will not factor into your analysis here.

(1) a. ɟaci$^{23.1}$ ji^{3} su-wa$^{2.2}$ gbala$^{2.4}$
Djatchi will tree-DEF climb
'Djatchi will climb a tree.'

b. ɔ^{3} ji^{3} dʒa^{31} li^{3}
3SG will coconuts eat
'They.SG will eat coconuts'

(2) a. ɲɔkpa$^{3.3}$ touri$^{1.1.3}$ ji^{3} lɛtrɪ$^{3.2}$ kɔpa$^{3.23}$ na^{3}
who Touri will letter send Q
'To whom will Touri send a letter?'

b. bɛba$^{2.2}$ touri$^{1.1.3}$ ji^{3} ɟaci$^{23.1}$ kɔpa$^{3.23}$ na^{3}
what Touri will Djatchi send Q
'What will Touri send to Djatchi?'

(3) a. bagʷɛ$^{3.1}$ ɔ^{3} ji^{3} kɔpɔ$^{3.232}$
paper he$_i$ will send.him$_k$
'It's a BOOK they.SG will send him (as opposed to a letter).'

b. touri$^{1.1.3}$ ɔ^{3} ji^{3} bagʷɛ$^{3.1}$ kɔpa$^{3.23}$
Touri 3SG will paper send
'They.SG will send TOURI a book (as opposed to Djatchi).'

Problem 8.2: Sentence-final particles in Mandarin

Consider the following sentences from Mandarin Chinese.

(1) a. Wǒ-men chī yòuzi.
1-PL eat pomelo
'We eat pomelo.'

b. Nǐ chī yòuzi ma?
2 eat pomelo MA
'Do you eat pomelo?'

c. Wǒ-men chī yòuzi ba.
1-PL eat pomelo BA
'Let's eat pomelo.'

(2) a. Zhāng Yíng jīntiān qù gōngyuán.
Zhang Ying today go park
'Zhang Ying's going to the park today.'

b. Zhāng Yíng jīntiān qù gōngyuán ma?
Zhang Ying today go park MA
'Is Zhang Ying going to the park today?'

c. Wǒ-men jīntiān qù gōngyuán ba.
1-PL today go park BA
'Let's go to the park today.'

(3) a. Zhào-lǎoshī gǎi zuòyè.
Zhao-teacher grade homework
'Professor Zhao grades homework.'

b. Zhào-lǎoshī gǎi zuòyè ma?
Zhao-teacher grade homework MA
'Does Professor Zhao grade homework?'

c. Wǒ-men gǎi zuòyè ba.
1-PL today grade homework BA
'Let's grade some homework.'

A. In a sentence or two each, describe the functions of *ma* and *ba* illustrated in the sentences above.

B. What syntactic position do *ma* and *ba* occupy?

C. Draw trees for (1a), (2b), and (3c).

Problem 8.3: Korean clause types

Consider the data that are included here below. And then answer the questions given at the end of the data set.

A. Based on the data, is Korean a head-final, head-initial, or mixed-headed language? If it is a mixed-headed language, indicate which heads are initial and which are final.

B. What are the interpretations/functions of *(n)ta*, *-nya*, *-ca*, and *-ela/ala*?

C. What syntactic position(s) are occupied by *(n)ta*, *-nya*, *-ca*, and *-ela/ala*?

(1) a.

지혜는	학교에	갔다.
cihyey-nun	hakkyo-ey	k-ass-ta.
Jihye-TOP	school-to	go-PST-?

‘Jihye went to school.’

b.

지혜는	학교에	갔냐?
cihyey-nun	hakkyo-ey	k-ass-nya?
Jihye-TOP	school-to	go-PST-?

‘Did Jihye go to school?’

(2) a.

도서관에	같이	가자.
tosekwan-ey	kathi	ka-ca.
library-to	together	go-?

‘Let’s go to the library together.’

b.

도서관에	같이	가라.
tosekwan-ey	kathi	k-ala.
library-to	together	go-?

‘Go to the library together!’

(3) a.

비가	온다.
pi-ka	o-nta.
rain-NOM	come-?

‘It’s raining.’ (*lit.* ‘Rain is coming.’)

b.

* 비가	온다았.
pi-ka	o-nta-ass.
rain-NOM	come-?-PST

Intended meaning: ‘It rained.’

c.

비가	오냐?
pi-ka	o-nya?
rain-NOM	come-?

‘Is it raining?’ (*lit.* ‘Is rain coming?’)

(4) a. 티라미수를 먹자!
thilamiswu-lul mek-ca!
tiramisu-ACC eat-?
'Let's eat tiramisu!'

b. * 티라미수를 먹었자!
thilamiswu-lul mek-ess-ca!
tiramisu-ACC eat-PST-?
Intended meaning: 'Let's have eaten tiramisu!'

(5) a. 네 이름을 적어라!
ney irum-ul cek-ela.
your name-ACC write.down-?
'Write down your name!'

b. * 네 이름을 적었어라!
ney irum-ul cek-ess-ela.
your name-ACC write.down-PST-?
Intended meaning: 'Have written down your name!'

(6) 협곡은 아름다웠다.
hyepkok-un alumtaw-ess-ta.
canyon-TOP beautiful-PST-?
'The canyon was beautiful.'

(7) a. 이 수업은 어렵냐?
i swuep-un elyep-nya?
this class-TOP difficult-?
'Is this class difficult?'

b. 아니, 이 수업은 어렵지 ⊠⊠.
ani, i swuep-un elyep-ci anh-ta.
no this class-TOP difficult-CI NEG-?
'No, this class is not difficult.'

c. * 아니, 수업은 이 어렵지 ⊠⊠.
ani, swuep-un i elyep-ci anh-ta.
no class-TOP this difficult-CI NEG-?
Intended: 'No, this class is not difficult.'

(8) a. 눈이 내리면 눈사람을 만들자.
nwun-i nayli-myen nwun-salam-ul mantul-ca.
snow-NOM fall-if snow-person-ACC make-?
'If it snows, let's make a snowman.'

b. * 면 눈이 내린다 눈사람을 만들자.
myen nuwn-i nayli-nta nwun-salam-ul mantul-ca.
if snow-NOM fall-DECL snow-person-ACC make-?
Intended meaning: 'If it snows, let's make a snowman.'

c. 눈이 내린 후에 눈사람을 만들자.
nwun-i nayli-n hu-e nwun-salam-ul mantul-ca.
snow-NOM fall-PST.COMP after-at snow-person-ACC make-?
'After it snows, let's make a snowman.'

(9) a. 학생이었을 때
haksayng-i-ess-ul ttay
student-be-PST-COMP time
'when I was a student'

b. 학생일 때
haksayng-i-l ttay
student-be-COMP time
'when I am a student'

c. * 학생을이었 때
haksayng-ul-i-ess ttay
student-COMP-be-PST time
Intended meaning: 'when I was a student'

Problem 8.4: Korean politeness

This problem expands directly on Exercise 8.3.

A. Korean has grammaticized formality. The examples above were in the plain style. Consider the following data in the polite style. Does your analysis from Exercise 8.3 extend to these data? If not, how could you change it to fit?

(1) a. 학교에 가요.
hakkyo-ey k-a-yo
school-to go-??-POL
'I am going to school.'
or (with appropriate intonation) 'Are they going to school?'
or 'Let's go to school.'
or 'Go to school!'

b. 학교에 갔어요.
hakkyo-ey k-ass-e-yo
school-to go-PST-??-POL
'I went to school.'
or (with appropriate intonation) 'Did they go to school?'
not *'Let's have gone to school.'
nor *'Have gone to school!'

Problem 8.5: Tuki A′-movements

The following data are drawn from Biloa (2013)

Tiki is a Bantu language (Niger-Congo) that is spoken in Cameroon. Consider the data below, and arrive at an analysis of the Tuki questions and emphasis constructions in (2) and (3), respectively. Draw trees that are fully annotated with appropriate features (per our current model) for the sentences in (2b) and (3e).

Glossing: AGR = agreement; PST = past tense (there are multiple distinct past tenses, which is not relevant to this assignment); FUT = future. Some glossing is simplified for the purposes of this assignment.

(1) a. Iyere a-mú-éná vâdzú.
teacher AGR-PST1-see children
'(The) teacher saw the children.'

b. Vâdzú vá-mú-eńá iyere.
children AGR-PST1-see teacher
'(The) children saw a/ the teacher.'

c. Para a-mú-dwǐ okutu.
priest AGR-PST1-baptize woman
'The /a priest baptized the /a woman.'

d. Iyere a-mú-úná imgbémé ná ndzáná.
teacher AGR-PST1-kill lion in forest
'The /a teacher killed a/the lion in the/a forest.'

(2) a. Áné ódzú Putá a-nu-bǎnám?
who FOC Puta AGR-FUT1-marry
'Who will Puta wed?'

b. Áté aye Putá a-ḿa-námbá?
what FOC Puta AGR-PST2-cook
'What did Puta cook?'

c. Ni ówú Putá o-endám n(a) adongo?
when FOC Puta AGR-goes to village
'When does Puta go to the village?'

d. Ówáte ówú Putá a-m(á)-iba mǒni?
why FOC Puta AGR-PST2-steal money
'Why did Puta steal the money?'

(3) a. Abongo a-má-kóséna agéé wáá yěndze ídzó.
Abongo AGR-PST2-buy.for wife his house yesterday
'Abongo bought his wife a house yesterday.'

b. Abongo ódzú a-má-kóséna agéé wáá yěndze ídzó.
Abongo FOC AGR-PST2-buy.for wife his house yesterday
'It is Abongo who bought his wife a house yesterday.'

c. Yěndze aye Abongo a-má-kóséna agéé wáá ídzó.
house FOC Abongo AGR-PST2-buy.for wife his yesterday
'It is a house that Abongo bought for his wife yesterday.'

d. Agéé wáá ódzú Abongo a-má-kóséna yěndze ídzó.
wife his FOC Abongo AGR-PST2-buy.for house yesterday
'It is his wife that Abongo bought a house yesterday.'

e. Ídzó ówú Abongo a-má-kóséna agéé wáá yěndze.
yesterday FOC Abongo AGR-PST2-buy.for wife his house
'It is yesterday that Abongo bought his wife a house.'

Problem 8.6: Yoda's syntax

In the film series *Star Wars* and resulting collection of fan fiction and novelizations, Yoda is a wise character who is a leader of the Jedi, a mythical group of peace-keeping warriors. Yoda's style of speech is relatively famous for its idiosyncracy. As a general caveat, a fair amount of Yoda's speech is just normal English syntax. But Yoda also has notable divergences from typical English patterns, which (while atypical) nonetheless are reminiscent of familiar grammatical structures, which is likely why Yoda does not sound to us like he is speaking complete gibberish. Our interest is less in why the many various writers of Star Wars decided to have Yoda speak the way he does, and instead more in why these artistic choices have the emotive and aesthetic effect on us that they do: Yoda's grammar is distinctively non-typical for English (and hence interesting) but it is nonetheless entirely interpretable, meaning that it creatively and judiciously breaks/bends grammatical structure of English without completely ignoring them.

Give an analysis of Yoda's syntax based on the examples below, using our grammatical model of English as we have developed it up until this point. Additional context is provided with some examples, but only work to analyze the bolded sentences. Choose 2 illustrative examples to draw trees for.

(1) a. **Away put your weapon**. I mean you no harm. (*The Empire Strikes Back*)
b. For eight hundred years have I trained Jedi. **My own counsel will I keep** on who is to be trained! (*The Empire Strikes Back*)
c. **This one a long time have I watched**. (*The Empire Strikes Back*)
d. **A Jedi craves not these things**. (*The Empire Strikes Back*)
e. If you leave now, **help them you could**. (*The Empire Strikes Back*)
f. Remember, a Jedi's strength flows from the Force. But beware. Anger, fear, aggression. **The dark side are they**. (*Return of the Jedi*)
g. Luke, when gone am I… **the last of the Jedi will you be**. (*Return of the Jedi*)
h. Mudhole? Slimy? **My home this is**. (*The Empire Strikes Back*)
i. It is finished. **No more training you require**. (*Return of the Jedi*)
j. Once you start down the dark path, forever will it dominate your destiny. **Consume you, it will**. (*The Empire Strikes Back*)
k. **On many long journeys have I gone**. (*Yoda: Dark Rendezvous: Star Wars Legends*)

l. Death is a natural part of life. Rejoice for those around you who transform into the Force. **Mourn them do not**. Miss them do not. Attachment leads to jealousy. The shadow of greed, that is. (*Revenge of the Sith*)

m. You think Yoda stops teaching, just because his student does not want to hear? **A teacher Yoda is**. Yoda teaches like drunkards drink, like killers kill. (*Yoda: Dark Rendezvous*)

n. **Clear your mind must be** if you are to discover the real villains behind this plot. (*Attack of the Clones*)

o. Only the Dark Lord of the Sith knows of our weakness. If informed the senate is, **multiply our adversaries will**. (*Attack of the Clones*)

p. If no mistake you have made, losing you are. **A different game you should play**. (*Shatterpoint* (a Star Wars novel))

q. Soon will I rest, yes, forever sleep. **Earned it I have**. Twilight is upon me, soon night must fall. (*Return of the Jedi*)

r. Named must your fear be before **banish it you can**. (*Dark Lord trilogy*)

s. When you look at the dark side, **careful you must be**. For the dark side looks back. (*Yoda: Dark Rendezvous*)

t. When **900 years you reach**, **look as good you will not**. (*Return of the Jedi*)

Identifying Puzzles 8.1: *Wh*-questions without movement in Lubukusu

All data from (or adapted from) Wasike (2007)

In this chapter we saw examples from Lubukusu (Bantu, Kenya) where *wh*-phrases (and other kinds of focused phrases) move to the left periphery and co-occur with a complementizer. But this is not the only way to form a *wh*-question in Lubukusu. Consider the examples below:

(1) a. Wafula a-la-kula siitabu? **Lubukusu**
Wafula AGR-FUT-buy book
'What will Wafula buy?'

b. Wafula a-la-kula siina?
Wafula AGR-FUT-buy what
'What will Wafula buy?'

c. Siina ni-syo Wafula a-la-kula?
what COMP-AGR Wafula AGR-FUT-buy
'What will Wafula buy?'

(2) a. Wamalwa a-a-funa luusaala **Lubukusu**
Wamalwa AGR-PST-break stick
'Wamalwa broke a stick.'

b. Wamalwa a-a-funa siina?
Wamalwa AGR-PST-break what
'What did Wamalwa break?'

c. Siina ni-syo Wamalwa a-a-funa?
what COMP-AGR Wamalwa AGR-PST-break
'What was it that that Wamalwa broke?'

(3) a. Babaana ba-kha-kule siitabu waaena? **Lubukusu**
children AGR-FUT-buy book where
'Where will children buy the book?'
b. waaena ni-o babaana ba-kha-kule siitabu
16where COMP-AGR children AGR-FUT-buy book
'Where will children buy the book?'

(4) a. Ba-ba-ana ba-kha-kule si-tabu liina? **Lubukusu**
2-2-child 2SM-FUT-buy 7-book 5when
"When will children buy the book?"
b. Liina ni-lwo ba-ba-ana ba-kha-kule si-tabu?
5when COMP-5AGR 2-2-child 2SM-FUT-buy 7-book
"When will children buy the book?"

Lay out the ways in which these kinds of data are a challenge to the model of *wh*-questions that we developed in this chapter.

You are not responsible for attempting to solve this: we will tackle patterns like this together later in the book. It is sufficient to articulate clearly the ways in which these data present a puzzle to be solved, based on our model as it has been developed to this point.

Identifying Puzzles 8.2: Korean embedded CPs

The data below are instances of embedded clauses in Korean, built on the problem set in Exercise 8.3. These data pose a puzzle for the account as we've sketched it so far in this chapter.

(1) a. 태린은 지혜가 학교에 갔냐고 물어봤다.
thaylin-un cihyey-ka hakkyo-ey k-ass-nya-ko mulepw-ass-ta.
Taerin-TOP Jihye-NOM school-to go-PST-?-COMP ask-PST-?
'Taerin asked if Jihye went to school.'
b. 선생님께서 도서관에 같이 가라고 하셨다.
sensayng-nim-kkeyse tosekwan-ey kathi ka-la-ko ha-sy-ess-ta.
teacher-HON-NOM.HON library-to together go-?-COMP do-HON-PST-?
'The teacher told us to go to the library together.'
c. 지혜가 티라미수를 먹자고 했다.
cihye-ka thilamiswu-lul mek-ca-ko ha-yss-ta.
Jihye-NOM tiramisu-ACC eat-?-COMP do-PST-?
'Jihye said we should eat tiramisu (together).'
d. * 유령이 이름을 적었어라고 했다.
yulyeng-i irum-ul cek-ess-ela-ko ha-yss-ta.
ghost-NOM name-ACC write.down-PST-?-COMP do-PST-?
Intended meaning: 'The ghost told me to have my name written down.'

e. 아이는 협곡이 아름답다고 외쳤다.
ai-tul-un hyepkok-i alumtaw-ess-ta-ko oychy-ess-ta.
child-pl-top canyon-nom beautiful-pst-?-comp shout-pst-?
'The children shouted that the canyon was beautiful.'

f. * 아이는 협곡이 아름답고다 외쳤다.
ai-tul-un hyepkok-i alumtaw-ess-ko-ta oychy-ess-ta.
child-pl-top canyon-nom beautiful-pst-comp-? shout-pst-?
Intended meaning: 'The children shouted that the canyon was beautiful.'

Explain why these data are not yet explainable on the model that we have. (An approach that can account for this will be developed in Chapter 13.)

To be clear, you are not responsible for attempting to solve this: we will tackle patterns like this together later in the book. It is sufficient to articulate clearly the ways in which these data present a puzzle to be solved, based on our model as it has been developed to this point.

9

Feature valuation: Case and agreement

Chapter outline

This chapter is devoted to **Case**[1] and **agreement**, two instances of feature valuation in syntax. Case is a property of noun phrases, marking a noun phrase's grammatical role within a sentence. Not every human language expresses this property audibly in terms of morphological distinctions. In languages with relatively poor case morphology, like English, this guidepost function of case is less evident; case arguably carries out its function indirectly by regulating the surface order of noun phrases (specifically, which links in a noun phrase chain can be expressed overtly). We discuss the canonical configurations within which case features are assigned. We show how this is implemented theoretically via the Agree operation, a syntactic operation that allows for features to be shared between elements in the syntax. We show how the Agree operation accounts for both agreement patterns and case patterns.

A first look at case 9.1

9.1.1 The basic functions of case

In order to understand the role of case in human language, it is useful to consider languages in which constituent order is not as fixed as it is in English. In German, for instance, unlike English, the subject of an ordinary declarative clause doesn't need to precede the verb, as shown in (1) and (2) (the structure of German sentences that allows this flexible word order is interesting but aside from the current point). The Japanese examples show a similar flexibility in word order between subjects and objects, though the verb is final in these sentences. In the examples, boldface indicates the subject, and italics indicates the object.

(1) German:

a. **Der Mann** sieht *den Hund*.
the man sees the dog
'The man sees the dog.'

b. *Den Hund* sieht **der Mann**.
the dog sees the man
same as (1a), not the same as (2a)

(2) a. **Der Hund** sieht *den Mann*.
the man sees the dog
'The dog sees the man.'

b. *Den Mann* sieht **der Hund**.
the man sees the dog
same as (2a), not the same as (1a)

(3) Japanese:

a. カエルが ハエを 見る。
kaeru-ga *hae-o* miru.
frog-NOM fly-ACC sees
'The frog sees the fly.'

b. ハエを カエルが 見る。
hae-o **kaeru-ga** miru.
fly-ACC frog-NOM sees
same as (3a), not the same as (4a)

(4) a. ハエが カエルを 見る。
hae-ga *kaeru-o* miru.
fly-NOM frog-ACC eats
'The fly sees the frog.'

b. カエルを ハエが 見る。
kaeru-o **hae-ga** miru.
frog-ACC fly-NOM sees
same as (4a), not the same as (3a)

Since German speakers can't rely on constituent order to identify subjects and objects, how is it possible for them to keep track of which noun phrase expresses which grammatical relation (and more generally, which sentence expresses which meaning)? The answer is that grammatical relations are encoded in German by various morphological case forms. In particular, the subjects of finite clauses in German appear in a particular form called the nominative case, whereas objects generally appear in accusative case. (5) gives a morphological analysis of the noun phrases in (1) and (2). The parallel Japanese distinction between nominative and accusative is illustrated in (6).

(5) a. d -er Mann, d -er Hund **German**
the -NOM man the -NOM dog

b. d -en Mann, d -en Hund
the -ACC man the -ACC dog

(6) a. カエル が ハエ が **Japanese**
kaeru -ga, hae -ga
frog -NOM fly -NOM

b. カエル を ハエ を
kaeru -o, hae -o
frog -ACC fly -ACC

Notice that in (5), the distinction between nominative and accusative case is marked once in German: on the head of the noun phrase (the determiner). Japanese likewise marks case a single time in a noun phrase, but on a morpheme that attaches to the entire DP rather than via inflection on the determiner.

In some languages, redundant case marking on the determiner and the noun is the norm. This is illustrated for modern Greek in (7).

(7) Modern Greek:

a. **O** **άνδρ-ας** βλέπει *τ-ο* *σκύλο.*
O **andr-as** vlepi *t-o* *skil-o.*
the.NOM man-NOM sees the-ACC dog-ACC
'The man sees the dog.'

b. **O** **σκύλ-ος** βλέπει *τ-ον* *άνδρ-α.*
O **skil-os** vlepi *t-on* *andr-a.*
the.NOM dog-NOM sees the-ACC man-ACC
'The dog sees the man.'

Finally, case can be marked solely on the noun, as in Latin.

Background Info

In the chart below we provide some descriptors of some widely-attested cases that nominals appear in cross-linguistically, and the parallel constructions in English.[2] Keep in mind that there are **many** more cases than these, and this list heavily favors cases attested in Indo-European languages (as a starting point of the discussion).

Case	Grammatical function	English forms	English example
Nominative	marks subjects of finite clauses	*we, I, they*	***We*** *arrived home.*
Accusative	marks direct object of transitive verbs	*us, me, them*	*Our dog greeted* ***us****.*
Dative	marks indirect objects (ex. recipients, goals)	*us/to us, me/to me*	*Our dog gave* ***us*** *kisses.*
Genitive	marks possessor of another noun	*'s, of*	***Maisha's*** *book; The pages* ***of the book****.*

(8) Latin:

a. **Fēmin-a** *can-em* videt.
woman-NOM dog-ACC sees
'The woman sees the dog.'

b. **Can-is** *fēmin-am* videt.
dog-NOM woman-ACC sees
'The dog sees the woman.'

But a single case morpheme marking an entire DP is highly typical cross-linguistically. We illustrate with Japanese here:

(9) Japanese:

a. 元気な 先生が かわいい 猫を 見る。
genkina sensei-ga *kawaii neko-o* miru.
lively teacher-NOM cute cat-ACC sees
'The lively teacher sees the cute cat.'

b. かわいい 猫が 元気な 先生を 見る。
kawaii neko-ga *genkina sensei-o* miru.
cute cat-NOM lively teacher-ACC sees
'The cute cat sees the lively teacher.'

To summarize the discussion in this section: noun phrases can be case-marked in various ways, including marking on the nominal, marking on determiners or modifiers, and tone-marking (among other things). But regardless of the particular pattern, case marking has the same basic purpose: it audibly expresses a noun phrase's grammatical function in a sentence.

9.1.2 Inherent case and structural case

In many languages, a noun phrase's morphological case depends not only on its function in the entire sentence, but also on the head that it is most closely associated with. For instance, in German, the object in a sentence typically appears in the accusative, as we have already seen. But individual verbs are able to assign their own particular non-accusative case to objects on an idiosyncratic basis. The typical, systematic/rule-based case we call structural case, determined by the grammatical structures themselves. We use the term inherent case to refer to lexically-determined cases in any particular instance. In German verbs sometimes assign inherent case to objects, which is typically dative,[3] depending on the verb. Some examples are shown in (10) and (11).

(10) German:

a. ✓{ d-en Hund, d-ie Frau } unterstützen (accusative)
the-ACC dog the-ACC woman support
'to support the { dog, woman }'

b. *{ d-em Hund, d-er Frau } unterstützen (dative)
the-DAT dog the-DAT woman support

(11) German:

a. *{ d-en Hund, d-ie Frau } helfen (accusative)
the-ACC dog the-ACC woman help
'to support the { dog, woman }'

b. ✓{ d-em Hund, d-er Frau } helfen (dative)
the-DAT dog the-DAT woman help

Verbs typically determine the case of the object, but this case is not always the standard/default case. So in the examples above, *unterstützen* 'support' assigns the typical accusative to its object, whereas *helfen* 'help' assigns the dative. So structural case is determined solely by the syntactic structures: it is predictable based on structure (a transitive sentence produces a nominative and an accusative argument in German, for example). But inherent case is determined more idiosyncratically by particular lexical items: in the examples above, *helfen* 'help' assigns dative case where we might otherwise expect accusative case.

Inherent case in Latin is illustrated in (12). As in German, each particular verb determines the case of its object, but in Latin, the choice of case ranges over three cases—dative, accusative, and ablative.

(12) Latin:

a. { fēmin -ae, *fēmin -am, *fēmin -ā } { sub- venīre, suc- currere } (dative)
woman -DAT -ACC -ABL under- come under- run
'to help the woman'

b. { fēmin -am, *fēmin -ae, *fēmin -ā } ad- iuvāre (accusative)
woman -ACC -DAT -ABL to- support
'to support the woman'

c. { fēmin -ā, *fēmin -ae, *fēmin -am } fruī (ablative)
woman -ABL -DAT -ACC enjoy
'to enjoy the company of the woman'

Japanese likewise has inherent case alongside structural cases. Accusative case in (13a) is the structural case for objects, but some verbs assign dative case to their objects (13b), and others assign **comitative** case (13c) (meaning something happened *with* someone/something else).

(13) Japanese:

a. 同級生 { を *に *と } 気遣う (accusative)
dōkyūsei { -o, *-ni, *-to } kizukau
classmate -ACC -DAT -COM worry.over
'to worry over a classmate'

b. 同級生 { *を に *と } 惚れる (dative)
dōkyūsei { *-o, -ni, *-to } horeru
classmate -ACC -DAT -COM fall.for
'to fall in love with a classmate'

c. 同級生 { *を *に と } 付き合う (comitative)
dōkyūsei { *-o, *-ni, -to } tsukiau
classmate -ACC -DAT -COM date
'to date a classmate'

In German and Latin, prepositions resemble verbs in determining the case of their complement. In German, prepositions assign accusative, dative, or (sometimes) the genitive; in Latin, they assign ablative or accusative.

(14) a. German:

durch d-ie Tür, bei d-er Kirche, während d-es Krieges
through the-ACC door, by the-DAT church, during the-GEN war
'through the door, by/near the church, during the war'

b. Latin:

dē sapienti-ā, ad rīp-am
about wisdom-ABL, to shore-ACC
'about wisdom, to the shore'

Japanese doesn't show this same pattern of case-marking inside PPs, but the patterns from German and Latin above are quite broadly attested crosslinguistically. The example from Tamil (Dravidian; India, Sri Lanka) in (15) shows how different postpositions (in Tamil P heads come after their complements) assign dative, accusative, and even nominative case.

(15) Spoken Tamil (Schiffman 1999, 37, 38, 43):

கோயிலுக்கு பக்கத்துலெ ஏழு மணி வரெக்கும் என்னெ தவிர
kooyil-ukku pakkattle, eeṟu maṇi-∅ varekkum, en-ne tavira
temple-DAT near, seven o'clock-NOM until, 1SG-ACC besides
'near the temple, until 7:00, besides me'

Finally, in German, Latin, and Tamil certain adpositions (a blanket term for pre- and postpostions) can assign more than one case. For the German and Latin, the accusative marks direction, and the other case (dative in German, ablative in Latin) marks location. In the Tamil example in (18), the same postposition can select a noun phrase in the dative or oblique cases, yielding slightly different interpretations.

(16) German:

a. in { d-ie, *d-er } Bibliothek schicken
in the-ACC the-DAT library send
'to send into the library'

b. in { d-er, *d-ie } Bibliothek arbeiten
in the-DAT the-ACC library work
'to work in the library'

(17) Latin:

a. in { bibliothēc -am, *bibliothēc -ā } mittere
in library -ACC -ABL send
'to send into the library'

b. in { bibliothēc -ā, *bibliothēc -am } labōrāre
in library -ABL -ACC work
'to work in the library'

(18) Spoken Tamil (Schiffman 1999, 39, 41) :

a. மரத்து மேலெ
marattu-∅ meele
tree-OBL atop
'on top of the tree (in contact with the tree)'

b. மரத்துக்கு மேலெ
marattu-kku meele
tree-DAT atop
'above the tree (no contact with the tree)'

9.2 Case-features

The next two sections introduce some general concepts and conditions that enable us to derive the distribution of case forms in English (which we focus on) and other languages. We begin by introducing the notion of a **Case-feature**.

Consider the contrast between (19) and (20):

(19) a. ✓ They will help her.

b. ✓ She will help them.

(20) a. * Them will help she.
Intended meaning: (19a)

b. * Her will help they.
Intended meaning: (19b)

Why are the sentences in (20) ungrammatical? The answer is that noun phrases in English are subject to the requirements in (21).

(21) a. Subject pronouns of finite clauses appear in the nominative.

b. Object pronouns appear in the objective.

As is evident, both of the subjects in (20) are objective (accusative) forms, and both of the objects are nominative forms. Each of the sentences in (20) therefore maximally violates the requirements in (21). Clearly this is only a restriction on pronouns specifically (in English) because full lexical DPs show no morphological distinctions based on grammatical function. So the DPs *the teacher* and *the children* in (22) show no morphological distinctions whether they are subjects or objects.

(22) a. The children will celebrate the teacher.

b. The teacher will celebrate the children.

Now compare the examples in (19) and (20) with those in (23).

(23) a. You will help her.

b. She will help you.

In Mainstream U.S. English, the pronouns *they* and *she* exhibit distinct forms for the nominative and objective, whereas *you* doesn't (we say that *you* is a syncretic form, in that one form realizes multiple different case functions). But because case syncretism between the nominative and the objective is not complete in English (in other words, because at least some pronouns still have distinct forms for the two cases), we will treat *you* as a nominative form in (23a), equivalent to *they* and *she*, but as an objective form in (23b), equivalent to *them* and *her*. Moreover, we will treat the noun phrase *my big brother* as a nominative form in (24a) and as an objective (accusative) form in (24b) for exactly the same reason.

(24) a. My big brother will help her.

b. She will help my big brother.

In order to disambiguate instances of case syncretism like *you* and *my big brother*, it is useful to associate noun phrases with **Case-features**. Each Case-feature has a value that is selected from among all of the various morphological case forms in that language (regardless of whether the case forms are expressed synthetically or analytically). In English, for instance, a Case-feature can assume one of three values: nominative, objective (accusative), possessive (genitive). In Russian, a Case-feature has a choice among six values (nominative, genitive, dative, accusative, locative, instrumental).

(25) Russian Case: (Timberlake 2004)

душа duša, ‘soul’					
Nominative	**Genitive**	**Dative**	**Accusative**	**Locative**	**Instrumental**
душá dušá	души́ duší	душé dušé	дýшу dúšu	душóй dušój	душé dušé

If we need to explicitly represent a noun phrase's Case-feature, we can do so by means of labels as in (26) and (27).

(26) a. [$_{\text{DP-NOM}}$ They] will help [$_{\text{DP-OBJ}}$ her].
b. [$_{\text{DP-NOM}}$ You] will help [$_{\text{DP-OBJ}}$ her].
c. [$_{\text{DP-NOM}}$ My big brother] will help [$_{\text{DP-OBJ}}$ her].

(27) a. [$_{\text{DP-NOM}}$ She] will help [$_{\text{DP-OBJ}}$ them].
b. [$_{\text{DP-NOM}}$ She] will help [$_{\text{DP-OBJ}}$ you].
c. [$_{\text{DP-NOM}}$ She] will help [$_{\text{DP-OBJ}}$ my big brother].

This is a useful perspective because English clearly *does* care about Case distinctions, but it is only realized morphologically on pronouns, and even then only on specific pronouns. This leads to a conclusion that Case is present syntactically, but is only sometimes realized morphologically.

You may also have noticed that we have begun to use a nuanced annotation: We refer to the abstract properties of Case (the proposed Case features being valued in the syntax) with capitalized "Case", what is sometimes called **abstract Case**; we refer to the morphological forms of nominals with lower-case "case", sometimes called **morphological case**. This distinction sometimes matters a lot in advanced analyses of syntactic patterns, though for the most part we won't dwell on it in this text. But when you see capitalization differences, it is because we refer to the abstract features as "Case", but the overt morphological forms as "case."

9.3 Case-feature valuation

In § 9.1.1, we said that the purpose of Case is to encode a noun phrase's grammatical function in the sentence. In order to implement this insight in structural terms, we will require each noun phrase to be associated with a syntactic head that **assigns a value** for a Case-feature to that noun phrase.[4] In English, Case assigners are either verbs or prepositions (with very few exceptions), but there are languages (usually ones with rich overt case morphology) where the set of Case-assigning heads includes adjectives and nouns. The formalism that we will use for assigning values to features assumes that the assigner enters the syntax prespecified, with the feature possessing a value, and the DP enters the syntax with an unvalued feature, which we denote with a blank (underlined) space.

(28) Feature formalism
  a. I°: [Case: NOM]
  b. DP: [Case: __]

What (28) denotes is that the head I possesses a Case-feature that is lexically-specified with a value. Since finite I° in most languages is assumed to be the assigner of nominative Case (as we will see below), we assume it is lexically specified with a valued Case-feature. On the other hand, we don't want DPs to be lexically specified with a value for their Case-feature, since which Case they have varies based on what role the DP plays in the sentence. Instead, we assume they have a Case-feature that lacks an inherent value, but will be **valued** (that is, be "assigned a value") in the course of deriving a syntactic structure.

This fits nicely with the idea of different heads assigning different Cases. It also immediately raises the question (at least for a syntactician!) of whether the paths that Case-features take on their journey from head to noun phrase are constrained in any way: how does a DP's Case-feature get a value? There are in fact precise structural configurations by which features are valued: traditionally syntacticians have identified three licit paths, or **Case assignment configurations**: the **head-to-complement** configuration, the **spec-head** configuration, and the **head-to-subpart-of-complement** configuration. Recall from Chapter 5 that 'Spec' (read as 'speck') refers to the specifier node in the X′ schema of phrase structure. The configurations are illustrated in Table 9.1; nodes that are irrelevant for Case assignment are omitted.

The remainder of the section discusses concrete examples. At the end of this section in § 9.4.2, we will introduce a feature valuation operation (called **Agree**) which is the main modern approach to feature valuation. Agree offers one operation that can account for all of the feature valuation configurations noted in (9.1) (and more!).

9.3.1 The head-to-complement configuration

The first Case assignment configuration is the simplest in the sense that it contains the fewest number of nodes. In English, it is used in connection with the assignment of objective case by P° or V°, as in (29). The first two examples are simple. The third shows a more complex structure with two heads, each of which assign Case to its DP complement.

(29) a. expecting us
  b. with them
  c. expecting a storm with high winds

Example (30) gives the associated structures, with the endpoints of the path highlighted. (The higher DP positions in (30a), (30c) are irrelevant here.)

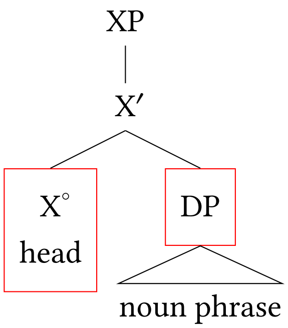
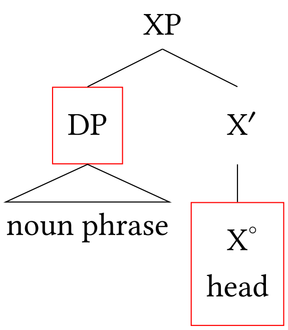
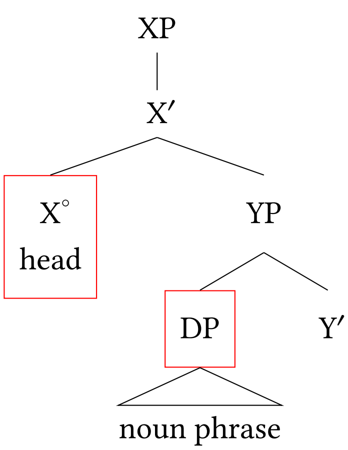
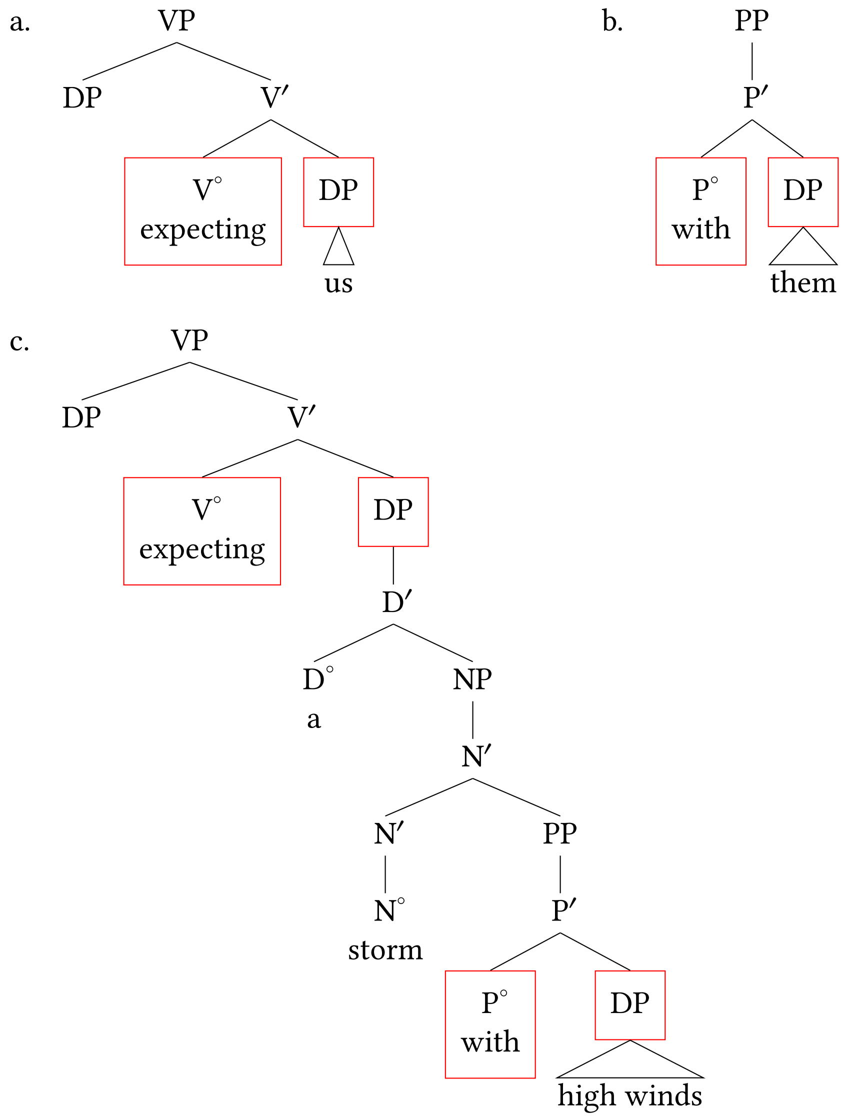
Configurations for valuation of Case-features (first approximation, to be adjusted)			
Tree	**Name(s)**	**Locality**	**Relation**
	head-to-complement	local	DP and X° are siblings
	head-to-specifier, spec-head	local	DP is the specifier of X°
	head-to-subpart-of-complement	nonlocal	X° c-commands DP

Table 9.1: Summary of Case assignment configurations.

(30)



In each of these instances we can assume that the licensing head (V° and P°, respectively) bears a valued Case-feature [Case: OBJ]. The DPs enter the derivation with an unvalued Case-feature [DP: __], which is then valued with objective Case in this configuration.

9.3.2 The head-to-specifier configuration

As we have just seen, heads can assign Case to their complements. Given the X′ schema, we might expect heads also assign Case to their other argument position in the spec-head configuration. Indeed, since subjects of sentences start out life as specifiers of verbs, this seems at first blush like an attractive way to assign Case to sentential subjects. According to this idea, what assigns nominative Case to *they* (or more precisely, to its trace in Spec,VP) in sentences like (31) is the finite verb *laughed* in (31a) and its nonfinite counterpart *laugh* in (31b).

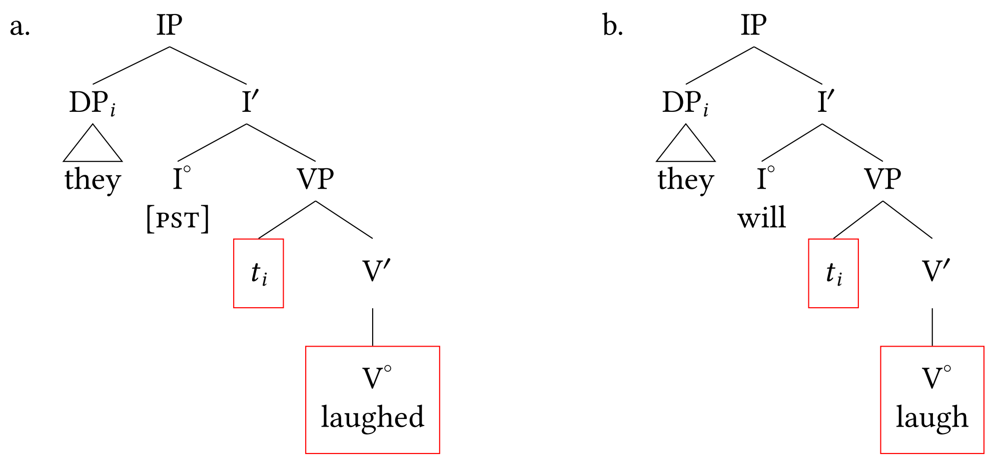
Word of Caution

In what follows, it is important to distinguish carefully between finite clauses and finite verbs. See the appendix to Chapter 2 (§ 2.9: Appendix: Finiteness in English) for detailed discussion.

(31) a. ✓ They laughed. (finite clause, finite verb)
b. ✓ They will laugh. (finite clause, nonfinite verb)

Although we will end up rejecting this approach, let us play it out for the moment in order to show why it is unsatisfactory. Example (32) shows the structures just described.

(32)



Finite clause, finite verb
(to be revised!)

Finite clause, nonfinite verb
(to be revised!)

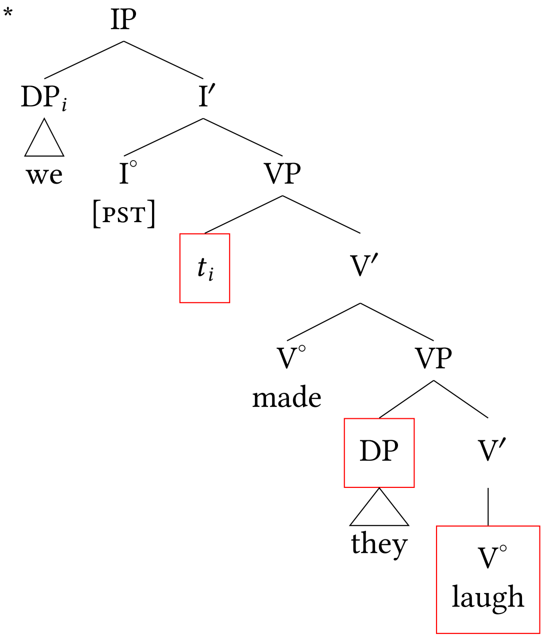
Now if verbs were able to assign nominative Case regardless of their finiteness, nonfinite *laugh* should be able to do so to *they* when it is the subject of non-finite clauses, for example in either small clauses or infinitives.

(33) a. * We made [they laugh]. (small clause)
Intended meaning: We made them laugh.

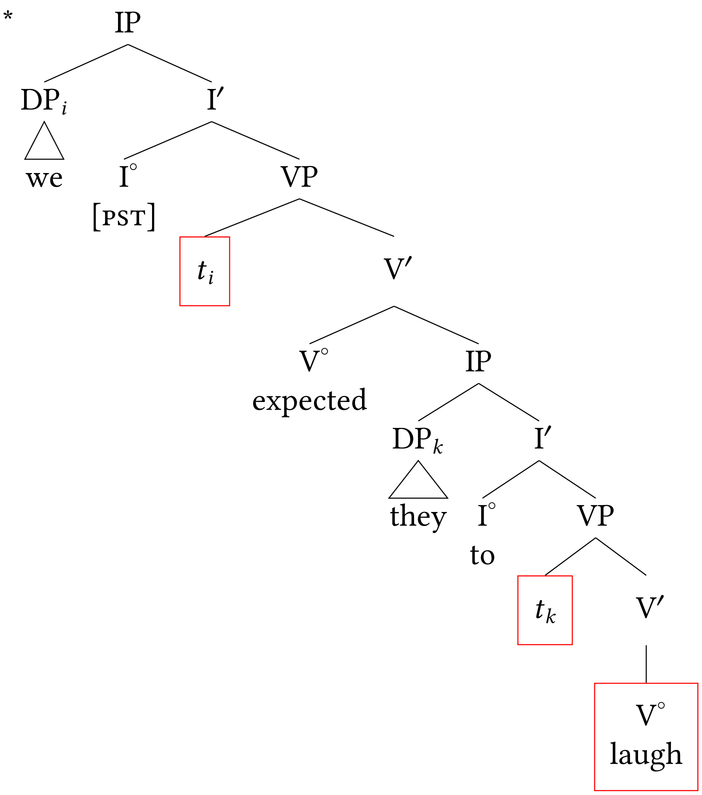
b. * We expected [they to laugh]. (*to* clause)
Intended meaning: We expected them to laugh.

The structures for these sentences are illustrated in (34). (In (34b), the lower subject moves on from Spec(VP) to Spec(IP), but that can't be the reason for the ungrammaticality, because that movement is grammatical in (32).) In each of these instances (modeling ungrammatical sentences) the embedded subject cannot appear in a nominative form.

(34) a. *

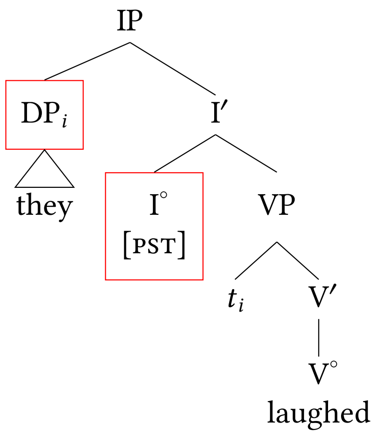


b. *

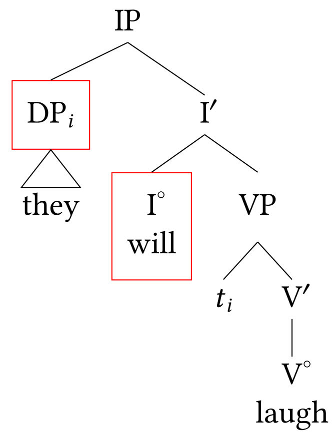


But these structures are impossible (and hence the sentences ungrammatical), a fact impossible to attribute to semantic anomaly, since the intended meaning can be expressed grammatically by simply replacing the nominative by the objective, as in (33). We must therefore reject the idea that it is V° that assigns nominative Case. However, we don't have to give up the idea that nominative Case is assigned in the spec-head configuration. All we need to do is revise our idea about which head it is that assigns Case, and have finite I° be the Case-assigner. This then gives (35), which supersedes (32).

(35) a.



b.



Finite clause, finite verb
(revised, final)

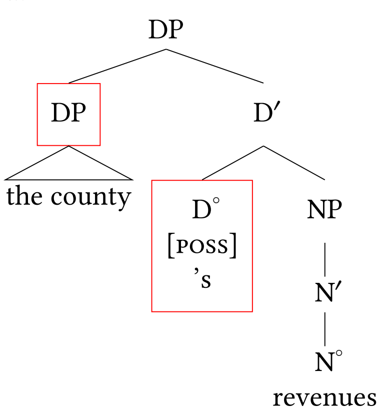
Finite clause, nonfinite verb
(revised, final)

The contrast between (31) and (33) now follows straightforwardly since I° is finite in (31), but missing or nonfinite in (33).[5]

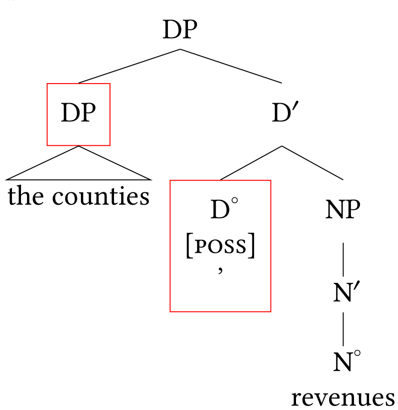
Nominative Case is not the only Case in English to be assigned in the spec-head configuration; possessive Case is another. Here, the Case-assigning head is the possessive determiner (*'s* or its silent plural variant). Three examples, along with their structures, are given in (36) and (37). For the possessive pronoun, we assume that the specifier and the head combine in the morphology as the synthetic form *their*.

(36) a. the county's revenues
b. the counties' revenues
c. their revenues

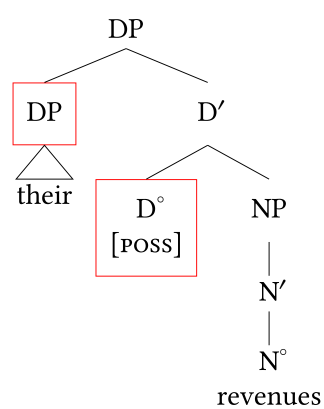
(37) a.



b.



c.



Extra Info

In possessive constructions, there are two DPs: an embedded one (the possessor) and the main one (the entire noun phrase containing both the possessor and the thing possessed). (37) highlights Case assignment to the possessor DP. The containing DP gets Case from some other head.

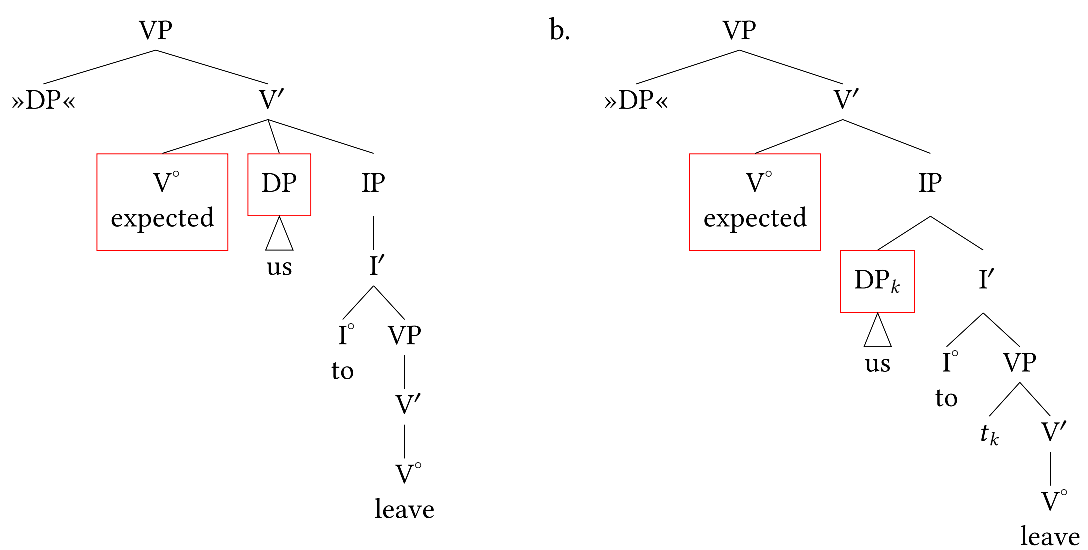
9.3.3 The head-to-subpart-of-complement configuration

The final Case assignment configuration is where a head values a phrase that is a subpart of its complement; we will consider a particularly common instance in which the valued phrase is in the specifier of the complement of the relevant head. This example is more unusual than the first two in two respects. First, as already mentioned in connection with (9.1), it is the only configuration in which a maximal projection intervenes between the Case-assigning head and the noun phrase. Second, in English, it allows V to assign objective Case to a directly following noun phrase in sentences like (38).

(38) They expected [us to leave].

Sentences of this type were at one time called **Exceptional Case-Marking (ECM)** sentences, because the subject of the embedded (non-finite) clause gets objective Case from the main clause verb. Over time, opinions have gone back and forth over exactly how exceptional "Exceptional Case-Marking" is. At first glance, it seems like we could avoid having to postulate this third configuration by giving (38) the main-clause VP structure in (39a) rather than that in (39b). We treat *to* as a nonfinite counterpart of *would* (compare the paraphrase in (40)).

(39) a. b.



Not like this

Like this

But there are several reasons to reject (39a). First, (38) has roughly the same meaning as (40).

(40) They expected that we would leave.

In both sentences, what is expected is a proposition (a state of affairs). This proposition is expressed as a single IP in (40), as indicated by the underlining, so that there is a direct correspondence between semantic and syntactic constituenthood. This parallel is preserved in (39b), but not in (39a).

A second, related, argument for rejecting (39a) comes from sentences containing expletive *there*. Recall from Chapter 4, § 4.5.1 that expletive *there* is licensed as the subject of a clause containing a verb of (coming into) existence. If we treat the DP following *expect* in ECM sentences as a subject, as we do in (39b), then (41a) is expected to be grammatical on a par with (41b).

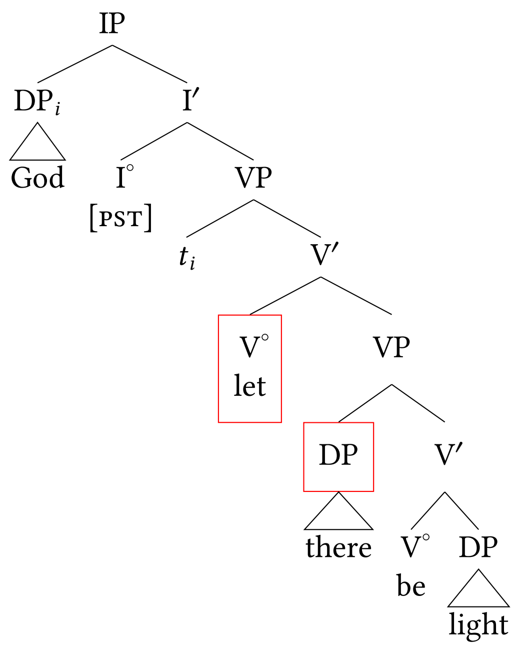
(41) a. They expect there to be a recession.
b. They expect that there will be a recession.

By contrast, treating that same DP as a complement of the matrix verb, as in (39a), would require complicating the licensing condition on expletive *there*. And of course, even if we were successful in revising the licensing condition, we would still forfeit the simple representation of the parallel between (38) and (40) just discussed. An additional conceptual argument is that the structure in (39a) violates the X′ schema and hence the binary branching hypothesis.

The term "ECM" is often restricted to sentences like (38), where the complement of the Case-assigning head is a nonfinite IP. In a slightly wider sense, the term can be used in connection with small clauses like (80a) from Chapter 4, repeated here as (42) and given the structure in (43).

(42) God let [*there* be light].

(43)



The Agree operation 9.4

9.4.1 Agreement in morphosyntax

Case is far from the only instance of feature valuation in syntax. It is quite common in the morphosyntax of the world's languages for a head to be valued for features on another phrase. Perhaps the most common instance of this is subject agreement on verbs, where a verb is inflected to match features of the subject of the sentence.

(44) English subject agreement

a. The dogs run-∅ fast.

b. The dog run-**s** fast.

English is relatively impoverished in its agreement properties; many languages have much richer inflection, both in terms of agreement appearing on more heads in the syntax, and in more instances, but also in terms of agreement displaying more features. In Swahili verbal elements have a subject marker (SM) that shows agreement with the subject of the sentence. Multiple verbs can have a subject marker and the subject marker agrees in noun class with the subject. In the examples below *Wekesa* (a personal name) is a class 1 noun, and *ng'ombe* 'cow' is a class 9 noun.

(45) Swahili subject agreement

a. Wekesa **a**-li-kuwa **a**-me-enda nyumba-ni.
1Wekesa 1SM-PST-be 1SM-PFV-go house-LOC
'Wekesa had gone home.'

b. Ng'ombe **i**-li-kuwa **i**-me-rudi shamba-ni.
9cow 9SM-PST-be 9SM-PFV-return field-LOC
The cow had returned to the field.

The most common features of nominals to be included in inflectional constructions like these are gender (noun class), number, and person (see Chapter 2 for an introduction to these). Syntacticians refer to these collectively in a bundle as *φ*-features (sometimes written phi-features; this is the Greek letter *φ*, which we pronounce as "fie", rhyming with *lie*).

Agreement in *φ*-features occurs on a wide variety of grammatical categories (cross-linguistically speaking). Many languages have concord within DPs where adnominal modifiers like adjectives, numerals, prepositions, and more inflect to match the *φ*-features of the head noun. We illustrate this widespread pattern with Swahili here.

(46) Swahili adjectives agree with head nouns (Wilson 1970)

a. m-tu m-zuri
1-person 1-nice/good
'a nice person'

b. wa-tu wa-zuri
2-person 2-nice/good
'nice people'

c. m-ti m-dogo
3-tree 3-small
'a small tree'

d. mi-ti mi-dogo
4-tree 4-small
'small trees'

(47) Swahili prepositions agree with head nouns (Carstens 2001; Wilson 1970)

a. ki-tabu ch-a mw-alimu
7-book 7-of 1-teacher
'the/a teacher's book'

b. rafiki y-a mw-alimu
9friend 9-of 1-teacher
'the/a teacher's friend'

Likewise, numerals in Swahili agree with their head nouns (reminder, the Arabic numerals in glosses are glosses for noun classes here).

(48) Swahili numerals agree with head nouns (Wilson 1970)

a. wa-tu wa-wili
2-people 2-two
'two people'

b. mi-ti mi-wili
4-trees 4-two
'two trees'

c. vi-azi vi-wili
8-potatoes 8-two
'two potatoes'

Likewise, other elements in clausal syntax can bear agreement. For example, in some languages complementizers can agree with subjects. In Kinande (spoken in the Democratic Republic of Congo) the complementizer agrees with the matrix subject, whereas in West Flemish (spoken in Belgium) the complementizer agrees with the embedded subject.

(49) Kinande complementizer agreement[6] (Baker 2008)

a. Mo-ba-nyi-bwire ba-ti Kambale mo-a-gul-ire eritunda.
AFF-2AGR-me-tell 2AGR-that 1Kambale AFF-1AGR-buy fruit
'They told me that Kambale bought fruit.'

b. Mo-n-a-layirire Kambale in-di a-gule amatunda.
aff-1SGAGR-PST-convince 1Kambale 1SG.AGR–that 1AGR-should.buy fruits
'I convinced Kambale that he should buy fruits.'

(50) West Flemish complementizer agreement (van Koppen 2017; Haegeman 2000)

a. K peinzen da-∅ / da-n dienen student nen buot gekocht eet.
I think that-3SG / that-3PL that student a boat bought has
'I think that that student has bought a boat.'

b. K peinzen da-n / *da-∅ die studenten nen buot gekocht een
I think that-that-3PL / that-3SG those students a boat bought have
'I think that those students have bought a boat.'

In Gweno (spoken in Tanzania), the negation morpheme at the end of a sentence agrees with the subject of the sentence.

(51) Gweno Negation Agreement (Dunlap 2024)

a. Ku-a-ghenda ufuva pfo
2SG.SM-PRES-go market NEG.2SG
'You don't go to the market.'

b. Thi-tha-ghenda ufuva nyi
1SG.NEG.SM-PRES.NEG-go market NEG.1SG
'I don't go to the market'

We could go on endlessly on this point: the variety of agreement patterns in the world's languages is both astonishing and fascinating. The takeaway lesson here is that there is clearly a need in our theory for a mechanism to value features on some element based on the features of a separate element. In the model we are building we will use the Agree operation; we describe how this is used to capture agreement in φ-features § 9.4.2, and then below in § 9.4.3 we discuss how the Agree operation can also be used to explain valuation of Case-features, and its relationship to DP licensing.

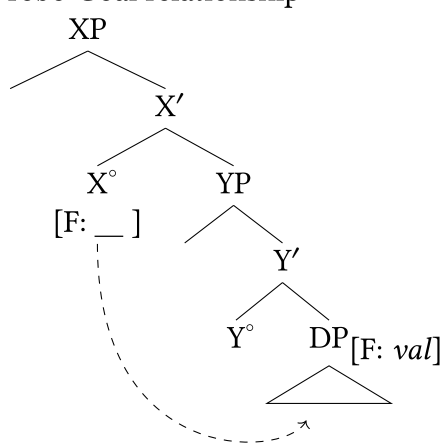
9.4.2 Agreement via Agree

Inflection based on φ-features is implemented syntactically using the Agree operation. So a Probe bearing some (valued) φ-features searches its c-command domain for a Goal that bears unvalued φ-features. Upon finding the closest such Goal, the Goal's φ-features are valued. Agree is fundamentally an operation where a valued feature on one projection gives a value to an unvalued feature on another projection.

(52) Agree
- A **Probe** is specified with a feature (F) is lacking a specified value, which it will seek a value for.
- A Probe searches its c-command domain for a matching feature F that contains a value.
- The closest instance of Feature F (the **Goal**[7]) values the unvalued feature F on a Probe.

As a reminder, a head A c-commands another head B if B is the sibling of A, or contained within (that is, dominated by) the sibling of A. We schematize this as in (53), where unvalued features are annotated with empty underlining and features (as we've done throughout) are annotated inside square brackets. We tend to use arrows to illustrate the probing operation, but as with movement arrows, these are for visual convenience and are not an obligatory part of the formalism. We use dotted arrows to represent probing operations to visually distinguish probing from movement.

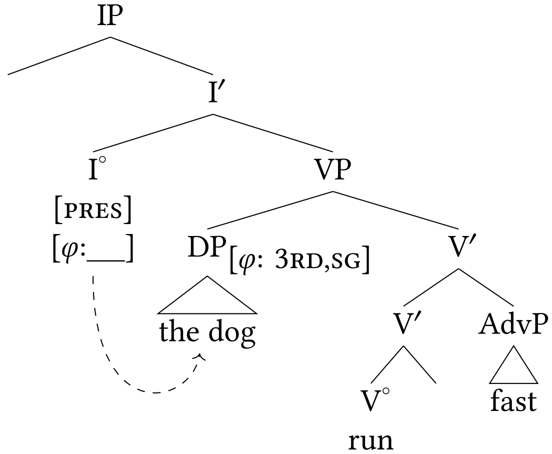
(53) Probe-Goal relationship:



We can look at a very simple example of the Agree operation at work in the example we started with in (44b). The initial position of the subject is presented in (54b) (following VPISH), and from that position the φ-Probe on I° searches its c-command domain for a DP with φ-features. It finds the DP *the dog*, which is third person singular.

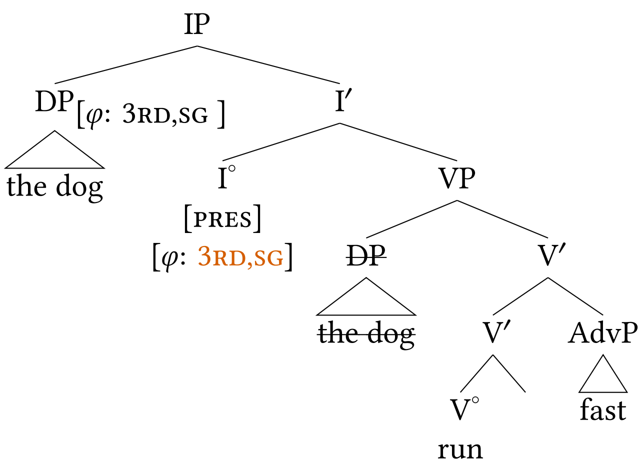
(54) a. The dog run-**s** fast.

b.



The result of this Agree operation is shown in (55). The previously-unvalued φ-features on I° are now valued with the features of the DP, and the DP has moved to Spec,IP (as is normal for English).

(55)



We don't draw or explain it here (see Chapter 12 for more details) but the tense and agreement features on I° appear morphologically on the main verb in English.

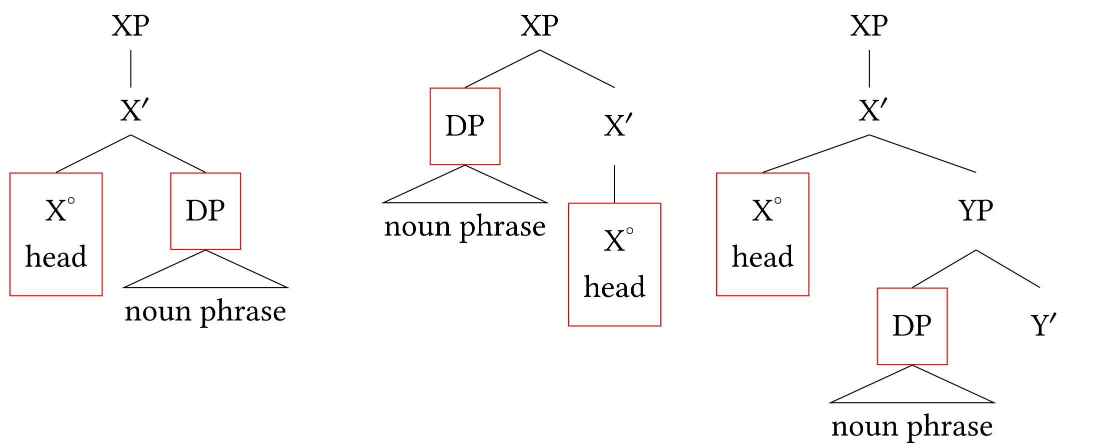
In contemporary research, the Agree operation is the standard mechanism to use to explain dependencies between two elements in the syntax. This includes the φ-features that we normally think of as "agreement" (or, conjugation), but also includes many other features within syntax. A prime example are Case-features, which we take up in the next section.

9.4.3 Case via Agree

The configurations we note above describe accurately major structural configurations that structural Case relations occur in, but as a theory or model of structural Case they are decidedly lacking—why should there be three different configurations for the same underlying operation? Example (56) repeats these configurations from above, slightly summarized, for reference.

(56) Case-assignment configurations, to be revised:

a. b. c.

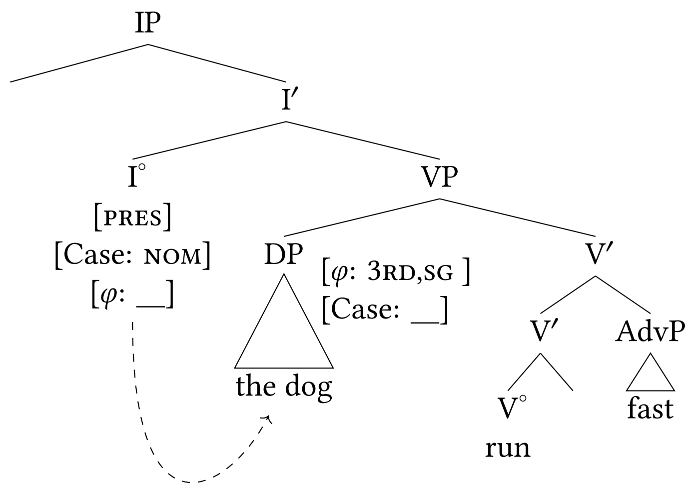


head-to-complement	*head-to-specifier*	*head-to-subpart-of-complement*
local, X° and DP siblings	local, DP is Spec,XP	nonlocal, X° c-commands DP

In fact, if we assume that the same Agree operation that accomplishes φ-agreement values Case-features, we can accommodate all three of these configurations. A head valuing its complement, or a phrase within its complement, straightforwardly follows from Agree: both of these phrases are c-commanded by the head. But how can a head value a phrase in its specifier? In fact, we just walked though an example of this in (55) above: as long as the spec-head relationship is the result of movement, we can assume that Agree also values Case via Agree, and the result of the Agree relation is the agreed-with phrase raising to the specifier of the head with the Probe, as in (55).

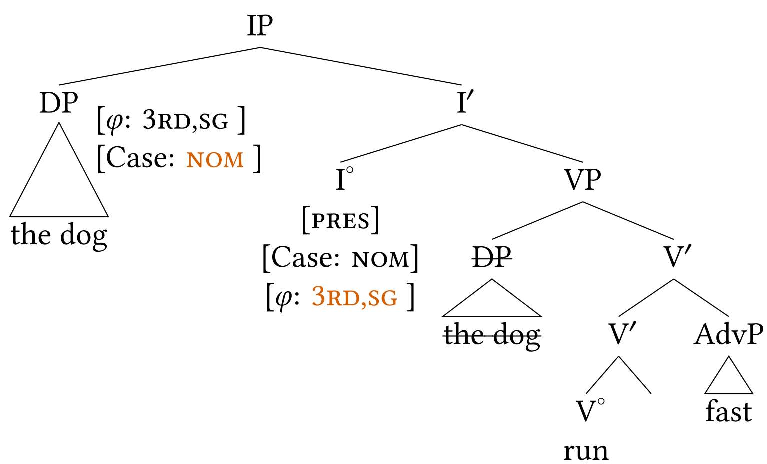
Case and φ-agreement show a key difference, however: the unvalued φ-features that are the Probe in an Agree relation are in the structurally higher position in an example like (54b). But if we add Case-features to the mix, as in (57), we can see that here the unvalued Case-features are c-command**ed** by the valued Case-features, instead of the other way around.

(57)



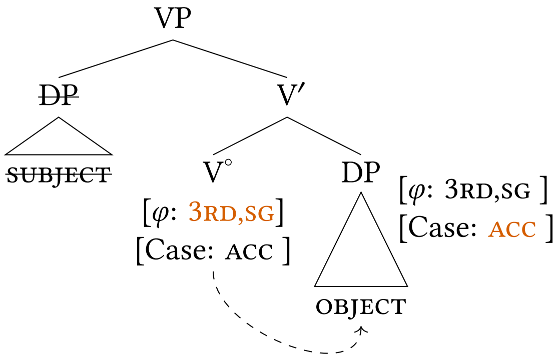
The typical assumption here is that Case-features and agreement features are essentially two sides of the same coin: Case is the realization of the Agree relation on the DP, and agreement is the realization of the Agree relation on the initiating head. So we assume that once an Agree relation has been established between a head and a phrase, all unvalued features shared between the two can be valued. So when the φ-Probe in (57) established an Agree relation with DP and values its φ-features, the unvalued Case-feature on the DP can also receive a value from the nominative feature on I°. So the result of the Agree relation in (57) is the configuration in (58).

(58)



What about accusative Case on objects? For now we can assume that there is also a φ-Probe on V° along with a valued Case-feature, so an Agree relation between V and some DP in its c-command domain results in valuation of Case on the object DP. The example in (59) assumes a third person singular object is Case-valued.

(59) Accusative Case valuation (first approximation, to be revised in Chapter 13)



English doesn't have agreement on verbs with objects, but it is in fact a common process across languages. (60) illustrates an example from Swahili, with the class 1 object agreement marker (glossed as 1ombelow) circled, which agrees with the class 1 object *mama*.

(60) Swahili object agreement

Wa-toto a-li-mw-ona mama.
2-children 2sm-pst-1om-see 1mama
'The children saw (their) mom.'

Therefore, we will assume the English (and all languages) share this agreement operation, which is sometimes evident in the case form of objects and sometimes marked via an agreement morpheme on the verb. We will modify the analysis of accusative Case licensing a bit in Chapter 13, but the core mechanics won't change.

This requirement for Case assignment to objects has traditionally been called the **Case Filter**, but we'll refer to it as the **DP Licensing Requirement**.

(61) DP Licensing Requirement:
All DP arguments must receive a Case value.

Especially for structural Cases, where DPs receive Case from some head in the syntax, DP licensing has become a central way in which syntacticians analyze and explain the distribution of DPs in sentences. That is to say, part of what explains why DP subjects or DP objects appear in the positions that they do in a particular construction in a particular language is how those DPs receive Case in those constructions. We can formalize this in the following way: all DP arguments can be assumed to enter the structure with an unvalued Case-feature ([Case: __]). The DP Licensing Requirement is essentially already encoded in the mechanics of Agree (unvalued features requiring a value), with the assumption that if a Case-feature fails to receive a value in the course of building a sentence, the sentence is ungrammatical. We will see how this plays out in various ways in what follows.[8]

You may be starting to find the tree-drawing overwhelming: there are so many features on each node of the tree! But we don't draw all of these features every time we draw a tree (because exactly that: it would quickly become overwhelming). In general, we only discuss the features and Agree operations that are relevant for whatever puzzle/construction is under discussion.

9.4.4 Agree and movement

The astute reader may be wondering: if Case is valued via Agree, why does, for example, finite I° in English appear to license its subject in its specifier, not in its c-command domain? What we assume in these instances is that some Probes require movement of the Goal after Agree has occurred (and, in fact, this happens with many Probes). Miyagawa (2010) calls this **Probe-Goal Union**, which is a fitting term for our purposes.

(62) **Probe-Goal Union**: A valued Goal moves to the specifier of the head bearing a Probe.

So when does a Goal undergo Probe-Goal Union, and when does it not? It doesn't appear that there is a large insight to be had here; some Goals move to their Probes, and others don't, and it appears to be fairly idiosyncratic when which occurs. The grammar of any particular language (or construction within a language) is consistent, but there doesn't appear to be a deep insight to be had about UG: some Agree relations in human languages are linked with movement, others are not. So children learning a language will simply have to learn that movement happens in some instances. This is fine—lots of aspects of language are learned and are not a part of UG. But we do want UG to contain mechanisms to encode those facts about language. Here we annotate a Probe as surrounded by large bullets if it also triggers movement (a formulation borrowed from Müller 2010).

(63) Probe that triggers Probe-Goal Union:
• [φ: __] •

This requirement of a phrase to move to a specifier position is often termed an EPP requirement, or an EPP quality. **EPP** used to stand for something that is now no longer in our theory[9] but can be thought of now as a requirement for an Extra Peripheral Position (as suggested by David Pesetsky). To our knowledge, EPP qualities encoded in syntax don't link with deep insights about UG, but rather are language-specific components of mental grammar that are learned by children. The existence of EPP qualities is much too common to be a coincidence, however, suggesting that UG must contain a way for children to encode this in their mental grammars. Similarly, the precise features included in a Probe seem to be matters of language-specific learning, but the nature of Probes in general (i.e., the Agree operation) appears to be a consistent component of UG.

There may also be times where a position seems to require that a phrase appear there independently of whether the phrase participated in an Agree relation, a sort of generalized EPP requirement. Syntacticians tend to posit [EPP] features on heads to encode these circumstances; these are instances that simply require some XP to be moved or added to a specifier position.

Arguably, English Spec,IP is one of these generalized EPP circumstances. Expletive subjects suggest this to us: there are instances where the subject that triggers agreement does not move to Spec,IP.

(64) a. There are children running in the park.
b. There is a child running in the park.

These sentences require the expletive subject *there* to be grammatical, but agreement is controlled by the low subject (and the low subject is Case-licensed as well). The requirement that an overt subject to occur in that position, even if it is an expletive, suggests that we need to place an [EPP] feature on I°. Many such requirements can in fact be reduced to PGU requirements on Probes, however, and this is generally the preferred route, since all an [EPP] feature does is annotate that something must occur in a position, it doesn't offer a deep or explanatory account of *why* that is so.

Different case/agreement alignments 9.5

A short comment is in order about kinds of morphological ase patterns that are distinct from the sorts that are described in the languages above. The patterns described in this section constitute an empirical puzzle in need of explanation based on the model that we've built thus far, but are included here to ensure your exposure to something closer to the true range of case-marking patterns in human languages. The nominative-accusative case alignment is a common pattern cross-linguistically, but it is not the only common pattern. In some languages, case forms appear to track much more closely to thematic roles rather than tracking grammatical functions. In Georgian, for example, agentive subjects are marked with the case form *-m(a)*, in both transitive sentences like (65a) and intransitive sentences like (65d). In contrast, theme arguments are marked with the distinct case form *-i*, whether that theme is an object as in the transitive in (65a), or a subject as in the unaccusative intransitive in (65b). We simply gloss these forms as AGENT and THEME here for the moment for the sake of clarity.

(65) Georgian case-marking (Butt 2006, 157)[1]

a. Nino-m Ceril-i daCera. **Georgian**
Nino-AGENT letter-THEME wrote
'Nino wrote a letter.'

b. Kar-i gaiɣo.
door-THEME opened
'The door opened.'

c. * Kar-ma gaiɣo.
door-AGENT opened
'The door opened.'

d. Nino-m imɣera.
Nino-AGENT sang
'Nino sang.'

Beyond that, there are many languages that have a case alignment that (like nominative-accusative) is not strictly linked with thematic role, but which nonetheless does not align with notions of 'subject' and 'object' that are familiar from European languages like German, Greek,

1. Glosses in these examples are adjusted and simplified for expositional purposes.

and Latin, or from East Asian languages like Korean and Japanese. In nominative-accusative languages, accusative is essentially a special case that is assigned in transitive constructions to the object: both transitive and intransitive sentences assign nominative to their subjects. The ergative-absolutive case alignment assigns a special case in transitive sentences, but it is assigned to the agentive subject (this is called ergative case). Transitive objects and intransitive subjects are both assigned the same case (called absolutive case). In Tongan (Austronesian, Tonga) absolutive case is marked with the *ʻa* morpheme on the intransitive subject in (66a) and the transitive object in (66b); Tongan marks ergative case (*ʻe*) on transitive subjects as shown in (66b).

(66) Tongan case-marking (Otsuka 2005)

a. Naʻe tangi ʻa Sione.
   PST cry ABS Sione
   'Sione cried.'

b. Naʻe maʻu ʻe Sione ʻa e ika.
   PST get ERG Sione ABS the fish
   'Sione got the fish.'

We see similar patterns emerge in agreement as well. The Mayan language Chol (Mexico) does not inflect nominals with case, but its agreement forms on verbs follow a ergative/absolutive pattern (in some tenses/aspects).[10] In these examples in (67) the suffixed agreement forms are absolutive, agreeing either with the transitive object in (67a) or the intransitive subject in (67b). The prefixed agreement is reserved for transitive subjects, as in (67a).

(67) Chol ergative-absolutive agreement (Coon 2013, 55)

a. Tyi a-k'el-e-yoñ.
   PFV 2ERG-watch-TV-1ABS
   'You watched me.'

b. Tyi ts'äm-i-yoñ.
   PFV bathe-ITV-1ABS
   'I bathed.'

This is just the tip of the iceberg with respect to the complexity of morphological case and agreement patterns cross-linguistically. Nominative-accusative and ergative-absolutive alignments are the predominant patterns, but there are fine-grained variations of these in a vary array of languages (not to mention distinctions in inherent case patterns, agreement on grammatical categories other than verbs, and other kinds of variety that are found).

Accounting for these kinds of case and agreement alignment require some minor modifications of the case valuation system that we sketched as part of our model here. And as is the case with language grammars, poking into the details of any individual language's grammar reveals a range of interesting nuances and sub-patterns. So the simple sketch of Agree offered in this chapter doesn't (on its own) explain every pattern of case and agreement in the world's languages. This small subsection is simply intended to open the door a small crack to show you that there is much more complexity to be evaluated. But even in analyses of these other kinds of patterns, you find the same Agree operation employed as a mechanism for feature valuation, just with differences employed in the implementation (e.g., which features make up Probes, where

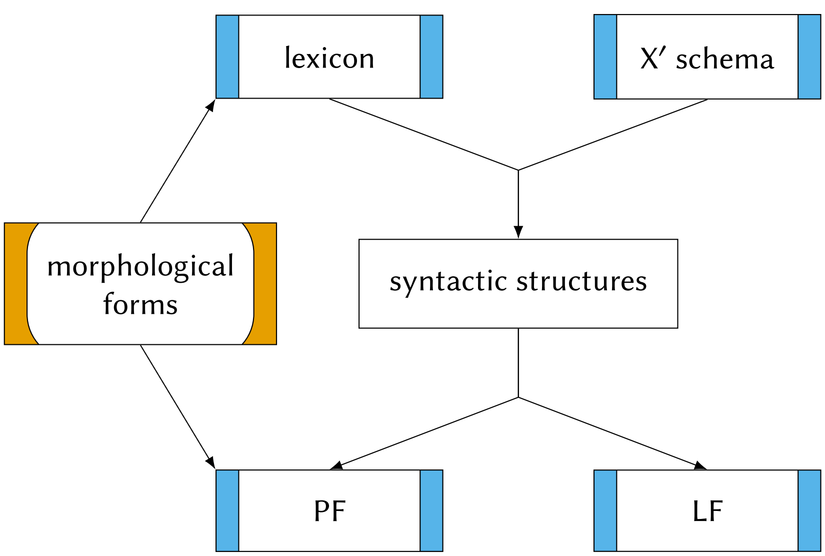
Probes are located, etc). The core theory that we sketched here remains a part of the UG toolkit for grammatical analysis.

State of the conceptual architecture 9.6

A model that includes feature valuation in the syntax also places some requirements on the overall conceptual architecture of the model. Specifically, we can't assume that all morphological forms are established from the start of syntax; rather, what form a verb takes, or a noun phrase takes, depends on the features it acquires over the course of building a sentence (e.g., case or agreement via Agree, inflection of verb forms via tense, etc). So at least some morphological forms are decided upon (and inserted into a sentence) after that sentence is already built. This is referred to as **Late Insertion** of forms. The model that assumes Late Insertion is called **Distributed Morphology (DM)**, which is the predominant view of the morphology-syntax interface. Within DM, some morphological forms are lexical, joining the X′ schema at the beginning of the derivation of a sentence. For example, lexical items like *cat* and *car* will always be realized as *cat* and *car*. But others are inserted based on how a derivation proceeds. So a verb like *run* will be realized as *run* in some syntactic contexts (ex., a future tense I°), but will take another morphological form in other syntactic contexts (e.g., *ran* under a past tense I°). In this way, morphology is "distributed": some of it seems to be lexical, whereas some other forms are dependent on the results of syntax.

We see a model emerging that is often referred to as the 'T-model' or the 'Y model' because of its visual correspondence to a T or Y (excluding the morphology portion we have added here).[11]

(68)



Phonological Form (PF) is where pronunciation is sorted out. Morphological forms are inserted here, but a lot of phonology depends on the morphological forms of elements, so Late Insertion must precede other aspects of phonology. Logical Form (LF) is roughly semantics, that is, the interface with concepts and meaning.

This conceptual architecture provides organizing principles for our theorizing. As established here, syntax is built from lexical items, and therefore the properties of lexical items can affect syntax. But phonology (PF) follows syntax, so the properties of phonology are *not* expected to affect syntax. This does appear to be true: we see no rules in the world's languages like "move a DP to Spec,IP if it begins with a /p/". Likewise, PF and LF are built from the output of syntax, but are not directly dependent on each other. This tracks with the dissociation of pronunciation and meaning (languages don't share particular sounds with particular meanings, except due to historical connection or contact between the languages). This discussion of the model is by no means exhaustive, but it's sufficient for our current purposes.

New Empirical Observations

- morphological case-marking of nominals
  - nominative-accusative alignment
  - ergative-absolutive alignment
- inherent vs. structural case
- agreement/inflection in φ-features

Components of the Theoretical Model

- Agree operation
- valued and unvalued features
- Case and agreement features
- morphological case vs. abstract Case features
- φ-features (phi-features)
- DP licensing
- Late Insertion of morphological forms

Notes 9.7

1. We will eventually in this chapter follow the standard convention in the field by capitalizing "Case" where it refers to an abstract syntactic feature and leaving it lowercase where it refers to a morphological realization of these features; in the leadup to that conclusion, we simply use lowercase "case."

2. This chart was originally inspired and based on a similar chart on Wikipedia but has been very heavily edited.

3. A very few German verbs assign a third case, the genitive. These are felt to be archaic by present-day speakers, so we ignore them here.

4. In addition to Case assignment, which holds between a head in a sentence and a noun phrase, there is also the phenomenon of case concord, where constituents within a noun phrase all share the same Case. If case concord exists in English, it is invisible, but we know it exists in other languages from examples like (7). The mechanisms for Case assignment that we discuss here don't (directly) account for case concord: for now, we will simply assume that Case is assigned to an entire DP and subconstituents receive Case by virtue of being a part of the a Case-licensed DP.

5. The higher finite I° in the clauses in (34) can't assign Case to the lower subject, since the relationship between the two is nothing like the requisite spec-head configuration. Instead, it assigns nominative to the higher subject (its "own" subject, as it were).

6. Glosses for the Kinande data are simplified for expositional purposes.

7. Be careful not to confuse this with the thematic role GOAL!

8. There are various caveats here that will become evident to the student continuing on in syntax, but which go beyond what an introductory text requires.

9. Within Government and Binding Theory, EPP stood for Extended Projection Principle: see Haegeman (1994) for a textbook discussion of Government and Binding Theory.

10. In fact, Chol is like many languages in showing split ergativity, displaying ergative-absolutive patterns in some instances but not in others. A full discussion of ergative-absolutive case and agreement goes far beyond what we can tackle in this textbook, but an interested reader can read about these patterns in Coon (2013).

11. It is also sometimes referred to as an inverted T-model or inverted Y-model.

Exercises and problems 9.8

Exercise 9.1: Practice with Case assignment

Explain how each of the DPs in the sentences below are assigned Case, per the model built in this chapter. Give a tree for each sentence.

(1) a. Three children chased their dog in the park.
  b. The puppy slept for eight hours.
  c. The students will be reading all week.
  d. The rains came early this year.
  e. My brother is cooking dinner.

(2) a. There are three children chasing their dog in the park
  b. It is raining.
  c. I want the children to be quiet.

Exercise 9.2: Practice with agreement

For each of the examples in (1), explain (using Agree formalism) the derivation of the agreement forms on the auxiliary verbs. For the purposes of this assignment, assume that the auxiliary verbs arise on I°.

(1) a. The children are leaving.
  b. The boy is running very fast.
  c. There are three raccoons fighting in the yard.

Exercise 9.3: Case assignment in infinitives

A. Table 9.2 summarizes the Case assignment facts for English discussed in the chapter. A few values remain to be filled in based on the evidence in (1).

Head	**Case**	**Configuration**
Adj	N/A	
D, possessive	POSS	spec-head
I, finite	NOM	spec-head
I, nonfinite (*to*)		
N	N/A	
P	OBJ	head-comp
V	OBJ	head-comp, head-spec

Table 9.2: Incomplete summary of case assignment

(1) a. * { Us, we, our } to leave now would not be helpful.
b. For { ✓ us, *we, *our } to leave now would not be helpful.

B. Why is there no entry (not even a "N/A" entry) for ordinary (= nonpossessive) D?

Exercise 9.4: Subjects of non-finite clauses

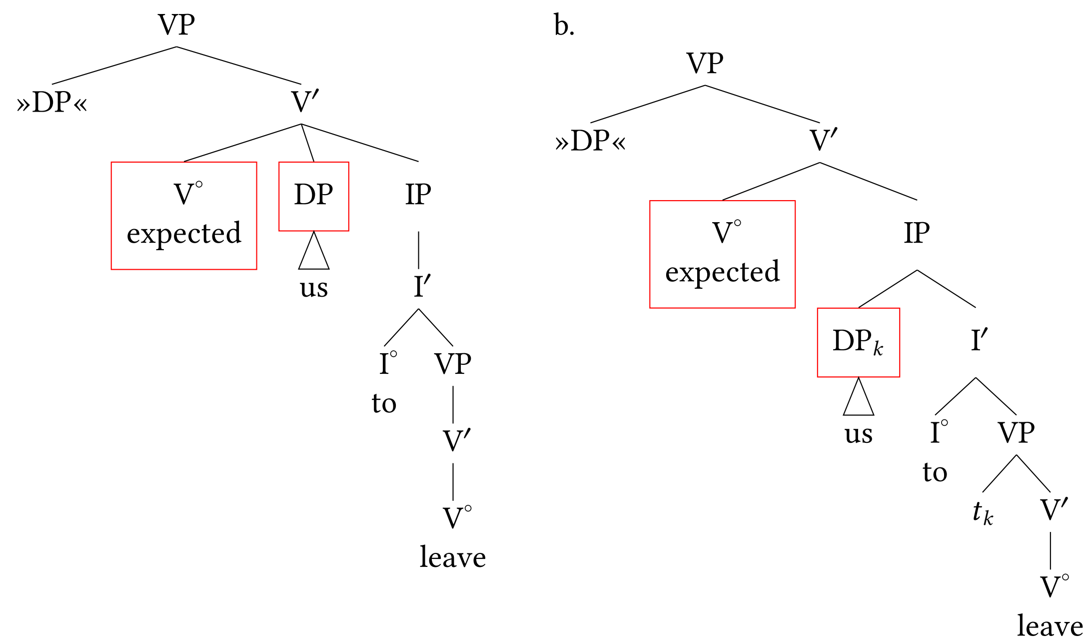
According to the analysis in the text, why are the sentences in (1) ungrammatical? You don't need to be concerned with the structure of the entire sentence; just treat the underlined part as an IP headed by nonfinite I.

(1) a. * They claim <u>to they understand Hegel</u>.
b. * They claim <u>they to understand Hegel</u>.

Exercise 9.5: Exceptional Case-marking

How is the thematic relation between *leave* and its argument *us* represented in (1b)? How about in (1a)? (Note: these trees were given earlier in (39).)

(1) a. b.



Exercise 9.6: More practice with Case assignment

A. Build structures for the sentences in (1). You can use triangle notation for all DPs. Assume that *for* can take IP complements. Assume further that *for* clauses can be subjects of sentences; in other words, PPs can substitute in Spec(VP) and move to Spec(IP).

(1) a. That should be them now.
b. They waited for us.
c. We waited for there to be a sale.
d. For daycare to be available for the children would be convenient.
e. For daycare to be available would be convenient for the parents.
f. I suspect the class to be difficult.

B. How do each of the DPs in (1) get Case? Which Case is assigned by what head in what configuration? Group similar patterns together.

Exercise 9.7: Null subjects and agreement

In many languages it is possible to leave a DP subject out of a sentence, potentially producing a sentence that consists of nothing other than a verb. We illustrate from Tuki (Bantu, Cameroon) in (1), but there is a vast number of unrelated languages across the world that have similar patterns.

The interpretation of null subjects tends to be of a pronominal subject, that is, a discourse-familiar referent that is referred to but not explicitly mentioned. In this chapter we assumed that agreement morphemes arise from unvalued features on a head (I° for subject agreement on verbs) that receive a value via Agree in the course of building the structure of a sentence. For this exercise, assume the Agree formalism is consistent for all subject agreement forms on verbs.

A. What is the subject of the sentences in (1b) and (1d)?

B. Draw a tree that specifies the agreement operation for (1b).

(1) Tuki null subjects (Biloa 2013, 61)

a. Mbárá a-bangáḿ.
Mbara AGR-cries
'Mbara cries'

b. A-bangáḿ.
AGR-cries
'They.SG cry.' or 'He/she cries.'

c. Vádzu vá-bangáḿ.
children AGR-cry
'Children cry.'

d. Vá-bangáḿ.
AGR-cry
'They.PL cry.'

Exercise 9.8: Non-Case-assigning heads

In the chapter, we stated that nouns and adjectives aren't Case-assigners in English. Provide evidence for that statement. One piece of evidence for each category is sufficient.

Exercise 9.9: Agreement and Case in Lubukusu

Data here are adapted from Wasike (2007). Glosses are adjusted for our current purposes.

Instructions: φ-feature agreement is often talked about as the flip side of Case assignment—shown on verb heads instead of DPs. Here, consider a variety of inflectional properties of Lubukusu (*wh*-agreement especially). Question: Is φ-feature agreement necessarily linked with Case assignment? Explain why or why not.

Background: Background: Lubukusu is a Bantu language of the Luyia subgroup that is spoken in Kenya. Lubukusu has a robust noun class system: noun classes are glossed with cardinal numbers (1, 2, 3, 4, etc). You don't have to explain anything about noun classes, though they do show what is agreeing with that, which is sometimes helpful to know. AGR = agreement morphemes, AGR glosses are accompanied with a noun class gloss based on what the noun class is of the subject that they agree in; PST = past tense; PRS = present tense; PFV = perfective aspect (translated similar to past tense); all verbs in Tiriki necessarily end in a vowel—sometimes this vowel is part of another affix, at other times we simply gloss it as FV (final vowel).

(1) a. Nafula a-kha-singa chii-ngubo kalaa.
1Nafula 1AGR-PRS-wash 10-clothes slowly
'Nafula is washing clothes slowly.'

b. Baba-ana ba-kha-singa chii-ngubo kalaa.
2-children 2AGR-PRS-wash 10-clothes slowly
'The children are washing the clothes slowly.

c. Nangila a-a-tekhela Wekesa kama-kaanda.
1Nangila 1AGR-PST-cook.for 1Wekesa 6-beans
'Nangila cooked beans for Wekesa.'

d. Baba-ana ba-a-tekhela Wekesa kama-kaanda.
2-children 2AGR-PST-cook.for 1Wekesa 6-beans
'Nangila cooked beans for Wekesa.'

(2) a. Wekesa ni-ye baba-aana ba-a-tekhela kama-kaanda.
1Wekesa COMP-1AGR 2-children 2AGR-PST-cook.for 6-beans
'It was Wekesa that the children cooked beans for.'

b. Kama-kaanda ni-k-o Nangila a-a-tekh-el-a Wekesa.
6-beans COMP-6AGR 1Nangila 1AGR-PST-cook-appl-fv 1Wekesa
'It was beans that Nangila cooked for Wekesa.'

(3) a. Nafula a-kha-singa chii-ngubo a-rieena?
1Nafula 1AGR-PRS-wash 10-clothes 1AGR-how
'How is Nafula washing clothes?'

b. Baba-ana ba-kha-singa chii-ngubo ba-rieena?
2-children 2AGR-PRS-wash 10-clothes 2AGR-how
'How are the children washing clothes?'

(4) a. Wafula a-a-loma a-li Nekesa a-a-rema kumu-saala.
1Wafula 1AGR-PST-say 1AGR-COMP 1Nekesa 1AGR-PST-cut 3-tree
'Wafula said that Nekesa cut the tree.'

b. Baba-ana ba-a-loma ba-li Nekesa a-a-rema kumu-saala.
2-children 2AGR-pst-say COMP-2AGR 1Nekesa 1-pst-cut 3-tree
'The children said that Nekesa cut the tree.'

Exercise 9.10: Case assignment inside subconstituents

This exercise extends Exercise 6.9.

A. If you have not already done so for Exercise 6.9, build structures for the underlined DPs in (1). Here, you can use triangle notation for both constituent DPs (*Kim* and *incompetents*).

(1) a. We dislike Kim's impulsive hiring of incompetents.
b. We dislike Kim impulsively hiring incompetents.

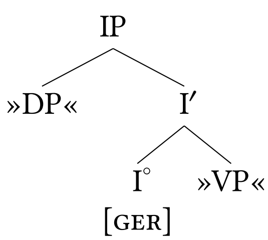
B. How do the DPs contained within the underlined phrase get Case? Which Case is assigned by what head in what configuration?

Exercise 9.11: Gerunds

This exercise extends Exercise 9.3, A.

English has a silent nonfinite gerund head ([GER]) with the elementary tree in (1).

(1)



The VP complement of this head must be headed by a verb form in -*ing*.

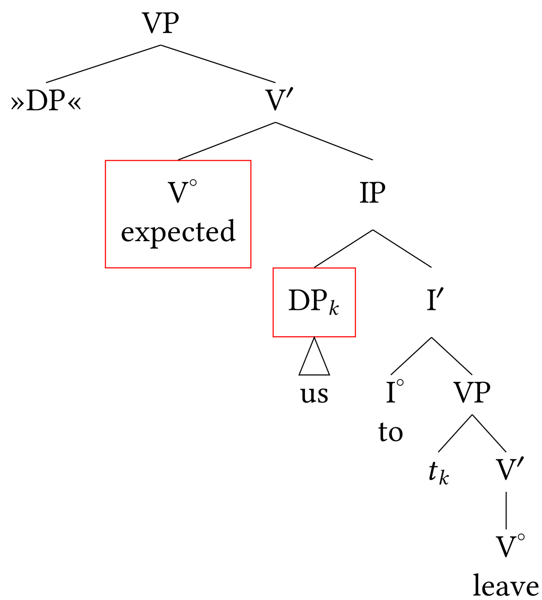
(2) { ✓Them, *they, *their } being late, we cancelled the meeting.

Task: Based on the evidence in (2), determine this head's Case assignment properties.

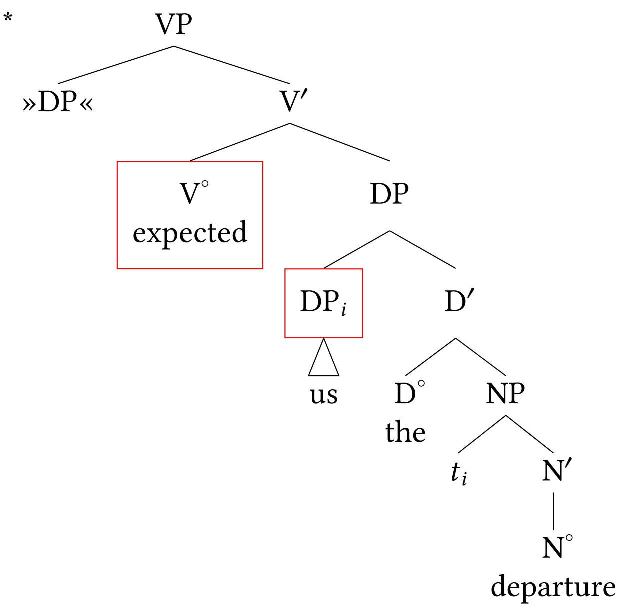
Problem 9.1: Complements of *expect*

A. The verb assigns Case to *us* in the head-spec configuration in both (1a) (= (39b)) and (1b), but (1b) is completely ungrammatical. Why is that? The intended meaning of (1b) is *expected our departure.*

(1) a. ✓

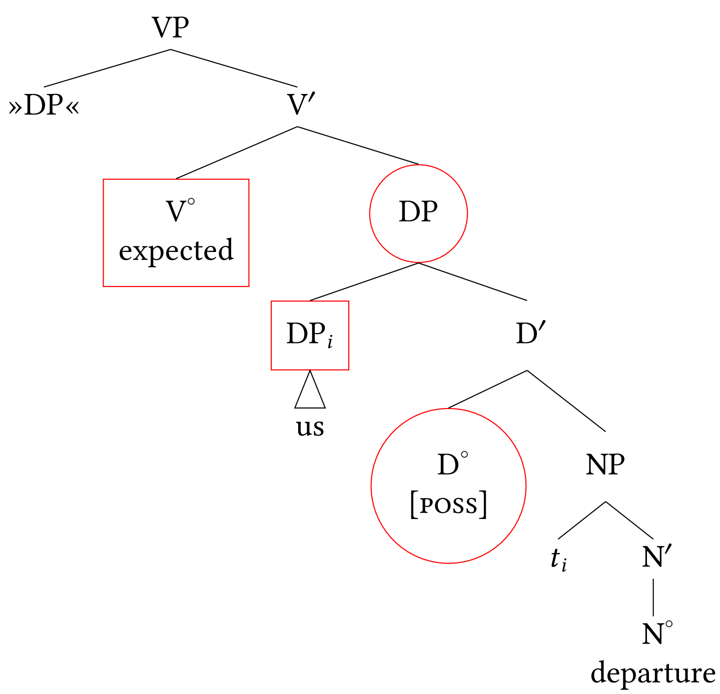


b. *



B. In (2), V assigns objective Case to the lower DP *us* in the head-spec configuration, as indicated by the boxes, and the possessive head assigns possessive Case to the higher DP, as indicated by the circles. Recall that possessive Case is invisible on full DPs. Again, the intended interpretation is *expected our departure.* But again, as in (1b), the result is completely ungrammatical. What has gone wrong?

(2) *



Problem 9.2: Archaic English PPs

This exercise extends Exercise 9.3, A.

English has (now very archaic) PPs of the type in (1). Assume that *there* is a DP, just like expletive *there* (even though the *there* at issue here is referential).

(1) thereafter (= after that), therefrom (= from that), therewith (=with that)

What would you suggest as the elementary tree shared by all of the relevant prepositions, and in what configuration do they assign Case to *there*?

Note: There are several solutions.

Problem 9.3: Welsh subject positions

These data are drawn from Roberts (2005), Awbery (1976), Sproat (1985), and Borsley, Tallerman, and Willis (2007).

Welsh is a Celtic language that is spoken in Wales (a part of the United Kingdom). Recall these basic Welsh examples from Problem 7.1 in Chapter 7. Here we are considering what the Case assignment properties of Welsh must be.

(1) Diflanodd y dyn.
disappeared the man
'The man disappeared.'

(2) Gwelodd y dyn y ci.
saw the man the dog
'The man saw the dog.'

(3) Gwêl y dyn y ci.
sees the man the dog
'The man sees the dog.'

(4) Gwelai y dyn y ci.
was.seeing the man the dog
'The man was seeing the dog.'

(5) Gall Ifor ddarllen llyfrau.
can Ifor reading books
'Ifor can read books.'

(6) Dylai Gwyn ddisgrifio'r llun.
ought Gwyn describe the picture
'Gwyn ought to describe the picture.'

(7) Prynodd Elin dorth o fara.
buy.PST.3S Elin loaf of bread
'Elin bought a loaf of bread.'

(8) Gwnaeth Elin brynu torth o fara.
do.PST.3S Elin buy loaf of bread
'Elin bought a loaf of bread.'

(9) Ddaru Elin brynu torth o fara.
PST Elin buy loaf of bread
'Elin bought a loaf of bread.'

(10) Ddaru Megan fynd adre.
PST Megan go home
'Megan went home.'

Tasks:

- Explain what structural configuration nominative Case must be assigned in for Welsh.
- Draw a tree of (5).

Identifying Puzzles 9.1: More on Welsh nominative Case

This puzzle expands on Problem 9.3. After solving that, now consider these additional examples that contain auxiliaries and aspect morphemes. Assume that aspect morphemes are introduced on their own functional heads, similar to English auxiliaries.

(1) Mae'r dyn yn gweld y ci.
is.the man in (ASP) seeing the dog
'The man sees the dog.'

(2) Mae'r dyn wedi gweld y ci.
is.the man after (ASP) seeing the dog
'The man sees the dog.'

(3) Bydd y dyn yn gweld y ci.
will.be the man in (ASP) seeing the dog
'The man will see the dog.'

(4) Mae ef ar weld y ddrama.
is he on (ASP) seeing the play
'He is about to see the play.'

(5) Mae Elin wedi prynu torth o fara.
be.PRES.3SG Elin PFV buy.INF loaf of bread
'Elin has bought/is buying a loaf of bread.'

(6) Mae Elin yn prynu torth o fara.
be.PRES.3SG Elin PROG buy.INF loaf of bread
'Elin has bought/is buying a loaf of bread.'

(7) Mae Rhiannon yn cysgu rwan.
be.PRES.3SG Rhiannon PROG sleep.INF now
'Rhiannon is sleeping now.'

(8) Mae Rhiannon wedi mynd adre.
be.PRES.3SG Rhiannon PERF go.INF home
'Rhiannon has gone home.'

(9) Ddaru Megan ddim mynd adre.
PST Megan NEG go.INF home
'Megan didn't go home.'

<u>Answer these questions</u>

1. What challenge do these examples pose to the analysis of Welsh nominative Case from Problem 9.3?
2. Give as precise a statement as you can about what position/positions Welsh subjects are licensed in.
3. Discuss the ways this does (or does not) fit naturally with the model developed in this chapter.

Identifying Puzzles 9.2: Assigning Case, edge cases

The underlined noun phrases do not fit our Case assignment configurations discussed above:

(1) a. We discussed the proposal <u>the very next week</u>.
b. We kept their books <u>five years</u>.
c. We don't read the Times <u>every single day</u>

There are two ways to think about these noun phrases: either they are not assigned Case, or they are assigned Case in some way that is non-obvious. Build a hypothesis for both options.

1. Assuming they are not assigned Case, why are these noun phrases exceptionally not required to be assigned Case? What is different about these vs. the ones we discussed in the chapter?
2. Assuming that they ARE assigned Case: can you think of a way of assigning Case to the underlined phrases in (1) without introducing a new Case assignment configuration?

Identifying Puzzles 9.3: Case-assigning determiners

In English, there are no Case-assigning determiners without a specifier. Is such a determiner possible in principle?

Identifying Puzzles 9.4: Is Case in a one-to-one relationship with DPs?

In the examples discussed in the text, there is always a one-to-one correspondence between Case-assigning heads and noun phrases. What would happen if we relaxed this condition?

A. In particular, give an example of a linguistic expression that would be grammatical if we allowed a single head (say, V) to assign Case to two noun phrases in the configurations that we have discussed?

B. Conversely, give an example of a linguistic expression that would be grammatical if we allowed a single noun phrase (or more precisely, a single chain) to get Case from more than one head?

This is speculative, of course-I–the intention is to work out what the logical possibilities are for languages based on our model as we have developed it here.

Identifying Puzzles 9.5: Multiple subjects in Japanese

Explain why the constructions below are a puzzle for our model as it has been developed thus far.

Note: Topic marking alternations have been suppressed from these data for simplicity, but be aware students who are native speakers of Japanese may find some of these sentences pragmatically odd without context.

(1) a. 象が 鼻が 長い。
zou-ga hana-ga nagai.
elephant-NOM nose-NOM long
'Elephants have long noses.' (*lit.* 'Elephants' noses are long.')

b. タンポポが 成長が 速い。
tanpopo-ga seichō-ga hayai.
dandelion-NOM growth-NOM fast
'Dandelions grow quickly.' (*lit.* 'Dandelions' growth is fast.')

c. 山本先生が 息子が 足が 痛い。
Yamamoto-sensei-ga musuko-ga ashi-ga itai.
Yamamoto-teacher-NOM son-NOM foot-NOM painful
'Professor Yamamoto's son's foot hurts.'

10 A-movement of subjects: Passives and more

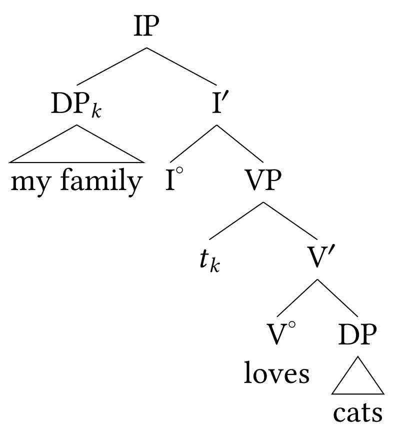
Chapter outline

In this chapter we discuss passivization, which involves movement of a non-subject to become a subject: we discuss the movement as an instance of movement to an argument position (or A-position), also known as A-movement. This is a pattern of movement that we have already encountered but which we have not been specific about before now. There are a number of ways that this movement is non-identical to the A′-movements (movement to non-argument positions) and head-movements we have discussed up until now. A-movement includes movement of a subject from Spec,VP to Spec,IP, but also includes movements from other positions to Spec,IP. This chapter is just an initial introduction: Chapter 11 discusses another significant construction centered on an A-movement that further demonstrates these properties. This chapter is fairly short, focusing on the existence of A-movements and some fundamentals about the properties of passives. Chapter 13 is where we fulfill the promises of this chapter and give a full formal treatment of passives.

10.1 A-movement of subjects in transitives

At this point we have seen various instances of movement: for example, we have seen *wh*-movement of *wh*-phrases to Spec,CP. We referred to this as movement to a non-argument position (an A′-position), which is also known as A′-movement. *Wh*-movement was one instance of A′-movement, but it also includes movements like focus movement and topicalization. We have also seen movement of heads, specifically I°-to-C° in questions as well. But even before that we had concluded that the initial positions of subjects is in Spec,VP, even though (in many instances) they move to Spec,IP.

(1)



We have become comfortable with these patterns, but there are relevant aspects of these patterns that perhaps seem so obvious that they easily escape attention. Consider the examples of *wh*-questions in (2), where it is possible to move either a *wh*-subject or a *wh*-object to Spec,CP.

(2) a. Who will ___ instruct the children?
b. Who will the teachers instruct ___ ?

Wh-movement, and A′-movement more generally, is comfortable moving relatively long-distance. So in (2b) the *wh*-object moves over the subject to Spec,CP. But this is not true of all kinds of syntactic movements. Consider the baseline sentence in (3a), which has the underlying partial structure in (3b).

(3) a. The teachers will instruct the children.
b. [$_{\text{IP}}$ will [$_{\text{VP}}$ the teachers instruct the children]]

Based on (3b), we expect movement to Spec,IP. But this movement is distinctly different than the *wh*-movement to Spec,CP. Whereas *wh*-movement allows an object to hop over a subject, the parallel instance of an object hopping over a subject to fill Spec,IP is ungrammatical.

(4) a. *The children will the teachers instruct.
b. [$_{\text{IP}}$ The children will [$_{\text{VP}}$ the teachers instruct ___]] .

(5) **A-movement** (movement to an argument position)

DP movement to a Case-licensing head (that is, an argument position).

As we will see, only DP arguments can participate in this movement; non-arguments are typically incompatible with this kind of movement.[1]

Wh-movement to Spec,CP is an instance of what we called A′-movement (A-bar movement), what we can think of as non-argument movement, since A′-movement is not restricted to argument DPs the way that A-movement is and A′-movement is not movement to a Case-licensing position.

(6) a. The children will leave tomorrow.
b. When will the children leave ___ ?

(7) a. The children are playing in the park.
b. Where are the children playing ___ ?

So we can see that it's possible to A′-move adjunct *wh*-phrases to Spec,CP without trouble. Attempting to A-move these adjunct phrases to the canonical subject position in Spec,IP is uniformly unacceptable. The examples in (8) are failed attempts to A-move adjunct phrases to Spec,IP. (Examples (8b) and (8c) entertain different potential positions of the subject when it doesn't move to Spec,IP, but both are ungrammatical.)

(8) a. * Tomorrow will the children leave.
b. * All day will the children play.
c. * All day will the children play.

These different locality profiles suggest different motivations for A-movement than for A′-movement. Whereas A-movement appears to share a connection with Case-licensing (and is more local), A′-movement generally seems linked to some kind of discourse property (ex., *wh*-questions, emphasis via focus or topic), and has a less strict locality profile.

(9) **A′-movement** (movement to a non-argument position)

Movement of arguments or adjuncts to a non-Case-licensing head. This movement is typically to Spec,CP and generally occurs for some discourse purpose.

Neither this definition of A′-movement (movement to non-argument positions) in (9) nor the definition of A-movement (movement to argument positions) in (9) capture every instance of A-movement and A′-movement cross-linguistically, but they are good enough for our current purposes. We get more precise about the distinctions between these different movement profiles (A vs. A′) when we discuss Binding Theory in Chapter 15.

The point of this discussion is that even though we see XPs of various sorts moving in different kinds of constructions, these movements are not all identical, and show different profiles (i.e., they show different collections of properties). This chapter discusses some major instances of A-movement in addition to the VPISH-induced movement discussed in this section; we introduce A-movement of derived subjects in passives in § 10.2–§ 10.3 and we discuss A-movement in unaccusative predicates in § 10.5. Chapter 11 discusses Raising constructions, another major instance of A-movement (i.e. movement to an argument position).

Characteristics of the passive 10.2

Passive constructions are a consistent instance of A-movement cross-linguistically. In English, as in many other languages, active sentences like (10a) have passive counterparts like (10b).

Background Info

In the remainder of this section, grammatical functions are indicated above the boxed element and thematic roles are indicated below. The following abbreviations are used:

SBJ—subject, OBJ—(direct) object, A—agent, THM—theme

(10) a. $\underset{\text{A}}{\overset{\text{SBJ}}{\boxed{\text{We}}}}$ approved $\underset{\text{THM}}{\overset{\text{OBJ}}{\boxed{\text{them}}}}$.

b. $\underset{\text{THM}}{\overset{\text{SBJ}}{\boxed{\text{They}}}}$ were approved $\underset{\text{A}}{\overset{\text{PP}}{\boxed{\text{(by us)}}}}$.

Passivization has a number of effects. First and foremost, the agent argument, which is expressed as the subject of the active sentence, appears in the passive as an optional *by*-phrase. Second, the now vacant subject position is taken over by the theme argument. In other words, the agent and theme arguments are linked to different **grammatical relations** in the passive than in the active. If we think of subject as outranking object on a hierarchical scale of grammatical relations (as is often assumed), then the passive **demotes** the agent argument and **promotes** the theme argument. Third, passive past participles, unlike their homonymous active counterparts, can't assign objective Case.

(11) a. ✓ We approved them. (active)

b. * It_{EXPL} was approved them (by us). (passive)

Intended meaning: 'They were approved (by us).'

In English, the passive is expressed analytically by a combination of the past participle and auxiliary *be*. Other languages allow the passive to be expressed synthetically, as illustrated in (12) for Korean, in (13) for Latin, and in (14) for Swahili.

(12) **Korean**

a. Active:

SBJ	OBJ	
철수가	계획을	허락하였다.
Chelswu-ka	kyehoyk-ul	helak-ha-yessta.
Chelswu-NOM	plans-ACC	approve-ACT-PST
A	THM	

'Chelswu approved the plans.'

b. Passive:

SBJ	PP	
계획이	(철수에 의하여)	허락되었다.
Kyehoyk-i	Chelswu-eyuyhaye	helak-toy-yessta.
plans-NOM	Chelswu-by	approve-PASS-PST
THM	A	

'The plans were approved (by Chelswu).'

(13) **Latin**

a. Active:

SBJ	OBJ	
Puer-ī	port-am	claud-unt.
boy-NOM.PL.M	door-ACC.SG.F	close-3PL.PRS
A	THM	

'The boys are closing the door.'

b. Passive:

SBJ	PP	
Port-a	(a puer-īs)	claud-it-ur.
door-NOM.SG.F	by boy-ABL.PL.M	close-3SG.PRS-PASS
THM	A	

'The door is being closed (by the boys).'

(14) **Swahili**

a. Active:

SBJ		OBJ
mw-anafunzi	a-li-som-a	ki-tabu.
1-student	1SM-PST-read-FV	7-book
A		THM

'The student read the book.'

b. Passive:

SBJ		PP
ki-tabu	ki-li-som-w-a	na mw-anafunzi.
7-book	7SM-PST-read-PASS-FV	by 1-student
THM		A

'The book was read by the student.'

In these languages, it is the bound morphemes *-toy-*, *-ur*, and *-w-* (respectively) that result in the effects of passivization mentioned above. As in English, the grammatical relations of the agent and theme arguments differ in the active and the passive, and passive verb forms cannot assign the Case that active verb forms can. So in Korean and Latin passives, the THEME subjects are nominative (NOM) case, and accusative (ACC) case becomes unacceptable.

(15) a. * 계획을 (철수에 의하여) 허락되었다.
Kyehoyk-ul (Chelswu-eyuyhaye) helak-toy-essta.
plans-ACC Chelswu-by approve-PASS-PST
Intended meaning: 'The plans were approved (by Chelswu).'

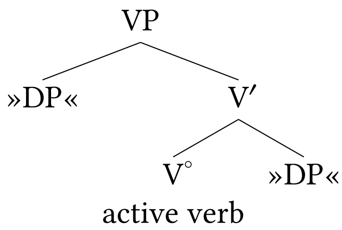
b. * Port-am (a puer-īs) claud-it-ur.
door-ACC.SG.F by boy-ABL.PL.M close-3SG.PRS-PASS
Intended meaning: 'The door is being closed by the boys.'

In Swahili, there is no overt case-marking of nominals, so the form of the agent argument and the theme argument does not change between the active and passive sentences.

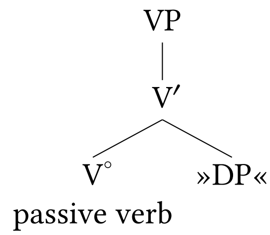
A movement analysis of the passive 10.3

10.3.1 Passive of simple sentences

Our analysis of the passive is based on the premise that theme arguments originate in the same structural position in both the active and the passive. This means that the elementary trees for active and passive participles both contain a complement position. However, the relevant elementary trees differ in two important ways. First, in the active, the agent argument is obligatorily **linked to** (= expressed in) Spec,VP, whereas in the passive, it is linked to an optional adjunct *by*-phrase. We will visually underline this by omitting Spec,VP in our elementary trees for passive participles. Second, passive participles in English (and passive verbs more generally) cannot assign objective Case; recall the ungrammaticality of (11b). (We will generally not represent this property in the elementary tree explicitly.) The elementary trees we propose for active and passive participles are thus as shown in (16).

(16) a. 

Active verb: assigns Case

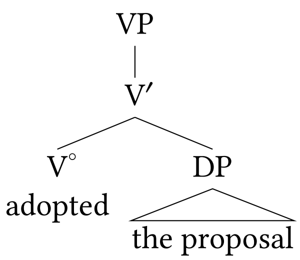
b. 

Passive verb: cannot assign Case

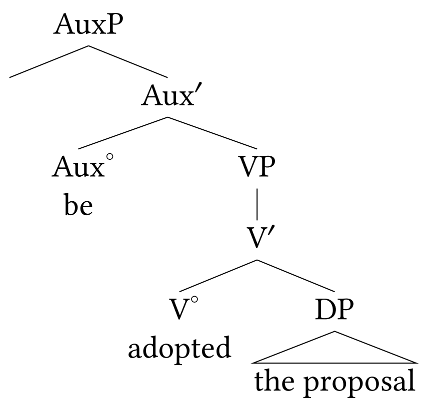
In what follows, we illustrate the derivation of a passive sentence like (17).

(17) The proposal was adopted.

First, we substitute the theme argument *the proposal* in the elementary tree for the passive main verb in (16b). This yields (18a). Then, we substitute (18a) as the complement of the passive auxiliary verb *be*, as in (18b). In order to distinguish the auxiliary verb (*be*) from the main verb (*adopted*), we give it the syntactic category Aux. Finally, we substitute the resulting structure as the complement of I, as in (18c).[2]

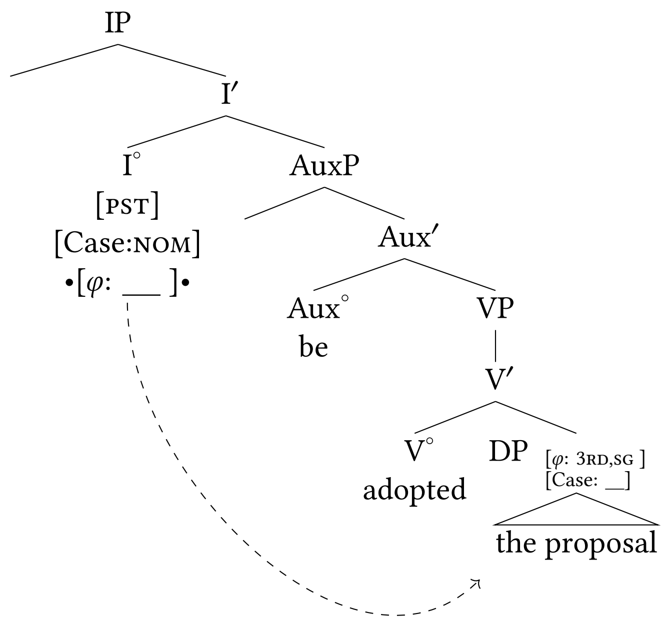
(18) a. 

Substitute theme argument in (16b)

b. 

Substitute (18a) as complement of passive auxiliary

c.

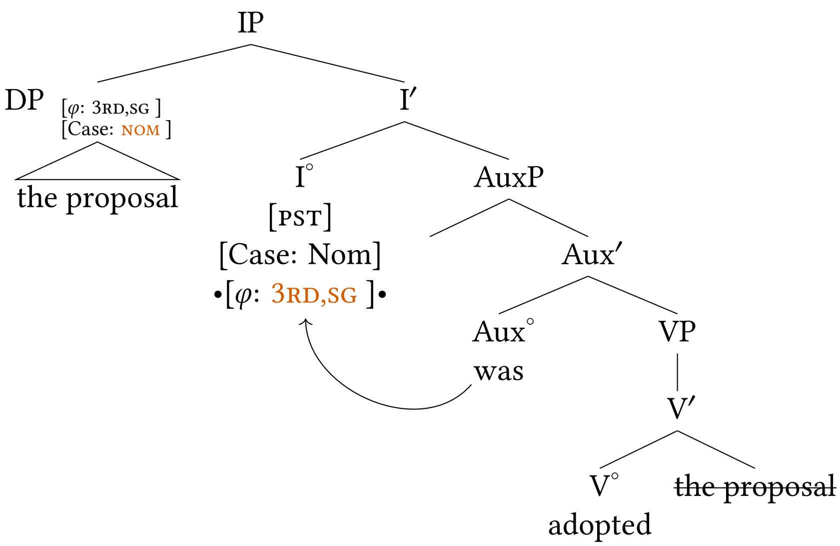


Substitute (18b) as complement of I°

Notice that the φ-probe is designated (with bullets) as a Probe that will undergo Probe-Goal Union: the Goal of Agree with the Probe will move to Spec,IP. This also allows us to stop annotating Spec,IP as a substition node (which we had been doing like this: »DP«) since that observation is now captured by the properties of the φ-probe.[3]

Because of the inability of the passive participle V° *adopted* to assign objective Case (which we discuss in more depth in Chapter 13), the THEME argument cannot receive Case in the complement position. Since every DP must receive Case (per the DP Licensing Condition: (61) of Chapter 9), the THEME argument can only be licensed by I°. In this instance, the absence of the AGENT argument means that the THEME argument is the closest to the φ-probe on I°. This Probe finds the THEME argument that is the complement of V° as a Goal, and the theme DP undergoes Probe-Goal Union, raising to Spec,IP. The resulting final structure is shown in (19).

(19)



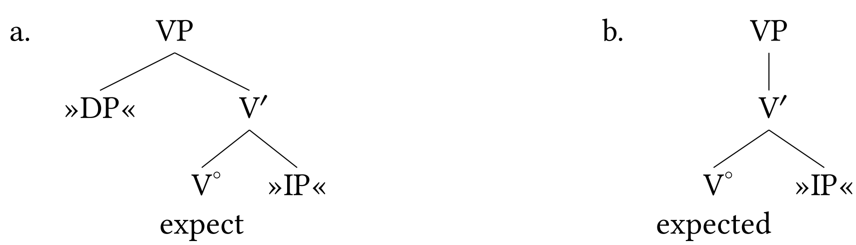
10.3.2 Passive of ECM sentences

This section focuses on the passive of ECM verbs like *expect,* which were introduced in Chapter 9. (20a) shows an active ECM verb sentence, and (20b) shows the corresponding passive.

(20) a. ✓ Your folks expect you to call.
b. ✓ You are expected to call.

The elementary tree for *expect* in (20a) is given in (21a). In accordance with the previous discussion, the elementary tree for the passive participle *expected* is as in (21b). The difference between the two trees is analogous to that between the trees in (16); the only difference is the syntactic category of the complement (DP in the Case of ordinary verbs, IP in the Case of ECM verbs). As in (16b), the elementary tree (21b) has no agentive argument in its specifier, and the verb lacks the ability to assign objective Case.

(21)

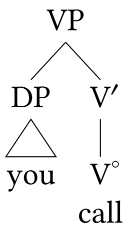


Active ECM verb: assigns Case

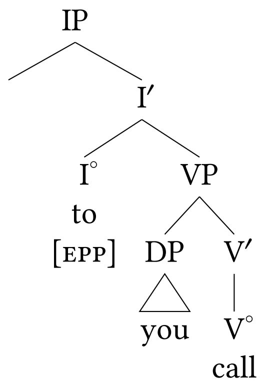
Passive ECM verb: cannot assign Case

In what follows, we illustrate the derivation of (20b). The derivation of the complement clause is shown in (22); we assume that the complement subject moves from Spec,VP to Spec,IP to provide the complement clause with a subject (recall the subject requirement discussed in Chapter 4, which we suggested is encoded by an EPP feature in § 9.4.4 of Chapter 9).

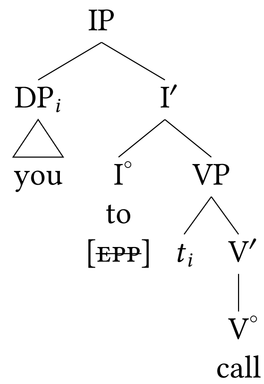
(22) a.



b.



c.



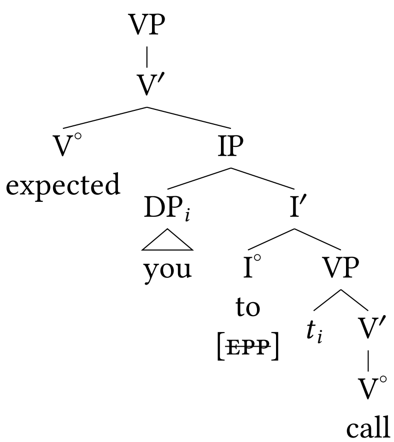
Complement VP

Substitute (22a) as complement of nonfinite I°

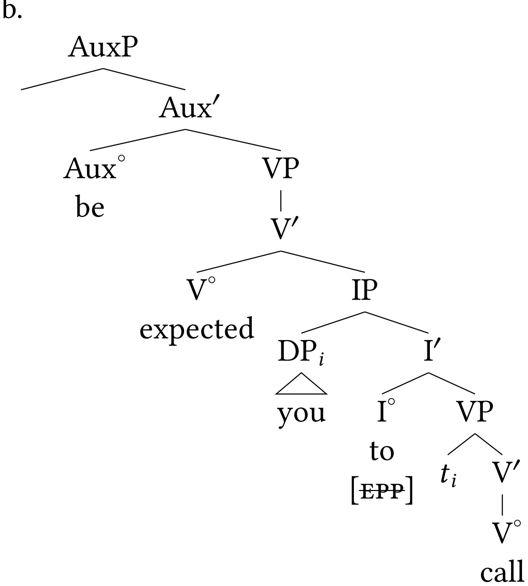
Move subject of complement clause

The subsequent steps of the derivation involving the matrix clause are as shown in (18). We omit the case and φ-features of the DPs in order to avoid visually cluttering the trees too much.

(23) a.



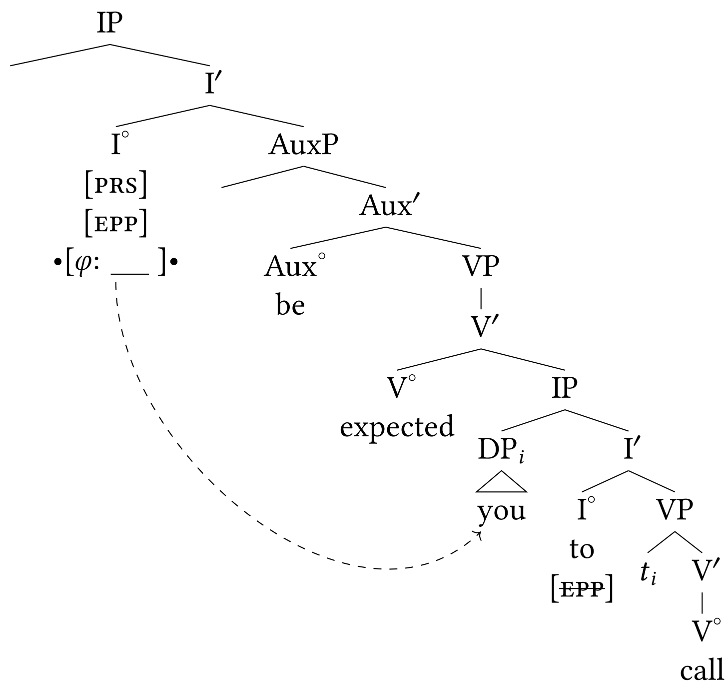
b.



Substitute (22c) as complement of passive participle of ECM verb

Substitute (23a) as complement of passive auxiliary

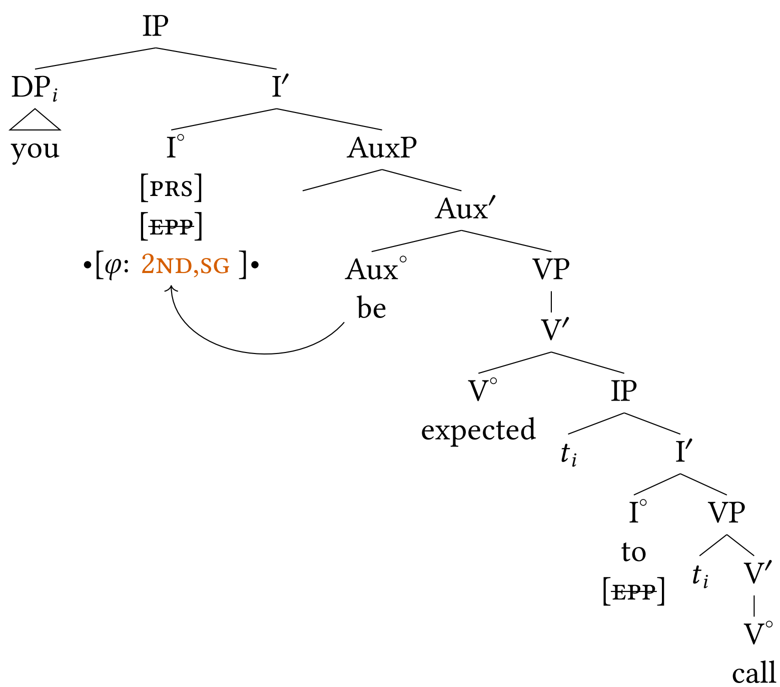
c.



Substitute (23b) as complement of matrix I°

In (22c), the passive participle of the ECM verb, being passive, cannot assign Case to the subject of the complement clause, which must therefore move to the nearest Case position. This is the matrix Spec,IP, where it receives nominative Case from finite I°. Again, the subject of the complement clause passes through Spec,AuxP on the way. The resulting structure is shown in (24).

(24)



10.3.3 Evidence for A-movement in passives: selectional restrictions

As we discussed in Chapter 7 passive constructions show interesting effects when considered alongside diagnostic properties for selection, including selectional restrictions and idiom chunks. We revisit those facts briefly here to show that the analysis of passives presented here accords with other properties that we've used to diagnose movement in other instances. Specifically, if THEME arguments of passives start their derivational life as the complement of the verb, we would expect that the selectional restrictions that verbs place on their objects would persist in passives, and we would expect that the verb-object V′ constituent would be a candidate for idioms that could persist in passive constructions.

Both of these predictions are upheld. As discussed previously, in English *kill* and *murder* have different selectional properties with respect to their objects (as shown in (25)). These same selectional restrictions hold of the derived subjects in (26), supporting the view that there is a derivational connection.

(25) a. The paramilitary murdered { ✓their parents, #the olive trees, #their houses }.
b. The paramilitary killed { ✓their parents, ✓the olive trees, #their houses }.

(26) a. { ✓their parents, #the olive trees, #their houses } were murdered by the paramilitary.
b. { ✓their parents, ✓the olive trees, #their houses } were killed by the paramilitary.

The same is evident for verb-object idiom chunks. The idioms in (27) maintain their idiomatic interpretations in the passive sentences in (28).

(27) a. They { *gave*, *paid* } hardly any <u>heed</u> to our proposal.
b. The Prime Minister *paid* due <u>homage</u> to the dead.
c. The government *keeps* close <u>tabs</u> on their operations.

(28) a. ✓ Hardly any heed was { *given*, *paid* } to our proposal.
b. ✓ Due homage was *paid* by the Prime Minister to the dead.
c. ✓ Close tabs were *kept* on their operations by the government.

10.4 Distinguishing active and passive sentences

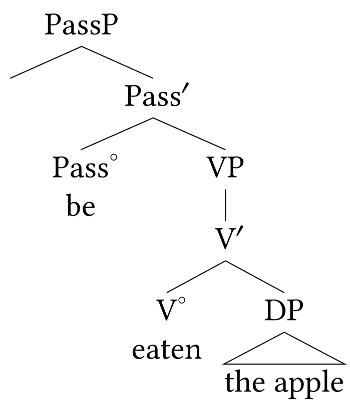
Empirically, it is easy to distinguish between active and passive sentences. Active sentences have overt agents and license accusative objects, whereas passive sentences suppress their agent and the thematic object is instead promoted to subject position. But what in our model accomplishes this? What creates an active versus a passive version of a verb in question? The short answer is that we don't yet have the full theoretical toolkit that we need in order to do this: we will be much closer by the end of Chapter 13. So at present we just discuss several of the salient issues.

We introduced an Aux above that appears in English passive sentences. In English, this is *be*: we use *be* for many functions in English, including progressive (*They are running*) and as a copula (*They are my friends*). But these are in fact not the same *be*: this is evident in (29), where a passive construction is in progressive aspect, and two instances of *be*-verbs appear.

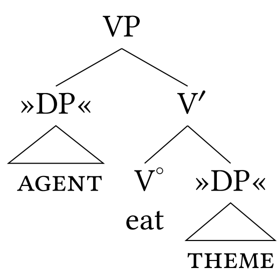
(29) The apples that I cut up for lunch are being eaten right now.

So if we want to be specific, we could call the auxiliary that occurs in passives its own separate thing, perhaps a Passive head (Pass°) that projects to a Passive Phrase (PassP). To simplify, we'll focus on a very simple passive sentence as in (30). We can imagine the tree structure of (30) would start something like in (31).

(30) The apple was eaten.

(31)


Having a separate projection for passives helps explain the dedicated auxiliary for passives. But note that there is more going on here: the verb *eat* is typically transitive, and as such we would typically expect an elementary tree for an *eat* VP that looks something like in (32), which notably includes an agent.

(32) Elementary tree for *eat*


But a hallmark of passive sentences is that agentive subjects are suppressed or absent, as in (31). But including a passive auxiliary doesn't clearly (in our model) have any good reason for suppressing the agentive subject inside the phrase below it. What does one thing have to do with the other?

A similar problem exists around Case in passives. As we saw above, in a passive sentence even though the subject is a theme argument, it is nonetheless marked with nominative Case, the Case normally reserved for subjects.

(33) a. They visited her.
b. She was visited.
c. * Her was visited.

This fits many aspects of our model: if the theme argument is the target of Agree from I° as part of the explanation for agreement with I° and movement to Spec,IP, we would in fact expect the result that it is marked with nominative Case. But this doesn't explain why accusative Case simply seems to disappear in passives. Nothing about adding a PassP for the passive auxiliary explains this directly.

We're not going to solve this right now: we will in Chapter 13. We simply want to point out that there are open questions. The lessons that we are taking from passives here is that they are another instance of movement, but specifically an A-movement with a landing site in Spec,IP (**not** an A′-movement landing in Spec,CP). And as we've noted above, A-movement has a different locality profile than A′-movement.

10.5 Unaccusatives: Inherently passive predicates

English has a class of verbs called inchoative or unaccusative predicates that have a passive flavor, but are not passive verbs.

(34) The vase dropped.

As with passive sentences, sentences containing these verbs have subjects that are themes rather than agents. However, unlike passives, they do not allow the expression of an agent, as shown in (35).

(35) * The vase dropped by the child. (ungrammatical on agentive reading of *by*-phrase)

This is despite the fact that the related transitive sentence *does* have a passive variant.

(36) a. The child dropped the vase.
b. The vase was dropped by the child.

So the intransitive *drop* does have a THEME subject, but it lacks other properties of the passive. In a sense, it feels *inherently* passive: a predicate that, as the usual manner of course, has a THEME subject and not an agentive subject. But is there any difference between an intransitive with a THEME subject and one with an AGENT subject?

In fact, many languages do show a difference like this. One famous example is the *ne*-morpheme that cliticizes (attaches) onto the verb in Italian that can be used to mark quantified direct objects, pronominalizing the nominal inside the quantifier as in (37b). This discussion follows the argumentation laid out in Burzio (1986).[4]

(37) a. Gianni inviterà molti studenti. **Italian**
Gianni will.invite many students
'Gianni will invite many students.'

b. Gianni *(ne) inviterà molti.
Gianni of-them will.invite many
'Gianni will invite many of them.'

Subjects generally disallow use of *ne*-cliticization.

(38) a. Molti (*ne) inviteranno Gianni. **Italian**
many (*of-them) will.invite Gianni
'Many will invite Gianni.'

b. Molti (*ne) telefoneranno.
many (*of-them) will.telephone
'Many will telephone.'

In passive sentences, however, it is possible to use the *ne*-cliticization with the derived subject of the passive.

(39) a. Molti esperti saranno invitati. **Italian**
many experts will.be invited.
'Many experts will be invited.'

b. Ne saranno invitati molti.
of-them will.be invited many.
'Many of them will be invited.'

Burzio (1986) concludes that *ne*-cliticization is only possible with underlying direct objects of the verb, even if they become subjects in passives. But this raises a puzzle when it comes to intransitive predicates. As we might expect, the intransitive verb to telephone (call) is impossible with *ne*-cliticization.

(40) a. Molti esperti telefoneranno. **Italian**
Many experts will.telephone.
'Many experts will telephone.'

b. * Ne telefonano molti.
of-them telephone many
'Many of them telephone.'

But some intransitive verbs behave differently; the verb *arrive*, in contrast, allows *ne*-cliticization.

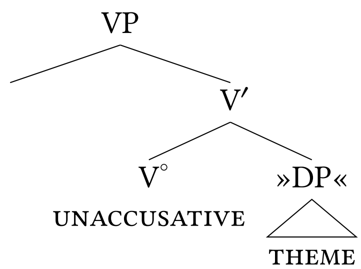
(41) a. Molti esperti arriveranno. **Italian**
Many experts will.arrive.
'Many experts will arrive.'

b. Ne arrivano molti.
of-them arrive many
'Many of them arrive.'

In short, the subject of *arrive* behaves as if it is underlyingly a direct object, not a subject. Looking at thematic roles, this is less surprising: the subject of intransitive *telephone/call* is agentive, plausibly an AGENT. The subject of *arrive* has no such requirement: a train can arrive, a storm can arrive, the day of a party can arrive. So *arrive* instead behaves like its subject is a THEME. In this sense, then, some predicates behave like they are inherently passive, taking a THEME object and A-moving that object to canonical subject position.

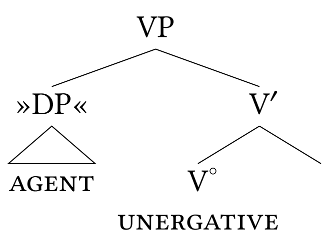
Predicates that behave in this manner are called unaccusative predicates: they take a theme argument but don't license accusative Case on the theme, so it becomes the subject of the sentence. We can capture the inherently passive quality of unaccusative predicates by treating them with a core similarity: their subjects originate in the canonical positions for THEME subjects, as complement to the verb. These predicates don't have an agent argument.

(42) Unaccusative predicate elementary tree



This contrasts with an intransitive predicate that does take an agent argument, e.g. *sing*. Here we would expect our agent to appear in the canonical position for agents, which for us at this point is Spec,VP. We refer to these predicates as unergative, for reasons that are arcane enough to not be worth spelling out.

(43) Unergative predicate elementary tree



In typical instances, neither the theme argument nor the agent argument are Case-licensed in their base position: both must necessarily be a Goal of Agree with I°, receive nominative Case, and in English raise to Spec,IP. So on the surface, unaccusative and unergative verbs tend to behave similarly in most sentences. But some languages have grammatical constructions (like *ne*-cliticization) that distinguish between them.

Are distinctions between unaccusatives and unergatives universal? It depends what is meant by that question. The specific grammatical properties that distinguish between them are certainly not universal. The *ne*-marking quirk of Italian is just that: a quirk of Italian. English (for example) has no parallel construction. And not all languages have grammatical properties that distinguish these classes of predicates (at least, it has not yet been demonstrated to be the case). But nothing about our model requires that a language have a grammatical property that shows variation between unaccusatives and unergative predicates: we can still analyse them as having these distinctive underlying VP structures. And we do this because there are in fact a large variety of unrelated languages that **do** show grammatical properties that distinguish between unaccusative and unergative predicates. It is not necessarily universal, but it is much too common to be a coincidence. So we will maintain the view that UTAH holds, and that even in intransitive predicates there are different underlying structures for predicates that take a THEME argument and those that take an AGENT argument.

Conclusions 10.6

Passive constructions and unaccusative constructions are a key reason why linguists necessarily distinguish between thematic roles and grammatical roles: mismatches between the two are readily found. THEME arguments are canonically direct objects, but by no means must they be direct objects: they can be subjects in passives and in unaccusatives. Likewise, agent arguments are typically subjects, but not always so (for example, agent arguments appear in a *by*-phrase in a passive sentence).

If THEMES are possible subjects, why are they not typical subjects? that is, why does the THEME never get promoted to subject in the presence of an agent? Another way of saying this is to ask: why are agents canonical subjects? These facts fall out naturally from our assumptions about UTAH and the locality of probing from I°. UTAH claims that agents are structurally higher than themes (universally). And if A-movement to subject will find the most local DP, the structurally higher DP will be the one that is the Goal of Agree by I°. So in any typical transitive construction, agents will be the grammatical subject. But this is effectively a conspiracy of various different underlying circumstances; there is no requirement in our theory that agents be subjects.

We have used passives and unaccusatives to exemplify A-movement. We previously had seen A-movement of agent arguments from Spec,VP to Spec,IP, but this chapter has shown that VPISH-related subject movement is part of a larger class of movements (which we call A-movements). Therefore, we now have two profiles of phrasal movement: movement to argument positions (A-movement) vs. movement to non-argument positions (A′-movement). As a reminder, A versus A′ is an unfortunate replication of terminology—here it is NOT a reference to intermediate bar-levels in X′-syntax. Shorthand for us at present is A′-movement is to Spec,CP for discourse purposes, and A-movement is to Spec,IP for Case purposes.

New Empirical Observations

- Passives do not assign accusative Case.
- Passives place the same selectional restrictions on their derived subjects that their active equivalents place on their objects.

Components of the Theoretical Model

- A-movement (movement to Case-licensing positions)
- A-movement vs. A′-movement (different profiles of movement)
- locality of probing (I° probes for the closest DP)
- unaccusatives vs. unergatives
- reaffirmation of UTAH principles

Notes 10.7

1. There are principled exceptions to this statement, like the cross-linguistic existence of locative inversion constructions. In general, though, A-movement is restricted to DPs in a way that A′-movement is not.

2. As we will see in Chapter 12, auxiliary *be* and *have* move from their original position to I°, where they form a complex verb with the tense morpheme. As this movement is irrelevant for present purposes, we ignore it here and throughout the chapter.

3. And the EPP quality of IP in English: see the discussion in § 9.4.4 of Chapter 9.

4. The specific data presentation/structure in this sequence is drawn from Pesetsky (2003), though they directly parallel what is presented in Burzio (1986), with minor adjustments.

Exercises and problems 10.8

Exercise 10.1: Basic practice with passives

Draw trees for the passive sentences in (1). For each one, explain (in prose) how Case plays a role. Draw triangles over DPs, PPs, and the bracketed phrases.

(1) a. the window was broken by the baseball.
b. Bald eagles were protected [as an endangered species.]
c. *Instructions for making wine:* Next, the grapes are crushed.
d. The onions were chopped by the line chef.
e. The votes were being counted.

Exercise 10.2: Basic practice with intransitives

Draw trees for the intransitive sentences in (1). For each one, explain (in prose) how Case plays a role. Draw triangles over DPs, PPs, and the bracketed phrases.

(1) a. The ball rolled down the hill.
b. The choir sang beautifully.
c. The cat jumped very high.
d. The cat slept peacefully.
e. The storm arrived earlier [than predicted.]
f. The butter melted in the pan.

Exercise 10.3: Japanese voice

Examine the following Japanese sentences.

A. Label the subject, object (if there is one), agent/cause, and patient in each of these sentences.

B. Which sentences are active, and which are passive?

C. How are active and passive sentences marked in Japanese? (*Hint:* the passive marking has two similar-looking allomorphs in these data.)

Note: Topic marking alternations have been suppressed from these data for simplicity, but be aware students who are native speakers of Japanese may find some of these sentences pragmatically odd without context.

(1) 蚊が 僕を 刺した
Ka-ga boku-o sashi-ta.
mosquito-NOM 1SG-ACC prick-PST
'A mosquito bit me.'

(2) 僕が 蚊に 刺された
Boku-ga ka-ni sas-are-ta.
1SG-NOM mosquito-by prick-ARE-PST
'I was bitten by a mosquito.'

(3) 大風が 木を 吹き倒した
Oo-kaze-ga ki-o fuki-taoshi-ta.
big-wind-NOM tree-ACC blow-down-PST
'A strong wind blew the tree down.'

(4) 木が 大風に 吹き倒された
Ki-ga oo-kaze-ni fuki-taos-are-ta.
tree-NOM big-wind-by blow-down-ARE-PST
'The tree was blown down by a strong wind.'

(5) 悠ちゃんが カエルを 捕まえた
Yū-chan-ga kaeru-o tsukamae-ta.
Yū-DIM-NOM frog-ACC catch-PST
'Yū caught a frog.'

(6) カエルが 悠ちゃんに 捕まえられた
Kaeru-ga Yū-chan-ni tsukamae-rare-ta.
frog-NOM Yū-by catch-RARE-PST
'The frog was caught by Yū.'

Exercise 10.4: Tiriki voice

Examine the following Tiriki (Bantu, Kenya) sentences.

A. Label the subject, object (if there is one), agent/cause, and patient in each of these sentences.

B. Which sentences are active, and which are passive?

C. How are active and passive sentences marked in Tiriki?

Glossing: Cardinal numbers mark noun class. SM = subject marker; PST = past tense; FV = final vowel (marks mood and other properties); AGR = agreement

(1) a. I-suna y-a-um-a va-ana.
9-mosquito 9SM-PST-bite-FV 2-child
'A mosquito bit the children.'

b. Va-ana va-a-lum-w-a na i-suna.
2-child 2SM-PST-bite-?-FV by 9-mosquito
'The children were bitten by a mosquito.'

(2) a. I-mbutsa i-ndulu y-a-huts-a ma-tuma hasi.
9-wind 9AGR-strong 9SM-PST-knock.down-FV 6-maize down
'A strong wind blew/knocked the maize down'

b. Ma-tuma ka-a-huts-w-a hasi na i-mbutsa i-ndulu.
6-maize 6SM-PST-knock.down-?-FV down by 9-wind 9AGR-strong
'The maize was blown/knocked down by a strong wind.'

(3) a. Inganji y-a-kumil-a li-shele.
1Inganji 1sm-pst-catch-fv 5-frog
'Inganji caught a frog.'

b. Li-shele ly-a-kumil-w-a na Inganji.
5-frog 5sm-pst-catch-?-fv by 1Ingaji
'The frog was caught by Inganji.'

(4) a. I-nzukha y-a-lond-a va-ana.
9-snake 9sm-pst-chase-fv 2-child
'A snake chased the children.'

b. Va-ana va-a-lond-w-a na i-nzukha.
2-child 2sm-pst-chase-?-fv by 9-snake
'The children were chased by a snake.'

Exercise 10.5: Clarifying concepts

Which, if any, of the following sentences are true? Support your conclusions with evidence.

(1) a. All subjects are agents.
b. All agents are subjects.
c. All subjects get nominative Case.
d. All noun phrases that get nominative Case are subjects.

Exercise 10.6: Structures for *get*-passives

Build structures for the sentences in (1).

(1) a. The puppeteers might get arrested.
b. The puppeteers got arrested.

Exercise 10.7: Expletives and passives

Explain the grammaticality pattern in (1), including how expletive *there* is licensed. Do noun phrases (DPs) differ from clauses (CPs, IPs) with respect to Case assignment? Assume for simplicity that expletive *it* substitutes directly into Spec,IP.

(1) a. It was suspected that there was a problem with the O-ring.
b. That there was a problem with the O-ring was suspected.
c. There was suspected to be a problem with the O-ring.
d. * There to be a problem with the O-ring was suspected.

Problem 10.1: Passives and ECM A

The active sentence in (1) has two conceivable passive counterparts in (2): the grammatical (2a) and the completely ungrammatical (2b) (intended to have the same meaning as (2a)). Why is (2b) ungrammatical?

(1) The administration will expect *us* to hire **them**.

(2) a. ✓ *We* will be expected to hire **them**.
b. * { **They**, **them** } will be expected { we, *us* } to hire.

Problem 10.2: Passives and ECM B

Consider the following judgments for the sentences with *expect* in (1) and (2).

(1) a. ✓ They expected there to be a solution.
b. ✓ There was expected to be a solution.

(2) a. * They expected for there to be a solution.
b. * There was expected for to be a solution.

Now consider the corresponding judgments for sentences with *want* in (3) and (4).

(3) a. ✓ They wanted there to be a solution.
b. * There was wanted to be a solution.

(4) a. ✓ They wanted for there to be a solution.
b. * There was wanted for to be a solution.

A. The judgments for the sentences with *expect* are as expected given our discussion in this and the previous chapter. The judgments for the sentences with *want* can be made to be consistent with the discussion as well. How so?

B. For some speakers, (2a) is grammatical. The remaining judgments remain the same. Revise your answers to (A) to account for this pattern of judgments.

Problem 10.3: Passives and ECM C

A. Build structures for the sentences in (1) and (2).

(1) a. We expect them to make headway.
b. We expect headway to be made.
c. Headway is expected to be made.

(2) a. The media expect the guerrillas to release the journalist.
b. The media expect the journalist to be released by the guerrillas.
c. The journalist is expected to be released by the guerrillas.

B. One of the sentences in (2) is structurally ambiguous. Which one is it, and why?

Identifying Puzzles 10.1: Restrictions on passives

There is a pattern in the follow examples regarding what arguments can be passivized or not. Give an empirical description of what is happening in these examples based on concepts from this textbook, being as precise as possible. Then, discuss the ways in which this impacts the theory of passives we have presented in this chapter. What is well-explained, and what is not (yet) well-explained? You don't have to have a complete analysis of the pattern; rather, explain with precision the puzzle that this raises that requires an explantion.

(1) ✓ The carpenter measured the boards.

(2) ✓ The boards were measured by the carpenter.

(3) a. ✓ The tickets cost 100 dollars.
    b. ✓ The concert lasted two hours.
    c. ✓ The carpenter measures 5′10″.
    d. ✓ Louis XIV reigned 72 years.
    e. ✓ The trip took two hours.

(4) a. * 100 dollars are cost by the tickets.
    b. * Two hours were lasted by the concert.
    c. * 5′10″ is measured by the carpenter.
    d. * 72 years were reigned by Louis XIV.
    e. * Two hours were taken by the trip.

11 Subject control versus raising

Chapter outline

There are recurring classes of matrix predicates cross-linguistically where one argument seems to be shared between the main clause and an embedded clause (ex., *Preston seems to be happy*, *Preston hopes to be happy*). Syntacticians have found these classes of predicates to be interesting puzzles that also are good testing grounds for underlying properties of our model, such as Case-licensing, locality of probing, and A-movement. This chapter overviews the two predominant areas that these discussions have centered on: *raising constructions* and *control constructions*.

Overviewing complement clauses 11.1

So far in this book, we have come across three types of clausal complements, summarized in the following table. (By 'clausal complement', we mean any complement that contains a subject and a predicate.) The clausal complement is bracketed, and the complement's I head, if any, is in boldface.

Type of complement	I is...	Matrix	Complement
Finite complement	modal or tense	We heard	[that the children **[PST]** danced].
ECM complement	nonfinite *to*	We expected	[the children **to** dance].
Small clause	absent	We saw	[the children dance].

Table 11.1: Summary of clausal complements up to this point

Despite their superficial diversity, the complement clause types in Table 11.1 all have one property in common—the presence of an overt subject (here, *the children*). But English also has nonfinite complement clauses where a subject is understood, but not expressed overtly. For instance, *dance*, the verb in the apparently subjectless complement clauses in (1), has an understood agent.

(1) a. Subject control:
The children agreed [to dance].

b. Raising:
The children seemed [to dance].

In both sentences, this agent is interpreted as being identical to the referent of the matrix subject *the children*. Yet unlike the second row of Table 11.1, where the matrix clause and the complement clause each have their own subject (*we, the children*), the sentences in (1) contain only a single overt subject, the one in the matrix clause. In this chapter, we argue that the nonfinite complements in (1) contain a structural subject position, just as in Table 11.1, which is filled by a silent element, and we argue further that this silent element is not the same in the two examples. Rather, we distinguish between **subject control** and **raising** (sometimes called subject-to-subject raising), as already indicated by the labels in (1).

Background Info

Some of the discussion here uses terms like *reference*, *refer*, and *corefer* (among others). If these are not clear from the glossary and our explanations here, you can consult § 15.1 for a more complete exposition.

In a subject control sentence like (1a), the complement subject position is filled by a silent pronominal element PRO, which corefers with the referent of the matrix subject. PRO has been proposed as a special kind of pronoun that only occurs in very particular positions (namely, subjects of nonfinite clauses). In other words, we give (1a) the structure in (2a), by analogy to that in (2b). (The numerical subscripts on *the children*, *PRO*, and *they* represent coreference, meaning that they all refer to the same real-world entity (here, the children).)

(2) a. [The children]$_1$ agreed [∅ PRO$_1$ to dance].
  b. [The children]$_1$ agreed [that they$_1$ would dance].

Extra Info

The idea behind the term 'subject control' is that the matrix subject fixes, or controls, the reference of PRO. Notice that the parallel between PRO and overt pronouns in (2) is not complete. PRO in (2a) must corefer with the matrix subject, whereas the overt pronoun in (2b) needn't. This is succinctly summarized by the notation in (i).

(i) a. [The children]$_1$ agreed [∅ PRO$_{1,*2}$ to dance].
  b. [The children]$_1$ agreed [that they$_{1,2}$ would dance].

It's for this reason that only sentences where the clausal complement is nonfinite can count as instances of subject control.

Raising sentences differ from subject control sentences in that the matrix subject does not start out in the matrix clause, but rather moves there from the complement clause. Omitting unnecessary details for the moment, the derivation is illustrated in (3).

(3) a. ______________ seemed [the children to dance].
  b. [The children]$_i$ seemed [t_i to dance].

The assumption that the matrix subject position starts out empty is supported by the fact that when *seem* takes a finite counterpart, that same position is filled by expletive *it*.

(4) It$_{\text{EXPL}}$ seemed [that the children danced].

Extra Info

Analogously to subject control, for a sentence to count as an instance of raising, the complement clause must be nonfinite, as in (3), since it is only then that the complement subject raises into the matrix clause. (4) contains the same matrix predicate as (3), but it is not an instance of raising because the complement subject remains within its clause.

A note on terminology. We will refer to the class of (Fregean) predicates to which *agree* belongs as **subject control predicates**. Similarly, we refer to the class of predicates like *seem* as **raising predicates**, that is, predicates whose subjects raise out of their complement clause.

These two different analyses for classes of predicates that otherwise (often) appear very similar on the surface creates a range of predictions: we would expect, for example, that raising predicates consistently behave like they don't assign a thematic role to their subject, whereas control predicates would behave like they do. And due to that, we would expect control predicates to place selectional restrictions on their subjects in ways that raising predicates do not. In short, proposing that these two have different structures should be diagnosable: in what follows, we show that it is.

Subject control 11.2

11.2.1 Evidence for two clauses

This section relies on the notion of selectional restrictions introduced in Chapter 7. Here, we will use the concept to show that subject control sentences contain two separate clauses, each with their own subject. We begin by showing that in finite-complement counterparts of subject control sentences like (5), which incontrovertibly contain two clauses, both the matrix and the complement verbs (or predicates more generally) impose separate selectional restrictions on their respective subjects. (For simplicity, we omit referential indices in what follows; unless otherwise noted, the intended interpretation is always the one where the complement subject corefers with the matrix subject.)

(5) The children agreed that they would dance.

Consider (6), where we have taken care to satisfy the selectional requirements of the complement clause (*wet* selects some physical object as its argument, and *get* imposes no further selectional restrictions of its own). We can therefore be sure that the acceptability contrast in (6) is due to the selectional restriction imposed by the matrix verb *agree*, which selects humans as subjects.

(6) a. The children agreed [that they would get wet].
    b. # The { trees, rocks } agreed [that they would get wet].

Conversely, in (7), we have taken care to satisfy the selectional restriction imposed by *agree*. Here, the acceptability contrast is due to the selectional restrictions imposed by the complement predicates. In particular, *elapse* selects stretches of time, and *evaporate* selects liquids. Neither selects humans.

(7) a. The children agreed [that they would speak Twi].
    b. # The children agreed [that they would { elapse, evaporate }].

If subject control sentences contain two clauses, as we are proposing, each with their own subjects, they ought to pattern analogously to (6) and (7), and this is in fact exactly what we find in (8) and (9).[1]

(8) a. The children agreed [PRO to get wet].
    b. # The { trees, rocks } agreed [PRO to get wet].

(9) a. The children agreed [PRO to speak Twi].
b. # The children agreed [PRO to { elapse, evaporate }].

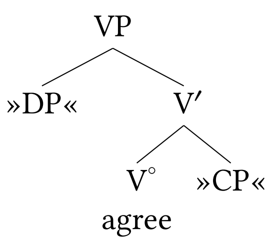
One last thing. Not all subject control predicates have a finite-complement variant.

(10) a. The children tried [PRO to learn Twi].
b. * The children tried [that they would learn Twi].

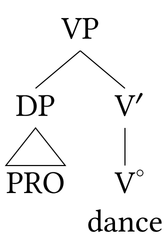
But even for such subject control predicates, we assume a biclausal structure with a PRO subject for the lower clause. This is because, just as in (8) and (9), the subject control verb and the lower predicate impose separate selectional restrictions on their respective subjects.

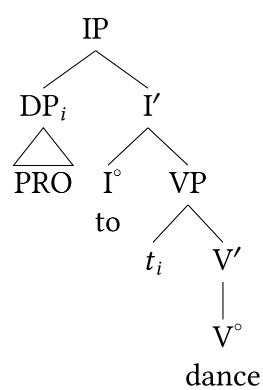
11.2.2 Deriving subject control sentences

Given the facts just discussed, it comes as no surprise that the syntactic representation of subject control sentences is straightforwardly analogous to that of their finite complement counterparts. The elementary tree for *agree* is the same for both cases and is given in (11).

(11)


Substituting a finite CP complement headed by *that* at the CP substitution node would yield structures for sentences like (5). Substituting a nonfinite CP complement headed by a silent complementizer yields structures for subject control sentences like (1a). In what follows, we illustrate the derivation of (1a) in relative detail, though we omit many features (ex., case/agreement) that are not central to our current points.

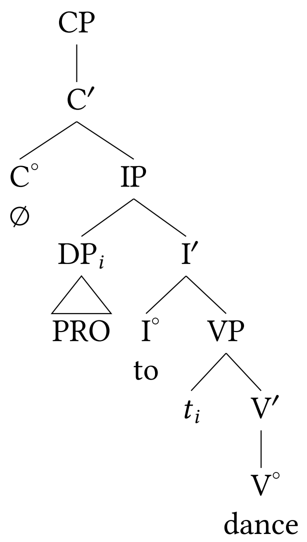
(12) a.


b.


Substitute PRO in specifier position of lower verb

Substitute (12a) as complement of *to* and move subject

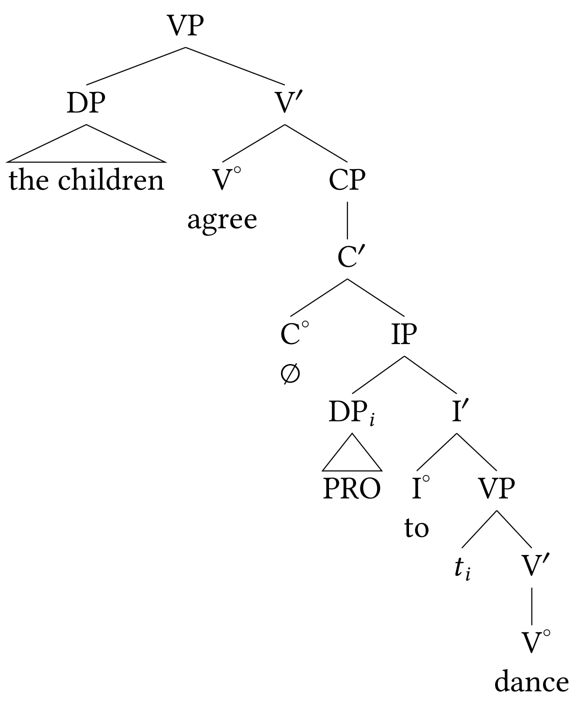
c.



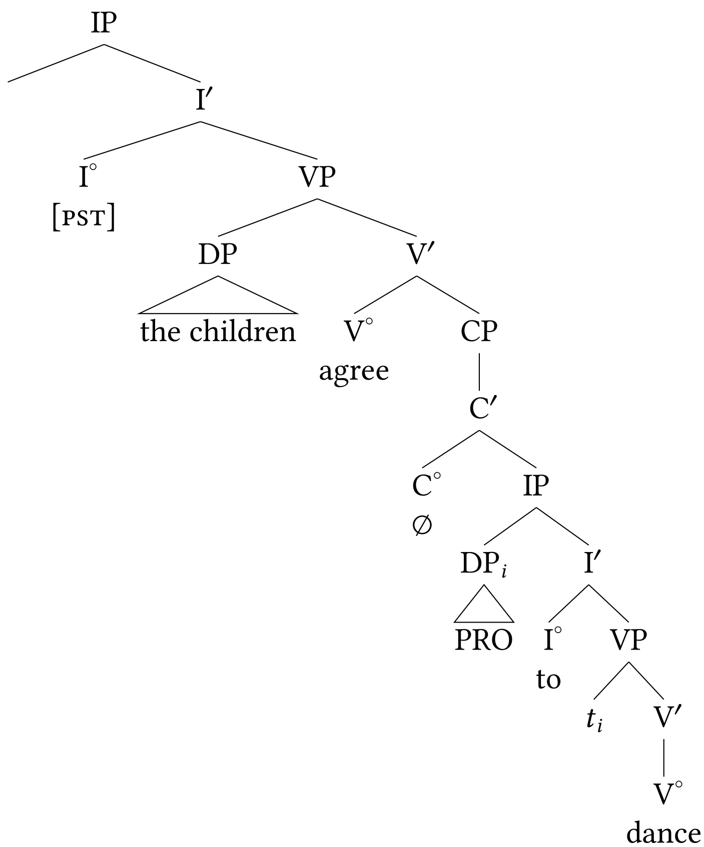
Substitute (12b) as complement of silent C°

Substituting the structure in (12c) as the complement of the control verb yields (13a), which in turn becomes the complement of matrix I°, yielding (13b).

(13) a.

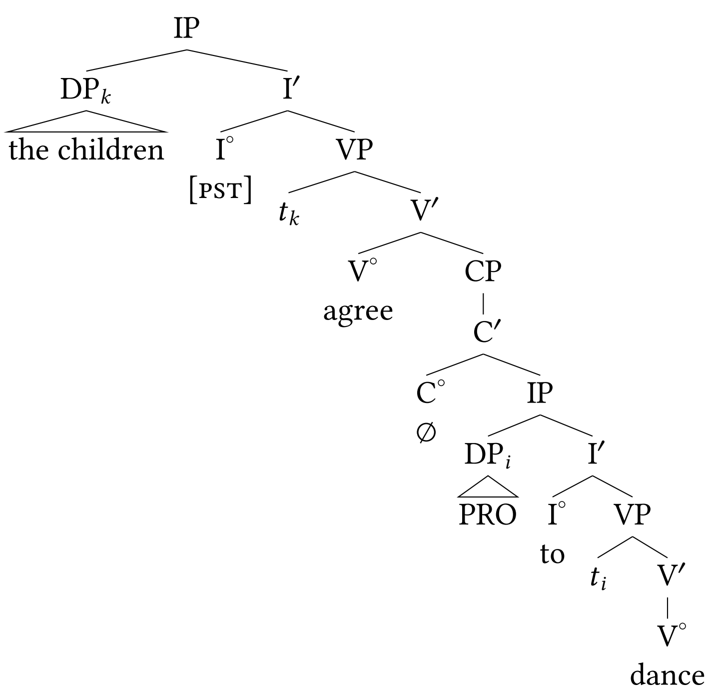


b.

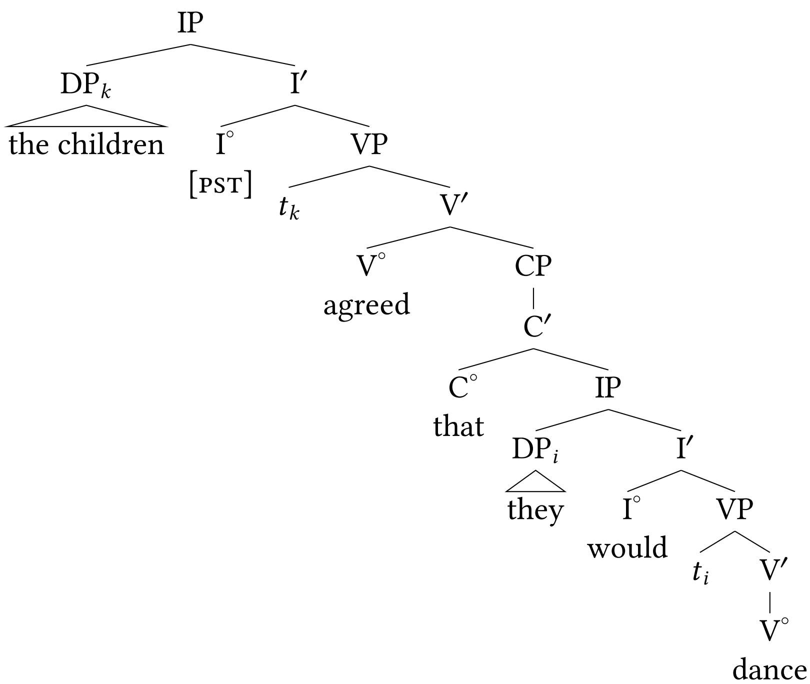


Finally, moving the matrix subject yields (14a). The structurally analogous tree for the finite complement counterpart of (14a) (= (5)) is shown for comparison in (14b). As is evident, the trees differ only in a few terminal nodes.[2]

(14) a.



b.



Raising 11.3

11.3.1 A "what if" detour

Let us turn now to raising sentences like (1b), repeated here as (15).

(15) The children seemed [to dance].

At first glance, it seems as if we could treat such sentences as instances of subject control. But that would leave us without an explanation for the contrast in (16)—in particular, for the grammaticality of (16b).

(16) a. * There agreed to be a problem.
b. ✓ There seemed to be a problem.

The analysis in the previous section does correctly rule out (16a), to which we assign the structure indicated schematically in (17).

(17) * There agreed [PRO to be a problem.]

(17) is ruled out for two reasons. First, expletive *there* fails to satisfy the selectional restriction of *agree*, which, as we saw earlier, selects humans as subjects. Second, and conversely, expletive

there is itself not licensed; it neither occupies the specifier position of a verb of (coming into) existence nor can it have originated in such a position.

It is true that the sentence contains the *there* licenser *be*, but *there* never substitutes into its specifier position. The predicate whose specifier position *there* does substitute into, namely *agree*, is not a *there* licenser.

(18) * There agreed some students.

So (17) correctly rules out (16a) as ungrammatical. But by the same token, if we give (16a) the analogous structure in (19), we incorrectly expect (16b) to be ungrammatical on a par with (16a).

(19) Incorrect structure: There seemed [PRO to be a problem].

The reason is that *there* in the representation in (19), once again, is not licensed, as shown by the ungrammaticality of (20).

(20) * There seemed a problem.

It is important to understand that although *agree* is a subject control predicate and *seem* is not, both verbs share the property of not being *there* licensers.

11.3.2 Nonthematic subject positions

At this point, notice that the representations in (17) and (19) are both ruled out because *there* isn't licensed in the matrix clause. However, only in (17) is the matrix verb's selectional restriction violated. It turns out that a crucial difference between *agree* and *seem* is that *seem* imposes no selectional restrictions. We can see this by replacing *agree* in (8) with *seem*, as we do in (21); the acceptability contrast between (8a) and (8b) then disappears.

(21) a. The children seemed to get wet.
b. The { trees, rocks } seemed to get wet.

Another noteworthy property of *seem* is that its specifier position is not (and, in fact, must not be) associated with any thematic role. It is true that *seem* takes an argument: what for lack of a better term we will call the proposition argument (its complement clause).[3] However, for some reason, this argument cannot substitute into the specifier position, as we clearly see in the finite complement counterparts of raising sentences.[4]

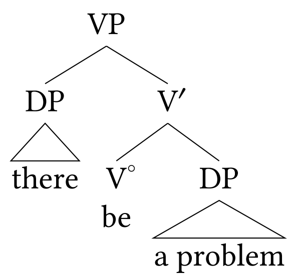
(22) a. It seems that the problem was too hard.
b. * That the problem was too hard seems.

To summarize: the subject position of *seem* is semantically deficient in the sense that it is associated neither with selectional restrictions nor with a thematic role. We will call such a subject position **nonthematic**.

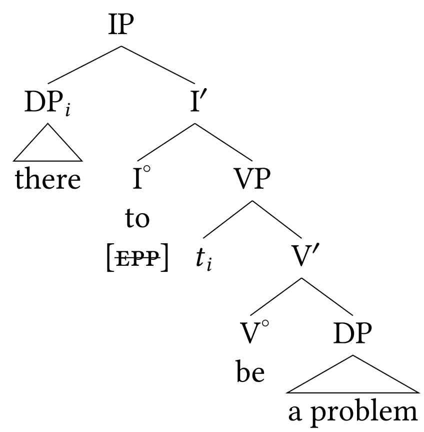
11.3.3 Deriving raising sentences

Of course, despite being superfluous from a semantic point of view, nonthematic subject positions are nevertheless syntactically obligatory (recall the subject requirement, § 4.1 and § 9.4.4). This makes possible the following analysis of subject raising sentences, which we illustrate by deriving (16b). We begin by deriving the complement clause. Note how the eventual matrix subject *there* is licensed in (23a) as the specifier of main verb *be*. We assume that the subject moves from Spec(VP) to Spec(IP) in (23b) in order to provide the complement clause with a subject in accordance with the subject requirement, satisfying the EPP.

(23) a.



b.

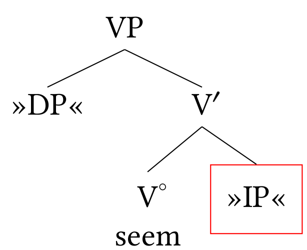


Substitute eventual matrix subject as specifier of lower verb

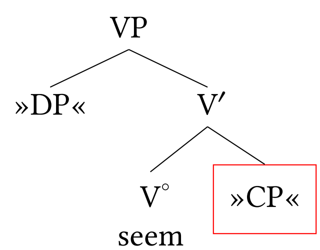
Substitute (23a) as complement of *to* and subject movement

We now substitute (23b) into the elementary tree for *seem* in (24a).

(24) a.



b.



Elementary tree for nonfinite complement, cf. (29a)

Elementary tree for finite complement, cf. (29b)

Before proceeding with the derivation, a few words about the elementary trees in (24) are in order. First, notice that both elementary trees in (24) contain a specifier position. Though semantically unnecessary, as discussed above, this position is motivated by the obligatoriness of expletive *it* in small clauses.

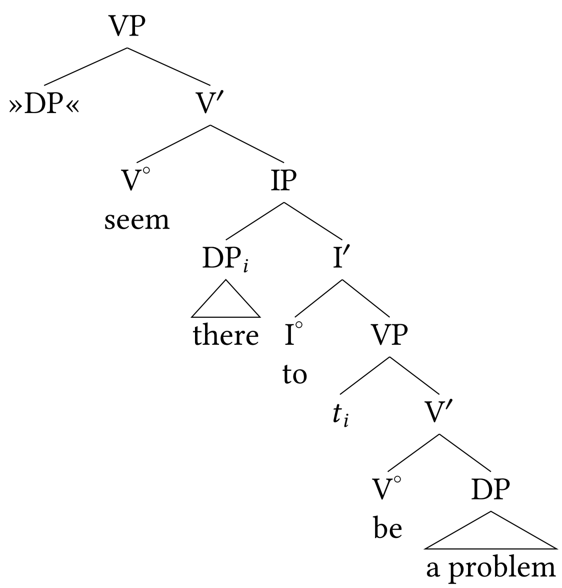
(25) a. They made [$_{\mathrm{VP}}$ it seem that there was a problem].

b. * They made [$_{\mathrm{VP}}$ seem that there was a problem].

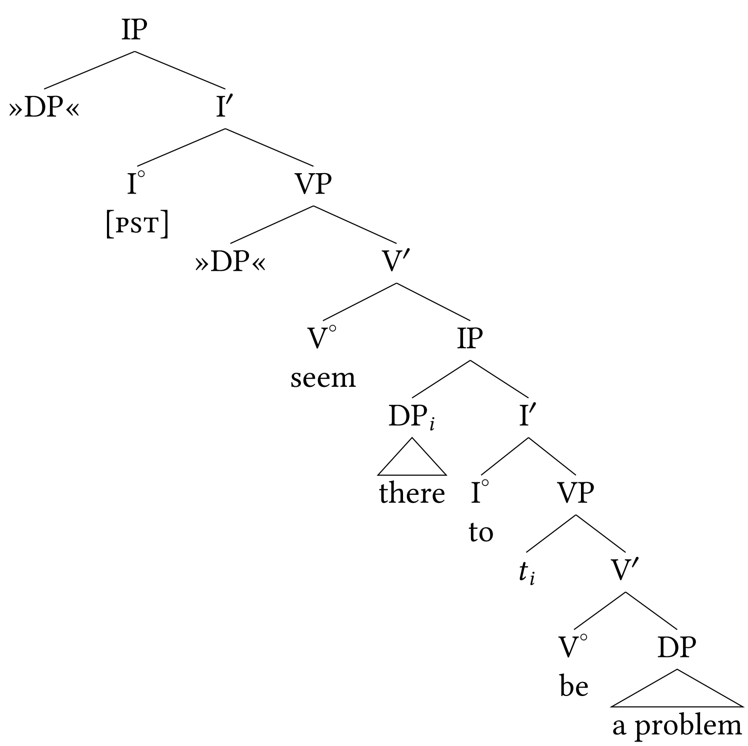
Second, the syntactic category of the clausal complement is IP in (24a), whereas it is CP in (24b), which is the elementary tree we would use in order to derive the finite complement counterpart of (16b) (*It seemed that there was a problem*).

Substituting the clause in (23b) as the complement of the elementary tree for raising *seem* in (24a) yields (26a), which in turn becomes the complement of the matrix I° element, yielding (26b).

(26) a.



b.



At this point in the derivation, the option arises in principle of substituting expletive *it* in the matrix Spec(VP) and moving it to the matrix Spec(IP). In fact, this is what we would do if the complement of *seem* were finite. In the case of a nonfinite complement, however, this step yields the ungrammatical (27).

(27) * It$_{\text{EXPL}}$ seemed there to be a problem.

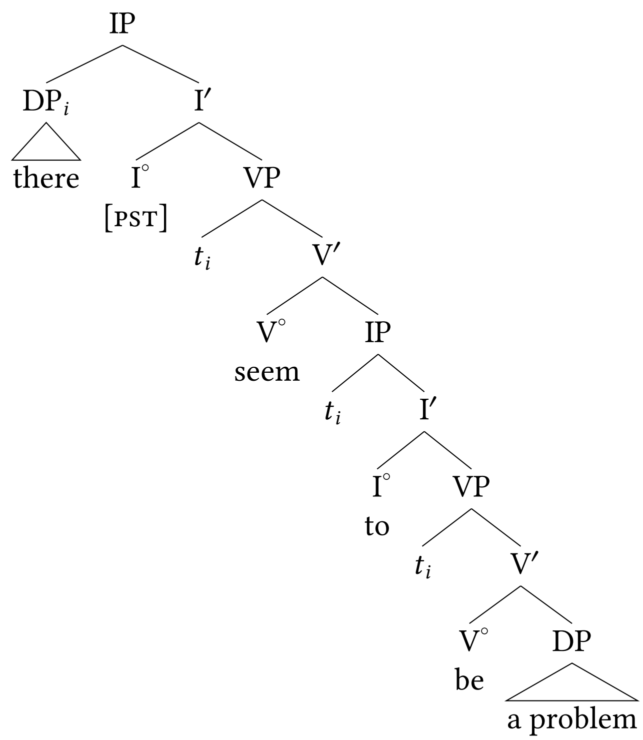
Why is (27) ungrammatical? The equally ungrammatical (28a) provides a hint; the ungrammaticality of the unambiguously objective case form *them* is reminiscent of (11b) in Chapter 10, repeated here as (28b).

(28) a. * It$_{\text{EXPL}}$ seemed them to dance.
b. * It$_{\text{EXPL}}$ was approved them (by us).

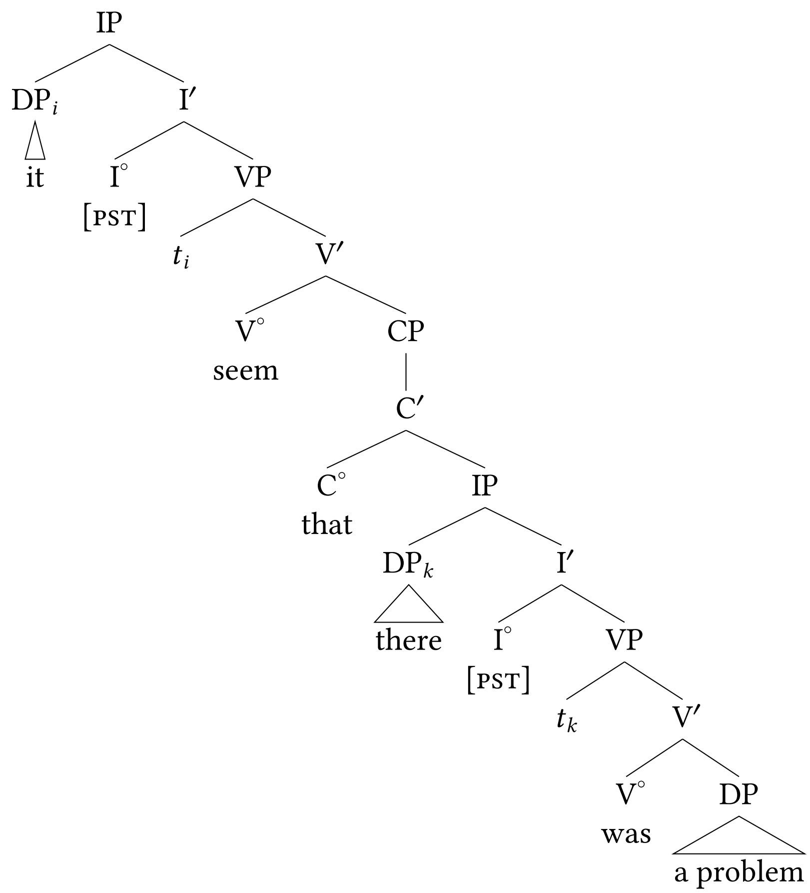
The parallel ungrammaticality of all three sentences leads us to conclude that *seems* is deficient in not being able to assign accusative case, similar to unaccusative verbs. (Recall that it is also deficient in not licensing a thematic specifier position.) Thus, both *there* in (27) and *them* in (28) don't receive case. If matrix I° is the only case-assigning head in the matrix clause, only one argument will be licensed by a matrix predicate with *seem*: the (raised) subject.

Since the complement-clause subject cannot get case within its own IP, it moves on via the matrix Spec(VP) to the matrix Spec(IP), as shown in (29a). Here, it can finally get case from finite I in the spec-head configuration. It is this movement of the subject out of the complement clause in (29a) that we refer to as **raising**. For comparison, the structure of the finite complement counterpart is shown in (29b); here, each of the two subjects gets nominative case from its own finite I.[5]

(29) a.



b.



More nonthematic subjects 11.4

In distinguishing between subject control and raising in this chapter, we have relied heavily on the grammaticality or ungrammaticality of sentences containing expletive *there*. As it turns out, expletive *there* is not the only **nonthematic** subject (= subject that is not associated with a thematic role). In this section, we present two further instances of nonthematic subjects: **subject idiom chunks** and **weather *it***.

11.4.1 Subject idiom chunks

In § 7.5.3, we introduced the notion of idiom chunks—phrases with interpretations that are noncompositional (i.e., not made up out of the meanings of their parts). In order to be interpreted idiomatically, an idiom must enter the structure as a constituent (since its interpretation is lexically stored as a unit, it must appear in the structure as a unit to receive that interpretation). For our current purposes we use idioms not just that are V′ constituents, but which are entire VP constituents, crucially containing a subject. In the idioms in (30) the relevant subject is underlined.

(30) a. <u>The cat</u> is out of the bag.
b. <u>The fur</u> will fly.
c. <u>The jig</u> is up.
d. <u>The pot</u> is calling the kettle black.
e. <u>The shit</u> hit the fan.

Idiom chunks share two important properties with expletive *there* for our current diagnostic purposes. First, just as expletive *there* must be licensed as the specifier of a verb of existence, the subjects in (30) have whatever idiomatic force they have only in connection with the rest of the idiom, but not otherwise.[6] In particular, the subject of the idiom chunk must start out as the specifier of the verb in the idiom. For instance, neither *cat* in (30a) nor *pot* in (30d) have a metaphorical sense of *secret* or *hypocrite*, respectively, in other syntactic contexts. So sentences like those in (31) have only literal interpretations.

(31) a. The cat is safe with them.
(only literal meaning; can't have the idiomatic meaning: 'The secret is safe with them.')
b. # Your neighbor sounds like a real pot.
(literal meaning is semantically deviant; also can't have the idiomatic meaning: 'Your neighbor sounds like a real hypocrite.')

Second, presumably because they are not interpreted literally, subject idiom chunks don't seem to refer or be associated with any thematic role, and so they can occupy the nonthematic subject position of raising predicates. As a result of these two properties, contrasts as in (32) are expected.

(32) a. # The cat agreed [PRO to be out of the bag].
b. [The cat]$_i$ seems [t_i to be out of the bag].

(32a) is ruled out both on a literal and an idiomatic reading. *Agree* selects humans as subjects and is therefore incompatible with *the cat* either as a literal or as an idiomatic (nonthematic) subject. In addition, *the cat* isn't licensed as an idiom chunk in the representation in (32a) because it is never a specifier of the lower verb (*be*, the verb heading the rest of the idiom). By contrast, both readings, and in particular the idiomatic one, are possible in (32b). This is expected, since the matrix subject originates in Spec(VP) of the complement clause, forming a constituent with the rest of the idiom.

Note that examples like (33) do not invalidate the diagnostic value of subject idiom chunks in distinguishing between subject control and raising predicates.

(33) The cat wanted [$_{CP}$ PRO to be out of the bag]. (only literal interpretation)

Here, the selectional restrictions that *want* imposes on its subject are less strict than in the case of *agree*. Since the subject of *want* isn't required to be human (but only to have a reasonably well-developed nervous system), the sentence is grammatical, unlike (32a). However, unlike in (32b), the matrix subject position isn't nonthematic and the matrix subject doesn't move out of the lower clause, so the sentence has only a literal interpretation. This is why a parallel sentence using an idiom with an inanimate subject as in (34) has no literal interpretation in the real world: literal *shit* can't want things.

(34) #The shit wanted [$_{CP}$ PRO to hit the fan.]

Examples like this show us that *want* places selectional restrictions on its subject.

11.4.2 Weather *it*

The third type of nonthematic subject is weather *it*, the subject of verbs of precipitation.

(35) ✓ It is { hailing, pouring, raining, sleeting, snowing }.

As with subjects of idiom chunks and their predicates, the licensing relationship between weather *it* and their predicates is mutual: not only is weather *it* licensed by weather verbs, but the weather verbs are in turn themselves licensed by weather *it*, as shown in (36).

(36) * The { air, atmosphere, environment, precipitation, sky, weather } is { hailing, pouring, raining, sleeting, snowing }.

Given the nonthematic character of weather *it*, contrasts as in (37) are expected.

(37) a. # It decided to rain at night.
b. It seems/appears to rain at night.

11.4.3 Summary

We have seen that expletive *there* is licensed as the subject of verbs of (coming into) existence. In a similar way, subjects of idioms and their predicates stand in a mutual licensing relationship, as does weather *it* with predicates denoting precipitation. Because of their special licensing requirements, none of these subjects is licensed by subject control predicates. Conversely, subject control predicates select humans as subjects, so that their selectional restrictions are not met by nonthematic subjects. By contrast, raising predicates like *seem* and *appear* neither interfere with the licensing of nonthematic subjects (which takes place in a lower clause) nor do they impose selectional restrictions that nonthematic subjects cannot meet. It is precisely because of their semantic deficiency or neutrality that they are able to act as grammatical catalysts, allowing licensing relations that are normally confined to the same clause to extend across an IP boundary.

As we have seen, the special properties of nonthematic subjects make them useful diagnostics for distinguishing subject control predicates from raising predicates. The relevant judgment patterns are summarized in Table 11.2; for completeness, we also include the judgments for manner adverbs discussed in connection with *promise*.

	Subject control	Raising
Expletive *there*	*	✓
Subject idiom chunk	# (or only literal)	✓ (idiomatic or literal)
Weather *it*	# (or metaphorical)	✓
It expletive in main clause	*	✓

Table 11.2: Judgments of subject control and raising sentences with nonthematic subjects.

New Empirical Observations

- Control predicates select for an external argument (i.e., a subject).
- Raising predicates do not select for an external argument (i.e., a subject).

Components of the Theoretical Model

- PRO is the embedded subject in a control construction.
- Subjects in raising constructions raise (move) to the matrix clause from the embedded clause.
- There exists nonthematic subject positions (EPP/Subject requirement), generally filled by an expletive or a moved phrase.
-

Notes 11.5

1. It might occur to a careful reader that an alternative approach to the facts in (8) and (9) is possible, according to which control sentences contain a single subject, which must simultaneously satisfy the selectional restrictions of both the higher and the lower verbs. Regardless of whether such an approach might be worked out for English, we do not adopt it, since it cannot be extended to subject control universally. In particular, the approach in question fails for Icelandic.

   In contrast to English (and most other languages), Icelandic has certain verbs whose subjects appear in some non-nominative case (genitive, dative, or accusative), even in finite clauses. The analysis of these so-called 'quirky case' subjects is beyond the scope of this textbook, but it is well established that they are true subjects (despite the lack of subject-verb agreement) (see Zaenen, Maling, and Thráinsson 1985; Sigurðsson 1991, and the many references therein). (i) gives examples of Icelandic finite clauses with an ordinary nominative subject and with a quirky case subject. The subjects are in **boldface**. Note that the underlined quantifiers agree in case with the subjects; this fact will be important directly. (How exactly this case agreement is implemented is irrelevant for present purposes.)

(i) a. Ordinary nominative subject:

Strákarn-ir	komust	all-ir	í	skóla.	
boys-DEF.NOM	got	all-NOM.PL.M	in	school	

'The boys all got to school.' (= Sigurðsson's (6a))

b. 'Quirky dative' subject:

Strákarn-um	leiddist	öll-um	í	skóla.
boys-DEF.DAT	was.bored	all-DAT.PL.M	in	school

'The boys were all bored in school.' (= Sigurðsson's (6c))

   (ii) shows that the clauses in (i) can be embedded under a subject control verb (here, *vonast til* 'hope for'). As in English, the subject of the embedded clauses is silent, but note that the quantifiers continue to exhibit the same case that they did in (i).

(ii) a. Embedded ordinary subject:

Strákarn-ir	vonast	til	að	PRO	komast	all-ir	í	skóla
boys-DEF.NOM	hope	for	COMP		get	all-NOM.PL.M	in	school

'The boys hope to all get to school.' (= Sigurðsson's (8a))

b. Embedded quirky subject:

Strákarn-ir	vonast	til	að	PRO	leiðast	ekki	öll-um	í	skóla
boys-DEF.NOM	hope	for	COMP		be.bored	not	all-DAT.PL.M	in	school

'The boys hope to all not be bored in school.' (= Sigurðsson's (8c))

In particular, in (ii.b), the quantifier must appear in the dative. From this, we conclude that the silent subject of the lower clause in (ii.b) is assigned quirky dative case in (ii.b), just as it is in the finite clause in (i.b). The fact that the matrix and embedded subjects are each assigned a different case in (ii.b) provides conclusive evidence that control constructions are indeed biclausal, since a single noun phrase cannot be assigned more than one case (even if it were to satisfy more than one selectional restriction at the same time).

2. Let us point out that we have not addressed an important question—namely, how PRO gets case. In English, PRO and overt noun phrases are in complementary distribution (at least almost perfectly so). In fact, the complementary distribution is not perfect. There is a bit of overlap in English—for instance, in the subject position of gerunds.

(i) a. ✓ [PRO going out with them] would bother me.
b. ✓ [{ Kim's, Kim } going out with them] would bother me.

In other words, the positions where PRO can appear are ones from which overt noun phrases are barred, and vice versa. It has therefore been proposed that PRO receives a case unique to it—so-called null case—which is assigned by nonfinite I° in the spec-head configuration. However, there is evidence from languages like Icelandic (see Note 1) and German that PRO can be assigned the same case features as overt subjects are, which contradicts the null case idea. We will not attempt to resolve the issue here, and we will ignore it in what follows. In many ways, PRO is a special kind of element in our theory, though there is a large research program attempting to derive the properties of PRO from more fundamental properties of the theory (Boeckx, Hornstein, and Nunes 2010, e.g.).

3. For simplicity, we ignore the verb *seem*'s optional experiencer argument (*It seems <u>to me</u> that you've solved the problem*), since it has no effect on our conclusions.

4. Clear evidence that the specifier position at issue is the Spec(VP) associated with *seem* (and not, say, some higher specifier position, such as Spec(IP)) comes from the grammaticality contrast in (i).

(i) a. They made [it seem [that the problem was hard]].
b. * They made [[that the problem was hard] seem].

5. We refrain from drawing tense lowering here to focus attention on the raising structures themselves: see § 12.2 for discussion of tense lowering.

6. The licensing relation between subject idiom chunks and their predicates is actually even tighter than that between expletive *there* and its licensers. Existential *there* must be licensed by verbs of (coming into) existence, but these verbs can occur independently of existential *there*. In other words, the licensing relationship is only one-way. By contrast, neither the subjects nor the predicates of idiomatic sentences can be interpreted idiomatically independently of each other. So in this case, as also for weather *it* and predicates of precipitation, the licensing relationship is mutual.

Exercises and problems 11.6

Exercise 11.1: Explaining grammaticality patterns with raising predicates

A. (1) is grammatical, but both sentences in (2), with the same intended meaning, are completely ungrammatical. Why is that?

(1) It seems that Jackie has solved the problem.

(2) a. * It seems Jackie to have solved the problem.
b. * Jackie seems that has solved the problem.

B. Repeat (A) for (3) and (4).

(3) It seems that they like caviar.

(4) a. * There seems that they like caviar.

b. * They seem that they like caviar.
c. * They$_i$ seem that t_i like caviar.
d. * Caviar$_i$ seems that they like t_i.

Exercise 11.2: Basic practice distinguishing raising and control A

A. Using the examples in (1)-(3) in Exercise 11.4 as a model, determine whether the underlined predicates in (1) are subject control or subject raising predicates.

(1) a. They failed to be on time.
b. They aspire to graduate early.
c. They remembered to do the laundry.
d. They are hesitant to move.
e. They are destined to get the job.
f. They are about to hit the jackpot.

B. Build structures for the sentences in (1).

Exercise 11.3: Basic practice distinguishing raising and control B

Using the examples below as a model, determine whether the verbs in (4) are raising predicates, subject control predicates, or neither. In each example below, the evidence takes the form of grammaticality judgments about the example sentences, and the conclusions are annotated below each example. For the purposes of this exercise, use only active verb forms.

(1) ✓ There chanced to be an opening
→ **Conclusion**: *Chance* is a raising verb.

(2) a. * There resolved to be a solution.
→ **Conclusion**:*Resolve* is not a raising verb.
b. ✓ They resolved to solve the problem.
→ **Conclusion**: *Resolve* is a subject control verb.

(3) a. * There is possible to be a solution.
→ **Conclusion**: *Possible* is not a raising adjective (despite the grammaticality of *It is possible that there is a solution.*).
b. * They are possible to solve the problem.
→ **Conclusion**: *Possible* doesn't allow a nonfinite complement, so it is not a subject control adjective either.

(4) agree, aspire, attempt, be, beg, cease, choose, claim, come, commence, continue, dare, demand, deserve, desire, determine, elect, end up, endeavor, expect, fail, forget, happen, have, hope, intend, look, mean, need, neglect, plan, pledge, prefer, presume, pretend, proceed, prove, purport, remember, request, start, strive, swear, tend, train, try, volunteer, vow, wish, yearn

Exercise 11.4: Nonverbal raising and control

Subject control predicates and raising predicates can belong to other syntactic categories than V. The predicates in (1) are adjectives, those in (2) are participles, and *about* in (3) is a preposition. Repeat (11.3) for these predicates. It's not always clear whether the participles are verbs or adjectives. If in doubt, treat them as adjectives; the exact category is not crucial for the purposes of the exercise.

Note: Subject control predicates and raising predicates that aren't verbs need to be integrated into the sentence using some form of the verb *be*. See Exercise 11.2, (1d)–(1f) for examples.

(1) afraid, anxious, apt, certain, content, eager, ecstatic, evident, fortunate, glad, happy, hesitant, liable, likely, lucky, possible, ready, reluctant, sorry, sure, unlikely

(2) bound, delighted, destined, determined, embarrassed, excited, fated, going, inclined, itching, jonesing, prepared, scared, (all) set, supposed, thrilled

(3) about

Exercise 11.5: Case in raising and control

A. What verb class does *prove* belong to in (1a) and in (1b)?

(1) a. They proved <u>there</u> to be an error in the calculation.
b. <u>There</u> proved to be an error in the calculation.

B. Which case does each instance of *there* get? How can you tell? What's the case-assigning head, and what's the case assignment configuration?

C. Build the trees for both sentences in (1).

Exercise 11.6: Selectional restrictions in control and raising

A. As mentioned in the text, certain subject control predicates, like *try*, cannot take finite complements. Provide evidence that in subject control sentences containing these predicates, both the matrix predicate and the complement predicate impose their own selectional restrictions, thus motivating a biclausal analysis.

B. Explain the acceptability contrast in (1).

(1) a. The children seemed to learn Twi.
b. # The children seemed to { elapse, evaporate }.

C. Show that *threaten* is both a subject control verb and a raising verb.

Exercise 11.7: Comparing different kinds of A-movement

Superficially, raising looks quite different than ordinary subject movement, the simple passive, and the passive of ECM verbs, but it shares properties with all three. Complete Table 11.3 to better understand the similarities and the differences. By 'local movement' is meant movement where the DP head of the movement chain doesn't leave its IP.

Name	Raising	Passive of ECM	Ordinary subject movement	Simple passive
Tail of chain	Spec(VP), low			
Intermediate links (if any)	Spec(IP), low; Spec(VP), high			
Head of chain	Spec(IP), high			
Case assigner	Finite I			
Case configuration	Spec-head			
Local?	No			

Table 11.3: Case assignment in raising and other constructions

Problem 11.1: Control in indirect questions

English indirect questions can be finite, as in (1), or nonfinite, as in (2).

(1) a. ✓ They know who they should invite.
   b. ✓ They know which topic they should talk about.
   c. ✓ They know who should speak.

(2) a. ✓ They know who to invite.
   b. ✓ They know which topic to talk about.
   c. * They know who to speak.

A. Using what you've learned in this chapter, build structures for (2) that are as parallel as possible to those for (1a). For simplicity, build a full structure just for (2a) and only structures for the indirect questions in (2b), (2c).

B. Why is (2c) ungrammatical?

Problem 11.2: Diagnosing control predicates

A. You have been asked to review an article for the linguistics journal *Glossa* by Professor A.B.C. Gerneweis, who concludes on the basis of the contrast in (1) that *volunteer* is a control verb. What is wrong with the argument?

(1) a. ✓ They volunteered to do the job.
    b. * There volunteered to do the job.

B. In your review, you graciously provide the conclusive evidence that Professor Gerneweis failed to provide.

Problem 11.3: Ambiguity in raising and control

A. The sentence in (1) is ambiguous. Explain, taking into account the facts in (2).

(1) They were determined to be U.S. citizens.

(2) a. ✓ They were very determined to be U.S. citizens.
    b. ✓ They were determined to be U.S. citizens by the INS.
    c. * They were very determined to be U.S. citizens by the INS.

B. Build the trees for the two interpretations of (1).

12 The verb raising parameter

Chapter outline

As we saw in Chapter 5, tense in English can be expressed in one of two ways. The future tense is expressed by *will*, which precedes the verb and is a free morpheme; that is, it can be separated from the verb and stand alone.

(1) a. * We watch *will* that show.
b. ✓ We *will* never watch that show.
c. Q: Will you watch that show?
A: ✓ Yes, we *will*.

By contrast, the past tense is expressed by a bound morpheme, ordinarily the suffix -*ed*, which combines with the verb to form a morphologically complex word.[1]

(2) a. ✓ We watch-*ed* that show.

b. * We-*ed* never watch that show.

c. Q: Did you watch that show?
A: *Yes, we -*ed*.

This dual expression of tense is typical of the Germanic language family, to which English belongs. In all of these languages, the future is expressed **analytically** (as two separate words), whereas the past is mostly expressed **synthetically** (as a single morphologically complex word).[2]

Treating sentences as projections of I°, as introduced in Chapter 5, provides a structural locus for free tense morphemes and is therefore straightforwardly compatible with the analytic expression of tense. The analogous semantic contribution of free and bound tense morphemes, in both English and other languages, is a good reason to extend the IP analysis to sentences with synthetic tense forms. But the extension does raise the question of how tense and the verb combine to form a complex word when tense is expressed synthetically. Either in the syntax or in the morphology, the tense features must be combined with the verb in order to be pronounced. In this chapter, we present evidence that in some languages, the verb moves up to I° before the structure is handed over to PF (where morphology determines its pronunciation), whereas in other languages, the verb remains *in situ* (that is, it does not move). Instead, tense moves down to the verb in the morphology itself. We will refer to the choice between V-to-I verb movement in the syntax and I-to-V tense lowering in the morphology as the **verb raising parameter**. As we will show, the parametric variation can be detected on the basis of the relative order of inflected verbs and certain adverbs.[3]

In the remainder of the chapter, we first contrast verb raising in French with tense lowering in English. Our description of English also includes discussion of a closely related and important topic in the grammar of the modern language: the *do*-support that is obligatory in negated sentences (*They don't like okra* vs. **They not like okra*). We then briefly review the parametric variation attested in the modern Scandinavian languages and the loss of verb raising in all but one of them, Icelandic.

12.1 Verb raising: V moves to I in the syntax

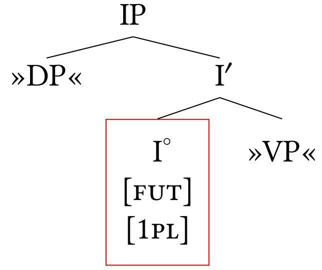
12.1.1 The future tense in French

In certain languages, the verb moves up to I° in the syntax. One such language is French, and we begin our discussion of the verb raising parameter by considering the future tense in French, which is formed by attaching suffixes to a verb's infinitive form.

Future tense of *chanter* 'sing'		**Present tense of *avoir* 'have'**	
je chanter-**ai**	'I will sing'	j'**ai**	'I have'
tu chanter-**as**	'you.SG will sing'	tu **as**	'you.SG have'
il, elle chanter-**a**	'he, she will sing'	il, elle **a**	'he, she has'
nous chanter-**ons**	'we will sing'	nous **avons**	'we will have'
vous chanter-**ez**	'you.PL will sing'	vous **avez**	'you.PL have'
ils, elles chanter-**ont**	'they will sing'	ils, elles **ont**	'they have'

Table 12.1: Future tense formation in French

As is evident from Table 12.1, the future tense affixes are nearly identical to the present tense forms of the verb *avoir* 'have', the only difference being that the affixes are truncated in the first and second person plural by comparison to the full two-syllable forms of *avoir*. This correspondence suggests that the future tense in French developed via a semantic shift from 'they have to V' to 'they will V'.[4] In addition, and more immediately relevant for the present discussion, the originally free forms of *avoir* were reanalyzed as bound morphemes.[5] The analytic roots of the synthetic French future tense thus indicate that the two ways of expressing tense (analytic or synthetic) are not just semantically parallel, but that they are also morphologically more closely related than might appear at first glance. Example (3) shows the elementary tree for the French future suffix in the first-person plural, which ends up being spelled out as -*ons* in the morphology.

(3)


The morphology of French therefore must have a rule that looks something like (4), which says that the combination of an I° with future tense and first plural features together is pronounced as the -*ons* suffix when occuring on a regular verb.

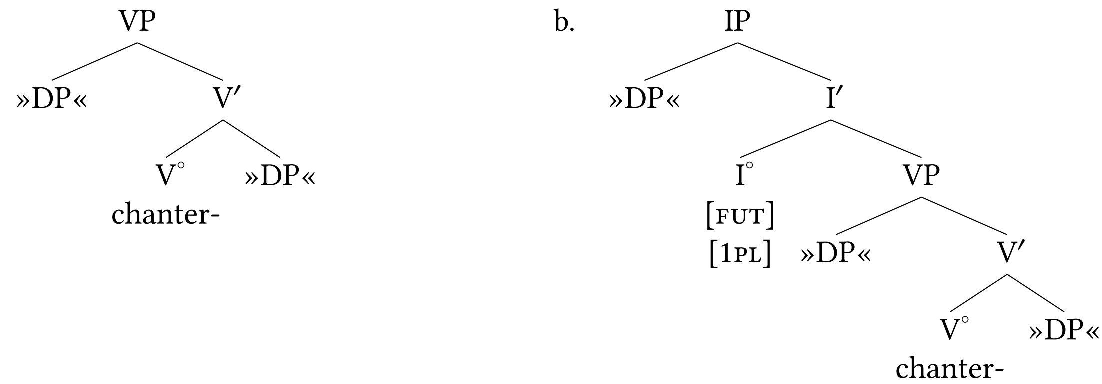
(4) I° + [FUT] + [1PL] ⟷ /-ons/ / *verb*___

Given elementary trees like (3), sentences like (5) can be derived as follows.

(5) Nous chanter -ons une chanson.
we sing -FUT.1PL a song
'We will sing a song.'

We begin with the elementary tree for the verb *chanter* in (6a) and substitute it as the complement of (3) to yield the structure in (6b).

(6) a. b.

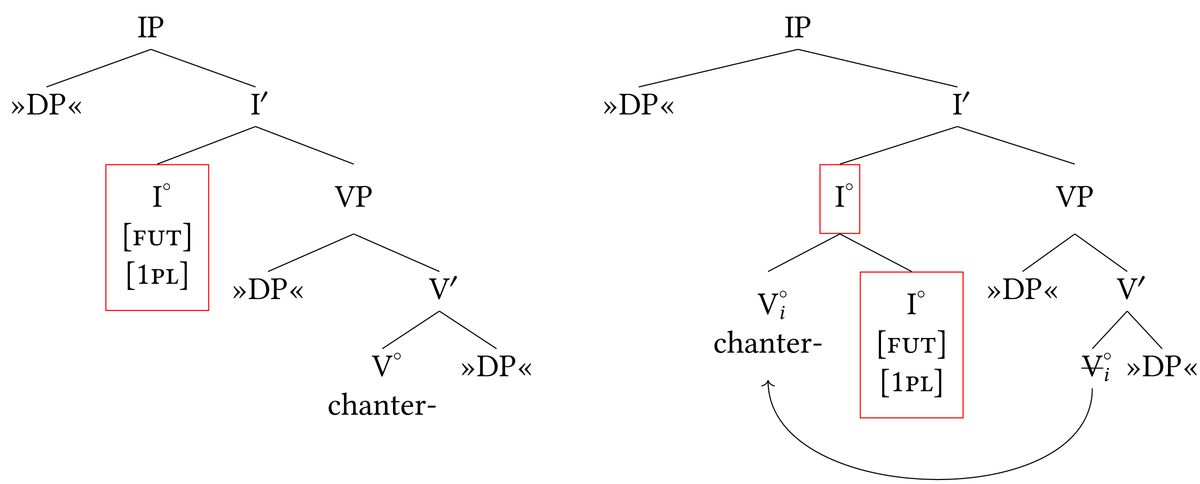


Elementary tree for *chanter*

Substitute (6a) in elementary tree (3)

The verb then moves to I′. Because I° is already filled, the verb cannot directly substitute in I′, but must adjoin to it, as shown in (7). The **head movement** that we invoke here is exactly the same formal operation that was introduced in Chapter 8, § 4 in connection with subject-aux inversion.

(7) a. b.

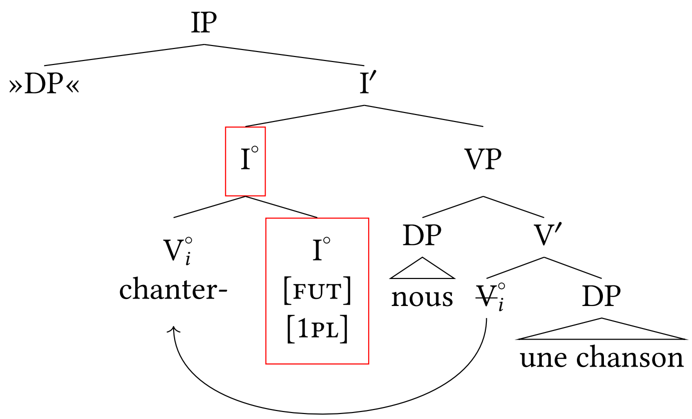


Select I° as target of adjunction

Left-adjoin V° to I°

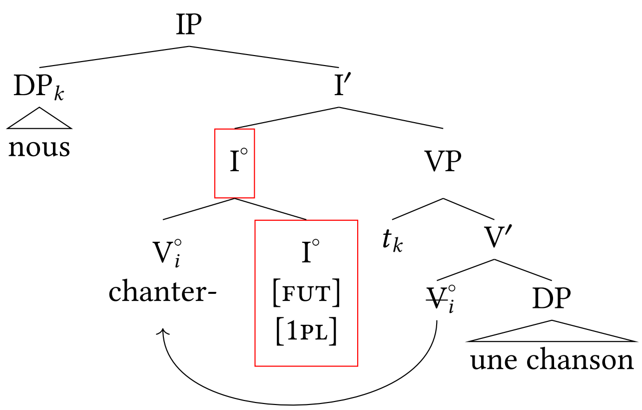
The remaining steps of the derivation, shown in (8), are identical to the ones that would be required to derive the corresponding English sentence *We will sing a song*.

(8) a.



Substitute subject and object

b.



Move subject

Finally, the structure in (8b) is handed over to the morphological component, where the combination of *chanter* and the future tense affix is spelled out as *chanterons* according to the morphological rule in (4).

12.1.2 The order of diagnostic adverbs and verbs in French

Extra Info

In reading the following section, bear in mind that our focus is not on the distribution of adverbs in and of itself. In particular, we are not claiming that all, or even most, adverbs left-adjoin to V′ in French; in fact, there are many that right-adjoin, and there are many that adjoin to different syntactic positions (both in French, and cross-linguistically). Rather, the idea is to use the particular subset of adverbs that must left-adjoin to V′ as a diagnostic tool to determine the structural position of finite verbs in French.

The facts of French presented so far are consistent with a verb raising analysis, but do not provide conclusive evidence in favor of it. In other words, nothing in what we have said so far prevents the French verb from remaining in situ and not combining with tense until the morphology. In this section, we present conclusive evidence in favor of the verb raising analysis that is based on the order of verbs and certain diagnostic adverbs (Emonds 1978).

As illustrated in (9)–(11), there are certain adverbs in French (underlined) that ordinarily precede the main verb of a sentence (in boldface), rather than follow it. (Strictly speaking, *à peine* is a PP; what is relevant for the purposes of the argument is not its syntactic category, but rather its syntactic distribution.) We have circled the auxiliary in each example below.

(9) a. ✓ Elle avait <u>à peine</u> **travaillé** trois heures.
she had hardly worked three hours
'She had hardly worked three hours.'

b. ✓ Mon ami a <u>complètement</u> **perdu** la tête.
my friend has completely lost the head
'My friend completely lost his head.'

c. ✓ J' avais <u>presque</u> **oublié** mon nom.
I had almost forgotten my name
'I had almost forgotten my name.'

(10) a. * Elle avait **travaillé** <u>à peine</u> trois heures.

b. * Mon ami a **perdu** <u>complètement</u> la tête.

c. * J' avais **oublié** <u>presque</u> mon nom.

(11) a. * Elle avait **travaillé** trois heures <u>à peine</u>.

b. * Mon ami a **perdu** la tête <u>complètement</u>.

c. * J' avais **oublié** mon nom <u>presque</u>.

Word of Caution

As highlighted by the grammaticality contrast in (i), constraints in adverb placement in one language don't necessarily carry over to their translation equivalents in another.

(i) a. * ... **perdu** la tête <u>complètement</u> = (11b)

b. ✓ ... **lost** one's head <u>completely</u>

The negative adverb *jamais* 'never' and the simple negative marker *pas* 'not' behave alike in this regard in French.[6]

(12) a. ✓ Les enfants d' aujourd'hui (n') ont <u>jamais</u> **écouté** la radio.
the children of today NE have never listened.to the radio
'Today's children have never listened to the radio.'

b. * Les enfants d'aujourd'hui (n') ont **écouté** <u>jamais</u> la radio.

c. * Les enfants d'aujourd'hui (n') ont **écouté** la radio <u>jamais</u>.

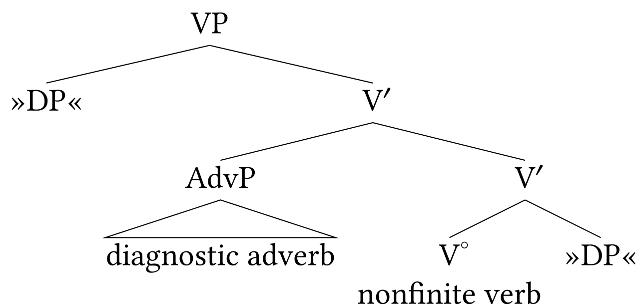
(13) a. ✓ Je (n') ai pas **fait** la vaisselle.
I NE have not done the dishes
'I haven't done the dishes.'

b. * Je (n') ai **fait** pas la vaisselle.

c. * Je (n') ai **fait** la vaisselle pas.

The word order facts in (9)–(13) follow straightforwardly if the adverbs in question must adjoin to the left of V′, as shown schematically in (14), rather than to the right.

(14)



But now consider an unexpected fact: when the main verb of the sentence is finite, the obligatory adverb-verb order that we have just seen is ungrammatical.

(15) a. * Elle à peine **travailler-a** trois heures.
she hardly work-FUT.3SG three hours
Intended meaning: 'She will hardly work three hours.'

b. * Mon ami complètement **perdr-a** la tête.
my friend completely lose-FUT.3SG the head
Intended meaning: 'My friend will completely lose his head.'

c. * Je presque **oublier-ai** mon nom.
I almost forget-FUT.1SG my name
Intended meaning: 'I will almost forget my name.'

d. * Les enfants d' aujourd'hui (ne) jamais **écouter-ont** la radio.
the children of today NE never listen.to-FUT.3PL the radio
Intended meaning: 'Today's children will never listen to the radio.'

e. * Je (ne) pas **fer-ai** la vaisselle.
I NE not do-FUT.1SG the dishes
Intended meaning: 'I won't do the dishes.'

Instead, the adverb must follow the verb, although it still cannot follow the entire V′.

(16) a. ✓ Elle **travaillera** à peine trois heures.

b. ✓ Mon ami **perdra** complètement la tête.

c. ✓ J'**oublierai** presque mon nom.

d. ✓ Les enfants d'aujourd'hui (n')**écouteront** jamais la radio.

e. ✓ Je (ne) **ferai** pas la vaisselle.

(17) a. * Elle **travaillera** trois heures à peine.
b. * Mon ami **perdra** la tête complètement.
c. * J'**oublierai** mon nom presque.
d. * Les enfants d'aujourd'hui (n')**écouteront** la radio jamais.
e. * Je (ne) **ferai** la vaisselle pas.

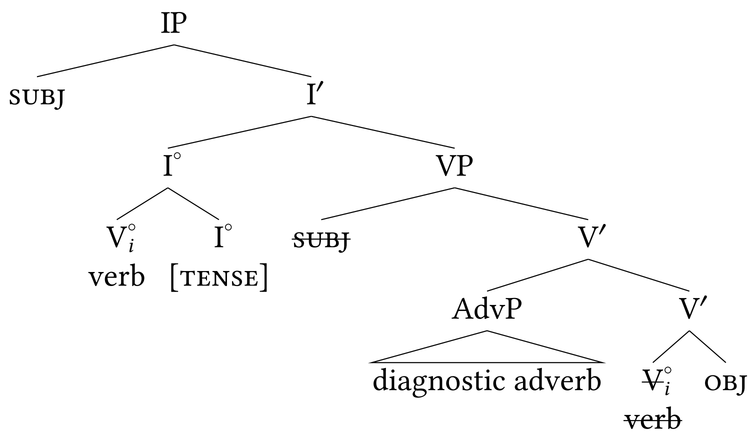
Table 12.2 summarizes all the facts just presented.

Finiteness of main verb	AdvP > verb …	verb > AdvP …	verb > XP > AdvP
nonfinite, as in (9)–(14)	✓	*	*
finite, as in (15)–(17)	*	✓	*

Table 12.2: Adverb placement in French by finiteness of main verb

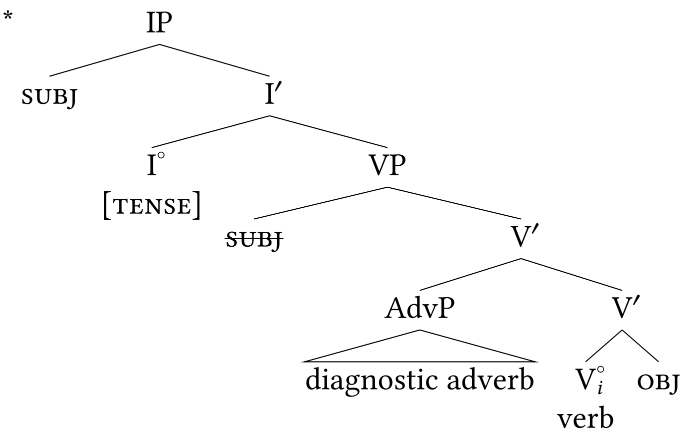
As already noted, the adverb placement facts for nonfinite verbs in the first row are expected under the assumption that diagnostic adverbs obligatorily left-adjoin to V′. This assumption also explains the rightmost judgment for finite verbs (the blue star in the undecorated box in the second row). The judgments highlighted in red dots, which are the opposite of their solid green counterparts in the row above, are puzzling at first glance. But they too follow straightforwardly if we assume that finite verbs obligatorily move to I in French, as in (18a).

(18) a. ✓



Verb raising yields finite V > Adv (grammatical in French)

b. *



No verb raising yields Adv > finite V (ungrammatical in French)

If French did not require finite verbs to move to I, as in the hypothetical scenario represented in (18b), it is difficult to see how the contrast between the solid green and the dotted red cells in Table 12.2 could be derived in a principled way.

As (19)–(21) show, the adverb facts for other inflected verbs (simple tenses) are parallel to those for the future tense (compare (15)–(17)), and it is therefore natural to extend the verb raising analysis to them as well.

(19) a. * Elle à peine **travaill-ait** trois heures.
she hardly work-IMPF.3SG three hours
Intended meaning: 'She used to hardly work three hours.'

b. * Mon ami complètement **perd** la tête.
my friend completely lose.PRS.3SG the head
Intended meaning: 'My friend is completely losing his head.'

c. * Je presque **oublie** mon nom.
I almost forget-PRS.1SG my name
Intended meaning: 'I am almost forgetting my name.'

d. * Les enfants (ne) jamais **écout-aient** la radio.
the children NE never listen.to-IMPF.3PL the radio
Intended meaning: 'The children never listened to the radio.'

e. * Je (ne) pas **fais** la vaisselle.
I NE not do-PRS.1SG the dishes
Intended meaning: 'I don't do the dishes.'

(20) a. ✓ Elle **travaillait** à peine trois heures.

b. ✓ Mon ami **perd** complètement la tête.

c. ✓ J'**oublie** presque mon nom.

d. ✓ Les enfants d'aujourd'hui (n')**écoutaient** jamais la radio.

e. ✓ Je (ne) **fais** pas la vaisselle.

(21) a. * Elle **travaillait** trois heures à peine.

b. * Mon ami **perd** la tête complètement.

c. * J'**oublie** mon nom presque.

d. * Les enfants d'aujourd'hui (n')**écoutaient** la radio jamais.

e. * Je (ne) **fais** la vaisselle pas.

In concluding our discussion of French, we draw your attention to the fact that verb movement, just like other instances of movement, allows us to accommodate mismatches between an item's expected position given its thematic or semantic relations and the position in which it is pronounced. In the case at hand, assuming verb movement allows us to maintain a simple generalization concerning these particular diagnostic adverbs (they obligatorily left-adjoin) regardless of the finiteness of the verb they modify. Perhaps even more importantly, we can maintain—regardless of verb finiteness or presence of adverbs—the idea encoded in the X′ schema that verbs and their complements are siblings. Nonfinite verbs presents no difficulty in this regard. But even in the case of finite verbs, where the verb-complement adjacency expected under sisterhood is interrupted by an intervening adverb, the expected structural relation is preserved through the trace of the verb.

Tense lowering: I lowers to V in the morphology 12.2

12.2.1 The order of diagnostic adverbs and verbs in English

Having established that French exhibits verb raising in the syntax, we now investigate the corresponding English facts, using exactly the same tool that we used in French—namely, the position of diagnostic adverbs. As in French, certain adverbs in English obligatorily precede nonfinite verbs.

(22) a. They will { almost, hardly, never } **fail**.

b. They have { almost, hardly, never } **failed**.

(23) a. * They will **fail** { almost, hardly, never }.

b. * They have **failed** { almost, hardly, never }.

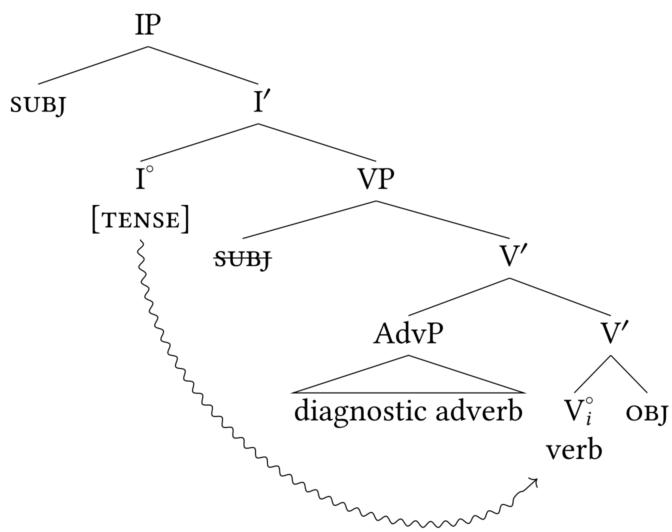
But unlike in French, these adverbs precede the main verb of a sentence even when the verb is finite. The word order Adv > Verb in (24a) is grammatical: this word order was precisely what was ungrammatical in French. In contrast, the word order Verb > Adv that was grammatical in French is ungrammatical in English (24b).

(24) a. They { almost, hardly, never } **failed**.

b. * They **failed** { almost, hardly, never }.

The ungrammaticality of (24b) means that the verb raising analysis that is successful for French is exactly wrong for English. Instead, English verbs always appear to be in the same position. We can conclude that, in English, all verbs remain in situ in the syntax and any tense morphemes lower and adjoin to V in the morphology. The input to the morphology is therefore (18b), repeated here as (25), and the order of the adverb and the verb remains unchanged after tense lowering. We mark tense lowering here with a squiggly arrow to help distinguish it from the other arrows we have been using (solid for movement, dashed for Agree).

(25)



Morphological tense lowering yields Adv > finite V
(grammatical in English regardless of finiteness of V)

This is the only lowering movement that we've seen or will see in this text, and (as we'll discuss below) it only happens in select instances. We can view it as a morphological fix that happens after syntax (on the PF branch of the inverted Y in the Y-model).

12.2.2 *Do*-support in English

We now turn to an apparently idiosyncratic and quirky consequence of the fact that English lacks verb raising—namely, the presence of the auxiliary *do* (*do*-support) that is necessary in sentences negated with *not*. In order to highlight the conditions under which *do*-support takes place, we will contrast sentences containing *not* with ones containing *never*.

In informal spoken English, *never* can function as a sentence negator equivalent to simple 'not', without its literal meaning of 'not ever'.

(26) Q: Did you get a chance to talk to Tom last night?
A: Nope, I *never* got a chance.

But despite their functional equivalence in contexts like (26), *not* and *never* differ from each other in a striking way: *not* obligatorily triggers *do*-support, whereas *never* doesn't. (All forms of *do* in this section are to be read without emphatic stress.)

(27) a. * They *not* **applied**.
 b. ✓ They { did *not*, did*n't* } **apply**.

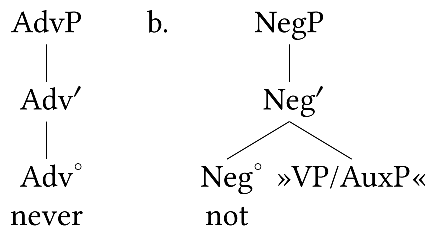
(28) a. ✓ They *never* **applied**.
 b. * They did *never* **apply**.

In order to explain this puzzling fact, we present an analysis of *do*-support that relies on two main ideas: first, that *never* and *not* are integrated into the structure of English sentences in different ways, and second, that tense (and syntactic heads more generally) can lower in the morphology, though only under certain structural conditions.[7]

A syntactic difference between *never* and *not*

As shown in (29), *never* is intransitive and does not select for a phrase that is on the main spine of the clause; therefore we will only ever find it as an adjunct in a sentence. Negation with *not*, on the other hand, selects for a VP or an AuxP, implying that it does sit on the main spine of the clause. We will show that this distinction is instructive about the properties of verbs and tense in English.

(29) a. b.



We present two pieces of evidence for this distinction. The first comes from **negative inversion**. (30a) shows an ordinary negative sentence, and (30b) shows its negative inversion counterpart, where the negative constituent (in boldface) has moved to the beginning of the sentence, and modal (in italics) precedes the subject (underlined). (We will give the structure for such a negative inversion sentence shortly; for the moment, it is sufficient to understand that (30b) is structurally analogous to the corresponding direct *wh*-question *What would they accept more happily?*)

(30) a. They *would* accept **no present** more happily.
 b. **No present** *would* they accept more happily.

An important property of this construction is that the material preceding the modal must be a maximal projection. Thus, while the full DP *no present* can precede *would* in (30b), its head, the negative determiner *no*, cannot undergo negative inversion on its own—see (31).

(31) * **No** *would* they accept **present** more happily.

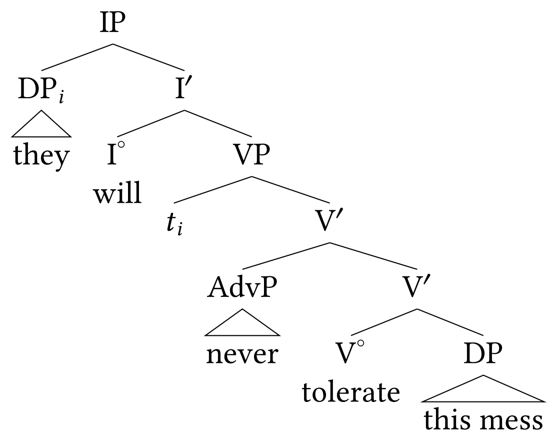
Bearing in mind this fact about negative inversion, consider the canonical and negative inversion sentences in (32).

(32) a. They *will* **never** tolerate this mess.

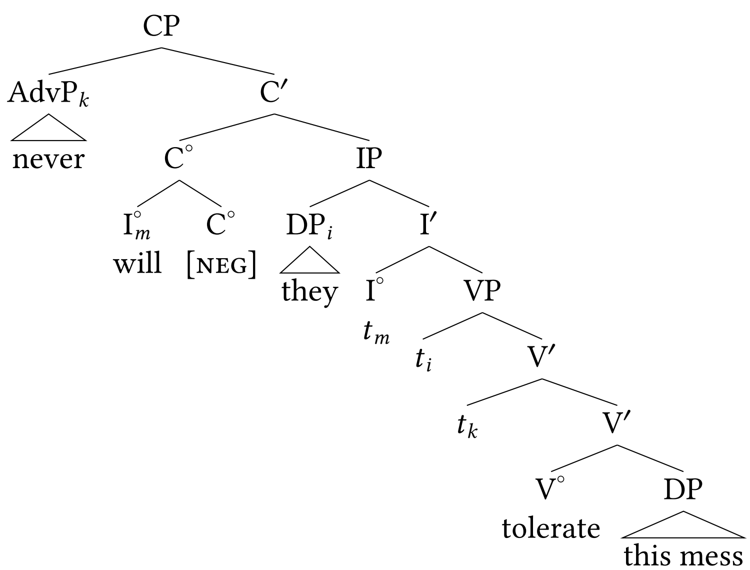
b. **Never** *will* they tolerate this mess.

Example (33) gives the structures for (32a) and (32b), which has the same structure as the corresponding *wh*-question *When will they tolerate this mess?* For present purposes, what is important is that *never* is a maximal projection adjoined to the clause and therefore a suitable candidate for negative inversion.

(33) a.



b.

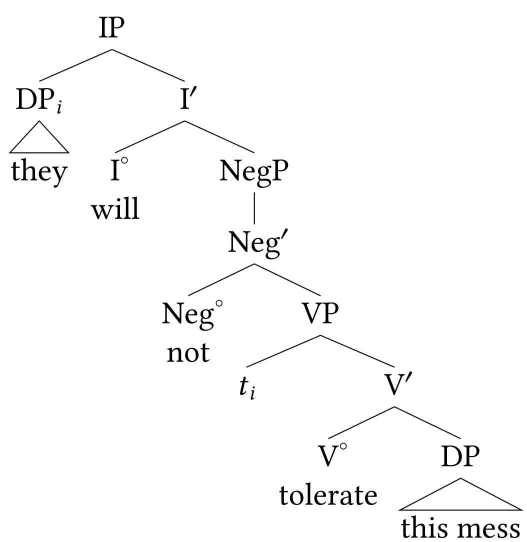


Now consider the *not* variant of (32a) in (34).

(34) They *will* **not** tolerate this mess.

Under the reasonable assumption that I° can take either NegP or VP as complements, (34) has the structure in (35).

(35)



In this structure, *not* on its own is not a maximal projection adjoined to the structure, and so *not*, like *no* but unlike *never*, should not be able to undergo negative inversion. (NegP is of course a maximal projection, but as it is on the main spine of the clause, it also contains the entire VP.) This expectation is confirmed by the ungrammaticality of (36).

(36) * **Not** <u>they</u> *will* tolerate this mess.

A second piece of evidence for the status of *not* (and its variant *n't*) as a transitive head comes from the fact that it optionally adjoins to I°, forming a complex head that can exhibit morphological irregularities. For instance, *n't* moves to combine with *will* and spells out the irregular form *won't*. Moreover, various non-mainstream English varieties allow the combination of *n't* with various forms of the aspectual auxiliaries *be* and *have* to be spelled out as *ain't*. Such irregularity is the hallmark of two heads combining (whether in the syntax or in the morphology).

A constraint on tense lowering in the morphology

We turn now to the second piece of our solution to the puzzle presented by the contrast between (27) and (28), repeated here as (37) and (38).

(37) a. * They *not* **applied**.
b. ✓ They { did *not*, did*n't* } **apply**.

(38) a. ✓ They *never* **applied**.
b. * They did *never* **apply**.

The idea is that tense lowering in the morphology is subject to the **locality condition** in (39).

(39) When a head A lowers onto a head B in the morphology, A and B must be in a **local** relation, in the sense that no projection of a head distinct from A and B **intervenes** on the path of branches that connects A and B.

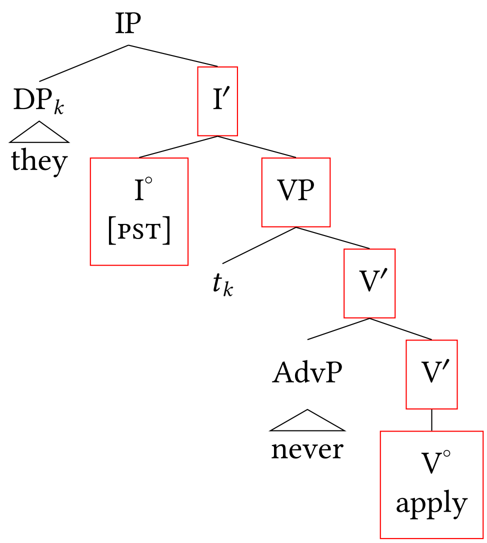
The notion of **intervene** is defined as in (40).[8]

(40) An element C, C distinct from (projections of) A and B, **intervenes** between A and B if and only if (some projection of) A dominates C and (some projection of) C dominates B.

It is important to understand that intervention is defined not in terms of linear precedence, but in terms of the structural relation 'dominate'. This means that the place to look for whether the locality condition in (39) is satisfied or violated is not the string of terminal nodes beginning with A and ending with B, but the path of branches that connects A with B in the tree. A visual clue to this kind of structural configuration is this: are all of the relevant heads (A,B,C) on the main spine of the clause? If so, then C intervenes between A and B.

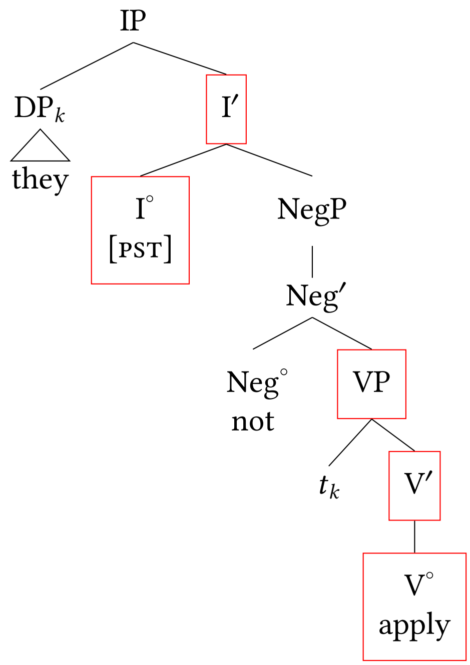
The structure for (38a) is given in (41). In this structure, tense lowering is consistent with the locality condition in (39), since adjoining *never* at V′ results in the adverb head being too embedded in the tree to intervene between I° and V°. (In other words, AdvP isn't on the path from I° to V°, indicated by the boxed nodes.)

(41)

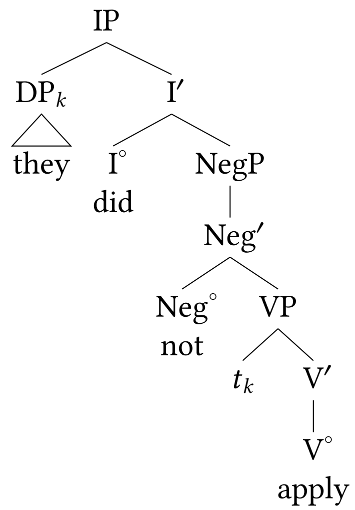


By contrast, in the structure in (42a) tense-lowering would violate the locality condition because the path between I° and V° is interrupted by NegP and Neg′. As a result, only the *do*-support variant of (42a) is grammatical, which is shown in (42b). It's true that NegP and Neg′ intervene between I° and V° in (42b) as well, but forms of *do* are free morphemes. Therefore, unlike tense affixes, they don't need to undergo tense lowering onto V° to form a well-formed morphological word. Since (39) is a constraint on tense lowering, not a constraint on syntactic trees in general, (42b) does not violate it.

(42) a.



b.



12.3 The loss of verb raising in Scandinavian

The modern Scandinavian languages are descendants of Old Norse (the language of the Vikings), which split into two main branches with self-explanatory names: Mainland Scandinavian (Danish, Norwegian, Swedish) and Insular Scandinavian (Faroese, Icelandic). Old Norse had verb raising, something which is preserved in Icelandic, is in its very final stages in Faroese, and has been lost in the standard variants of the Mainland Scandinavian languages. The examples in this section are from or modeled after those in Holmberg and Platzack (1995) and Platzack (1988).

The Icelandic facts are illustrated in (43). As is usual with examples from non-English Germanic languages, we give examples in the form of subordinate clauses in order to avoid complications associated with aspects of main clause word order. As the examples show, Icelandic patterns with French.

Extra Info

The Icelandic characters eth (capital Ð, lowercase ð) and thorn (capital Þ, lowercase þ) represent the voiced and voiceless 'th' sounds in *this*, *eth* (IPA [ð]) and *thin*, *thorn* (IPA [θ]), respectively. The Danish character ø and the Swedish character ö stand for the same mid-front rounded vowel (IPA [ø]) (more or less like the vowel in 'day', except that the lips are rounded as for 'do').

(43) Icelandic:

a. ✓ að Jón **keypti** { ekki, aldrei, raunverulega } bókina
that Jón bought not never actually book.DEF
'that Jón { didn't buy, never bought, actually bought } the book'

b. * að Jón { ekki, aldrei, raunverulega } **keypti** bókina

Examples (44) and (45) illustrate the corresponding facts for modern Danish and Swedish. As is evident, the pattern is the converse of the Icelandic one.

(44) Danish:

a. * at Ulf **købte** { ikke, aldrig, faktisk } bogen
that Ulf bought not never actually book.DEF
Intended meaning: 'that Ulf { didn't buy, never bought, actually bought } the book'

b. ✓ at Ulf { ikke, aldrig, faktisk } **købte** bogen

(45) Swedish:

a. * att Ulf **köpte** { inte, aldrig, faktiskt } boken
that Ulf bought not never actually book.DEF
Intended meaning: 'that Ulf { didn't buy, never bought, actually bought } the book'

b. ✓ att Ulf { inte, aldrig, faktiskt } **köpte** boken

The loss of verb raising in the Mainland Scandinavian languages has been the subject of detailed investigation (Falk 1993; Platzack 1988; Sundquist 2003). For instance, in Swedish, the earliest examples of tense lowering are from the late 1400s. During a transitional period from 1500 to 1700, both verb raising and tense lowering are attested, sometimes even in the same text (as in examples (46b) and (47b)).

(46) a. at Gudz ord **kan** ey vara j honom
that God's word can not be in him
'that God's word cannot be in him'

b. när thet **är** ey stenoghth
when it is not stony
'when it is not stony'

(47) a. om den dristigheten än **skulle** wara onågigtt uptagen
if that boldness yet would be amiss taken
'if that boldness would yet be taken amiss'

b. om annar sywkdom ey **krenker** nokon
if another illness not ails someone
'if someone isn't afflicted with another illness'

Finally, after 1700, verb raising in Swedish dies out completely in the standardized variety. However, it is preserved in at least one regional dialect, that of Älvdalen.

(48) Älvdalen Swedish:

a. ✓ um du **for** int gar ita ia firi brado
if you get not done this before breakfast
'if you don't get this done before breakfast'

b. ✓ fast die **uar** int ieme
if they were not home
'if they weren't home'

c. ✓ ba fo dye at uir **uildum** int fy om
just because that we would not follow him
'just because we wouldn't follow him'

Faroese, spoken on the Faroes, a group of islands roughly midway between Iceland and mainland Europe, is at the very tail end of losing verb raising (Heycock, Sorace, and Hansen 2010; Heycock et al. 2011). Speakers do not ordinarily produce verb raising sentences, but when asked to give grammaticality judgments, many speakers accept both word orders in (49), characterizing the verb raising variant in (49a) as archaic.

(49) Faroese:

a. Verb raising (archaic):
Hann spur, hvi tad **eru** ikki fleiri tilikar samkomur.
he asks why there are not more such gatherings
'He asks why there aren't more such gatherings.'

b. Tense lowering (vernacular):
Hann spur, hvi tad ikki **eru** fleiri tilikar samkomur.

In concluding this section, we draw your attention to the absence of *do*-support in (44b), (45b), and (47b). As the availability of negative inversion in (50a) shows,[9] the translation counterparts of *not* have a different status in Scandinavian than they do in English. They are ordinary intransitive adverbs, and therefore tense lowering is possible in modern Mainland Scandinavian. (Icelandic negative inversion in (50b) is shown for completeness; the verb would be able to move up to I° even if negation were a head on the main spine of the clause.)

(50) a. Swedish:
Inte *vet* jag var hon bor.
not know I where she lives
'I don't know where she lives.'

b. Icelandic:
Ekki *veit* ég hvar hún býr.
not know I where she lives

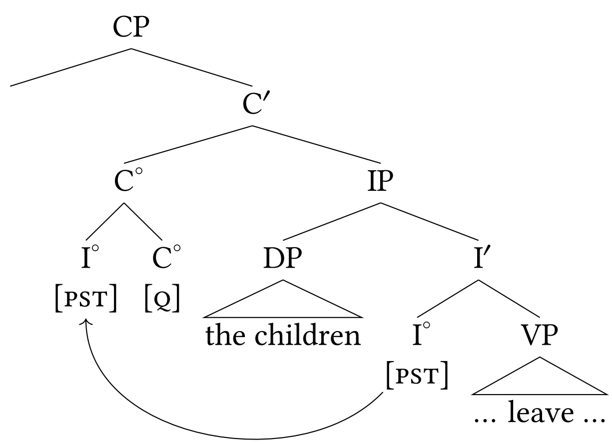
Differences in verb movement within a language 12.4

Syntacticians often speak broadly about whether verbs raise (or not) in a given language, as these properties do tend to typify languages as a whole. But even within languages there can be differences between different kinds of verbal elements. English is an excellent example. As we discussed above, main verbs in English do not raise to I°, which explains word order with the relevant diagnostic adverbs. It also explains other patterns, such as *do*-support in questions.

(51) a. The children left.
b. Did the children leave?

As we discussed in Chapter 8, the subject-auxiliary inversion in English is explained via I-to-C movement. When this happens in a past tense sentence, the past tense is realized with *do*-support in C, leaving the main verb no longer inflected for tense.

(52)



We can assume here that the inflected *do* auxiliary is the morphological way of expressing past tense in English when it is separated from a verb.

But not all verbs in English behave like this. (53) and (54) show that in questions, the auxiliary verbs raise to C°, and *do*-support is unacceptable.

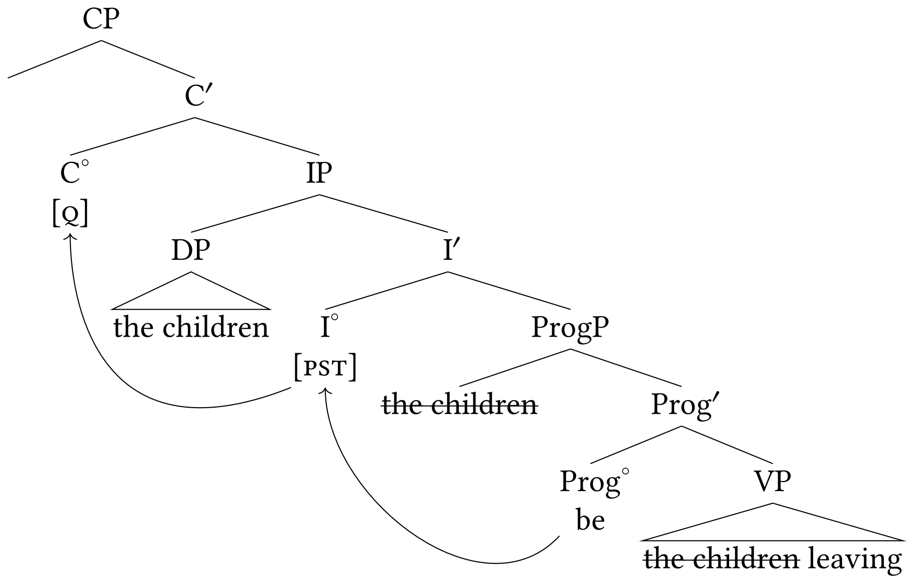
(53) a. The children were leaving.
b. Were the children leaving?
c. *Did the children be leaving?

(54) a. The children had left.
b. Had the children left?
c. *Did the children have left?

This suggests that when I-to-C movement occurs in English questions, the auxiliaries *be* and *have* are in fact in I°, and therefore move to C°. This is of course the opposite pattern of lexical main verbs in English, which don't raise to I$^\circ$ and also don't raise to C$^\circ$. The conclusion is that English auxiliary verbs do raise to I$^\circ$, but main verbs do not.

(55) a. Were the children leaving?

b.



Verb raising cross-linguistically 12.5

As is true of the position of subjects cross-linguistically, the syntactic height of verbs is a major point of cross-linguistic variation across languages. In the discussion above we focused on some major European languages (English, French, Scandinavian varieties). In those instances, surface word order appears to be quite similar between the languages at first glance, but we see important differences emerge when we explore details of the positions of main verbs, auxiliaries, adverbs, and negation. As it turns out, these lessons about verb raising are quickly generalizable to other languages where the positions of major constituents of a sentence diverge significantly from what we've seen above.

12.5.1 Verb raising with VSO word order

As we discussed in Chapter 7, a significant portion (though still a minority) of the world's languages have a VSO basic word order. We repeat the examples here, which come from a wide variety of unrelated languages: Niuean (Austronesian, Niue Island and New Zealand), Irish (Celtic, Ireland), Turkana (Nilotic, Kenya) , Nandi (Nilotic, Kenya) and the Mayan languages Q'anjob'al (Guatemala/Mexico) and Ch'ol (Mexico). In the following examples, the verb is **bolded in red**, the subject *italicized in green*, and the object underlined in black.

(56) a. **To lagomatai** *he ekekafo* a ia. **Niuean**
FUT help *ERG doctor* ABS him
'The doctor will help him.' (Seiter 1980, 29)

b. **D'oscail** *Seán* an doras. **Irish**
open.PST *Sean* the door
'Sean opened the door.' (Guilfoyle 2000, 66)

c. **ɛ-mɪn-a** *ŋesì* ŋa-atuk' **Turkana**
3-love-A *he* cows
'He loves cows.' (G. Dimmendaal 1983, 62)

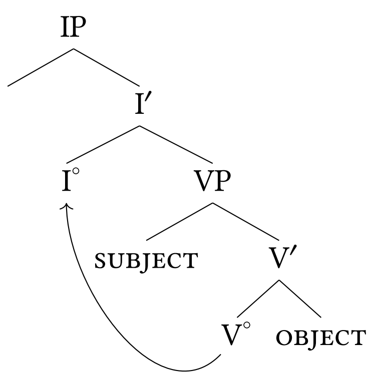
d. **ki:ri:p** *la:kwɛ:t* ɑrtɛ̂:t **Nandi**
watched *child* lamb
'The child watched the lamb.' (Creider and Creider 1983, 3)

e. **Max y-il** *no' tx'i'* naq Lwin. **Q'anjob'al**
PFV A3-see *CLF dog* CLF Pedro
'The dog saw Pedro.'
(Baquiax Barreno et al. 2005, 169, via Clemens and Coon 2018, 240)

f. **Tyi i-kuch-u** *aj-Maria* jiñi si'. **Ch'ol**
PFV A3-carry-SS *CLF-Maria* DET wood
'Maria carried the wood.' (Clemens and Coon 2018, 240)

We suggested in Chapter 7 that V-to-I movement was responsible for VSO word order, paired with the assumption that subjects remained lower in the structure than I° (presumably in Spec,VP).

(57) Schematic derivation of VSO word order:

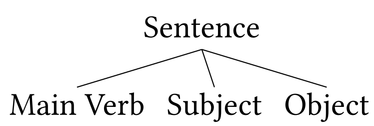


At the time this was introduced in Chapter 7, this head movement of V° moving to I° was a novel proposal to deal with a very distinct word order. But this chapter shows that there is nothing “exotic” about verb raising—it is a thoroughly mundane property of human language syntax that occurs across a broad range of languages. In a language like French its presence is often obscured by movement of subjects to Spec,IP. In languages like those in (56), however, the lack of subject movement makes the position of the verb more salient.

In at least some VSO languages there are still constructions that illustrate the underlying structure of the verb phrase, however. In Irish, for example, the canonical word order is VSO. But some constructions have auxiliary forms that occupy I°, in which case we see the predicted underlying order appear where subjects precede verbs.[10]

(58) Tá [Séamas ag oscailt an dorais]. **Irish auxiliary construction**
is James open.PROG the door-GEN
‘James is opening the door.’ (Doyle 2001, 67)

Again, this is exactly what we expect to occur if VSO word order is generated by movement of verbs to I°, rather than some other fundamental distinction in clause structure. One could attempt to argue, for example, that word order in VSO languages is fundamentally different than other languages: perhaps there is not X′-syntax, or perhaps X′-syntax is the wrong approach to word order in human languages. We could think instead that Irish doesn't have a verb phrase or complex hierarchy of sentence constituents, something like (59):

(59)


But this would not predict the shift in word order between main verbs and subjects that occurs in examples like (58).

Likewise, there are constructions in Irish that are arguably small clauses, that is, instances where a predicate selects for a truncated complement clause structure that arguably lacks I°:

(60) a. Tháinig siad abhaile agus [iad ag amhrán]. **Irish small clause adjunct**
came they home and them sing.PROG
‘They came home singing.’ (Doyle 2001, 67)

b. Chonaic me [Máire ag rince]. **Irish small clause complement**
saw I Mary dance.PROG
‘I saw Mary dancing.’ (Doyle 2001, 67)

c. Chuala sí [Séamas ag oscailt an dorais]. **Irish small clause complement**
heard she James open.PROG the door-GEN
‘She heard James opening the door.’ (Doyle 2001, 68)

In these constructions, like in the auxiliary constructions above, we see subjects preceding main verbs instead of following them. This is directly predicted by a verb raising account: in instances where V° doesn't raise to I°, we see a word order that is *not* VSO (either because I° is occupied by some other element like an auxiliary, or because I° is not present, as in a small clause). Facts like these are much harder to explain on an approach that assumes a lack of underlying structure, as in (59).

12.5.2 Verb raising in an agglutinative SVO language

We see a similar kind of pattern arise in Tiriki, a Bantu language spoken in Western Kenya. Tiriki's canonical word order is SVO, and like other Bantu languages discussed in this volume, the cardinal numbers in glosses represent noun classes (a form of grammatical gender). They show you what is agreeing with what, but aside from that you can ignore them. Subject agreement and tense morphemes are generally thought to arise on I°. This of course raises the important question that we've been investigating in this chapter: does inflection lower onto the verb, or does the verb raise to inflection?

In the examples below, we can see select adverbs (circled) that can appear at the right edge of a transitive sentence, but can also appear intervening between the verb and its object. As with the French patterns above, and with the VSO patterns above, a V-X-O word order raises a fundamental puzzle for X′-theoretic analyses of verb phrases (without additional theoretical mechanisms like movement). If the verb and its object are siblings, and structures are binary branching, how is it that it's possible for XPs (subjects, adverbs) to intervene between the verb and the object?

(61) a. Musa a-tuy-i mu-saala khavere
1Musa 1SM-hit-PST 3-tree twice
'Moses hit the tree twice.'

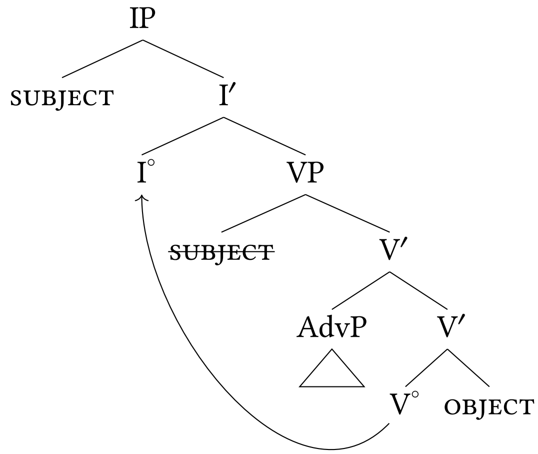
b. Musa a-tuy-i khavere mu-saala
1Musa 1SM-hit-PST twice 3-tree
'Moses hit the tree twice.'

(62) a. Amukotso a-tekh-i ma-kanda mwisaangaala
1Amukotso 1SM-cook-PST 6-beans in.happiness
'Amukotso cooked beans happily.'

b. Amukotso a-tekh-i mwisaangaala ma-kanda
1Amukotso 1SM-cook-PST in.happiness 6-beans
'Amukotso cooked beans happily.'

At this point, of course, this is non-mysterious. Tiriki must have verb raising like French and like Irish. Assuming the adverb is adjoined to VP, a right-adjunction would result the adverb assuming sentence-final position, while a left-adjunction results in the adverb preceding the object. The V-Adv-Obj word order is then derived via V° raising to I°.

(63) Tiriki verb raising (S V Adv O word order)



12.5.3 Verb raising and changes in word order

In some instances, significant differences in word order within a language can be attributed to verb raising. We illustrate here from Vata, a Kru language spoken in Côte d'Ivoire. Simple sentences like those in (64) could readily give the impression that Vata is an SVO language.[11]

(64) Vata (Koopman 1984, 27–28)

a. ń lé bǐ sákǎ́
I eat now rice
'I am eating rice right now.'

b. ń lì sákǎ́
I eat-PFV rice
'I ate rice.'

In other constructions, however, the word order is noticeably different, instead looking more like an object-verb language (more specifically, an SAuxOV language). Each of the examples below use auxiliaries, either for tense/aspect or for negation.

(65) Vata (Koopman 1984, 28)

a. wá lā mɔ̆́ dlǎ́
they AUX.PFV him kill
'They have killed him.'

b. ń kǎ́ná gòlí mlǐ́ pùtū sà.
I AUX.FUT my mounds in grass remove
'I will clear the weed from my mounds.'

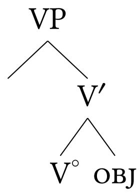
(66) Vata (Koopman 1984, 31–32)

a. ɔ́ ɔ̆́ gbā vātàwì.
s/he NEG speak Vata-tongue
'S/he does not speak Vata.'

b. à ní-là sákā̋lì.
we AUX.NEG-yet rice eat
'We have not yet eaten rice.

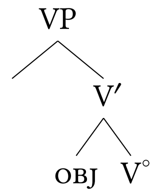
One approach to analyzing this pattern would be that the VP in Vata flip-flops between being head initial (in examples like (64)) and being head-final (like the examples in (65)–(66)).

(67) a.



Hypothetical head-initial Vata VP (not final)

b.

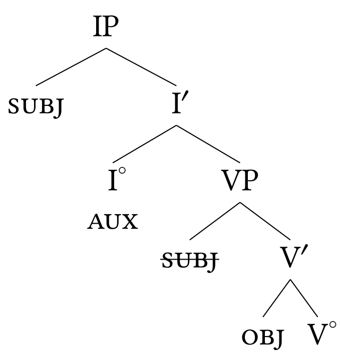


Hypothetical head-final Vata VP (not final)

We would then need to claim that I° without an auxiliary selects for the head-initial VP, and auxiliaries select for the head-final VP. But such a connection is essentially arbitrary: our model would just as easily allow auxiliaries to select for the head-initial VP.

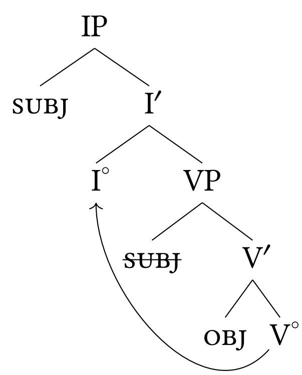
A simpler approach is that Vata has verb-raising except for when an Aux is in I°; that is, we could assume that an auxiliary in I° blocks verb movement. This allows us to further assume that VP in Vata is always head-final. When I° is not occupied, V-to-I movement produces SVO word order. When I° is occupied by an Aux, V stays in place in the head-final VP, resulting in Subj-Aux-O-V word order.

(68) a.



Vata S-Aux-O-V word order (no verb movement, final)

b.



Vata SVO word order (V-to-I movement, final)

Adverbs: a side comment that is also the main point 12.6

The main point of this chapter is to demonstrate that there is variation across languages in the structural height of verbs. In order to show this, we relied on particular kinds of adverbs that were useful in diagnosing the differences between languages (and between structures within a single language) with respect to the position of verbal elements. As we noted above, not just any adverb will do: particular adverbs in each language were useful for diagnosing these verb movement distinctions. Notably, the adverbs that are consistently useful for this task have roughly equivalent translations (ex., *often*). Why? As it turns out, adverbs of shared semantics also share a structure across the world's languages. That is to say, adverbs appear in consistent positions in a sentence based on their meaning.

(69) A summary of the proposed crosslinguistic adverb hierarchy:
discourse-oriented > evaluative > epistemic > subject-oriented (> Neg) > manner
(Ernst 2014, 109)

Discourse-oriented adverbs are speaker-oriented (like epistemic and evaluative adverbs) because they provide a speaker's assessment of the sentence they are uttering, particularly about the act of saying the sentence itself. So in (70) the speaker is saying that the utterance itself is frank, or brief, or honest.

(70) Discourse-oriented adverbs in English (Ernst 2014, 110)
  a. Frankly, she would never be caught dead in an SUV.
  b. Briefly, their solution was to tighten the frame and readjust the cylinder.
  c. Honestly, why would the management try to do such a thing?

Other speaker-oriented adverbs like epistemic and evaluative adverbs also reflect a speaker's assessment of a sentence. But rather than commenting on the act itself of saying the sentence, epistemic adverbs comment on the relative likelihood or evidence for a sentence, and evaluative adverbs provide a speaker's assessment of the information communicated in the sentence.

(71) Epistemic and evaluative adverbs in English (Ernst 2014, 110)
  a. Epistemic:
    i. Modal: *probably*, *certainly*, *possibly*, *maybe*, *definitely*, *necessarily*, *perhaps*
    ii. Evidential: *obviously*, *clearly*, *evidently*, *allegedly*
  b. Evaluative: *(un)fortunately*, *mysteriously*, *tragically*, *appropriately*, *significantly*

Consistently cross-linguistically, discourse adverbs behave structurally higher than evaluative adverbs (72), and evaluative adverbs behave structurally higher than epistemic adverbs (73).

(72) a. **Frankly**, Albert has **unfortunately** bought defective batteries.
b. ***Unfortunately**, Albert has **frankly** bought defective batteries.

(73) a. Albert **unfortunately** has **probably/obviously** bought defective batteries.
b. *Albert **probably/obviously** has **unfortunately** bought defective batteries. (Ernst 2014, 110)

Epistemic and evaluative adverbs are generally thought to adjoin in the IP-domain.

We won't walk through all the categories. Instead, we'll just illustrate structurally low adverbs: the manner adverbs that describe a subset of the activities described by a verb phrase (roughly speaking, the *manner* that the event occurred in). That is to say, (74a) describes not all the events where the orchestra plays the soft sections, but only the subset of those events where the orchestra plays the soft sections loudly.

(74) a. This orchestra plays even the soft sections **loudly**.
b. The committee arranged all of our affairs **appropriately**.
c. She faced her fears **bravely**. (Ernst 2014, 111)

Manner adverbs are generally thought to adjoin in the VP domain, and are commonly referred to as "low adverbials" for this reason.

A key finding is that it is far from only English or European languages that exhibit this hierarchy of adverbials. In fact, the adverb hierarchy is stunningly well-attested across a broad range of languages that are unrelated geographically, genetically, and with respect to their grammatical properties. One example comes from Tiriki, a Bantu language spoken in Kenya. An epistemic adverbial like *haundi* 'perhaps/maybe' appears at the beginning of a sentence (or between the subject and the verb), as shown in (75). But manner adverbs are disallowed in this position, appearing instead postverbally (76). The parenthesized checkmarks and asterisks show other possible and impossible positions of the adverb.

(75) <u>Tiriki epistemic adverb</u>:

Haundi	Injira (✓)	a-la-sinza	(*) i-chirishi (*)
maybe	1Injira	1SM-FUT-slaughter	9-bull

'Maybe Injira will slaughter a bull.'

(76) <u>Tiriki manner adverb</u>:

(*) Salano	(*) a-chend-i	(✓) hango	**mm-bwiivisi**.
1Salano	1SM-walk-PST	home	in-secrecy

'Salano walked home stealthily.'

Even languages which at first seem to challenge this generalization nonetheless end up supporting it. An interesting finding in this respect is how in Malagasy, the order of adverbs we describe above is inverted. This difference is evident in the difference between the manner and frequency adverbs between the Malagasy example and its English translation in (77).

(77) Malagasy manner and frequency adverbs

Manasa	**tsara**	ny	lambany	**foana**	i	Ketaka.
wash	well	DET	clothes	always	DET	Ketaka

'Ketaka **always** washes his clothes **well**.' (Pearson 2000, 331)

But in fact, while the linear order appears reversed in (77), it is *consistently* reversed. In other words, the relative order of the different classes of adverbs is (largely) maintained in Malagasy, it just comes in the opposite order. In fact, in Malagasy the direct linear order is partially replicated, and partially inverted. (79) shows a portion of Cinque's (1999) proposed universal adverb hierarchy, and (79) shows the adverb order that is observed in Malagasy.

(78) Italian adverb hierarchy (partial):

1		2		3		4		5		6
generalemente	»	*già*	»	*più*	»	*sempre*	»	*complemente*	»	*bene*
generally		already		anymore		always		completely		well

(Cinque 1999)

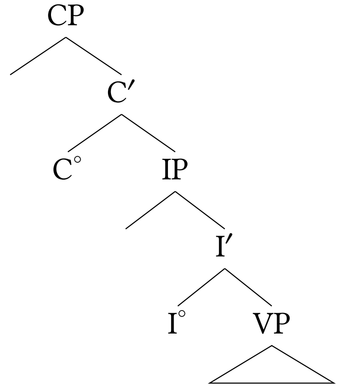
(79) Malagasy adverb hierarchy:

1		2				6		5		4		3
matetika	»	*efa*	»	**VERB**	»	*tsara*	»	*tanteraka*	»	*foana*	»	*intsony*
generally		already				well		completely		always		anymore

(Pearson 2000, 345)

This is clearly a different pattern from Italian/English/Tiriki, but suspiciously so: the postverbal sequence of adverbs is directly inverted from the Italian ones. So while we can't say that a specific linear order of adverbs is universal, there is clearly something universal about the hierarchy in general—even when it is not directly replicated, the hierarchy still seems to be asserting itself.[12] An intro textbook is not the place for a complete discussion of the derivation of the adverb facts, but the general idea is that adverbs attach in positions that match their semantics. So, manner adverbs describe the manner in which the event described by a verb phrase occurs, and we find them attaching at VP levels. Discourse adverbs describe a speaker's stance towards an entire sentence, so we find them attaching at the CP level. So while linear order itself is not universal (which is no surprise given when we've learned thus far about syntax), the core functional hierarchy of a clause *does* appear to be universal, as presented in (80).

(80)



A hasty conclusion from the preceding discussion might be that verb-initial languages automatically invert the adverb order. But this is not the case, as in some instances verb-initial languages appear to retain the (uninverted) adverb hierarchy. One illustration here comes from Kipsigis, where manner adverbs appear at the right edge of the sentence, and epistemic adverbs appear to their left, though still right of the verb. So while Malagasy adverb order inverts the English expectations, Kipsigis has a similar adverb order to English (epistemic » manner).

(81) Kipsigis adverb positions (Bossi and Diercks 2019)

Koo-∅-jiiɑsi	ɪnɛ	lɑɑgook	saanɪɪt	kuʧoogi.
PST-3PL-break	definitely	children	plate	quickly

'The children definitely broke the plate quickly.'

So despite both being verb-initial languages, Kipsigis and Malagasy have underlying differences such that the word order of adverbs is inverted in Malagasy but is not in Kipsigis.

The takeaway lesson here is that there is a consistent hierarchy of adverbs across languages that is consistent with the core functional hierarchy of the clause (roughly, CP > IP > VP). This conclusion can feel like it comes out of nowhere: we often think of the "important" parts of the grammar of a sentence as the predicate and arguments: is the language SVO? SOV? VSO? And this is of course relevant, but when thinking about universal properties of human languages, the order of verbs, objects, and subjects is in fact one of the more variable properties of languages. If we're looking for the most stability, it (perhaps ironically?) comes from optional elements in any given sentence: adverbs and adverbial modifiers. Beyond being interesting on their own account, adverbs are extraordinarily helpful to syntacticians precisely because they are so consistent in their hierarchical structure. If we are attempting to diagnose the position of elements in the syntax, adverbs are some of the most reliable elements that can help us do that. This is why adverbs were so helpful for understanding the variable positions of verbs in different languages, and why we needed to identify specific diagnostic adverbs in order to do that.

12.7 Conclusions

The conclusion of this chapter is that verbs are pronounced at positions of different structural height in different languages. This is true in instances where the effects on word order are transparent (comparing English SVO against Kipsigis VSO, for example). But even in languages like French or Tiriki that share SVO word order with English, there are subtle but persistent differences in the positions of adverbs that are best explained by the proposal that verbs in French and Tiriki (and in many other languages) raise up out of the VP to I°.

Notably, there is no clear theory-internal reason for *why* verbs move. That is to say, we have provided clear and consistent empirical evidence that is well-analyzed by verb movement, but nothing in our system explains from a conceptual perspective why verbs move. In our estimation, the field has not yet arrived at any deeper explanation. For example, we might think that there would be a crosslinguistic link where all verbs that move share some property P, and all verbs that don't move lack property P: this would suggest a deeper explanation for the movement. But to our knowledge nobody has (as of yet) identified such a property. So instead the presence or absence of verb movement at present has a status in our model not dissimilar to head-final vs. head-initial phrases: UG does not place a requirement on either, but both are available, and languages are free to evolve or devolve either pattern.

It is important to note that this conclusion does *not* abandon the existence of VPs: VPs still exist, but verbs sometimes move out of them. Patterns like those in Irish and French demand this conclusion, because there are certainly instances where verbs appear where we think I° would be located, but some constructions in those same languages (ex., small clauses or auxiliary constructions) create conditions where the verb does not move, and we instead see a VP appearing similarly to how we would expect based on how our model has been developed to this point. So while VPs still exist, constituents can move out of them, obscuring their existence.

New Empirical Observations

- adverb positions are consistent crosslinguistically
- specific classes of adverbs can be used as diagnostic adverbs for verb positions
- verbs behave like they are in different positions in different languages
- in some languages, verbs are in different positions in different constructions

Components of the Theoretical Model

- head movement (via head-adjunction) is available for verbs
- verbs move to higher heads
- V-to-I movement is a common form of verb movement
  - consistently in some languages
  - variably in other languages (based on the particular construction)
- V-to-I movement can generate VSO word order, and can explain other kinds of word order alternations

Notes 12.8

1. In what follows, we focus on the past tense since the present tense is not overtly marked in English. The *-s* of the third person singular expresses subject agreement rather than present tense (Kayne 1989).

2. Yiddish and the southern German dialects from which it developed are exceptions in this regard. In these languages, the synthetic simple past has been completely replaced by the analytic present perfect (Middle High German *ich machte* 'I made' > Yiddish *ikh hob gemakht*, lit. 'I have made').

3. It has been proposed that verb raising is correlated with the 'strength' of a language's subject-verb agreement morphology (the number of distinct person-number endings in the verbal paradigm for, say, the present tense). We do not present this proposal here, as it is not clear that the correlation holds up under close scrutiny (J. Bobaljik 2002; Heycock and Sundquist 2017; Heycock and Wallenberg 2013).

4. A comparable shift occurred in English from 'they have to V' to 'they must V'. Such semantic shifts, with concomitant changes in morphological status (see Note 5), are very common across languages.

5. Such reanalysis might be the source of much, if not all, inflectional morphology. In many cases, especially in languages that are not written, the sources of the inflections would be obscured by further linguistic changes, primarily phonological reduction. Consider, for instance, the development of the future tense in Tok Pisin, an English-based contact language that originated in the 1800s and that has become the national language of Papua New Guinea. In current Tok Pisin, particularly among speakers who learn it as a first language, the future marker is the bound morpheme *b-*. We are fortunate to have written records of Tok Pisin from the late 1800s, and so we happen to know that this morpheme is the reflex of the adverb phrase *by and by*, which the earliest speakers of Tok Pisin used to express future tense. The modern form of the marker developed via reduction (*by and by* > *baimbai* > *bai* > *b-*). Without written records, the derivation of the modern form would be speculation at best.

6. Historically, the negative marker in French was *ne*, and *jamais* and *pas* (an intensifier literally meaning 'step') had no negative force of their own. Modern English has comparable intensifiers, as in *I don't want to do it <u>one bit</u>, <u>at all</u>.* In the course of the history of French, *ne*, being phonologically weak, was often elided in speech, and *jamais* and *pas* were reanalyzed as carrying negative force. In modern French, *ne* is characteristic of the formal language, and in some spoken varieties, such as Montreal French, *ne* hardly ever occurs. In the present discussion, we disregard *ne*, treating it as a semantically meaningless particle and glossing it as NE.

7. *Do*-support and the syntax of negation raises some thorny problems, and no completely satisfactory analysis of it exists as yet. So although our analysis is adequate to explain the contrast between (27) and (28), it is not intended to solve many other puzzles that have been discovered in connection with these phenomena.

8. A similarly-functioning definition could be written in terms of closest c-command instead of via domination.

9. For some reason, Danish *ikke* 'not' cannot invert, perhaps because it cannot bear prosodic stress.

10. Glossing for the Irish examples follows the style of McCloskey (1996).

11. We have adjusted some of the transcriptions from Koopman (1984) here, largely for readability. We have transcribed the [−ATR] vowels using the typical current IPA symbols. We have also adjusted the notation of tones: we use a double accent for high tones (a̋), a single accent for mid-high tones (á), a simple overbar for mid tones (ā), and a grave accent for low tones (à). Various glosses are also adjusted to the Leipzig glossing conventions.

12. Pearson (2000) provides an intriguing explanation for how the hierarchy of adverbs gets inverted in Malagasy; the interested reader is referred to that paper.

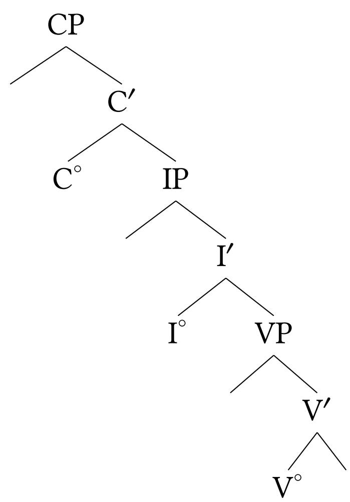
Exercises and problems 12.9

Exercise 12.1: Head-adjunction

We now have a variety of constructions that include head movement, or movement of the head of a phrase to a c-commanding head of a phrase. This exercise will practice drawing out the precise structure of what the result of head movement (i.e., head adjunction) looks like. We usually don't draw this full detail into a tree in order to keep things more visually simple, but this is an exercise in working out the full details.

The questions below ask you to draw modifications of the tree in (1).

(1)



Draw trees for the scenarios below, drawing in full detail the resulting structure of the heads.

A. A construction based on (1) where V° moves to I°.

B. A construction based on (1) where I° moves to C°.

C. A construction based on (1) where V° moves to I° and then I° moves to C°.

Exercise 12.2: Tree-drawing practice, with attention to V positions A

As the chapter discusses, in English main verbs do not vary with respect to their position with adverbs. We analyzed this as a result of English main verbs not undergoing any movement. But the examples in (1) show that the progressive auxiliary *be* and the frequency adverbial *always* do show different word orders with respect to each other.

(1) a. Children should **always** be having adventures.
    b. Children are always having adventures.

A. Propose a position for *always*.

B. Propose an explanation for the difference in word order between *always* and progressive *be* in the examples in (1).

C. Draw a tree for (1b).

Exercise 12.3: The position of English tense

A. Build a structure for (1). (Don't build structures for the material in parentheses.)

(1) They didn't only write the letter (but they sent it).

B. Now build a structure for (2), making sure that it is consistent with the locality constraint on head movement from the chapter.

(2) They not only wrote the letter (but they sent it).

C. There turn out to be two structures for (2). They are topologically distinct, but there is no semantic difference between them. What's the difference between the structure that you came up in (B) and the alternative?

Exercise 12.4: Japanese auxiliary constructions

Exercise written by Xiaoxing Yu

Examine the Japanese data below and answer the following questions.

A. Where is tense marked in each of these sentences?

B. Is there an analysis of these data that reasonably incorporates head-to-head movement? If so, what is it?

C. Draw trees for (3) and (4). You may triangle over DPs and assume case- and topic-marking particles are DP-internal.

Background Info

TE is a suffix that connects a verb to following auxiliaries in Japanese; you do not have to explain its presence. BEN abbreviates 'benefactive', which means the action specified by the main verb is done for someone's benefit. CMPL abbreviates 'completive', which means the event specified by the main verb is irreversible or done to completion (and in Japanese, often also means it is regrettable). TRL abbreviates 'trial', which means the action specified by the main verb is done just to try it out.

(1) お兄ちゃんが 弁当を 作った。
Onii-chan-ga bentō-o tsukut-ta.
big.brother-DIM-NOM boxed.lunch-ACC make-PST
'My big brother packed lunch.'

(2) お兄ちゃんが 弁当を 作って くれた。
Onii-chan-ga bentō-o tsukut-te kure-ta.
big.brother-DIM-NOM boxed.lunch-ACC make-TE AUX.BEN-PST
'My big brother packed lunch for me.'

(3) みかんが 腐った。
Mikan-ga kusat-ta.
orange-NOM rot-PST
'The orange has gone bad.'

(4) みかんが 腐って しまった。
Mikan-ga kusat-te shimat-ta.
orange-NOM rot-TE AUX.CMPL-PST
'(Oh no,) the orange has completely gone bad.'

(5) 先週 納豆を はじめて 食べた。
Senshū nattō-o hajimete tabe-ta.
last.week fermented.soybeans-ACC for.the.first.time eat-PST
'I ate fermented soybeans for the first time last week.'

(6) 先週 納豆を はじめて 食べて みた。
Senshū nattō-o hajimete tabe-te mi-ta.
last.week fermented.soybeans-ACC for.the.first.time eat-TE AUX.TRL-PST
'I tried eating fermented soybeans for the first time last week.'

Exercise 12.5: English *do*-support in questions

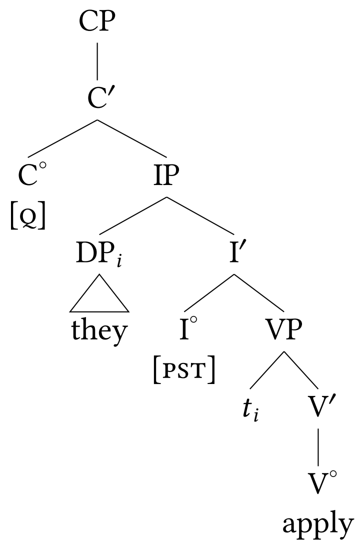
A. The chapter discusses *do*-support in negative sentences. In Modern English, *do*-support is also obligatory in the formation of direct questions from sentences with finite main verbs, as shown in (1) and (2).

(1) They will apply. → ✓ Will they apply?

(2) a. They applied. → * Applied they?
b. They applied. → ✓ Did they apply?

Given the basic structure in (3), can you explain the contrast in (2)?

(3)



B. Under what circumstances would the question in (2a) be grammatical?

C. Build the structure for (4).

(4) Won't they apply?

Exercise 12.6: Shakespearean English: negation

Build structures for the following expressions from Shakespearean English. What is different between this variety of English (now extinct) and the contemporary Mainstream English discussed in this chapter?

(1) a. They not edify mens' eares.
b. We not fear to tempt your laughter.

Problem 12.1: English *too*

In both (1) and (2), the (a) examples are unambiguous, with an "also" reading associated with the agent argument. The (b) examples are also unambiguous, with a "contrary to expectation" reading. The (c) examples are structurally ambiguous between an "also" reading associated with the activity and an "also" reading associated with the theme argument.

(1) a. They too will read the novel.
(They—in addition to an unspecified set of readers—will read the novel.)
b. They will too read the novel.
(Contrary to expectation, they will read the novel.)
c. They will read the novel too.
(They will—in addition to an unspecified set of activities—read the novel.)
(They will read—in addition to an unspecified set of readable things—the novel.)

(2) a. They too read the novel.
b. They did too read the novel.
c. They read the novel too.

A. Provide the two elementary trees for *too* required to account for the above facts. For clarity, you can distinguish between *too*$_{\text{also}}$ and *too*$_{\text{contrast}}$.

B. How are the elementary trees from (A) integrated into larger structures?

C. Given your answers to (A) and (B), why doesn't (2a) have a "contrary to expectation" reading? (After all, *too* immediately precedes *read* in (2a) just as it does in (1b) and (2b).)

D. Given your answers to (A) and (B), why don't (2a) or (2b) have an "also" reading associated with the activity?

Problem 12.2: English emphasis movement and tense

Explain the facts in (1) and (2). Ignore the first conjunct and the conjunction *and*.

They said they would finish the project, and …

(1) ✓ finish the project they { will, did }.

(2) a. * { will, did } finish the project they.
b. * finished the project they.

Problem 12.3: Lopit word order

Data here are drawn from Moodie and Billington (2021).

Lopit is a Nilotic language spoken in South Sudan. The data below provide evidence for the basic word order in Lopit including verbs, auxiliaries, and negation (negative auxiliaries).

A. Explain the canonical word order of Lopit based on the examples in (1) and draw a tree for (1f) that represents an initial hypothesis about the underlying structure of Lopit declarative sentences.

B. Explain the differences in word order with negative auxiliaries and other auxiliary verbal elements, and draw a tree for (2b).

C. Explain what outstanding questions these data raise (if any) for your analysis.

Notes:

- You can take note of the inflection on verbal forms and it can factor into your outstanding questions, but you don't have to shape your whole analysis around it.
- Some glosses are simplified from the original source for the purposes of this assignment.
- Glossing notes: IPFV = imperfective aspect (an ongoing event); PFV = perfective aspect (a completed event); NOM = nominative case; ABS = absolutive case; PER = persistive affix; AGR = agreement affix; SG = singular; PL = plural; Q = question marker; SEQ = sequential marker; SBO = subordinate marker; COMP = complementizer; NEG = negation

Lopit basic word order

(1) a. Ɛ-ìl:á xɔ́t:ɔ́ɲí ìŋɛ̂.
AGR-wash.IPFV mother.NOM baby.ABS
'The mother washes the baby.' (256)

b. Ɛ-ŋîbá xàbʊ̀ ŋɔ̀lɛ́ʔ.
AGR-arrive.PFV chief.NOM yesterday
'The chief arrived yesterday.'

c. Ó-dótò móítéʔ.
AGR-leave.IPFV morning
'He leaves tomorrow.'

d. Á-lâ-dáxá náŋ ŋìrìjà.
AGR-PER-eat 1SG.NOM food.ABS
'I'm still eating food.'

e. Ɔ-xɔ́ɲ mùnù xìŋɔ̀xʊ̂.
AGR-bite snake.NOM dog.ABS
'The snake bit the dog.'

f. É-xì-rjè íɲɛ́ mùnù.
AGR-PFV-tread.on 3SG.NOM snake.ABS
'He trod on a snake.'

g. É-j:ò xítò.
AGR-cry.IPFV child.NOM
'The child is crying.'

Lopit word order with negation/auxiliaries

(2) a. Á-wú náŋ à=tòrít.
AGR-go 1SG.NOM to=Torit
'I'm going to Torit.'

b. Ìɲá náŋ l-á-wú à=tòrít.
NEG 1SG.NOM SBO-AGR-go to=Torit
'I'm not going to Torit.'

(3) a. A-íwús-ó náŋ ʃaì tò=kùbaìjà ìn:àŋ.
AGR-drink-IPFV 1SG.NOM tea.ABS with=cup.ABS this
'I drink tea with this cup.'

b. Íɲà náŋ l-a-íwús-ò ʃaî.
not.be 1SG.NOM SBO-AGR-drink-IPFV tea.ABS
'I'm not drinking tea.'

(4) a. Á-ij:én náŋ xìɟɔ̀ ò-lòt-ú íɲɛ́.
AGR-know 1SG.NOM COMP AGR-go 3SG.NOM
'I know that he is coming.'

b. Á-ij:én náŋ xìɟɔ̀ íɲà íɲɛ́ ò-lòt-ú.
AGR-know 1SG.NOM COMP not.be 3SG.NOM SBO-AGR-go
'I know that he is not coming.'

(5) a. … x-ɔ̀-xɔ́ɲ mùnù ɪ̀ɲɛ̀ dɛ̂=xɛ̀ɟʊ̀k.
SEQ-AGR-bite snake.NOM 3SG.ABS on=leg.ABS
'… and the snake bit him on the leg.'

b. … xɔ̀ɟɔ́ mùnù x-ɔ̀-xɔ́ɲ ɪ̀ɲɛ̀ dɛ̂=xɛ̀ɟʊ̀k.
and.then snake.NOM SEQ-AGR-bite 3SG.ABS on=leg.ABS
'…and then the snake bit him on the leg.'

Problem 12.4: Lubukusu phrase structure and adverbs

Data and judgments from Aggrey Wasike

The data set below shows the possible (and impossible) syntactic positions of various adverbs in Lubukusu. As noted in this chapter, we evaluated positions of constituents using particular 'diagnostic adverbs' because adverbs tend to adjoin to specific positions in the clause based on their semantics. This problem set asks you to analyze Lubukusu phrase structure using these adverbs.

1. There are two categories of adverbs that appear in the data below.
   - What are the syntactic properties of these two categories?
   - Each category of adverb shares a unifying semantic factor. Can you identify it?
2. What position do the adverbs adjoin to?
3. What are the syntactic positions of subjects and verbs in Lubukusu?
4. Draw trees for (4b) and (5a) (put triangles over all DPs, AdvPs, and PPs)

(1) *sisangafu* 'happily'

a. Wafula a-tekh-ile ka-ma-kanda sisangafu
1Wafula 1SM-cook-PFV 6-6-beans happily
'Wafula happily cooked beans.'

b. Wafula atekhile sisangafu kamakanda
1Wafula 1SM-cook-PFV happily 6-6-beans
'Wafula happily cooked beans.'

c. * Wafula sisangafu atekhile kamakanda
1Wafula happily 1SM-cook-PFV 6-6-beans
'Wafula happily cooked beans.'

d. * Sisangafu Wafula atekhile kamakanda
happily 1Wafula 1SM-cook-PFV 6-6-beans
'Wafula happily cooked beans.'

(2) *ekhabi endayi* 'fortunately', 'by good luck'

a. E-khabi e-ndayi Wafula a-ba a-tekh-ile ka-ma-kanda
9-luck 9-good 1Wafula 1SM.PST-be 1SM-cook-PFV 6-6-beans
'Fortunately/by good luck Wafula had cooked beans.'

b. ?? Wafula ekhabi endayi aba atekhile kamakanda
1Wafula 9-luck 9-good 1SM.PST-be 1SM-cook-PFV 6-6-beans
'Fortunately/by good luck Wafula had cooked beans.'

c. * Wafula aba atekhile ekhabi endayi kamakanda
1Wafula 1SM.PST-be 1SM-cook-PFV 9-luck 9-good 6-6-beans
'Fortunately/by good luck Wafula had cooked beans.'

d. * Wafula aba atekhile kamakanda ekhabi endayi
1Wafula 1SM.PST-be 1SM-cook-PFV 6-6-beans 9-luck 9-good
'Fortunately/by good luck Wafula had cooked beans.'

(3) *wakana* 'perhaps', 'maybe'

a. Wakana Wafula a-lakat-a e-khafu
perhaps 1Wafula 1SM-will.slaughter-FV 9-cow
'Perhaps/maybe Wafula will slaughter a cow.'

b. ? Wafula wakana a-lakat-a e-khafu
1Wafula perhaps 1SM-will.slaughter-FV 9-cow
'Perhaps/maybe Wafula will slaughter a cow.'

c. ?? Wafula a-lakat-a wakana e-khafu
1Wafula 1SM-will.slaughter-FV perhaps 9-cow
'Perhaps/maybe Wafula will slaughter a cow.'

d. ?? Wafula a-lakat-a e-khafu wakana
1Wafula 1SM-will.slaughter-FV 9-cow perhaps
'Perhaps/maybe Wafula will slaughter a cow.'

(4) *kalaa* 'slowly'

a. Ba-a-sakhulu ba-nywe-changa ka-ma-lwa kalaa
2-2-elder 2SM-drink-HAB 6-6-beer slowly
'Elders usually drink beer slowly.'

b. Ba-a-sakhulu ba-nywe-changa kalaa ka-ma-lwa
2-2-elder 2SM-drink-HAB slowly 6-6-beer
'Elders usually drink beer slowly.'

c. * Ba-a-sakhulu kalaa ba-nywe-changa ka-ma-lwa
2-2-elder slowly 2SM-drink-HAB 6-6-beer
'Elders usually drink beer slowly.'

d. * Kalaa ba-a-sakhulu ba-nywe-changa ka-ma-lwa
slowly 2-2-elder 2SM-drink-HAB 6-6-beer
'Elders usually drink beer slowly.'

(5) *buung'ali* 'definitely'

a. Buung'ali Wekesa a-a-ly-a chii-ng'eeni.
definitely 1Wekesa 1SM-PST-eat-FV 10-fish
'Wekesa definitely ate the fish.'

b. ? Wekesa buung'ali a-a-ly-a chii-ng'eeni.
1Wekesa definitely 1SM-PST-eat-FV 10-fish
'Wekesa definitely ate the fish.'

c. ? Wekesa a-a-ly-a buung'ali chii-ng'eeni.
1Wekesa 1SM-PST-eat-FV definitely 10-fish
'Wekesa definitely ate the fish.'

d. ? Wekesa a-a-ly-a chii-ng'eeni buung'ali.
1Wekesa 1SM-PST-eat-FV 10-fish definitely
'Wekesa definitely ate the fish.'

(6) *sifwi* 'stealthily'

a. Wafula ol-ile engo sifwi
1Wafula 1SM.arrive-PFV 16.home stealthily
'Wafula arrived home stealthily.'

b. Wafula ol-ile sifwi engo
1Wafula 1SM.arrive-PFV stealthily 16.home
'Wafula arrived home stealthily.'

c. * Wafula sifwi ol-ile engo
1Wafula stealthily 1SM.arrive-PFV 16.home
'Wafula arrived home stealthily.'

d. * Sifwi Wafula ol-ile engo
stealthily 1Wafula 1SM.arrive-PFV 16.home
'Wafula arrived home stealthily.'

Problem 12.5: Guébie word order

Data provided by Hannah Sande based on work with Sylvain Bodji and Agodio Badiba Olivier.

Guébie is a Kru language spoken in Côte d'Ivoire. Answer the following questions based on the Guébie data provided below. Consider the questions in sequence, as they provide guidance to ease the process of sorting through the data set.

1. What is the basic word order of Guébie with respect to subjects, verbs, objects, and auxiliaries? Are there alternative word orders? If so, what are they?
2. What kinds of morphosyntactic contexts affect the word order? What morphosyntactic contexts do each kind of word order occur in?
3. Start with what you understand to the be the more straightforward cases. What analysis of positions of verbs, subjects, and objects can explain these facts?
   (a) Are there movements?
   (b) When do the movements occur?
   (c) What moves, and where does it move from/to?
4. For your analysis, give a short (1–3 sentences) statement of what the puzzle is, give a short statement (2–4 sentences) of what analysis explains the puzzle, and draw trees of each of

the two sentences in example (6). (If being done as a class exercise, draw the trees, but statements may only be thought out, not written out.)

Background Info

Abbreviations used:

SG = singular, PL = plural, IRR = irrealis, IPFV = imperfective, PFV = perfective, NEG = negative, PROG = progressive, GEN = genitive, MDL = middle

Tone marking:

Tone is marked with a four-number system where the tonal marking follows the word. Here 4 represents the highest tone and 1 the lowest. Contour tones are marked with two tones within a syllable boundary, and syllable boundaries are marked with periods.

Extra Info

The information below is not needed for the exercise. It is provided for the curious reader.

Ne is glossed as NEG.PFV below, but new data collected after these examples were compiled suggests it is solely a negative morpheme (NEG) and does not expone tense or aspect. We have elected to keep the original gloss here, but the interested reader is directed to Sande (2022, 27).

(1) a. ɔ^{3} li^{3} dʒa^{31}
3SG eat.IPFV coconuts
'They.SG eat coconuts'

b. ɔ^{3} li^{3} dʒa^{31}
3SG eat.PFV coconuts
'They.SG ate coconuts (recently)'

(2) a. ɔ^{3} wa^{1} touri$^{1.1.3}$
3SG like.IPFV Touri
'They.SG like Touri'

b. ɔ^{13} touri$^{1.1.3}$ wa^{2}
3SG.NEG Touri like
'They.SG don't like Touri.'

(3) a. ɔ^{3} ji^{3} dʒa^{31} li^{3}
3SG will.IPFV coconuts eat
'They.SG will eat coconuts'

b. e^{4} ji^{3} dʒa^{31} lilije$^{2.2.1}$ kotʃi$^{42.3}$
I will coconuts eat.PROG start
'I will start eating coconuts (regularly)'

(4) a. * dʒa^{31} li^{3} ju^{4}
coconuts eat.PFV boy
Intended: 'A/The boy ate coconuts'

b. * touri$^{1.1.3}$ wa^{2} ɔ^{13}
Touri like 3SG.NEG
Intended: 'They.SG don't like Touri.'

(5) a. kɔgʊlɪɲɔ$^{4.2.2.2}$ pi^{3} saka$^{3.3}$ (k)uɓə^{31}
farmer cook.PFV rice yesterday
'Yesterday a/the farmer cooked rice'

b. * (k)uɓə^{31} pi^{3} kɔgʊlɪɲɔ$^{4.2.2.2}$ saka$^{3.3}$
yesterday cook.PFV farmer rice
'Yesterday a/the farmer cooked rice'

(6) a. ɟaci$^{23.1}$ ji^{3} su-wa$^{2.2}$ gbala$^{2.4}$
Djatchi will tree-DEF climb
'Djatchi will climb a tree.'

b. ɟaci$^{23.1}$ gbala$^{2.4}$ su-wa$^{2.2}$
Djatchi climb.PFV tree-DEF
'Djatchi climbed the tree.'

(7) a. ɟaci$^{23.1}$ la^{3} touri$^{1.1.3}$ jokuni$^{2.3.3}$
Djatchi NEG.PFV Touri visit
'Djatchi has not visited Touri.'

b. ɔ^{3} ne^{4} a^{3} kubə$^{4.21}$ jokuni$^{2.3.3}$
3SG NEG.PFV 1PL yesterday visit
'They.SG did not come visit us yesterday.'

c. ɔ^{3} pja^{32} dibo$^{2.1}$ ma^{1} ɔ^{3} ne^{4} u^{2} li^{3}
3SG buy.PFV plantain but 3SG NEG.PFV 3SG.OBJ.CL eat
'They.SG bought a plantain but didn't eat it.'

(8) a. * ɟaci$^{23.1}$ ji^{3} la^{3} touri$^{1.1.3}$ jokuni$^{2.3.3}$
Djatchi will NEG.PFV Touri visit
Intended: 'Djatchi will not have visited Touri.'

b. * ɟaci$^{23.1}$ la^{3} ji^{3} touri$^{1.1.3}$ jokuni$^{2.3.3}$
Djatchi NEG.PFV will Touri visit
Intended: 'Djatchi will not have visited Touri.'

Problem 12.6: English auxiliary structure

This is an exercise to arrive at a more complete understanding of the functional structure of the clause in Mainstream U.S. English. Judgments are offered here based on the textbook authors' grammars, but if your English grammar is different, you may answer the questions based on your own grammar (just be sure to specify what judgments are different in your response).

A. What is the general structure of modals and auxiliaries in English? That is, what order do they occur in? Where in a tree might you propose that these auxiliaries live? For now, set aside the participle morphology on the main verb.

(1) a. Preston should run every day this week.
b. Preston should be running every day this week.
c. * Preston should is running every day.
d. Preston should have been running every day this week.
e. * Preston should been have running every day.
f. Preston has run every day this week.
g. Preston should have run every day this week.
h. * Preston should has run every day this week.

B. How can the distribution of tense morphology in the preceding examples and the following examples be explained (in terms of syntactic structures)?

(2) a. Elena runs every day.
b. Elena is running every day.
c. Elena has run every day.
d. Elena has been running every day.
e. * Elena is have running every day.
f. * Elena been has running every day.

C. Given your analysis of the preceding data, how do you explain the patterns in (3a)–(3f)? That is, why are multiple auxiliary-type-elements acceptable sometimes but not others?

(3) a. Ketanji could be running every day.
b. Ketanji could have run every day.
c. * Ketanji could had run every day.
d. * Ketanji has is running every day.
e. * Ketanji has was running every day.
f. * Ketanji can might run every day.

D. In the following data are a collection of yes/no questions. What basic movement operation occurs in yes/no questions? Can the following data be explained given the analysis of auxiliaries that you established above?

(4) a. Should Sonia run on her injured leg?
b. Should Sonia be running on her injured leg?
c. * Should be Sonia running running on her injured leg?
d. Sonia is running right now.
e. Is Sonia running right now?
f. Sonia runs every day.
g. * Runs Sonia every day?
h. Does Sonia run every morning?
i. Sonia ran every day last week.
j. Did Sonia run this morning?
k. * Ran Sonia this morning?
l. * Did Sonia ran this morning?
m. Sonia has run every day this week so far.

n. Sonia has been running all week.
o. Has Sonia run today?
p. Has Sonia been running?
q. * Has been Sonia running?
r. * Been Sonia has running?
s. * Did Sonia should be running?
t. * Did Sonia should been have running?
u. Should Sonia have been running?

Problem 12.7: English main verbs vs. auxiliary verbs

This exercise extends Exercise 12.6. As in Exercise 12.6, the judgments offered here are based on the textbook authors' grammars, but if your English grammar is different, you may answer the questions based on your own grammar (just be sure to specify what judgments are different in your response).

A. Consider the following uses of *have* and *be*. What can you conclude about these structures, based on the conclusions from Exercise 12.6?

- How is the possessive *have* the same/different than the *have* discussed in Exercise 12.6?
- What about the copula *be*? How is it the same or different than the *be* discussed in Exercise 12.6?
- What kind of structure do you want to give to copular *be*?

(1) a. Jackson is a teacher.
b. Is Jackson a teacher?
c. * Does Jackson be a teacher? *Mainstream U.S. English, to mean the same as (1b)*
d. Jackson has a pencil.
e. * Has Jackson a pencil? *in Mainstream U.S. English*
f. Does Jackson have a pencil?
g. The fire hydrant is red.
h. Is the fire hydrant red?
i. * Does the fire hydrant be red?
j. The store has a purple door.
k. * Has the store a purple door? *in Mainstream U.S. English*
l. Does the store have a purple door?

Problem 12.8: VP pro-forms in English with *do*-support

A. Explain the grammaticality contrast in (2), assuming the judgments as given. If necessary, invent a new syntactic category for *so* to belong to.

(1)		A (challenging B):	You're lying; you didn't go to the movies.
(2)	a.	B (responding to the challenge):	I did *so* go to the movies.
	b.	B (responding to the challenge):	* I *so* went to the movies.

B. Some speakers accept (2b) as a response to (1). How does the grammar of such speakers differ from the grammar of speakers with the contrast in (2)?

Problem 12.9: English ellipsis

English, like many languages, has a process of ellipsis that eliminates a discourse-familiar constituent from a sentence. We can assume that ellipsis is a PF process that leaves a discourse-familiar syntactic constituent unpronounced. This means that a constituent exists in the syntax, but is not pronounced. Each example is somewhat lengthy to create a viable ellipsis context, but we have bracketed the sentence you are to analyse in each example.

Notably, ellipsis is not an instance of not mentioning something, but instead it is an intentional silence that is specifically meaningful. In (1), the question can be answered in (1b), an answer that has a specific meaning that native speakers know: it means "I will bring a bag lunch today." To represent this, we can also write out the unpronounced constituent, represented with strikeout as in (1c).

(1)
 a. Q: Will you bring a bag lunch today?
 b. A: [I will.]
 c. A: [I will ~~bring a bag lunch today~~.]

In this exercise we use the strikeout notation to make the unpronounced constituent clear.

A. First, based on (2) arrive at an analysis of what constituent is being elided in these data.

B. Draw a tree for (2c-ii). State (or annotate on your tree) which node is elided.

C. Second, consider the additional evidence in (3). These sentences are in past tense, and show a slightly different pattern. Explain the difference with (2) and why it occurs.

D. Draw a tree for (3b-ii). State (or annotate on your tree) which node is elided.

(2)
 a. The children will read all afternoon, and [I will ~~read all afternoon~~, too].
 b. The guests might arrive this afternoon; I know that [Preston definitely will ~~arrive this afternoon~~].
 c. i. Q: Should Khydence go home early today?
  ii. A: [She should ~~go home early today~~].

(3)
 a. I ate two breakfasts this morning, and [my dog did ~~eat two breakfasts this morning~~, too].

b. i. Person A: You ran an entire marathon?!?
   ii. Person B: Yup! [I did ~~run an entire marathon~~].

c. i. Q: Did they prep the kitchen to cook dinner?
   ii. A: [They did ~~prep the kitchen to cook dinner~~].

13

Decomposing VP and CP

Chapter outline

The biggest storyline in this chapter is that we (finally) offer an explanation for ditransitive verbs. The solution, crucially, includes complicating a structure (the verb phrase) which has to this point been relatively uncomplicated for us. We will propose a decompositional approach, where the structure that we used to call VP is in fact composed of additional, more specific projections than just VP alone. We show how this decompositional approach quickly extends to a variety of other properties of the verbal domain across languages. We conclude by showing that CP, in addition to VP, is amendable to a similar kind of decomposition.

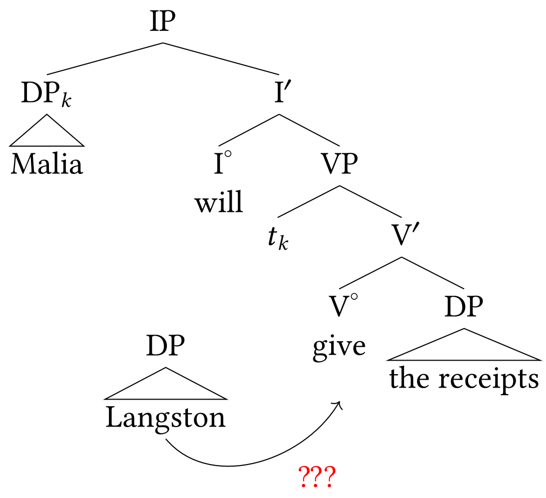
Introduction 13.1

The main lesson of this chapter is that some domains of the clause that we have been treating as simple are not as simple as they have seemed up until now. The downside is that we have to adjust our understanding of how human language syntax works (it counts as a downside only because changing our minds about things is hard); the upside is that we can explain a much larger range of empirical properties of human languages. Some of these empirical properties are problems for our model that you may have noticed by now, and some will most likely be new to you. For example, in Chapter 5, § 3.2.1, we mentioned the **binary-branching hypothesis**—the idea that syntactic nodes have at most two children. At first glance, this hypothesis is threatened by the (probably universal) existence of **double object sentences** in natural language, illustrated for English in (1).

(1) Malia will give Langston the receipts.

In such sentences, the verb appears to be associated with three semantic arguments (AGENT, RECIPIENT, THEME), and it looks like the RECIPIENT (*Langston*) and the THEME (*the receipts*) must both be represented as complements of the verb. This is the view that underlies traditional grammatical terminology, where the verb is said to be ditransitive and to take two objects, the indirect object (the RECIPIENT) and the direct object (the THEME). If objects of verbs are complements of verbs (an assumption that has worked nicely for us), ditransitive predicates as in (1) would suggest that verbs can have two complements. But the X′ schema only allows a single complement of a verb.

(2)



Do ditransitive predicates break the binary-branching X′ schema? We will show here that they do not. The answer that we will arrive at is that the X′ schema is still viable, but that there is more structure inside the VP domain than we have been assuming.

We will start with an initial look at English that suggests we may need more structures inside VP than we have originally assumed (§ 13.2). But we don't fully explain English at first, because there are other languages that demonstrate the relevant properties of ditransitives more transparently: we discuss Swahili as an example in § 13.2.2, where a morpheme called an *applicative* is used to add objects to a verb. We show how this can be extended to English double object constructions, and even to lexical ditransitives like *give*, which lack overt applicative morphemes but nonetheless show parallel behavior. We then discuss the implications for UTAH and the consistent positions of arguments of particular thematic roles.

This is the decompositional approach to the verb phrase: What we originally have treated as a simplex unit (VP) is actually composed of multiple syntactic projections, so VP is decomposed into a complex structure. Section 13.3 discusses additional evidence for this decompositional approach to the verb phrase domain, and § 13.4 discusses how DPs are case-licensed in this context, as the new structures affect how we originally assigned case to objects.

Section 13.6 steps outside the verb phrase domain, instead looking at CP: the shared insight is that, just like VP, a domain that we previously treated as a simple, single projection may in fact be composed of more complex syntactic structure.

Decomposing VP 13.2

13.2.1 The causative alternation

English has a class of verbs expressing movement or other changes of state that are known as **inchoative** verbs. These verbs are intransitive, and their subject is a THEME argument. (In this respect, inchoative verbs resemble the passive of ordinary verbs, but recall from Chapter 10 that unlike passives, they are incompatible with an agentive *by* phrase.) Some examples are given in (3) and (4); the theme argument is in **boldface**.

(3) a. **The apple** dropped.
    b. **The ball** rolled (down the hill).
    c. **The boat** sank.

(4) a. **The water** { boiled, froze }.
    b. **The butter** melted.
    c. **The door** opened.

Many (though not all) inchoative verbs also have a transitive use, where the subject is an AGENT or CAUSE initiating the change of state, and the theme argument appears in postverbal position, as illustrated in (5) and (6).

(5) a. The children dropped **the apple**.
    b. The children rolled **the ball** (down the hill).

c. The { explosion, sailors } sank **the boat**.

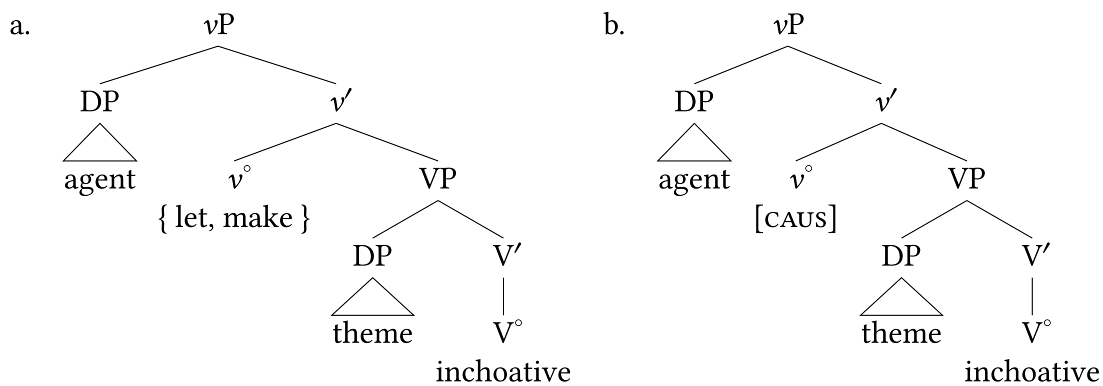
(6) a. The chemist { boiled, froze } **the water**.
b. The chef melted **the butter**.
c. The { janitor, wind, key } opened **the door**.

The alternation between the intransitive and transitive uses of these verbs is known as the **inchoative-causative alternation** (or causative alternation for short).

The causative meaning of the transitive variants suggests giving them the same structure as ordinary causative sentences like (7b).

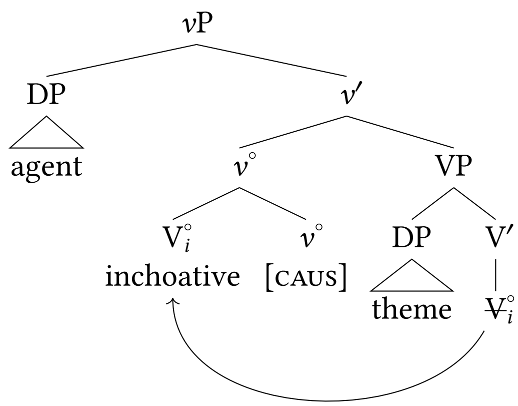
(7) a. The apple dropped.
b. The children { let, made } the apple drop.

The structure for (7b) is given in (8a); the verbs expressing causation (*let* or *make*) take a small clause complement headed by the intransitive inchoative verb. In (8b), the analog to *let* or *make* is an abstract (= silent) verbal head [CAUS]. To distinguish this silent verbal head from typical lexical verbs, we will adopt the standard convention and call it "little *v*" (v°, which projects to a *v*P). You may also notice that the theme object is in a slightly different position than we've previously assumed: we'll see the reasons for that below.

(8)


The structural parallel in (8) captures the semantic parallel between (7a) and (7b), but the word order is not yet right: in (7a), the verb precedes the theme. This is a garden-variety mismatch between where an element is interpreted and where it is pronounced, and we solve it in the usual way: with movement. Specifically, the inchoative verb head-moves and adjoins to the causative verbal head, as shown in (9).

(9)



The complex verbal head is spelled out as a homonym of the simple inchoative head. This suggests that our conclusions about English verbs from Chapter 12 were too hasty; it is not that English verbs do not move, but instead that English verbs don't move past *v*°.

This discussion leaves some questions open: specifically, what happens with UTAH (assignment of thematic roles to uniform positions) when the THEME argument is here in the specifier of VP, when previously we placed it in the complement of V°? We will cover this below in § 13.2.6. The main lesson we want to take from this initial exploration of decomposing VP (breaking down VP into more specialized projections) is that our agentive subjects can reasonably be analyzed as entering the structure higher than VP. Here, we assume *v*P is the projection where agentive subjects join the verbal domain.

13.2.2 Applicatives as an introduction to ditransitive structure

A vast number of languages have overt derivational morphology on verbs that shows the difference between predicates with one object and predicates with multiple objects.[1] The conclusions that languages like this force us into ends up being informative to us in our analysis of ditransitives cross-linguistically. Crucially here, the conclusion from above (that agentive subjects enter the structure in *v*P) opens the door to a more expansive understanding of how other verbal arguments enter the structure.

Applicatives are morphemes that add an object to a sentence. The most familiar sort of applicatives add what we often think of as an **indirect object**, a RECIPIENT or a BENEFACTIVE argument.[2] Swahili is a Bantu language spoken across East Africa as a *lingua franca*. In the Swahili examples below, the applicative morpheme *-e/-i* in the (b) examples adds a RECIPIENT/BENEFACTIVE argument to the sentence (the different vowel quality in the applicative morphemes is a result of a vowel harmony process unrelated to our concerns here).

(10) a. Wa-zazi wa-li-som-a ki-tabu.
2-parents 2SM-PST-read-FV 7-book
'The parents read a book.'

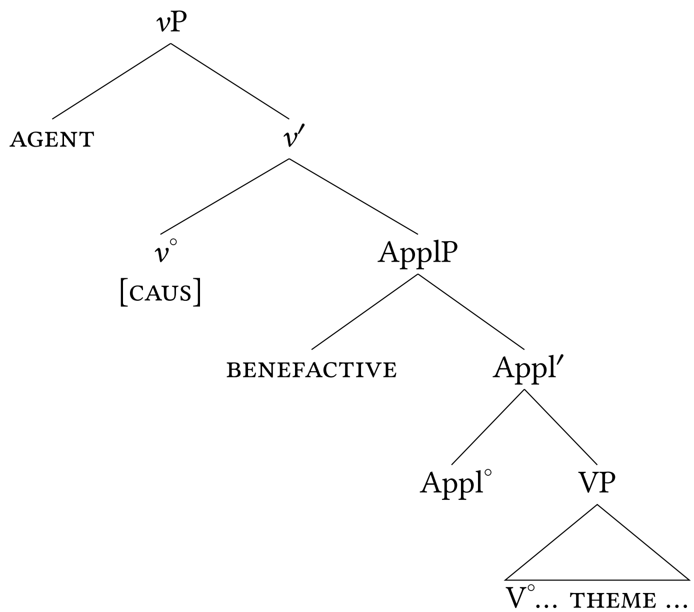
b. Wa-zazi wa-li-som-e-a Maisha ki-tabu.
2-parents 2SM-PST-read-APPL-FV Maisha 7-book
'The parents read Maisha a book.'

(11) a. M-pishi a-li-pik-a ch-akula ch-a asabuhi
1-cook 1sm-pst-cook-fv 7-food 7-of 9.morning
'The chef cooked breakfast.'

b. M-pishi a-li-pik-i-a wa-toto ch-akula ch-a asabuhi
1-cook 1sm-pst-cook-appl-fv 2-children 7-food 7-of 9.morning
'The chef cooked the children breakfast.'

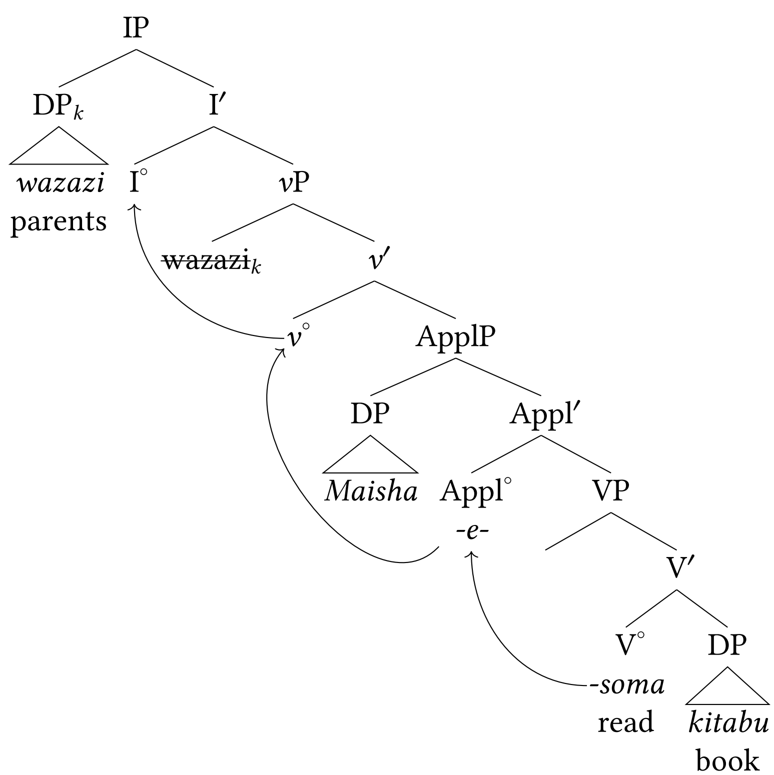
Our approach thus far in this text generally assumes that when identifiable morphemes perform specific semantic functions, we give those morphemes a functional projection in the syntax. In the story we have been developing thus far in this chapter, *v*° is a functional projection that introduces an AGENT argument to the sentence in its specifier. Here, we have a morpheme that introduces a RECIPIENT to the sentence. Extending the logic we've used above, we can assume an applicative head (Appl°) which introduces a RECIPIENT/BENEFACTIVE argument in its specifier.

(12)



Syntacticians working on Bantu languages assume head movement past *v*° (and subjects move to Spec,IP), with the result that the verb head picks up the applicative head/morpheme on the way to its final landing position. Example (13) illustrates how this analysis works for the Swahili double object construction in (10b).

(13)



We can see how a decompositional approach to the verb phrase can then be extended in productive ways to account for valency-changing verbal morphology within the verb phrase: morphemes that introduce changes to argument structure can readily be analyzed as heads with their own projections, introducing their own arguments to the syntax. Part of what enables an approach like this where a separate projection (ApplP) introduces the RECIPIENT/BENEFACTIVE argument is that the agentive subject is also introduced by a separate projection: where previously we were attempting to put all arguments of a predicate into VP, by letting go of that assumption we have space for all the arguments of a clause.

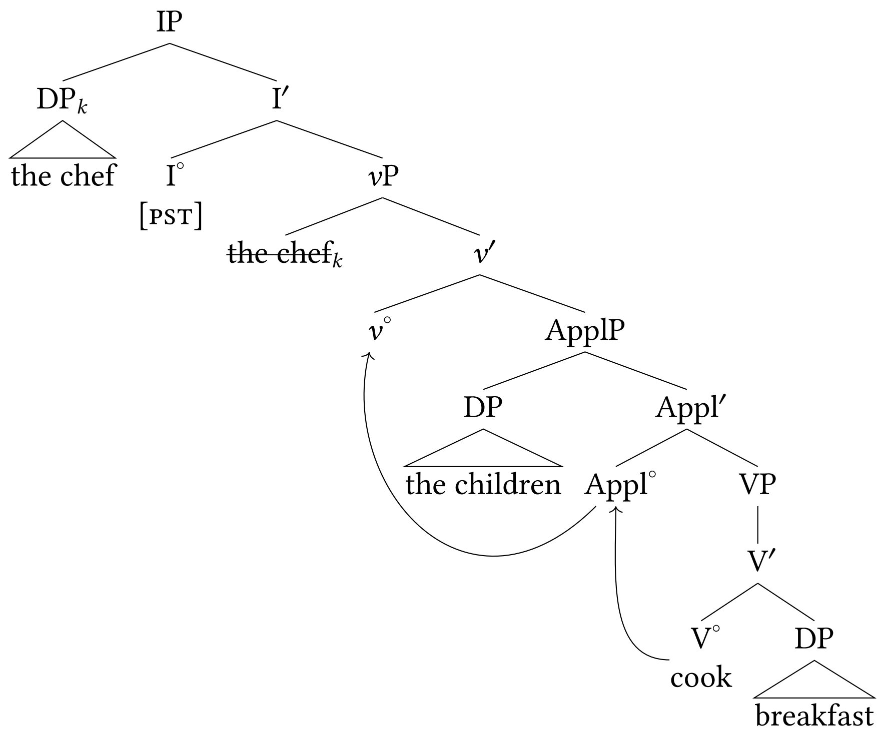
13.2.3 Applicatives in English

As you may have noticed, the Swahili applicative constructions may have unfamiliar morphology to an English speaker, but the direct translations of the sentences into English are thoroughly unproblematic.

(14) a. The parents read a book.
b. The parents read **Maisha** a book.
a. The chef will cook breakfast.
b. The chef will cook **the children** breakfast.
a. Her son will bake a birthday cake.
b. Her son will bake **her** a birthday cake.

In fact, quite like Swahili, English can readily add benefactive arguments to transitive sentences to form ditransitive sentences. The only difference with Swahili in this respect is that there is no overt verbal morpheme in English distinguishing those sentences. But we can adopt the same approach that Swahili makes available to us: perhaps benefactive arguments are typically introduced by ApplPs, even in languages lacking overt applicative morphology like English. This would suggest that a sentence like (14b) would have a structure like (15).

(15)



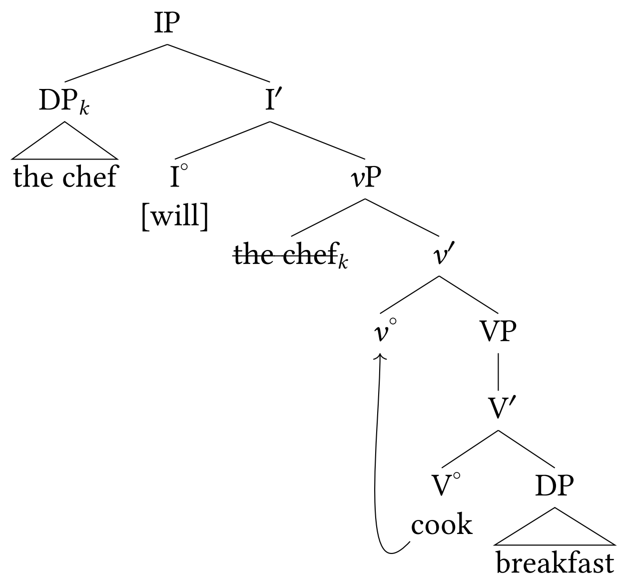
The applicative head allows us the benefit of assuming a stable argument structure for verbs like *read*, *cook*, and *bake*: these are in fact typical transitive verbs taking an AGENT subject and a THEME object. The availability of an applicative structure is what allows for predicates that are normally monotransitive (with a single object) to also at times occur with two objects in a ditransitive construction. The only difference in this respect between Swahili and English, then, is that Appl° does not have an overt morpheme in English, but it does in Swahili.

13.2.4 Little *v*P in monotransitive sentences

So what happens in simple monotransitive sentences, with only one object? We didn't use little *v*° previously for those, do we need to do so now? If we want to maintain our stance thus far that thematic roles are assigned to consistent positions in the syntax, the position of AGENT/CAUSE subjects in Spec,*v*P in the ditransitives above ought to also be the position of AGENT/CAUSE subjects in monotransitive sentences. So we will assume that little *v*° is always present.

(16) a. The chef will cook breakfast.

b.



You may have noticed that the THEME object is located in a different position for the analysis of inchoatives above. We will be more specific about the position of THEME objects below; for now we simply want to point out that we will generalize the analysis of little *v*° to claim that it always introduces agentive subjects.

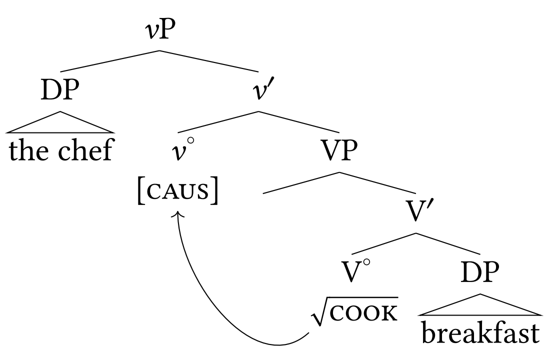
13.2.5 The decomposition of lexical ditransitives

There are various consequences of this approach. Specifically, what we once thought of as a simple instance of a lexical item (a verb) is in fact a complex element, composed of multiple pieces of syntax. So instead of just a single verb *cook* being a V°, in this section we've seen that *cook* can either be *v*° + V°, or *v*° + Appl° + V°. It's important that we don't just say that the English word *cook* is V° alone; on our approach here VP itself contains no AGENT, but English speakers instinctively know that *cook* requires both an AGENT and a THEME.

We will assume, therefore, that the lexical roots of verbs are represented as just that: roots that are not yet fully verbs. We use the square root symbol to represent this.[3]

(17) a. The chef will cook breakfast.

b.



English speakers' knowledge of English must therefore include knowledge of how to pronounce various combinations of syntactic structures and verb roots. This we're actually familiar with from Chapter 9. We know that the pronunciation of syntactic structures (phrases and sentences) depends on what makes up those structures; previously, we saw this with different case/agreement features resulting in different morphological forms being spelled out. As we did before, we assume there are spellout rules that specify what combinations of features and syntactic elements results in what pronounced form.

(18) Spellout rules for *cook*:
$v^\circ + \sqrt{\text{COOK}} \longleftrightarrow$ *cook*
$v^\circ + \text{Appl}^\circ + \sqrt{\text{COOK}} \longleftrightarrow$ *cook*

Selected verbs in English require a ditransitive structure, that is, they are lexically specified as being necessarily ditransitive. These lexical ditransitives can be explained if there is only a pronunciation for the root $\sqrt{\text{GIVE}}$ when it combines with an applicative; there is no monotransitive (applicative-less) option like there is for *cook* in (18).

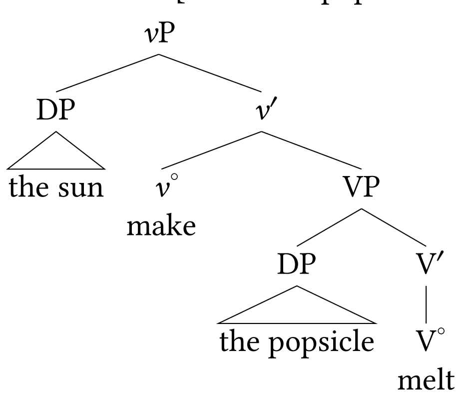
(19) Spellout rule for *give*:
$v^\circ + \text{Appl}^\circ + \sqrt{\text{GIVE}} \longleftrightarrow$ *give*

13.2.6 The status of UTAH

The Uniform Theta Assignment Hypothesis (UTAH) is the component of our model that ensures that thematic roles are assigned to consistent positions in the syntax. Before this chapter, this meant that subjects entered the syntax in Spec,VP and theme objects entered the syntax as siblings of V°. The particulars of this have changed, however. AGENT/CAUSE subjects are now introduced in Spec,*v*P; applicative projections now introduce RECIPIENT/BENEFACTIVE arguments. But we're faced with a conundrum for theme arguments: in the inchoative constructions we began with, THEMES were necessarily introduced in Spec,VP, so that they would precede the main verb when it remained in that position.

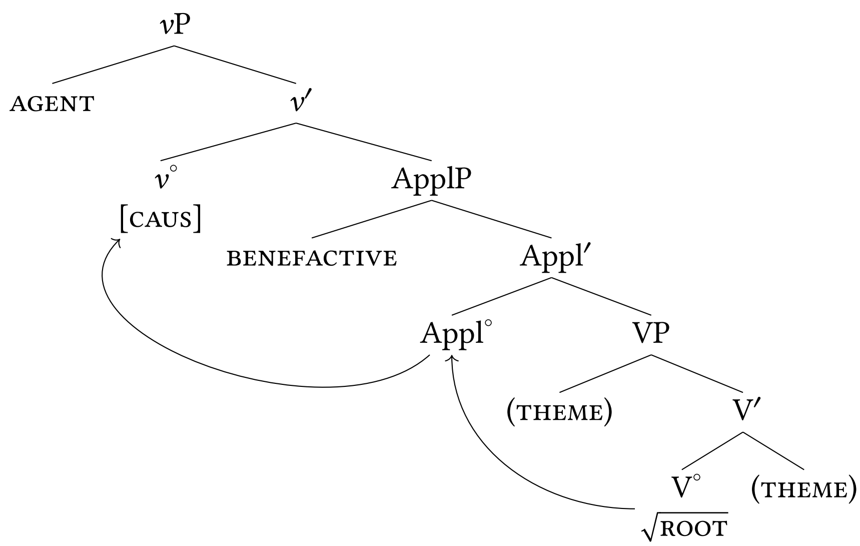
(20) a. The sun will [make the popsicle melt].

b.

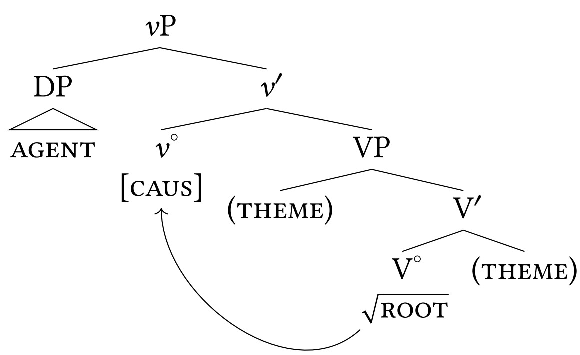


This contrasts, however, with the complement-of-the-verb position we've been assuming generally is the position of THEMES. We won't completely resolve this question in this textbook, but for our purposes what appears to be the case is that VP introduces THEMES. VP has a special status in syntactic structures as the only argument-introducing head that is a lexical projection and can also sit at the bottom of the entire sentence. Because *v*P and ApplP are functional structure atop VP, the only position they have for a new argument is their specifier; on the other hand, VP can in principle introduce arguments in either its complement or its specifier. We will hence assume that, having both of these positions available, VP can (in principle) introduce a THEME argument in either its complement position or in its specifier.

(21) UTAH illustrated (in a ditransitive)



(22) UTAH illustrated (in a monotransitive)



13.3 Additional evidence for decomposition of VP

We have already seen overt morphological evidence from Swahili suggesting a decompositional approach to the verb phrase domain: some languages have overt morphology for adding arguments to verbs phrases. However, English also suggests that this decompositional approach is valuable. We show two arguments here.

13.3.1 Decomposition of VP matches facts from idioms

Recall from our discussions in Chapter 7 that **idioms** are expressions whose meaning does not follow straightforwardly from the individual parts, as in (23). (The idiomatic meaning can sometimes be traced back to an etymological source, but even if that is possible, the source is often unknown to most of the idiom's users.)

(23) a. red tape 'bureaucratic difficulties'
b. the Big Apple 'New York City'
c. kick the bucket 'die'
d. let the chips fall where they may 'disregard the consequences of one's actions'

As we've discussed previously (see Chapter 7 and 11), idioms have idiosyncratic meanings based on their parts, and therefore must be memorized (lexical) knowledge. A result of this line of thinking is that idioms are pre-established chunks of syntax, which in our model we take to be constituents. The examples in (23) trivially satisfy this constraint: *red tape* is an NP, *the Big Apple* is a DP, and *kick the bucket* and *let the chips fall where they may* are instances of V′.

Section 7.4.3 of Chapter 7 discusses the conclusion that idioms necessarily enter the syntax as a unit, as a constituent. The motivation for this is to account for the absence of theoretically possible idioms like the made-up example in (24), where *blue* and *hopping*, though adjacent, don't form a constituent.

(24) a. They've bred a strain of *blue hopping* drosophila.
Intended meaning: 'They've bred a strain of drosophila that is unusually large.'
b. The great apes all have *blue hopping* brains.
Intended meaning: 'The great apes all have unusually large brains.'
c. She's a *blue hopping* child for her age.
Intended meaning: 'She's an unusually large child for her age.'

This conclusion, as a baseline, would seem to suggest that idioms consisting of discontinuous chunks should not exist. But at first glance that appears to be contradicted by idioms like those in (25).

(25) a. <u>give</u> someone <u>the creeps</u> 'make someone uneasy'
b. <u>throw</u> someone <u>to the wolves</u> 'sacrifice someone'

The way that we dealt with this previously was to suggest that idioms may start as a unit in the syntax, but movement may obscure that. This is how we explained why the idiomatic verb phrase *the shit hit the fan* in (26) can appear discontinuous, with an intervening inflectional element in each example in (27).

(26) The shit hit the fan.
Figurative interpretation: There is suddenly a lot of trouble, or a serious argument.

(27) a. The shit will hit the fan.

b. The shit might hit the fan.
c. The shit could hit the fan.
d. The shit hit the fan.
e. The shit could have been hitting the fan, for all we knew.

We can apply a similar line of reasoning to the apparently discontinuous idioms in (25). Just as the decompositional analysis of the verb phrase domain allows us to preserve the binary-branching hypothesis in the face of *prima facie* counterevidence, it also allows us to preserve the locality constraint on idioms in the face of apparently discontinuous idioms. This is because the complex *v*P/VP analysis allows us to say that what is idiomatic in (25) are the underlined instances of V′ in (28).

(28) a. [CAUS] someone ... $\underline{[\ \sqrt{\text{GIVE}}\ ]\text{ the creeps}}$
b. [CAUS] someone $\underline{[\ \sqrt{\text{THROW}}\ ]\text{ to the wolves}}$

This evidence suggests that there is a level of representation (pre-movement) where the V° and the direct object are a constituent to the exclusion of the indirect object. This constituency does not hold up after movement, of course, but given that it is the case pre-movement, the structure allows for these kinds of idioms that are constituents underlyingly but are discontinuous on the surface.

13.3.2 Decomposition of VP matches facts from adverbial modification

We can see a similar argument emerge from modification of verb phrases with the adverbial *again* (Dowty 1979; Stechow 1995). Example (29) gives an example where there is ambiguity as to the meaning of *again*:

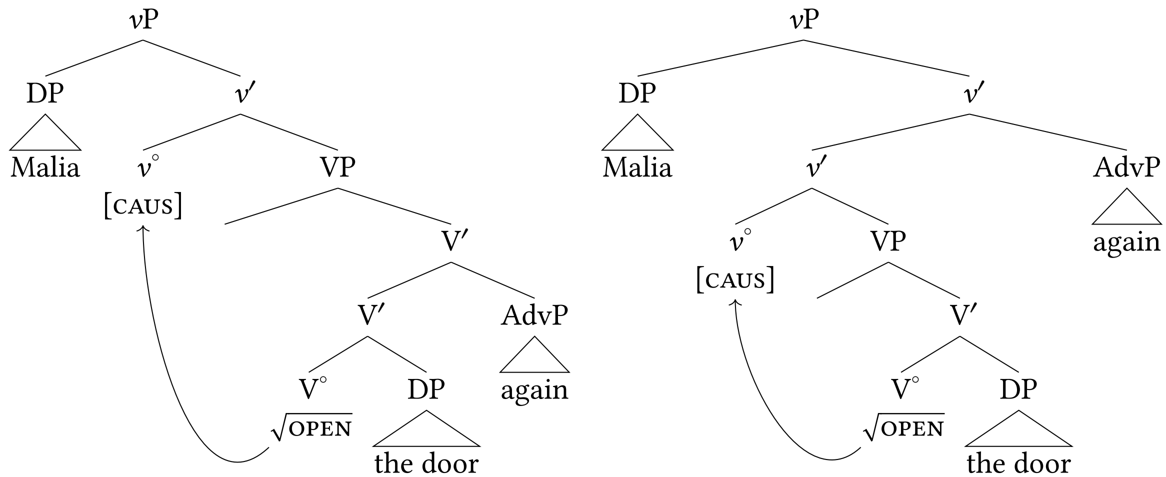
(29) Malia opened the door again.
a. **Repetitive reading**: Malia has opened the door twice.
b. **Restitutive reading**: Malia only opened the door once, but the door was previously open (perhaps the wind blew it shut, or some other person closed it).

The ambiguity here lies in whether the agentive action is considered to have happened again or not. On the repetitive reading, the agentive action by Malia happened at least twice (with the result of the action—the door being open—happening twice as well). The second reading is one where Malia's action only happened once, but the result itself is what may be recurring here (the door's open state), which we can call the restitutive reading (the door being restored to its previous state).

Our approach to modification via adjuncts to this point would suggest that different readings are the result of different attachment points of the adjunct. But a simplex VP structure like we have used up until this chapter would give us no option for structural difference between these readings.

On the other hand, if we have complex *v*P structures where the CAUSE/AGENT is introduced in a separate projection from the theme, there is a natural explanation for the ambiguity of *again*: it is a structural ambiguity based on different attachment sites for the adjunct.

(30) a. b.



Restitutive reading of *again*: the door is open multiple times, but Malia acted once

Repetitive reading of *again*: Malia acted multiple times to open the door

The ambiguity of *again* in (29) finds a natural explanation as a structural ambiguity in a decomposed *v*P: repetitive readings are instances where *again* modifies *v*P, and restitutive readings are those where *again* is adjoined to VP.

There is always a cost and a benefit to any justified adjustment to a scientific model. Here, the cost is complexity. All other things being equal, a simpler analysis (a simpler model) is the better model—this is the famous Occam's Razor. But here we are showing that all other things are not equal. First, a simplex binary-branching VP always had no explanation for ditransitive predicates, despite ditransitive predicates being normal in human language. But as we are showing, this approach also has promise to explain other intersecting properties of language, such as inchoative/causative alternations, apparently discontinuous idioms, and repetitive/restitutive readings of *again*. So syntacticians have concluded that complex verb phrases are in fact justified.

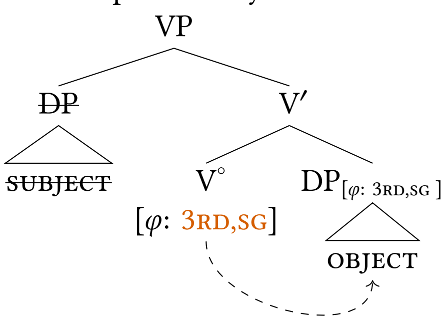
Case in ditransitive sentences 13.4

Recall from Chapter 9 that we use the concept of (abstract) Case licensing to explain the distribution of noun phrases in sentences: different kinds of syntactic constructions (and, different languages!) prefer their DPs in different positions, and abstract Case licensing is how we encode/explain this in our model.

(31) DP Licensing Requirement:
All DP arguments must receive a Case value.

Previously, we said that I° assigns nominative Case to subjects, that (crucially for our current discussion) V° assigns accusative case to objects, and that both do so via the Agree operation.

(32) The Chapter 9 story for case in VP



This chapter raises multiple questions. First, we now have structures that can accommodate ditransitive verbs. But how are RECIPIENT/BENEFACTIVE arguments Case-licensed? Second, we have complicated our structure of the verb phrase generally, which will allow us another way to explain Case-licensing even in non-ditransitive contexts. Throughout this section we will use passive sentences as a diagnostic to facilitate the discussion.

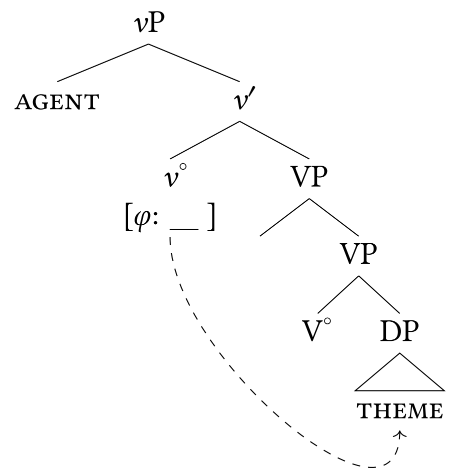
13.4.1 Licensing objects in transitives

We are currently equipped to tackle a fundamental property of passive verbs: the connection between suppressing an agent and losing the ability to license accusative case on an object. Examples (33c) and (33d) attempt constructions that don't promote the THEME to subject position, and both are ungrammatical.

(33) a. The children popped the balloons.
b. The balloons were popped.
c. * Were popped the balloons.
d. * It was popped the balloons.

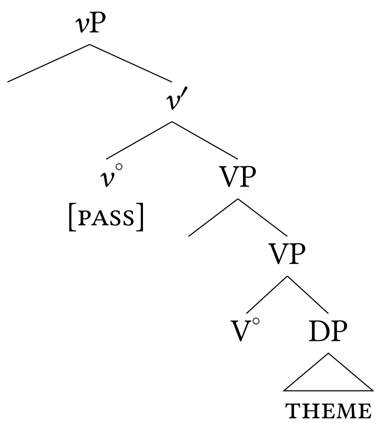
Originally noted by Luigi Burzio, this was known as Burzio's generalization: verbs that do not assign an external thematic role do not license case on an object.[4] We can capture this straightforwardly by placing the case-licensing features for objects on the same head that introduces the external thematic role: $v°$. If we assume that $v°$ bears a φ-probe and undergoes an Agree operation with a φ-bearing DP in its c-command domain, we can likewise assume that the case feature on the DP is licensed by that same Agree operation, parallel to how Agree with I° licenses subjects.

(34) Active *v*P



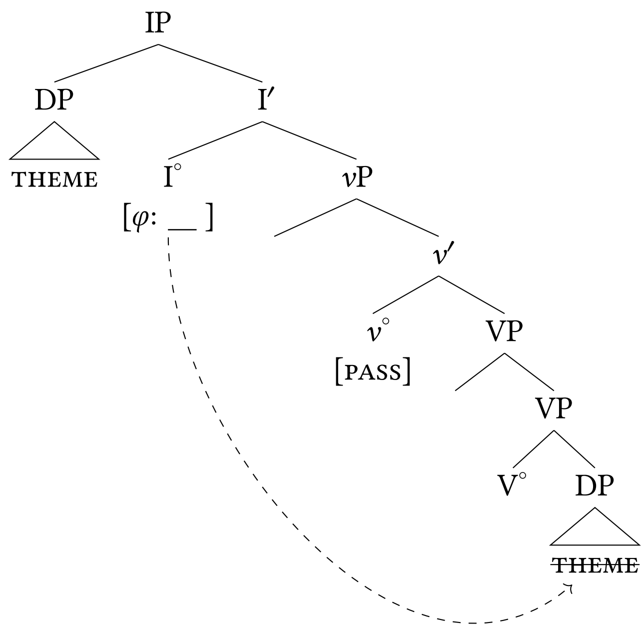
What happens, then, in a passive sentence? The fundamental difference between a passive and an active sentence can be represented by assuming passives and actives use different $v°$ heads. We'll label the passive $v°$ as [PASS], and it does not bear a φ-probe and it does not assign a thematic role to its specifier.

(35) Passive *v*P



So how does the THEME argument get licensed in a passive sentence? The *v*P itself doesn't contain the resources to do so. But as we know in passives like (33b), the THEME in these instances becomes the subject of the sentence. Therefore, as we discussed in Chapter 9, in passive sentences the THEME argument (underlyingly the complement of the verb) becomes the subject. In English, this means that it raises to Spec,IP after agreeing with I°, receiving nominative case from I°.

(36)



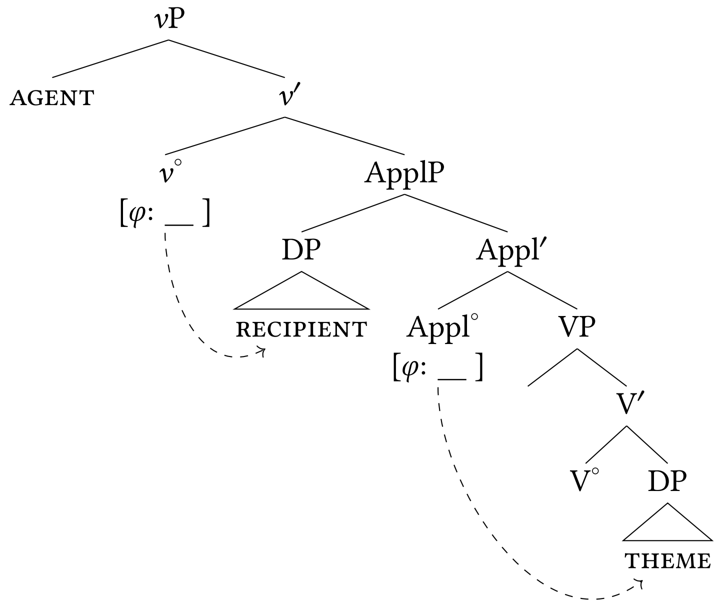
In this tree we keep things abstract, ignoring the presence of an auxiliary in English passives (many languages have no auxiliary verb in their passive constructions; this is a quirk of English).

This account of object licensing does several things for us. It provides a straightforward link between external (agentive) thematic roles and object case licensing (Burzio's generalization), accounting straightforwardly for passives. It also takes the role of object case licensing away from lexical verbs, which is desirable because (as we see with passives) the structural case licensing of objects can vary with the same lexical verb, based on the kind of syntactic construction. An account with a case-licensing φ-probe on v° accounts for this.

13.4.2 Licensing objects in ditransitives

We have now incorporated ditransitive constructions into our model via Applicative heads. This is largely motivated by languages that have overt applicative morphology on verbs in double object constructions (like Swahili), but the same theoretical components can be utilized in a language like English. A key of a ditransitive construction is that an additional thematic role is introduced (typically a RECIPIENT/BENEFACTIVE), and likewise the construction can also structurally license an additional object. We assume that the Appl° head accomplishes both of these. As is typical, the agentive subject introduced by v° is licensed by I°. The φ-probe on v° probes and finds the most local DP, in this instance the RECIPIENT object. The THEME object is now licensed structurally by the φ-probe on Appl°.

(37) Ditransitive (active) *v*P



This does make predictions, of course: in a passive construction, the object that should lose its Case-licensing and need to be licensed by I° is the object that is licensed by *v*°, namely, the RECIPIENT DP. The THEME DP is licensed by Appl° and shouldn't be affected by the passive. This prediction is confirmed here:[5]

(38) a. The chef cooked the children breakfast.
b. The children were cooked breakfast.
c. *? Breakfast was cooked the children.

13.4.3 Licensing objects in unaccusatives/inchoatives

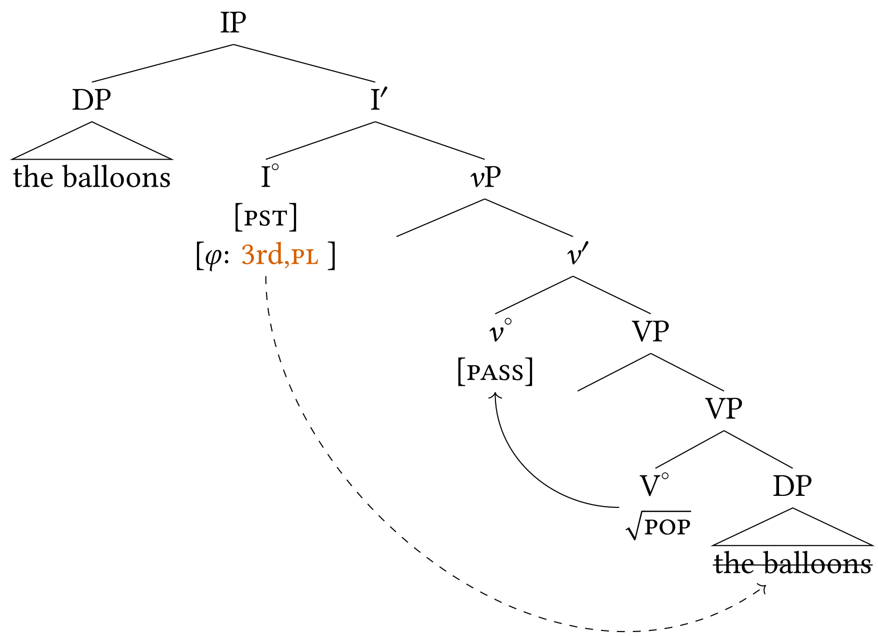
Recall that there are intransitive constructions where the single argument of the predicate is the subject of the sentence, but carries the thematic role of THEME.

(39) a. The chair broke.
b. The rain fell.

Based on what we've concluded in this chapter, we expect this THEME argument to originate inside the VP. But these DPs end up as canonical subjects in Spec,IP, bearing nominative Case. How is this so? Researchers have generally treated these as essentially inherently passive predicates: they are predicates that do not assign accusative case to their internal argument, and do not assign an external thematic role (adhering to Burzio's Generalization).

(40) a. The balloons popped.

b.



The assumption here is one which (thankfully) matches the name of the phenomenon: unaccusative predicates are those that inherently don't assign accusative Case to their internal arguments. In the framework we've sketched here, this means that the $v°$ that occurs with unaccusatives is the passive $v°$, with no external argument and without the ability to assign accusative, which in our model means there is no φ-probe on $v°$. The only Case licenser in a sentence like (40a) is finite I°, so the THEME argument is the Goal of an Agree operation initiated by the probe on I°.

Decomposing lexical ditransitive predicates 13.5

There is a common cross-linguistic pattern that English also realizes which may be on your mind. Many ditransitive sentences with two DP objects have a counterpart in which the order of the RECIPIENT (**red, bold**) and THEME (*blue, italics*) arguments is reversed and the RECIPIENT is expressed as a PP rather than as a DP.

(41) a. Malia gave **Langston** *the receipts*.
  b. Malia gave *the receipts* **to Langston**.

At first glance, the DP-PP sentences seem to be completely synonymous with their DP-DP double object counterparts and to stand in a one-to-one correspondence with them. Indeed, early on in generative grammar, it was held that any DP-PP ditransitive sentence could be transformed into a DP-DP double object sentence by an operation known as Dative Shift (in many languages, RECIPIENTS are marked by dative case morphology or dative case particles). However, certain semantic restrictions on the two sentence types have led this view to be abandoned (Green 1974; Oehrle 1976; Jackendoff 1990). For instance, RECIPIENTS in DP-DP double object sentences, but not in DP-PP ditransitive sentences, are constrained to be animate.

(42) DP-DP ditransitive sentence:
  a. Malia sent Langston the receipts.
  b. * Malia sent the post office box the receipts.

(43) DP-PP ditransitive sentence:
  a. Malia sent the receipts to Langston.
  b. Malia sent the receipts to the post office box.

This effect is so strong that noun phrases that can be interpreted as inanimate in a DP-PP ditransitive sentence are coerced into an animate interpretation in the corresponding DP-DP double object sentence, if that is possible. For instance, in (44b), *Philadelphia* cannot be interpreted as a location, as is possible in (44a), though it can be interpreted metonymically as 'people at the Philadelphia office' (of an organization/corporation).[6]

(44) a. Malia sent the receipts to Philadelphia.
  (ambiguous between location and metonymy reading)
  b. Malia sent Philadelphia the receipts.
  (only metonymy reading)

What the facts in (42)–(44) suggest is that ascribing exactly the same thematic role (that of RECIPIENT) to the first DP in a double object sentence and to the PP in a DP-PP ditransitive sentence is not quite correct. Rather, the PP headed by *to* denotes a path or direction along which the THEME moves, and the complement of *to* denotes the path's endpoint or **goal**, which can be either a RECIPIENT at that location, as in (43a), or a pure location, as in (43b). We give the structures that we are assuming shortly.

This move of carefully distinguishing between RECIPIENTS and LOCATIONS is supported by some patterns using the predicates *get* and *go*: *get* is a transitive predicate that takes a RECIPIENT subject and a THEME complement, whereas *go* occurs with a THEME subject and a LOCATION as a complement. There is a parallel between (42)–(44) on the one hand and the corresponding simple *get* and *go* sentences in (45) and (46) on the other. It is also possible to replace *get* by *have* or *receive* without changing the judgments; the point is that verbs that require RECIPIENT arguments behave differently than those that simply require a GOAL argument, confirming this analysis of the ditransitive sentences on the same grounds.

(45) Parallel to DP-DP ditransitive sentence:
  a. Langston got the receipts.
  b. * The post office box got the receipts.
  c. Philadelphia got the receipts.
  (only metonymy reading)

(46) Parallel to DP-PP ditransitive sentence:

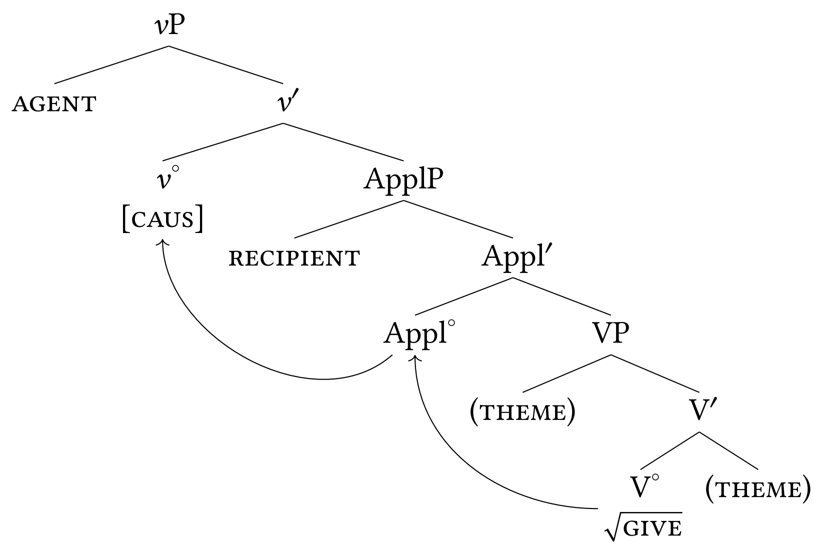
a. The receipts went to Langston.

b. The receipts went to the post office box.

c. The receipts went to Philadelphia.
(ambiguous between metonymy and location reading)

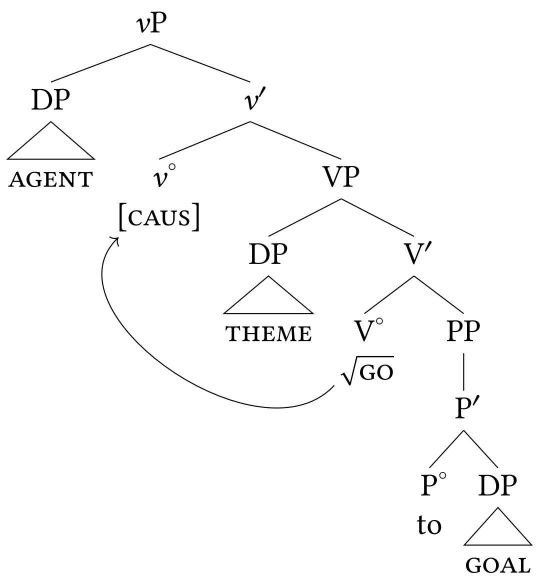
Schematic structures for (45) and (46) are given in (47). As we saw in the preceding sections, Appl° assigns a RECIPIENT thematic role to the DP it introduces in its specifier, and the THEME DP is introduced inside the VP (we draw it as a complement, though it could readily go in either complement or specifier position of VP). In contrast, the DP-PP construction requires the THEME to enter in its specifier and the GOAL PP is the complement of V°. We explicitly draw the [CAUS] feature on *v*° in these trees to re-emphasize how the lexical meaning of *give* is derived. For each, we show the morphological rule explaining that the lexical predicate *give* in English can realize two different structures: in one the RECIPIENT argument is added via an applicative, and in the other the GOAL argument is added as a PP.

(47) a.



DP-DP double object structure
$v^{\circ} + \text{Appl}^{\circ} + \sqrt{\text{GIVE}} \longleftrightarrow$ *give*

b.



DP-PP ditransitive structure
$v^{\circ} + \mathrm{Appl}^{\circ} + \sqrt{\textsc{go}} \longleftrightarrow$ *give*

In the DP-PP ditransitive examples presented so far, the GOAL PP complement is headed by a transitive P. The X′ schema leads us to expect that semantically suitable intransitive heads should also be able to serve as complements of *go*, and this expectation is borne out in (48), where *here* and *there* are plausibly pro-forms for PPs, P heads without arguments of their own. Both *here* and *there* can appear in DP-PP ditransitives.

(48) a. Malia sent the receipts { here, there }.
b. The receipts went { here, there }.

Notice that *here* and *there* unambiguously refer to LOCATIONS or to PATHS with LOCATIONS as endpoints (or GOALS). Therefore, neither (48a) nor (48b) have metonymy readings, in contrast to (44a) and (46c), respectively. Moreover, the ill-formedness of (49a) and (49b) follows from the fact that these sentences violate the selectional restriction imposed by *get* on its specifier (compare the absence of location readings in (44b) and (46b)).[7,8]

(49) a. * Malia sent { here, there } the receipts.
b. * { Here, there } got the receipts.

Decomposing CP 13.6

The decompositional approach to the verb phrase that we have adopted here probably raises many questions for you—if VP is actually composed of lots of syntactic structure, does that mean that kind of analysis is possible all over the place? Short answer: yes. That can be intimidating, but it also is simply reflective of how complex human language actually is. We conclude this chapter by moving outside the verb phrase, looking up the tree at CP. As we will see, there is good reason to think that there is actually complex syntactic structure at multiple levels. We have already seen this in the inflectional domain, where we see projections for auxiliary verbs, negation, and other elements proliferate at times. But one area that we have continued to treat with the utmost simplicity is the CP domain, composed of (up to this point) a simplex CP.

Rizzi (1997) pointed out a variety of Italian facts that make it challenging to claim that the CP domain is always as simple as a sole CP projection. To understand the argument, we'll consider the two kinds of emphasis constructions we first encountered in Chapter 8: topicalization and focus movement. In Italian topicalization, the topic phrase (the element being talked about) is at the left edge of the sentence, and there is a pronoun inside the sentence that refers to it (similar to the English translation).

(50) Italian topicalization
Il tuo libro, lo ho letto
DET your book, it I.have read
'Your book, I have read it.'

Focus movement in Italian is similar but not identical: there is prosodic emphasis on the focused phrase at the left edge of the sentence, and no matching pronoun lower in sentence. Focalization, as in English, yields a contrastive reading.

(51) Italian focus movement
IL TUO LIBRO ho letto (, non il suo)
DET your book I.have read not DET his
'Your book I read (, not his).'

Based on this evidence, we could assume that both focused phrases and topic phrases are in Spec,CP in Italian. But Rizzi (1997) shows that in embedded clauses, different constructions make this simplex CP impossible to maintain.[9]

(52) Italian complementizers
a. *che*, roughly 'that'
- precedes finite embedded clauses
- precedes topic/focus XPs
- [$_{CP}$ *che* topic/focus [$_{IP}$ *sentence*]]

b. *di*, roughly ‘of’
   - precedes non-finite embedded clauses
   - follows topic/focus XPs
   - [$_{CP}$ topic/focus *di* [$_{IP}$ *sentence*]]

We can see these basic patterns illustrated in (53), where the embedded CP in each instance is introduced by either *che* or *di* (depending on the finiteness of the embedded clause).

(53) a. Credo [$_{CP}$ $\boxed{\text{che}}$ loro apprezzerebbero molto il tuo libro.]
I.believe that they would.appreciate much DET your book
‘I believe that they would appreciate your book very much.’

b. Credo [$_{CP}$ $\boxed{\text{di}}$ apprezzare molto il tuo libro.]
I.believe of to.appreciate much DET your book
‘I believe of to appreciate your book very much.’

When we add topicalization to the mix, however, we see how it interacts with these complementizers. The examples in (54) show that *che* is grammatical when it precedes a topicalized phrase, but ungrammatical when it follows that phrase. All the following Italian data are drawn from Rizzi (1997, 288ff).

(54) a. Credo $\boxed{\text{che}}$ **il tuo libro**, loro lo apprezzerebbero molto.
I.believe that DET your book they it would.appreciate much
‘I believe that your book, they would appreciate it a lot.’

b. * Credo, **il tuo libro**, $\boxed{\text{che}}$ loro lo apprezzerebbero molto.
I.believe that DET your book they it would.appreciate much
Attempted: I believe, your book, that they would appreciate it a lot.

This is already somewhat problematic from our treatment of CP thus far. If topicalized phrases are in Spec,CP, they ought to precede a complementizer element. But here, the topicalized phrase *follows* the complementizer. This is further complicated by the fact that all complementizers do not behave the same; the *di* complementizer is precisely the opposite, *following* the topicalized phrase instead of preceding it.

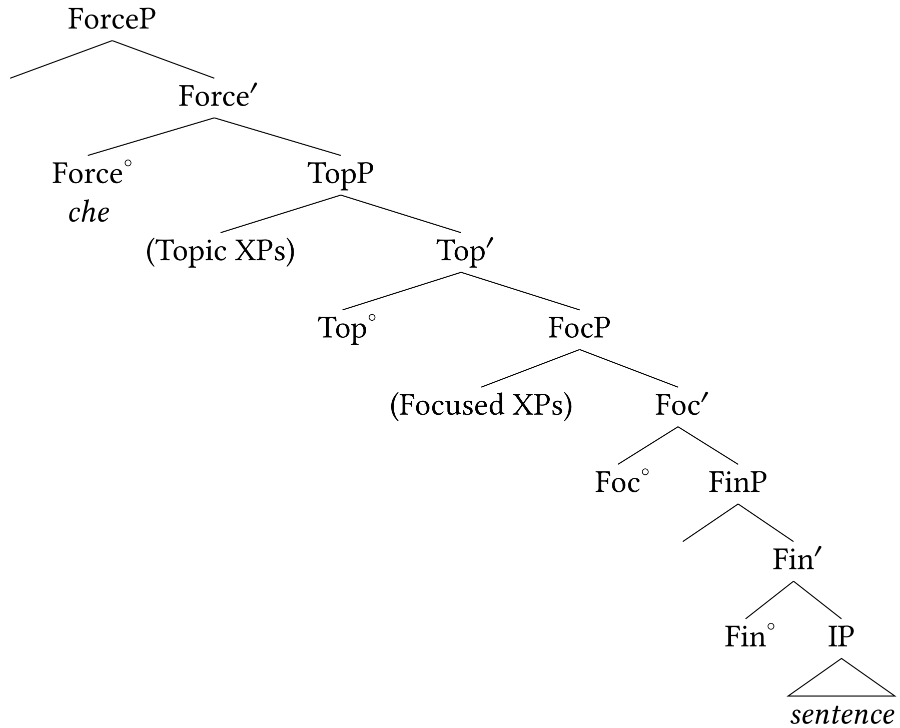
(55) a. * Credo $\boxed{\text{di}}$ **il tuo libro**, apprezzar-lo molto.
I.believe of DET your book to.appreciate-it much
Attempted: ‘I believe of your book to appreciate it a lot.’

b. Credo **il tuo libro**, $\boxed{\text{di}}$ apprezzar-lo molto.
I.believe DET your book of to.appreciate-it much
‘I believe your book of to appreciate it a lot.’

So what we find is that topicalized phrases in Italian, which presumably are in a consistent position in the left periphery of the clause, vary with respect to which complementizers they occur with. The *che* complementizer precedes topics, and the *di* complementizer follows topics. The same facts hold for focused elements and other kinds of left peripheral elements. We don’t walk through the full range of empirical facts here, but Rizzi (1997) concludes that the left periphery of sentences has a maximal structure looking something like (56); this is what was formerly our

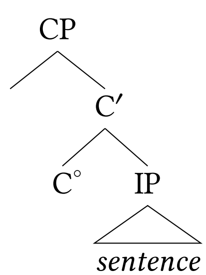
CP. He labeled the highest projection in the CP domain (where *che* appears) as 'Force' (where sentence force is marked, e.g. declaratives vs. questions) and he marked the lowest projection as Fin(iteness), as it interfaces with IP.

(56) Structure of the left periphery (formerly CP)



Rizzi proposed that this complex CP only emerged when strictly necessary, and otherwise a simplex CP occurs. So we will nonetheless regularly draw tree structures that look like (57), unless we are otherwise compelled to by the evidence.

(57)



Rizzi's proposals have turned out to be a best case scenario for a theoretician: proposing a structure based on indirect evidence to account for a complicated set of facts, which are then later confirmed by overt, direct evidence from unrelated languages. We've already seen that Gungbe (a Niger-Congo language spoken in Nigeria and Benin) has overt morphology for topic and focus projections. Previously, we said these realize C° heads, with their associated topicalized/focused phrase in Spec,CP.

(58) a. Dàn lɔ́ yà, kòfí hù-*(ì)
snake the TOP Kofi kill.PFV-3SG
'As for the snake, Kofi killed it.'

b. Ɛtɛ́$_k$ *(wɛ̀) Sɛ́ná xìá t_k? **Object information question**
what *(FOC) Sena read-PRF
'What did Sena read?'

c. Wémàk *(wɛ̀) Sɛ́ná xìá t_k. **Object focus**
book *(FOC) Sena read-PRF
'Sena read A BOOK.'

What we didn't see previously is that it is possible to embed these constructions in an embedded clause introduced by a complementizer. This is shown for a focus construction in (59b) and a topic construction in (60b).

(59) a. Ùn lɛ̀n ɖɔ̀ Sɛ́ná xìá wémà lɔ́
I think.PFV COMP Sena read.PFV book DET
'I think that Sena read the specific book.'

b. Ùn lɛ̀n [$_{ForceP}$ ɖɔ̀ [$_{FocP}$ wémà lɔ́ wɛ̀ Sɛ́na xìá t_k]]
I think.PFV COMP book DET FOC Sena read.PFV
'I think that Sena read THE SPECIFIC BOOK.'

(60) a. Dàn lɔ́ yà Kòfí hù ì
snake DET TOP Kofi kill-PFV it
'As for the specific snake, Kofi killed it.'

b. Ùn ɖɔ́ [$_{ForceP}$ ɖɔ̀ [$_{TopP}$ dàn lɔ́ yà Kòfí hù ì]]
I say.PFV COMP snake DET TOP Kofi kill.PFV it
'I said that as for the specific snake, Kofi killed it.'

This immediately causes problems for a simplex CP, but is straightforwardly explained if there is a complex left periphery as proposed by Rizzi, and the complementizers in the examples below sit in Force°, the same position as Italian *che*.

Conclusion: exploding projections everywhere? 13.7

This chapter has allowed us to handle nuanced complexity in both the low domain of the clause (we can call it the VP/verbal domain) and the high domain of the clause (we can call it the CP domain, or the left periphery of the clause). Some of this complexity you may have been aware of before (for example, the need to account for ditransitive verbs); some of it may have been new to you. But either way, parts of our theory that we were previously simple are now more complex. Though it's worth nothing that the model itself hasn't really become more complex—our basic tools for building structures (the X′ schema, feature valuation by Agree, etc.) all remain the same. The change is that structures that we were treating as a single projection in the syntax, we are now treating as if they were multiple projections.

An important question that may be on your mind is: just how expansive is this decomposition enterprise? Are we going to keep taking individual projections and exploding them into a multitude of more complex ones? The short answer is 'no': for the most part, we are arriving at something close to the maximal complexity of clause structure. But in terms of how our model works, the answer is a qualified 'yes': if the evidence suggests that there exists another projection in a clause structure, then we posit one! It does not much affect our overall model or approach, but some trees for some sentences might be somewhat more complex than one might have expected.

It does raise questions about how to draw specific trees. We don't need to draw the whole structure in (56) to replace every CP we ever draw: we would never accomplish anything if we were doing this. (56) represents the proposed maximal structure our former CP can take; in many constructions in many languages, there is no need to posit that level of complexity. So drawing a simplex CP where it is merited remains our standard practice. The complex left periphery is invoked where it appears to be necessary. What's actually going on in speaker's minds? That's an important question that we don't know all the answers to yet. Rizzi originally said that CP is simplex (unexploded) CP unless the particular construction demanded more complexity, and most syntacticians treat it like that. We treat lots of things like this—we know where negation goes in a tree when it's present, but in a non-negated sentence we don't draw negation. Same for various kinds of auxiliaries, and other particles/constructions—we know where they go in the tree when they're present, but we treat them as absent when they seem to be absent. The idea is that CP behaves in a similar way.

The *v*P, in contrast, dependably behaves as if it's present: in our model as we've developed it in this chapter, *v*° encodes whether or not a sentence is active or passive, it case-licenses objects, and it introduces AGENT thematic roles. Those functions are happening in most sentences, and therefore we consistently expect *v*P to be present. Syntacticians still don't always *draw* *v*P in a sentence—sometimes we want our trees to be smaller because we're using them to communicate things about structure that is not inside *v*P, meaning we simplify what *v*P looks like (similar to drawing a triangle over a phrase).

So we can conclude that even simple sentences in a language can have a very complex structure, with lots of hierarchical projections. Is human language necessarily this way? Do all languages share the exact same functional projections? What is actually universal as part of UG, and what is not? There has historically been a split in the field in this regard: some people propose a universal 'map' of functional projections that is hard-coded into UG. Otherwise propose that UG is simpler, and the kinds of projections that emerge are posited by children as they learn the languages that they are learning. We are inclined towards the latter explanation, but proponents of this approach are left to explain why there is such strong cross-linguistic consistency in the hierarchy of functional projections. For the most part, these are empirical questions that are still being worked out. If a sequence of functional heads turns out to be universal, it's plausibly part of UG. If it turns out NOT to be universal, then it's clearly not.

New Empirical Observations

- ditransitive sentences
- applicative morphemes
- topic and focus constructions in the left periphery of a sentence

Components of the Theoretical Model

- decomposing simple projections to complex domains
- AGENTS introduced in little *v*P
- RECIPIENTS introduced in ApplP
- accusative case and voice are encoded on v°, via Agree in the usual way
- CP projections: ForceP, TopP, FocP, FinP

Notes 13.8

1. The languages reported here (and the pattern is common) do have a select number of lexical ditransitives, verbs like *give* and *teach* that inherently take two objects. In some languages these show fundamental distinctions from ditransitives derived from applicatives; in others, they behave roughly the same. The world of ditransitive syntax is expansive and we don't attempt to describe the full scope of variation between languages. If you find yourself wanting more, you should take more syntax classes!

2. But, we should note, there is fascinating variety and fine-grained variation between languages in the behavior of applicatives. A good introduction is Jeong (2007).

3. The approach presented here is a mildly modified (and not fully precise) version of lexical decomposition as is standard in Distributed Morphology (Halle and Marantz 1993; McGinnis 2017; J. D. Bobaljik 2017).

4. While this remains a robust crosslinguistic generalization, it is not a literal universal: some languages have impersonal passive constructions where an external argument is removed but an internal argument remains. Burzio's Generalization is certainly too accurate to be a coincidence, so there is something foundational about the link between external arguments and licensing objects, even if it is not irreducibly part of Universal Grammar.

5. There are languages which treat their objects in ditransitives (somewhat) symmetrically, allowing either to be passivized in a passive construction. Researchers have posited solutions to these puzzles, but it takes us afield from our main tasks here. Part of the solution is that even in languages that can passivize either object, these same asymmetries tend to appear in corners of their grammars, suggesting that something like this hierarchical account exists even in those cases (Holmberg, Sheehan, and van der Wal 2019; Newman 2021).

6. Metonymy is the traditional term for various types of figurative language use, notably including the one relevant here, where an expression that literally refers to a location is used to refer instead to a group of people typically

located there. Common examples include *the White House* (broadly 'the U.S. executive', more narrowly 'the U.S. president along with close staff'), *the Kremlin* ('the Russian government'), *Westminster* ('the U.K. parliament'), and so on.

7. The alternation in (i)—specifically, the well-formedness of (b)—is only apparently problematic for what we say in the text.

   (i) a. ✓ Amy sent the mail { back, off }.
   b. ✓ Amy sent { back, off } the mail.

   Back and *off* are so-called **particles**, which can behave like ordinary PPs, as in (a), but also more like bound affixes, as in (b). A detailed analysis of the syntax of particles is beyond the scope of this chapter, but evidence for their differing syntactic status in (i) comes from contrasts like (ii).

   (ii) a. ✓ Amy sent the mail right { back, off }.
   (cf. ✓ right to the dean)
   b. * Amy sent right { back, off } the mail.

8. From what we have said so far, it is clear that not every DP-PP ditransitive sentence has a double object (DP-DP) counterpart. Specifically, DP-PP ditransitive sentences with GOALS that are LOCATIONS rather than RECIPIENTS have no DP-DP double object counterpart. However, since GOALS are not required to be pure locations, but can instead be RECIPIENTS, it might still be the case that every DP-DP double object sentence has a DP-PP ditransitive counterpart. But this turns out not to be true either. The reason is that in a DP-PP ditransitive structure, the preposition *to* imposes a semantic requirement on the THEME: namely, that the referent of the THEME argument travel (or at least be able in principle to travel) to the referent of the GOAL denoted by the complement of *to* along some path. By contrast, THEMES in DP-DP double object sentences, which lack *to*, aren't subject to such a requirement. For instance, since it is perfectly possible for ideas or migraines to be the result of certain causes, the DP-DP double object sentences in (i) are acceptable.

   (i) <u>DP-DP Double object sentence</u>:

   a. The scandal gave the reporter an idea.
   b. Flickering lights give some people a migraine.

   (ii) <u>DP-PP ditransitive sentence</u>:

   a. * The scandal gave an idea to the reporter.
   b. * Flickering lights give a migraine to some people.

   The reason that the corresponding DP-PP ditransitive sentences in (ii) are unacceptable is that the idea and the migraine are conceptualized as arising within somebody's mind or brain as the result of a cause, but without having traveled there along some path. One way of putting this in terms of thematic roles is to say that the argument introduced by Appl° in these instances is an experiencer rather than an ordinary RECIPIENT. As expected, the simple *get* and *go* sentences in (i) and (ii) are parallel to (iii) and (iv).

   (iii) <u>Parallel to DP-DP double object sentence</u>:

   a. The reporter got an idea.
   b. Some people get a migraine headache.

   (iv) <u>Parallel to DP-PP ditransitive sentence</u>:

   a. * An idea went to the reporter.

b. * A migraine headache goes to some people.

Contagious diseases, incidentally, seem not to be conceptualized as traveling along a path. Instead, they are conceptualized as spreading (occupying their original location in addition to the new location). This explains the contrast between (v) and (vi).

(v) a. Lisa gave Joel a bad cold.
b. Joel got a bad cold.

(vi) a. * Lisa gave a bad cold to Joel.
b. * A bad cold went to Joel.

9. Aspects of this summary were inspired by and adapted from a class handout on the topic from Paul Hagstrom.

Exercises and problems 13.9

Exercise 13.1: Practice with ditransitive verbs A

Draw trees for the following sentences, triangling over DPs, PPs, and AdvPs.

(1) Joan gave Antonio a new blanket.
(2) Angel sends Liam a letter every month.
(3) Mateo should put the cheesecake on the table.
(4) The leak in the hull eventually sank the ship.
(5) The baker might burn the focaccia.
(6) Warm spring days will melt the ice in due time.
(7) The birthday party guests sang me a song.
(8) The professor's students could write her a poem.
(9) Kingston built his family a house on the hill.

Exercise 13.2: Practice with ditransitive verbs B

A. Suggest semantic decompositions for the bolded verbs in the following sentences.

(1) The builders **laid** new flooring in the bathroom.

(2) Paolo already **fed** the cat dinner.

(3) My time teaching first-graders **taught** me patience.

(4) Grandma **showed** me photos of Mom when she was a little girl.

(5) Please **notify** me of any changes to the document.

B. Study the sentence pairs below. Answer in one or two sentences: what do the bolded verbs have in common? You do not need to explain *why* this phenomenon occurs, only *what* the phenomenon itself is.

(6) a. They **set** the glass on the coaster.
    b. * They **set** the coaster the glass.

(7) a. You can **plug** your computer into the outlet.
    b. * You can **plug** the outlet your computer.

(8) a. The fox **recounted** its life story to the crow.
    b. * The fox **recounted** the crow its life story.

(9) a. The phonologist **reported** her findings to the conference attendees.
    b. * The phonologist **reported** the conference attendees her findings.

Exercise 13.3: English ditransitive ambiguities

A. Provide a paraphrase for the two interpretations of the newspaper headline in (1) and the VP shell structure corresponding to each paraphrase.

(1) Lawyers Give Poor Free Legal Advice

B. Repeat (A) for (2).

(2) I will <u>make you my favorite sandwich</u>.

C. Repeat (A) for the two salient interpretations of the punchline in (3).

(3) Q: What did the Zen master say to the guy at the hotdog stand?
    A: Make me one with everything.

Exercise 13.4: English *put*

Consider the discussion of decomposing lexical ditransitives in § 13.5. Based on the evidence in (1)-(2), what is the analysis of English *put*? Draw a tree for (1a) and write out a morphological spellout rule for *put* (similar to how we do for *give* in the chapter).

(1) a. Amy put the books on the shelf.
    b. * Amy put the shelf the books.

(2) a. Amy put the books there.
    b. * Amy put there the books.

Exercise 13.5: Object control versus ECM

Using (1) and (2) as a model, determine whether the verbs in (3) are ECM verbs or object control verbs (or neither!). Use only active verb forms. For the purpose of this exercise, do not be concerned about how you would semantically decompose any object control verbs that you find.

Judgments for *assume*:

(1) a. ✓ I assumed the meeting to be over.
    b. ✓ I assumed there to be a problem.
    c. * I assumed the meeting that it was over.
    d. ✓ I assumed that there was a problem.

Conclusion for *assume*:

The grammaticality of (1b) shows that *assume* is an ECM verb. The judgments in (1c)–(1d) are not required, but they corroborate the ECM analysis.

Judgments for *convince*:

(2) a. ✓ I convinced my friend to take the job.
    b. * I convinced there to be a problem.
    c. ✓ I convinced my friend that he should take the job.
    d. * I convinced that there was a problem.

Conclusion for *convince*:

The ungrammaticality of (2b) shows that *convince* is an object control verb. The judgments in (2c)- (2d) are not required, but they corroborate the object control analysis.

(3) acknowledge, advise, allow, anticipate, ask, beg, blackmail, challenge, command, consider, corral, dare, deem, determine, discover, encourage, enjoin, expect, fear, find, forbid, get, help, instruct, invite, know, order, perceive, permit, predict, pressure, prompt, prove, provoke, remind, report, request, require, tell, tempt, urge, warn

Exercise 13.6: Literal and figurative readings with ditransitive predicates

Explain the pattern of interpretations in (1) and (2), or as much of it as possible. Assume the judgments as given even if they are not your own.

(1) a. The finger { got, went } to the patient. (unambiguously literal)
 b. The surgeon gave the finger to the patient. (unambiguously literal)
 c. The surgeon got the finger to the patient. (unambiguously literal)

(2) a. The patient got the finger (from the surgeon). (ambiguous between literal and idiomatic reading)
 b. The surgeon gave the patient the finger. (ambiguous between literal and idiomatic reading)
 c. The surgeon got the patient the finger. (unambiguously literal)

Exercise 13.7: Practice with decomposing predicates

A. *Get* is one of the most versatile verbs in English. In addition to taking DP and PP/AdvP complements, it also takes AdjP complements. Give the VP shell structures for the sentences in (1).

(1) a. The children got wet.
 b. The rain got the children wet.

B. Give VP shell structures for the sentences in (2). As much as possible, avoid assuming silent heads.

(2) a. The red cloth enraged the bull.
 b. The new fiscal policies enriched the king.
 c. The news saddened me.

C. Build VP shell structures for the sentences in (3).

(3) a. They kept the president's arrival almost completely secret.
 b. They kept the president's arrival a complete secret.

Exercise 13.8: Causative predicates in English

Modern English has causative verbs that are morphologically related to either other verbs, as in (1a), or to other syntactic categories, as in (1b).

(1) a. drink, fall, lie (as in 'lie down', not 'prevaricate'), rise, sit
 b. full, gold

The causatives are not spelled out as homonymous with the same form, but at one point in history the two forms were related by a regular phonological rule (specifically, ablaut or umlaut—see the Wikipedia entries for these terms for more information). But historical sound changes has since obscured those regularities. Over time, the meaning of the originally causative form has also sometimes drifted away from a strict causative meaning. Can you figure out the causative verbs in question?

Hint: Consult etymonline (see a word's "Related entries") or the Oxford English Dictionary Online.

Problem 13.1: English-learning children's (in)transitives

A. Sentences like (1) are ungrammatical in adult English.

(1) You're giggling me!
Intended meaning: 'You're making me giggle!'

But many children go through a stage of producing such sentences with activity verbs such as *dance*, *giggle*, *laugh*, and others. What is the difference between the children's grammar and the adult grammar?

B. If necessary, modify your answer to (A) in light of the judgments of (2)–(4) for adult English.

(2) a. The horses { walked, trotted, galloped, jumped }.
b. The trainers { walked, trotted, galloped, jumped } the horses.

(3) a. The athletes { walked, ran, marched } around the track.
b. The coach { walked, ran, marched } the athletes around the track.

(4) a. * At some point, all new parents want to sleep their baby.
b. ✓ I have one set to show the desktop, and another to <u>sleep</u> the display. ... Hit Command-Option-H to Hide Others. This works in almost any app, letting you quickly <u>disappear</u> all apps other than the frontmost one.
c. ✓ They were known as the 'desaparecidos', because the Argentinian junta would 'disappear' them—for instance, by throwing them out of planes over the Andes.

Problem 13.2: Explaining (missing) patterns of interpretation

Why does (1b) lack a 'fetch' interpretation?

(1) a. They got the package.
Interpretation 1: ✓ They received the package.
Interpretation 2: ✓ They fetched the package.
b. We got them the package.
Interpretation 1: ✓ We made it come about that they received the package.
Interpretation 2: * We made it come about that they fetched the package.

Problem 13.3: Apparent gaps in paradigms of transitivity alternations

Unlike *get* in (1), verbs of transfer like *hand*, *kick*, *throw*, and others don't participate in the causative alternation, as shown in (2). Why might this be so?

(1) a. ✓ The package got to Julie.
    b. ✓ We got the package to Julie.

(2) a. * The ball { handed, kicked, threw, ... } to Julie.
    b. ✓ We { handed, kicked, threw, ... } the ball to Julie.

Problem 13.4: Persian light verbs

The data here come from Harley (2017) and Folli, Harley, and Karimi (2005).

Persian (Indo-Iranian) is a largely verb-final language that expresses many verbal concepts via complex predicates composed of a lexical element and what is often termed a *light verb*. The approach to complex verb phrases adopted in this chapter provide a toolkit for analyzing Persian complex predicates.

For the examples below, come up with an analysis for the structures given. Draw tree structures for the sentences in (3). You can ignore the morpheme *be* and the morpheme glossed EZ (for *ezâfe*) for the purposes of this exercise.

(1) a. تیم ما آن‌ها را شکست داد.
    Tim-e mâ unâ-ro shekast dâd.
    team-EZ we they-râ defeat gave
    'Our team defeated them.'

  b. تیم ما از آن‌ها شکست خورد.
    Tim-e mâ az unâ shekast xord.
    team-EZ we of they defeat collided
    'Our team was defeated by them.' (Literally(ish): our team encountered defeat from them.)

(2) a. مینو بچه را کتک زد.
    Minu bachcha-ro kotak zad.
    Minu child-râ beating hit
    'Minu hit the child.'

  b. بچه را کتک خورد.
    Bachche kotak xord.
    child beating collided
    'The child got hit.'

(3) a. آب به جوش آمد.
Âb be jush âmad.
water BE boil came
'The water boiled'

b. نیما آب را به جوش آورد.
Nimâ âb-ro be jush âvard.
Nima water-râ BE boil brought
'Nima boiled the water.'

(4) a. هما به گریه افتاد.
Homâ be gerye oftâd.
Homa BE crying fell
'Homa started to cry.'

b. هما را به گریه انداخت.
Nimâ Homâ-ro be gerye andâxt.
Nima Homa-râ BE crying dropped
'Nima made Homa (start to) cry.'

(5) a. میز را تمیز کردم.
Miz-o tamiz kard-am.
table-râ clean made-1SG
'I cleaned the table.'

b. میز تمیز شد.
Miz tamiz shod.
table clean became
'The table got/became clean.'

Problem 13.5: Figurative readings with ditransitive: *give an earful*

The example in (1) uses the idiom *give an earful*, which means something like 'to angrily tell someone what you think about something.' We have assumed throughout this text that idioms must be constituents (at some level of representation) to retain their idiomatic interpretation. Given the decomposition of VP that we've undertaken in this chapter, to what extent does this confirm or disconfirm our assumptions about idiom constituency?

(1) Tennessee coach Pat Summitt, the longtime head of the Lady Vols, gives an earful to Alexis Hornbuckle during their win over Texas Tech.
(Daily Pennsylvanian, 28 March 2005, p. 9)

14

Additional properties of A′-movement

Chapter outline

In Chapter 8, we showed that *wh*-phrases in questions and various other constructions move from their **canonical** position (the position where they are interpreted with respect to their thematic role) to Spec(CP). We also showed that this is one among a variety of movements with a similar set of characteristics, which the field refers to as A′-movements (movement to a non-argument position). In this chapter we expand our understanding of A′-movements in various ways. First, we discuss various constructions that (sometimes invisibly) display properties of A′-

movement; we begin with relative clauses, and then discuss *wh*-in-situ constructions. This directs us to a discussion of a major theoretical perspective, the Copy Theory of Movement.

Also, we consider some significant constraints on A′-movement. Specifically, while A′-movement is able to occur at quite long distances, it is in fact constrained. We describe a number of so-called **islands for movement** (domains which are impossible to move a syntactic element out of), and discuss ways that human language navigates these constraints.

Relative clauses 14.1

Across the world's languages, it is typical for a language grammar to contain a strategy for converting a CP into a structure that can modify a noun phrase. (1) shows two examples that are similar to each other in English, where the bracketed DP contains not just a determiner and a noun (familiar DP material) but also an entire additional embedded clause. these bracketed clauses modifying nouns are called relative clauses. (1) shows two strategies for relative clauses in English: one which uses a *wh*-pronoun, and one which uses a *that* complementizer.

(1) a. Alex said that they bought [$_{\text{DP}}$ a feast [which they plan to devour ___.]]
  b. Alex said that they bought [$_{\text{DP}}$ a feast [that they plan to devour ___.]]

A key property to observe is that *devour* in English is a transitive verb that always requires its object. Nonetheless, the sentences in (1) show no overt object of devour; there is a gap there instead. But in relative clauses (like in questions) the interpretation of the gap is not in question. Here, we know that the thing that is being planned to be devoured is the *feast*. So relative clauses leave a gap inside of them, which is interpreted as the same element as the head noun that the relative clause is used to modify. This section explores the properties of relative clauses and shows how they parallel the other A′-constructions that we have previously explored (such as *wh*-questions).

14.1.1 *Wh*-relative clauses

As (2) and (3) show, there is a striking parallel in English between *wh*-relative clauses and indirect questions: both are introduced by *wh*-phrases. In each of them we also annotate a gap that appears inside the clause: whatever thematic position the *wh*-phrase relates to is empty in a question and in a relative clause.

(2) Indirect question:
  a. We can't remember *who* ___moved in next door.
  b. We can't remember *who(m)* you met ___.
  c. We can't remember *where* you met them ___.

d. We can't remember *which* you prefer ___.
e. We can't remember *whose parents* you met ___.

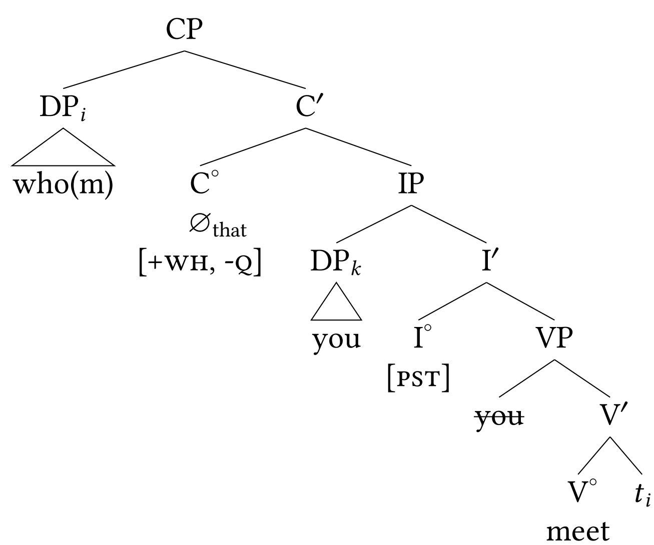
(3) Relative clause:

a. the people *who* ___moved in next door.
b. the people *who(m)* you met ___.
c. the place *where* you met them ___.
d. the movie *which* you prefer ___.
e. the friend *whose parents* you met ___.

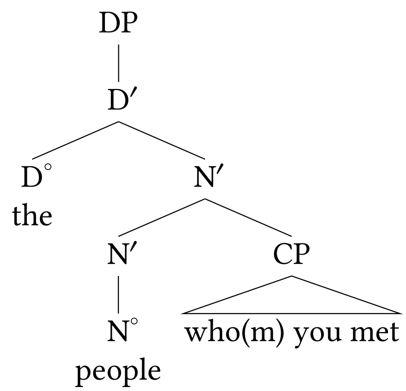
This parallel follows straightforwardly if we assume that *wh*-relative clauses are structurally parallel to indirect questions. The structure of the relative clause in (3b) is given in (4a): a *wh*-phrase raises from its base position to Spec,CP of a C° bearing a *wh*-feature (but, crucially, not a Q feature, as relative clauses are not questions). In contrast to indirect questions, which are ordinarily complements, relative clauses are always modifiers. We therefore integrate the relative clause into the surrounding syntactic structure by adjunction, as in (4b) (where we collapse the internal structure of the relative clause for simplicity).

Note that we annotate C° in relative clauses as [+WH, -Q]; recall from Chapter 8 that C° heads are where our clause-typing features are located. Relative clauses do have *wh*-properties and therefore are [+WH]. On the other hand, they are not questions, and therefore are marked as [-Q]. As with all such components of our theory, we mark them here as we are establishing our model, but we don't necessarily overtly draw the relevant features into every tree unless relevant for the discussion that is underway.

(4) a.



b.



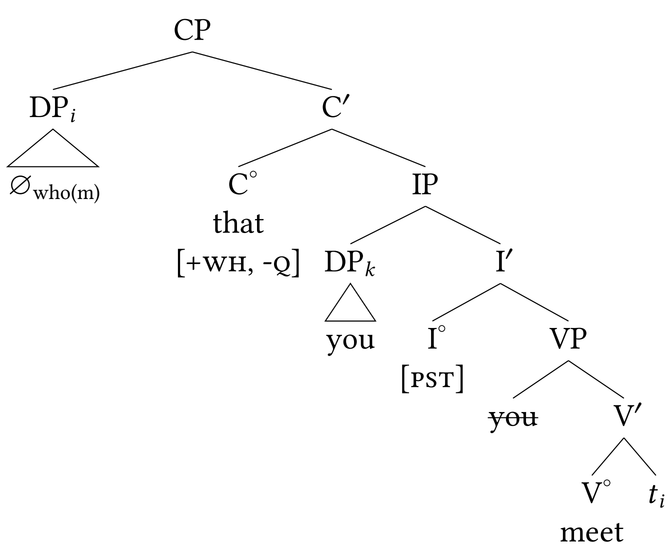
14.1.2 *That* relative clauses

As we noted above, in addition to *wh*-relative clauses, English also has *that* relative clauses, as illustrated in (5).[1]

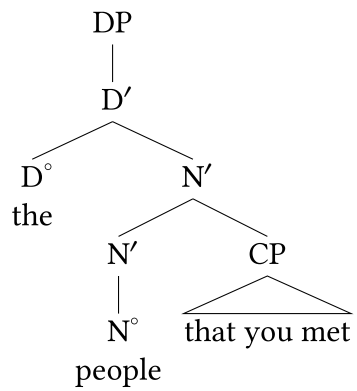
(5) a. the people *that* ___moved in next door.
b. the people *that* you met ___.
c. the place *that* you met them ___.
d. the movie *that* you prefer ___.

Structurally, *that* relative clauses are parallel to *wh*-relative clauses: they share a distribution, and they show the same gap inside the relative clause that *wh*-relative clauses have. But in contrast to *wh*-relative clauses, there is an overt complementizer *that* in the *that* relative clauses, and the *wh*-phrase is silent. The structures corresponding to the ones in (4) are given in (6).

(6) a.



b.



14.1.3 Zero relative clauses

Given the availability of silent *wh*-elements and silent complementizers in English, we would expect to find relative clauses that are not introduced by any overt element at all. Such **zero relative clauses** (also known as **contact relative clauses**) are indeed possible in English, as shown in (7).[2]

(7) a. the people ____ you met.
b. the place ____ you met them.
c. the movie ____ you prefer.

Given the grammaticality of (7), we would expect the zero relative counterparts of subject relative clauses like (7a)–(7b) to be fully grammatical and acceptable, but speakers tend to reject examples like (7c), especially when they are presented in isolation.

(8) a. ✓ The people *who* moved in next door were from New York.
b. ✓ The people *that* moved in next door were from New York.
c. ?? The people moved in next door were from New York.

Nevertheless, structurally analogous examples, as in (9), are attested in some varieties of North American English.

(9) a. Everybody ____ lives in the mountains has an accent all to theirself. (Christian and Wolfram 1976, front matter)
b. Three times a day some nurse ____ looks like Pancho Villa shoots sheep cum into my belly. (Hiaasen 1995, 248–249)

A number of such examples are documented in a wide variety of English varieties.[3] The proper analysis of zero subject relative clauses is still being debated and goes beyond the scope of this textbook, so we don't attempt to resolve it here.

14.1.4 Doubly marked relative clauses

Finally, given the discussion so far, we would expect to find the converse of zero relative clauses—namely, ones with an overt *wh*-element in Spec(CP) combined with an overt complementizer, as in (10).

(10) the people *who(m) that* you met

Such doubly marked relative clauses are judged to be unacceptable in mainstream varieties of modern English. However, just like doubly marked indirect questions, they are attested in Middle English, as shown in (40) (once again, the examples are from the Penn-Helsinki Parsed Corpus of Middle English).

(11)
a. thy freend *which that* thou has lorn (cmctmeli.m3, 218.C1.31)
   ‘your friend that you have lost’
b. the conseil *which that* was yeven to yow by the men of lawe and the wise folk (cmctmeli.m3, 226.C2.373)
   ‘the counsel that was given to you by the men of law and the wise people’
c. the seconde condicion *which that* the same Tullius addeth in this matiere (cmctmeli.m3, 228.C1.429)
   ‘the second condition that this same Tullius adds in this matter’
d. for hire olde freendes *which that* were trewe and wyse (cmctmeli.m3, 237.C2.799)
   ‘for her old friends who were loyal and wise’
e. the fire of angre and of wratthe, *which that* he sholde quenche (cmctpars.m3, 308.C2.859)
   ‘the fire of anger and wrath, which he should quench’

Doubly marked relative clauses are also attested in vernacular varieties of other languages. (12) gives examples from Bavarian German.

(12) Bavarian (Bayer 1983, 23, (10a,b)):
a. der Hund *der wo* gestern d’ Katz bissn hod
   the dog who that yesterday the cat bitten has
   ‘the dog that bit the cat yesterday’
b. die Frau *dera wo* da Xaver a Bussl g’gem hod
   the woman who.dat that the a kiss given has
   ‘the woman that Xaver gave a kiss to’

So there may not be some deep insight to be had about the restriction in modern English against doubly-marked relative clauses, other than that modern English doesn’t allow them. Because human language in general clearly allows them. We saw this same conclusion in Chapter 8 when discussing A′-movements of various sorts: many languages have an overtly moved element (ex., a *wh*-phrase, a focus, a topic, etc) that co-occurs with a complementizer in C°, so whatever restriction occurs in modern Mainstream U.S. English in that regard is peculiar to this language at this time.

Constraints on movement 14.2

In all of the examples of A′-movement that we considered in Chapter 8, there was just a single CP: the *wh*-phrase was contained within the CP somewhere and moves to the edge of CP, into Spec,CP.

(13) $[_{CP}$ What$_i$ were they reading t_i] ?

But there are many instances where movement appears to be much longer distance, however. In (14) the *wh*-phrase originates as the object of the verb at the bottom of the embedded clause: the *wh*-object moves out of the embedded clause and moves to the edge of the main clause CP.

(14) $[_{CP}$ What$_i$ did he say $[_{CP}$ that they were reading t_i]] ?

But if you examine (15), it is evident that *wh*-phrases can move to the main clause CP out of an extended sequence of embedded clauses. Examples like those in (15) suggest that the structural distance that a *wh*-phrase can move is unlimited, often called **unbounded** (that is, unbounded apart from performance considerations such as limitations on memory: create too many more levels of embedding and it's hard to remember what you are talking about). Movement in (13) is called **local** movement, as distinct from **long-distance** movement in (14)–(15c) (also known as **nonlocal** movement).

(15) a. $[_{CP}$ What$_i$ does she believe
$[_{CP}$ that he said
$[_{CP}$ that they were reading t_i]]] ?
b. $[_{CP}$ What$_i$ are they claiming
$[_{CP}$ that she believes
$[_{CP}$ that he said
$[_{CP}$ that they were reading t_i]]]] ?
c. $[_{CP}$ What$_i$ do you think
$[_{CP}$ that they are claiming
$[_{CP}$ that she believes
$[_{CP}$ that he said
$[_{CP}$ that they were reading t_i]]]]] ?

Surprisingly, however, even very short-distance movement can be stunningly ungrammatical, as shown in (16).

(16) a. They read Jonathan's book.
b. * Whose$_i$ did they read t_i book? (cf. ✓ Whose book did they read?)

Accordingly, beginning in the 1960s, attempts were made to understand the conditions under which *wh*-movement is and isn't possible. Ross (1967) catalogued various contexts where *wh*-movement isn't possible, which he gave the descriptive blanket term **islands** (the idea behind the metaphor being that a *wh*-phrase is marooned—it can move within the island, but not escape off of it).

Ross's work represents significant empirical progress over what was known at the time about *wh*-movement, but from a theoretical point of view, islands are simply a list of syntactic contexts, sharply raising the question of whether it is possible to characterize them and to derive the constraints on movement associated with them in a principled way. This chapter is not intended to summarize the decades of work since Ross's work. Rather, we illustrate Ross's island constraints in § 14.2.1 and present some conclusions/motivations for the existence of island constraints (but we don't attempt to explain all of them in this introductory text).

14.2.1 Islands for movement

Ross (1967) first discovered a number of contexts out of which *wh*-movement is unacceptable. A lot of work has happened since then; the list of islands in (17) summarizes well documented patterns that occur across a broad range of languages.

(17) Islands for movement
- noun complement clauses,
- relative clauses,[4]
- indirect questions (*wh*-islands)
- possessive noun phrases
- coordinate structures
- sentential subjects
- adjunct clauses

Coordinate Structure Constraint

We will begin with the contraint against extracting a single element out of a coordinate structure (i.e., a conjunction: *X and Y*).

(18) a. The children read novels and comic books.
b. * What$_i$ did the children read novels and t_i?
c. * What$_i$ did the children read t_i and comic books?

This restriction is not simply about *and* needing some phonological material on either side of it. It is possible to extract out of an *about* PP, as in (19b).

(19) a. The children read novels about dolphins.
b. What$_i$ did the children read novels about t_i ?

(20) a. The children read novels about dolphins and comic books about superheroes.
b. * What$_i$ did the children read novels about t_i and comic books about superheroes?
c. * What$_i$ did the children read novels about dolphins and comic books about t_i?

One challenge to pondering island constraints is that sentences with movement out of islands are just so confusing. This is kind of the point—our language grammars can have long-distance movement (as the examples above in (15) show clearly), but long-distance movement out of these island configurations is clearly unacceptable. And it's tempting to say that the sentences "just don't make sense" (or something like that). But consider an example like (21):

(21) a. The customer ordered tiramisu and espresso.
b. * [Which dessert]$_i$ did the customer order t_i and espresso?
c. * [Which beverage]$_i$ did the customer order tiramisu and t_i?

It's completely reasonable to think that someone might know, for example, that the customer ordered a dessert and an espresso, but didn't know the particular dessert. We know this is reasonable because we can construct a scenario: *I know that the customer ordered a dessert and an espresso, but I don't know which dessert; which dessert did they order?* But if we attempt to ask it all in one sentence, as in (21b), it is clearly unacceptable. And in fact, such questions are unacceptable in a broad and unrelated group of the world's languages, making this kind of island constraint much too common to be a coincidence: there is clearly something deep about the nature of human language that constrains long-distance movement of this sort.

Complex Noun Phrase Constraint

Next, we illustrate the island character of noun complement clauses (CPs that are complements of Ns). First, recall from (14)–(15c) that long *wh*-movement out of *that*-complement clauses to verbs is grammatical. By contrast, *wh*-movement out of *that*-complements to nouns is not, as shown in (22) and (23). For clarity, in the next few examples, the heads associated with the complement clauses are underlined.

Extra Info

In the rest of this section and for much of the remaining chapter, we indicate unacceptability with asterisks, in accordance with Ross's and Chomsky's assumption that these sentences violate structural constraints. But when discussing approaches that attempt to derive such examples from extragrammatical considerations, we use '#' instead.

Noun complement:

(22) a. They made the claim [that they know the mayor].
b. * Who$_i$ did they make the claim [that they know t_i]?

(23) a. They mentioned the fact [that they had met the mayor].

b. * [Which public figure]$_i$ did they mention the fact [that they had met t_i]?

Particularly striking is the contrast between (22b) and (23b) on the one hand and the essentially synonymous examples in (24) on the other.

(24) a. ✓ Who$_i$ did they claim [that they know t_i]?

b. ✓ [Which public figure]$_i$ did they mention [that they had met t_i]?

It is likewise ungrammatical to move out of a relative clause; this shares the characteristic with the previous example in that both are instances of attempted extraction out of a CP embedded inside a DP (though relative clauses and complement clauses have distinct internal structure).

(25) Relative clause:

a. They met someone [who knows the president].

b. * [Which public figure]$_i$ did they meet someone [who knows t_i]?

There is no difference based on the strategy of relative clause formation: *that* relative clauses likewise are islands for movement:

(26) Relative clause:

a. They met someone [that knows the president].

b. * [Which public figure]$_i$ did they meet someone [that knows t_i]?

Both relative clauses and noun complement clauses form islands for extraction.

Sentential subjects

A sentential subject is an instance where the subject of a sentence is itself a CP (i.e., a sentence, hence the name). (27a) gives a basic example of a sentence with a sentential subject (in brackets). As (27b) shows, it is ungrammatical to extract a *wh*-phrase from inside a sentential subject.

(27) Sentential subject:

a. [That they personally know the president] seems improbable.

b. * Who$_i$ does [that they personally know t_i] seem improbable?

Wh-islands

(28) illustrates the island character of indirect questions, a configuration which is often referred to as a *wh*-island (because a *wh*-phrase in the embedded clause appears to turn the embedded clause into an island). In connection with (28b), it is important to distinguish the two instances of *wh*-movement: that of *which problem* and that of *how*. *Which problem* moves from its original position as complement of *solve* to the Spec(CP) of the complement clause. It is this movement—grammatical on its own—that creates an island for any further *wh*-movement, preventing *how* from moving "off island" to the Spec(CP) of the matrix clause.

Word of Caution

Be sure to interpret *How* in (28b) as modifying the complement verb *solve*, as indicated by the trace, not the matrix verb *forgotten*. In other words, a possible answer to (28b) on the attempted reading is *by Fourier analysis*, but not *by succumbing to Alzheimer's*.

Indirect question (*wh*-island):

(28) a. They have forgotten which problem$_k$ they should solve t_k by Fourier analysis.

b. * How$_i$ have they forgotten which problem$_k$ they should solve t_k t_i ?

Adjunct islands

Finally, we illustrate so-called adjunct islands: it is generally ungrammatical to extract out of a clause that is an adjunct to the main clause. (29a) shows an example of a bracketed adjunct clause modifying the main clause. (29b) attempts to question the object from inside the adjunct clause, but is wholly ungrammatical.

(29) a. The children ate lunch [while the dogs chased squirrels.]

b. What$_i$ did the children eat lunch [while the dogs chased t_i?]

This is a very high-level summary, and notably, none of these restrictions are *explained* here. But they are broadly recognized restrictions/constraints on movement constructions which have been attested in a broad array of the world's languages, suggesting that they tell us something about UG.

14.2.2 Explaining (some) constraints on movement

The field has yet to arrive at a unified explanation for all of the island contexts introduced in § 14.2.1 (so it is not clear whether or not there is a unified explanation).[5] And in fact, the emerging picture may in fact be that there is no unifying explanation for the range of islands above: they may in fact derive from different underlying properties of language. We offer an initial way to think about them here that point us to some relevant properties of human language, but a full exploration goes beyond the scope of this textbook.

Two possible derivations for long-distance *wh*-movement

When we consider examples of long-distance *wh*-movement like (14), repeated in (30), two possible derivations come to mind.

(30) What did he say that they were reading?

On the one hand, the *wh*-phrase might move in one fell swoop from the position where its thematic role is interpreted, however deeply embedded that is, to the sentence-initial Spec(CP) position, yielding a *wh*-movement with a single instance of movement, with one trace and one final landing site, as in (31).

(31) [$_{\text{CP}}$ What$_i$ [$_{\text{IP}}$ did he say [$_{\text{CP}}$ that [$_{\text{IP}}$ they were reading t_i]]]] ?

On the other hand, *wh*-movement might take place in several steps. The first step takes the moved constituent from its original position to the nearest Spec(CP), and each subsequent step takes it to the next higher Spec(CP). This derivation of (30), which involves two steps and involves two distinct traces of movement, is shown in (32).

(32) [$_{\text{CP}}$ What$_i$ [$_{\text{IP}}$ did he say [$_{\text{CP}}$ t_i that [$_{\text{IP}}$ they were reading t_i]]]] ?

This kind of derivation of long-distance *wh*-movement would allow the stable claim that *wh*-movement targets the CP domain, but all *wh*-movements would be relatively local, targeting their most local CP and moving step-by-step in that way.

Derivations as in (32) are known as **cyclic** (the idea being that each successively higher CP forms a separate cycle in the derivation of the entire sentence). Cyclic derivations decompose apparently nonlocal movement into sequences of local movement. Derivations as in (31), which do not decompose nonlocal movement, are accordingly known as noncyclic (often described as movement occurring in **one fell swoop**, using the English idiom meaning ‘all at once’).

Wh-islands show that *wh*-movement is cyclic

Given only grammatical instances of long-distance *wh*-movement like (14)–(15c), repeated here as (33), it is impossible to decide which of the two alternatives just presented is correct, or even whether a choice must be made between them. (The parentheses around traces are meant to annotate the question: does movement go through those positions or not?)

(33) a. [$_{\text{CP}}$ What$_i$ did he say [$_{\text{CP}}$ (t_i) that they were reading t_i]] ?
b. [$_{\text{CP}}$ What$_i$ does she believe [$_{\text{CP}}$ (t_i) that he said [$_{\text{CP}}$ (t_i) that they were reading t_i]]] ?
c. [$_{\text{CP}}$ What$_i$ are they claiming [$_{\text{CP}}$ (t_i) that she believes [$_{\text{CP}}$ (t_i) that he said [$_{\text{CP}}$ (t_i) that they were reading t_i]]]] ?
d. [$_{\text{CP}}$ What$_i$ do you think [$_{\text{CP}}$ (t_i) that they are claiming [$_{\text{CP}}$ (t_i) that she believes [$_{\text{CP}}$ (t_i) that he said [$_{\text{CP}}$ (t_i) that they were reading t_i]]]]] ?

However, the existence of syntactic islands forces us to choose the cyclic alternative. For instance, consider the derivation of the ungrammatical question in (28b), repeated as (34). (Remember to interpret *how* in (34) as modifying *solve*, not *forgotten*, as indicated by the trace.)

(34) * How$_i$ have they forgotten [$_{\text{CP}}$ which problem they should solve t_i] ?

If *wh*-movement were able to occur in one fell swoop, there is no obvious reason why long-distance *wh*-movement in (34) ought to be ungrammatical. But the ungrammaticality of the question can be made to follow if we assume that *wh*-movement is cyclic, moving in multiple shorter steps through more local positions. The conclusion that the field has reached is that *wh*-movement cannot move out of a finite clause without passing through the edge of that clause.[6] The idea is that the relevant position where *wh*-phrases land in the embedded CP is already occupied in (34), and moving a lower phrase past that position is unacceptable.

The current way this is typically modeled in syntactic theory is to claim that finite CPs are phases, and phases constitute barriers to movement unless movement occurs through their edge. Spec,CP is thought of using the metaphor of an "escape hatch"—you can't just move directly out of a finite CP, but you can move to its edge and then continue to further move from that edge-of-CP position. There is a large amount of theoretical work on phases and their predecessors in the theory (which were called *bounding nodes* in *subjacency*, and *barriers*, in various different iterations of generative syntactic theory). The key insight, however, is that while it is possible to move long-distance over multiple CPs, you must use the edge of those CPs as escape hatches: finite CPs cannot simply be passed over by movement. In the absence of an intermediate landing site, *wh*-movement is ruled out as ungrammatical. So the problem with a sentence like (34) is that *how* did not move through the edge of the embedded CP.

Relative clauses pose the same problem

Recall the analysis of relative clauses that we proposed in § 14.1: they are *wh*-constructions involving *wh*-movement to the edge of the relative clause CP.

(35) a. I was just talking to the friend [$_{\text{CP}}$ who$_k$ I was visiting t_k in Paris]
b. Where$_i$ was I just talking to the friend [$_{\text{CP}}$ who$_k$ I was visiting t_k t_i]

Our analysis proposed this to also be the case even in relative clauses where *wh*-phrases are not overt.

(36) a. I was just talking to the friend [$_{\text{CP}}$ $\varnothing_{\text{who}}$ that I was visiting $\varnothing_{\cancel{\text{who}}}$ in Paris]
b. Where$_i$ was I just talking to the friend [$_{\text{CP}}$ $\varnothing_{\text{who}}$ that I was visiting $\varnothing_{\cancel{\text{who}}}$ t_i]

Relative clauses, then, are amenable to an analysis of their island effects based on the same kind of intervention effect: the typical pathway for moving a *wh*-element out of a CP is blocked by another *wh*-phrase.

14.2.3 Open questions: explaining other islands

Despite there being a clear logic to the *wh*-islands being based in a locality constraint, there are a number of examples of islands that don't seem to have anything to do with intermediate *wh*-phrases blocking movement out of a finite CP.

(37) a. That Katanji like lattes is unsurprising. Sentential subject island
b. * What$_k$ is that Katanji likes t_k unsurprising?

(38) a. Elena cooked dinner after Sonia prepared the ingredients. Adjunct clause island
b. * What$_k$ did Elena cook dinner after Sonia prepared t_k?

These are examples of constructions that are consistent islands for movement, but which don't seem to have anything to do with an intervening *wh*-element: the embedded CPs (sentential subject, adjunct) don't have any other *wh*-element inside of them. So whatever it is that makes these islands must be something other than an intermediate *wh*-phrase blocking of *wh*-movement through the edge of that CP.

One idea that has been advanced is that movement is only possible out of an embedded clause that is complement to the main clause predicate. Adjunct clauses and sentential subjects are excluded in this instance, as none of them are complements to the main clause predicate.[7] But even that doesn't explain all the remaining instances of islands. The attempted extraction from a coordinate structure, for example, doesn't seem to have anything to do with an occupied Spec,CP blocking an escape hatch, or with moving out of a non-complement CP (these don't include crossing CPs at all).

(39) Coordinate structure constraint
a. The students love stories about heroic heroes and about villanous villains.
b. * What$_k$ do the students love stories about heroic heroes and about t_k?

So while there do appear to be some potential generalizations (locality/minimality problems induced by occupied escape hatches, extraction from non-complements), there are still open questions.

14.2.4 Takeaway lessons from constraints on movement

So what can we conclude about the properties of A′-movement more generally? First, it is clear that A′-movement can be long distance, though it does appear to proceed cyclically through local CP edges. Second, A′-movement is not unbounded—there are many domains that restrict movement dependencies out of them; the ultimate source of these island constraints is still under investigation. But the existence of islands for movement is much too common in the world's languages to be a coincidence, suggesting something relatively deep about language and/or cognition that generates them.

Wh-in-situ: *wh*-questions without *wh*-movement 14.3

14.3.1 *Wh*-scope

For the purposes of this section, it is convenient to introduce the notion of ***wh*-scope**. For simplicity, we restrict our attention for the moment to questions, and define the notion as in (40).

(40) **Scope** refers to the level of structure that an interpretation applies to.

In the context of *wh*-elements, the scope of a *wh*-phrase in a question is identifiable with respect to some full answer to the question. The *wh*-phrase's scope in the question is the IP that corresponds to the root in the full answer.

In (41), the *wh*-phrases are underlined in red, and their scope is in **bold**.[8]

(41) a. Question: When$_i$ did [$_{\text{IP}}$ **the visitors arrive** $\boldsymbol{t_i}$] ?
Full answer: [$_{\text{IP}}$ The visitors arrived at noon].

b. Question: Who$_i$ do [$_{\text{IP}}$ **the parents think that the children saw** $\boldsymbol{t_i}$] ?
Full answer: [$_{\text{IP}}$ The parents think that the children saw the teacher].

Notice a distinct difference with indirect questions, where the scope of the *wh*-phrase is the embedded clause and not the matrix clause.

(42) a. I wonder why the teachers are so grouchy today.
b. I wonder how they will finish the work.

wh-phrases like *why* and *how* in (42) have embedded scope and not matrix scope: the matrix clause is a declarative statement, so the why question is only interpreted inside and in the context of the matrix declarative.

It is useful to think of the *wh*-phrases in (41) and (42) as marking their scope in the sense that they move from inside their scope to the edge of it. In structural terms, a *wh*-phrase moves to the specifier of the lowest projection that dominates its scope. We will refer to this position as the *wh*-phrase's **scope-marking position** or **scope position**.

14.3.2 *Wh*-in situ

In English, *wh*-phrases generally come to occupy their scope position by *wh*-movement. In other languages, however, *wh*-phrases do not undergo visible *wh*-movement, but instead remain **in situ** (that is, in the position where the *wh*-phrase enters the derivation of the question, whether by substitution or adjunction). Such ***wh*-in situ languages** include Chinese, Hindi, Korean, and Japanese, among many others. Example (43) gives the Japanese counterparts of (41).[9]

Once again, the *wh*-phrases are underlined in red, and their scope indicated in **bold**. As is evident, a *wh*-phrase in a *wh*-in situ language is contained within its scope, rather than moving to the edge of it. (We assume that the question marker か *-ka* is the overt counterpart to the silent complementizer that we have been assuming for English questions; this is why it is represented outside of the IP, in the clause-final position that is expected given the head-final character of Japanese phrase structure.)

Extra Info

The abbreviations used in the glosses are: accusative (ACC), locative (LOC), nominative (NOM), plural (PL), present tense (PRS), question (Q), topic (TOP).

(43) a. Question:

[$_{\text{IP}}$ **見学者は いつ 着きました**] か。
[$_{\text{IP}}$ **kengakusha-wa itsu tsuki-mashita**] ka?
visitor-TOP when arrive-PST Q
'When did the visitors arrive?'

Full answer:

[$_{\text{IP}}$ 見学者は 昼に 着きました] 。
[$_{\text{IP}}$ kengakusha-wa hiru-ni tsuki-mashita] .
visitor-TOP noon-at arrive-PST
'The visitors arrived at noon.'

b. Question:

[$_{\text{IP}}$ **両親は 子供たちが 誰を 見た と 思います**] か。
[$_{\text{IP}}$ **ryōshin-wa kodomo-tachi-ga dare-o mi-ta to omoi-masu**] ka?
parents-TOP child-PL-NOM who-ACC see-PST that think-PRS Q
'Who do the parents think that the children saw?'

Full answer:

[$_{\text{IP}}$ 両親は 子供たちが 先生を 見た と 思います] 。
[$_{\text{IP}}$ ryōshin-wa kodomo-tachi-ga sensei-o mi-ta to omoi-masu] .
parents-TOP child-PL-NOM teacher-ACC see-PST that think-PRS
'The parents think that the children saw the teacher.'

The notable point is that all of these *wh*-phrases do indeed have matrix scope: the main-clause CP in these instances is has the force of a *wh*-question. These are not instances where the scope of the *wh*-phrase is somehow interpreted as a lower structure: these are matrix questions (main-clause questions).

Wh-in-situ raises questions for our account thus far. If CPs designated as *wh*-questions are identified as such via [+Q, +WH], then on our assumptions up until now, *wh*-phrases ought to be attracted to Spec,CP; why does that not happen here? As we will see, *wh*-in-situ opens the door for us to make another significant theoretical modification within our model. Specifically, we can start to think of movement not as relocating a constituent, but as adding an additional representation of that constituent within a syntactic structure.

A Copy Theory of Movement 14.4

14.4.1 A Copy Theory of Movement for *wh*-in-situ

Since one aim of generative syntax is to find the most general theory possible that captures what is shared among all languages, it has been proposed to derive *wh*-in situ questions by *wh*-movement—contrary to superficial appearances. This becomes possible by slightly revising our ideas about movement. Instead of saying that movement leaves a trace, let us say that movement is an instance of copying a phrase, and putting that copy in some other position in the structure. So what we have been representing as in (44a), we can now represent as in (44b).

(44) a. What$_k$ did the children read t_k?
     b. What$_k$ did the children read ~~what~~$_k$?

(45) Copy Theory of Movement:
     Movement in language is an instance where some X° or XP is copied and the copy is placed in some other position in the structure.
     a. Copies are placed in higher positions than the original element.
     b. As a general rule, only one copy is pronounced.

Under the Copy Theory of Movement, we have to explain why language is not full of the same XPs repeated over and over again; here we simply stipulate that normally only one copy is pronounced, but advanced work in syntax has worked out much more precise theories, but are more detailed than would be helpful for us in an intro textbook.[10] One upside of this theory is that it allows us to readily explain the existence of *wh*-movement constructions and *wh*-in-situ constructions: we can simply say that Universal Grammar allows variation with respect to which copy is pronounced in a movement operation. In English, it is the highest copy of *wh*-movement that is pronounced;[11] in *wh*-in situ languages like Japanese, it is the lowest copy. In what follows, we indicate silent copies by striking them through.

(46) a. [$_{\text{CP}}$ Which movie did [$_{\text{IP}}$ the children watch ~~which movie~~]] ?
     b. The parents asked [$_{\text{CP}}$ which movie [$_{\text{IP}}$ the children watched ~~which movie~~]].

(47) a. [$_{\text{CP}}$ ~~どの 映画を~~ [$_{\text{IP}}$ 子供たちは どの 映画を 見ました] か] 。
     [$_{\text{CP}}$ ~~dono eiga-o~~ [$_{\text{IP}}$ kodomo-tachi-wa dono eiga-o mi-mashi-ta] ka] .
     child-PL-TOP which movie-ACC see-PST Q
     ‘Which movie did the children watch?’
     b. 両親は [$_{\text{CP}}$ ~~どの 映画を~~ [$_{\text{IP}}$ 子供たちが どの 映画を 見た] か]
     ryōshin-wa [$_{\text{CP}}$ ~~dono eiga-o~~ [$_{\text{IP}}$ kodomo-tachi-ga dono eiga-o mi-ta] ka]
     parents-TOP child-PL-NOM which movie-ACC see-PST Q

聞いた。
kii-ta.
ask-PST
'The parents asked which movie the children watched.'

On this approach, both *ex situ wh*-constructions and *in situ wh*-constructions are in fact movement constructions. In one, we see the result of the movement overtly in the surface pronunciation (*ex situ*); in the other, we don't see the movement in the pronunciation (*in situ*).

Therefore, what we have previously considered to be "traces" of movement (annotated as *t*) are, under the Copy Theory of Movement, unpronounced copies of moved elements. That said, many syntacticians often still use traces to annotate trees and examples because they are visually unobtrusive; in these instances, a trace is therefore interpreted to stand in for an unpronounced copy of a moved element.

Word of Caution

For convenience, we will continue to use the terms "*wh*-movement" and "*wh*-in situ" to refer to languages, questions, and so on, in which the highest and lowest copies of a *wh*-phrase are pronounced, respectively.

14.4.2 Pronouncing multiple copies in *wh*-movement

Independent evidence in favor of the Copy Theory of Movement comes specific constructions in specific languages where multiple copies of a single constituent are pronounced. For instance, some speakers of German reject the examples in (48), while accepting the corresponding ***wh*-copy questions** in (49) as grammatical. These are here marked as % to recognize that they are not fully grammatical for all German speakers.

(48) a. % [$_{\text{CP}}$ Wen glauben die Besucher, [$_{\text{CP}}$ dass sie gesehen haben?
who-ACC believe the visitors that they seen have
'Who do the visitors believe that they saw?'

b. % [$_{\text{CP}}$ Wen glaubst du, [$_{\text{CP}}$ dass Martin meint, [$_{\text{CP}}$ dass du magst?
who-ACC believe you that Martin thinks that you like
'Who do you believe that Martin thinks that you like?'

(49) a. % [$_{\text{CP}}$ Wen glauben die Besucher, [$_{\text{CP}}$ wen sie gesehen haben?
who-ACC believe the visitors who-ACC they seen have
'Who do the visitors believe that they saw?'

b. % [$_{\text{CP}}$ Wen glaubst du, [$_{\text{CP}}$ wen Martin meint, [$_{\text{CP}}$ wen du magst?
who-ACC believe you who-ACC Martin thinks who-ACC you like
'Who do you believe that Martin thinks that you like?'

The overt repetition of the *wh*-phrase in (49) lends strong support to the idea that *wh*-phrases move through Spec(CP) in long-distance movement. In addition, it supports the idea that grammars can differ as to which copies of movement are pronounced. For the speakers of German who speak the (49) variety, multiple copies of the moved *wh*-phrase may be pronounced.

14.4.3 Partial *wh*-movement

Certain languages exhibit yet a further variant of *wh*-movement in which the *wh*-phrase undergoes *wh*-movement, but to a position lower than its scope position. The scope position itself is occupied by a distinct *wh*-phrase (generally the language's counterpart of *what*). This phenomenon, known as **partial *wh*-movement**, is illustrated for German in (50).[12] The true (= contentful) *wh*-phrase is underlined in red, and the scope-marking *wh*-phrase is **in blue boldface**, which can be considered a *wh*-expletive, filling an obligatory position (in German).

(50) a. **Was** glauben die Besucher, wen $\emptyset_{dass}$ sie ~~wen~~ gesehen haben?
what believe the visitors who-ACC they seen have
'Who do the visitors believe that they saw?'

b. **Was** glauben die Besucher, mit wem $\emptyset_{dass}$ sie ~~mit wem~~ gesprochen haben?
what believe the visitors with who-DAT they spoken have
'Who do the visitors believe that they talked with?'

For comparison, (51) shows the full *wh*-movement counterparts of (50), where the *wh*-phrase functions as a scope marker on its own.[13]

(51) a. Wen glauben die Besucher, ~~wen~~ $\emptyset_{dass}$ sie ~~wen~~ gesehen haben?
who-ACC believe the visitors they seen have
'Who do the visitors believe that they saw?'

b. Mit wem glauben die Besucher, ~~mit wem~~ $\emptyset_{dass}$ sie ~~mit wem~~ gesprochen haben?
with who-DAT believe the visitors they spoken have
'Who do the visitors believe that they talked with?'

The scope of the *wh*-phrase needs to be indicated in one of the two ways just presented. The absence of scope marking results in severe ungrammaticality, as shown in (52).

(52) a. * Glauben die Besucher, wen $\emptyset_{dass}$ sie ~~wen~~ gesehen haben?
believe the visitors who-ACC they seen have
Intended meaning: (51a)

b. * Glauben die Besucher, mit wem $\emptyset_{dass}$ sie ~~mit wem~~ gesprochen haben?
believe the visitors with who-DAT they spoken have
Intended meaning: (51b)

14.4.4 Resumptive pronouns as pronunciation of multiple copies

The *wh*-questions above are not the only instances where we find overt pronominal forms where we might have expected traces (i.e., unpronounced lower copies) to occur. One classic example comes from resumptive pronouns. Resumptive pronouns are instances where pronouns occur where there might otherwise be expected to be a gap.[14] (53) gives an example where a pronoun appears in the expected gap of a relative clause.

(53) There are guests who I am curious about what they are going to say.

Mainstream U.S. English doesn't regularly permit sentences like (53), though some speakers do allow them (including authors of this text). Other languages require them; the resumptive pronouns are circled in the examples below.

(54) a. Irish (McCloskey 1990)
an ghirseach ar ghoid na síogaí í
the girl COMP.PST stole the fairies her
'the girl who the fairies stole away'

b. Hebrew (Borer 1984)
ha-ʔiš še- raʔ ʔoto
the-man COMP see.PST.AGR him
'the man that I saw'

There are a wide variety of analyses of such constructions depending on the language, but at least one is that the lower pronoun is the pronunciation of the lower copy of the *wh*-chain. At the very least, constructions like those in (54) show overtly the validity of an approach that assumes mental representations of syntactic content at the positions where we place unpronounced copies/traces. This fits broadly with our model assuming consistent argument positions in syntax (UTAH), but it is always nice to find overt confirmation. But it is especially conducive to an approach like the Copy Theory of Movement, wherein lower copies may either be pronounced directly (as in the *wh*-copying constructions above) or pronounced in a reduced pronominal form, as with resumptive pronouns.

14.4.5 Pronoun copying in A′-movement

We see additional kinds of patterns crosslinguistically that are likewise readily analyzed as pronunciation of multiple copies in a resumption construction. One example comes from Seereer, an Atlantic language from the Niger-Congo language family that is spoken in Senegal in West Africa. Sereer has SVO word order in (neutral) declarative sentences (55). Questions move *wh*-phrases to the left periphery (56).

(55) Jegaan a jaw-'-a maalo fe. Seereer
Jegaan 3SBJ cook-ST-DV rice DET
'Jegaan cooked rice.' (Baier 2018, 6)

(56) Xar$_i$ Jegaan a jaw-'-u t_i ?
what Jegaan 3SBJ cook-PST-EXT
'What did Jegaan cook?' (Baier 2018, 6)

Baier (2018) glosses the final suffix on verbs as either DV (default vowel) or as EXT (for *extraction*), which is a form that appears on the verb in the context of A′-extraction. As Baier notes, the final suffix *-u* appears in various sorts of A′-movement, not just *wh*-questions.

(57) Final suffix *-u*

a. Focus fronting: (Baier 2018, 7)

Maalo Mataar a jaw-u.
rice Mataar 3SBJ cook-EXT
'Mataar cooked RICE.'

b. Relative clause: (Baier 2018, 7)
Maalo fe [$_{CP}$ Mataar a ci'-uu-n-a].
rice DET Mataar 3SBJ cook-EXT-3OBJ-REL
'the rice that Mataar gave him'

Notably, long-distance *wh*-movement in Seereer leaves visible morphology at the edges of the CPs that it moves through. Baier calls these pronominal forms Left Edge Resumptives (LERs). In the examples below we have circled the LERs.

(58) a. Xar$_i$ xalaat-o [$_{CP}$ yee ten$_i$ Jegaan a ga'-u t_i]?
what think-2SG.SBJ.EXT COMP 3SG Jegaan 3SBJ see-EXT
'What do you think Jegaan saw ?'

b. Xar$_i$ xalaat-o [$_{CP}$ yee ten$_i$ Yande a lay-u [$_{CP}$ yee ten$_i$ Jegaan
what think-2SG.SBJ.EXT COMP 3SG Yande 3SBJ say-EXT COMP 3SG Jegaan
a jaw-u t_i]?
3SBJ cook-EXT
'What do you think Yande said Jegaan cooked ?' (Baier 2018, 12)

Here, we have to assume the kind of complex CPs that we introduced at the end of Chapter 13. The complementizer *yee* occupies a head slightly higher than the position that *wh*-phrases move through.

As Baier demonstrates, different kinds of *wh*-phrases leave different kinds of LERs, including *maaga* 'where', *neen* 'how', and *den* 'who (pl)'.

(59) a. Tam$_i$ a xalaat-u [$_{CP}$ yee *(maaga$_i$) ret-o t_i]?
where 3SBJ think-EXT COMP LER go-2SG.SBJ.EXT
'Where does he think you went?'

b. Nam$_i$ xalaat-o [$_{CP}$ yee *(neen$_i$) otew oxe a war-u acek
how think-2SG.SBJ.EXT COMP LER woman DET 3SBJ kill-2SG.SBJ.EXT chicken
ale t_i]?
DET
'How do you think the woman killed the chicken?' (Baier 2018, 14)

(60) Aniini foog-o [$_{CP}$ yee *(den$_i$) ndet-u Dakar t_i]?
who.PL think-2SG.SBJ.EXT COMP LER.3PL go.PL-EXT Dakar
'Who all do you think went to Dakar?' (Baier 2018, 14)

Several things are notable here. One, these are instances where the putative lower copies of the *wh*-phrase are not produced in their exact form, but are instead realized as pronominal resumptives. And in long-distance movement, these copies are again exactly where we expect to see traces of movement if long-distance movement occurs successive-cyclicly through the left edge of CPs. Here we assume a slightly more complex CP, with projections both for the *yee* complementizer and also below that for *wh*-phrases to move through, but this fits with our expectations of a decompositional CP (see § 13.6 of Chapter 13).

Baier (2018) argues that these are in fact movement constructions: they are prohibited out of islands. This is demonstrated for an adjunct island in (61) and a relative clause in (62).

(61) Adjunct island:

* Xar$_i$ ret-o Dakar [$_{CP}$ balaa ten$_i$ Jegaan a jik-u *t*$_i$]?
what go-2SG.SBJ.EXT Dakar before LER Jegaan 3SBJ buy-EXT
Intended 'What did you go to Dakar before Jegaan bought ?' (Baier 2018, 18)

(62) Complex noun phrase:

* Xar$_i$ ga'-o [$_{DP}$ otew oxe [$_{CP}$ ten$_i$ jik-na *t*$_i$]]?
what see-2SG.SBJ.EXT woman DET 3SG buy-C$_{REL}$
Intended: 'What did you see the woman who bought ?' (Baier 2018, 18)

14.4.6 Child English

Although adult Mainstream US English exhibits neither *wh*-copy questions or partial *wh*-movement, both types of questions have been reported for child English (Thornton 1995, 187), as illustrated in (63) and (64).

(63) a. Who do think who Grover wants to hug?
b. Who do you think who's in there, really, really, really?

(64) a. What do you think what Cookie Monster eats?
b. What do you think what the baby drinks?

Interpreting these effects is not straightforward, of course: how/why do children mistakenly do this? It is only a mistake, of course, because the adult English they are learning doesn't possess such structures. But linguists take the emergence of such structures in child language to show what kinds of grammatical hypotheses children are entertaining for their languages. The occasional use of such structures by English-learning children—even when it is not in the adult English they are learning from—suggests that something about UG and/or human cognition is 1) inclined to put *wh*-phrases at intermediate stages of movement, and 2) includes multiple copies of *wh*-phrases in the mental grammatical representation of *wh*-questions.

So it is notable that the precise position that *wh*-copies show up in (non-adult-like) child speech is precisely the same structural positions where they appear cross-linguistically, which is precisely the position that we think intermediate movement steps of long-distance *wh*-movement targets: Spec,CP of the embedded CPs. Not all languages repeat their *wh*-phrases, but when we do see it, we see it precisely in the position that our model posits as an escape hatch for moving out of finite CPs.

New Empirical Observations

- relative clauses share A′-properties with other kinds of A′-movement
- there are constraints on movement, even apparent long-distance movement. These constraints are called **islands**
  - noun complement clauses
  - relative clauses
  - indirect questions (*wh*-islands)
  - coordinate structures
  - sentential subjects
  - adjunct clauses
- *wh*-in-situ
- some languages have constructions in which multiple copies of *wh*-elements are realized (*wh*-copying and resumption)

Components of the Theoretical Model

- phases explain some aspects of islands
  - CP is a phase
  - DP is a phase
  - movement out of a phase occurs through the specifier of the phase (an "escape hatch")
- Copy Theory of Movement

Notes 14.5

1. The *that* counterpart to (3e) is missing; it would look as in (i).

   (i) the friend ___'s parents that you met

   The intended interpretation is as in (3e). In other words, (i) might go on as in (ii.a), but not as in (ii.b), where the relative clause is *that you met*, not the intended ___*'s parents that you met.*

   (ii) a. The friend [___'s parents that you met] is coming over for dinner.
   b. The friend's parents [that you met] are coming over for dinner.

2. Once again, the counterpart to (3e) is missing for reasons analogous to those in Note 1; just delete *that* in the examples there.

3. (65)–(68) give further examples from several varieties of English, classified by linguistic environment.

(65) Existential *there* clause:

a. There's a shortcut ___ takes you to the shops. (Henry 1995, 125)

b. "Thanks for the hurricane, there's a hundred fifty thousand houses in Dade County ___ need new roofs", he began. (Hiaasen 1995, 110–111)

c. … it might be worthwhile to mention that there's a train ___ leaves Pangbourne, I know, soon after five … (Jerome 1889, 207)

(66) *It* cleft:

a. It was John ___ told us about it. (Henry 1995, 125)

b. 'Tis grace ___ hath brought me safe thus far ("Amazing Grace")

c. It was the quickness of the hand ___ deceived the eye. (Jerome 1889, 221)

(67) Copular construction:

a. John is the person ___ could help you with that. (Henry 1995, 125)

b. You're the second guy this month ___ wants to take out trade in this bizzare fashion. (Wagner 1986, 119)

c. He's the one ___ inspected the damn things. (Hiaasen 1995, 5)

(68) Introduction of discourse entity in object position:

a. I met a man ___ can speak five languages. (Henry 1995, 125)

b. how come we have … a pink-haired punk granddaughter ___ got the manners of a terrorist? … Wears somethin' ___ makes the garage door flap up? (Wagner 1986, 81)

4. Ross (1967) treated relative clauses together with noun complement clauses, subsuming the noun phrases containing them under the category of complex noun phrases. We follow more modern treatments by distinguishing the two cases.

5. Boeckx (2012) makes this clear, and to our estimation no conclusions in the field since then have changed this conclusion.

6. Chomsky arrived at and modified this conclusion a number of times over a wide range of work, ex., Chomsky (1973), Chomsky (1977), Chomsky (1986a), and Chomsky (2001)).

7. For discussion of these approaches, see Huang (1982) and Stepanov (2007).

8. In questions with a single *wh*-phrase, the scope of the *wh*-phrase coincides with the question's ground (recall the discussion of *it* clefts in Chapter 3, § 3.1.4). However, the notions of scope and ground aren't identical, as they don't coincide in more complicated questions with multiple *wh*-phrases (questions of the type *Who saw what?*).

9. In Japanese, subjects and topics of sentences are marked by が *-ga* and は *-wa*, respectively. Topics are generally restricted to matrix clauses. When subjects function as topics, as they often do, topic marking overrides subject marking (in other words, the sentence contains no *ga*-marked phrase). When nominative case marking is at issue, linguists use subordinate clauses to avoid the masking effect of topic marking, but we are not interested in case marking here, so we are free to use matrix clauses.

10. A prominent example of work about which copies are realized in a movement chain is Nunes (2004); for a curious student, that is a good place to start.

11. Even languages that ordinarily require *wh*-movement allow *wh*-in situ under certain circumstances. For instance, English **echo questions** commonly exhibit *wh*-in situ (see Chapter 8 for examples), and all but one *wh*-phrase is required to remain in situ in English multiple *wh*-questions like (i).

(i) a. Who ~~who~~ saw what?

b. * Who what ~~who~~ saw ~~what~~?

12. A classic reference for partial *wh*-movement is McDaniel (1989), which discusses the phenomenon in German and Romani. Lutz, Müller, and Stechow (2000) is a collection of papers providing a cross-linguistic survey of partial *wh*-movement and *wh*-scope marking more generally.

13. As noted in the previous section, not all German speakers accept long-distance *wh*-movement as in (48).

14. The example in (53) is from McCloskey (2017), who cites Prince (1990), who attributed the examples to Tony Kroch, an author of this text. This section is being authored after his passing so we can't confirm these with the source, though they are also grammatical for Diercks.

Exercises and problems 14.6

Exercise 14.1: Basic practice with relative clauses

Draw trees of the noun phrases in (1), all of which contain relative clauses. You can draw triangles over DPs, PPs, and AdvPs contained inside the relative clauses.

(1) a. the dog that I saw this morning
  b. the student who arrived in the classroom first
  c. the server who brought us the menus
  d. the coffee shop where I work
  e. the ex-boyfriend that I really need to call
  f. the car which I hit
  g. the tree that I planted

Exercise 14.2: Basic practice with relative clauses: pied-piping

Build structures for the noun phrases in (1).

(1) a. the house [in which you live]. (pied piping)
  b. the house [which you live in]. (preposition stranding)

Exercise 14.3: Practice with islands for movement

The example in (1) is a simple object question, which we have analyzed as movement of the *wh*-phrase from the complement of the verb to Spec,CP.

(1) Who did the children visit ___?

If this is an instance of syntactic movement, we expect it to be constrained by islands for movement. Construct sentences that show that the dependency between the *wh*-phrase and a gap are in fact movement using islands for movement. For each of the islands in (2), invent an attempted *wh*-question similar to (1) which places the gap (the base position of the *wh*-phrase) inside an island. If done precisely, the resulting sentences will all be ungrammatical in English (but you can do this without being a native speaker of English; if anything, it might be easier without native speaker judgments interfering).

(2) a. *wh*-island
b. complex DP island (noun complement clauses, relative clauses)
c. adjunct island
d. coordinate structure
e. sentential subjunct

We model one example of a *wh*-island in (3). Here the pronounced *wh*-phrase is in the matrix clause, but its base position is inside an island for movement (a CP with a *wh*-phrase already in its specifier). The result is ungrammatical, which is what is expected for *wh*-movement.

(3) * Who do you wonder [$_{\text{CP}}$ why the children visited ___] ?

Come up with another kind of *wh*-island for your answers above, and also come up with examples of the other islands, using the discussion in the chapter for guidance. The goal is to practice constructing islands for diagnostic purposes, because islands are often used to diagnose whether a syntactic dependency behaves like movement or not.

Exercise 14.4: Using islands for analytical purposes

According to the analysis of relative clauses presented in this chapter, *that* relative clauses like (1) have the structure in (2a). But since the *wh*-phrase is silent, an alternative analysis is possible in principle, according to which the *wh*-phrase remains in its original position, as shown in (2b).

(1) the people *that you met*

(2) a. the people [$_{\text{CP}}$ ~~who~~$_i$ that [$_{\text{IP}}$ you met t_i]] (movement)
b. the people [$_{\text{CP}}$ that [$_{\text{IP}}$ you met ~~who~~$_i$]] (no movement)

Consider the evidence given in (3). Based on this evidence, which analysis in (2) is preferable? Explain. We use trace notation to represent the interpreted position of the location phrase. So the sentence is grammatical if asking where the visit occurred, but ungrammatical if asking where the original meeting occurred.

(3) a. Where$_k$ did you visit t_k the people that you met?
b. * Where$_k$ did you visit the people that you met t_k?

Exercise 14.5: Structural ambiguities with relative clauses

A. The noun phrase (1) is structurally ambiguous. Paraphrase or otherwise distinguish between the two interpretations.

(1) the cat that ate the rat that ate the cheese

B. Build the structure for the first relative clause, the unambiguous portion (*the cat that ate the rat*).

C. Build structures for the two interpretations, indicating clearly which structure goes with which interpretation. For simplicity, you can use triangle notation for the relative clauses.

Exercise 14.6: Ambiguity of embedded clauses

A. Build structures for the two readings of (1).

(1) What would you like the power to do?

B. Briefly explain how your structures from (A) are consistent with subjacency.

Exercise 14.7: Complex embeddings

The following quote (from *The Prime of Miss Jean Brodie*) exists in several different forms. The one in (1) is the most abstruse.

(1) It's the kind of thing that people who like that kind of thing like.

A. Build the full structure for the relative clause *who like that kind of thing*. Here and throughout, assume for simplicity that *of* phrases are complements.

B. Build the structure for the relative clause *that people who like that kind of thing like*. Use triangle notation for the embedded relative clause that you've already built in (A).

C. Build the structure for the remainder of the sentence (that is, for the matrix sentence excluding both relative clauses).

D. Describe how the structures in (A)–(C) fit together.

Exercise 14.8: Structural ambiguity with relative clauses

The sentences in (1) and (2) are structurally ambiguous. Build partial structures for them, as indicated by the slashes, and explain how those structures fit together to yield the various readings. Focus on the intended reading and the most salient funny reading, but feel free to address other unintended readings as well.

In (2), you will need to come up with a structure for the gerund clause introduced by meeting by extending material covered in Chapter 11.

Treat *freak accident*, *North Koreans* and *Oval Office* as single words.

(1) Gibson plays a man
/ who develops the ability to understand
/ what women are thinking
/ after a freak accident.
(Source: https://www.imdb.com/title/tt0207201/news?year=2000, accessed 18 March 2012)

(2) I remember meeting a mother of a child
/ who was abducted by the North Koreans
/ right here
/ in the Oval Office.
(Source: https://www.slate.com/id/76886/, accessed 8 April 2010)

Problem 14.1: Locality of movement in relative clauses

Consider the sentence in (1a) and the simplified subpart of it in (1b).

(1) a. Already Agassiz had become interested in the rich stores of the extinct fishes of Europe, especially those of Glarus in Switzerland and of Monte Bolca near Verona, of which at that time, only a few had been critically studied.
(Encyclopædia Britannica Online, Agassiz, (Jean) Louis (Rodolphe). Accessed 27 August 1999.)

b. the fishes of Europe, of which only a few had been studied

A. Does the relative clause in (1b) respect our islands for movement, or is this an exceptional instance? Explain.

B. Is *of Europe* a complement or an adjunct of *fishes*? How can you tell?

C. Does *had* in (1b) undergo V-to-I raising? How can you tell?

Problem 14.2: An English relative clause variant

Build the full structure for the noun phrase in (1). For the purposes of this exercise, treat *has* as a modal.

(1) a product that's time has come

Hint: Some speakers have reanalyzed *that* in relative clauses as a relative pronoun instead of a complementizer.

Problem 14.3: Relative clauses with nonfinite clauses

Provide as insightful a description of the following facts as you can. What patterns do you notice? Can you state any generalizations?

(1)	a.	✓	a person who		we can hire
	b.	✓	a person with whom		we can consult
	c.	✓	a person who(m)		we can consult with
	d.	✓	a person who		can fix the sink
(2)	a.	*	a person who		to hire
	b.	✓	a person with whom		to consult
	c.	✓	a person who(m)		to consult with
	d.	*	a person who		to fix the sink
(3)	a.	✓	a person	____	that we can hire
	b.	*	a person with	______	that we can consult
	c.	✓	a person	____	that we can consult with
	d.	✓	a person	____	that can fix the sink
(4)	a.	?	a person	______	for us to hire
	b.	*	a person with	______	for us to consult
	c.	?	a person	______	for us to consult with
	d.	%	a person	______	for to fix the sink
(5)	a.	✓	a person		we can hire
	b.	*	a person with	______	we can consult
	c.	✓	a person	____	we can consult with
	d.	%	a person		can fix the sink
(6)	a.	*	a person		us to hire
	b.	*	a person with	______	us to consult
	c.	*	a person	____	us to consult with
(7)	a.	✓	a person		to hire

b. * a person with ______ to consult
c. ✓ a person ____ to consult with
d. ✓ a person to fix the sink

Problem 14.4: Swahili relative clauses

The data in this problem include data from and adapted from Ngonyani (1999), Keach (1980), and Edelsten, Kula, and Marten (2010).

The textbook gives analyses for multiple types of relative clauses in English, those with relative pronouns and those with a complementizer:

a. the movie which I am seeing this weekend
b. the movie that I am seeing this weekend

Like English, Swahili has multiple strategies for forming relative clauses, which have an influence on word order and agreement facts.

Task:

1. Come up with a clear statement of what the puzzle is, i.e., a description of the Swahili relative clause facts, and what needs to be explained.
2. Provide an analysis that solves the puzzle.
3. Draw trees for the DPs containing relative clauses in (1c) and in (1e).
4. There are multiple analyses possible that both fit our data and fit our theory. If you notice more than one possible analysis, you can describe them both, but you may not have sufficient data to distinguish between them.

Notes:

- REL = relative agreement ; SA = subject agreement, OA = object agreement, PST = past tense; FUT = future tense; numerals = noun classes
- Do not be concerned at present with explaining the presence of object agreement (OA). We will address these kinds of constructions later on in the semester, but at present, assume that OA comes in on V with the verb root.
- The future tense in Swahili has two allomorphs (*-ta-* and *-taka-*) – you are not responsible for explaining their distribution: it's a purely morphological process.
- Assume that subject agreement arises on Infl. This is the result of unvalued features on Infl being valued by the subject.
- Some of the examples here include full sentences, some are just DPs. But most contain DPs with relative clauses. Because you are only responsible for analyzing the relative clauses here, be sure that you focus your investigation properly.
- Class 16 is a locative noun class agreement, agreeing with a locative DP.

(1) a. Wa-vulana wa-ta-i-nunua pombe hapa.
2-boys 2SA-FUT-9OA-buy 9beer 16here
'The boys will buy beer here.'

b. wa-vulana amba-o wa-ta-i-nunua pombe hapa
2-boys amba-2REL 2SA-FUT-9OA-buy 9beer 16here
'the boys who will buy beer here'

c. pombe amba-yo wa-vulana wa-ta-i-nunua hapa
9beer amba-9REL 2-boys 2SA-FUT-9OA-buy LOC16
'the beer which the boys will buy here'

d. * pombe wa-vulana wa-taka-yo-i-nunua hapa
9beer 2-boys 2SA-FUT-9REL-9OA-buy LOC16
'the beer which the boys will buy here'

e. pombe wa-taka-yo-i-nunua wa-vulana hapa
9beer 2SA-FUT-9REL-9OA-buy 2-boys LOC16
'the beer which the boys will buy here'

(2) a. M-geni a-li-ona gari.
1-guest 1SA-PST-see 5car
'The guest saw a car.'

b. gari a-li-lo-ona m-geni
5car 1SA-PST-5REL-see 1-guest
'the car that the guest saw'

c. * gari m-geni a-li-lo-ona
5car 1-guest 1SA-PST-5REL-see

d. gari amba-lo m-geni a-li-ona
5car amba-5REL 1-guest 1SA-PST-see
'the car that the guest saw'

(3) a. M-tu a-li-kwenda.
person 1SA-PST-go
'The person went.'

b. m-tu amba-ye a-li-kwenda
person amba-1REL 1SA-PST-go
'the person that went.'

c. m-tu a-li-ye-kwenda
person 1SA-PST-1REL-go
'the person that went.'

(4) a. Mw-alimu a-li-wa-somea wa-nafunzi ki-tabu.
1-teacher 1SA-PST-2OA-read.to 2-students 7-book
'The teacher read the book to the students.'

b. ki-tabu a-li-cho-wa-somea mw-alimu wa-nafunzi
7-book 1SA-PST-7REL-2OA-read.to 1-teacher 2-students
'The book that the teacher read to the students'

c. * ki-tabu amba-cho a-li-wa-somea mw-alimu wa-nafunzi
7-book amba-7REL 1SA-PST-2OA-read.to 1-teacher 2-students
'The book that the teacher read to the students'

d. * ki-tabu mw-alimu a-li-cho-wa-somea wa-nafunzi
7-book 1-teacher 1SA-PST-7REL-2OA-read.to 2-students
'The book that the teacher read to the students'

(5) a. vi-tunguu wa-li-vyo-vi-nunua wa-talii soko-ni
8-onions 2SA-PST-8REL-8OA-buy 2-tourists 9market-LOC
'the onions which the tourists bought in the market'

b. Soko-ni pa-li-po-jaa wa-tu pa-li-kuwa mji-ni
5market-LOC 16SA-PST-16REL-be.full 2-people 16SA-PST-be 3town-LOC
'The market that was full of people was in the town.'

c. Soko-ni tu-li-po-kwenda pa-li-jaa wa-tu
5market-LOC 1PLSA-PST-16REL-go 16SM-PST-be.full 2-people
'The market that we went to was full of people.'

d. vi-tunguu amba-vyo ni-li-vi-nunua soko-ni
8-onions amba-8REL 1SGSA-PST-8OA-buy 9market-LOC
'the onions which I bought in the market'

e. hapa amba-po wa-vulana wa-ta-pa-nunua bia
here16 amba-16REL 2-boys 2SA-FUT-16OA-buy 9beer
'here where the boys will buy beer'

Problem 14.5: Lopit questions

Data here are drawn from Moodie and Billington (2021).

Lopit is a Nilotic language spoken in South Sudan. The canonical word order in Lopit is VSO. This problem set explores the properties of questions in Lopit. This problem set can be completed on its own, but Exercise 12.3 in Chapter 12 has additional relevant data.

A. Based on the data in (1), arrive at an initial analysis of the basic word order of Lopit.

B. Examine the Lopit questions in (2), and come to an analysis of yes/no questions and *wh*-questions. Draw trees of (2a) and (3a).

C. Explain what outstanding questions these data raise (if any) for your analysis.

Notes:

- You can take note of the inflection on verbal forms and it can factor into your outstanding questions, but you don't have to shape your whole analysis around it. Case is glossed on noun phrases but you do not have to explain it at all (it is its own separate puzzle).
- Some glosses are simplified from the original source for the purposes of this assignment.
- Glossing notes: IPFV = imperfective aspect (an ongoing event); PFV = perfective aspect (a completed event); NOM = nominative case; ABS = absolutive case; PER = persistive affix; AGR = agreement affix; AGR = agreement; SG = singular; PL = plural; Q = question marker; SEQ = sequential marker; SBO = subordinate marker; COMP = complementizer; NEG = negation

Basic word order in Lopit

(1) a. Ɛ-ìl:á xɔ́t:ɔ́ɲí ìŋɛ̂.
AGR-wash.IPFV mother.NOM baby.ABS
'The mother washes the baby.' (256)

b. Á-lâ-dáxá náŋ ŋìɾìjà.
AGR-PER-eat 1SG.NOM food.ABS
'I'm still eating food.'

c. Ɔ-xɔ́ɲ mùnù xìŋɔ̀xʊ̂.
AGR-bite snake.NOM dog.ABS
'The snake bit the dog.'

d. É-xì-rjè íɲɛ́ mùnù.
AGR-PFV-tread.on 3SG.NOM snake.ABS
'He trod on a snake.'

e. É-j:ò xítò.
AGR-cry.IPFV child.NOM
'The child is crying.'

f. Á-ísó náŋ ìjè bùk.
AGR-give 1SG.NOM 2SG.ABS book.ABS
'I give you the book.'

Questions in Lopit

(2) a. Aí-wòló náŋ ìjè ŋɔlɛ́ʔ.
AGR-see.PFV 1SG.NOM 2SG.ABS yesterday
'I saw you yesterday.'

b. X-aí-wòló náŋ ìjè ŋɔlɛ́ʔ?
Q-AGR-see.PFV 1SG.NOM 2SG.ABS yesterday
'Did I see you yesterday?'

c. X-í-mòr íjé nàŋ?
Q-AGR-insult.N 2SG.NOM 1SG.ABS
'Did you insult me?'

(3) a. X-ɪ-ígɛ́m-á íjé ɲò?
Q-AGR-work-IPFV 2SG.NOM what
'What are you doing?'

b. X-ì-fíjá íjé ɲò?
Q-AGR-clean-IPFV 2SG.NOM what
'What are you cleaning?'

c. X-a-ísjéɾé náŋ ŋaì dómí?
Q-AGR-give 1SG.NOM who knife.ABS
'To whom did I give the knife?'

Problem 14.6: Finite complement clauses

At first glance, the structure in (1a) seems preferable to that in (1b) on the grounds that it is simpler in the sense of postulating fewer nodes. It is standardly argued, however, that the structure in (1b) with a silent complementizer is preferable. Provide evidence for the standard view.

(1) a. They believe [$_{\text{IP}}$ we have read *War and Peace*].
  b. They believe [$_{\text{CP}}$ $\varnothing_{\textit{that}}$ [$_{\text{IP}}$ we have read *War and Peace*]].

Problem 14.7: Preposing *though* in English

This exercise extends Exercise 8.7 in Chapter 8. Refer to that exercise for information concerning *though* preposing.

A. The examples in (1)—and in particular, the *though* clauses—are completely acceptable. (The remaining material is given to provide context; it should be ignored for the purposes of the exercise.)

(1) a. ✓ Though I wonder why the problem seems difficult, my friends don't wonder—they know exactly.
  b. ✓ Though the students enjoy problems which are difficult, they can't always solve them completely.

But the corresponding *though* preposing variants in (2) are not.

(2) a. * Difficult though I wonder why the problem seems, ...
  b. * Difficult though the students enjoy problems which are, ...

Explain the ill-formedness of the *though* clauses in (2) in terms of islands for movement.

B. Given our expectations about islands for movement, is the acceptability judgment for (3b) expected? Explain.

(3) a. Though we regret that the problem seems difficult, we will assign it anyway.
  b. * Difficult though we regret that the problem seems, ...

Problem 14.8: Japanese focus particle *-koso*

The following problem and its data are adapted from Sano (2000).

One of the functions of the particle こそ *-koso* in Japanese is to express contrastive **focus**, such as in the following sentence.

(1)

太郎は	藍子に	[$_{\text{CP}}$	女子学生と	お茶こそ	飲んだ	と	]	言った
Tarō-wa	Aiko-ni	[$_{\text{CP}}$	joshi-gakusei-to	ocha-koso	non-da	to	]	it-ta
Tarō-TOP	Aiko-DAT	[$_{\text{CP}}$	female-student-with	tea-FOC	drink-PST	C	]	say-PST

けれども、	お酒を	飲んだ	とは	言わなかった。
keredomo,	osake-o	non-da	to-wa	iw-anakat-ta.
though	sake-ACC	drink-PST	C-TOP	say-NEG-PST

‘Though Tarō said to Aiko that he drank tea with a female student, he didn’t say that he drank sake.’

In this sentence, *-koso* serves to emphasize the contrast between Tarō telling Aiko he drank tea and Tarō telling Aiko he drank sake. In English, we would say the same sentence with extra stress (indicated by underline) on the word ‘tea’; notice that *-koso* attaches precisely to its Japanese equivalent お茶 *ocha*.

This function of *-koso* is licensed by a concessive conjunction like けれども *-keredomo* and が *-ga* ‘but, although’. In (1), this licensing extends across a CP boundary. In fact, this licensing can even extend across two CP boundaries, as in (2).

(2)

僕は	[$_{\text{CP}}$	太郎が	藍子に	[$_{\text{CP}}$	女子学生と	お茶こそ	飲んだ	と	]
Boku-wa	[$_{\text{CP}}$	Tarō-ga	Aiko-ni	[$_{\text{CP}}$	joshi-gakusei-to	ocha-koso	non-da	to	]
I-TOP	[$_{\text{CP}}$	Tarō-NOM	Aiko-DAT	[$_{\text{CP}}$	female-student-with	tea-FOC	drink-PST	C	]

言った	と	]	思う	けれども、	お酒を	飲んだ	と	言った	とは	思わない。
it-ta	to	]	omou	keredomo,	osake-o	non-da	to	it-ta	to-wa	omow-anai.
say-PST	C	]	think	though	sake-ACC	drink-PST	C	say-PST	C-TOP	think-NEG

‘Though I think Tarō said to Aiko that he drank tea with a female student, I don’t think he said that he drank sake.’

However, the *-koso*-concessive licensing relationship doesn’t extend across all kinds of syntactic boundaries. Consider the data below and answer the following questions:

A. What other syntactic phenomenon have we seen that features long-distance relationships like this one?

B. For each example sentence in (3)–(5), name the boundary that the *-koso*-concessive licensing relationship fails to extend across.

C. What other phenomenon have we seen that parallels the inability of *-keredomo* to license *-koso* in (3)–(5)?

D. What important difference exists between the phenomena named in (A) and (D) and the phenomena observed with *-koso* here?

(3) ?* 太郎は 藍子に [$_{\text{DP}}$ 女子学生と お茶こそ 飲んだ 事実を] 話した
Tarō-wa Aiko-ni [$_{\text{DP}}$ joshi-gakusei-to ocha-koso non-da jijitsu-o] it-ta
Tarō-TOP Aiko-DAT [$_{\text{DP}}$ female-student-with tea-FOC drink-PST fact] tell-PST
けれども、 お酒を 飲んだ ことは 話さなかった。
keredomo, osake-o non-da koto-wa hanas-anakat-ta.
though sake-ACC drink-PST thing-TOP tell-NEG-PST
'Though Tarō told Aiko (the fact) that he drank tea with a female student, he didn't tell her that he drank sake with the student.'

(4) * 太郎は [$_{\text{CP}}$ 次郎と お茶こそ 飲む] 女子学生を 見た
Tarō-wa [$_{\text{CP}}$ Jirō-to ocha-koso nom-u] joshi-gakusei-o mi-ta
Tarō-TOP [$_{\text{CP}}$ Jirō-with tea-FOC drink-NPST] female-student-ACC see-PST
けれども、 お酒を 飲む 女子学生は 見なかった。
keredomo, osake-o nom-u joshi-gakusei-wa mi-nakat-ta.
though sake-ACC drink-NPST female-student-TOP see-NEG-PST
'Though Tarō saw a female student who drinks tea with Jirō, he didn't see a female student who drinks sake.'

(5) * 太郎は [$_{\text{CP}}$ 誰と お茶こそ 飲んだ か] 言った けれども、 誰と
Tarō-wa [$_{\text{CP}}$ dare-to ocha-koso non-da ka] it-ta keredomo, dare-to
Tarō-TOP [$_{\text{CP}}$ who-with tea-FOC drink-PST Q] say-PST though sake-ACC
お酒を 飲んだ とは 言わなかった。
osake-o non-da to-wa iw-anakat-ta.
drink-PST thing-TOP say-NEG-PST
'Though Tarō said who he drank tea with, he didn't say who he drank sake with.'

Problem 14.9: Hungarian word order

Hungarian is a Uralic language spoken in Hungary.

A. Read over the data and data descriptions below.

B. Propose a structure explaining the core word order patterns described here about the nature of the preverbal subject position in Hungarian.

C. Draw a tree structure for (1) and (2a).

Given the small data set, there are a variety of solutions that are consistent with the data. Be as precise as you can, given the data, but there may be multiple viable analyses that you cannot decide between.

Hungarian data patterns

Example (1) gives a Hungarian sentence with so-called neutral order.[1]

(1) Emöke látta Attilá-t tegnap este.
    saw -ACC yesterday evening
    'Emöke saw Attila last night.'

In such neutral sentences, which are felicitous as answers to the question *What happened?*, the subject occupies clause-initial position and precedes the verb. By contrast, in *wh*-questions like (2), the clause-initial position is occupied by the *wh*-phrase, and the subject follows the finite verb.

(2) a. Ki-t látott Emöke tegnap este?
    who-ACC saw yesterday evening
    'Who(m) did Emöke see last night?'

  b. Mikor látta Emöke Attilá-t?
    when saw -ACC
    'When did Emöke see Attila?'

It is noteworthy that answers to such questions preserve the constituent order of the question when the answer is intended as an exhaustive answer to the question. In fact, under such an interpretation, the variant without subject-verb inversion is ungrammatical.

(3) a. Attilá-t látta Emöke tegnap este.
    -ACC saw yesterday evening
    'Emöke saw Attila (and no one else) last night.'

  b. Tegnap este látta Emöke Attilá-t.
    yesterday evening saw -ACC
    'Emöke saw Attila last night (and only then).'

(4) a. * Attilá-t Emöke látta tegnap este.

  b. * Tegnap este Emöke látta Attilá-t.

It is ungrammatical to ask a question when other material appears before the verb.

(5) a. * Ki-t tegnap este látta Emöke?
    who-ACC yesterday evening saw
    Intended meaning: 'Who(m) did Emöke see last night?'

  b. * Mikor Attilá-t látta Emöke?
    when -ACC saw
    Intended meaning: 'When did Emöke see Attila?'

Consider the examples below as well, paying particular attention to the word order in the embedded clauses. (Hungarian is a so-called null subject language, so the subjects of finite clauses can be silent, as they are in the matrix clauses in the following examples.)

(6) a.

Azt	mondják,	hogy	Attilá-t	látta	Emöke	tegnap	este.
it	say.3PL	that	-ACC	saw		yesterday	evening

‘They say (it) that Emöke saw (only) Attila last night.’

b.

Azt	mondják,	hogy	tegnap	este	látta	Emöke	Attilá-t.
it	say.3PL	that	yesterday	evening	saw		-ACC

‘They say (it) that Emöke saw Attila last night (only).’

(7) a.

Azt	kérdeztük,	hogy	Attilá-t	látta-e	Emöke	tegnap	este.
it	asked.1PL	that	-ACC	saw-Q		yesterday	evening

‘We asked (it) whether Emöke saw (only) Attila last night.’

b.

Azt	kérdeztük,	hogy	tegnap	este	látta-e	Emöke	Attilá-t.
it	asked.1PL	that	yesterday	evening	saw-Q		-ACC

‘We asked (it) whether Emöke saw Attila last night (only).’

(8) a.

Azt	kérdeztük,	hogy	ki-t	látott	Emöke	tegnap	este.
it	asked.1PL	that	who-ACC	saw		yesterday	evening

‘We asked (it) who(m) Emöke saw last night.’

b.

Azt	kérdeztük,	hogy	mikor	látta	Emöke	Attilá-t.
it	asked.1PL	that	when	saw		-ACC

‘We asked (it) when Emöke saw Attila.’

15

Binding theory

Chapter outline

This chapter is devoted to binding theory, the part of syntactic theory that is concerned with how the interpretation of anaphoric noun phrases is constrained by syntactic considerations. For the purposes of binding theory, it is useful to distinguish several types of noun phrases: **full noun phrases** (*the question*, *the student that asked the question,* and so on), **ordinary pronouns** (*I*, *you*, *they*, and so on), **reflexive pronouns** (*myself*, *yourself*, *themselves*, and so on) and **reciprocal pronouns** (*each other*).

We begin by presenting background concepts concerning reference and related notational conventions. We then summarize the predominant approach to conditions on coreference between pronouns, reflexives, and referential noun phrases: Chomsky's (1981) Binding Theory, reviewing c-command and then addressing Principles A, B, and C. We conclude by discussing ways in which Binding Theory is useful diagnostically to understand unrelated patterns in syntax.

Reference and related notions 15.1

15.1.1 Reference

The preeminent function of noun phrases like *Benjamin Franklin, my two cats, the king of France, Santa Claus,* or *colorless green ideas* is to **refer**—that is, to stand for a particular entity, the **referent**. As the examples show, referents can be entities in the actual world, entities in some possible world, or even entities that could not possibly exist in principle. One of the characteristic features of human language is the absence, in general, of a one-to-one relation between noun phrases and referents. On the one hand, it is possible to use different noun phrases to refer to the same referent. The classic example for this is the fact that the expressions *the morning star* and *the evening star* both have the same referent, the planet Venus (which is not a star at all!). Some families are nickname rich, so the same person could be referred to by a variety of names (e.g. one person could be legally named *Abigail* but also called *Abby*, *Max*, and *Lulu*). On the other hand, the same noun phrase can have different referents. For instance, *my two cats*, used either by the same person at different points in time or by different persons at the same point in time, can refer to distinct feline individuals. Similarly, *my current checking account balance* can refer to vastly differing amounts of money (depending on who says it, and/or when they say it).

In general, then, determining the intended referent of an expression requires recourse to a particular discourse context (who is speaking when, to whom, and so on). The interpretation of certain expressions, however, is particularly context-dependent. The expressions in question are pronouns. For instance, it is perfectly natural to introduce a new topic in a conversation with a friend using (1a) (provided that the speaker and the friend have in common a unique (or at least uniquely salient) acquaintance by the name of Vanessa). But replacing the proper noun *Vanessa* with a pronoun, as in (1b), in the same context is decidedly odd.

(1) a. I ran into Vanessa the other day.
    b. I ran into her the other day.

But if Vanessa has already been mentioned in the discourse, then (1b) is perfectly felicitous, as illustrated in the mini-discourse in (2).

(2) A: Have you seen Vanessa recently?
    B: I ran into her the other day.

Pronouns, then, in contrast to ordinary noun phrases, are **referentially dependent** on some **antecedent** in the discourse that clarifies the referent. The term 'antecedent' is potentially misleading. Since it is derived etymologically from Latin *ante-cedens* 'one who walks before', an antecedent might reasonably be expected to be required to precede a referentially dependent expression. However, contrary to this expectation, precedence is neither a necessary nor a sufficient condition for antecedenthood. In a sentence like (3), for instance, *Zelda* does not precede *she,* yet can still serve as the pronoun's antecedent.

(3) If she calls, tell Zelda that the package arrived.

Conversely, *Zelda* precedes *her* in both examples in (4), but is unable to serve as the antecedent of *her* in (4b) (in other words, (4b) cannot mean 'Zelda likes herself').

(4) a. Zelda believes that nobody likes her.
b. Zelda likes her.

A less misleading term for the notion of antecedent might be 'referential anchor', but we will continue to use the standard term 'antecedent'.

15.1.2 Coreference

A discourse will often contain more than one possible antecedent for a pronoun. For instance, in (5), *he* can refer to either *Preston* or *Victor.*

(5) Preston told Victor that he needed some time off.

In any particular discourse context, of course, one interpretation may well be favored over the other, given what is known about Preston, Victor, their respective work loads, and so on. In case the antecedent for *he* must be explicitly specified in an unambiguous way, this can be achieved as in (6).

(6) a. Preston told Victor that he, Preston, needed some time off.
b. Preston told Victor that he, Victor, needed some time off.

When (5) is given the interpretation in (6a), *he* and *Preston* are said to **corefer**. On the alternative interpretation in (6b), it is *he* and *Victor* that corefer.

15.1.3 Coindexing

It is convenient to introduce a general notational device to represent coreference relations. Let us begin by associating any noun phrase with a **referential index**. In the literature, it is standard to use the letters of the alphabet as indices, beginning with *i* (for 'index': indices [ˈɪndəsiːz] is the plural for index). But as we already use *i*, *j*, *k*, … as indices with another function (associating traces/copies with their associated moved elements), we adopt the convention in this chapter of using the natural numbers as referential indices (ex., 1, 2, 3, etc). Specifically, in order to indicate an interpretation in which two expressions refer to the same discourse entity (that is, in which they corefer), we assign the same index to both expressions. The two expressions are then said to be **coindexed**. On the other hand, in order to indicate an interpretation in which two expressions refer to distinct discourse entities (that is, in which they do not corefer), we assign distinct indices to each of the two expressions. Such expressions are said to be **contraindexed**. In neither case are the specific indices important: the choice of indices is arbitrary. All that matters for this annotation is that DPs bearing the same index refer to the same real-world entity, and DPs bearing different indices refer to different real-world entities. That is, both indexings in (7) represent the interpretation in (6a), and both indexings in (8) represent the interpretation in (6b).

The (a) examples use the numerals the way we will in this chapter as indices, and the (b) examples use Greek letters as indices (chosen based on which ones looked most fancy).

(7) a. Preston$_1$ told Victor$_2$ that he$_1$ needed some time off.
b. Preston$_\Omega$ told Victor$_\Psi$ that he$_\Omega$ needed some time off.

(8) a. Preston$_5$ told Victor$_6$ that he$_6$ needed some time off.
b. Preston$_\Sigma$ told Victor$_\Delta$ that he$_\Delta$ needed some time off.

The point is that the indices themselves aren't meaningful: we could choose any symbol that we want here. The point is that the same index on two elements indicates that those elements corefer, and different indices on two elements indicates that those elements do not share a referent.

(9) gives a further possible indexing for the sentence in (5). Of course, in any particular discourse, this indexing is felicitous only if a discourse entity with the index 3 (say, Preston's brother Landon) has already been mentioned.

(9) Preston$_1$ told Victor$_2$ that he$_3$ needed some time off.

15.2 The puzzle: restrictions on coreference

Take good care to distinguish between reference and indexing. Reference relations are actual linguistic relations that we have intuitions about. For instance, we have the intuition that *Preston* and *him* can corefer in (10a), but not in (10b).

(10) a. Preston thinks that everyone admires him.
b. Preston admires him.

The assignment of indices, on the other hand, is a notational device that is intended to represent arbitrary reference relations; the indices themselves have no independent linguistic or psychological status. As a result, it is perfectly possible to assign referential indices to noun phrases so as to represent interpretations of a sentence that could be imagined to be possible, but which are in fact ungrammatical. Two such ungrammatical indexings are illustrated in (11).

(11) a. * Preston$_1$ admires him$_1$.
b. * He$_1$ admires Preston$_1$.

This makes a notational point: it's possible to assign indices in a way that generates an ungrammatical sentence, since the annotation just marks a pattern of (attempted) coreference. But it also makes an empirical point: a referential DP and a pronoun cannot corefer when occurring in the configurations in (11).

Notice that the proposition that both (11a) and (11b) are trying to express is not inherently semantically anomalous. It can be expressed perfectly grammatically as in (12).

(12) Preston$_1$ admires himself$_1$.

Notice furthermore that what makes the sequences in (11) ungrammatical is the index assignment. The same sequences of words as in (11) are grammatical sentences when associated with the indices in (13).

(13) a. ✓ Preston$_1$ admires him$_2$.
b. ✓ He$_1$ admires Preston$_2$.

In other words, the grammaticality of a sequence is always determined with respect to a particular interpretation.

It is often convenient to combine the information in (11) and (13) as in (14). The examples in (14) list multiple possible coreference possibilities on *him* and *himself*.

(14) a. ✓ Preston$_1$ admires him$_{2, *1}$.
b. ✓ He$_{2, *1}$ admires Preston$_1$.
c. ✓ Preston$_1$ admires himself$_{1, *2}$.

Why can't sentences like (11a) or (11b) express the proposition that is expressed grammatically in (14c)? Such questions are precisely the puzzle that **binding theory** seeks to answer. Before we can discuss an analysis of this puzzle, though, we need to review a core concept.

15.3 Reintroducing c-command

A central component of Binding Theory is the structural relation called **c-command**. This was first introduced in § 11 of Chapter 3, but has not been a central part of our discussion of our model up until this point, and as such we re-introduce it here.

We will use the puzzle from Exercise 3.3 to do so: consider the baseline sentences in (15), the first an active transitive sentence and the second a passive version of that sentence.

(15) a. Jay read seven books last night.
b. Seven books were read last night.

The puzzle here centers on the English expression *any*. As (16) shows, this is ungrammatical when modifying the object of our baseline sentence.

(16) * Jay read any books last night.

It is possible to have *any* in that position, however, if the sentence is negated as in (17). So something about negation appears to license English *any*. This is why *any* is in a class of elements called negative polarity items, items that are licensed by negative polarity.

(17) Jay didn't read any books last night

But not just any instance of negation in a sentence licenses *any*. So the passive sentence in (18) is ungrammatical, despite being negated (we saw a grammatical parallel of this without *any* in (15b) above).

(18) * Any books weren't read last night.

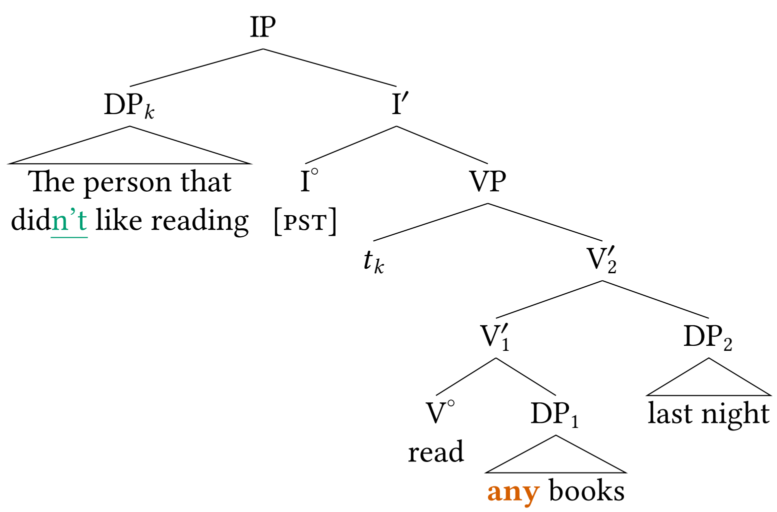
A first pass at this discussion might lead us to conclude that negation must linearly precede *any* in order for *any* to be licensed. But examples like (19) show that this first attempt at a generalization is insufficient; (19b) contains a negation that linearly precedes *any*, but is still ungrammatical.

(19) a. The person that didn't like reading read seven books last night.
b. * The person that didn't like reading read **any** books last night.

So linear precedence is insufficient. What relationship is it that must hold between negation and *any* that is present in (17), but is absent in (19b) and in (18)? The answer is **c-command**:

(20) A **c-commands** B if and only if
a. neither A nor B dominates the other, and
b. the first branching node that dominates A also dominates B.

We can use the simplified tree structure in (21) (sketching the structure of (19a)) to walk through this definition of c-command. So when looking for c-command relationships between two nodes, crucially neither node can dominate the other. So *any* in (21) is not c-commanded by the DP that contains it, because the DP node (DP_1) dominates *any*. Likewise, both V′ nodes, VP, I′, and IP all dominate *any*, and therefore cannot c-command *any*.

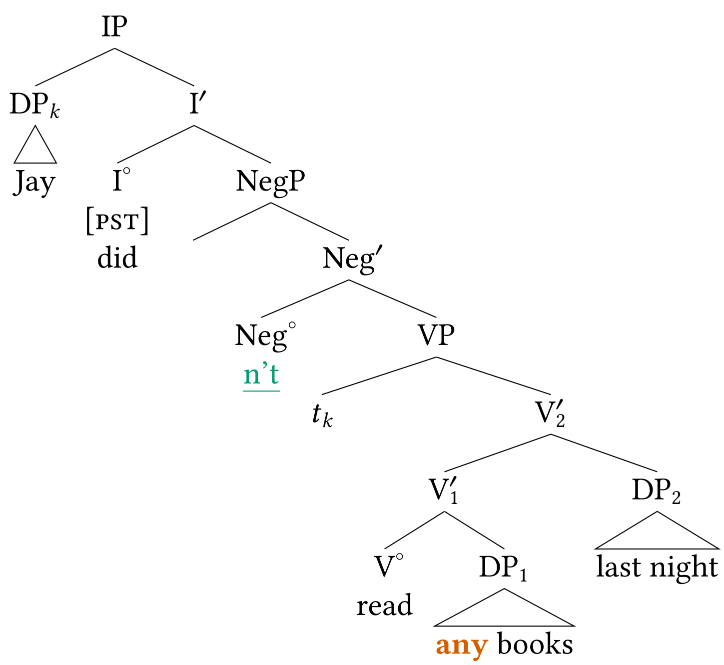
(21)


In contrast, consider the second condition in (20): for A to c-command B, the first branching node dominating node A also dominates B. We can evaluate this for the V° node in (21). The first branching node dominating A is V'_1, and V'_1 dominates *any* (i.e. *any* is contained within V'_1). And, crucially, V° itself does not dominate *any*. Therefore we can say that in (21), V° c-commands *any*. In contrast, V° does not c-command DP_2, but DP_2 c-commands V°.

There are various ways to conceptualize c-command that might be helpful. One is to strictly visualize it—trace your finger from the node in question up to its parent, and then down the other branch—that first node (where your finger started) c-commands anything down the second branch. The familial metaphor may also be helpful. Symmetric c-command is the sibling relationship: two XPs c-command each other. Asymmetric c-command is the uncle/aunt relationship (or great-uncle/aunt, or great-great-uncle/aunt). So in our example, DP_1 is a parent of *any* (or grandparent, depending on the internal structure of DP_1). V° is the sibling of DP_1, and therefore V° is the aunt/uncle of *any*. In general syntacticians don't extend the familial metaphor of trees to this extent, but we mention it in case it helps to conceptualize what c-command is.

So, in the tree in (21), V° c-commands *any* and DP_2 c-commands *any* (but, of course, none of the sub-constituents of DP_2 do). Likewise, I° c-commands *any* and DP_k c-commands *any*, but none of the sub-constituents of DP_k c-command *any*. This gives us a path to explaining the distribution of *any*: **negation must c-command *any***.[2] In (21) the negation is a sub-constituent of DP_k and therefore does not c-command *any*. Compare that to (22), which depicts the structure of (17) above. This sentence is grammatical with *any*, and in this sentence Neg° does in fact c-command *any*: the first branching node that dominates Neg° is Neg′, and Neg′ also dominates *any*.

(22)



This analysis proves accurate: the examples below discuss the indomitable Pomona-Pitzer Sagehens football team. The negative adverb *never* can license *any* in object position, but if *never* is embedded inside a larger constituent, it no longer c-commands *any*, and the sentence becomes ungrammatical.

(23) a. The Sagehens never lost **any** games before today.
b. * The Sagehens always lost any games before today.

(24) a. * The never-beaten Sagehens lost any games before today.
b. * The always-beaten Sagehens lost **any** games before today.

This is an extended discussion of the negative polarity item *any*, but the purpose is to familiarize you with the notion of c-command. C-command plays a central role in binding theory, which we turn to now.

Chomsky's (1981) Binding Theory 15.4

The standard binding theory of Chomsky (1981) was developed mainly based on the pattern of facts from English, but it has proven to hold up quite well cross-linguistically. Chomsky's binding theory contains three conditions (also known as principles), which govern the distribution of reflexive and reciprocal pronouns (Principle A), ordinary pronouns (Principle B), and referential noun phrases (Principle C), respectively. We present each of these principles in turn.

15.4.1 Principle A

Consider the English binding facts in (25). These examples use a self-reflexive (*herself*). Self-reflexives are instances of what are often called **anaphors**, a class that also includes reciprocals like *each other*. In (25) we see that a self-reflexive in object position readily is coreferential with its subject, but not with a subconstituent of that subject.

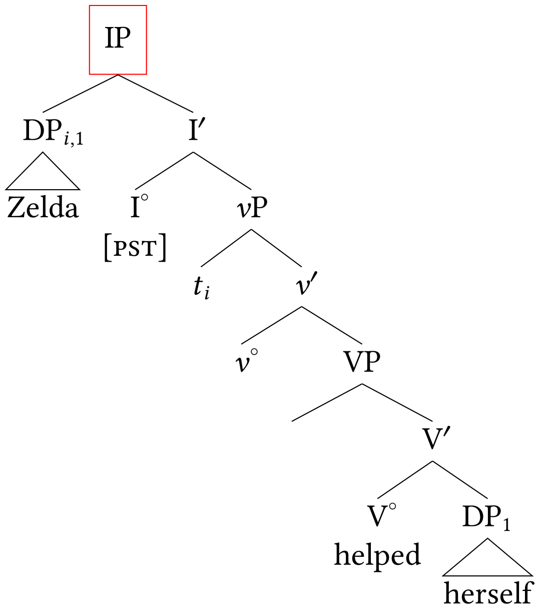
(25) a. [Zelda]$_1$ helped [herself]$_1$.
b. [Zelda's sister]$_1$ helped [herself]$_1$.
c. * [Zelda]$_1$'s sister helped [herself]$_1$.

Word of Caution

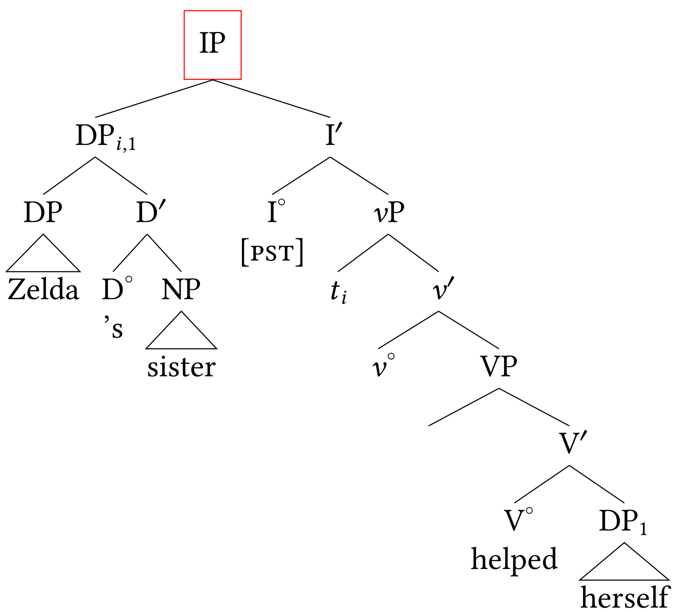
The term **anaphor** in Chomsky's usage refers to reflexive and reciprocal pronouns. This usage is potentially confusing because in general linguistic usage, anaphora refers to referential dependence regardless of morphological form. In other words, ordinary pronouns and even full noun phrases can count as anaphors in this wide sense. Here, we will use the restricted sense of the term, in keeping with Chomsky's usage. But you will often see an expression referred to as *anaphoric*, which often means that it picks up its reference from context in some way, not that it is specifically a reflexive or a reciprocal.

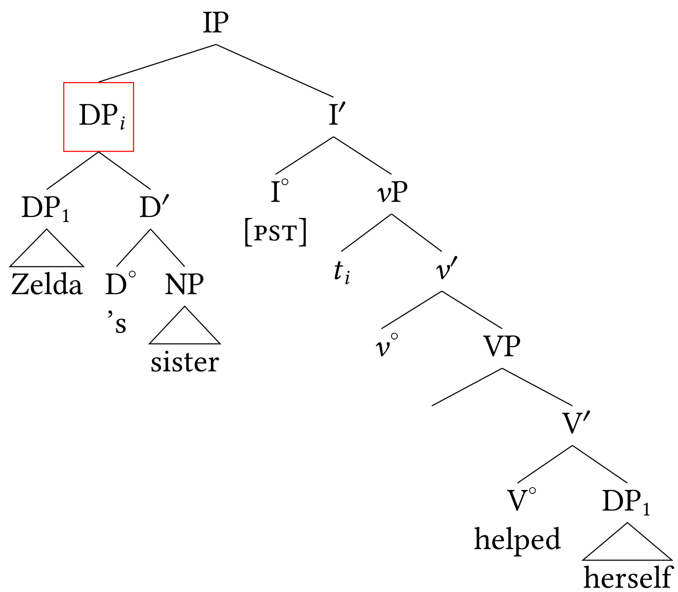
Chomsky (1981) derives the grammaticality pattern in (25) on the basis of c-command, defined above in (20) and explained in § 15.3. The structures for the sentences in (25) are shown in (26). The branching node dominating the intended antecedent is highlighted by a box. Notice now that the red boxed nodes dominates that anaphor in (26a) and (26b), but not in (27). In other words, the anaphor is c-commanded by the intended antecedent in (26a) and (26b), but not in (27).

(26) a.



b.



(27)


These configurational relations suggest the condition on reflexives (and in fact, anaphors in general) in (28).

(28) Principle A (to be revised):
An anaphor must be c-commanded by a coindexed antecedent.

(28) is generally expressed more succinctly as in (29), where the notion of binding is defined as in (30).

(29) An anaphor must be **bound**. (to be revised)

(30) A **binds** B if and only if
  a. A c-commands B, and
  b. A and B are coindexed.

As it turns out, (29) is a necessary but not sufficient condition on anaphors in English, since it fails to account for complex sentences like (31).

(31) Landon$_1$ thinks that Malia$_2$ will help { herself$_2$ / *himself$_1$ }.

The grammaticality of the variant with *herself* is unproblematic (*herself* is bound by *Malia*). But given (29), the ungrammaticality of the variant with *himself* is surprising, because *himself* is bound by *Landon*.

We therefore need to specify a **binding domain**, a locality domain within which an anaphor must be bound. A fully complete definition of binding domains is fairly complex; we will offer an initial version here that we will work our way up to it in several steps, motivating each revision of the definition in turn.[3]

(32) Principle A (final version):

An anaphor must be bound within its binding domain.

(33) Binding domain (to be revised):

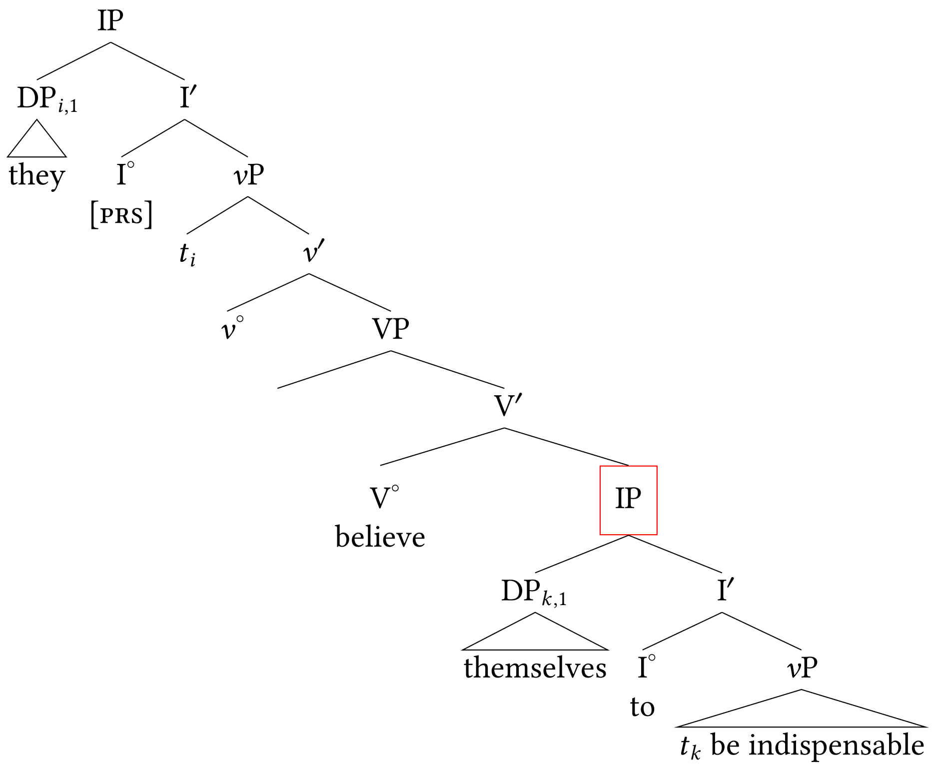
The binding domain for an expression is the lowest IP that contains that expression.

The formulation in (33) is a good initial step—in our example from (31), this restricts the possible (grammatical) antecedent of the embedded object to the subject of the embedded clause, which is within the lowest IP containing the self-phrase. Attempting to have the main clause subject be the antecedent of the embedded self-reflexive violates (32) on the assumption of the binding domain in (33), because the main clause subject is not within the lowest IP that contains the anaphor.

To continue to address our understanding of the binding domain, however, consider (34a), which we can assume to have the structure in (34b).

(34) a. They$_1$ believe themselves$_1$ to be indispensable.

b.



Let's consider in detail why (34) poses a difficulty for the definition in (33). As is evident from (34b), the lowest IP that contains the anaphor is the complement IP (indicated by the box). But this complement IP contains no antecedent for the anaphor, so (34a) is incorrectly predicted to be ungrammatical. But it IS grammatical. So we need a definition of binding domains that can account for both (31) and (34a).

(35) Binding domain (second attempt, to be revised again):

The binding domain for an expression is the lowest **finite** IP that contains that expression.

The redefinition in (35) excludes non-finite IPs, as in (34a), but includes tensed/agreeing IPs as in (31). This therefore ensures that the embedded IP in (34a) does not constitute a binding domain, and only the main clause IP does, explaining why (34a) is grammatical. In contrast, the embedded IP in (31) is itself finite, and therefore a binding domain, explaining why Principle A must be satisfied by an antecedent inside the embedded clause.

As it turns out, we're still not done yet. (35) correctly explains (31) and (34a), but is unable to account for contrasts as in (36).

(36) a. ✓{ She, they, I } bought Preston$_{1}$'s book about himself$_{1}$.
  b. * Preston$_{1}$ bought { her, their, my } book about himself$_{1}$.

In both sentences, the lowest finite IP that contains *himself* (the only IP in the sentence) also contains *Preston*, in accordance with the definition of binding domain in (35). The ungrammaticality of (36b) is therefore unexpected.

A first step in accounting for such contrasts lies in making reference to DPs in addition to IPs in the definition of governing category, as in (37).

(37) Binding domain (to be revised):

The binding domain for an expression is the lowest finite IP <u>or DP</u> that contains that expression.

As a final step (for our current purposes), we can ask what natural class of elements includes finite IPs and DPs. The answer is the **phase**, which includes CPs (which contain finite IPs) and DPs. Previously we saw phases constraining *wh*-movement: movement out of a phase is ungrammatical unless it passes through the edge of the phase (see § 14.2.2 of Chapter 14). Here we can see that they also map onto the relevant domain for calculating binding.

(38) Binding domain (to be revised):

The governing category for an expression is the phase.

An intrepid reader may still yet turn up some interesting details that challenge this generalization, and we don't mean to imply that there aren't apparent exceptions to this that would need to be sorted out. But this is a good enough definition that captures the majority of cases.[4]

15.4.2 Principle B

We turn now to the distribution of pronouns.

Word of Caution

As noted earlier, reflexive and reciprocal pronouns are combined in Chomsky's usage under the rubric of anaphors. This leaves the unqualified term 'pronoun' to refer to ordinary personal pronouns (*I*, *you*, *he*, and so on). Unless otherwise noted, we follow this usage below.

Replacing the anaphors in (25) by pronouns yields the sentences in (39).

(39) a. * [Zelda]$_1$ helped [her]$_1$.
b. * [Zelda's sister]$_1$ helped [her]$_1$.
c. ✓ [Zelda]$_1$'s sister helped [her]$_1$.

The resulting grammaticality pattern, which is the converse of that in (25), suggests the condition in (40).

(40) Principle B (to be revised):
A pronoun must be **free**, that is, not bound by an antecedent.

As in the case of our first formulation of Principle A, the formulation of Principle B in (40) is not yet quite adequate. This time, however, the condition errs on the side of caution, incorrectly ruling out grammatical sentences like (41).

(41) John$_1$ thinks Mary will help him$_1$.

Since *John* c-commands *him* (along with everything else in the sentence), (40) incorrectly leads us to expect (41) to be ungrammatical.

The fact that the antecedent and the pronoun belong to different clauses in (41) suggests reformulating (40) to incorporate the concept of a binding domain, as in (42).

(42) Principle B (final):
A pronoun must be free <u>in its binding domain</u>.

This definition is the inverse of Principle A, which captures the empirical observation that anaphors and pronouns are in complementary distribution. As above, we assume the binding domain to be the phase (finite CPs and DPs, though we haven't discussed DPs as phases much in this text).

15.4.3 Principle C

Consider the sentences in (43). Keep in mind that these are grammatical sentences if the pronouns refer to someone else, but we are specifically evaluating the reading where the pronoun and the embedded nominal *Arianna* are coreferent.

(43) a. She$_{2,*1}$ treats Arianna$_1$ well.
b. She$_{2,*1}$ claims that they treat Arianna$_1$ well.
c. She$_{2,*1}$ claims that we know that they treat Arianna$_1$ well.

It is tempting to blame the ungrammaticality of these sentences on something about the pronoun, since one reading of the pronoun is acceptable, but another is not. But if we view this through the perspective of Principle B, in each of these sentences the pronoun *she* is not bound (it is co-indexed with *Arianna*, but not c-commanded by *Arianna*). So the problem with these sentences cannot be a Principle B violation.

In order to rule out sentences like (43a), Binding Theory contains a third principle that governs the distribution of full lexical noun phrases (referred to in Chomsky's usage as **R(eferential)-expressions**).

(44) Principle C: (final)
R-expressions must be free.

Principle C is reminiscent of Principle B, but differs from it in that the anti-binding requirement is absolute (= independent of a binding domain), as required by the ungrammaticality of (43b) and (43c).

The examples in (43) would all be called instances of **cataphora**, instances of referential dependence where the antecedent follows the referentially dependent element. It is worth pointing out explicitly that not all instances of cataphora are ungrammatical. There are many examples where the referentially dependent element precedes the antecedent that are grammatical. For example, in (45) the pronoun *he* linearly precedes the coindexed nominal *Langston*; crucially, *he* does not c-command *Langston*, since it is embedded in an adjunct clause. Sentences like these are grammatical, as expected under our Binding Principles; since neither element is bound, neither Principle B nor Principle C is violated.

(45) [If he_1 calls,] tell Langston_1 I'll be back in an hour.

Word of Caution

The term 'anaphora' is generally used to refer to referential dependence, subsuming both cataphora and what might be called strict anaphora, where the antecedent precedes the referentially dependent element. According to the metaphor underlying the terms, discourse flows along like a stream, with temporally earlier elements located upstream from later ones. The antecedent is then upstream of the referentially dependent element in strict anaphora (< Greek ἀνα *ana* 'up(stream)'), but downstream in cataphora (< Greek κατά *kata* 'down(stream)').

A final point needs to be made about Principle C. We pointed out earlier that Principle C is an absolute requirement in the sense that it makes no reference to binding domains. It is absolute in a further sense as well: it makes no reference to whether the binder is a referentially dependent element. Thus, Principle C rules out (46) on a par with (43a).

(46) *? Arianna_1 treats Arianna_1 well.

It has been argued, however, that given the right discourse conditions, sentences like (47), which are structurally parallel to (46), are in fact acceptable. Two naturally-occurring examples of the same type are given in (48).

(47) Nobody likes Oscar. Even Oscar_1 doesn't like Oscar_1.

(48) a. Phil_1 said that if he_2 came by, Phil_1 would show him_2 around.
(in an email from manager@ling.upenn.edu to beatrice@ling.upenn.edu)

b. Luke_1 thinks that everyone has as much integrity as Luke_1 has.
(overheard at the White Dog Cafe, Philadelphia, PA, 17 Feb 2001)

But equivalent sentences in which the full noun phrase is bound by a referentially dependent element are ungrammatical.

(49) * Nobody likes Oscar. Even he_1 doesn't like Oscar_1.

(50) a. * He_1 said that if { he_2 / Sean_2 } came by, Phil_1 would show him_2 around.
b. * He_1 thinks that everyone has as much integrity as Luke_1 has.

Although judgments regarding contrasts of the type illustrated by (47)–(50) can be delicate, it seems reasonable to weaken Principle C to include reference to the status of the binder, as is done in (51).

(51) Hypothetical weakened version of Principle C:
Full noun phrases must not be bound by a referentially dependent element.

That said, the sentences in (47) tend to have some kind of special emphatic quality that might suggest that the original version of Principle C is in effect. For our purposes, the principles as we stated them above are sufficient (and most research relies on these statements as described here). But there is an entire universe of research literature on Binding Theory and the relevant patterns of coreference. Some well-studied languages like English, German, Italian, and Spanish have extensive work on them. Most languages, even languages with robust reference grammars written about them, don't have nearly the same level of detailed syntactic research. There remains much work to be done, and we don't mean to imply that Binding Theory and the binding principles noted here are the final word on the issue. But the principles as stated above capture a vast amount of the empirical properties of binding effects in the world's languages.

Binding Theory as a diagnostic 15.5

The properties of binding in the grammar of human languages is obviously a puzzle in its own right, that deserves its own exploration and explanation. But because the patterns captured by Principles A, B, and C are so reliable and consistent, they actually allow us to use binding theoretic principles as diagnostics in the service of other investigations. Specifically, as Binding Theory makes clear, c-command is a central component of binding properties, and c-command is a very specific structural configuration. So we can use the availability (or not) of binding configurations to diagnose c-command relationships between different elements in the syntax.

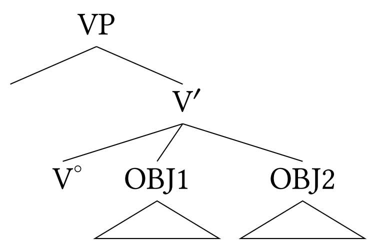
15.5.1 Ditransitives

Binding theory (and related phenomena) played a major role in the theoretical developments that led to the decomposition of the verb phrase into a more complex structure, as we explored in Chapter 13. In the absence of the complex VP approach, ditransitives are troublesome for the X′ schema: if heads have a single complement that is typically their object, how is it possible to have two objects, as in ditransitive verbs?

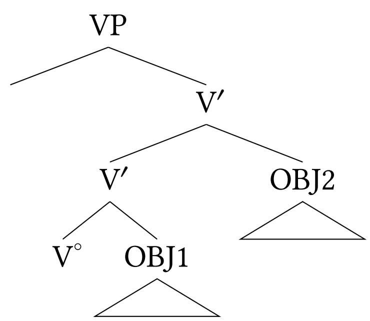
(52) Maisha gave Langston a gift.

We will consider three main alternatives that were under consideration when this was first being worked on: one is a departure from binary branching, allowing for two complements of the verb (53a). The second is an adjunction structure (53b), and the third in (53c) is something closer to the structure we adopted in Chapter 13. There is more complexity to the VP domain than these structures suggest (as we saw in Chapter 13), but we'll use the structure in (53c) as a proxy for any structure where OBJ1 c-commands OBJ2.

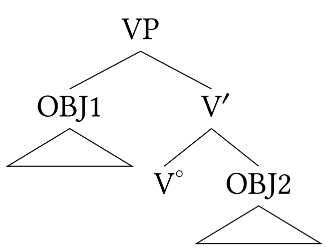
(53) a.



b.



c.



Crucially, each of these have different c-command relationships between the DP objects in a ditransitive. In (53a) the DPs are in a mutual c-command relationship (they each c-command the other); in (53b) OBJ2 asymmetrically c-commands OBJ1, and in (53c) and other structures like it (such as an applicative structure), OBJ1 c-commands OBJ2. This means that operations or constructions that are sensitive to c-command make different predictions for each of the structures above. Binding is one of these, and it played an important role in the development of complex VPs as a theoretical model for ditransitive verbs (Barss and Lasnik 1986; Larson 1988).

As we saw above, reflexives and reciprocals are subject to Principle A, meaning they must be c-commanded by their antecedent. So in a OBJ1-OBJ2 ditransitive predicate, what are the c-command relationships between the two predicates? (54) shows that OBJ1 can have the antecedent for a self-reflexive in OBJ2, but not the other way around.

(54) a. I showed Khydence herself (in the mirror).
b. * I showed herself Khydence (in the mirror).

This suggests an asymmetrical c-command relationship between the DP objects (ruling out the ternary structure in (53a)). And (54) behaves as if OBJ1 c-commands OBJ2, not the other way round; this is consistent with the structure in (53c) and not in (53b).

Reciprocals like *each other* are also anaphors, expected to be subject to Principle A. In the examples below, *the other* is the anaphor that must be bound within its binding domain.

(55) a. I showed each student the other's paper.
b. * I showed the other's friend each student.

This shows us the exact same conclusion as above: OBJ1 behaves like it c-commands OBJ2, but not the other way around: it appears to be an asymmetric c-command relationship.

Another diagnostic is a cousin of the binding principles we referred to above: quantifier-variable binding. In an example like (56), there is a 'bound' reading where the pronoun *their* is interpreted in light of the quantifier: for each child, they love their own mother. We show this reading via co-indexing. The non-bound reading is always available (every child loves some person's mother).

(56) [Every child]$_k$ loves their$_k$ mother.

Crucially, for a bound reading to be possible, the quantifier must c-command the variable. This allows is to create a similar scenario in ditransitive constructions to test the c-command relationships between objects. The examples below make one object a quantifier, and the other contains a variable.

(57) a. I gave [every worker]$_k$ their$_k$ paycheck.
b. * I gave its$_k$ owner [every paycheck]$_k$.
c. [Every paycheck]$_k$ was given to its$_k$ owner.
(based on Larson 1988, (3b))

The example in (57c) is included to show that the meaning in (57b) is perfectly natural; it is the particular configuration of quantifier and variable in (57b) that make it uninterpretable. This pattern confirms what we saw above: if variables must be c-commanded by the quantifier in order to get the relevant reading, these data suggest that OBJ1 asymmetrically c-commands OBJ2.

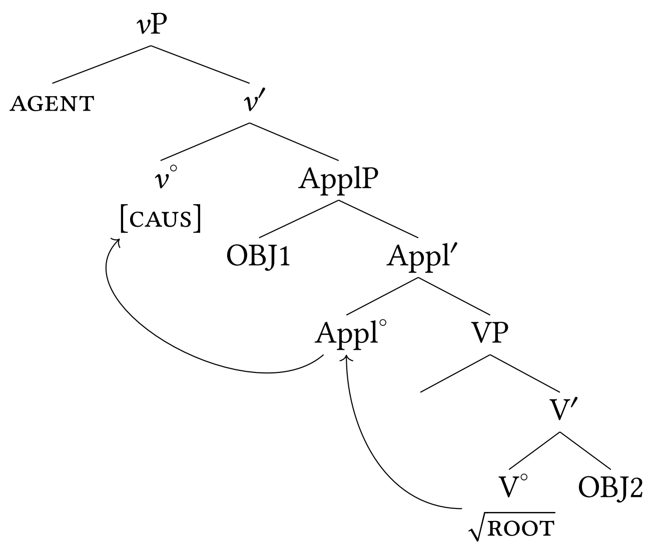
In § 15.3 above we used negative polarity items (NPIs) like *any* to understand c-command, because NPIs need to be c-commanded by negation. We can likewise use that relationship to diagnose the structure of ditransitives.

(58) a. I showed no one anything.
b. * I showed anyone nothing.
(Larson 1988, 337)

This is again consistent with the conclusion from each example above: OBJ1 asymmetrically c-commands OBJ2. This is consistent with the structure in (53c), and inconsistent with either of the other structures.

There are two main take-away lessons here. First is that these patterns reconfirm the conclusions of Chapter 13 regarding the internal structure of the verb phrase; it is always comforting to find converging evidence from different empirical domains for an analytical conclusion. Our structures in the field have evolved since we used structures like (53c); the predominant approach currently uses structures that look like (59). The crucial fact in the current discussion is that this current model, like (53c), is a structure where OBJ1 asymmetrically c-commands OBJ2.

(59)



Second (and more relevant to the current chapter): in addition to patterns of binding and coreference being their own puzzle, one of the major roles of Binding Theory and Binding Principles is in diagnosing structure in the course of investigating other properties of language. A syntactic diagnostic is any property of language that behaves predictably enough that we can use it to understand *other* properties of language. Principles A, B, and C are just that: very consistent properties of language.

15.5.2 Distinguishing movement types

In this textbook we have also distinguished two different kinds of movement: A-movement is movement to an argument position (that we described as a Case-licensing position), whereas A′-movement is movement to a non-argument position: it is independent of Case and is instead typically linked with discourse properties, often typified by movement to the CP-domain. And we've previously seen some properties distinguishing these kinds of movements beyond just the target of movement. For example, A′-movement, like the *wh*-movement in (60b), can be quite long-distance, skipping over intermediate (non-*wh*) arguments. In contrast, a parallel situation attempting to A-move an embedded object to main-clause subject position is ungrammatical.

(60) a. I think that the children want to visit Grandpa.
b. Who$_{\mathrm{k}}$ do you think that the children want to visit t_k ?

(61) a. The children$_{\mathrm{k}}$ seem t_k to want to visit Grandpa.
b. * Grandpa$_{\mathrm{k}}$ seems the children to want to visit t_k .

There are additional syntactic properties that are part of these separate profiles of movement (A versus A′). As it turns out, a major way that A-movement and A′-movement are distinguished is on the basis of binding configurations.

(62) A- versus A′-movement
a. A-moved phrases bind from their landing position.
b. A′-moved phrases bind from their base position.

We can see this by looking at particular configurations of A-movement and A′-movement. Recall from Chapter 11 that *seem*-type verbs are *raising* verbs, where the embedded subject moves from the embedded clause to the matrix clause.

(63) a. Preston$_k$ seems to himself$_k$ [$_{\mathrm{IP}}$ ~~Preston~~ to be winning.]
b. * Preston$_k$ seems to him$_k$ [$_{\mathrm{IP}}$ ~~Preston~~ to be winning.]

Notably, these judgments are inverted when the subject is left in the lower clause:

(64) a. * It seems to himself$_k$ that Preston$_k$ is winning.
b. It seems to him$_k$ that Preston$_k$ is winning.

This is a quintessential property of A-movement: it creates new positions for binding, so when the subject moves to Spec,IP in the matrix clause, binding for that DP is calculated from that position.

A′-movement behaves distinctly different in this respect. In (65a) the pronoun in subject position cannot co-refer with *Arianna*. This is in accordance with Principle C: *Arianna* is an R-expression, and can never be bound. Co-indexing *Arianna* with the pronoun (which also c-commands *Arianna*) results in a Principle C violation. Key for our purposes here is that A′-moving the phrase containing *Arianna* to Spec,CP in (65b) does not significantly change these judgments: it is still quite difficult to get a reading of the pronoun co-referring with *Arianna*.

(65) a. Arianna$_1$ sent over the pictures of herself$_1$.

b. [Which pictures of herself$_1$]$_k$ did Arianna$_1$ send over [~~which pictures of herself$_1$~~]$_k$?

This is notable because in its overt position in (65b), *herself* is not c-commanded by its antecedent (*Arianna*). Based on Principle A, we would expect (65b) to be ungrammatical: but it is not. Instead, *herself* is grammatical in just the same way that it is in (65a). The finding of syntacticians has been that this is a consistent property of A′-movement. A′-moved XPs participate in binding as if they hadn't moved: they bind from their lower position and not their higher position. So in our example in (65b), the DP *which pictures of herself* is calculated for binding purposes as if it is in its base position, not its moved position. This is often referred to as **reconstruction**, on the metaphor that a moved phrase (for binding purposes, perhaps at LF) reconstructs back into its trace position. The copy theory of movement allows us to say that the original copy of the moved phrase is the one that participates in binding, but we still refer to this as reconstruction.

This is just a very simple introduction to a much more complex topic. The point for our current purposes is to again illustrate the role of binding properties in natural language syntax. For syntacticians as researchers, the properties of binding are so predictable and consistent cross-linguistically that binding itself becomes a toolkit for investigating other phenomena, whether attempting to diagnose the structural relationship between phrase, or understanding whether a moved phrase A-moved or A′-moved.

New Empirical Observations

- reference, coreference, and coindexing
- There are restrictions on coreference.

Components of the Theoretical Model

- c-command (review)
- binding (coreference + c-command)
- bound and free noun phrases
- Binding Theory
- binding domains
  - Principle A: Anaphors must be bound within their binding domains
  - Principle B: Pronouns must be free within their binding domains
  - Principle C: Full noun phrases must be free

Notes 15.6

1. Thanks to Eva Banik for the Hungarian examples in this section. For detailed discussions of relevant facts, see also Kiefer and Kiss (1994) and Kiss (2002).

2. There's a long history of work on negative polarity items (NPIs) that shows that negation is not the only licensing environment for NPIs: a closer (but still not comprehensive) generalization is to say that downward entailing (DE) environments license NPIs, of which negation is one such DE environment. But negation *does* need to be in a c-command relationship with an NPI to license it, so that portion of our discussion is accurate and sufficient for our purposes. But we wanted to note here that the semantics literature contains a lengthy and interesting literature on licensing of NPIs. See Chierchia (2013) for an overview.

3. As is the case with any introductory text, an interested reader should consult the relevant literature for more details (Safir 2004; Reuland 2011; Charnavel and Sportiche 2016).

4. A significant exception is so-called 'logophoric' uses of anaphors (also called 'exempt' as they appear to be exempt from Principle A), exceptional instances where anaphors appear to be bound long-distance.

   (66) John$_i$ said to Mary that nobody would doubt that physicists like himself$_i$ were a godsend. (Kuno 1987, 125)

   It does appear that exempt anaphora have different syntactic properties, though it can be tricky to disentangle them from anaphora that are subject to Principle A (for recent overviews of the issues, see Charnavel and Sportiche 2016; Charnavel 2019, 2020)).

Exercises and problems 15.7

Exercise 15.1: Basic practice with binding conditions A

For each sentence in (1), explain how each argument DP satisfies Binding Theory (be precise about which binding principles are being discussed and why)

(1) a. [The children]$_k$ are chasing [each other]$_k$ all through the neighborhood.
    b. [The boss]$_k$ said that they$_k$ are leaving early today.
    c. [The boss]$_k$ said that they$_i$ are leaving early today.
    d. She$_k$ adores [the coach]$_k$.
    e. Mike$_k$ said that Preston needs to visit him$_k$.
    f. [The guest speaker]$_k$ introduced herself$_k$ because her host was out sick.
    g. [Preston]$_k$'s mom loves him$_k$.

Exercise 15.2: Basic practice with binding conditions B

Each sentence in (1) is ungrammatical for a Binding Theoretic reason. Explain why each sentence is ungrammatical, being specific about all relevant components of Binding Theory.

(1)
a. * [The coach]$_k$ supports her$_k$.
b. * She$_k$ really likes Maisha$_k$.
c. * Mike$_k$ said that Preston needs to correct himself$_k$.
d. * She$_k$ said that Skylar really likes Maisha$_k$.
e. * [Preston]$_k$'s mom loves himself$_k$.
f. * Langston$_k$ said that himself$_k$ is hungry.

Problem 15.1: Norwegian anaphors A

Data are drawn from Hellan (1988).

A. Describe the ways that the Norwegian forms *seg* and *seg selv* are similar to / different from English *X-self.*

B. To what extent does Chomsky's Binding Theory cover these Norwegian forms?

Like English, Norwegian also distinguishes between pronouns with forms containing *selv* 'self' and ones without. As (1) and (2) show, *selv* is in complementary distribution with zero (that is, where a form with *selv* can occur, a form without *selv* cannot, and vice versa).

(1)
a. ✓ Jon$_1$ beundrer seg selv$_1$.
admires SEG self
b. * Jon$_1$ beundrer seg$_1$.
'Jon$_1$ admires himself$_1$.'

(2)
a. * Jon$_1$ bad oss beundre seg selv$_1$.
asked us admire SEG self
b. ✓ Jon$_1$ bad oss beundre seg$_1$.
'Jon$_1$ asked us to admire him$_1$.'

(3)
a. ✓ Jon$_1$ snakket om seg selv$_1$.
talked about SEG self
b. * Jon$_1$ snakket om seg$_1$.
'Jon$_1$ talked about himself$_1$.'

(4)
a. * Jon$_1$ bad oss snakke om seg selv$_1$.
asked us talk about SEG self
b. ✓ Jon$_1$ bad oss snakke om seg$_1$.
'Jon$_1$ asked us to talk about him$_1$.'

(5) Jon$_1$ så seg selv$_1$ sitte i stolen.
saw SEG self sit in chair.DEF
'Jon$_1$ saw himself$_1$ sit in the chair.'

(6) a. ✓ [$_{IP}$ Jon$_1$ bad oss forsøke å få deg til å snakke pent om seg$_1$].
asked us try to get you towards to talk nicely about SEG
'Jon$_1$ asked us to try to get you to talk nicely about him$_1$.'

b. * Jon$_1$ var ikke klar over at [$_{IP}$ vi hadde snakket om seg$_1$].
was not aware over that we had talked about SEG
'Jon$_1$ was not aware that we had talked about him$_1$.'

Problem 15.2: Norwegian anaphors B

Data and descriptions here are drawn from Hellan (1988).

This problem set builds on Problem 15.1.

In addition to the distinction between elements with and without *selv*, Norwegian distinguishes between the two third-person singular forms *seg* and *ham*. Like *seg*, *ham* occurs on its own as well as before *selv*.

(1) Jon$_1$ snakket om { seg / *ham } selv$_1$.
talked about SEG HAM self
'Jon$_1$ talked about himself$_1$.'

(2) a. Vi snakket med Jon$_1$ om { ham / *seg } selv$_1$.
we talked with about HAM SEG self
'We talked with Jon$_1$ about himself$_1$.'

b. Vi fortalte Jon$_1$ om { ham / *seg } selv$_1$.
we told about HAM SEG self
'We told Jon$_1$ about himself$_1$.'

(3) Vi gjorde Jon$_1$ glad i { seg / *ham } selv$_1$.
we made fond of SEG HAM self
'We made Jon$_1$ fond of himself$_1$.'

A. Describe the empirical similarities and differences between *seg selv* (from Problem 15.1) and *ham selv*.

B. Chomsky's Binding Theory as presented here won't provide a toolkit for explaining these differences, but you do have a toolkit given the concepts we've learned in the textbook so far. As precisely as you are able, explain where *ham selv* appears as compared to where *seg selv* appears.

Glossary

asterisk (*) indicates a sentence's syntactic ill-formedness (ungrammaticality)

(1) * The had bastard the nerve confront me to

Enclosing material that is preceded by an asterisk with parentheses indicates that including the material in parentheses is ungrammatical. Thus, (2) abbreviates the examples in (3).

(2) a. The (*those) cats like treats.
    b. My cats like(*s) treats.

(3) a. i. The cats like treats.
       ii. * The those cats like treats.
    b. i. My cats like treats.
       ii. * My cats likes treats.

On the other hand, prefixing an asterisk to material that is enclosed in parentheses indicates that the parenthesized material is obligatory. Thus, (4) abbreviates the examples in (5).

(4) a. They consumed *(dinner).
    b. My cat like*(s) treats.

(5) a. i. * They consumed.
       ii. They consumed dinner.
    b. i. * My cat like treats.
       ii. My cat likes treats.

78, 224

{ } curly brackets Curly brackets enclose alternatives. For instance, (6) abbreviates the two examples in (7).

(6) They do { not, so } like your brother.

(7) a. They do not like your brother.

b. They do so like your brother.

Curly brackets may be combined with ( ) parentheses. , 561

( ) parentheses Parentheses enclose optional elements. For instance, (8) abbreviates the two examples in (9).

(8) They do (not) like your brother.

(9) a. They do like your brother.
b. They do not like your brother.

Parentheses may be combined with { } curly brackets. For instance, (10) abbreviates the examples in (11).

(10) They do (not, so) like your brother.

(11) a. They do like your brother.
b. They do not like your brother.
c. They do so like your brother.

, 561

% percent A percent sign indicates that an example is accepted as grammatical by some speakers, but rejected by others.

(12) % We use a gas stove anymore; anymore we use a gas stove.
(chiefly Midwest, but found throughout the United States except New England; earliest recorded examples from Northern Ireland)

253

 # pound sign indicates a sentence's semantic or pragmatic ill-formedness.

(13) # Colorless green ideas sleep furiously.

A-movement Movement of a DP to a Case-licensing head (that is to say, an argument position), which typically occurs in connection with some core grammatical function. In general, only DP arguments can participate in A-movement. Examples include movement of subjects to Spec,IP and movement of theme arguments to Spec,IP in passives. , 565, 575

absolutive case A case form that canonically marks objects of transitive sentences and subjects of intransitive sentences in an ergative-absolutive case alignment. 357, 568

abstract Case Refers to the proposed abstract Case-features on nominals that must be valued or assigned. Traditionally capitalized in the syntax literature in contradistinction with lowercase morphological case. 337, 575

accusative case A case form that canonically marks objects of transitive sentences in a nominative-accusative case alignment. 145, 331, 563, 575

adjunct A constituent that is both sibling to an intermediate projection and child of an intermediate projection. Often associated with modifiers which add descriptive information to another constituent. For instance, *on Tuesdays* in *We eat tuna on Tuesdays* is an adjunct. 187, 579, *see also* complement & specifier

adposition Any P, regardless of whether it is head-initial or head-final. See Chapter 6 for examples. 229, *see also* postposition

agent A thematic role assigned to a (perceivedly) conscious or sentient argument that brings about a state of affairs by its own will. For example, in the sentence *The extraterrestrials abducted my cows*, the agent is *the extraterrestrials*. 140, 581, *see also* cause & instrument

Agree operation The operation within current generative syntactic theory that allows the properties/features of one element to be determined by another element in a sentence (for example, verbs agreeing in person and number with their subject). The Agree operation consists of a Probe (which seeks out a value for particular features) and the Goal, the phrase bearing those features that values the features on the Probe. 349

agreement The sharing of one or more features between two (or more) parts of a sentence. A common example is subject-verb agreement, where a verb is inflected differently according to features of its subject such as person, number, or gender/noun class. 329, 569

algorithm An explicit, step-by-step procedure for solving a problem or accomplishing a task.

Examples: instructions for installing a water filter or for filing your income taxes; a pesto recipe; a knitting pattern.

applicative A form of a verb (or a verbal derivational morpheme) that increases its number of arguments (valency) by one. One way of conceiving of this is to say that an applicative adds an argument by promoting one of its adjuncts to an argument. Often the promoted adjunct is a recipient, benefactive, instrument, or location. For examples see the discussion in Chapter 4, § 4.1.3. 137

argument From the point of view of formal logic, an argument is an input to a predicate (in the formal logic sense). From the point of view of syntax, specifically X′ theory, an argument is a linguistic expression occupying the specifier or complement position of a head. Because a predicate can have more semantic arguments than the X′ schema provides, semantic arguments are not necessarily expressed as syntactic arguments. For discussion, see Chapter 4, § 4.1 and Chapter 5, § 3. Conversely, not all syntactic arguments are semantic arguments; see Chapter 4, §§ 4.5.1, 4.1 for examples. 53, 132, 177, 561, 562, 564, 567, 568, 582, *see also* expletive

argument array A list of semantic arguments associated with a predicate. As we use the term in this book, the list is unordered. The semantic arguments in the array are mapped onto (that is, associated with) positions in syntactic structure. As discussed in more detail in Chapter 9, it is possible for the same argument array to be associated with more than one structure. Conversely, the same structure can be associated with more than one argument array. 281

auxiliary *do* See Chapter 3, § 2.5.2

A′-movement Movement of a DP to a non-Case licensing head, which typically occurs for some discourse purpose. Put another way, A′-movement is movement to a non-argument position. It is pronounced "A-bar movement," and is not related to the bar-levels of the X′-schema (an unfortunate overlap in terminology). Examples include *wh*-, topic, and focus movement. , 575

base-generated Base-generation refers to the idea that some phrases have not moved to their surface position, and are instead pronounced in precisely the places they first entered the tree. So if a phrase is base-generated in CP, that means its original position in the tree is in CP (and therefore that it didn't move to CP from some other position). 312

benefactive A thematic role assigned to an argument that receives something good or otherwise benefits from an event. For example, in the sentence *Janelle's big brother packed her lunch*, *Janelle* is the benefactive argument. 467, 562, *see also* recipient

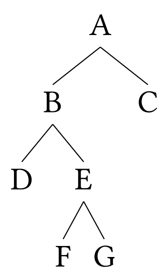
binding theory A component of syntactic theory concerned with how syntax constrains the interpretation of noun phrases referring to entities previously referred to in the conversation (so-called anaphoric noun phrases). See Chapter 15 for a more thorough discussion. 541

bound morpheme A morpheme that cannot stand alone, but must form part of a larger word, like plural *-s*, *un-*, or *-ness*. In contrast, free morphemes, like *cat* or *happy*, can stand alone. A trailing hyphen indicates that a bound morpheme is a prefix; a leading hyphen indicates a suffix. , 569

Burzio's generalization A correlation according to which verbs that are not associated with a thematic role for an external argument (passives, unaccusatives) are unable to assign accusative case. For more discussion, see Chapter 13. 478

c-command A syntactic constituent is said to c-command its siblings and all nodes dominated by its siblings. In the example below the node labeled C c-commands the nodes B, D, E, F, and G. Node C does not c-command A.

(14)



116, 541

c-selection Refers to *categorial* selection, wherein syntactic heads often specify what kind of syntactic constituent they combine with, ex., a DP or a PP. 256, 579

case A property of noun phrases that marks their grammatical role within a sentence; often capitalized as "Case" when understood to refer to an abstract syntactic feature, as opposed to a morphologically overt distinction. Common examples include the nominative case and the accusative case. 329, 567

case concord The sharing of Case features within a noun phrase. 360

Case Filter 354, 567, *see* DP Licensing Requirement

causative A form of a verb that increases its number of arguments (valency) by one by adding a second agent argument. The second agent *causes* or *permits* the first agent to do the action.

Examples:

(15) a. 아이가 옷을 입었다. **Korean**
Ai-ka os-ul ip-ess-ta.
child-NOM clothes-ACC put.on-PST-DECL
'The child put on clothes.'

b. 아빠가 아이한테 옷을 입혔다.
Appa-ka ai-hanthey os-ul ip-hy-ess-ta.
dad-NOM child-DAT clothes-ACC put.on-PST-DECL
'Dad put clothes on the child.'

137

cause A thematic role assigned to a (perceivedly) nonsentient argument or one without will that brings about a state of affairs. For example, in the sentence *The sun melted all the snow*, *the sun* is the cause. 140, *see also* agent, instrument & causative

clause A constituent that contains a subject, possibly silent (in **boldface**), and a predicate (in *italics*). Clauses can be subdivided into **ordinary clauses** and **small clauses**. Ordinary clauses can be further subdivided into **finite clauses**, which can stand alone, and **nonfinite clauses**, which can't. All ordinary clauses contain an Infl element (underlined)—a modal, auxiliary, silent tense morpheme, or the nonfinite marker *to*. Small clauses differ from ordinary clauses in lacking an Infl element.

(16) a. **Our friends** *must be in Cancun by now.* Finite clause
b. **Bill** *has never seen a racoon.*
c. **They** *are our friends.*
d. **Bill** *[past] arrived.*

(17) a. (John seems) ____ *to be having problems.* Nonfinite clause
b. (I expect) ______ *to know tomorrow.*
c. (I expect) **them** *to know tomorrow.*
d. (We consider) **them** *to be our friends.*

(18) a. (They made) **us** *do it.* Small clause
b. (We consider) **them** *our friends.*

183,

closed class A syntactic category into which new words are not easily added. Functional categories are generally closed classes. Closed classes stand in contradistinction with open classes. 42, 575

competence The abstract knowledge about language as a formal system that humans carry in their heads, as opposed to the physical sounds and signs of language that users actually speak, sign, and mentally process (performance). See also the similar but distinct contrast between E-language and I-language. 267

complement A constituent that is sibling to a lexical projection (i.e., a head) and child of an intermediate projection. Often "completes" the meaning of the lexical projection it is siblings with. For example, *blue cheese* is a complement to *eats* in the sentence *Stella often eats blue cheese*. 176, 579, *see also* adjunct & specifier

complementary distribution If two syntactic elements never appear in the same context, they are said to be in complementary distribution. For example, the English infinitive particle *to* only appears where the verb does not take any inflection (ex., *-s* or *-ed*), so infinitive *to* and verbal inflection are in complementary distribution. 67

complementizer A word or morpheme that turns an independent sentence into the complement of a matrix verb, or an adjunct to a main clause. Examples in English include *whether*, *if*, *that*, *because*, and *while*. 183, 569

compound tense A tense that is expressed analytically, for instance, the English present progressive *she is singing* or the French *passé composé elle a chanté* 'she sang, she has sung'. Compound tenses contrast with simple tenses, which are formed synthetically, like the English simple past *she sang* or the French *imparfait elle chantait* 'she used to sing, she was singing'.

Compound tenses in one language can correspond functionally to simple tenses in another language, and vice versa. For instance, the English compound present progressive corresponds to a French simple present, whereas the English simple past often corresponds to a French *passé composé*. , 579

constituent A unit of grammatical structure. Evidence for certain constituents comes from constituenthood tests. In tree structures, constituents are represented as nodes. 93

control verb A subclass of the verbs that take *to* infinitive complements, superficially similar to raising verbs. In contrast to the subject of a raising verb, however, the subject of a control verb starts out in the same clause as the control verb itself, undergoing local A-movement to subject position, but not raising out of an embedded clause. Accordingly, the subject of the control verb's nonfinite complement is not a trace of the matrix subject, but rather a separate silent pronoun PRO, which is coindexed with the matrix subject. For more discussion, see Chapter 11. , 578

Copy Theory of Movement The Copy Theory of Movement is a particular perspective in generative syntax about the representation of movement within the model. It says that any movement construction does not relocate a syntactic element within a structure, but instead copies that element and then adds that copy into the structure as well. So instead of our model saying that movement consists of a moved element and a trace in its original position (as in the first example below), instead movement is as in second example, where the original position of the moved element still contains the original version of that element, it is now simply an unpronounced copy.

(19) a. Who$_k$ are you going to visit t_k today?
b. Who$_k$ are you going to visit ~~who~~$_k$ today?

We generally represent unpronounced copies with strikeout (~~like this~~), though it is not uncommon to still use traces to annotate movement in trees and examples because of the visual simplicity of traces. , 569, 575, 581

count noun See Chapter 3, § 2.3.1

cyclic A derivation that decomposes apparently nonlocal movement into a sequence of local movements is said to be cyclic. Each successive step in the sequence of movements is correspondingly called a cycle. In contradistinction to movement in one fell swoop. The term cyclic is also used more generally to refer to iterative syntactic operations, like the Merge operation in modern syntactic theory. 510, 575

dative case A case form that canonically marks indirect objects in a sentence. 145

declarative A kind of sentence or clause that canonically expresses a statement. Examples:

(20) a. *I like calico kittens quite a lot.*
b. *The band might not play next week.*
c. *Friends ought to stick together.*

Note that declarative is a clause type (and is associated with certain morphosyntactic properties). Statements can also be made more indirectly using interrogative and imperative clause types, for example.

(21) a. *Did you know that sea turtles love pancakes?* (interrogative)
b. *Recall that every pair of integers sums to an integer.* (imperative)

Possible declarative equivalents of these sentences are as below.

(22) a. *Sea turtles love pancakes.*
b. *Every pair of integers sums to an integer.*

, 572, 573

demotion 375, *see* grammatical relation

determiner A syntactic category that includes the definite article *the,* the indefinite article *a* and its variant *an,* the demonstratives *this* and *that,* and ordinary and reflexive pronouns. English also has a silent determiner, marked by _____ in (23) below, which resembles *some* in that it can be used with both mass nouns and plural count nouns.

(23) a. I see _____ rice on the table.
b. I heard _____ lions out in the bush last night.

203, 205

direct question See Chapter 8, § 8.1.2

Distributed Morphology The predominant generative model of the syntax-morphology interface (Halle and Marantz 1993). In Distributed Morphology (DM), some morphological forms are lexical (items like *heel* and *hill*, which don't change in realization), but others are inserted during the derivation (for instance, *sing* is realized as *sang* under a past-tense I°), "distributing" the implementation of morphology across different parts of the generative linguistic architecture. DM crucially assumes Late Insertion of morphological forms, wherein specific morphological forms are inserted late in the derivation, after the features of the syntactic structure are determined. 358

***do*-support** A phenomenon in English in which an unstressed auxiliary *do* is required to form certain kinds of questions and negative sentences. Compare the following examples with and without the insertion of an auxiliary *do*:

(24) a. Do killer whales eat krill?
(*asking for information*)
b. Killer whales eat krill?
(*asking for confirmation, or expressing surprise; never a neutral request for information*)

(25) a. I { do not, don't } like mannequins.
b. * I like not mannequins.
c. * I not like mannequins.

See Chapter 12, § 12.2.2 for a detailed analysis of this phenomenon. 427

DP Licensing Requirement The requirement that all noun phrases which are also arguments receive a Case value. Traditionally called the Case Filter. 354

dual A grammatical number that indicates two of a noun. Appears in most Arabic varieties, Sanskrit, and Yup'ik, for example. See the discussion in Chapter 2, § 2.3.1. 49

E-language Stands for *externalized* language, and is used to refer to language as used by people: actual spoken and perceived sentences, languages as we refer to them out in the world (Swahili, Korean, etc), and other kinds of language-in-use, for instance, properties that fall under sociolinguistics, psycholinguistics, etc. E-language stands in contrast with I-language. 28, 565, 572, 576

empirical Concerning or pertaining to data. In the context of this textbook, the relevant data have to do with the grammatical status or the interpretation of phrases and sentences. When you are asked to give an empirical argument, your argument must be based on judgments concerning phrases or sentences, not on purely conceptual considerations like simplicity, economy, theory-internal consistency, and so on.

EPP The requirement that a phrase move to a specifier position. Can be thought of as a requirement for an Extra Peripheral Position. The acronym originated from the concept of the Extended Projection Principle, but is no longer used to mean this in the same way as it originally was (see Haegeman 1994 for a summary). 355

ergative case A case form that canonically marks the agentive subject of a transitive sentence in an ergative-absolutive case alignment. 357, 568

ergative-absolutive case alignment A system of marking case relations that differs distinctly from nominative-accusative case alignment. In the canonical form of this case system, subjects of intransitive predicates and objects of transitive predicates are marked with the same case (called absolutive case) and subjects of transitives (generally AGENTS) are marked with a distinct case form (called ergative case). 357, 561, 567, 575

event An action or a state and the participants in that action or state. In formal semantics, this concept is instead referred to with the term "eventuality" (and the term "event" denotes an action, specifically excluding states), but this textbook uses the term "event" to cover both actions and states. 133, 581

evidence Here and throughout the book, we will use the term 'evidence' to refer to empirical arguments for a conclusion, rather than in a more liberal sense in which it includes conceptual arguments based on theoretical virtues such as simplicity, economy, theory-internal consistency, and so on. In other words, for us, the collocation 'empirical evidence' is redundant.

Evidence consists of a linguistic expression (a sentence or phrase), along with a grammaticality judgment (✓ or *).

For instance, we might ask whether the underlined noun phrase in (26a) is a complement or an adjunct of the italicized verb. The grammaticality of (26b) is evidence that it is an adjunct (given assumptions spelled out in Chapter 5).

(26) a. They *partied* the entire weekend.
b. ✓ They *did so* the entire weekend.

What about the status of the same string in (27a)? In this case, it is the ungrammaticality of (27b) that is evidence for the expression being a complement.

(27) a. They *enjoyed* the entire weekend.
b. * They *did so* the entire weekend.

35, 121, 170, 171

exceptional case marking (ECM) Describes a configuration where the subject of a nonfinite complement clause receives structural accusative case from the matrix clause. Also sometimes referred to as "raising-to-object."

experiencer A thematic role assigned to an argument that undergoes a sensory, cognitive, or emotional experience. For example, in the sentence *The suddenness of the news caught Kaviya by surprise*, *Kaviya* is the experiencer. 140, 581

expletive (from Latin *explere* 'to fill out') A syntactic argument that isn't a semantic argument. Sometimes also referred to as a 'dummy pronoun' or a 'pleonastic pronoun.' English examples are *it* and *there*, as in *It is raining* and *There are children playing in the park*. See Chapter 4, § 4.5.1 for more discussion. 4

feature Characteristics of a syntactic constituent which may take on a finite number of values. Features are usually annotated inside [square brackets]. Examples include gender , number, and person for nouns and noun phrases and clause type (declarative or interrogative) for complementizers. The specific values a feature can take on vary by language; for example, English generally distinguishes declarative, interrogative, and imperative clause types, while Korean distinguishes a fourth propositive/cohortative clause type. 49, 181, 307, 561, 562

feature-driven Refers to an analytical approach wherein operations in syntax don't happen freely, but they are triggered, or initiated, by a feature in the syntax. 307

finite Finite verbs are verbs that are inflected for tense and person-number agreement. Finite clauses are clauses that can stand alone. See Chapter 3, § 2.9 for more discussion. 421

floating quantifier Certain quantificational modifiers of nouns are able to "float" away from the noun that they modify. In English this includes *all* and *both*. Floating quantifiers are used to diagnose syntactic movement. 262

flout Refers to a phenomenon where speakers may choose to use aspects of languages in ways that intentionally depart from their canonical uses, for interpretive effect. Often used to discuss flouting Gricean maxims to create implicatures within pragmatics. 257

focus A subcategory of emphasis constructions. Focused phrases tend to be contrastive with some other relevant set of alternatives. So *I ATE an apple* is interpreted to mean that I ate an apple, as opposed to doing something else with the apple. 98, 310

formative A general term for an abstract meaning unit. Like morphemes, formatives can be pronounced or silent, but they needn't be minimal meaning units.

free morpheme A morpheme that can stand alone, like *cat* or *happy*. In contrast, bound morphemes, like plural *-s*, *un-*, or *-ness*) are part of a larger word. 417, 563

full noun phrase Any noun phrase that is not an ordinary pronoun or reflexive pronoun.

Examples: *John*, *the boy next door*, *the dog that ate the homework*, *a lame excuse*, *the problem with them*, and *Annabelle's confidence in herself*.

As the last two examples show, a full noun phrase can contain an ordinary or reflexive pronoun; it just can't entirely consist of one.

functional category A syntactic category that serves purely grammatical functions, without much semantic (meaningful) content, for example determiner, complementizer, and inflection. Less strictly, also refers to lexical items in such categories, such as *any*, *whether*, and *might*. 41, 564, 573

gap A gap is a pre-theoretical way of referring to the empty position in an extraction construction where a moved element would typically be located.

(28) a. What did you eat ___ for lunch?
b. The friend who I always like to visit ___ in Paris.

In our model gaps are traces (or unpronounced copies, within the Copy Theory of Movement). 500

gender Grammatical gender is a set of mutually exclusive word classes for nouns and pronouns. In many languages, the genders of pronouns correspond to the biological sexes, but the gender assignment for nouns is typically more arbitrary. This is illustrated in (29) for German, a language with three genders: masculine (M), feminine (F), and neuter (N).

(29)
a. die Wache, das Mädchen
the.F sentinel, the.N girl
'the sentinel (male or female), the girl'

b. der Hund, die Katze
the.M dog, the.F cat
'the dog (male or female), the cat (male or female)'

c. der Tisch, die Tasse, das Fenster
the.M table, the.F cup, the.N window
'the table, the cup, the window'

English doesn't have gender as a grammatical feature, since the so-called genders of pronouns like *he* and *she* simply correspond to the sex of the person being referred to. In languages with so-called "noun class" systems, grammatical genders (i.e. noun classes) are not correlated with human sex/gender in any way. Many Bantu languages have 15-20 noun classes, for example, with classes that correlate to humans, to inanimates, and locations, but no correlation to human genders. , 568

goal A thematic role assigned to an argument that is the endpoint of a path. For example, in the sentence *Last summer I flew to Korea to see my grandmother*, *to Korea* is the goal. 141, *see also* location

government Traditionally used to refer to the requirement of certain verbs and prepositions for their complements to appear in a particular case form. In generative grammar, the sense of the term has been extended to refer to the structural relation between a head and its complement.

grammatical relation The grammatical relations in a sentence are listed in (30); see Chapter 4, § 4.4 for more discussion. They are often conceived of in a ranking: formally in some theoretical approaches, informally in others.

(30) subject — direct object — indirect object — object of preposition

promotion ⟵ ⟶ demotion

Various syntactic operations can change the grammatical relation of a noun phrase. Depending on whether a noun phrase moves "up" or "down" the hierarchy in (30), the noun phrase is said to be **promoted** or **demoted**. For instance, the passive in English demotes the subject of an active sentence to the object of the preposition *by*. In addition, it promotes the object to subject. 143, 375

ground The part of a sentence that contains background information that is presupposed, already part of the pool of "common knowledge" shared by both speakers, or part of the set of topics already under discussion. For instance, *my dog* is the ground in the sentence *My dog likes to run about in the summer*, since the existence of the speaker's dog is presupposed, commonly known by both participants in the conversation, and likely already under discussion. 98

head The term *head* has three different meanings in syntactic theory. First, in the theory of phrase structure, the term refers to the syntactic and semantic core of a phrase. In X′ theory, the particular theory of phrase structure used in this book, a head projects an intermediate and maximal projection, along with optional arguments.

Second, the head of a movement chain is simply the highest element in the chain. In the case of verb movement, the head in the chain sense happens to be a head in the X′ sense. But when a maximal projection moves, the head of the chain is a maximal projection.

Third, the term *head* can refer to the noun that is modified by a relative clause. Given our analysis of relative clauses as involving *wh*-movement, and hence a movement chain (see Chapter 13), this usage is potentially especially confusing. In the relative clauses in (31), the head of the relative clause in this third sense is *italicized*, whereas the head of the chain formed by movement of the relative clause is underlined.

(31) a. the *book* [$_{\text{CP}}$ [$_{\text{DP}}$ which]$_i$ I am reading t_i.]
b. the *author* [$_{\text{CP}}$ [$_{\text{DP}}$ whose book]$_i$ I like t_i best.]

, 571, 578

head movement Movement of a syntactic head to adjoin to another, higher syntactic head. One English example is the movement of modals and auxiliary *do* to C° in forming yes-no questions—for details, see Chapter 8, § 8.2. 280, 301, 420, 575

homograph One of two or more linguistic forms that are spelled alike, but different in function or meaning or in pronunciation.

Example:

bank ‘river bank’ and *bank* ‘financial institution’ (different meaning)

read (infinitive) and *read* (participle) (different pronunciation)

, 571, 582, *see also* homophone, homonym & word

homonym Can refer to either homograph or homophone.

homophone One of two or more linguistic forms that are pronounced alike, but distinct in function or meaning or in spelling.

Examples:

bank ‘river bank’ and *bank* ‘financial institution’ (same spelling, different meaning)

reed and *read* (infinitive); *red* and *read* (participle) (related meaning, different spelling)

63, 64, 571, 582, *see also* homograph, homonym & word

hypercorrection The psychological process of constructing a non-prescribed linguistic form by analogy to a prescribed form. Also the resulting form itself. Example: *They feel badly*, by analogy to *They drive badly*. The prescribed form is *They feel bad*. In *They drive badly*, *badly* modifies *drive*, which is an ordinary verb. In *They feel bad*, on the other hand, *feel* is a linking verb, and *bad* is a predicative adjective (it predicates a particular property of the subject of the sentence; in other words, it attributes the property to the subject).

They feel badly is acceptable in prescriptions for so-called "standard" English only if it is the verb *feel* that is being modified; that is, if the sentence is intended to mean *They have a bad sense of touch.*

As always, note that the prescribed "standard" form is not a cognitive or linguistic status, but a social move to privilege one form of speaking over others. 30

I-language Stands for *internalized* language, and is used to refer to mental grammar, i.e. the knowledge that speakers have about their own languages. I-language stands in specific contrast with E-language. 28, 565, 567, 576

idiom A sequence of words whose fixed, specific, and nonliteral meaning cannot be derived (more or less straightforwardly) from its parts. For example, in English, *red* refers to a color and *herring* to a kind of fish, but the expression *red herring* has the fixed, specific, and nonliteral meaning 'misleading clue'. This meaning cannot be derived directly from the words in the expression without knowing the associated story. 269

imperative A kind of sentence or clause that canonically expresses a command or request. Examples:

(32) a. *Come downstairs this instant!*
b. *Please don't be so petty.*
c. *Just let me know when you're free.*

Note that imperative is a clause type (and hence is associated with, in English, a certain form of the verb). Commands or requests can also be made using declarative and interrogative clause types, for example.

(33) a. *I'd really appreciate you helping out with the cows.* (declarative)
b. *Would you mind giving me another week?* (interrogative)

Possible imperative equivalents of these sentences are as below.

(34) a. *Come help out with the cows.*
b. *Give me another week.*

152, 566, 569, 573

indirect question See Chapter 8, § 8.1.2

inherent case Inherent case refers to case forms that are idiosyncratically based on lexical items. For example, some verbs or prepositions assign a particular case form to their objects that is not the normal object case form in that language. Inherent case is sometimes characterized as being attached to a particular thematic role as well: for example, in some languages EXPERIENCERS are expressed with dative case even when they are the subject of a sentence. 333

instrument A thematic role assigned to a (perceivedly) nonsentient argument or one without will that is used to bring about a state of affairs. For example, in the sentence *The chef julienned the celery with their new knife*, *their new knife* is the instrument. 140, 562, *see also* agent & cause

interrogative A kind of sentence or clause that canonically expresses a question. Examples:

(35) a. *Who's running for Taiko Club president this semester?*
b. *Did Basil know his portrait was aging in Dorian's stead?*
c. *Will the final exam be cumulative?*

Note that interrogative is a clause type (and hence is associated with, in English, a certain intonation and certain kinds of syntactic movements). Questions can also be posed using declarative and imperative clause types, for example.

(36) a. *Sorry, I can't quite seem to remember your name.* (declarative)
b. *Remind me where we decided to go for dinner.* (imperative)

Possible interrogative equivalents of these sentences are as below.

(37) a. *Sorry, what's your name?*
b. *Where did we decide to go for dinner again?*

, 566, 572

intransitive Traditionally used of verbs that take no object. We use the term in a more general sense to refer to any syntactic category that takes no complement. For instance, the *italicized* heads are transitive in the (a) examples, but intransitive in their (b) counterparts.

(38) a. We have *eaten* the pizza. verb
b. We have *eaten*.

(39) a. They crawled *underneath* the table. preposition
b. They crawled *underneath*.

(40) a. I like *that* radio. determiner
b. I like *that*.

135, *see also* transitive

Late Insertion The idea that some morphological forms are decided on and inserted into a sentence *after* the sentence is already built by other syntactic operations. 358, 567

left periphery Refers to the structurally high parts of a clause (the CP-domain) with left-branching heads and specifiers. 309

lexeme 14, 573, *see* word

lexical ambiguity The association of a single word with more than one lexeme. 14, 583

lexical category A syntactic category that has some semantic (meaningful) content, for example noun, adjective, or verb. Less strictly, also refers to lexical items in such categories, such as *cat*, *flashy*, or *hunt*. Appears in contradistinction with the concept of a functional category. 41, 575

lexical knowledge Linguistic knowledge that must be memorized and cannot be derived from rules or broader patterns. For example, the fact that *cat* is a noun and the fact that it refers a particular class of furry, four-legged, whiskered animals are both pieces of lexical knowledge; a child who has never heard the word *cat* cannot deduce these facts from rules they have previously encountered, or based on the sounds of the phonemes themselves. Rather, these properties of *cat* must simply be memorized by a learner of English. The abstract collection of lexical knowledge is called the lexicon. 265

lexicon The abstract collection of linguistic knowledge that must be memorized and cannot be derived from rules or broader patterns. Can be thought of as a mental "dictionary" that holds not just the meanings of words, but also their syntactic (and other linguistic) properties. 265, 574, *see* lexical knowledge

linking verb Also known as **copular verb**. One of a class of verbs including the copula *be*, as well as (among others) *appear*, *become*, *feel*, *get*, *grow*, *look*, *prove*, *seem*, *smell*, *sound*, *taste*, and *turn*, when used as in (41).

(41) a. They { are, appear, became, grew, look, proved, seem } competent.
b. It { feels, looks, smells, sounds, tastes } fine.
c. He { became, got, grew } old.
d. It turned rancid.

2, 571, *see also* hypercorrection

location A thematic role assigned to arguments that are points or regions in space. For example, in the sentence *The gerbils congregated underground*, *underground* is the location. 141, 562, 576, *see also* path & goal

mass noun See Chapter 3, § 2.3.1

matrix A structure that contains another structure. Typically used in the collocation 'matrix clause,' referring to a CP or IP that contains another CP or IP. 183, 565

measure A thematic role assigned to an argument that expresses extension along some dimension (length, duration, temperature, and so on). For example, in the sentence *The gymnast flipped a full seven hundred and twenty degrees before landing gracefully*, *a full seven hundred and twenty degrees* is the measure. 142

morpheme A minimal meaning-bearing element.

Words are not necessarily the smallest meaning-bearing elements in a language. For instance, *cats* is a single word, but consists of two morphemes, *cat* and the plural morpheme *-s*. *Unhappiness* consists of the three morphemes *un-*, *happy*, and *-ness*.

Morphemes can be **free** or **bound**. Free morphemes (like *cat* and *happy*) can stand alone, whereas bound morphemes (like *-s*, *un-*, and *-ness*) cannot. A trailing hyphen indicates that a bound morpheme is a prefix; a leading hyphen indicates a suffix. 137, 563, 569, *see also* bound morpheme & free morpheme

morphological case Refers to the overtly marked morphological changes in nominals that arise in different syntactic positions. For instance, the English first-person singular pronoun (usually) appears as *I* when occurring as the subject of a sentence, but appears as *me* when occurring as the object of a verb or preposition, so we say that *I* is in the "nominative case" (or "subjective case") and *me* is in the "accusative case" (or "objective case"). Traditionally spelled with lowercase in the syntax literature in contradistinction with uppercase abstract Case. 337, 561

movement In syntactic theory, constituents often move from one position to another. The presence of movement in syntactic theory reflects the assumption that human syntactic knowledge sometimes represents constituents in multiple positions. Examples include A-movement, A′-movement, raising, head movement, and subject movement. Newer approaches often theorize that movement of constituents is actually copying of constituents; see Copy Theory of Movement for more. 261, 273, 561, 562, 565, 569, 571, 577, 580, 581

negative polarity item A syntactic element that can typically only occur together with negative words such as *no*, *never*, *hardly*, and *barely*; the negative word is said to *license* the negative polarity item. Often abbreviated as NPI. In the examples below, NPIs are italicized and their licensers are underlined.

(42) a. I *(<u>never</u>) see you *anymore*.
b. She *(<u>hardly</u>) *ever* comes to the office.
c. *(<u>Not</u>) just *anyone* would make a good mayor.

542

nominal Of or relating to a noun or its projections (N, N′, NP); more generally, of or relating to a noun phrase (DP). 204

nominative case A case form that canonically marks subjects of transitive sentences and subjects of intransitive sentences in a nominative-accusative case alignment. 145, 331, 563, 575

nominative-accusative case alignment A system of marking case relations that differs distinctly from ergative-absolutive case alignment. Subjects of transitives and subjects of intransitives are marked with the same case form (called nominative case) and objects of transitives are marked with a distinct case (accusative case). 356, 561, 568, 575

OED Oxford English Dictionary

one fell swoop A derivation in which nonlocal movement occurs in a single step is said to occur in one fell swoop. This is in contrast to movements that occur via a series of local movements (as in a cyclic derivation). The name comes from Shakespeare's *Macbeth*, where it describes a bird of prey taking an entire nest of chicks for prey in one single terrible ("fell") swoop. 510, 566

open class A syntactic category into which new words may be added relatively easily. Lexical categories are generally open classes. Open classes stand in contradistinction with closed classes. 42, 564

ordinary pronoun Synonymous with "personal pronoun." The following table lists the English ordinary pronouns.

Person	Number	Subject forms	Object forms	Possessive forms	
				Short	Long
1	SG	I	me	my	mine
2		you	you	your	yours
3		he, she, they, it	him, her, them, it	his, her, their, its	his, hers, theirs, its
1	PL	we	us	our	ours
2		you	you	your	yours
3		they	them	their	theirs

Table 15.1: English personal pronouns

147, 569, *see also* full noun phrase

path A thematic role assigned to an argument that connects locations. For example, in the sentence *The maenads danced through the woods*, *through the woods* is the path. 141, 570, *see also* goal

performance The actual language that humans speak, sign, and mentally process, as opposed to the abstract knowledge about language as a formal system that they have. See also the similar but distinct distinction between I-language and E-language. 267, 565

phase A phase in current generative syntactic theory refers to a portion of syntactic structure that is built and then transferred to LF and PF, after which it cannot be further modified. It demarcates chunks of the syntax where structure is built. This text doesn't properly engage phases, the interested reader is recommended to consult Citko (2014). 511

phi-feature A feature of a noun or noun phrase; examples include noun class/gender, number, and person. Also written "φ-feature". 52, *see also* feature

Plato's problem "How do we know so much with so little evidence?" The idea comes from a philosophical tradition that posits innate knowledge, and explores the apparent gap between our knowledge and our experiences that resulted in that knowledge. Applied to language, we appear to have more grammatical knowledge about our language(s) than the input that we learned from would suggest. The idea is that language reveals innate properties of human cognition. Its application to language acquisition is called the argument from the poverty of the stimulus. 16, *see also* poverty of the stimulus

postposition A head-final P. Coined in order to avoid the expression 'head-final preposition', which offends the etymologically-inclined as a contradiction in terms. See Chapter 6 for examples. 229, *see also* adposition

poverty of the stimulus The fact that it appears that children are exposed to a smaller amount of natural language data than would be necessary to explain the extent of the grammatical knowledge that they have. Generative linguistics proposes Universal Grammar as the solution to this puzzle. 16, 576, *see also* Plato's problem

predicate The term 'predicate' in linguistics has two distinct (though related) senses, what we will call the **subject-predicate** sense and the **predicate-argument** sense.

The subject-predicate sense derives from traditional logic, where propositions are divided into two parts, the subject and the predicate, and the predicate is what is affirmed (or denied) as an attribute of the subject.

In the history of formal logic, this original sense was generalized to include relations missing more than a single argument. In the resulting predicate-argument sense, in linguistics, the term 'predicate' refers to a head that expresses a logical relation. Typically, predicates in this sense are verbs, but other types of heads can function as predicates in this sense as well.

The two senses are illustrated in (43) and (44). The predicate is underlined; notice that the two senses can pick out the same expression, as in the examples (43b) and (44b).

(43) a. Bill gives money to charity. — Subject-predicate sense
b. Bill swims.

(44) a. Bill gives money to charity. — Predicate-argument sense
b. Bill swims.
c. Sheila's criticism of the plan.

132, 562, *see also* argument

pro-form It is convenient to group together pronouns and expressions like *do so* and *one* as pro-forms (< Latin *pro* 'instead of').

Note that the traditional term 'pronoun' is misleading, since pronouns substitute for entire noun phrases, not just for nouns (the fact that Latin has no articles may explain the traditional term). A more accurate term for pronouns would therefore be pro–noun phrases; however, we continue to use the traditional term, at least where there is no danger of confusion.

promotion 375, 562, *see* grammatical relation

proposition An expression in language of something that is either true or false: a potential fact. Put another way, it can be thought of as a complete thought. So a single noun phrase like *a cat* is not a complete thought: it is not something that can be true or false and therefore is not a proposition. But the sentence *I saw a cat* can potentially be true or false, and therefore is a proposition.

The same proposition can be expressed by different linguistic forms. Conversely, the same linguistic form can express different propositions.

Examples:

Same meaning, different form: *Aliens have abducted Eleanor. Eleanor has been abducted by aliens.*

Same form, different meaning: *We last saw Eleanor an hour ago* (spoken on December 11, 2000 at 3 PM vs. on January 14, 2002 at 5 AM). 133

raising The movement of a subject from an embedded clause to a matrix clause. , 565, 575, 578, *see also* control verb & raising verb

raising verb A subclass of the verbs that takes *to* infinitive complements, superficially similar to control verbs. Raising verbs lack a specifier position and fail to assign case. In contrast to the subjects of a control verb, the subject of a raising verb starts out as the subject of that verb's nonfinite complement clause and becomes the matrix subject by subject raising—hence, the name of the class. For more discussion, see Chapter 11. , 565

recipient A thematic role assigned to an argument that receives something, good or bad. For example, in the sentence *I gave my mother a call*, *my mother* is the recipient. 141, 562, 581, *see also* benefactive

reflexive pronoun The English reflexive pronouns are easy to identify because they all contain a form of the morpheme *self*, as laid out in the following table.

	1	**2**	**3**
SG	myself	yourself	himself herself itself
PL	ourselves	yourselves	themselves

Table 15.2: English reflexive pronouns

, 569, *see also* full noun phrase

relative clause A relative clause is a kind of embedded clause that is used to modify a nominal. The standard formulation involves a gap within the relative clause, with the gap corresponding to the head of the relative clause.

(45) a. The small puppy [that ___tore up the couch].
b. The couch [that the small puppy tore up ___].

Many languages involve multiple related-but-distinct strategies for forming relative clauses. For example, English has relative clauses with *that*-complementizers and relative clauses that use wh-phrases.

(46) a. The small puppy [which ___tore up the couch].
b. The couch [which the small puppy tore up ___].

Generative syntax models analyze relative clauses as instances of *wh*-movement of an overt or covert operator (a *wh*-phrase) from the thematic position inside the relative clause to Spec,CP of the relative clause (creating the gap). Relative clauses tend to have properties that parallel a language's other A'-properties. 500

resumptive pronoun A pronoun that shares a referent and original position with the syntactic position of a phrase that has been extracted or is left-dislocated. For instance, in the sentence *As for ice cream, it's not my favorite*, *it* is a pronoun that "resumes" the noun *ice cream.* 312

s-selection Refers to *semantic* selection. Lexical items often place semantic requirements of various sorts on their arguments, and s-selection refers to these kinds of requirements. For example, a predicate might require an animate subject, or a human subject. , 579

scope The level of syntactic structure that the interpretation of a given syntactic element applies to is said to be that element's scope. If a syntactic element B is in the scope of another element A, A is said to take scope over B. For example, in (47a), the negative *not* takes scope over the modal *can*, yielding the paraphrased interpretation in (47b). In (48a), the modal *can* takes scope over the negative *not*, yielding the paraphrased interpretation in (48b).

(47) a. The orangutan **can**not read the book.
b. Paraphrase: It is not the case that the orangutan **can** read the book.
(The orangutan is illiterate, or the book is out of reach.)

(48) a. The orangutan **can** NOT read the book.
b. Paraphrase: **It is possible** for the orangutan to not read the book.
(The orangutan is free to choose NOT to read the book.)

100, 513

selectional restrictions Lexical items place different kinds of requirements on the elements they combine with: we refer to these requirements as selectional restrictions. Selectional restrictions can be about syntactic requirements (c-selection) or semantic requirements (s-selection), but used on its own the term tends to refer to s-selection. 256

simple tense A tense that is expressed synthetically, for instance, the English simple past *she sang* or the French *imparfait elle chantait* 'she used to sing, she was singing'. Simple tenses contrast with compound tenses, which are formed analytically, like the English present progressive *she is singing* or the French *passé composé elle a chanté* 'she sang, she has sung'.

Simple tenses in one language can correspond functionally to compound tenses in another language, and vice versa. For instance, the English simple past corresponds to a French *passé composé*, whereas the English compound present progressive often corresponds to a French simple present. 425, 447, 565

small clause A clause containing predication without any inflection (in English, morphemes indicating tense and person information). See the examples below, in which the small clauses are italicized.

(49) He found *the turtleneck a tad too posh.*
(cf. *The turtleneck was a tad too posh.*)

(50) I watched *the snowflakes dance on the wind.*
(cf. *The snowflakes danced on the wind.*)

(51) I consider *them my closest friend.*
(cf. *They are my closest friend.*)

See Chapter 4, § 4.5.3 for more. 151

specifier A constituent that is sibling to an intermediate projection and child of a maximal projection. Also see complement, adjunct. 177

spellout rule Rules that specify how combinations of lexical, morphological, and syntactic features are realized morphologically (i.e., pronounced). 472

stative A form of a verb that changes its meaning from an action to a state. Often associated with perfects, from which they can develop (for instance in Ancient Greek, Middle Egyptian).

Example:

(52) a. ドアが 開く。 **Japanese**
Doa-ga ak-u.
door-NOM open-NPST
'The door opens.'

b. ドアが 開いている。
Doa-ga ai-tei-ru.
door-NOM open-STATIVE-NPST
'The door is open.'

137

structural case Structural case refers to case forms that are assigned to nominals in particular structural configurations, such as sentential subject or object of a transitive verb. 333

subcategorization Lexical items are said to be subcategorized according to the syntactic properties of their complement. For instance, verbs can be subcategorized according to whether their elementary tree has a complement or not, according to the syntactic category of the complement (CP, DP, IP, PP, and so on), according to other syntactic properties of the complement, (for instance, finiteness, case, and so on).

subject See Chapter 4, § 4.4.1 9, 564, 565, 577, 578, 580

subject movement Movement of a subject within a single clause (as opposed to subject raising). Usually, the term refers to the movement of a subject from Spec(VP) to Spec(IP). But the term can be used more generally to include movement from Spec(NP) to Spec(DP). , 575

subject raising , 580, *see* raising

subject requirement The requirement that subjects appear in sentences. This requirement appears to be purely syntactic (that is, independent of semantic considerations). 147

subject-aux inversion The process that forms a yes-no question from the corresponding declarative sentence. In declarative sentences containing a modal, an auxiliary, or main verb *be*, subject-aux inversion consists in switching the order of the subject and that element (highlighted by italics below).

(53) a. He *should* apply to both schools. Modal
b. *Should* he apply to both schools?

(54) a. The guy she met last night *is* coming along. Auxiliary *be*
b. *Is* the guy she met last night coming along?

(55) a. They *do* have to clean their room. Auxiliary *do*
b. *Do* they have to clean their room?

(56) a. The mail *has* come. Auxiliary *have*

b. *Has* the mail come?

(57) a. They *are* superbly qualified. Main verb *be*

b. *Are* they superbly qualified?

97, 301

syncretic When referring to case forms, this refers to an instance where a single case form corresponds to more than one case function (ex., the same case form being used for both dative and accusative functions). 336

syntactic category Words are classified into syntactic categories based on their syntactic distributions, that is, what positions they appear in in sentences. Examples include both categories that correspond to traditional "part of speech" categories (ex., nouns, verbs, adjectives, and determiners), as well as categories that have no counterpart (ex., inflection, complementizer). 21, 564, 569, 573, 575

syntactic island A configuration in which movement is blocked. 107

tense A linguistic category associated with temporal reference (what is the relation of the time of the event under discussion to the time of speaking?) as well as with aspect (is the speaker's focus on the event's inception, completion, duration, repetition, general truth, and so on?). , 569, *see also* finite

thematic role A specific kind of role in an event; specific thematic roles appear consistently across languages. Examples include agent, theme, and recipient. Within the generative syntactic framework these are often referred to as θ-roles (theta-roles), with specific properties associated with them. , 562–564, 568, 570, 572, 574, 576, 578, 581

theme A (catch-all) thematic role assigned to an argument that undergoes motion, a change of state, or is most affected by an event. For example, in the sentence *The architect built a new library*, the theme is *a new library*. 142, 581

theta role A thematic role assigned to particular arguments of a predicate, generally notated in small caps. Examples include AGENT, THEME, and EXPERIENCER. Also written θ-role. 281, 582

trace A trace is an annotation strategy to show movement in a syntactic analysis. The trace (t) marks the original position of an element, where an element moved from. A trace is often attached to a subscript to make clear what element moved from that trace (so t_k is coindexed with *who* in the example below).

(58) Who$_k$ are you going to visit t_k today?

In earlier versions of generative syntactic theory, traces were theoretical entities with their own properties and principles attached to them. In current versions of generative syntactic theory, traces are simply an annotation strategy for unpronounced copies of moved elements (see the Copy Theory of Movement). 274, 569

transitive Traditionally used of verbs that take a single object. We use the term in a more general sense to refer to any syntactic category that takes a single complement. 135, *see* *also* intransitive

unaccusative Unaccusative predicates lack an external (i.e. agentive) argument, generally being intransitive predicates with a THEME subject. 387

unergative Unergative predicates lack an internal (object) argument, generally being intransitive predicates with an AGENT subject. 387

Uniform Theta Assignment Hypothesis (UTAH) The hypothesis (due to Baker 1988) that Universal Grammar specifies that each theta role is assigned to a particular structural position. 281, 388

valency The number of syntactic arguments a verb combines with. See Chapter 4, § 4.1.3 for further discussion. 135, 562, 564

VP-Internal Subject Hypothesis The hypothesis within generative syntax that the subjects of sentences originate inside the verb phrase. Put another way, all arguments of verbs have an original position inside the verb phrase even if they move to another position during the course of building the structure of the sentence. 261, 582

VPISH See VP-Internal Subject Hypothesis. 261

***wh*-question** A question that asks for some kind of lexical information. So named because in English, they often (but not always) begin with words spelled with an initial *wh-* like *who*, *what*, *when*, *where*, *why*, or *which*. Like yes-no questions, they can less canonically be felicitously answered with expressions of uncertainty, probability, or inference. For example,

(59) a. Q: When did you last see Mom?

b. A: { Last year , Two days ago , 3 June 2003 , Not sure , # Yes , # No }.

(60) a. Q: How did you get to school?

b. A: { By bike , The bus , I walked , Can't recall , # Yes , # No }.

See Chapter 8, § 8.1.1 for more. 293

word The term 'word' has at least three distinct meanings.

1. An **orthographic word** is a particular sequence of written characters. *Bank* in the sense of 'river bank' and 'financial institution' are the same orthographic word, as are *read* as an infinitive (*to read a book*), as a past tense form (*I read the book*), and as a past participle (*I have read the book*).

   See also homograph, homophone.

2. A **word form** is a linguistic form of a particular grammatical type. For instance, *cut* can be an infinitive (*to cut*), a present tense form (*Whenever I stuff envelopes, I cut myself*), a past tense form (*Yesterday, I cut myself*), or a past participle (*I have cut myself again*). In this case, four distinct word forms are associated with a single orthographic word.

3. A **lexeme** is an abstract meaning unit that can subsume several different word forms. For instance, the lexeme *be* subsumes the eight word forms *am*, *are*, *be*, *been*, *being*, *is*, *was*, and *were*. The lexeme *cut* subsumes the three word forms *cut*, *cuts*, and *cutting*.

In lexical ambiguity, the same orthographic word is associated with more than one lexeme, as in the case of *bank* 'river bank' and *bank* 'financial institution'.

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yes-no question A question that can only be answered felicitously with "yes", "no", or less canonically some expression of uncertainty, probability, or inference. For example,

(61) a. Q: Did Dad make pancakes for breakfast?
b. A: i. { Yes , No , Maybe , I'm not sure , I don't know , Sure smells like it }.
ii. # { Waffles , Bananas , Mom , Lunch , Dinner }.

See Chapter 8, § 8.1.1 for more. 293, 580, 582

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